Chapter 3

Algebra

Follows largely Lang with some Bourbaki.

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3.1 Magmas and Monoids

Definition 3.1.1 (Magma)
*Let X be a set. A **law of composition** on $X \times X$ is a function*

$$\cdot : X \times X \to X$$

and we typically write the composition of $x, y \in X$ as either

$$x \cdot y$$

or xy, or $x + y$ in the commutative case.

*A pair $(X, \cdot)$ consisting of a set X and law of composition on X is called a **magma**.*

Definition 3.1.2 (Magma/Monoid)
*A **magma** $(X, \cdot)$ is said to be*

- ***associative** if $(x \cdot y) \cdot z = x \cdot (y \cdot z)$*
- ***commutative** if $x \cdot y = y \cdot x$ for all $x, y \in X$*
- ***unital** if there exists $e \in X$ such that $e \cdot x = x \cdot e = x$ for all $x \in X$. Such an e is called an **identity**.*
- *a **monoid** if it is both **associative** and **unital***

Proposition 3.1.3 (Identity is Unique)
A magma $(X, \cdot)$ has at most one element e such that

$$x \cdot e = e \cdot x = x$$

for all $x \in X$.

Definition 3.1.4 (Invertible / Monoid)
*Let $(X, \cdot)$ be a unital magma. An element $x \in X$ is **invertible** if there exists $y \in X$ such that*

$$x \cdot y = y \cdot x = e$$

Proposition 3.1.5 (Inverses are unique)
Let $(X, \cdot)$ be a monoid. If $x \in X$ is invertible then its inverse is unique and denoted x^{-1}.

Proof. Suppose that $xy = xy' = e = yx = y'x$. Then

$$xy = e \implies y'(xy) = y'e = y' \implies (y'x)y = y' \implies y = y'$$

■

Definition 3.1.6 (Homomorphism)
*Let $(X, \cdot)$, $(Y, \cdot)$ be magmas. Then a function $\varphi : X \to Y$ is said to be a **magma homomorphism** if it satisfies*

$$\varphi(x_1 \cdot x_2) = \varphi(x_1) \cdot \varphi(x_2) \quad \forall x_1, x_2 \in X$$

*If $(X, \cdot)$ and $(Y, \cdot)$ are unital then φ is **unital** if*

$$\varphi(e_X) = e_Y$$

*If $(X, \cdot)$ and $(Y, \cdot)$ are monoids then φ is a **monoid morphism** if it satisfies both these conditions.*

Proposition 3.1.7 ($\mathbb{N}$ is an initial object)
Let $(X, \cdot)$ be a monoid and $x \in X$. Then there is a unique monoid morphism

$$x^{(-)} : (\mathbb{N}, +) \to (X, \cdot)$$

such that

$$x^1 = x$$

Furthermore if $\varphi : (X, \cdot) \to (Y, \cdot)$ is a monoid morphism then

$$\varphi(x^n) = \varphi(x)^n$$

for all $x \in X$ and $x \in \mathbb{N}$.

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Proposition 3.1.8 ($\mathbb{Z}$ is an initial object)
Let $(X, \cdot)$ be a monoid and $x \in X$ be an invertible element. Then there is a unique monoid morphism

$$x^{(-)} : (\mathbb{Z}, +) \to (X, \cdot)$$

such that

$$x^1 = x$$

and

$$x^{-n} \text{ is the inverse of } x^n$$

Furthermore if $\varphi : (X, \cdot) \to (Y, \cdot)$ is monoid morphism then

$$\varphi(x^n) = \varphi(x)^n$$

for all $x \in X$ invertible and $n \in \mathbb{Z}$.

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3.2 Groups

Definition 3.2.1 (Group)
*A **group** is a **monoid** $(G, \cdot)$ in which every element is invertible.*

*A group G is said to be **abelian** if the binary operation is **commutative**. In this case we typically write the group operation additively*

$$g + h$$

Definition 3.2.2 (Subgroup, Normal Subgroup)
*A **subgroup** $H \leq G$ is a subset with the following properties*

- $e_G \in H$
- $x, y \in H \implies xy \in H$
- $x \in H \implies x^{-1} \in H$

*A subgroup H is said to be **normal** in G if in addition it satisfies*

$$gHg^{-1} \coloneqq \{ghg^{-1} \mid g \in G\} = H$$

for all $g \in G$. NB it is easily verified that in an abelian group every subgroup is normal.

Proposition 3.2.3 (Subgroup is a group)
Let H be a subgroup of $(G, \cdot)$ then $(H, \cdot|_{H \times H})$ is a group.

Example 3.2.4
$\mathbb{Z}$ is an abelian group under addition. The subgroups are of the form $n\mathbb{Z}$.

Definition 3.2.5
*Let $(G, \cdot)$ and $(H, \cdot)$ be groups. A function $\varphi : G \to H$ is a **group homomorphism** if*

- $\varphi(e_G) = e_H$
- $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$

*Define the **image** of φ to be*

$$\mathrm{Im}(\varphi) = \{\varphi(g) \mid g \in G\}$$

*and the **kernel** to be*

$$\ker(\varphi) \coloneqq \{g \in G \mid \varphi(g) = e_H\}$$

It may be verified that $\mathrm{Im}(\varphi)$ is a subgroup of H and $\ker(\varphi)$ is a normal subgroup of G.

Proposition 3.2.6 (Raise to the n-th power)
Let $g \in G$ be a group element. Then there exist a unique group homomorphism

$$g^{(-)} : (\mathbb{Z}, +) \to (G, \cdot)$$

satisfying

$$g^1 = g$$

In other words such that

$$g^0 = e_G$$

$$g^{n+m} = g^n \cdot g^m \quad \forall n, m \in \mathbb{Z}$$

Proposition 3.2.7
Let $g \in G$ be a group element. Then

$$(g^n)^m = g^{nm}$$

for all integers $n, m \in \mathbb{Z}$.

Definition 3.2.8 (Order of an element)
*For $g \in G$ define the **order** of g to be $o(g) \coloneqq \inf\{n \geq 0 \mid g^n = e\}$ where $\inf \emptyset = \infty$.*

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*We say g has **finite order** if $o(g) \neq \infty$.*

Definition 3.2.9 (Subgroup generated by an element)
*The **subgroup generated** by an element $g \in G$ is defined to be $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$*

It may be shown that when g has finite order n we have

$$\langle g \rangle = \{e, g, ..., g^{n-1}\}$$

and in particular $\#\langle g \rangle = o(g)$.

Proposition 3.2.10 (Cosets)
Let H be a subgroup of G. The following is an equivalence relation on G

$$g_1 \underset{H}{\sim} g_2 \iff g_1 g_2^{-1} \in H$$

and the equivalence classes are precisely the sets of the form

$$gH = \{gh \mid h \in H\} = [g]_{\underset{H}{\sim}}$$

*for some $g \in G$. Such an equivalence class is called a **right coset** and we denote the set of right cosets by*

$$G/H$$

*Define the **index** of H in G by $[G : H] := \#G/H$. When H is finite each equivalence class has order $\#H$.*

*We say $\{g_i \in G\}_{i \in I}$ is a set of **coset representatives** for H if the corresponding equivalence classes $\{[g_i]\}_{i \in I}$ are pairwise disjoint and cover G.*

Proof. It's trivial to show that $\underset{H}{\sim}$ is an equivalence relation (precisely because H is a subgroup). Therefore by (2.1.6) the equivalence classes form a partition which we denote G/H.

We claim that $[g_1] = g_1 H$. Then $g_2 \in [g_1] \iff g_1 \underset{H}{\sim} g_2 \iff g_2 \underset{H}{\sim} g_1 \iff g_2 g_1^{-1} \in H \iff g_2 \in g_1 H$, which shows that the sets are equal.

The translation map $\psi_g : G \to G$ given by $g' \to gg'$ is bijective (for it has a two-sided inverse equal to $\psi_{g^{-1}}$). So in particular restricts to a bijective map $H \to gH$. This shows that all the cosets have the same order. ∎

Example 3.2.11
$d\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ of index d. A set of coset representatives are $\{0, 1, ..., d-1\}$.

Corollary 3.2.12 (Lagrange's Theorem)
Let $H \leq G$ be a subgroup then

$$\#G = [G : H] \times \#H$$

More generally if $K \leq H$ then

$$[G : K] = [G : H][H : K]$$

Example 3.2.13
$d\mathbb{Z} \subseteq e\mathbb{Z} \iff d \mid e$ and $[e\mathbb{Z} : d\mathbb{Z}] = e/d$.

Proposition 3.2.14
Let $g \in G$ be an element of finite order. Then

$$o(g) \mid \#G$$

Furthermore

$$g^n = e \iff o(g) \mid n$$

Proof. The first statement follows because the order $o(g)$ equals the order of the subgroup $\langle g \rangle$ generated by g.

Let $m = o(g)$ then by the division algorithm $n = qm + r$ for some $r < m$. Then $e = g^n = g^{qm} g^r = (g^m)^q g^r = e^q g^r = g^r$. By minimality we have $r = 0$ and $m \mid n$ as required. ∎

Proposition 3.2.15 (Quotient Group)
Let N be a normal subgroup G. Then the set of cosets

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$$G/N$$

forms a group under the binary operation

$$g_1N \cdot g_2N \to (g_1g_2)N$$

with identity eN.

a) *There is a canonical surjective group homomorphism*

$$\pi : G \to G/N$$

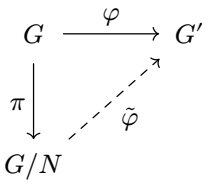
$$g \to gN$$

with kernel N.

b) *Let $N \subseteq H$ be a subgroup then define the correseponding subgroup of G/N*

$$H/N \coloneqq \pi(H) = \{hN \mid h \in H\}.$$

c) *Let $\varphi : G \to G'$ be a homomorphism with $N \subseteq \ker(\varphi)$, then there exists a unique homomorphism $\tilde{\varphi}$ making the diagram commute*



such that

a) $\mathrm{Im}(\varphi) = \mathrm{Im}(\tilde{\varphi})$
b) $\ker(\tilde{\varphi}) = \ker(\varphi)/N$

Corollary 3.2.16 (Isomorphism Theorem)
Let $\varphi : G \to H$ be a group homomorphism, then there is a canonical isomorphism

$$G/\ker(\varphi) \xrightarrow{\sim} \mathrm{Im}(\varphi)$$

Proposition 3.2.17 (Correspondence Theorem)
Let $\pi : G \to G'$ be a surjective homomorphism with $\ker(\varphi) = N$ then there is a bijective correspondence of subgroups

$$\begin{aligned} \{H \leq G \mid N \subseteq H\} &\longleftrightarrow \{H' \leq G'\} \\ H &\to \pi(H) \\ \pi^{-1}(H') &\leftarrow H' \end{aligned}$$

which preserves index, that is

$$[G' : H'] = [G : H]$$

Furthermore $\#H' = [H : N]$.

Proposition 3.2.18
Suppose $N \subset G$ and $K \subset N$ are normal subgroups. Then $K \subset G$ is normal and there is a canonical isomorphism

$$(G/N)/(K/N) \xrightarrow{\sim} G/K$$

given by

$$\overline{g+N} \to g + K$$

3.2.1 Cyclic Groups

Definition 3.2.19
*A group G is said to be **cyclic** if there is a surjective group homomorphism*

$$(\mathbb{Z}, +) \longrightarrow (G, \cdot)$$

*equivalently if there is $g \in G$ such that $\langle g \rangle = G$. Such a g is called a **generator** for G.*

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Proposition 3.2.20
Consider the additive group $(\mathbb{Z}, +)$. Then

a) *Every subgroup is of the form $d\mathbb{Z}$ for $d \geq 0$ and is itself cyclic*
b) *When $d > 0$, then $Z/d\mathbb{Z}$ has a complete set of coset representatives*

$$S := \{0, 1, ..., d-1\}$$

c) *In particular $[Z : d\mathbb{Z}] = d$ when $d > 0$*
d) *$d\mathbb{Z} \subseteq e\mathbb{Z} \iff e \mid d$ and in this case $[e\mathbb{Z} : d\mathbb{Z}] = \frac{d}{e}$*

Proof. We prove each in turn

a) By (2.2.6) every subgroup is of the form $d\mathbb{Z}$. Multiplication map $[d] : \mathbb{Z} \to d\mathbb{Z}$ shows it is itself cyclic.
b) By the division algorithm (2.2.5) S is a complete set. Given $i, j \in S$ we note that $|i - j| < d$. And $i \underset{d}{\sim} j \implies d \mid |i - j| \implies |i - j| = 0 \implies i = j$. Therefore the set S consists of distinct coset representatives.
c) This is clear from the previous step
d) The first equivalence is clear. By (3.2.12)

$$[\mathbb{Z} : d\mathbb{Z}] = [\mathbb{Z} : e\mathbb{Z}][e\mathbb{Z} : d\mathbb{Z}]$$

and the result follows.

■

Proposition 3.2.21
Let G be a cyclic group. Then

- *If G is infinite it is isomorphic to $\mathbb{Z}$*
- *If G is finite it is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ for some $n > 0$*

Proof. By the previous Proposition the kernel of the homomorphism $\mathbb{Z} \to G$ is of the form $n\mathbb{Z}$ for $n \geq 0$. By (...) G is isomorphic to $\mathbb{Z}/n\mathbb{Z}$. When $n = 0$ this is canonically isomorphic to $\mathbb{Z}$.

By the previous Proposition $\mathbb{Z}/n\mathbb{Z}$ is finite for $n > 0$ and therefore if G is not finite we must have $n = 0$. Similarly If G is finite then we must have $n > 0$. ■

We analyse the structure of finite cyclic groups in more detail. First recall the definition of **Euler's Totient Function**

Definition 3.2.22 (Euler Totient Function)
Define the function

$$\phi(n) = \#\{0 < d \leq n \mid (d, n) = 1\}$$

Proposition 3.2.23 (Finite Cyclic Groups)
Consider a finite cyclic group G of order n. Then

a) *The order of g^r is $\frac{n}{n,r}$ where $0 < r \leq n$.*
b) *There are $\phi(n)$ generators*
c) *For every $d \mid n$ there is a unique subgroup of order $\frac{n}{d}$ given by $\langle g^d \rangle$, which is cyclic.*
d) *For $d \mid n$ there are precisely $\phi(d)$ elements of order d*
e) *There are precisely d elements of order dividing d*

Proof. We prove each in turn

a) $(g^r)^s = e_G \iff g^{rs} = e_G \overset{(3.2.14)}{\iff} n \mid rs \overset{(2.2.13)}{\iff} \frac{n}{n,r} \mid s$. Therefore g^r has order $\frac{n}{n,r}$ as required.
b) Note h is a generator iff $o(h) = n$. So g^r is a generator iff $(n, r) = 1$ by the previous step. As $G = \{g, g^2, ..., g^n\}$ the result follows by definition of the totient function.
c) Recall there is a canonical surjective morphism $\pi : \mathbb{Z} \to G$ with kernel $n\mathbb{Z}$ and $\pi(1) = g$. By (3.2.17) the subgroups H of G correspond bijectively to subgroups H' of $\mathbb{Z}$ containing $n\mathbb{Z}$, preserving the index. By (3.2.20) these are of the form $H' = d\mathbb{Z}$ for $d \mid n$, which correspond under π to subgroups $H = \langle g^d \rangle$. Further $[G : \langle g^d \rangle] = [\mathbb{Z} : d\mathbb{Z}] = d$ whence $\#\langle g^d \rangle = \frac{n}{d}$. By definition $\langle g^d \rangle$ is cyclic.
d) Let $G[d]$ be the unique (cyclic) subgroup of order d. If h has order d then $\langle h \rangle$ has order d, and therefore by uniqueness is equal to $G[d]$. In particular $h \in G[d]$. Therefore by the previous part there are $\phi(d)$ elements of order d

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e) Suppose h has order $e \mid d$. Both G and $G[d]$ contain a unique subgroup of order e and therefore by uniqueness this is simply $G[e] \subseteq G[d]$. Similarly by uniqueness $G[e] = \langle h \rangle$. Therefore $h \in G[d]$. Conversely suppose $h \in G[d]$ then $o(g) \mid d$ by (3.2.14). Therefore $G[d]$ consists of all the elements of order dividing d.

■

Corollary 3.2.24
Let n be a positive integer then

$$n = \sum_{d|n} \phi(d)$$

Proof. Consider a cyclic group G of order n. Every element has order dividing n so the result follows from the previous Proposition by partitioning the group G into subsets consisting of elements of equal order. ■

For an abelian group G define the following subgroup

$$G[d] := \{g \in G \mid g^d = e\}.$$

We have shown for a cyclic group that $\#G[d] = d$ whenever $d \mid n$ and it is empty otherwise. We claim that this can be used to characterize cyclic groups. NB the following is adapted from this stackexchange answer

Proposition 3.2.25 (Characterization of cyclic group)
Let G be a finite abelian group such that $\#G[d] \leq d$ for all $d \mid n$. Then G is cyclic.

Proof. Let $n = \#G$ and G_d be the subset of elements of order exactly d. Then we wish to show that G_n is non-empty as any element of this set will be a generator. We actually show that $\#G_d = \phi(d) > 0$ whenever $d \mid n$.

Note that $G_d \subseteq G[d]$. If it's non-empty then for any $y \in G_d$, we have $\langle y \rangle$ is a subgroup of $G[d]$ of order d. As $\#G[d] \leq d$ we have $G[d] = \langle y \rangle$ is cyclic of order d. In other words G_d is equal to the set of generators for $G[d]$. By the previous Proposition $G[d]$ has $\phi(d)$ generators. We conclude that for all $d \mid n$ we have G_d is either empty or of order $\phi(d)$.

Therefore

$$n = \sum_{d|n} \#G_d \leq \sum_{d|n} \phi(d) = n$$

Therefore we must have equality everywhere and $\#G_d = \phi(d)$ as required. ■

Example 3.2.26
Let $G = \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ of order p^2. Then $G[p] = G$ so $\#G[p] = p^2 > p$.

3.2.2 Group Actions

Definition 3.2.27 (Group Action)
Let G be a group and S a set. A group action of G on S is a map

$$G \times S \longrightarrow S$$

$$(g, s) \longrightarrow g \cdot s$$

such that

- $es = s$
- $g(hs) = (gh)s$

Definition 3.2.28 (Faithful group action)
A group action G on S is faithful if

$$gs = s \quad \forall s \in S \Longrightarrow g = e$$

Definition 3.2.29 (Free group action)
A group action G on S is free if

$$g \neq e \Longrightarrow gs \neq s \quad \forall s$$

Definition 3.2.30 (Orbit/Stabilizer)
Let G be a group with an action on S and $s \in S$. Define the stabilizer subgroup

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$$G_s := \{g \in G \mid gs = s\}$$

and the orbit

$$Gs := \{gs \mid g \in G\}$$

Proposition 3.2.31 (S is disjoint union of orbits)
Let G be a group with an action on S. Then the following is an equivalence relation

$$s \sim t \iff gs = t \text{ some } g \in G$$

and the equivalence classes are precisely the orbits of elements of S under G. Further S is the disjoint union of orbits.

Remark 3.2.32
An action is free if and only if $G_s = \{e\}$ for all $s \in S$.

Proposition 3.2.33 (Orbit-Stabilizer Theorem)
Let G be a group with an action on S. Given an element $s \in S$ there is a natural bijection

$$G/G_s \longrightarrow Gs$$

between the cosets of G_s and the orbit Gs. In particular when G is finite

$$\#G = \#(Gs) \times \#G_s$$

and when the action is free

$$\#G = \#(Gs)$$

3.2.3 Symmetric Group

Definition 3.2.34 (Symmetric Group)
Let S_n denote the set of permutations (bijections) of $\{1, ..., n\}$.

*Permutations $\sigma, \tau \in S_n$ are called **disjoint** if the supports are disjoint. Note disjoint permutations commute.*

Definition 3.2.35 (Cycle)
Let $i_1, ..., i_r \in J_n$ be an ordered r-tuple, the permutation which maps

$$i_k \to \begin{cases} i_{k+1} & k < r \\ i_1 & k = r \end{cases}$$

*is denoted by $(i_1\, i_2 \ldots i_r)$ and called a **cycle**.*

*A cycle with two elements $(i\, j)$ is called a **transposition**. Finally an **adjacent transposition** is one of the form $(i\, i+1)$.*

Proposition 3.2.36
Let $\sigma \in S_n$. Then σ may be represented by

a) a product of disjoint cycles, which is unique up to permutation of cycles.
b) a product of transpositions, the number of which is unique modulo 2
c) There is a well-defined group homomorphism

$$\varepsilon : S_n \to \{-1, 1\}$$

such that

$$\varepsilon((i\, j)) = -1$$

and $\varepsilon(1) = 1$.
d) A cycle σ of length r satisfies $\varepsilon(\sigma) = (-1)^{r+1}$

Proposition 3.2.37 (Adjacent Transpositions Generate the Symmetric Group)
Every $\sigma \in S_n$ is a product of adjacent transpositions $(i\, i+1)$, $1 \le i < n$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Every permutation is a product of transpositions, so it suffices to write each transposition $(i\, j)$ with $i < j$ as a product of adjacent ones, which we do by induction on $j - i$. If $j - i = 1$ there is nothing to show. If $j - i > 1$ then

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$$(i\,j) = (j-1\,j)(i\,j-1)(j-1\,j)$$

since conjugating $(i\,j-1)$ by $(j-1\,j)$ replaces $j-1$ with j, and $(i\,j-1)$ is a product of adjacent transpositions by induction. ∎

3.2.4 Shuffle Permutations

Definition 3.2.38
Let $p+q=n$ be positive integers. We define the subgroup of ***shuffle permutations***

$$\mathrm{Sh}(p,q) \coloneqq \{\sigma \in S_n \mid \sigma(1) < ... < \sigma(p) \text{ and } \sigma(p+1) < ... < \sigma(p+q)\}$$

Similarly if $p+q+r=n$ are positive integers define the subgroup

$$\mathrm{Sh}(p,q,r) \coloneqq \{\sigma \in S_n \mid \sigma(1) < ... < \sigma(p) \text{ and } \sigma(p+1) < ... < \sigma(p+q) \text{ and } \sigma(p+q+1) < ... < \sigma(p+q+r)\}$$

Let $u : [1,p] \to [1,n]$ and $v : [1,q] \to [1,n]$ be injective maps such that $u([1,p]) \cap v([1,q]) = \emptyset$. Define $u \star v \in S_n$ by

$$(u \star v)(i) \coloneqq \begin{cases} u(i) & 1 \le i \le p \\ v(i-p) & p < i \le p+q \end{cases}$$

If u, v are order preserving then $u \star v \in \mathrm{Sh}(p,q)$. Every shuffle permutation is of this form.

Lemma 3.2.39
Let $X, Y \subset \mathbb{N}$ be a finite sets of order n. There exists a unique order isomorphism

$$u : X \to Y$$

Proposition 3.2.40
Let $p+q=n$ be positive integers. There is a bijection

$$\mathrm{Sh}(p,q) \to \{u : [1,p] \to [1,n] \text{ order preserinvg}\}$$

$$\sigma \to \sigma|_{[1,p]}$$

Proof. We provide an explicit inverse. For given u the set $Y \coloneqq [1,n] \setminus u([1,p])$. As u is order preserving it is injective, whence $\#Y = n - \#u([1,p]) = n-p = q$. Therefore there is an order isomorphism $v : [1,q] \to Y$ and we may define the shuffle permutation $\sigma \coloneqq u \star v$. ∎

Proposition 3.2.41
Let $p+q=n$ be positive integers. There is an injective group homomorphism

$$\begin{aligned} S_p \times S_q &\to S_n \\ (\sigma_1, \sigma_2) &\to \sigma_1 \star (\sigma_2 + p) \end{aligned}$$

The image is the set $\sigma \in S_n$ for which $\sigma([1,p]) = [1,p]$ (equivalently $\sigma([p+1,q]) = \sigma([p+1,q])$.

Proposition 3.2.42
Let $p+q=n$ be positive integers. There is a bijection

$$\mathrm{Sh}(p,q) \to S_n / (S_p \times S_q)$$

Furthermore $\#\,\mathrm{Sh}(p,q) = \frac{(p+q)!}{p!q!}$. By a similar argument $\#\,\mathrm{Sh}(p,q,r) = \frac{(p+q+r)!}{p!q!r!}$.

Proof. Let $\sigma \in S_n$. Then there exists an order isomorphism $u : [1,p] \to \sigma([1,p])$ and $v : [1,q] \to \sigma([p+1,p+q])$. Define $\sigma_1(i) \coloneqq u^{-1}(\sigma(i))$ and $\sigma_2(j) \coloneqq v^{-1}(\sigma(j+p))$. Then we may verify that $\sigma_1 \in S_p$, $\sigma_2 \in S_q$ and

$$(u \star v) \circ (\sigma_1 \star (\sigma_2 + p)) = \sigma$$

so the map is surjective.

Suppose $\sigma, \sigma' \in \mathrm{Sh}(p,q)$ are such that

$$\sigma = \sigma' \circ (\sigma_1 \star (\sigma_2 + p))$$

for $\sigma_1 \in S_p$ and $\sigma_2 \in S_q$. If $\sigma_1 \neq e$ then there exists $1 \le i < j \le p$ such that $\sigma_1(j) < \sigma_1(i)$. Then by monotonicity of σ' we have $\sigma(j) = \sigma'(\sigma_1(j)) < \sigma'(\sigma_1(i)) = \sigma(i)$, which contradicts monotonicity of σ. So we conclude $\sigma_1 = e$ and similarly $\sigma_2 = e$. Therefore $\sigma = \sigma'$, and the map is injective. By the Orbit-Stabilizer theorem we have

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$$\#\operatorname{Sh}(p,q) = \#S_n / \#\big(S_p \times S_q\big)$$

from which the result follows. ∎

Definition 3.2.43 (Circulant Permutation)
Let n be an integer, and write $\rho_n \in S_n$ for the permutation

$$\{1,...,n\} \to \{2,3,...,n,n+1\}$$

Note that $\varepsilon(\rho_n) = (-1)^{n+1}$.

Proposition 3.2.44 (Symmetry of Shuffling)
Let $p+q=n$ be positive integers. Then there is a bijection

$$\begin{aligned} \operatorname{Sh}(p,q) &\to \operatorname{Sh}(q,p) \\ \sigma &\to \sigma \circ \rho^p_{p+q} \end{aligned}$$

Note that $\varepsilon\big(\rho^p_{p+q}\big) = (-1)^{pq}$ by (3.2.43).

Proposition 3.2.45 (Associativity of Shuffling)
The following map is a bijection

$$\begin{aligned} \pi : \operatorname{Sh}(p+q,r) \times \operatorname{Sh}(p,q) &\to \operatorname{Sh}(p,q,r) \\ (\sigma_1,\sigma_2) &\to \sigma_1 \circ (\sigma_2 \star (1_r + p + q)) \end{aligned}$$

Similarly so is

$$\begin{aligned} \pi' : \operatorname{Sh}(p,q+r) \times \operatorname{Sh}(q,r) &\to \operatorname{Sh}(p,q,r) \\ (\sigma_1,\sigma_2) &\to \sigma_1 \circ \big(1_p \star (\sigma_2 + p)\big) \end{aligned}$$

Proof. It is straightforward to verify that $\pi(\sigma_1,\sigma_2)$ is injective and satisfies the ordering properties. Therefore π is well-defined and we claim it is surjective.

Given $\sigma \in \operatorname{Sh}(p,q,r)$ let $u : [1,p+q] \to \sigma([1,p+q])$, $v : [1,r] \to \sigma([p+q+1,p+q+r])$ be the unique order isomorphisms. Define $\sigma_1 = u \star v$ and $\sigma_2 := u^{-1} \circ \sigma|_{[1,p+q]}$. We may verify that $\sigma_1 \in \operatorname{Sh}(p+q,r)$ and $\sigma_2 \in \operatorname{Sh}(p,q)$, and furthermore that $\pi(\sigma_1,\sigma_2) = \sigma$. By counting π is injective.

For the second case let $u : [1,p] \to \sigma([1,p])$, $v : [1,q+r] \to \sigma([p+1,p+q+r])$ be the unique order isomorphisms. Define $\sigma_1 = u \star v$ and $\sigma_2(j) := v^{-1}(\sigma(j+p))$. The verification follows as before. ∎

3.2.5 Totally Ordered Abelian Group

Definition 3.2.46 (Ordered Abelian Group)
*An abelian group $(G,+)$ together with a total order $\leq$ is an **ordered abelian group** if it satisfies*

$$x \leq y \implies x + z \leq y + z \quad \forall x,y,z \in G$$

Define $G^+ := \{g \in G \mid (0 \leq g) \wedge (g \neq 0)\}$ and $G^- := \{g \in G \mid (g \leq 0) \wedge (g \neq 0)\}$.

3.2.6 Sums in an Abelian Group

Proposition 3.2.47 (Finite Sums in an Abelian Group)
Let G be an abelian group. Define $\psi_0 := 0 \in G$ (the value on the one-point set G^0) and, for $n \geq 1$, maps $\psi_n : G^n \to G$ recursively by

$$\psi_n(x_1,...,x_n) = x_1 + \psi_{n-1}(x_2,...,x_n)$$

so that in particular $\psi_1 = 1_G$. Then each ψ_n is invariant under permutations, that is $\psi_n\big(x_{\sigma(1)},...,x_{\sigma(n)}\big) = \psi_n(x_1,...,x_n)$ for every $\sigma \in S_n$. Moreover:

a) $\psi_{m+r}(x_1,...,x_m,y_1,...,y_r) = \psi_m(x_1,...,x_m) + \psi_r(y_1,...,y_r)$ for $m,r \geq 0$
b) $\psi_n(0,...,0) = 0$ and $\psi_{n+1}(x_1,...,x_n,0) = \psi_n(x_1,...,x_n)$ for $n \geq 0$
c) $\psi_n(x_1+y_1,...,x_n+y_n) = \psi_n(x_1,...,x_n) + \psi_n(y_1,...,y_n)$ for $n \geq 0$
d) if $\varphi : G \to H$ is a group homomorphism to an abelian group H then $\psi_n(\varphi(x_1),...,\varphi(x_n)) = \varphi(\psi_n(x_1,...,x_n))$
e) $\psi_n(-x_1,...,-x_n) = -\psi_n(x_1,...,x_n)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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We show that each ψ_n is invariant under permutations by induction on n; for $n \le 1$ there is nothing to show. Since S_n is generated by the adjacent transpositions (3.2.37) it suffices to check invariance under $(i\,i+1)$ for $1 \le i < n$, and let $n \ge 2$ with ψ_{n-1} invariant.

If $i \ge 2$ the transposition only exchanges two entries of $(x_2, ..., x_n)$, so ψ_n is unchanged since ψ_{n-1} is invariant. If $i = 1$ then

$$\begin{aligned}\psi_n(x_2, x_1, x_3, ..., x_n) &= x_2 + (x_1 + \psi_{n-2}(x_3, ..., x_n)) && \text{by the recursion}\\ &= x_1 + (x_2 + \psi_{n-2}(x_3, ..., x_n)) && \text{by associativity and commutativity in } G\\ &= \psi_n(x_1, x_2, x_3, ..., x_n) && \text{by the recursion}\end{aligned}$$

For a) we use induction on m. For $m = 0$ both sides are $\psi_r(y_1, ..., y_r)$, as $\psi_0 = 0$. For $m \ge 1$, by the recursion, the induction hypothesis and associativity,

$$\psi_{m+r}(x_1, ..., x_m, y_1, ..., y_r) = x_1 + (\psi_{m-1}(x_2, ..., x_m) + \psi_r(y_1, ..., y_r)) = \psi_m(x_1, ..., x_m) + \psi_r(y_1, ..., y_r)$$

Next, $\psi_n(0, ..., 0) = 0 + \psi_{n-1}(0, ..., 0) = 0$ by induction on n, and by a) with $r = 1$ (so $\psi_1(0) = 0$) $\psi_{n+1}(x_1, ..., x_n, 0) = \psi_n(x_1, ..., x_n) + 0$, which proves b). The remaining properties are proved by induction on n, the case $n = 0$ being clear ($\psi_0 = 0$, and $\varphi(0) = 0$ for d)). For c), writing $x' = (x_2, ..., x_n)$ and $y' = (y_2, ..., y_n)$,

$$\psi_n(x + y) = (x_1 + y_1) + \psi_{n-1}(x' + y') = (x_1 + \psi_{n-1}(x')) + (y_1 + \psi_{n-1}(y')) = \psi_n(x) + \psi_n(y)$$

using the induction hypothesis and commutativity. For d), $\psi_n(\varphi(x)) = \varphi(x_1) + \psi_{n-1}(\varphi(x')) = \varphi(x_1) + \varphi(\psi_{n-1}(x')) = \varphi(\psi_n(x))$ since φ is additive. Finally e) is d) for the homomorphism $x \mapsto -x$ of G. ∎

Definition 3.2.48 (Sum of a Finite Family)
Let G be an abelian group, written additively. For a family $(g_i)_{i \in I}$ in G indexed by a finite set I with $n \ge 0$ elements, define

$$\sum_{i \in I} g_i := \psi_n\left(g_{e(1)}, ..., g_{e(n)}\right)$$

for any enumeration of I, that is a bijection $e : \{1, ..., n\} \to I$, where ψ_n is as in (3.2.47). In particular the sum over the empty set is $\psi_0 = 0$. If $g_i = 0$ for all $i \in I$ then $\sum_{i \in I} g_i = 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

If $e, e' : \{1, ..., n\} \to I$ are two enumerations of I then $\sigma := e^{-1} \circ e' \in S_n$ and $g_{e'(k)} = g_{e(\sigma(k))}$, so the sum is independent of the enumeration since ψ_n is invariant under permutations by (3.2.47). If $g_i = 0$ for all $i \in I$ then every entry of $\psi_n\left(g_{e(1)}, ..., g_{e(n)}\right)$ is 0, so the sum is 0 since $\psi_n(0, ..., 0) = 0$ by the same proposition. ∎

Proposition 3.2.49 (Sum over a Disjoint Union)
Let G be an abelian group and $(g_i)_{i \in I \cup J}$ a family in G indexed by the union of two disjoint finite sets I and J. Then

$$\sum_{i \in I \cup J} g_i = \sum_{i \in I} g_i + \sum_{i \in J} g_i$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $|I| = m \ge 0$ and $|J| = r \ge 0$, and let $e_I : \{1, ..., m\} \to I$ and $e_J : \{1, ..., r\} \to J$ be enumerations. Since I and J are disjoint, $e : \{1, ..., m + r\} \to I \cup J$ with $e(k) = e_I(k)$ for $k \le m$ and $e(k) = e_J(k - m)$ for $k > m$ is an enumeration of $I \cup J$, and by (3.2.47) (the identity for ψ_{m+r}) and independence of the enumeration (3.2.48) $\sum_{i \in I \cup J} g_i = \psi_{m+r}\left(g_{e(1)}, ..., g_{e(m+r)}\right) = \psi_m\left(g_{e_I(1)}, ..., g_{e_I(m)}\right) + \psi_r\left(g_{e_J(1)}, ..., g_{e_J(r)}\right) = \sum_{i \in I} g_i + \sum_{i \in J} g_i$. ∎

Definition 3.2.50 (Sum of an Almost Zero Family)
Let G be an abelian group, written additively. Let $(g_i)_{i \in I}$ be a family in G indexed by an arbitrary set I for which all but finitely many g_i are zero, that is whose ***support*** $\operatorname{supp}(g) := \{i \in I \mid g_i \neq 0\}$ *is finite. Define*

$$\sum_{i \in I} g_i := \sum_{i \in \operatorname{supp}(g)} g_i$$

using the sum over the finite set $\operatorname{supp}(g)$ from (3.2.48). Then $\sum_{i \in I} g_i = \sum_{i \in J} g_i$ for every finite subset $J \subseteq I$ containing $\operatorname{supp}(g)$. In particular, if I is finite this is the sum of the finite family $(g_i)_{i \in I}$ from (3.2.48) (take $J = I$).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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Let $J \subseteq I$ be finite with $\operatorname{supp}(g) \subseteq J$. Then J is the disjoint union of $\operatorname{supp}(g)$ and $J \setminus \operatorname{supp}(g)$, and $g_i = 0$ for every $i \in J \setminus \operatorname{supp}(g)$, so by (3.2.49) and (3.2.48) (a sum of zero terms is 0)

$$\sum_{i \in J} g_i = \sum_{i \in \operatorname{supp}(g)} g_i + \sum_{i \in J \setminus \operatorname{supp}(g)} g_i = \sum_{i \in \operatorname{supp}(g)} g_i$$

which is the defining value. If I is finite we may take $J = I$. ∎

Proposition 3.2.51 (Sums of Almost Zero Families)
Let G be an abelian group and $(g_i)_{i \in I}$, $(h_i)_{i \in I}$ families in G indexed by the same set I, with all but finitely many terms zero. Then $(g_i + h_i)_{i \in I}$ and $(-g_i)_{i \in I}$ are again families with all but finitely many terms zero, and

a) $\sum_{i \in I}(g_i + h_i) = \sum_{i \in I} g_i + \sum_{i \in I} h_i$
b) $\sum_{i \in I}(-g_i) = -\sum_{i \in I} g_i$, and consequently $\sum_{i \in I}(g_i - h_i) = \sum_{i \in I} g_i - \sum_{i \in I} h_i$
c) if $\varphi : G \to H$ is a group homomorphism to an abelian group H (for instance multiplication by an element r of a ring R on the additive group $G = R$) then $(\varphi(g_i))_{i \in I}$ again has all but finitely many terms zero, and $\sum_{i \in I} \varphi(g_i) = \varphi\left(\sum_{i \in I} g_i\right)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $J \subseteq I$ be a finite subset containing every i with $g_i \neq 0$ or $h_i \neq 0$. For $i \notin J$ we have $g_i + h_i = 0$, $-g_i = 0$ and $\varphi(g_i) = \varphi(0) = 0$, so all three families have all but finitely many terms zero, and J is a valid choice of finite subset in the definition of all of the sums involved. Therefore

$$\sum_{i \in I}(g_i + h_i) = \sum_{i \in J}(g_i + h_i) = \sum_{i \in J} g_i + \sum_{i \in J} h_i = \sum_{i \in I} g_i + \sum_{i \in I} h_i$$

$$\sum_{i \in I}(-g_i) = \sum_{i \in J}(-g_i) = -\sum_{i \in J} g_i = -\sum_{i \in I} g_i$$

$$\sum_{i \in I} \varphi(g_i) = \sum_{i \in J} \varphi(g_i) = \varphi\left(\sum_{i \in J} g_i\right) = \varphi\left(\sum_{i \in I} g_i\right)$$

using that finite sums are additive and commute with negation and with group homomorphisms (3.2.47). Finally $g_i - h_i = g_i + (-h_i)$, so the last claim of b) follows from a) and the first part of b). ∎

Proposition 3.2.52 (Sums over a Partition)
Let G be an abelian group and $(g_i)_{i \in I}$ a family in G with all but finitely many terms zero. If $I = \bigsqcup_{\lambda \in \Lambda} I_\lambda$ is a partition of I into disjoint subsets then each $(g_i)_{i \in I_\lambda}$ and the family $\left(\sum_{i \in I_\lambda} g_i\right)_{\lambda \in \Lambda}$ again have all but finitely many terms zero, and

$$\sum_{i \in I} g_i = \sum_{\lambda \in \Lambda} \sum_{i \in I_\lambda} g_i$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $J \subseteq I$ be finite containing every i with $g_i \neq 0$, and let $\Lambda_J := \{\lambda \in \Lambda \mid I_\lambda \cap J \neq \emptyset\}$, which is finite. For $\lambda \in \Lambda_J$ the set $J \cap I_\lambda$ is a valid choice for the subfamily, so $\sum_{i \in I_\lambda} g_i = \sum_{i \in J \cap I_\lambda} g_i$, while for $\lambda \notin \Lambda_J$ all terms of the subfamily are zero and $\sum_{i \in I_\lambda} g_i = 0$. So $\left(\sum_{i \in I_\lambda} g_i\right)_{\lambda \in \Lambda}$ is almost zero with Λ_J a valid choice, and since J is the disjoint union of the finitely many $J \cap I_\lambda$, $\lambda \in \Lambda_J$, induction on $|\Lambda_J|$ using (3.2.49) gives

$$\sum_{i \in I} g_i = \sum_{i \in J} g_i = \sum_{\lambda \in \Lambda_J} \sum_{i \in J \cap I_\lambda} g_i = \sum_{\lambda \in \Lambda} \sum_{i \in I_\lambda} g_i$$

∎

Lemma 3.2.53 (Reindexing and Iterated Sums)
Let G be an abelian group.

a) If $(g_i)_{i \in I}$ has all but finitely many terms zero and $\sigma : I' \to I$ is a bijection, then $\left(g_{\sigma(i')}\right)_{i' \in I'}$ has all but finitely many terms zero and $\sum_{i' \in I'} g_{\sigma(i')} = \sum_{i \in I} g_i$.
b) If $\left(g_{(i,k)}\right)_{(i,k) \in I \times K}$ has all but finitely many terms zero then so do each row $\left(g_{(i,k)}\right)_{i \in I}$ and $\left(\sum_{i \in I} g_{(i,k)}\right)_{k \in K}$, and

$$\sum_{(i,k) \in I \times K} g_{(i,k)} = \sum_{k \in K} \sum_{i \in I} g_{(i,k)}$$

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and likewise with the roles of i and k exchanged.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

a) Let $J \subseteq I$ be finite containing every i with $g_i \neq 0$. Then $\sigma^{-1}(J)$ is finite and contains every i' with $g_{\sigma(i')} \neq 0$, so the reindexed family has all but finitely many terms zero, and

$$\sum_{i' \in I'} g_{\sigma(i')} = \sum_{i' \in \sigma^{-1}(J)} g_{\sigma(i')} = \sum_{i \in J} g_i = \sum_{i \in I} g_i$$

since σ restricts to a bijection $\sigma^{-1}(J) \to J$ and finite sums do not depend on the ordering of the terms.

b) Apply (3.2.52) to the partition $I \times K = \bigsqcup_{k \in K} I \times \{k\}$. The block $\left(g_{(i,k')}\right)_{(i,k') \in I \times \{k\}}$ is the reindexing of the row $\left(g_{(i,k)}\right)_{i \in I}$ along the bijection $i \mapsto (i,k)$, so by a) the rows have all but finitely many terms zero, with $\sum_{(i,k') \in I \times \{k\}} g_{(i,k')} = \sum_{i \in I} g_{(i,k)}$. Hence the partition formula reads $\sum_{(i,k) \in I \times K} g_{(i,k)} = \sum_{k \in K} \sum_{i \in I} g_{(i,k)}$, and $\left(\sum_{i \in I} g_{(i,k)}\right)_{k \in K}$ has all but finitely many terms zero. The other order is the same argument applied to the partition $I \times K = \bigsqcup_{i \in I} \{i\} \times K$. ∎

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3.3 Rings and Modules

3.3.1 Commutative Rings

Definition 3.3.1 (Ring)
A ring consists of a triple $(A, +, \cdot)$ where A is a set and $+$ and $\cdot$ are laws of composition ("additive" and "multiplicative" respectively) such that the following holds

- $(A, +)$ *is an* ***abelian group****, whose identity element we refer to as* 0_A*.*
- $(A, \cdot)$ *is a* ***monoid****, whose identity element we refer to as* 1_A
- $+$ *and* $\cdot$ *satisfy the* ***distributive property****, that is for all* $x, y, z \in A$

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

$$(x + y) \cdot z = x \cdot z + y \cdot z$$

For $x \in A$ we write the additive inverse as $-x$, and abbreviate multiplication $x \cdot y =: xy$.

We say that A is a ***zero-ring*** *(or trivial) if $0_A = 1_A \iff A = \{0\}$.*

A is ***commutative*** *if in addition $xy = yx$ i.e. $(A, \cdot)$ is abelian.*

Example 3.3.2
The set of integers $\mathbb{Z}$ (see the Integers subsection) is the canonical example of a ring with operations of addition and multiplication.

Definition 3.3.3 (Subring)
A ***subring*** *of a ring A is a subset B such that*

- $0_A, 1_A \in B$
- $x \in B \implies -x \in B$
- $x, y \in B \implies x + y \in B$
- $x, y \in B \implies x \cdot y \in B$

Then $(B, +|_{B\times B}, \cdot\,|_{B\times B})$ is a ring.

Definition 3.3.4 (Multiplicative set)
A subset $S \subset A$ is said to be ***multiplicative*** *if*

- $1 \in S$
- $x, y \in S \implies xy \in S$

Further it is said to be ***saturated*** *if in addition*

$$x, y \in S \iff xy \in S$$

Definition 3.3.5 (Integral Domain)
A commutative ring A is said to be an ***integral domain*** *if it is not a zero-ring and it is* cancellative*, that is*

$$ab = ac, a \neq 0 \implies b = c.$$

Definition 3.3.6 (Reduced)
A commutative ring A is said to be ***reduced*** *if for all $a \in A$*

$$a^n = 0 \implies a = 0$$

Definition 3.3.7 (Unit / Group of Units)
An element $0 \neq a$ of a ring A is called a ***unit*** *if it has a two-sided multiplicative inverse.*

For A not a zero-ring, the set of units $A^\star$ forms a group under multiplication, called the ***group of units****.*

Definition 3.3.8 (Field)
A ***field*** *K is a commutative non-zero ring such that every non-zero element is a unit, so that $K^\star$ is a group under multiplication and $K^\star = K \setminus \{0\}$.*

Proposition 3.3.9
Every subring of a field K is an integral domain.

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Proof. Suppose $A \subset K$ is a subring. Suppose that $a, b \in A$ such that $ab = 0$. Suppose $a \neq 0$ then $a^{-1}ab = 0 \implies b = 0$. ∎

Note we have the implications

Corollary 3.3.10
Let A be a ring then we have the following implications

$$\textbf{field} \implies \textbf{integral domain} \implies \textbf{reduced}$$

Definition 3.3.11 (Ring homomorphism)
*A **ring homomorphism** $\varphi : A \to B$ is a mapping which is both a multiplicative (monoid) and additive (group) homomorphism*

- $\varphi(0_A) = 0_B$
- $\varphi(1_B) = 1_B$
- $\varphi(x + y) = \varphi(x) + \varphi(y)$
- $\varphi(xy) = \varphi(x)\varphi(y)$

*The **kernel** of φ is defined to be*

$$\ker(\varphi) = \{a \mid \varphi(a) = 0_B\}$$

Definition 3.3.12 (Ideal)
*A (two-sided) **ideal** $\mathfrak{a}$ of a ring A is a subset of A which is an additive subgroup and closed under multiplication by A :*

- $0_A \in \mathfrak{a}$
- $x, y \in \mathfrak{a} \implies x + y \in \mathfrak{a}$
- $x \in \mathfrak{a} \implies -x \in \mathfrak{a}$
- $x \in \mathfrak{a}, a \in A \implies ax, xa \in \mathfrak{a}$

*$\mathfrak{a}$ is said to be **proper** if $\mathfrak{a} \neq A$.*

Lemma 3.3.13 (Proper ideal)
An ideal $\mathfrak{a}$ is proper if and only if $1 \notin \mathfrak{a}$ if and only if $\mathfrak{a} \cap A^\star = \emptyset$.

Alternatively $\mathfrak{a} = A$ if and only if $1 \in \mathfrak{a}$ if and only if $\mathfrak{a} \cap A^\star \neq \emptyset$.

Proposition 3.3.14
Let $\varphi : A \to B$ be a ring homomorphism, then

a) The kernel $\ker(\varphi)$ is a two-sided ideal of A
b) The image $\varphi(A)$ is a subring of B
c) φ is injective if and only if $\ker(\varphi) = \{0\}$

Proposition 3.3.15 (Krull's Theorem)
Let A be a ring and $\mathfrak{a}$ a proper ideal. Then it is contained in a proper maximal ideal $\mathfrak{m}$.

In particular any non-unit $a \notin A^\star$ is contained in a maximal ideal.

Proposition 3.3.16 (Criteria to be a Field)
Let A be a ring. Then the following are equivalent

a) A is a field
b) A is not the zero-ring and the only proper ideal is $\{0\}$.

Proof. a) $\implies$ b). By definition A is not the zero-ring. Let $\mathfrak{a}$ be a proper ideal. By (3.3.13) $\mathfrak{a} \cap A^\star = \emptyset \implies \mathfrak{a} = \{0\}$ as required.

b) $\implies$ a). Suppose $0 \neq a \in A$ then the ideal $Aa = (a)$ is either $\{0\}$ or A. However $a = 1 \cdot a \in (a)$ which implies $(a) \neq \{0\}$ and therefore $(a) = A$. In particular there exists $a^{-1} \in A$ such that $a^{-1}a = 1_A$ and a is invertible. ∎

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3.3.2 Sums in a Ring

Proposition 3.3.17 (Distributivity for Sums in a Ring)
Let A be a ring, $r \in A$ and $(a_i)_{i\in I}$ a family in A with all but finitely many terms zero. Then $(ra_i)_{i\in I}$ and $(a_i r)_{i\in I}$ again have all but finitely many terms zero, and

$$r\sum_{i\in I} a_i = \sum_{i\in I} ra_i \qquad \left(\sum_{i\in I} a_i\right) r = \sum_{i\in I} a_i r$$

If $(b_j)_{j\in J}$ is another such family then $(a_i b_j)_{(i,j)\in I\times J}$ has all but finitely many terms zero, and

$$\left(\sum_{i\in I} a_i\right)\left(\sum_{j\in J} b_j\right) = \sum_{(i,j)\in I\times J} a_i b_j = \sum_{i\in I}\sum_{j\in J} a_i b_j$$

and likewise with the order of summation exchanged.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By the distributive property $x \mapsto rx$ and $x \mapsto xr$ are homomorphisms of the additive group of A, so the first claims are (3.2.51) for these homomorphisms. If $a_i b_j \neq 0$ then $a_i \neq 0$ and $b_j \neq 0$, so $a_i b_j$ vanishes outside a product of two finite sets. Write $b := \sum_{j\in J} b_j$. Then by the first part applied to b and then to each a_i

$$\left(\sum_{i\in I} a_i\right) b = \sum_{i\in I} a_i b = \sum_{i\in I} a_i \sum_{j\in J} b_j = \sum_{i\in I}\sum_{j\in J} a_i b_j = \sum_{(i,j)\in I\times J} a_i b_j$$

where the last equality is (3.2.53). The other order of summation is identical. ∎

3.3.3 Modules I

Definition 3.3.18 (Module)
*Let A be a ring. A **left A-module** $(M, +, \cdot)$ is an abelian group $(M, +)$ together with a "multiplication" operation*

$$\cdot : A \times M \to M$$

which satisfies the following properties

- $(a \times_A a') \cdot x = a \cdot (a' \cdot x)$
- $(a +_A a') \cdot x = a \cdot x + a' \cdot x$
- $a \cdot (x + y) = a \cdot x + a \cdot y$

*Similarly a **right A-module** $(M, +, \cdot)$ is an abelian group $(M, +)$ together with a multiplication operation*

$$\cdot : M \times A \to M$$

- $(a \times_A a') \cdot x = (x \cdot a') \times_A a$
- $(a +_A a') \cdot x = a \cdot x + a' \cdot x$
- $a \cdot (x + y) = a \cdot x + a \cdot y$

*Considering the first property M is a left A-module iff it is a right A^{op} module in the obvious way. Therefore in the usual case that A is commutative the concepts coincide and we may speak simply of an **A-module**, though we almost always write the action on the left.*

Similarly almost all results for left A-modules carry over unchanged for right A-modules. In this case we may simply by refer to A-modules rather than state the result for both cases separately.

Definition 3.3.19 (Submodule)
*Let $(M, +, \cdot)$ be a left A-module. Then a subset $N \subset M$ is called an A-**submodule** if*

- *N is a subgroup of $(M, +)$*
- *$m \in N, a \in A \implies am \in N$*

Then $(N, +|_{N\times N}, \cdot|_{A\times N})$ is a left A-module. Similar definition applies for a right A-module.

Definition 3.3.20 (Module homomorphism)
Let $(M, +, \cdot), (N, +, \cdot)$ be left A-modules. A function $f : M \to N$ is an A-module homomorphism if

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- *It is an (additive) group homomorphism* $(M,+) \to (N,+)$.
- *It is A-linear;* $\forall a \in A, m \in M \quad f(a \cdot m) = a \cdot f(m)$

It may be verified that f *is bijective if and only if it's an isomorphism. In this way we have the following categories*

- A-**Mod** *the category of modules over a* commutative *ring* A
- ${}_A$**Mod** *the category of left* A*-modules*
- **Mod**$_A$ *the category if right* A*-modules*

Definition 3.3.21 (Kernel and Image)
The ***kernel*** *of a module homomorphism* f *is given by*

$$\ker(f) := \{m \in M \mid f(m) = 0\}$$

and the ***image*** *is given by*

$$\mathrm{Im}(f) = f(M)$$

Example 3.3.22 (Trivial Examples)
A ring A *is a left* A*-module over itself, denoted* A_s.

Definition 3.3.23 (Restriction of Scalars)
Let $\varphi : A \to B$ *a ring homomorphism and* M *a* B*-module. Then we may consider* M *as an* A*-module in the obvious way. Denote this by* $[M]_\varphi$.

Proposition 3.3.24 (Submodules constitute a lattice)
Let M *be an* A*-module then the collection* $\mathrm{SubMod}(M)$ *of* A*-submodules form a complete sub-lattice of* $\mathcal{P}(M)$ *with meet and join given by*

$$\bigwedge_{i\in I} N_i = \bigcap_{i\in I} N_i$$

and (the ***internal sum****)*

$$\bigvee_{i\in I} N_i = \bigcap_{N_i \subseteq N \leq M} N =: \sum_{i\in I} N_i = \left\{ \sum_{j\in J} n_j \;\middle|\; n_j \in N_j \quad \#J < \infty \right\}$$

Moreover it is the image of the closure operator $\langle - \rangle : \mathcal{P}(M) \to \mathcal{P}(M)$ *given by*

$$\langle X \rangle = \bigcap_{X \subseteq N} N = \left\{ \sum_j a_j x_j \;\middle|\; x_j \in X \right\}$$

Proof. The A-submodules of M naturally form a Moore family of subsets of M. By (2.1.41) they form a complete sub-lattice with the given form of meet and join. Furthermore it is the image of the given closure operator. The only non-trivial statement is the explicit form of $\sum_{i\in I} N_i$ TODO. ∎

Lemma 3.3.25
Let M *be a module. Then*

a) $\langle \bigcup_{i\in I} X_i \rangle = \sum_{i\in I} \langle X_i \rangle$
b) $\langle \bigcup_{i\in I} N_i \rangle = \sum_{i\in I} N_i$
c) $N_1 \subseteq N_2 \implies N_1 + N_2 = N_2$

Proof.
a) This follows from (2.1.44) applied to the closure operator $\langle - \rangle$
b) This follows from a) because $N_i = \langle N_i \rangle$
c) This follows from b) because $N_1 \cup N_2 = N_2$

∎

Definition 3.3.26 (Hom Sets)
A ***module homomorphism*** $\varphi : M \to N$ *is an additive group homomorphism which commutes with the* A *action*

$$\varphi(am) = a\varphi(m) \quad \forall a \in A \, m \in M$$

Denote the abelian group of A*-module homomorphisms*

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$$\mathrm{Hom}_A(M,N)$$

and the endomorphism ring

$$\mathrm{End}_A(M) \coloneqq \mathrm{Hom}_A(M,M)$$

When A is commutative then these have natural A-module and A-algebra structures respectively.

3.3.4 Operations on Ideals

For this section we assume A is a commutative ring.

Definition 3.3.27 (Product of ideal and module)
Let M be an A-module and $\mathfrak{a} \triangleleft A$ an ideal. Define

$$\mathfrak{a}M = \langle \mathfrak{a} \cdot M \rangle = \left\{ \sum_{i=1}^{n} a_i m_i \;\middle|\; a_i \in \mathfrak{a} \quad m_i \in M \right\}$$

Proposition 3.3.28 (Lattice of Ideals)
Let A be a ring and $\mathcal{I}(A)$ the set of ideals. Then $\mathcal{I}(A)$ forms a complete lattice ordered by inclusion with join and meets given by

$$\bigwedge_{i\in I} \mathfrak{a}_i = \bigcap_{i\in I} \mathfrak{a}_i$$

and

$$\bigvee_{i\in I} \mathfrak{a}_i = \bigcap_{\mathfrak{a}_i \subseteq \mathfrak{a}} \mathfrak{a} =: \sum_i \mathfrak{a}_i \coloneqq \left\{ \sum_i a_i \;\middle|\; a_i \in \mathfrak{a}_i \right\}$$

This induces a corresponding closure operator

$$\langle - \rangle : \mathcal{P}(A) \to \mathcal{I}(A)$$

given by

$$\langle X \rangle \coloneqq \bigcap_{X \subseteq \mathfrak{a}} \mathfrak{a} = \left\{ \sum_j a_j x_j \;\middle|\; a_j \in A \quad x_j \in X \right\}$$

Proposition 3.3.29
Let A be a ring and $\mathfrak{a}_i$ a family of ideals. Then

$$\langle \bigcup_{i\in I} \mathfrak{a}_i \rangle = \sum_{i\in I} \mathfrak{a}_i$$

Definition 3.3.30 (Product of ideals)
The product of two ideals $\mathfrak{ab}$ is

$$\mathfrak{ab} = \left\{ \sum_{i=1}^{n} a_i b_i \;\middle|\; a_i \in \mathfrak{a} \quad b_i \in \mathfrak{b} \right\}$$

and is itself an ideal.

Definition 3.3.31 (Coprime)
*We say two elements x, y of a commutative ring A are **co-prime** if $(x,y) = (1) \iff ax + by = 1$ for some $a, b \in A$*

We say a family of ideals $\{\mathfrak{a}_i\}_{i\in I}$ are co-prime if $\sum_{i\in I} \mathfrak{a}_i = A$.

Definition 3.3.32 (Principal Ideal)
*A **principal ideal** is an ideal generated by a single element*

$$(a) \coloneqq \langle \{a\} \rangle = Aa$$

Lemma 3.3.33
A principal ideal (a) is proper if and only if $a \notin A^\star$

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Definition 3.3.34 (Maximal Ideal)
*An ideal $\mathfrak{m} \triangleleft A$ is **maximal** if it is both proper and not contained in another proper ideal.*

Definition 3.3.35 (Prime Ideal)
*An ideal $\mathfrak{p} \triangleleft A$ is **prime** if it is both proper and satisfies the following property*

$$xy \in \mathfrak{p} \implies x \in \mathfrak{p} \vee y \in \mathfrak{p}$$

Definition 3.3.36 (Radical Ideal)
*An ideal $\mathfrak{a} \triangleleft A$ is **radical** if it satisfies the following property*

$$x^n \in \mathfrak{a} \implies x \in \mathfrak{a}$$

Proposition 3.3.37 (Maximal ideals exist)
Let A be a ring and $\mathfrak{a} \triangleleft A$ a proper ideal. Then it is contained in some maximal ideal $\mathfrak{m}$.

In particular there always exists a maximal ideal by considering $\mathfrak{a} = (0)$.

Proof. Simple application of Zorn's Lemma. ∎

Proposition 3.3.38 (Properties of prime ideals)
Let $\mathfrak{p}$ be a prime ideal and $\mathfrak{a}$, $\mathfrak{b}$ be ideals then the following are equivalent

a) $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$
b) $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$
c) $\mathfrak{ab} \subseteq \mathfrak{p}$

Proof. a) $\implies$ b) Follows because $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{a}$

b) $\implies$ c) Follows because $\mathfrak{ab} \subseteq \mathfrak{a} \cap \mathfrak{b}$

c) $\implies$ a) If $\mathfrak{a} \nsubseteq \mathfrak{p}$, then choose $a \in \mathfrak{a} \setminus \mathfrak{p}$. By hypothesis $a\mathfrak{b} \subseteq \mathfrak{p}$ and since $\mathfrak{p}$ is prime $\mathfrak{b} \subseteq \mathfrak{p}$. ∎

Corollary 3.3.39 (Ideal version of primality)
Let $\mathfrak{p}$ be a proper ideal. Then $\mathfrak{p}$ is prime if and only if the following condition holds for all ideals $\mathfrak{a}$, $\mathfrak{b}$

$$\mathfrak{ab} \subseteq \mathfrak{p} \implies \mathfrak{a} \subseteq \mathfrak{p} \text{ or } \mathfrak{b} \subseteq \mathfrak{p}$$

In particular for all $k > 0$ we have

$$\mathfrak{a} \subseteq \mathfrak{p} \iff \mathfrak{a}^k \subseteq \mathfrak{p}$$

Proof. One direction has been shown in (3.3.38). Conversely suppose $fg \in \mathfrak{p}$ then apply the condition to the ideals (f) and (g) we find $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. ∎

Lemma 3.3.40 (Prime ideals are meet-prime)
Let $\mathfrak{p}$ be a prime ideal. Then

$$\bigcap_{i=1}^{n} \mathfrak{a}_i \subseteq \mathfrak{p} \implies \mathfrak{a}_i \subseteq \mathfrak{p} \text{ some } i = 1...n$$

in other words $\mathfrak{p}$ is meet-prime in the lattice of ideals.

Proof. Suppose $\mathfrak{a}_i \nsubseteq \mathfrak{p}$ for all i then there exists $x_i \in \mathfrak{a}_i \setminus \mathfrak{p}$. Then $x_1...x_n \in \bigcap_{i=1}^{n} \mathfrak{a}_i \subseteq \mathfrak{p}$ by hypothesis, so by primality $x_i \in \mathfrak{p}$ for some i, a contradiction. ∎

Lemma 3.3.41 (Generate prime ideals)
Let A be a ring, S a multiplicative set and $\mathfrak{b} \triangleleft A$ such that $\mathfrak{b} \cap S = \emptyset$ then

$$\mathcal{I} = \{\mathfrak{a} \mid \mathfrak{b} \subseteq \mathfrak{a} \quad \mathfrak{a} \cap S = \emptyset\}$$

has a maximal element, which is prime.

Proof. Since $\mathfrak{b} \in \mathcal{I}$ it is non-empty. By Zorn's Lemma it has a maximal element, $\mathfrak{p}$. We claim it is prime, for suppose $xy \in \mathfrak{p}$ and $x, y \notin \mathfrak{p}$. Then by maximality $\mathfrak{p} + (x)$ and $\mathfrak{p} + (y)$ intersect S. Therefore S intersects $(\mathfrak{p} + (x))(\mathfrak{p} + (y)) \subseteq \mathfrak{p}$, a contradiction. ∎

Definition 3.3.42 (Minimal prime)
*Let A be a ring and $\mathfrak{a} \triangleleft A$ a proper ideal. A prime ideal $\mathfrak{p}$ is a **minimal prime over** $\mathfrak{a}$ if it contains $\mathfrak{a}$, and every other such prime ideal contains $\mathfrak{p}$.*

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We say it is simply a ***minimal prime*** *if it is minimal over* (0).

Proposition 3.3.43 (Prime ideals are chain complete)
Let $\{\mathfrak{p}_i\}_{i\in I}$ *be a* ***chain*** *of prime ideals, then* $\bigcap_i \mathfrak{p}_i$ *and* $\bigcup_i \mathfrak{p}_i$ *are prime ideals.*

Proof. By (3.3.37) $\bigcup_i \mathfrak{p}_i$ is an ideal, and it's easily verified to be prime. Clearly $\bigcap_i \mathfrak{p}_i$ is an ideal. Suppose $a, b \notin \bigcap_i \mathfrak{p}_i$ then $a \notin \mathfrak{p}_j$ and $b \notin \mathfrak{p}_k$ with $j \leq k$. Then $b \notin \mathfrak{p}_j$, and $ab \notin \mathfrak{p}_j$ by primality, whence $ab \notin \bigcap_i \mathfrak{p}_i$. ■

Corollary 3.3.44 (Minimal primes exist)
Let A *be a ring and* $\mathfrak{a} \lhd A$ *be a proper ideal contained in a prime ideal* $\mathfrak{p}$*. Then there exists a minimal prime over* $\mathfrak{a}$ *contained in* $\mathfrak{p}$*.*

In particular there always exists a minimal prime over $\mathfrak{a}$ *and every prime ideal contains a minimal prime ideal.*

Proof. We may use (3.3.43) together with Zorn's Lemma. ■

Proposition 3.3.45
Let A *be a ring. Then the set* $\mathrm{Rad}(A)$ *of radical ideals forms a complete sub-lattice of the lattice of ideals* $\mathcal{I}(A)$*. This induces a closure operator*

$$\sqrt{-} : \mathcal{I}(A) \to \mathrm{Rad}(A)$$

given by

$$\sqrt{\mathfrak{a}} \coloneqq \bigcap_{\mathfrak{a}\subseteq\mathfrak{r}} \mathfrak{r} = \{x \mid x^n \in \mathfrak{a} \quad n > 0\}$$

The "join" is given by

$$\bigvee_{i\in I} \mathfrak{a}_i = \sqrt{\sum_i \mathfrak{a}_i}$$

In particular

a) $\mathfrak{a} \subseteq \sqrt{\mathfrak{a}}$
b) $\mathfrak{a} \subseteq \mathfrak{b} \implies \sqrt{\mathfrak{a}} \subseteq \sqrt{\mathfrak{b}}$
c) $\sqrt{\sqrt{\mathfrak{a}}} = \sqrt{\mathfrak{a}}$

Proof. The set of radical ideals is closed under arbitrary intersections (which are meets in the lattice $\mathcal{I}(A)$). Therefore by (2.1.41) it forms a complete sub-lattice with meet given by intersection of ideals.

It also shows that $\sqrt{-}$ as defined is a closure operator with image $\mathrm{Rad}(A)$, which demonstrates the required properties.

Finally we just need to show that $I' \coloneqq \{x \mid x^n \in \mathfrak{a} \quad n > 0\}$ is equal to $\sqrt{\mathfrak{a}}$. Firstly it's an ideal for if $x, y \in I'$ then $x^n \in \mathfrak{a}$ and $y^m \in \mathfrak{a}$, so we may show that $(x+y)^{n+m} \in \mathfrak{a}$ whence $x + y \in I'$. Similarly $a \in A$ and $x \in I'$ implies $(ax)^n = a^n x^n \in I'$. It's radical for suppose $x^m \in I'$ then $x^{mn} = (x^m)^n \in \mathfrak{a}$ by definition whence $x \in I'$. As it contains $\mathfrak{a}$ we find that $\sqrt{\mathfrak{a}} \subseteq I'$. Let $\mathfrak{r}$ be another radical ideal containing $\mathfrak{a}$ then $x \in I' \implies x^n \in \mathfrak{a} \implies x^n \in \mathfrak{r} \implies x \in \mathfrak{r}$. Therefore the reverse inclusion follows. ■

Proposition 3.3.46 (Prime Nullstellensatz)
The radical of an ideal satisfies

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{a}\subseteq\mathfrak{p}:\mathfrak{p}\text{ prime}} \mathfrak{p}$$

Further the intersection may be taken over all minimal *primes over* $\mathfrak{a}$*.*

Proof. Suppose $x \in \sqrt{\mathfrak{a}}$ and $\mathfrak{p} \supseteq \mathfrak{a}$. Then $x^n \in \mathfrak{p} \implies x \in \mathfrak{p}$. Therefore $\sqrt{\mathfrak{a}} \subseteq \bigcap_{\mathfrak{a}\subseteq\mathfrak{p}} \mathfrak{p}$. Conversely suppose $x \notin \sqrt{\mathfrak{a}}$ then $S \coloneqq \{1, x, x^2, \ldots\}$ is a proper multiplicative set such that $S \cap \mathfrak{a} = \emptyset$. By (3.3.41) there is a prime ideal $\mathfrak{p}$ containing $\mathfrak{a}$ which does not intersect S. Therefore $x \notin$ RHS as required. ■

Proposition 3.3.47 (Properties of Radical Ideals)
Let $\mathfrak{a}$, $\mathfrak{b}$ *be ideals then*

a) $\sqrt{\mathfrak{a}^k} = \sqrt{\mathfrak{a}}$ *for* $k > 0$
b) $\sqrt{\sum_i \mathfrak{a}_i} = \sqrt{\sum_i \sqrt{\mathfrak{a}_i}}$
c) $\sqrt{\mathfrak{a}} = A \iff \mathfrak{a} = A$
d) $\sum_i \mathfrak{a}_i = A \iff \sum_i \sqrt{\mathfrak{a}_i} = A$
e) $\sum_{i=1}^n \mathfrak{a}_i^{k_i} = A \iff \sum_{i=1}^n \mathfrak{a}_i = A \quad k_i > 0.$

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Proof.

a) This may be shown by direct calculation or combining (3.3.46) and (3.3.38).
b) This follows by applying (2.1.44) to the closure operator $\sqrt{-}$.
c) $\sqrt{\mathfrak{a}} = A \iff 1 \in \sqrt{\mathfrak{a}} \iff 1 \in \mathfrak{a} \iff \mathfrak{a} = A$
d) This follows from combining c) and b)
e) This follows from d), b) and a)

■

Lemma 3.3.48 (Ideal Finitely Generated by Nilpotents is Nilpotent)
Let A be a ring with ideals $\mathfrak{a}, \mathfrak{b}$ such that $\sqrt{\mathfrak{a}} \subseteq \mathfrak{b}$ and $\mathfrak{b}$ finitely generated. Then there exists an integer $n > 0$ such that $\mathfrak{b}^n \subseteq \mathfrak{a}$.

Definition 3.3.49 (Extended and contracted ideals)
*Let $\varphi : A \to B$ be a homomorphism and $\mathfrak{a}$ (resp. $\mathfrak{b}$) be an ideal of A (resp. B). Define the **contraction** (resp. **extension**) ideals as follows*

$$\mathfrak{b}^c := \varphi^{-1}(\mathfrak{b})$$
$$\mathfrak{a}^e := \varphi(\mathfrak{a})B := \langle \varphi(\mathfrak{a}) \rangle = \left\{ \sum_i b_i \varphi(a_i) \;\middle|\; a_i \in \mathfrak{a} \right\}$$

*An ideal is said to be **contracted** (resp. **extended**) if it is of the form $\mathfrak{b}^c$ (resp. $\mathfrak{a}^e$)*

Proposition 3.3.50 (Operations on ideals)
Let $\varphi : A \to B$ a ring homomorphism and $\mathfrak{a} \lhd A$, $\mathfrak{b} \lhd B$ ideals then

a) $\mathfrak{b}^c \lhd A$ and $\mathfrak{a}^e \lhd B$
b) $\mathfrak{b}^c$ proper if and only if $\mathfrak{b}$ is proper
c) $\mathfrak{b}^{ce} \subseteq \mathfrak{b}$ and $\mathfrak{a} \subseteq \mathfrak{a}^{ec}$
d) $\mathfrak{a}^{ece} = \mathfrak{a}^e$ and $\mathfrak{b}^{cec} = \mathfrak{b}^c$
e) $\mathfrak{b}^{ce} = \mathfrak{b} \iff \mathfrak{b}$ is an extended ideal $\iff \mathfrak{b} \subseteq \mathfrak{b}^{ce}$
f) $\mathfrak{a}^{ec} = \mathfrak{a} \iff \mathfrak{a}$ is a contracted ideal $\iff \mathfrak{a}^{ec} \subseteq \mathfrak{a}$
g) $\sqrt{\mathfrak{b}^c} = \left(\sqrt{\mathfrak{b}}\right)^c$
h) $\left(\sqrt{\mathfrak{b}^c}\right)^e \subseteq \sqrt{\mathfrak{b}}$ with equality when φ is surjective

When φ is surjective every ideal $\mathfrak{b} \lhd B$ is extended, and the contracted ideals are precisely the ideals containing $\ker(\varphi)$.

Proof. We prove each in turn

a)-c) Straightforward

d) By the previous step $\mathfrak{b}^{ce} \subseteq \mathfrak{b} \implies (\mathfrak{b}^{ce})^c \subseteq \mathfrak{b}^c$, similarly $\mathfrak{b}^c \subseteq (\mathfrak{b}^c)^{ec}$. The other relation is similar.

e)-f) These follow from c) and d)

g) $x \in \left(\sqrt{\mathfrak{b}}\right)^c \iff \varphi(x) \in \sqrt{\mathfrak{b}} \iff \varphi(x)^n \in \mathfrak{b} \iff \varphi(x^n) \in \mathfrak{b} \iff x^n \in \mathfrak{b}^c \iff x \in \sqrt{\mathfrak{b}^c}$

h) By c) and g) we find $\left(\sqrt{\mathfrak{b}^c}\right)^e = \left(\sqrt{\mathfrak{b}}\right)^{ce} \subseteq \sqrt{\mathfrak{b}}$. We will show that when φ is surjective every ideal is extended, in which case the equality follows from e).

Suppose that φ is surjective. Then by e) we only need to show that $\mathfrak{b} \subseteq \mathfrak{b}^{ce}$ for every ideal $\mathfrak{b}$. Let $y \in \mathfrak{b}$ then $y = \varphi(x)$, whence $x \in \mathfrak{b}^c$ and $y \in \mathfrak{b}^{ce}$. ■

Proposition 3.3.51
Let $\varphi : A \to B$ be a ring homomorphism and $\mathfrak{a}_1, ..., \mathfrak{a}_n$ ideals of A. Then

$$\left(\sum_{i=1}^n \mathfrak{a}_i\right)^e = \sum_{i=1}^n \mathfrak{a}_i^e$$

Similarly

$$\left(\prod_{i=1}^n \mathfrak{a}_i\right)^e = \prod_{i=1}^n \mathfrak{a}_i^e$$

Proof. Evidently $\varphi(\mathfrak{a}_i) \subseteq \varphi\left(\sum_{i=1}^n \mathfrak{a}_i\right) \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e$ and therefore $\mathfrak{a}_i^e \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e \implies \sum_{i=1}^n \mathfrak{a}_i^e \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e$.

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Similarly $\varphi\left(\sum_{i=1}^n \mathfrak{a}_i\right) \subseteq \sum_{i=1}^n \varphi(\mathfrak{a}_i) \subseteq \sum_{i=1}^n \mathfrak{a}_i^e$ whence the reverse inclusion follows.

The second statement follows similarly. ■

Corollary 3.3.52
Let $\varphi : A \to B$ be a ring homomorphism then extension and contraction constitute a monotone Galois connection

$$\{\mathfrak{a} \lhd A\} \longleftrightarrow \{\mathfrak{b} \lhd B\}$$

and therefore is order-preserving and satisfies the adjoint property

$$\mathfrak{a} \subseteq \mathfrak{b}^c \iff \mathfrak{a}^e \subseteq \mathfrak{b}$$

is satisfied.

Proof. Extension and contraction satisfy conditions of (2.1.50) by (3.3.50) c) and d) ■

Corollary 3.3.53
Let $\varphi : A \to B$ be a ring homomorphism then there is a order-preserving bijection between "contracted" and "extended ideals"

$$\{\mathfrak{a} \lhd A \mid \mathfrak{a} \text{ contracted}\} \longleftrightarrow \{\mathfrak{b} \lhd B \mid \mathfrak{b} \text{ extended}\}$$

which restricts to proper ideals.

Proof. We've shown that $\mathfrak{a}$ (resp. $\mathfrak{b}$) is contracted (resp. extended) if and only if the given maps are mutually inverse. Note that $\mathfrak{b}$ is proper implies $\mathfrak{b}^c$ is proper. Furthermore $\mathfrak{b}^c$ proper implies $\mathfrak{b}^{ce} \subseteq \mathfrak{b}$ is proper. Therefore it restricts to proper ideals. ■

Proposition 3.3.54 (Inverse image of maximal / prime ideals)
Let $\varphi : A \to B$ be a morphism then

- *$\mathfrak{q} \lhd B$ prime $\implies \varphi^{-1}(\mathfrak{q})$ prime*
- *$\mathfrak{n} \lhd B$ maximal and φ surjective $\implies \varphi^{-1}(\mathfrak{n})$ is maximal*

Proposition 3.3.55
Consider maps $\varphi : A \to B$ and $\psi : B \to C$ and an ideal $\mathfrak{a} \lhd A$. Then extension of ideals is transitive, that is

$$\psi(\varphi(\mathfrak{a})B)C = (\psi \circ \varphi)(\mathfrak{a})C$$

3.3.5 Quotient Rings

Proposition 3.3.56 (Quotient Ring)
Let $(A, +, \cdot)$ be a ring and $\mathfrak{a}$ an ideal. As $\mathfrak{a}$ is an additive subgroup we may consider the quotient group $(A/\mathfrak{a}, +)$. For an element $a \in A$ write $a + \mathfrak{a}$ for the coset $[a]_\mathfrak{a} \in A/\mathfrak{a}$. There is a well-defined multiplicative law of composition

$$\cdot : A/\mathfrak{a} \times A/\mathfrak{a} \to A/\mathfrak{a}$$

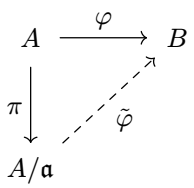
$$(a + \mathfrak{a}) \cdot (b + \mathfrak{a}) \to (a \cdot b + \mathfrak{a})$$

which makes $(A/\mathfrak{a}, +, \cdot)$ into a ring. Further there is a canonical surjective ring homomorphism

$$\pi : A \to A/\mathfrak{a}$$

with the following properties

- $\ker(\pi) = \mathfrak{a}$
- *Every morphism $\varphi : A \to B$ such that $\mathfrak{a} \subseteq \ker(\varphi)$, factors uniquely through π.*



- $\ker(\tilde{\varphi}) = \ker(\varphi)/\mathfrak{a}$
- *$\tilde{\varphi}$ is injective if and only if* $\ker(\varphi) = \mathfrak{a}$
- *$\tilde{\varphi}$ is surjective if and only if φ is surjective*

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For an ideal $\mathfrak{b} \supseteq \mathfrak{a}$ define the corresponding quotient ideal

$$\mathfrak{b}/\mathfrak{a} := \{b + \mathfrak{a} \mid b \in \mathfrak{b}\} = \pi(\mathfrak{b})$$

This induces a bijective, order-preserving correspondence of ideals

$$\{\mathfrak{b}' \lhd A/\mathfrak{a}\} \underset{\pi(-)}{\overset{\pi^{-1}(-)}{\rightleftarrows}} \{\mathfrak{b} \lhd A \mid \mathfrak{a} \subseteq \mathfrak{b}\}$$

under which maximal (resp. prime, radical) ideals of A containing $\mathfrak{a}$ correspond to maximal (resp. prime, radical) ideals of $A/\mathfrak{a}$.

Corollary 3.3.57 (Isomorphism Theorem)
Let $\varphi : A \to B$ be a ring homomorphism. Then this induces a canonical isomorphism

$$A/\ker(\varphi) \cong \varphi(A) \subset B$$

Corollary 3.3.58 (Second Isomorphism Theorem)
Let $\mathfrak{b}$,$\mathfrak{a}$ be ideals then there is a unique morphism making the diagram commute

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ {\scriptstyle\pi}\downarrow & \swarrow & \downarrow{\scriptstyle\pi} \\ A/(\mathfrak{a}+\mathfrak{b}) & \xrightarrow{\sim} & (A/\mathfrak{a})/((\mathfrak{a}+\mathfrak{b})/\mathfrak{a}) \end{array}$$

which is in fact an isomorphism. If $\mathfrak{a} + \mathfrak{b} \subseteq \mathfrak{c}$ this restricts to an isomorphism of $A/\mathfrak{b}$-modules

$$\begin{array}{ccc} \mathfrak{c} & \xrightarrow{\pi} & \mathfrak{c}/\mathfrak{a} \\ {\scriptstyle\pi}\downarrow & \swarrow & \downarrow{\scriptstyle\pi} \\ \mathfrak{c}/(\mathfrak{a}+\mathfrak{b}) & \xrightarrow{\sim} & (\mathfrak{c}/\mathfrak{a})/((\mathfrak{a}+\mathfrak{b})/\mathfrak{a}) \end{array}$$

Proposition 3.3.59 (Criteria for Maximal, Prime and Reduced)
Let $\mathfrak{a} \lhd A$ then $\mathfrak{a}$ is

- ***maximal** if and only if $A/\mathfrak{a}$ is a field*
- ***prime** if and only if $A/\mathfrak{a}$ is an integral domain*
- ***radical** if and only if $A/\mathfrak{a}$ is reduced*

Proof. Suppose $\mathfrak{a}$ is maximal then it is by definition proper so $A/\mathfrak{a}$ is not the zero-ring. By (3.3.56) then $A/\mathfrak{a}$ has no proper non-zero ideals and so by (3.3.16) is a field.

Conversely if $A/\mathfrak{a}$ is a field it is by definition not the zero ring, and so by (3.3.56) $\mathfrak{a}$ is proper. Furthermore by the same result $\mathfrak{a}$ is maximal.

Suppose $\mathfrak{a}$ is prime and $\overline{x} \cdot \overline{y} = 0$. Then by definition $\overline{x \cdot y} = 0 \implies x \cdot y \in \mathfrak{a} \implies x \in \mathfrak{a}$ or $y \in \mathfrak{a} \implies \overline{x} = 0$ or $\overline{y} = 0$. This shows that $A/\mathfrak{a}$ is an integral domain. The converse is similar. ∎

Corollary 3.3.60
Let $\mathfrak{a} \lhd A$ be a proper ideal, then the following implications hold

$$\textbf{maximal} \implies \textbf{prime} \implies \textbf{radical}$$

Proof. This follows by combining (3.3.10) and (3.3.59). ∎

Corollary 3.3.61 (Field Morphisms are injective)
Let $\varphi : k \to B$ be a homomorphism from a field to a non-zero ring. Then φ is injective.

Proof. $\ker(\varphi)$ is an ideal. As $\varphi(1_k) = 1_B$ and $0_B \neq 1_B$ then $\ker(\varphi) \neq k$. Since the only ideals are (0) and k we see $\ker(\varphi) = \{0\}$ and φ is injective. ∎

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3.3.6 Chinese Remainder Theorem

Proposition 3.3.62 (Chinese Remainder Theorem)
Let A be a commutative ring with ideals $\mathfrak{a}_1, ..., \mathfrak{a}_n$ which are pairwise coprime (i.e. $\mathfrak{a}_i + \mathfrak{a}_j = A$ for all $1 \leq i < j \leq n$). Then there is a canonical isomorphism

$$A/\bigcap_{i=1}^{n} \mathfrak{a}_i \longrightarrow A/\mathfrak{a}_1 \times ... \times A/\mathfrak{a}_n$$

Further we have the following identity

$$\bigcap_{i=1}^{n} \mathfrak{a}_i = \prod_{i=1}^{n} \mathfrak{a}_i$$

Proof. Consider the canonical ring homomorphism

$$\varphi : A \longrightarrow A/\mathfrak{a}_1 \times ... \times A/\mathfrak{a}_n.$$

Trivially it has kernel $\bigcap_{i=1}^{n} \mathfrak{a}_i$. Therefore by (3.3.56) it is sufficient to show it is surjective. First we show that $e_1 := (1, 0, ..., 0)$ is in the image of φ. For we have $x_i + y_i = 1$ for $x_i \in \mathfrak{a}_1$, $y_i \in \mathfrak{a}_i$ and $1 < i \leq n$. Take $f_1 = y_2 ... y_n \in \bigcap_{i=2}^{n} \mathfrak{a}_i$; since $y_i = 1 - x_i$, $f_1 = \prod_{i=2}^{n}(1 - x_i) \in 1 + \mathfrak{a}_1$ (as each $x_i \in \mathfrak{a}_1$), so combined with $f_1 \in \bigcap_{i=2}^{n} \mathfrak{a}_i$, $\varphi(f_1) = e_1$ as required.

Similarly there exists f_i such that $\varphi(f_i) = (0, ..., 1, ..., 0) =: e_i$. Finally for $\overline{x_1} \in A/\mathfrak{a}_1, ..., \overline{x_n} \in A/\mathfrak{a}_n$ then

$$\varphi(f_1 x_1 + ... + f_n x_n) = (\overline{x_1}, ..., \overline{x_n})$$

For the final statement we may assume by induction that

$$\bigcap_{i=1}^{n-1} \mathfrak{a}_i = \prod_{i=1}^{n-1} \mathfrak{a}_i$$

We have equations

$$x_i + y_i = 1 \quad x_i \in \mathfrak{a}_i, y_i \in \mathfrak{a}_n$$

Then

$$x_1 ... x_{n-1} = \prod_{i=1}^{n-1}(1 - y_i) \in 1 + \mathfrak{a}_n$$

so we may conclude

$$\bigcap_{i=1}^{n-1} \mathfrak{a}_i + \mathfrak{a}_n = A.$$

This then reduces to the case $n = 2$. Suppose $x \in \mathfrak{a} \cap \mathfrak{b}$, by assumption $1 = y + z$ for $y \in \mathfrak{a}$ and $z \in \mathfrak{b}$. Then $x = x(y + z) \in \mathfrak{ab}$, so $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{ab}$. The reverse inclusion is obvious. ∎

Proposition 3.3.63 (Chinese Remainder Theorem II)
Let A be a commutative ring with $\mathfrak{a}_1, ..., \mathfrak{a}_n$ co-prime ideals. For any positive integers $k_1, ..., k_n$ there is a canonical isomorphism

$$A/\bigcap_{i=1}^{n} \mathfrak{a}_i^{k_i} \longrightarrow A/\mathfrak{a}_1^{k_1} \times ... \times A/\mathfrak{a}_n^{k_n}$$

and

$$\bigcap_{i=1}^{n} \mathfrak{a}_i^{k_i} = \prod_{i=1}^{n} \mathfrak{a}_i^{k_i}$$

Proof. By (3.3.47) the ideals $\mathfrak{a}_1^{k_1}, ..., \mathfrak{a}_n^{k_n}$ are pairwise co-prime. Therefore we may apply (3.3.62). ∎

3.3.7 Irreducible and Reduced rings

We say an element x is nilpotent if $x^n = 0$. By (3.3.45) these form an ideal.

Definition 3.3.64 (Nilradical)
*Define the **nilradical** to be the set (ideal) of nilpotents*

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$$N(A) := \sqrt{(0)} \overset{(3.3.46)}{=} \bigcap_{\mathfrak{p}} \mathfrak{p}$$

Clearly A is reduced if and only if $N(A) = \{0\}$.

We also make the following definition

Definition 3.3.65 (Irreducible)
*Let A be a ring. We say A is **irreducible** if $N(A)$ is prime.*

The notion of irreducible ring is related to the notion of minimal primes

Proposition 3.3.66
A ring A is irreducible if and only if it has a unique minimal prime ideal. In this case it is equal to $N(A)$.

Proof. First we note that every prime ideal contains $N(A)$. If A is irreducible then by definition $N(A)$ is prime and it is therefore the unique minimal prime ideal.

Conversely if $\mathfrak{p}_0$ is the unique minimal prime ideal then by (3.3.44) it is contained in every prime ideal. Therefore (3.3.64) $N(A) = \bigcap_{\mathfrak{p}} \mathfrak{p} = \mathfrak{p}_0$ is prime and A is irreducible. ∎

Proposition 3.3.67 (Integral Domain $\iff$ Reduced and Irreducible)
Let A be a ring. Then the following are equivalent

a) A is an integral domain
b) A is reduced and has a unique minimal prime
c) (0) is prime
d) A is reduced and irreducible.

The following may be useful

Proposition 3.3.68
Let A be a ring then the sum of an invertible and nilpotent element is again invertible

$$A^\star + N(A) \subseteq A^\star$$

Proof. For if

$$a = u + n = u(1 - (u^{-1})(-n))$$

with $n \in N(A)$ and $u \in A^\star$. Then way may reduce to the case $1 - n$ and observe that

$$(1-n)^{-1} = \sum_{i=0}^{\infty} n^i$$

which by assumption is a finite sum. ∎

3.3.8 Algebra over a Commutative Ring

For what follows let A be a commutative ring.

Definition 3.3.69 (Algebra (over a commutative ring))
An algebra over A (or an A-algebra) is a pair (i_B, B) where B is a (not necessarily commutative) ring and $i_B : A \to B$ is a ring homomorphism.

We call i_B the structural morphism and write $a \cdot b := i_B(a)b$

Morphisms of A-algebras are the ring homomorphisms $\varphi : B \to C$ such that $\varphi \circ i_B = i_C$. This then constitutes a category $\mathbf{Alg}_A$.

If k is a field an algebra over k is referred to as a k-algebra.

Definition 3.3.70 (Sub-algebra)
Let (i_B, B) be an A-algebra. A sub-algebra C is a subring C of B for which

$$a \in A \quad c \in C \implies i_B(a)c \in C$$

Example 3.3.71 (Algebra over commutatve sub-ring)
If $A \subset B$ is a commutative sub-ring, then B is naturally a A-algebra.

The polynomial ring $A[X]$ is naturally an A-algebra

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Definition 3.3.72 (Algebra generated by a set)
Let B be an A-algebra. The collection of A-subalgebras forms a Moore family. Therefore by (2.1.41) there is a canonical closure operator

$$A[-] : \mathcal{P}(B) \to \mathrm{SubAlg}_A(B)$$

which we denote by $A[S]$ for $S \subset B$. A more explicit characterization when S is finite is given in the Polynomial Rings in Many Variables subsection. More generally we have

$$A[S] = \bigcup_{S' \subset S | S' \text{ finite}} A[S']$$

Proposition 3.3.73
Let $B = A[S]$ be an A-algebra and $\mathfrak{a}$ a sub-A-module of B. Then $\mathfrak{a}$ is an ideal if and only if

$$s \in S \implies s\mathfrak{a} \subseteq \mathfrak{a}$$

Proof. One direction is obvious. Suppose the condition given holds, and define

$$B' := \{b \in B \mid b\mathfrak{a} \subseteq \mathfrak{a}\}$$

Then clearly $S \subseteq B'$. It's easy to show that B' is a sub-A-algebra of B, so $B' = B$ and $\mathfrak{a}$ is an ideal. ∎

3.3.9 Finite-type Algebras

Definition 3.3.74 (Finite algebra)
An A-algebra B is finite *if it is finite as an A-module.*

Definition 3.3.75 (Finitely generated algebra)
An A-algebra B is finitely generated (or of finite type) if there exists an integer $n \in \mathbb{N}$ and a surjection of A-algebras

$$A[X_1, ..., X_n] \to B$$

the images of X_i are the generators.

Proposition 3.3.76
Let C be a finitely-generated B algebra and B a finitely-generated A-algebra. Then C is finitely-generated as an A-algebra.

Proof. Suppose $C = B[c_1, ..., c_n]$. Then by definition $c_i = F_i(b_1, ..., b_m)$ for some $F_i \in A[X_1, ..., X_m]$. Then we may take as a generating set

$$b_1^{\lambda_1} ... b_m^{\lambda_m}$$

where $\lambda \in \mathbb{N}^m$ runs through all the exponents of non-zero coefficients of the F_i. ∎

3.3.10 Bimodules

For applications it is often useful to have multiple rings acting on a single module ("bimodule"). For example an A-algebra B naturally has an action from both A and B. It may also allow us to generalise results which would otherwise only hold in the commutative case.

Definition 3.3.77 (Bimodule)
*Let A, B be rings. We say an abelian group M is a (A, B)-**bimodule** if it is both a left A-module and a right B-module for which the two actions commute*

$$a \cdot (m \cdot b) = (a \cdot m) \cdot b \quad \forall a \in A, b \in B, m \in M$$

We may denote this by ${}_AM_B$ in order to emphasise the actions. If homomorphisms are defined in the obvious way then these constitute a category ${}_A\mathbf{Mod}_B$. Recall that every abelian group is automatically both a left and right $\mathbb{Z}$-module with the following action for $N \in \mathbb{Z}$

$$N \cdot m = m \cdot N := \begin{cases} \underbrace{m + ... + m}_{N \text{ times}} & N \geq 0 \\ -(m \cdot (-N)) & N < 0 \end{cases}$$

Therefore we have the following generalisations

- *A right B-module is precisely a $(\mathbb{Z}, B)$-bimodule*

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- *A left A-module is precisely a $(A,\mathbb{Z})$-bimodule*
- *When A is commutative an A-module has a well-defined (A,A)-bimodule structure by defining*

$$a\cdot m\cdot a' := a'a\cdot m = aa'\cdot m$$

  *in other words A-**Mod** is (equivalent to) a full subcategory of ${}_A\mathbf{Mod}_A$.*
- *A ring A has an obvious (A,A)-bimodule structure by associativity of the multiplication operation*

We will understand by the notation N_B that N is a $(\mathbb{Z},B)$-bimodule and by ${}\{\}_A M$ that M is a $(A,\mathbb{Z})$-bimodule.

We generalize slightly the consideration of the third bullet point

Proposition 3.3.78 ((A,A)-Bimodule $\equiv$ A-module)
Let $\varphi : A\to B$ be a homomorphism of commutative rings and Z a B-module. Then Z is naturally a (B,A)-module with the following action

$$b\cdot m\cdot a := \varphi(a)b\cdot m \qquad (\star)\quad \forall a\in A, b\in B, m\in Z$$

Suppose M is another (B,A)-bimodule satisfying $(\star)$ then we may identify the (B,A)-bimodule homomorphisms and the left B-module homomorphisms.

$$\mathrm{Hom}(M,\,{}_BZ_A)\xrightarrow{\sim}\mathrm{Hom}({}_BM,\,{}_BZ)$$

In particular when $\varphi = 1_A$ and $B=A$ we may identify (A,A)-bimodules and A-modules, as well as the corresponding homomorphisms. Explicitly for M,Z A-modules these have well-defined (A,A)-bimodule structures and there is a bijection

$$\mathrm{Hom}({}_AM_A,\,{}_AZ_A)\xrightarrow{\sim}\mathrm{Hom}(M,Z)$$

Proof. We may show that the right A-module structure on Z is well-defined because B is commutative and using the associativity of the B-module action. Similarly the associativity of the B-module action ensures that the actions commute and so form a (B,A)-bimodule structure.

Suppose $\theta : M\to Z$ is a (left) B-module homomorphism. Then

$$\theta(m\cdot a)=\theta(\varphi(a)\cdot m)=\varphi(a)\theta(m)=\theta(m)\cdot a$$

so it is (B,A)-bilinear as required. The converse is clear, so we see that the sets are equal as subsets of the set of abelian group homomorphisms.

The case $\varphi = 1_A$ is immediate. ∎

We generalize the notion of "Hom-Set"

Definition 3.3.79 (Hom Functors)
Let A,B,C be arbitrary rings. Then we have an enriched hom functor for ${}_A\mathbf{Mod}_B$

$$\mathrm{Hom} : {}_A\mathbf{Mod}_B^{op}\times{}_A\mathbf{Mod}_B\to\mathbf{AbGrp}$$

In order to generalise the Tensor-Hom adjunction we consider the following functors ("right-linear" and "left-linear" respectively)

$$\mathrm{RHom} : {}_B\mathbf{Mod}_A^{op}\times{}_C\mathbf{Mod}_A\to{}_C\mathbf{Mod}_B$$
$$\mathrm{LHom} : {}_A\mathbf{Mod}_B^{op}\times{}_A\mathbf{Mod}_C\to{}_B\mathbf{Mod}_C$$

For example if $\psi\in\mathrm{RHom}({}_BM_C,\,{}_AN_C)$ then the (A,B)-bimodule action is defined to be

$$(a\psi b)(m) := a\psi(bm)$$

and similarly if $\varphi\in\mathrm{LHom}({}_AM_B,\,{}_AN_C)$ then

$$(b\varphi c)(m) := \varphi(mb)c$$

which we may verify satisfies the axioms of a bimodule. In order to standardize notation we may write Hom_A in place of RHom *and* LHom.

If $B=C=\mathbb{Z}$ then we recover the simple case (3.3.26).

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*If A is a commutative ring then we've observed that A-**Mod** is a full subcategory of ${}_A\mathbf{Mod}_A$ and the functors* RHom, LHom *become equal to the simple case (3.3.26) when restricted to this subcategory.*

3.3.11 Module Direct Product and Sum

Definition 3.3.80 (External Direct Product / Sum)
*Let A be a ring and $\{M_i\}_{i\in I}$ a family of A-modules. Define the **external direct product** as the set of ordered tuples indexed over I*

$$\prod_{i\in I} M_i \coloneqq \left\{ (m_i)_{i\in I} \;\middle|\; m_i \in M_i \right\}$$

*with the obvious module operations. The **external direct sum** is the subset of tuples for which all but finitely many elements are zero. We denote this as follows*

$$\bigoplus_{i\in I} M_i$$

Clearly when I is finite then these are equal.

Proposition 3.3.81 (Categorical Product)
Let $\{M_i\}_{i\in I}$ be a family of left A-modules and consider the family of projections

$$\pi_i : \prod_{i\in I} M_i \to M_i$$

For Z an (A,C)-bimodule there is a natural isomorphism of left C-modules

$$\begin{aligned} \mathrm{Hom}_A\left(Z, \prod_{i\in I} M_i\right) &\xrightarrow{\sim} \prod_{i\in I} \mathrm{Hom}_A(Z, M_i) \\ \theta &\to (\pi_i \circ \theta)_{i\in I} \end{aligned}$$

This in particular includes the case $A = C$ is commutative.

Proposition 3.3.82 (Categorical Coproduct)
Let $\{M_i\}_{i\in I}$ be a family of left A-modules and consider the family of inclusions

$$u_i : M_i \to \bigoplus_{i\in I} M_i$$

For Z a (A,C)-bimodule there is a natural isomorphism of right C-modules

$$\begin{aligned} \mathrm{Hom}_A\left(\bigoplus_{i\in I} M_i, Z\right) &\xrightarrow{\sim} \prod_{i\in I} \mathrm{Hom}_A(M_i, Z) \\ \theta &\to (\theta \circ u_i)_{i\in I} \end{aligned}$$

This in particular includes the case $A = C$ is commutative.

Corollary 3.3.83
Let $(M_i)_{i\in I}$ and $(N_j)_{j\in J}$ be families of left A-modules. Then there is an isomorphism of abelian groups

$$\begin{aligned} \mathrm{Hom}_A\left(\bigoplus_{i\in I} M_i, \prod_{j\in J} N_j\right) &\cong \prod_{(i,j)\in I\times J} \mathrm{Hom}_A(M_i, N_j) \\ \psi &\to (\pi_j \circ \psi \circ u_i)_{(i,j)} \end{aligned}$$

where $\pi_j : \prod_{j\in J} N_j \to N_j$ and $u_i : M_i \to \bigoplus_{i\in I} M_i$ are the canonical projections and injections respectively.

These maps are A-linear if A is commutative.

Definition 3.3.84 (Free Module)
An A-module M is

- ***free** if it is isomorphic to*

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$$\bigoplus_{i\in A} A =: A^{(I)}$$

for some indexing set I

- ***finite free*** *if it is free with respect to a* finite *indexing set I*

Under the isomorphism $M \to \bigoplus_{i\in I} A$ the set of elements $\{m_i\}_{i\in I}$ corresponding to the standard basis vectors e_i is called a ***basis*** *for M.*

We say that M has ***rank*** *n if there exists a basis of order n. NB rank is not necessarily unique.*

Proposition 3.3.85
Let M be an (A, C)-bimodule then there is a canonical isomorphism of right C-modules

$$\begin{aligned}\mathrm{Hom}_A(A, M) \cong \quad & M \\ \theta \to \quad & \theta(1_A)\end{aligned}$$

This in particular includes the case $A = C$ is commutative.

Proposition 3.3.86
Let M be a free left A-module with basis $(m_i)_{i\in I}$ and N an (A, C)-bimodule then there is an isomorphism of right C-modules

$$\begin{aligned}\mathrm{Hom}_A(M, N) \cong \quad & \prod_{i\in I} N \\ \theta \to \quad & (\theta(m_i))_{i\in I}\end{aligned}$$

This in particular includes the case $A = C$ is commutative.

3.3.12 Free Modules

Definition 3.3.87 (Faithful Module)
We say an A-module M is ***faithful*** *if*

$$am = 0 \quad \forall m \in M \implies a = 0$$

Definition 3.3.88 (Linearly Independent, Spanning and Basis)
Let M be an A-module and $S \subset M$ a set. We say S is

- ***spanning*** *if $\langle S\rangle = M$*
- ***linearly independent*** *if for every finite subset $\{s_1, ..., s_n\} \subseteq S$ with s_i distinct we have*

$$\sum_{i=1}^{n} a_i s_i = 0 \implies a_i = 0 \quad 1 \le i \le n$$

- *a* ***basis*** *if it is both spanning and linearly independent*

Definition 3.3.89 (Finite Module)
An A-module M is ***finite*** *if there exists a finite spanning set.*

Definition 3.3.90 (Minimal spanning set)
Let M be an A-module. Then $S \subset M$ is a ***minimal spanning set*** *if it generates M and no proper subset does so.*

Definition 3.3.91 (Free Module)
Let M be an A-module. We say that M is a ***free module*** *over A if it has a basis.*

Proposition 3.3.92 (Free A-module is an external sum of A)
An A-module M is free if and only if it is isomorphic to $\bigoplus_{i\in I} A$ for some I. The isomorphism is given by

$$\sum_{i\in I} a_i m_i \to (a_i)_{i\in I}$$

Definition 3.3.93 (Internal Direct Sum Module)
An A-module M is an ***internal direct sum*** *of submodules $\{M_i\}_{i\in I}$ the canonical mapping of the external direct sum*

$$\bigoplus_{i\in I} M_i \to M$$

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is an isomorphism.

Proposition 3.3.94
Let M be an A-module and $\{M_i\}_{i\in I}$ a family of submodules. Then the following are equivalent

a) $\sum_{i\in I} M_i$ is the internal direct sum of the family $\{M_i\}_{i\in I}$
b) The relation $\sum_{i\in I} m_i = 0$ implies $m_i = 0$ for all $i \in I$
c) For any $i \in I$ we have $M_i \cap \left(\sum_{k\neq i} M_k\right) = \{0\}$

Proposition 3.3.95
*We say that M_1, M_2 are **supplementary** submodules of M if M is the internal direct sum of M_1 and M_2.*

*We say that M_1 is a **direct factor** of M if it is supplementary to another submodule.*

3.3.13 Exact Sequences

Definition 3.3.96 (Vector space)
*If k is a field and V a k-module, then we say V is a **vector space** over k.*

Remark 3.3.97
We will see that every vector space is free and every k-submodule is a direct factor.

Proposition 3.3.98 (Kernel)
*Let $\varphi : M \to N$ be an A-module homomorphism, then the **kernel** of φ*

$$\ker(\varphi) \coloneqq \{m \in M \mid \varphi(m) = 0\}$$

is an A-submodule of M. Observe φ is injective iff $\ker(\varphi) = 0$.

Proposition 3.3.99 (Image)
Let $\varphi : M \to N$ be an A-module homomorphism then the image

$$\mathrm{Im}(\varphi) = \{\varphi(m) \mid m \in M\}$$

is an A-submodule of N.

Definition 3.3.100 (Quotient Module)
*Let $N \subseteq M$ be an A-submodule then define the **quotient module** M/N to be the quotient group with an action of A given by*

$$a(m + N) = (am + N)$$

When $N \subseteq P \subseteq M$ is a sequence of submodules then define the A-submodule P/N of M/N by

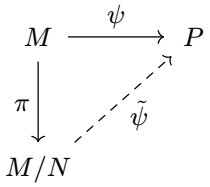
$$P/N \coloneqq \{p + N \mid p \in P\}$$

Proposition 3.3.101 (Quotient Module Properties)
Let $N \subseteq M$ be an A-submodule then there is a canonical surjective morphism

$$\pi : M \to M/N$$

with the following properties

a) $\pi(m) = m + N$
b) $\ker(\pi) = N$
c) Every homomorphism $\psi : M \to P$ such that $N \subseteq \ker(\psi)$, factors uniquely through π



Furthermore there is a bijection of A-submodules

$$\{P' \subseteq M/N\} \leftrightarrow \{P \mid N \subseteq P \subseteq M\}$$

given by $P' = P/N$. In the situation above $\ker(\tilde{\psi}) = \ker(\psi)/N$. In particular if $\ker(\psi) = N$ then $\tilde{\psi}$ is injective.

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Corollary 3.3.102
Let $\psi : M \to N$ be an A-module homomorphism, then this induces an isomorphism

$$M/\ker(\psi) \cong \mathrm{Im}(\psi)$$

Definition 3.3.103 (Exact Sequence)
*Let $N \xrightarrow{\varphi} M \xrightarrow{\psi} P$ be an sequence of A-module homomorphisms. We say it is **exact** if*

$$\mathrm{Im}(\varphi) = \ker(\psi)$$

It is equivalent to the following two conditions

a) $\psi \circ \varphi = 0$
b) $\psi(m) = 0 \implies m = \varphi(n)$ *for some* $n \in N$.

An exact sequence of the form

$$0 \to N \to M \to P \to 0$$

*is said to be **short-exact**.*

Remark 3.3.104
There are a few trivial observations

- $0 \to M \to N$ *is exact if and only if the map* $M \to N$ *is injective*
- $M \to N \to 0$ *is exact if and only if the map* $M \to N$ *is surjective.*

Proposition 3.3.105 (Isomorphism induced by short-exact sequence)
Let $N \subseteq M$ be a A-submodule then there is a canonical short-exact sequence

$$0 \to N \to M \to M/N \to 0$$

Conversely suppose we have a short exact sequence

$$0 \to N \xrightarrow{i} M \xrightarrow{\pi} P \to 0$$

then this induces an isomorphism

$$M/i(N) \cong P$$

If N is a submodule of M then we would simply write $M/N \cong P$.

Proposition 3.3.106 (Third Isomorphism Theorem)
Let $N \subseteq N' \subseteq M$ be a chain of modules then there is a short-exact sequence

$$0 \to N'/N \to M/N \to M/N' \to 0$$

which then induces an isomorphism

$$(M/N)/(N'/N) \cong M/N'$$

Proposition 3.3.107 (Product of ideal and quotient module)
Let N be a submodule of M and $\mathfrak{a} \lhd A$ an ideal. Then

$$\mathfrak{a}(M/N) = (N + \mathfrak{a}M)/N$$

Proposition 3.3.108 (Induced module)
Let M be an A-module and $\mathfrak{a}$ an ideal such that $\mathfrak{a}M = 0$, then M is naturally an $A/\mathfrak{a}$-module with action given by

$$\overline{a} \cdot m := a \cdot m$$

3.3.14 Dual Module

Proposition 3.3.109 (Dual Module)
Let M be a left (resp. right) A-module. Then the set

$$M^\vee := \mathrm{Hom}_A(M, A)$$

is canonically a right *(resp.* left*) A-module.*

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If M is a finite free left (resp. right) A-module with basis $\{v_1, ..., v_n\}$ then $M^\vee$ is a finite free right *(resp.* left*) A-module with basis $\{v_1^\vee, ..., v_n^\vee\}$ where these are the unique homomorphisms satisfying*

$$v_i^\vee(v_j) = \delta_{ij}$$

Moreover every basis of $M^\vee$ is of this form.

Proposition 3.3.110 (Hom-Set is free)
Let M be a finite free left A-module with basis $\{v_1, ..., v_n\}$ and N a (A, B)-bimodule. Then there is an isomorphism of right B-modules

$$\begin{aligned} \mathrm{Hom}_A(M, N) \cong & \bigoplus_{i=1}^{n} N \\ \theta \to & (\theta(v_i))_{i \in I} \end{aligned}$$

If A is commutative and N is a finite-free A-module with basis $\{w_1, ..., w_m\}$ then there is further an isomorphism of A-modules

$$\begin{aligned} \mathrm{Hom}_A(M, N) \cong & \bigoplus_{i,j=1}^{n,m} A \\ \theta \to & \left(w_j^\vee(\theta(v_i))\right)_{i,j} \end{aligned}$$

where $w_1^\vee, ..., w_m^\vee$ is the dual basis for $N^\vee$. In particular $\mathrm{Hom}_A(M, N)$ is a finite-free A-module with basis

$$\left\{w_j v_i^\vee\right\}_{i,j}$$

Definition 3.3.111 (Dual Functor)
Let A be a commutative ring and $\varphi : M \to N$ an A-module homomorphism. Define the dual homomorphism $\varphi^\vee : N^\vee \to M^\vee$ by

$$\varphi^\vee(\psi) := \psi \circ \varphi$$

This determines a contravariant functor

$$(-)^\vee : A\textbf{-Mod} \to A\textbf{-Mod}$$

Corollary 3.3.112 (Double Dual Natural Isomorphism)
Let A be a commutative ring and M a finite free A-module then the canonical A-module homomorphism

$$\begin{aligned} \eta : M \longrightarrow & \quad M^{\vee\vee} \\ x \to & \quad (\varphi \to \varphi(x)) \end{aligned}$$

is an isomorphism, which is natural in M.

Corollary 3.3.113
The contravariant functor $(-)^\vee : \mathbf{FiniteFreeMod}_A \to \mathbf{FiniteFreeMod}_A$ is an equivalence of categories and therefore full and faithful.

Proof. Use the dual isomorphism η together with (2.6.29) and (2.6.28). ∎

3.3.15 Matrices

For this section we assume that A is a commutative ring. In this context $A^n := A \times ... \times A$ is a finite free module with basis $e_1, ..., e_n$. Matrices are concrete realisations of linear maps of finite free modules.

Proposition 3.3.114
Let M, N be finite free A-modules with ordered bases $\mathcal{B} := \{v_1, ..., v_n\}$, $\mathcal{B}' := \{w_1, ..., w_m\}$ respectively. Let $\varphi : M \to N$ be an A-linear map and define $[\varphi]$ to be the unique $m \times n$ matrix satisfying the following condition

$$\varphi(v_i) = \sum_{j=1}^{m} [\varphi]_{ji} w_j \quad \forall i = 1...n$$

*(**notice sum over the first index**). Then it is also the unique matrix ensuring that the following diagram commutes*

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$$\begin{array}{ccc} M & \xrightarrow{\varphi} & N \\ \uparrow \sim & & \uparrow \sim \\ A^n & \xrightarrow{x \to [\varphi]x} & A^m \end{array}$$

where the bottom arrow is given by usual multiplication of a matrix by a column vector

Proposition 3.3.115 (Matrices as linear maps)
Let M, N be free A-modules with ordered bases $\mathcal{B} := \{v_1, ..., v_n\}$, $\mathcal{B}' := \{w_1, ..., w_m\}$ respectively. Then there are mutually inverse isomorphisms of A-modules

$$\begin{aligned} \mathrm{Mat}_{m\times n}(A) &\longleftrightarrow \mathrm{Hom}_A(M, N) \\ E &\to \hat{E} \\ [\varphi]^{\mathcal{B}}_{\mathcal{B}'} &\leftarrow \varphi \end{aligned}$$

where

$$\begin{aligned} \hat{E}\left(\sum_{i=1}^{n} \lambda_i v_i\right) &:= \sum_{j=1}^{m}\left(\sum_{i=1}^{n} E_{ji}\lambda_i\right) w_j \\ \varphi(v_i) &= \sum_{j=1}^{m} [\varphi]_{ji} w_j \end{aligned}$$

If we further consider a free A-module P with ordered bases $\mathcal{B}'' = \{u_1, ..., u_p\}$ then

$$\begin{aligned} \hat{E} \circ \hat{F} &= \widehat{EF} \\ [\psi \circ \varphi]^{\mathcal{B}}_{\mathcal{B}''} &= [\psi]^{\mathcal{B}'}_{\mathcal{B}''}[\varphi]^{\mathcal{B}}_{\mathcal{B}'} \end{aligned}$$

Observe that

$$[1_M]^{\mathcal{B}}_{\mathcal{B}} = I_n$$

Furthermore there is an isomorphism of A-algebras

$$\begin{aligned} \mathrm{End}_A(M) &\longleftrightarrow \mathrm{Mat}_{n,n}(A) \\ \varphi &\to \left(v_i^\vee(\varphi(v_j))\right)_{ij} \\ \sum_{ij} E_{ij} v_i v_j^\vee &\leftarrow E \end{aligned}$$

Corollary 3.3.116
Matrix multiplication is associative. In particular

$$(EF)v = E(Fv)$$

Proof. We may consider the free A-modules A^n, A^m and A^p with canonical bases. The result follows because function composition is associative and $\hat{\cdot}$ is injective. ∎

Corollary 3.3.117
There is an isomorphism of A-modules

$$\begin{aligned} \mathrm{Mat}_{m\times n}(A) &\longleftrightarrow \mathrm{Hom}_A(A^n, A^m) \\ E &\to (v \to Ev) \end{aligned}$$

and further an isomorphism of A-algebras

$$\mathrm{Mat}_{n,n}(A) \longleftrightarrow \mathrm{End}_A(A^n)$$

Corollary 3.3.118
Let M be a finite free A-module with bases $\mathcal{B}, \mathcal{B}'$ and $\varphi \in \mathrm{End}_A(M)$. Then φ is an isomorphism if and only if $[\varphi]^{\mathcal{B}}_{\mathcal{B}'} \in \mathrm{Mat}_{n\times n}(A)$ is invertible.

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Corollary 3.3.119 (Change of basis)
Let M be a finite free A-module and $\mathcal{B}, \mathcal{B}'$ bases then

$$[1_M]^{\mathcal{B}}_{\mathcal{B}'} = \left([1_M]^{\mathcal{B}'}_{\mathcal{B}}\right)^{-1}$$

and

$$[\varphi]^{\mathcal{B}'}_{\mathcal{B}'} = P[\varphi]^{\mathcal{B}}_{\mathcal{B}}P^{-1}$$

where

$$P := [1_M]^{\mathcal{B}}_{\mathcal{B}'}$$

is invertible.

Definition 3.3.120 (Transpose)
*Let E be an $m \times n$ matrix in A, then define the **transpose** of E to be the $n \times m$ matrix E^t where*

$$\left(E^t\right)_{ij} := E_{ji}$$

Proposition 3.3.121
Let M, N be finite-free A-modules with bases $\mathcal{B} = \{v_1, ..., v_n\}$ and $\mathcal{B}' = \{w_1, ..., w_m\}$. Let $\varphi : M \to N$ be an A-module homomorphism and $\varphi^\vee : N^\vee \to M^\vee$ the dual homomorphism then

$$[\varphi^\vee]^{\mathcal{B}'^\vee}_{\mathcal{B}^\vee} = \left([\varphi]^{\mathcal{B}}_{\mathcal{B}'}\right)^t$$

Similarly if E is an $m \times n$ matrix over A then

$$\hat{E}^\vee = \widehat{E^t}$$

where the right hand side is understood to be with respect to the dual bases.

Corollary 3.3.122
Let E, F be matrices then

$$(EF)^t = (FE)^t$$

3.3.16 Vector Spaces

Definition 3.3.123 (Vector Space)
*A **vector space** V over k is simply a k-module.*

*A k-submodule is referred to as a **subspace***

*A k-module homomorphism is referred to as a **linear map***

A vector space is finite-dimensional if it is finite as a k-module.

The main result on vector spaces is that bases exist and all have the same cardinality. Recall that $\langle\cdot\rangle$ is a closure operator. We show that $(V, \langle\cdot\rangle)$ determines a matroid so that we may appeal to results in the Matroids section. First we need to show that the notions of independence coincide

Proposition 3.3.124 (Equivalent definitions of linear independence)
Let V be a vector space and $S \subset V$. Then the following are equivalent

a) S is linearly independent
b) No proper subset $S' \subset S$ satisfies $\langle S' \rangle = \langle S \rangle$
*c) **Matroid Independence** $x \in S \implies x \notin \langle S \setminus \{x\} \rangle$*

Further S is independent if and only if every finite subset of S is.

Proof. b) $\iff$ c) This is b).

a) $\implies$ c). If $x \in \langle S \setminus \{x\} \rangle$ then it's clear that S is not linearly independent.

c) $\implies$ a). Suppose we have a linear relationship

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$$0 = \sum_{i=1}^{n} \lambda_i v_i \quad v_i \in S$$

By renumbering assume that $\lambda_1 \neq 0$, then rearrange to show $v_1 \in \langle S \setminus \{v_1\} \rangle$, contradicting the hypothesis. ∎

We show that $(V, \langle \cdot \rangle)$ satisfies the Exchange Property and therefore constitutes a matroid.

Proposition 3.3.125 (Exchange Property)
Let V be a vector space and $S \subset V$. Then

$$y \in \langle S \cup \{x\} \rangle \setminus \langle S \rangle \implies x \in \langle S \cup \{y\} \rangle$$

Proof. Suppose y is as given, then

$$y = \lambda x + \sum_i \lambda_i s_i \quad s_i \in S$$

The case $\lambda = 0$ is obvious. Otherwise we may rearrange to find

$$x = \lambda^{-1} y - \sum_i \lambda^{-1} \lambda_i s_i$$

whence $x \in \langle S \cup \{y\} \rangle$. ∎

Therefore we have the following

Proposition 3.3.126 (Vector Spaces are Free)
Every vector space has a basis, and in the finite-dimensional case every basis is finite of the same size. We denote this by $\dim_k V$.

More generally every linearly independent set is contained in a basis (so has order at most $\dim_k V$) and every spanning set contains a basis (so has order at least $\dim_k V$)

Proof. Follows from (2.3.7) and (2.3.12). The final statement follows from (2.3.13). ∎

Proposition 3.3.127 (Basis Criteria)
Let V be a vector space with $n = \dim_k V$ and $\mathcal{B} \subseteq V$. Then TFAE

a) $\mathcal{B}$ is a basis
b) $\mathcal{B}$ is linearly independent and $\#\mathcal{B} \geq \dim_k V$
c) $\mathcal{B}$ is spanning and $\#\mathcal{B} \leq \dim_k V$

Proof. The equivalence of a), b) and c) follows from (2.3.14). ∎

Proposition 3.3.128
A vector space $V = \{0\}$ if and only if $\dim_k V = 0$

Proposition 3.3.129 (Image of a basis)
Let $\varphi : V \to W$ be a linear map

a) If S is linearly-independent and φ is injective, then $\varphi(S)$ is linearly-independent
b) If Γ is spanning then (φ is surjective $\iff$ $\varphi(\Gamma)$ is spanning)
c) If $\mathcal{B}$ is a basis then (φ is an isomorphism $\iff$ $\varphi(\mathcal{B})$ is a basis and φ injective on $\mathcal{B}$)

Proof.

a) Suppose $\sum_i \lambda_i \varphi(s_i) = 0 \implies \varphi\left(\sum_i \lambda_i s_i\right) = 0$. As φ is injective this implies $\sum_i \lambda_i s_i = 0 \implies \lambda_i = 0$.
b) If φ is surjective then for $w \in W$ we have $\varphi(v) = w$ for some $v \in V$. By hypothesis $v = \sum_i \lambda_i \gamma_i$ and $w = \sum_i \varphi(\lambda_i)$. Conversely given $w \in W$ by hypothesis $w = \sum_i \lambda_i \varphi(\gamma_i) = \varphi\left(\sum_i \lambda_i \gamma_i\right)$ and φ is surjective as required.
c) Suppose φ is isomorphism, then it's surely injective on $\mathcal{B}$ and by a),b) $\varphi(\mathcal{B})$ is a basis. Conversely if $\varphi(\mathcal{B}) =: \mathcal{B}'$ is a basis then by b) φ is surjective. Suppose $\varphi(v) = 0$. Then by hypothesis $v = \sum_i \lambda_i v_i$ for $v_i \in \mathcal{B}$ and $0 = \varphi(v) = \sum_i \lambda_i \varphi(v_i)$. By hypothesis $\varphi(v_i)$ are distinct elements of the basis $\mathcal{B}'$ and therefore $\lambda_i = 0$ and $v = 0$. Therefore φ is injective and hence bijective.

∎

Corollary 3.3.130 (Dimension is an invariant)
Dimension is preserved under isomorphism. More generally for $\varphi : V \to W$ we have

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$$\varphi \text{ injective} \implies \dim_k V \leq \dim_k W$$

$$\varphi \text{ surjective} \implies \dim_k V \geq \dim_k W$$

Proposition 3.3.131
Let $W \subseteq V$ be finite-dimensional vector spaces then the dimension of the quotient module satisfies

$$\dim_k V/W = \dim_k V - \dim_k W$$

Proof. Let $\{v_1, ..., v_m\}$ be a basis of W, then there exists a basis $\{v_1, ..., v_m, v_{m+1}, ..., v_n\}$ of V containing the first by (3.3.126). We claim that

$$\{[v_{m+1}], ..., [v_n]\}$$

is a basis for V/W, and the result follows. For given $[v] \in V/W$ then

$$v = \sum_{i=1}^{n} \lambda_i v_i$$

since the basis is spanning. We have

$$v - \sum_{i=m+1}^{n} \lambda_i v_i \in W$$

therefore

$$[v] = \left[\sum_{i=m+1}^{n} \lambda_i v_i\right] = \sum_{i=m+1}^{n} \lambda_i [v_i]$$

and the given set is spanning. Similarly suppose

$$\sum_{i=m+1}^{n} \lambda_i [v_i] = 0$$

then by definition $\sum_{i=m+1}^{n} \lambda_i v_i \in W$. Therefore

$$\sum_{i=m+1}^{n} \lambda_i v_i = \sum_{i=1}^{m} \lambda_i v_i$$

and since v_i are linearly independent we must have $\lambda_i = 0$. ∎

Proposition 3.3.132 (Injective Criteria)
Let $\varphi : V \to W$ be a linear map then

$$\varphi \text{ injective} \iff \ker(\varphi) = \{0\} \iff \dim_k \ker(\varphi) = 0$$

Proof. Note for any linear map φ we have $\varphi(0) = 0$. Therefore φ injective clearly shows $\ker(\varphi) = \{0\}$. Conversely suppose $\ker(\varphi) = 0$ and $\varphi(v) = \varphi(w)$. Then $\varphi(v - w) = 0 \implies v - w = 0 \implies v = w$ as required. ∎

Definition 3.3.133 (Rank)
Let $\varphi : V \to W$ be a linear map then define

$$\operatorname{rank}_k(\varphi) := \dim_k(\operatorname{Im}(\varphi))$$

Proposition 3.3.134 (Surjective Criteria)
Let $\varphi : V \to W$ be a linear map with W finite-dimensional then

$$\varphi \text{ surjective} \iff \operatorname{rank}_k(\varphi) = \dim_k W$$

Proof. This follows directly from (2.3.15). ∎

Proposition 3.3.135 (Isomorphism Theorem / Rank-Nullity)
Let $\varphi : V \to W$ be a linear map then this induces an isomorphism

$$V/\ker(\varphi) \xrightarrow{\sim} \operatorname{Im}(\varphi)$$

in particular when V is finite-dimensional

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$$\dim_k V = \dim_k \ker(\varphi) + \operatorname{rank}_k(\varphi)$$

Corollary 3.3.136 (Isomorphism Criteria)
Let V, W vector spaces with W finite-dimensional. A linear map $\varphi : V \to W$ is an isomorphism if and only if any two *of the following are satisfied*

a) $\dim_k \ker(\varphi) = 0 \iff \varphi$ *injective*
b) $\dim_k V = \dim_k W$
c) $\operatorname{rank}_k(\varphi) = \dim_k W \iff \varphi$ *surjective*

Proof. The rank-nullity equation ensures that if any two hold the third is automatically satisfied. In this case φ is isomorphism as required. ∎

Corollary 3.3.137 (Endomorphism Isomorphism Criteria)
Let V be a finite-dimensional vector space and $\varphi : V \to V$ then TFAE

a) φ *is injective*
b) φ *is surjective*
c) φ *is an isomorphism*

Proof. For the equivalence of a), b) and c) we may use the previous result with $W = V$ and note $\dim_k W = \dim_k V$ is automatically satisfied. ∎

Proposition 3.3.138 (Internal Direct Sum)
Let U_1, U_2 be two subspaces of V then TFAE

a) $U_1 \cap U_2 = \{0\}$ *and* $V = U_1 + U_2$
b) Every $v \in V$ *may be written uniquely as* $u_1 + u_2$ *for* $u_i \in U_i$.

and we say $V = U_1 \oplus U_2$ is an ***internal direct sum*** *and U_2 is a* ***supplementary subspace*** *for U_1.*

Proposition 3.3.139 (Subspaces are direct factors)
Every subspace U has a supplementary subspace U' such that

$$V = U \oplus U'$$

Proof. Let $\mathcal{B}_1$ be a basis for U and extend to a basis $\mathcal{B}$ and define $\mathcal{B}_2 := \mathcal{B} \setminus \mathcal{B}_1$. Then it's easy to show that $U' = \langle \mathcal{B}_2 \rangle$ is a supplementary subspace. ∎

Proposition 3.3.140 (Dimension formula for direct sums)
Suppose $V = U_1 \oplus U_2$, $\mathcal{B}_1$ is a basis for U_1 and $\mathcal{B}_2$ is a basis for U_2. Then $\mathcal{B}_1 \cap \mathcal{B}_2 = \emptyset$ and $\mathcal{B}_1 \cup \mathcal{B}_2$ is a basis for V. In particular

$$\dim_k V = \dim_k U_1 + \dim_k U_2$$

3.3.16.1 Dual Space

Definition 3.3.141 (Dual Space)
Let V be a k-vector space and define the ***dual space*** *to be*

$$V^\vee := \operatorname{Hom}_k(V, k)$$

This is an abelian group and even a k-vector space under the obvious operations. The construction $V \to V^\vee$ determines a contravariant functor

$$(-)^\vee : \mathbf{Vect}_k \to \mathbf{Vect}_k$$

Definition 3.3.142 (Annihilator)
Let V be a vector space and $U \subseteq V$ a subspace. Define the ***annihilator*** *of U by*

$$U^\circ = \{\theta \in V^\vee \mid \theta(u) = 0 \quad \forall u \in U\}$$

This is a linear subspace of $V^\vee$.

Proposition 3.3.143 (Dimension formula for annihilators)
There is a canonical isomorphism by restriction

$$V^\vee / U^\circ \to U^\vee$$

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In particular when V is a finite-dimensional vector space then

$$\dim_k V = \dim_k U + \dim_k U^\circ$$

Proof. Let W be a supplementary subspace and consider the morphism $V = U \oplus W \xrightarrow{\pi_U} U$. Then $(\theta \circ \pi_U)|_U = \theta$ so the restriction map is surjective. Clearly the kernel is U°. The dimension formula follows from (3.3.135) and (3.3.109). ∎

Corollary 3.3.144 (Dual rank = rank)
Let $\varphi : V \to W$ be a linear map and $\varphi^\vee : W^\vee \to V^\vee$ then

$$\begin{aligned} \ker(\varphi^\vee) &= \operatorname{Im}(\varphi)^\circ \\ \operatorname{Im}(\varphi^\vee) &\subseteq \ker(\varphi)^\circ \end{aligned}$$

In the finite-dimensional case $\operatorname{Im}(\varphi^\vee) = \ker(\varphi)^\circ$ and

$$\begin{aligned} \dim_k \ker(\varphi^\vee) &= \dim_k W - \operatorname{rank}_k(\varphi) \\ \operatorname{rank}_k(\varphi^\vee) &= \operatorname{rank}_k(\varphi) \end{aligned}$$

Proof. Note $\ker(\varphi^\vee) = \operatorname{Im}(\varphi)^\circ$ and $\operatorname{Im}(\varphi^\vee) \subseteq \ker(\varphi)^\circ$ by the definitions.

Consider the finite-dimensional case. By (3.3.143)

$$\dim_k \ker(\varphi^\vee) = \dim_k \operatorname{Im}(\varphi)^\circ = \dim_k W - \operatorname{rank}_k(\varphi)$$

By rank-nullity applied to $\varphi^\vee$ and $\dim_k W = \dim_k W^\vee$ we deduce

$$\operatorname{rank}_k(\varphi^\vee) = \operatorname{rank}_k(\varphi).$$

By (3.3.143) and rank-nullity applied to φ

$$\dim_k \ker(\varphi)^\circ = \dim_k V - \dim_k \ker(\varphi) = \operatorname{rank}_k(\varphi).$$

Finally by (2.3.15) $\operatorname{Im}(\varphi^\vee) = \ker(\varphi)^\circ$. ∎

From this it follows that taking duals reflects and preserve isomorphisms

Corollary 3.3.145 ($(-)^\vee$ reflects isomorphisms)
Let $\varphi : V \to W$ be a linear map of finite-dimensional spaces then

a) φ is injective if and only if $\varphi^\vee$ is surjective
b) φ is surjective if and only if $\varphi^\vee$ is injective
c) φ is iso if and only if $\varphi^\vee$ is iso

Proof. Note by (3.3.144) we have $\operatorname{rank}_k(\varphi) = \operatorname{rank}_k(\varphi^\vee)$ and by (3.3.109) $\dim_k V^\vee = \dim_k V$.

φ is surjective $\iff \operatorname{rank}_k(\varphi) = \dim_k W = \operatorname{rank}_k(\varphi^\vee) \overset{(3.3.135)}{\iff} \dim_k \ker(\varphi^\vee) = 0 \iff \ker(\varphi^\vee) = \{0\}$

φ is injective $\iff \dim_k \ker(\varphi) = 0 \overset{(3.3.135)}{\iff} \operatorname{rank}_k(\varphi) = \dim_k V \iff \dim_k V^\vee = \operatorname{rank}_k(\varphi^\vee) \iff \varphi^\vee$ is surjective.

The last point may be deduced from the first two, or the fact that $(-)^\vee$ is full and faithful (3.3.113) and category-theoretic result (2.6.42). ∎

3.3.16.2 Bilinear Pairings

Definition 3.3.146 (Bilinear maps)
Let V, W be vector spaces a bilinear map ψ is a map

$$\psi : V \times W \to k$$

which is k-linear in each variable. We denote the set of bilinear maps as

$$\operatorname{Bilin}_k(V, W)$$

Proposition 3.3.147 (Matrix Representation)
Let V, W be finite-dimensional vector spaces with bases $\{v_1, ..., v_n\}$ and $\{w_1, ..., w_m\}$ then there is a canonical isomorphism

$$\begin{aligned} \operatorname{Bilin}_k(V, W) &\xrightarrow{\sim} \operatorname{Mat}_{n,m}(k) \\ \psi &\to \left(\psi(v_i, w_j)\right)_{ij} \end{aligned}$$

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In particular a bilinear map ψ is determined uniquely by the values $\psi(v_i, w_j)$.

Proposition 3.3.148 (Dual maps)
Let V and W be vector spaces, then there is a natural bijection

$$\begin{array}{ccccc} \mathrm{Mor}_k(V, W^\vee) & \leftrightarrow & \mathrm{Bilin}_k(V, W) & \leftrightarrow & \mathrm{Mor}_k(W, V^\vee) \\ \psi_L & \leftarrow & \psi & & \\ & & \psi & & \to \psi_R \end{array}$$

where

$$\psi_L(v)(w) = \psi(v, w) = \psi_R(w)(v)$$

When V, W are finite-dimensional then ψ_L is an isomorphism if and only if ψ_R is an isomorphism. In this case we say ψ is a perfect pairing. More generally

$$\mathrm{rank}_k(\psi_L) = \mathrm{rank}_k(\psi_R)$$

Proof. The bijections stated are obvious. One may show that $\psi_L = \psi_R^\vee \circ \eta_V$ where $\eta_V : V \to V^{\vee\vee}$ is the dual isomorphism. Therefore ψ_L is an isomorphism if and only if $\psi_R^\vee$ is an isomorphism, and by (3.3.145) if and only if ψ_R is an isomorphism. Since η_V is surjective we have $\mathrm{rank}_k(\psi_L) = \mathrm{rank}_k(\psi_R^\vee) = \mathrm{rank}_k(\psi_R)$, by (3.3.144). ∎

Proposition 3.3.149
Let $\psi : V \times V \to k$ be a bilinear map. Then the following are equivalent

a) ψ is a perfect pairing
b) ψ_L is an isomorphism
c) ψ_R is an isomorphism
d) ψ_L is injective
e) ψ_R is injective
f) ψ_L is surjective
g) ψ_R is surjective.

Proof. We may use the fact $\dim_k V = \dim_k V^\vee$ and (3.3.136) to show b) ⟺ d) ⟺ f) and c) ⟺ e) ⟺ g). We've already shown that a) ⟺ b) ⟺ c). ∎

Definition 3.3.150 (Orthogonal Complement)
Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces. Suppose $U \subset V$ is a subspace then define the ***orthogonal complement***

$$U^\perp := \{w \in W \mid \psi(v, w) = 0 \quad \forall v \in U\}$$

Proposition 3.3.151
Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces and $U \subset V$ a subspace. Then

$$\dim_k U + \dim_k U^\perp = \dim_k V$$

Indeed ψ_R induces an isomorphism $U^\perp \to U^\circ$.

Proof. We claim that $\psi_R(U^\perp) = U^\circ$. For if $w \in U^\perp$ then $\psi_R(w)(v) = \psi(w, v) = 0$ for all $v \in U$, and so $\psi_R(w) \in U^\circ$. Conversely given $\theta \in U^\circ$, as ψ_R is surjective, there is $w \in W$ such that $\psi_R(w) = \theta$. By definition $w \in U^\perp$ as required.

As ψ_R is injective then $\dim_k U^\perp \overset{(3.3.130)}{=} \dim_k U^\circ \overset{(3.3.143)}{=} \dim_k V/U = \dim_k V - \dim_k U$. ∎

Remark 3.3.152
In the case $V = W$, then it's not necessarily true that $U \cap U^\perp = \{0\}$, and so $U^\perp$ is not necessarily a complementary subspace.

The classic example is the perfect pairing on $\mathbb{R}^n$ induced by vDv^T for a real diagonal matrix D. Then it's true in general if and only if D is positive-definite.

Proposition 3.3.153 (Quotients are dual to subspaces)
Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces. Suppose $U \subset V$ is a subspace, then there is a canonical perfect pairing

$$\psi' : V/U \times U^\perp \to k$$

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given by

$$\psi'(v+U,w)=\psi(v,w)$$

Proof. The given map is well defined, for suppose $v_1+U=v_2+U$ then $v_1-v_2\in U\implies\psi(v_1-v_2,w)=0\quad\forall w\in U^\perp\implies\psi(v_1,w)=\psi(v_2,w)$ as required. It's clearly k-bilinear.

It's clear that ψ'_R is injective, because $\psi'_R(w)=0_{V/U}\implies\psi_R(w)=0_V\implies w=0$.

By the previous Proposition $\dim_k U^\perp=\dim_k V/U$. Therefore by (3.3.136) ψ'_R is an isomorphism and ψ' is perfect.∎

3.3.17 Inverse Limits

We record the basic construction of an inverse limit of a tower of rings, modules, or algebras, as a concrete model for the categorical inverse limit (2.6.129), together with its functoriality and its compatibility with products; these are used repeatedly to build completions later on.

Definition 3.3.154 (Inverse Limit of a Tower)
Let $\mathcal{C}$ be one of **Ring**, *A-***Mod** *(for a fixed ring A), or Alg_A (for a fixed ring A, so in particular* $\mathbf{Alg}_k$*) – each an algebraic category (2.6.184) (via the usual axioms; (2.6.185), (2.6.186), (2.6.187) respectively). Let $\cdots\twoheadrightarrow M_2\twoheadrightarrow M_1\twoheadrightarrow M_0$ be a sequence of morphisms $\rho_n:M_n\to M_{n-1}$ of $\mathcal{C}$ $(n\geq 1)$. Define*

$$\varprojlim_n M_n:=\left\{(m_n)_{n\geq0}\in\prod_{n\geq0}M_n\;\middle|\;\rho_n(m_n)=m_{n-1}\text{ for every }n\geq1\right\},$$

a sub-object of $\prod_n M_n$ in $\mathcal{C}$ (a subring, submodule, or subalgebra respectively, the ambient operations acting componentwise), with the evident projections $\lambda_n:\varprojlim_m M_m\to M_n$. Writing $\rho_{n,n}:=1_{M_n}$ and $\rho_{n,m}:=\rho_{m+1}\circ\cdots\circ\rho_n:M_n\to M_m$ for $n>m$ (composites of consecutive ρ_k's), every $(m_n)_{n\geq0}\in\varprojlim_n M_n$ satisfies the ***repeated compatibility***

$$\rho_{n,m}(m_n)=m_m\quad\text{for all }n\geq m,$$

not just for consecutive n,m: by induction on $n-m$, composing the defining relations $\rho_k(m_k)=m_{k-1}$ for $m<k\leq n$.

Lemma 3.3.155 (This Realises the Categorical Inverse Limit)
In the setting of (3.3.154), regard $I:=\mathbb{N}$ (usual order) as a directed preorder and the tower as an inverse system $F:I^{op}\to\mathcal{C}$, $F(n):=M_n$, with restriction maps $\rho_{n,m}$ as in (3.3.154). Then $\left(\varprojlim_n M_n,\{\lambda_n\}\right)$ is an inverse limit for F in the sense of (2.6.129).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\mathcal{C}$ is an algebraic category (2.6.184), in each of the three cases (as recalled in (3.3.154)). By (2.6.191), $\varprojlim_n F(n)$ exists in $\mathcal{C}$, with underlying set $\left\{(m_n)_n\in\prod_n M_n:\rho_{n,m}(m_n)=m_m\text{ for all }n\geq m\right\}$ and the ambient operations acting coordinatewise; by the repeated compatibility (3.3.154), this defining condition holds for all $n\geq m$ as soon as it holds for consecutive $n=m+1$, so this is exactly the sub-object $\varprojlim_n M_n$ of (3.3.154), with the same (componentwise) structure and projections. So $\left(\varprojlim_n M_n,\{\lambda_n\}\right)$ is this categorical inverse limit. ∎

Corollary 3.3.156 (Functoriality of the Inverse Limit)
Let $\mathcal{C}$ be as in (3.3.154), $(A_n,\rho_n)_{n\geq0}$ and $(B_n,\tau_n)_{n\geq0}$ towers in $\mathcal{C}$, and for each $n\geq0$ let $\sigma_n:A_n\to B_n$ be a morphism of $\mathcal{C}$ such that, for every $n\geq1$, the square

$$\begin{array}{ccc} A_n & \xrightarrow{\sigma_n} & B_n \\ {\scriptstyle\rho_n}\downarrow & & \downarrow{\scriptstyle\tau_n} \\ A_{n-1} & \xrightarrow[\sigma_{n-1}]{} & B_{n-1} \end{array}$$

commutes, i.e. $\tau_n\circ\sigma_n=\sigma_{n-1}\circ\rho_n$. Then

$$\varprojlim\sigma:\varprojlim_n A_n\longrightarrow\varprojlim_n B_n,\quad(a_n)_n\mapsto(\sigma_n(a_n))_n$$

is a well-defined morphism of $\mathcal{C}$; and if every σ_n is an isomorphism, so is $\lim_{\leftarrow}\sigma$.

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Proof. By (3.3.155), $\lim_{\leftarrow n} A_n$ and $\lim_{\leftarrow n} B_n$ (with their projections) are inverse limits, in the sense of (2.6.129), of the inverse systems $F, G : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ built from the two towers; the commuting squares above are exactly the naturality hypothesis of (2.6.132) for the family $\sigma = (\sigma_n)_n$ (composing squares for consecutive n gives it for the composite restriction maps $\rho_{n,m}$, $\tau_{n,m}$, $n > m$, as needed there). That corollary gives the induced morphism $\lim_{\leftarrow n} \sigma_n$, agreeing on each coordinate with σ_n by construction of the induced map on limits-as-sets (2.6.130) – that is, the map $(a_n)_n \to (\sigma_n(a_n))_n$ above – and an isomorphism whenever every σ_n is. ∎

Proposition 3.3.157 (Inverse Limits Commute with Products)
Let $\mathcal{C}$ be as in (3.3.154), K a set, and for each $k \in K$ a tower $(M_{k,n}, \rho_{k,n})_{n\geq 0}$ in $\mathcal{C}$. Then $\left(\prod_{k\in K} M_{k,n}\right)_{n\geq 0}$, with restriction maps $\prod_{k\in K} \rho_{k,n} : \prod_k M_{k,n} \to \prod_k M_{k,n-1}$, is again a tower in $\mathcal{C}$, and

$$\Phi : \prod_{k\in K} \lim_{\leftarrow n} M_{k,n} \longrightarrow \lim_{\leftarrow n} \left(\prod_{k\in K} M_{k,n}\right), \quad \Phi\big((x_k)_{k\in K}\big) := \Big(\big(x_{k,n}\big)_{k\in K}\Big)_{n\geq 0} \quad \big(x_k = (x_{k,n})_{n\geq 0}\big)$$

is a well-defined isomorphism of $\mathcal{C}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $F_k : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ for the inverse system of the tower $(M_{k,n})_n$, so $\left(\lim_{\leftarrow n} M_{k,n}, \{\lambda_{k,n}\}_n\right)$ realises $\lim_{\leftarrow} F_k$ (3.3.155), and likewise $\left(\lim_{\leftarrow n}\left(\prod_k M_{k,n}\right), \{\Lambda_n\}_n\right)$ realises $\lim_{\leftarrow}\left(\prod_k F_k\right)$, where Λ_n is projection onto the n-th coordinate. Since $\prod_{k\in K} \lim_{\leftarrow n} M_{k,n}$ (with the evident coordinate projections p_k) is the categorical product of the $\lim_{\leftarrow n} M_{k,n}$ in $\mathcal{C}$, (2.6.133) gives that $\lim_{\leftarrow n}\left(\prod_k M_{k,n}\right)$ exists (as it already does, here) and that there is a *unique morphism* Φ' (of $\mathcal{C}$, not assumed a priori to be an isomorphism) such that

$$\pi_{k,n} \circ \Lambda_n \circ \Phi' = \lambda_{k,n} \circ p_k \quad \text{for all } k \in K, n \geq 0,$$

where $\pi_{k,n} : \prod_{k'} M_{k',n} \to M_{k,n}$ is the product projection; and that this Φ' is an isomorphism.

The explicit Φ above satisfies exactly this defining property: for $(x_k)_k \in \prod_k \lim_{\leftarrow n} M_{k,n}$ (so $x_k = (x_{k,n})_n$), both $\pi_{k,n}\big(\Lambda_n\big(\Phi\big((x_k)_k\big)\big)\big)$ and $\lambda_{k,n}\big(p_k\big((x_k)_k\big)\big)$ equal $x_{k,n}$. By uniqueness of the morphism with this property, $\Phi = \Phi'$; as Φ' is an isomorphism, so is Φ. ∎

Corollary 3.3.158 (Cofinal Reindexing of a Tower)
Let $\mathcal{C}$ be as in (3.3.154), $(M_n, \rho_n)_{n\geq 0}$ a tower in $\mathcal{C}$, and $n_0 < n_1 < n_2 < \cdots$ a strictly increasing sequence in $\mathbb{N}$. Then, writing $\tilde{\rho}_N := \rho_{n_{N+1},n_N} : M_{n_{N+1}} \to M_{n_N}$ for the composite restriction map of (3.3.155), $\left(M_{n_N}, \tilde{\rho}_N\right)_{N\geq 0}$ is again a tower in $\mathcal{C}$, and

$$(m_n)_{n\geq 0} \mapsto \left(m_{n_N}\right)_{N\geq 0}$$

defines an isomorphism $\lim_{\leftarrow n} M_n \xrightarrow{\sim} \lim_{\leftarrow N} M_{n_N}$ of $\mathcal{C}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Regard $I := \mathbb{N}$ as a directed preorder and, per (3.3.155), the tower $(M_n)_n$ as an inverse system $F : I^{\mathrm{op}} \to \mathcal{C}$ realising $\lim_{\leftarrow n} M_n$. Let $I' := \{n_N : N \geq 0\} \subset I$, a full subcategory: it is directed (a subset of the totally ordered set I), and cofinal, as $n_N \to \infty$ so every $n \in I$ satisfies $n \leq n_N$ for some N (2.6.126). The order-preserving bijection $\sigma : \mathbb{N} \to I'$, $\sigma(N) := n_N$, is an isomorphism of the directed preorders $\mathbb{N}$ and I', and $F\sigma : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ is exactly the tower $\left(M_{n_N}, \tilde{\rho}_N\right)_N$ (its restriction map for $N < N+1$ is $F(\sigma(N) \to \sigma(N+1)) = \rho_{n_{N+1},n_N} = \tilde{\rho}_N$, and composites thereof match for $N < N'$); so, by (2.6.98) (applied to σ) together with (3.3.155) applied to the tower $\left(M_{n_N}, \tilde{\rho}_N\right)_N$, $\left(\lim_{\leftarrow n} M_n, \{\lambda_{n_N}\}_{N\geq 0}\right)$ realises $\lim_{\leftarrow i\in I'} F(i)$, and $\left(\lim_{\leftarrow N} M_{n_N}, \{\lambda'_N\}_{N\geq 0}\right)$ (the latter's own projections) also realises it; identifying these two cones over $F|_{I'^{\mathrm{op}}}$ via the (unique, by terminality) comparison isomorphism between them, which sends $(m_n)_{n\geq 0} \in \lim_{\leftarrow n} M_n$ (viewed as a cone-point of the first) to the compatible sequence $\left(m_{n_N}\right)_{N\geq 0} \in \lim_{\leftarrow N} M_{n_N}$ agreeing with it on projections $\lambda_{n_N} = \lambda'_N \circ$ (this isomorphism).

By (2.6.131) (as I' is cofinal in I), the canonical morphism of cones $\lim_{\leftarrow i\in I} F(i) \to \lim_{\leftarrow i\in I'} F(i)$ is an isomorphism, and by construction (restriction of the limit cone's own projections λ_n to $i = n_N \in I'$) it sends $(m_n)_{n\geq 0} \to \left(m_{n_N}\right)_{N\geq 0}$ under the identification of the previous paragraph. Composing the two isomorphisms of this and the previous paragraph gives the stated isomorphism $\lim_{\leftarrow n} M_n \xrightarrow{\sim} \lim_{\leftarrow N} M_{n_N}$, $(m_n)_n \to \left(m_{n_N}\right)_N$. ∎

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3.4 Tensor Products

3.4.1 Commutative Tensor Product

We consider the "Tensor Product" of modules over a single commutative ring as an important special case, and as a guide to the general bimodule case.

In order to streamline the (typically tedious) verification of standard properties we make heavy use of the notion of **representable bifunctor** (2.6.65) however all the results may be proved in an essentially identical "low-tech" way.

First we define the so-called "Internal Hom Functor"

Definition 3.4.1 (Internal Hom)
Let A be a commutative ring and M, N be A-modules. The set of A-linear homomorphisms $M \to N$ is itself an A-module. Therefore we have an enriched hom functor

$$\mathrm{Hom} : A\text{-}\mathbf{Mod}^{op} \times A\text{-}\mathbf{Mod} \to A\text{-}\mathbf{Mod}$$

where

$$(a \cdot \psi)(m) := a \cdot \psi(m) = \psi(a \cdot m)$$

and in particular we have a bijection

$$\mathrm{Mor}(M, N) \xrightarrow{\sim} \mathrm{Nat}(\mathrm{Hom}(N, -), \mathrm{Hom}(M, -))$$

Definition 3.4.2 (Bilinear Map)
Let M, N, Z be A-modules. Then a map

$$\psi : M \times N \to Z$$

*is **bilinear** if it satisfies*

- ***additive*** $\psi(m + m', n) = \psi(m, n) + \psi(m', n)$ *and* $\psi(m, n + n') = \psi(m, n) + \psi(m, n')$
- ***A-linear*** $\psi(am, n) = \psi(m, an) = a\psi(m, n)$ *for all* $a \in A$

This yields a bifunctor which we denote by

$$\mathrm{Bilin}(-, -; -) : (A\text{-}\mathbf{Mod} \times A\text{-}\mathbf{Mod})^{op} \times A\text{-}\mathbf{Mod} \to A\text{-}\mathbf{Mod}$$

Remark 3.4.3 (A-linearity)
The A-module structure on the image of Hom *and* Bilin *is only well-defined because A is commutative. For given $\psi : M \to N$ an A-linear map then for $(a \cdot \psi)$ to be A-linear we may verify this requires*

$$a'a\psi(m) = (a\psi)(a'm) = aa'\psi(m)$$

which in general does not hold unless A is commutative.

Proposition 3.4.4 (Existence of Tensor Product)
The bifunctor $\mathrm{Bilin}(-, -; -)$ *is representable by a functor*

$$\otimes_A : A\text{-}\mathbf{Mod} \times A\text{-}\mathbf{Mod} \to A\text{-}\mathbf{Mod}$$

for which there exists an isomorphism natural in M, N, Z

$$\Phi : \mathrm{Hom}(M \otimes N, Z) \cong \mathrm{Bilin}(M, N; Z)$$

For every pair M, N there is a distinguished bilinear map

$$i : M \times N \to M \otimes N$$

*through which every bilinear map $M \times N \to Z$ factors uniquely. We denote the **elementary tensor** by*

$$m \otimes n := i(m, n)$$

Further for morphisms $f : M \to M'$ and $g : N \to N'$ there is a unique morphism

$$(f \otimes g) : M \otimes N \to M' \otimes N'$$

such that

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$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

Proof. In light of (2.6.66) it is sufficient to show that $\text{Bilin}(M, N; -)$ is representable for fixed M, N and the rest of the properties follow, observing that $i = \Phi(1_{M \otimes N})$.

For let F be the free abelian group on $M \times N$ and let $S \subset F$ be the subgroup generated by elements of the form

- $(m + m', n) - (m, n) - (m', n)$
- $(m, n' + n) - (m, n') - (m, n)$
- $(am, n) - a(m, n)$
- $(m, n) - (m, an)$

Observe there is a canonical map $M \times N \to F$. Suppose that Z is an abelian group and $\psi : M \times N \to Z$ is a bilinear map. Then we may extend by linearity to a group homomorphism $\psi : F \to Z$. By construction $S \subseteq \ker(\psi)$ whence there is a unique group homomorphism $\hat{\psi} : F/S \to Z$ such that $\hat{\psi} \circ i = \psi$ where $i : M \times N \to F/S$ is the canonical inclusion. This shows that $(F/S, i)$ is a universal element for $\text{Bilin}(-, -; -)$ as we required and we denote this by $M \otimes N$. The representation exists as a functor of sets by (2.6.62), and Φ is clearly an isomorphism of A-modules because

$$\Phi(f + g) = (f + g) \circ i = f \circ i + g \circ i = \Phi(f) + \Phi(g)$$

and

$$\Phi(\lambda f) = (\lambda f) \circ i = \lambda(f \circ i) = \lambda \Phi(f)$$

■

Lemma 3.4.5 (Currying Lemma)
Let M, N, Z be A-modules then there are natural isomorphisms

$$\begin{array}{ccccccc} \text{Bilin}(M, N, Z) & \cong & \text{Hom}(M, \text{Hom}(N, Z)) & \cong & \text{Hom}(N, \text{Hom}(M, Z)) & \cong & \text{Bilin}(N, M, Z) \\ \psi & & m \to (n \to \psi(m, n)) & & n \to (m \to \psi(m, n)) & & (n, m) \to \psi(m, n) \end{array}$$

This gives the following important result (which may be seen as an adjoint relationship)

Proposition 3.4.6 (Tensor-Hom Adjunction)
Let M, N, Z be A-modules then we have the following natural isomorphism of functors

$$\begin{array}{rl} \text{Hom}(M \otimes N, Z) \cong & \text{Hom}(M, \text{Hom}(N, Z)) \\ & \theta \to (m \to (n \to \theta(m \otimes n))) \end{array}$$

We may also use the Currying Lemma (3.4.5) to demonstrate symmetry of the tensor product

Proposition 3.4.7 (Symmetry of Tensor Product)
There is a natural isomorphism

$$\begin{array}{rl} M \otimes N \cong & N \otimes M \\ m \otimes n \to & n \otimes m \end{array}$$

Proof. We observe that there is a natural isomorphism of functors which the tensor products represent

$$\text{Bilin}(M, N; Z) \cong \text{Bilin}(N, M; Z)$$

so the result follows from (2.6.68). ■

Proposition 3.4.8 (Associativity of Tensor Product)
There is a natural isomorphism of A-modules

$$\begin{array}{rl} (M \otimes N) \otimes P \cong & M \otimes (N \otimes P) \\ (m \otimes n) \otimes p \to & m \otimes (n \otimes p) \end{array}$$

Proof. We may make repeated use of the Tensor-Hom adjunction to exhibit natural isomorphisms

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$$\begin{aligned}\mathrm{Hom}((M\otimes N)\otimes P, Z) &\cong \mathrm{Hom}(M\otimes N, \mathrm{Hom}(P,Z))\\ &\cong \mathrm{Hom}(M, \mathrm{Hom}(N, \mathrm{Hom}(P,Z)))\\ &\cong \mathrm{Hom}(M, \mathrm{Hom}(N\otimes P, Z))\\ &\cong \mathrm{Hom}(M\otimes(N\otimes P), Z)\end{aligned}$$

The required isomorphism is then a consequence of (2.6.68). For explicit form set $Z = M\otimes(N\otimes P)$ and consider the identity map 1_Z. We see tracing back under these natural isomorphisms this corresponds to the given map. ■

3.4.2 Bimodule Tensor Product

To develop the tensor product in general it is convenient to work with bimodules. We make heavy use of the notion if bimodule hom-sets (3.3.79).

Definition 3.4.9 (Bilinear maps)
Consider the bimodules ${}_AM_B$, ${}_BN_C$. We say that a map

$$\psi : {}_AM_B \times {}_BN_C \to {}_AZ_C$$

*is (A,C)-**bilinear** if it satisfies all of the following conditions*

- ***additive*** $\psi(m+m', n+n') = \psi(m,n)+\psi(m',n)+\psi(m,n')+\psi(n,n')$
- ***balanced*** $\psi(mb,n) = \psi(m,bn)$
- ***A-linear*** $\psi(am,n) = a\psi(m,n)$
- ***C-linear*** $\psi(m,nc) = \psi(m,n)c$

Extending the notation from earlier we denote the set of additive, balanced and (A,C)-bilinear maps by $\mathrm{Bilin}(M,N;Z)$. *This determines a functor*

$$\mathrm{Bilin} : ({}_A\mathbf{Mod}_B \times {}_B\mathbf{Mod}_C)^{op} \times {}_A\mathbf{Mod}_C \to \mathbf{AbGrp}$$

where addition is defined pointwise. We may also consider the functor of only additive and balanced maps

$$\mathrm{Balan} : (\mathbf{Mod}_B \times {}_B\mathbf{Mod})^{op} \times \mathbf{AbGrp} \to \mathbf{AbGrp}$$

and the functor

$$\mathrm{RBilin} : ({}_B\mathbf{Mod}_C \times {}_C\mathbf{Mod}_D)^{op} \times {}_A\mathbf{Mod}_D \to {}_A\mathbf{Mod}_B$$

where we drop the B-linear requirement but also introduce an extra (A,B)-module structure on the maps, as follows

$$(a\psi b)(m,n) \coloneqq a\psi(bm,n)$$

Proposition 3.4.10 (Existence of Bimodule Tensor Product)
Let A, B, C be arbitrary rings

- *The bifunctor* $\mathrm{Balan}(M,N;-)$ *is represented by the **balanced tensor product***

  $$\otimes_B : \mathbf{Mod}_B \times {}_B\mathbf{Mod} \to \mathbf{AbGrp}$$

  with natural isomorphism for Z an Abelian group

  $$\begin{aligned}\Phi : \mathrm{Hom}(M_B \otimes_B {}_BN, Z) &\cong \mathrm{Balan}(M_B, {}_BN; Z)\\ \theta &\to \theta\circ i\end{aligned}$$

  where i is a universal balanced map

  $$i : M\times N \to M_B\otimes_B {}_BN$$

  *The tensor product is generated by **elementary tensors** of the form $m\otimes n \coloneqq i(m,n)$.*

- *The bifunctor* $\mathrm{Bilin}(M,N;-)$ *is represented by the balanced tensor product $M\otimes_B N$ with an additional (A,C)-bimodule structure given by*

  $$a\cdot(m\otimes n)\cdot c \coloneqq (a\cdot m)\otimes(n\cdot c)$$

  This determines a bifunctor

  $$\otimes_B : {}_A\mathbf{Mod}_B \times {}_B\mathbf{Mod}_C \to {}_A\mathbf{Mod}_C$$

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and a natural isomorphism of Abelian groups

$$\begin{aligned}\mathrm{Hom}({}_AM_B \otimes_B {}_BN_C, Z) &\cong \mathrm{Bilin}({}_AM_B, {}_BN_C; {}_AZ_C)\\ \theta &\to \theta \circ i\end{aligned}$$

In either case, if $f : M \to M'$ and $g : N \to N'$ are morphisms then there is a unique morphism

$$(f \otimes g) : (M \otimes N) \to (M' \otimes N')$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

Furthermore for maps $f' : M' \to M''$ and $g' : N' \to N''$ the following property is satisfised

$$(f' \otimes g') \circ (f \otimes g) = (f' \circ f) \otimes (g' \circ g)$$

Proof. In light of (2.6.65) we only need to show that the functors are representable for fixed M, N, as the functoriality will follow immediately by abstract nonsense.

We first show that $\mathrm{Balan}(-,-;-)$ is representable. For let F be the free abelian group on $M \times N$ and let $S \subset F$ be the subgroup generated by elements of the form

- $(m + m', n) - (m, n) - (m', n)$
- $(m, n' + n) - (m, n') - (m, n)$
- $(mb, n) - (m, bn)$

Observe there is a canonical map $M \times N \to F$. Suppose that Z is an abelian group and $\psi : M \times N \to Z$ is a balanced additive map. Then we may extend by linearity to a group homomorphism $\psi : F \to Z$. By construction $S \subseteq \ker(\psi)$ whence there is a unique group homomorphism $\hat{\psi} : F/S \to Z$ such that $\hat{\psi} \circ i = \psi$ where $i : M \times N \to F/S$ is the canonical inclusion. This shows that $(F/S, i)$ is a universal element for $\mathrm{Balan}(-,-;-)$ and we denote this by $M_B \otimes_B {}_BN$. Then (2.6.62) exhibits Φ as a natural isomorphism of sets, which may readily observed to be additive.

In order to define the (A, C)-bimodule structure in the second case let $\psi_a : M \to M$ denote multiplication by a and $\psi_c : N \to N$ multiplication by c. We note these are both B-module homomorphisms by definition so we may define the action by functoriality of the balanced tensor product case

$$\begin{aligned}a \cdot v &:= (\psi_a \otimes 1_N)(v) \quad \forall v \in M \otimes N\\ v \cdot c &:= (1_M \otimes \psi_c)(v) \quad \forall v \in M \otimes N\end{aligned}$$

These are B-linear by construction and we may demonstrate (A, C)-bimodule actions are associative and commute by using the functoriality of the tensor product

$$\begin{aligned}(\psi_{a'} \otimes 1_N) \circ (\psi_a \otimes 1_N) = (\psi_{a'} \circ \psi_a) \otimes 1_N &= (\psi_{a'a}) \otimes 1_N\\ (1_M \otimes \psi_c) \circ (1_M \otimes \psi_{c'}) = 1_M \otimes (\psi_c \circ \psi_{c'}) &= 1_M \otimes \psi_{cc'}\\ (\psi_a \otimes 1_N) \circ (1_M \otimes \psi_c) = \psi_a \otimes \psi_c &= (1_M \otimes \psi_c) \otimes (\psi_a \otimes 1_N)\end{aligned}$$

Further by construction the canonical map i is (A, C)-bilinear. Consider any (A, C)-bilinear map $\psi : M \times N \to Z$ then by the previous part there is a unique group homomorphism $\hat{\psi} : M \otimes_B N \to Z$ such that

$$\hat{\psi}(m \otimes n) := \psi(m, n)$$

which we see by construction (and linearity) is (A, C)-bilinear. It is clearly the unique such map so that we may conclude $\mathrm{Bilin}(-,-;-)$ has universal element $({}_AM_B \otimes_B {}_BN_C, i)$. The (2.6.62) exhibits Φ as a natural isomorphism of sets which may be readily exhibited to be additive. ∎

Lemma 3.4.11 (Currying Lemma)
Let M, N, P, Y, Z be bimodules. Then there is an isomorphism of abelian groups and (A, B)-bimodules respectively

$$\begin{aligned}\mathrm{Bilin}({}_AM_B, {}_BN_C; {}_AY_C) &\cong \mathrm{Hom}({}_AM_B, \mathrm{RHom}({}_BN_C, {}_AY_C))\\ &\cong \mathrm{Hom}({}_BN_C, \mathrm{LHom}({}_AM_B, {}_AY_C))\\ \mathrm{RBilin}({}_BN_C, {}_CP_D; {}_AZ_D) &\cong \mathrm{RHom}({}_BN_C, \mathrm{RHom}({}_CP_D, {}_AZ_D))\end{aligned}$$

contravariant in M, N, P and covariant in Y and Z.

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Proof. The maps on abelian groups are clear, namely given a bi-additive map ψ we have a well-defined homomorphism of abelian groups

$$m \to (n \to \psi(m,n))$$

The verification that the maps are well-defined and bijective is tedious but mechanical. ∎

Proposition 3.4.12 (Tensor-Hom Adjunction)
Let M, N, P, Y, Z be bimodules then

$$\begin{aligned}\mathrm{Hom}({}_AM_B \otimes {}_BN_C, {}_AY_C) &\cong \mathrm{Hom}({}_AM_B, \mathrm{RHom}({}_BN_C, {}_AY_C))\\ &\cong \mathrm{Hom}({}_BN_C, \mathrm{LHom}({}_AM_B, {}_AY_C))\\ \mathrm{RHom}({}_BN_C \otimes {}_CP_D, {}_AZ_D) &\cong \mathrm{RHom}({}_BN_C, \mathrm{RHom}({}_CP_D, {}_AZ_D))\end{aligned}$$

where the first is an isomorphism of Abelian Groups and the second of (A,B)-bimodules. Further the isomorphisms are contravariant in M, N, P and covariant in Y and Z.

Proof. This first two isomorphism follow directly from the Currying Lemma and universal property for bimodule tensor product.

The final isomorphism follows from the Currying Lemma if we show that there is an isomorphism

$$\begin{array}{ccc}\mathrm{RHom}({}_BN_C \otimes {}_CP_D, {}_AZ_D) & \xrightarrow{\sim} & \mathrm{RBilin}({}_BN_C, {}_CP_D; {}_AZ_D)\\ \downarrow & & \downarrow\\ \mathrm{Hom}(N_C \otimes {}_CP, Z) & \xrightarrow{\sim} & \mathrm{Balan}(N_C, {}_CP; Z)\end{array}$$

where the bottom is simply the universal property of the balanced tensor product and the top we require to also be (A,B)-bilinear. Recall the underlying sets are the same and the horizontal maps are given by $\theta \to \theta \circ i$ in both cases. It's clear that the map is well-defined and injective. Further it's surjective because if θ is (B,D)-bilinear on elementary tensors it is (B,D)-bilinear everywhere. Finally we need to show that the isomorphism is (A,B)-bilinear as follows

$$(a(\theta \circ i)b)(n,p) = a(\theta \circ i)(bn,p) = a\theta((bn) \otimes p) = a\theta(b(n \otimes p)) = (a\theta b)(n \otimes p)$$

∎

Proposition 3.4.13 (Associativity of Tensor Product)
Let ${}_AM_B$, ${}_BN_C$, ${}_CP_D$ be bimodules then there is a natural isomorphism of (A,D)-bimodules

$$\begin{aligned}(M \otimes_B N) \otimes_C P &\cong M \otimes_B (N \otimes_C P)\\ (m \otimes n) \otimes p &\to m \otimes (n \otimes p)\end{aligned}$$

Proof. This follows much as in the commutative case. Let Z be an (A,D)-bimodule. Then by the Tensor-Hom adjunction there are natural isomorphisms of Abelian groups

$$\begin{aligned}\mathrm{Hom}(({}_AM_B \otimes {}_BN_C) \otimes {}_CP_D, {}_AZ_D) &\cong \mathrm{Hom}({}_AM_B \otimes {}_BN_C, \mathrm{RHom}({}_CP_D, {}_AZ_D))\\ &\cong \mathrm{Hom}({}_AM_B, \mathrm{RHom}({}_BN_C, \mathrm{RHom}({}_CP_D, {}_AZ_D)))\\ &\cong \mathrm{Hom}({}_AM_B, \mathrm{RHom}({}_BN_C \otimes_C {}_CP_D, {}_AZ_D))\\ &\cong \mathrm{Hom}({}_AM_B \otimes_B ({}_BN_C \otimes_C {}_CP_D), {}_AZ_D)\end{aligned}$$

The required isomorphism is then a consequence of (2.6.68). For explicit form set $Z = M \otimes (N \otimes P)$ and consider the identity map 1_Z. We see tracing back under these natural isomorphisms this corresponds to the given map. ∎

3.4.3 Extensions of Scalars

We observe that for a ring homomorphism $\varphi : A \to B$ we have on B a natural (A,B)-bimodule and (B,A)-bimodule structure. Using the tensor product then we may use this to extend the coefficients from A to B. We first prove some elementary lemmas

Lemma 3.4.14 (Unit Hom)
Let $\varphi : B \to C$ be a ring homomorphism and Z a (A,C)-bimodule. Then there is a natural isomorphism of (A,B)-bimodules

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$$\begin{aligned}\mathrm{RHom}({}_BC_C, {}_AZ_C) &\cong {}_AZ_B\\ \theta &\to \theta(1)\\ z(-) &\leftarrow z\end{aligned}$$

and similarly let $\varphi : A \to B$ be a ring homomorphism and Y a (B,C)-bimodule then there is a natural isomorphism of (A,C)-bimodules

$$\begin{aligned}\mathrm{LHom}({}_BB_A, {}_BY_C) &\cong {}_AY_C\\ \theta &\to \theta(1)\end{aligned}$$

Proof. Observe

$$(a\theta b)(1) = a\theta(\varphi(b)) = a\theta(1)\varphi(b)$$

so the left to right map is an (A,B)-bimodule homomorphism. Further $\theta(c) = \theta(1)c$ so the maps are mutual inverses.■

Lemma 3.4.15 (Unit Hom (Commutative Ring Case))
Let $\varphi : A \to B$ be a homomorphism of commutative rings and Z a B-module. Then there is a natural isomorphism of B-modules

$$\begin{aligned}\mathrm{Hom}(B, Z) &\cong Z\\ \theta &\to \theta(1)\end{aligned}$$

Proposition 3.4.16 (Extension of Scalars)
Let M be an (A,C)-bimodule and $\varphi : A \to B$ a ring homomorphism. We may define a (B,C)-module

$$M_{(B)} \coloneqq {}_BB_A \otimes_A M$$

For Z a (B,C)-bimodule there is a natural isomorphism of abelian groups

$$\begin{aligned}\mathrm{Hom}\Big(M_{(B)}, Z\Big) &\xrightarrow{\sim} \mathrm{Hom}(M, {}_AZ_C)\\ \psi &\to m \to \psi(1 \otimes m)\\ b \otimes m \to b \cdot \phi(m) &\leftarrow \phi\end{aligned}$$

In otherwords we have an adjunction (2.6.69)

$${}_A\mathbf{Mod}_C \longleftrightarrow {}_B\mathbf{Mod}_C$$

Setting $C = \mathbb{Z}$ yields an adjunction for left modules

$${}_A\mathbf{Mod} \longleftrightarrow {}_B\mathbf{Mod}$$

In particular for M a left A-module we have an isomorphism of abelian groups

$$\mathrm{Hom}\Big(M_{(B)}, B\Big) \xrightarrow{\sim} \mathrm{Hom}_A(M, B)$$

which is B-linear when B is commutative.

Proof. By the tensor-hom adjunction (3.4.12) and (3.4.14) there is a natural isomorphism

$$\mathrm{Hom}(B \otimes_A M, Z) \cong \mathrm{Hom}(M, \mathrm{LHom}({}_BB_A, {}_BZ_C)) \cong \mathrm{Hom}(M, {}_AZ_C)$$

■

Proposition 3.4.17
Let $\varphi : A \to B$ be a homomormorphism of commutative rings and $\varphi : M \to N$ a homomorphism of A-modules. Then there is a unique B-module homomorphism $\varphi_{(B)} : M_{(B)} \to N_{(B)}$ such that

$$\varphi_{(B)}(1 \otimes m) = 1 \otimes \varphi(m)$$

Proof. The homomorphism $\varphi_{(B)} = 1_B \otimes \psi$ exists by functoriality of the tensor product (3.4.4). It clearly satisfies the given property, and as the tensor product is generated by elementary tensors it is evidentally unique. ■

Proposition 3.4.18 (Tensor Unit)
Let M be an (A,C)-bimodule then there is a natural isomorphism of (A,C)-bimodules

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$$\begin{aligned} A \otimes_A M &\cong M \\ a \otimes m &\to a \cdot m \\ 1 \otimes m &\leftarrow m \end{aligned}$$

Similarly there is a natural isomorphism of (A, C)-bimodules

$$M \otimes_C C \cong M$$

Proof. By (3.4.16) with $\phi = 1_A$ there is a natural isomorphism

$$\mathrm{Hom}(A \otimes_A M, -) \cong \mathrm{Hom}(M, -)$$

and so natural isomorphism follows from (2.6.68).

We may deduce that there is a natural isomorphism of $(C^{\mathsf{op}}, A^{\mathsf{op}})$-bimodules

$$C^{\mathsf{op}} \otimes_{C^{\mathsf{op}}} M^{\mathsf{op}} \cong M^{\mathsf{op}}$$

which amounts to a natural isomorphism of (A, C)-bimodules

$$M \otimes_C C \cong M$$

■

Proposition 3.4.19 (Transitivity of Extension of Scalars)
Let M be a (A, D)-bimodule and $\varphi : A \to B$, $\psi : B \to C$ homomorphisms of commutative rings. Then there is an isomorphism of (C, D)-bimodules

$$\begin{aligned} C \otimes_B (B \otimes_A M) &\xrightarrow{\sim} {}_C C_A \otimes_A M \\ c \otimes (b \otimes m) &\to (c\psi(b)) \otimes m \end{aligned}$$

Proof. By associativity (3.4.13) and (3.4.18) there is a natural isomorphism

$$C \otimes_B (B \otimes_A M) \cong (C \otimes_B B) \otimes_A M \cong C \otimes_A M$$

■

Proposition 3.4.20 (Transitivity of Extension of Scalars (Commutative Case))
Let M be an A-module and $\phi : A \to B$ and $\psi : B \to C$ homomorphisms of commutative rings. Then there is an isomorphism of C-modules

$$C \otimes_B (B \otimes_A M) \cong {}_C C_A \otimes_A M$$

Proof. Regarding M as an (A, A)-bimodule we have an isomorphism of (C, A)-bimodules by (3.4.19) which is a-fortiori a C-module isomorphism. ■

3.4.4 Tensor Product Commutes with Direct Sum

Proposition 3.4.21 (Tensor Product Commutes with Sum)
Let $(M_i)_{i \in I}$ be a family of (A, B)-bimodules and $(N_j)_{j \in J}$ a family of (B, C)-bimodules. Then there is an isomorphism of (A, C)-bimodules

$$\left(\bigoplus_{i \in I} M_i \right) \otimes_B \left(\bigoplus_{j \in J} N_j \right) \cong \bigoplus_{(i,j) \in I \times J} (M_i \otimes_B N_j)$$

Corollary 3.4.22
Suppose M is an (A, B)-bimodule and N is a (B, C)-bimodule and $M' \subseteq M$ and $N' \subseteq N$ are direct factors. Then the canonical map is injective

$$M' \otimes_B N' \hookrightarrow M \otimes_B N$$

Corollary 3.4.23
*Let N be **free** left A-module with basis $(n_i)_{i \in I}$ and M a (B, A)-bimodule. Then there is an isomorphism of left B-modules*

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$$M \otimes_A N \cong \bigoplus_{i \in I} M$$
$$m \otimes \sum_i a_i n_i \to (m \cdot a_i)_{i \in I}$$
$$\sum_i (m_i \otimes n_i) \leftarrow (m_i)_{i \in I}$$

When A is commutative then this is a (B, A)-bimodule isomorphism. Further when M is an A-module then this is an isomorphism of A-modules.

Proof. By (3.4.21) and (3.4.18)

$$\begin{array}{ccccc} M \otimes_A N \cong & M \otimes_A \left(\bigoplus_{i \in I} A \right) & \cong & \bigoplus_{i \in I} (M \otimes_A A) \cong & \bigoplus_{i \in I} M \\ m \otimes \sum_i a_i n_i \to & m \otimes (a_i)_{i \in I} \to & & (m \otimes a_i)_{i \in I} \to & (m \cdot a_i)_{i \in I} \end{array}$$

■

Corollary 3.4.24
If M, N are free A-modules with bases $\{m_i\}_{i \in I}$ and $\{n_j\}_{j \in J}$ then $M \otimes_A N$ is a free A-module with basis $\{m_i \otimes n_j\}_{(i,j) \in I \times J}$.

Corollary 3.4.25 (Free modules are flat)
Let N be a free left A-module and $i : M' \to M$ an injective map of (B, A)-bimodules. Then the corresponding map

$$i \otimes 1_N : M' \otimes_A N \to M \otimes_A N$$

is injective.

Corollary 3.4.26 (Extension of Scalars (Free Module))
*Let M be **free** left A-module with basis $\{m_i\}_{i \in I}$ and $\varphi : A \to B$ a ring homomorphism. Then there is a canonical isomorphism of left B-modules*

$$M_{(B)} \cong \bigoplus_{i \in I} B$$
$$b \otimes \sum_i a_i m_i \to (b\varphi(a_i))_{i \in I}$$

In particular $\{1 \otimes m_i\}_{i \in I}$ is a basis for $M_{(B)}$. Further when Z is a left B-module then there is an isomorpihsm of abelian groups

$$\begin{array}{ccc} \mathrm{Hom}_A(M, Z) & \cong & \mathrm{Hom}_B\left(M_{(B)}, Z\right) \cong \prod_{i \in I} Z \\ \theta & & \to \quad (\theta(m_i))_{i \in I} \end{array}$$

which is B-linear when B is commutative.

When M is a finite free A-module of rank n (with basis $\{v_i\}$) then $M_{(B)}$ is a finite free B-module of rank n (with basis $\{1 \otimes v_i\}$). When B is commutative, $M_{(B)}^\vee$ is also a finite free B-module of rank n.

Proof. The first isomorphism follows directly from (3.4.23). Then by (3.3.83) there is an isomorphism

$$\mathrm{Hom}\left(M_{(B)}, Z\right) \cong \mathrm{Hom}\left(\bigoplus_{i \in I} B, Z\right) \cong \prod_{i \in I} \mathrm{Hom}(B, Z) \cong \prod_{i \in I} Z$$

■

We may generalize (3.3.110)

Proposition 3.4.27 (Extension of Scalars (Hom-Set))
Let $\varphi : A \to B$ be a homomorphism of commutative rings, M a finite-free A-module and N a finite-free B-module. Then

$$\mathrm{Hom}_A(M, N) \cong \mathrm{Hom}_B\left(M_{(B)}, N\right)$$

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is a finite-free B-module. More precisely suppose M has basis $\{v_1, ..., v_n\}$ and N has basis $\{w_1, ..., w_m\}$ then $\mathrm{Hom}_A(M,N)$ *has basis*

$$\left\{w_j v_i^\vee\right\}_{i,j}$$

Proof. By (3.4.26) there are isomorphisms of B-modules

$$\begin{aligned}\mathrm{Hom}_A(M,N) \cong \mathrm{Hom}_B\left(M_{(B)}, N\right) \cong \bigoplus_{i=1}^{n} N \cong \bigoplus_{i=1}^{n}\bigoplus_{j=1}^{m} B\\ \theta \to \qquad 1\otimes\theta \to (\theta(v_i))_i \to \left(w_j^\vee(\theta(v_i))\right)_{i,j}\end{aligned}$$

Under this isomorphism the standard basis of the right hand side corresponds to the set $\left\{w_j v_i^\vee\right\}_{i,j}$ whence these constitute a basis. ∎

3.4.5 Tensor Product Exact Sequences

Proposition 3.4.28 (Tensor Product is Right Exact)
Consider an exact sequence of A-modules (i.e. such that β is surjective and $\ker(\beta) = \mathrm{Im}(\alpha)$*)*

$$M \xrightarrow{\alpha} N \xrightarrow{\beta} P \to 0$$

Then the corresponding sequence of A-modules

$$M \otimes_A Z \xrightarrow{\alpha\otimes 1_Z} N \otimes_A Z \xrightarrow{\beta\otimes 1_Z} P \otimes_A Z \to 0$$

is exact. Similarly the sequence

$$Z \otimes_A M \xrightarrow{1_Z\otimes\alpha} Z \otimes_A N \xrightarrow{1_Z\otimes\beta} Z \otimes_A P \to 0$$

is exact.

Proof. Trivially the submodule $\mathrm{Im}(\beta\otimes 1_Z)$ contains elementary tensors and therefore equals $P\otimes_A Z$. In otherwords $\beta\otimes 1_Z$ is surjective. Furthermore by functoriality $(\beta\otimes 1_Z)\circ(\alpha\otimes 1_Z) = (\beta\circ\alpha)\otimes 1_Z$ which is zero on elementary tensors and therefore zero everywhere. In particular $\mathrm{Im}(\alpha\otimes 1_Z) \subseteq \ker(\beta\otimes 1_Z)$. Consider a commutative diagram

$$\begin{array}{ccc} P\times Z & \xrightarrow{\psi} & (N\otimes_A Z)/\mathrm{Im}(\alpha\otimes 1_Z) \\ & \searrow^{\overline{\beta\otimes 1_Z}} & \downarrow \\ & & P\otimes_A Z \end{array}$$

where ψ is yet to be defined. For a given $p\in P$ and $z\in Z$ consider the subset

$$\mathcal{I}_{p,z} \coloneqq \{n\otimes z \mid n\in N, \beta(n) = p\} \subset N\otimes_A Z$$

Observe that any ψ making the diagram commute satisfies $\psi(p,z)\in\overline{\mathcal{I}_{p,z}}$, and we aim to show this set is a singleton.

As β is surjective $\mathcal{I}_{p,z}$ is non-empty. First we show that $x,y\in\mathcal{I}_{p,z} \implies x-y\in\mathrm{Im}(\alpha\otimes 1_Z)$. For if $\beta(n)=\beta(n')$ then $\beta(n-n')=0$ by definition $n-n'=\alpha(m)$ for some $m\in M$. It follows that $n\otimes z - n'\otimes z = \alpha(m)\otimes z = (\alpha\otimes 1_Z)(m\otimes z)$. This shows that $\overline{\mathcal{I}_{p,z}}$ is a singleton, whose element we define to be $\psi(p,z)$. Then ψ is the unique map such that

$$\psi(\beta(n), z) = \overline{n\otimes z}$$

for all $n\in N$ and $z\in Z$. Clearly ψ is A-bilinear and therefore there is a map $\sigma : P\otimes_A Z \to (N\otimes_A Z)/\mathrm{Im}(\alpha\otimes 1_Z)$ such that $\sigma(\beta(n)\otimes z) = \overline{n\otimes z}$. By linearity $\sigma\circ\overline{\beta\otimes 1_Z}$ is the identity which shows $\overline{\beta\otimes 1_Z}$ is injective. Therefore $\ker(\beta\otimes 1_Z) = \mathrm{Im}(\alpha\otimes 1_Z)$ as required (for $x\in\ker(\beta\otimes 1_Z) \implies (\beta\otimes 1_Z)(x) = 0 \implies \left(\overline{\beta\otimes 1_Z}\right)(\overline{x}) = 0 \implies \overline{x} = 0 \implies x\in\mathrm{Im}(\alpha\otimes 1_Z)$). ∎

Proposition 3.4.29 (Quotient of a Tensor Product)
Consider exact sequences

$$0 \to M' \xrightarrow{i} M \xrightarrow{\pi_1} M/M' \to 0$$

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$$0 \to N' \xrightarrow{j} N \xrightarrow{\pi_2} N/N' \to 0$$

Then there is an exact sequence

$$\begin{array}{rcccl} (M' \otimes N) \oplus (M \otimes N') & \xrightarrow{\alpha} & M \otimes N & \xrightarrow{\pi_1 \otimes \pi_2} & M/M' \otimes N/N' \to 0 \\ (m' \otimes n, m \otimes n') & \to & i(m') \otimes n + m \otimes j(n') & \to & \overline{m} \otimes \overline{n} \to 0 \end{array}$$

In particular there is a canonical isomorphism

$$(M \otimes N)/(M' \otimes N + M \otimes N') \cong M/M' \otimes N/N'$$

Proof. Observe $\pi_1 \otimes \pi_2 = \left(\pi_1 \otimes 1_{N/N'}\right) \circ (1_M \otimes \pi_2)$ by comparison on elementary tensors. By (3.4.28) each of these maps is surjective, and so the composite is surjective. Evidently $(\pi_1 \otimes \pi_2) \circ (i \otimes 1_N) = 0$ and $(\pi_1 \otimes \pi_2) \circ (1_M \otimes j) = 0$ by checking elementary tensors. This in particular means that $(\pi_1 \otimes \pi_2) \circ \alpha = 0$ where α is the first map. Therefore $\operatorname{Im}(\alpha) \subseteq \ker(\pi_1 \otimes \pi_2)$ and it suffices to demonstrate the reverse inclusion. Suppose $z \in \ker(\pi_1 \otimes \pi_2)$. This means precisely that $(1_M \otimes \pi_2)(z) \in \ker\left(\pi_1 \otimes 1_{N/N'}\right) = \operatorname{Im}\left(i \otimes 1_{N/N'}\right)$ by (3.4.28). Therefore $(1_M \otimes \pi_2)(z) = \left(i \otimes 1_{N/N'}\right)(y)$ for $y \in M' \otimes N/N'$. By (3.4.28) again $y = (1_{M'} \otimes \pi_2)(x)$ for $x \in M' \otimes N$ and $(1_M \otimes \pi_2)(z) = (i \otimes \pi_2)(x)$. Therefore $(1_M \otimes \pi_2)(z - (i \otimes 1_N)(x)) = 0$ and $z - (i \otimes 1_N)(x) = (1_M \otimes j)(w)$ for some $w \in M \otimes N'$. Therefore we conclude $z \in \operatorname{Im}(\alpha)$ as required. ∎

3.4.6 Vector Space Tensor Product

Recall that vector spaces are free, every linearly independent set may be extended to a basis and every subspace is a direct factor. This simplifies the structure of tensor product.

Proposition 3.4.30
Let V, W be k-vector spaces with subspaces V' and W'. Then the canonical k-module homomorphism

$$V' \otimes_k W' \to V \otimes_k W$$

is injective.

Proof. The homomorphism exists by (3.4.4). By (3.3.139) both V' and W' are direct factors. Therefore the map is injective by (3.4.22). ∎

Proposition 3.4.31
Let V, W be k-modules. Suppose $\{v_i\}_{i \in I}$ and $\{w_j\}_{j \in J}$ are linearly independent (resp. bases) then $\{v_i \otimes w_j\}_{(i,j) \in I \times J}$ is a k-linearly independent subset (resp. a k-basis) of $V \otimes_k W$.

Proof. The case that they are bases is simply (3.4.24). For the general case we may use (3.3.126) to extend to bases, then a-fortiori the given set, being a subset of a basis, is linearly independent. ∎

3.4.7 Algebra Tensor Product

Let A be a commutative ring and revert to the case of commutative tensor product described in (3.4.4). We show that given two A-algebras B, C the tensor product naturally forms an algebra. We first prove some preliminary results.

Lemma 3.4.32
Let B be a commutative A-algebra. Then there exists a unique homomorphism of A-modules

$$B \otimes_A B \to B$$

such that

$$b \otimes b' \to bb'$$

Proof. Multiplication is bilinear so the map exists by universal property. ∎

Lemma 3.4.33
Let B, C be commutative A-algebras and define the A-module $X \coloneqq B \otimes_A C$. Then there is a unique homomorphism of A-modules

$$m : X \otimes_A X \to X$$

such that

$$(b \otimes c) \otimes (b' \otimes c') \to (bb') \otimes (cc')$$

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Define $1_X := (1 \otimes 1)$ *then it satisfies*

a) $m(1_X \otimes x) = m(x \otimes 1_X) = x$
b) $m(x \otimes y) = m(y \otimes x)$
c) $m(x \otimes m(y \otimes z)) = m(m(x \otimes y) \otimes z)$
d) $m((x + y) \otimes z) = m(x \otimes z) + m(y \otimes z)$
e) $m(x \otimes (y + z)) = m(x \otimes y) + m(x \otimes z)$

for $x, y, z \in X$.

Proof. By associativity and commutativity of the tensor product there is an A-module isomorphism

$$X \otimes_A X \cong (B \otimes_A B) \otimes_A (C \otimes_A C)$$
$$(b \otimes c) \otimes (b' \otimes c') \to (b \otimes b') \otimes (c \otimes c')$$

composing with $m_B \otimes m_C$ where $m_B : (B \otimes_A B) \to B$ and $m_C : (C \otimes_A C) \to C$ are the maps given in (3.4.32), yields the required map m.

The properties may be verified on tensors of the form $x_i = (b_i \otimes c_i)$ and $y_j = \left(b'_j \otimes c'_j\right)$. The results follow from linearity, since $\left(\sum_i x_i\right) \otimes \left(\sum_j y_j\right) = \sum_{i,j}(x_i \otimes y_j)$. ∎

Proposition 3.4.34 (Algebra Tensor Product)
Let B, C *be commutative* A *algebras. Then the* A*-module* $B \otimes_A C$ *has a unique commutative ring structure with unit* $1 \otimes 1$ *and multiplication which satisfies*

$$\cdot : (B \otimes_A C) \times (B \otimes_A C) \to B \otimes_A C$$
$$(b \otimes c) \cdot (b' \otimes c') := (bb') \otimes (cc')$$

and may be extended by linearity. Further there is an ring homomorphism

$$i_A : A \to B \otimes_A C$$
$$a \to (a \otimes 1) = (1 \otimes a)$$

making $B \otimes_A C$ *into an* A*-algebra.*

Proof. Let $X := B \otimes_A C$ and consider the map $m : X \otimes_A X \to X$ defined in (3.4.33). Define $x \cdot y := m(x \otimes y)$ then the properties of m ensure that this satisfies the properties of a ring. ∎

Proposition 3.4.35 (Criteria to be an algebra homomorphism)
Let B, C, Z *be commutative* A*-algebras and let* $\phi : B \otimes_A C \to Z$ *be an* A*-module homomorphism. Then the following are equivalent*

a) ϕ *is an* A*-algebra homomorphism*
b) $\phi((b \otimes c) \cdot (b' \otimes c')) = \phi(b \otimes c) \cdot \phi(b' \otimes c')$

Similarly suppose $\psi : Z \to B \otimes_A C$ *is an* A*-module homomorphism and* Z *is generated as an* A*-module by* $S \subset Z$*. Then the following equivalent*

a) ψ *is an* A*-algebra homomorphism*
b) $\psi(ss') = \psi(s)\psi(s')$

Proof. In each case $a) \implies b)$ is obvious. For the converse we simply need to show that ϕ and ψ are in general multiplicative. This follows by linearity and because every element of the tensor product is a linear combination of elementary tensors. ∎

Proposition 3.4.36 (Functoriality)
Let $\phi : B \to B'$ *and* $\psi : C \to C'$ *be* A*-algebra homomorphisms. The* A*-module homomorphism*

$$\phi \otimes \psi : B \otimes_A C \to B' \otimes_A C'$$
$$b \otimes c \to \phi(b) \otimes \phi(c)$$

is an A*-algebra homomorphism.*

Proof. The A-module homomorphism exists by (3.4.4). It is in A-algebra homomorphism by (3.4.35). ∎

Proposition 3.4.37 (Tensor Product is Coproduct)
Let B, C *be commutative* A*-algebras then there are* A*-algebra homomorphisms*

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$$\begin{aligned} i_B : B &\to B \otimes_A C \\ b &\to b \otimes 1 \\ i_C : C &\to B \otimes_A C \\ c &\to 1 \otimes c \end{aligned}$$

which are natural in B and C respectively. In particular $B \otimes_A C$ is both a C-algebra and a B-algebra.

Furthermore for Z an A-algebra there is a bijection

$$\begin{aligned} \mathrm{AlgHom}_A(B \otimes_A C, Z) &\cong \mathrm{AlgHom}_A(B, Z) \times \mathrm{AlgHom}_A(C, Z) \\ \psi &\to (\psi \circ i_B, \psi \circ i_C) \end{aligned}$$

which is natural in Z.

Proof. The existence of i_B and i_C is easily demonstrated using universal property of tensor product. Observe that in general

$$(\psi \circ i_B)(b) = \psi(b \otimes 1_C)$$

$$(\psi \circ i_C)(c) = \psi(1_B \otimes c)$$

Furthermore the map is clearly well-defined. It's injective because if ψ and ψ' have the same image then $\psi(b \otimes 1) = \psi'(b \otimes 1)$ and $\psi(1 \otimes c) = \psi'(1 \otimes c)$ whence $\psi(b \otimes c) = \psi'(b \otimes c)$. By linearity they are everywhere equal.

To show that the map is a bijection we construct a two-sided inverse. For given A-algebra homomorphisms $f : B \to Z$ and $g : C \to Z$ then $(b, c) \to (f(b)g(c))$ is A-bilinear and so corresponds to an A-module homomorphism

$$\begin{aligned} \psi_{f,g} : B \otimes_A C &\to Z \\ b \otimes c &\to f(b)g(c) \end{aligned}$$

and by (3.4.35) this is an algebra homomorphism. Clearly $\psi \circ i_B = f$ and $\psi \circ i_C = g$. Conversely let $\psi : B \otimes_A C \to Z$ be an algebra homomorphism and we define $f(b) := \psi(b \otimes 1_C)$ and $g(c) := \psi(1_C \otimes c)$. Then $f(b)g(c) = \psi(b \otimes c)$ which shows that $\psi_{f,g}$ agrees with ψ on elementary tensors, and therefore is identically equal. This completes the demonstration that $\psi_{f,g}$ is a two-sided inverse to the given map.

Naturality in Z is straightforward. ∎

Proposition 3.4.38 (Extension of Scalars)
Let B, C be commutative A-algebras. Then define the C-algebra

$$B_{(C)} := C \otimes_A B$$

There is a natural isomorphism for any C-algebra Z

$$\begin{aligned} \mathrm{AlgHom}_C\left(B_{(C)}, Z\right) &\cong \mathrm{AlgHom}_A(B, Z) \\ \psi &\to \psi \circ i_B \end{aligned}$$

Proof. We consider the commutative diagram obtained from (3.4.37)

$$\begin{array}{ccc} \mathrm{AlgHom}_A\left(B_{(C)}, Z\right) & \xrightarrow{\sim} & \mathrm{AlgHom}_A(B, Z) \times \mathrm{AlgHom}_A(C, Z) \\ \cup & & \cup \\ \mathrm{AlgHom}_C\left(B_{(C)}, Z\right) & \xrightarrow{\sim} & \mathrm{AlgHom}_A(B, Z) \times \{i_{CZ}\} \end{array}$$

An A-algebra homomorphism $\phi : B_{(C)} \to Z$ is a C-algebra homomorphism precisely when $\phi \circ i_C = i_{CZ}$ so that the bottom arrow is well-defined and bijective as required. ∎

Proposition 3.4.39 (Transitivity of Extension of Scalars)
Let B, C be commutative A-algebras and D a commutative C-algebra. Then there is an isomorphism of D-algebras

$$\begin{aligned} D \otimes_C (C \otimes_A B) &\cong D \otimes_A B \\ d \otimes (c \otimes b) &\to (dc) \otimes b \end{aligned}$$

Proof. Let Z be a D-algebra then there is a natural isomorphism of functors

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$$\begin{aligned}\mathrm{AlgHom}_D(D \otimes_C (C \otimes_A B), Z) &\cong \mathrm{AlgHom}_C(C \otimes_A B, Z)\\ &\cong \mathrm{AlgHom}_A(B, Z)\\ &\cong \mathrm{AlgHom}_D(D \otimes_A B, Z)\end{aligned}$$

Consider the case $Z = D \otimes_A B$ and the identity map yields the required isomorphism (by the Yoneda Lemma). ∎

Proposition 3.4.40
Let B be a commutative A-algebra then there is a natural isomorphism of A-algebras

$$\begin{aligned}u_B : B &\xrightarrow{\sim} B_{(A)}\\ b &\to 1 \otimes b\\ ab &\leftarrow a \otimes b\end{aligned}$$

Furthermore for C a commutative A-algebra there is a commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{u_B} & A \otimes_A B \\ & \searrow^{u_B} & \downarrow{\scriptstyle i_A \otimes 1_B} \\ & & C \otimes_A B \end{array}$$

Proof. By (3.4.38) there is an bijection natural in Z

$$\begin{aligned}\mathrm{AlgHom}_A\left(B_{(A)}, Z\right) &\xrightarrow{\sim} \mathrm{AlgHom}_A(B, Z)\\ \psi &\to \psi(1 \otimes -)\end{aligned}$$

which by the Yoneda Lemma (2.6.58) yields the required isomorphism (given by the image of $1_{B_{(A)}}$). ∎

Proposition 3.4.41 (Structural Morphisms are Injective)
Let B, C be commutative A-algebras such that the structural morphisms are injective and A is a direct factor of B and C (e.g. if $A = k$ is a field). Then the canonical maps (3.4.37)

$$B \to B \otimes_A C$$

$$C \to B \otimes_A C$$

are injective.

Proof. By (3.4.22) the canonical map

$$B \otimes_A A \to B \otimes_A C$$

is injective. By (3.4.40) we have a canonical isomorphism $B \cong B \otimes_A A$ and the composite is simply the canonical map $B \to B \otimes_A C$. ∎

Proposition 3.4.42 (Extension of Ideals under Tensor Product)
Let B, C be commutative A-algebras and $\mathfrak{b} \lhd B$ and $\mathfrak{c} \lhd C$ ideals. Then $\mathfrak{b}^e$ (resp. $\mathfrak{c}^e$) is the image of the A-module $\mathfrak{b} \otimes_A C$ (resp. $B \otimes_A \mathfrak{c}$) in $B \otimes_A C$. Furthermore there is a canonical A-algebra isomorphism

$$(B \otimes_A C)/(\mathfrak{b}^e + \mathfrak{c}^e) \cong B/\mathfrak{b} \otimes_A C/\mathfrak{c}$$

Proof. Let $\phi : B \to B \otimes_A C$ be the structural morphism then for $b \in \mathfrak{b}$ we have $b \otimes c = (c \otimes 1)(b \otimes 1) = (1 \otimes c)\phi(b)$ whence $\phi(\mathfrak{b}) \subseteq \mathfrak{b} \otimes_A C \subseteq \mathfrak{b}^e$. Evidently $\mathfrak{b} \otimes_A C$ is an ideal and therefore we see $\mathfrak{b} \otimes_A C = \mathfrak{b}^e$ as required. The isomorphism follows from (3.4.29). ∎

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3.5 Localisation

Algebraically, localisation can be seen as enlargening a ring to include inverses. In terms of the ideal structure this means removing (proper) ideals which contain the newly inverted elements. Geometrically ideals correspond to points/subsets, so localisation may be viewed as reducing the set of interest.

Recall the definition of multiplicative set. Some rather canonical examples are as follows

Example 3.5.1
The set $S_f = \{1, f, f^2, ...\}$ is m.c. but not necessarily saturated. As an example consider $A = \mathbb{Z}$ and $S_n = \{1, n, n^2, ...\}$ for n composite. Then $pq \in S_n$ but $p \notin S_n$.

We denote the localisation of A at S_f by A_f (or alternatively $A[1/f]$).

Example 3.5.2
If $\mathfrak{p} \lhd A$ is a prime ideal, then $A \setminus \mathfrak{p}$ is a saturated multiplicative set. More generally, we show later that S is a saturated multiplicative set if and only if it's of the form

$$A \setminus \bigcup_i \mathfrak{p}_i$$

for some family of prime ideals. We denote the localisation of A at $\mathfrak{p}$ by $A_{\mathfrak{p}}$.

3.5.1 Rings

Definition 3.5.3 (Localisation of a ring)
Let A be a ring and S a multiplicative set. Define the set

$$S^{-1}A = \left\{ \frac{a}{s} \;\middle|\; a \in A \, s \in S \right\}$$

under the equivalence relation

$$\frac{a}{s} = \frac{b}{t} \iff u(at - bs) = 0 \quad \text{some } u \in S.$$

then this is a ring in the obvious way

Proposition 3.5.4
The set $S^{-1}A$ is a ring under the obvious ring operations. It is non-zero precisely when S is proper. There is a canonical homomorphism

$$i_S : A \to S^{-1}A$$
$$a \to \left[\frac{a}{1}\right]$$

a) $\ker(i_S) = \mathrm{Ann}(S)$
b) *$S^{-1}A$ is the zero-ring if and only if $0 \in S$ if and only if there exists $s, t \in S$ such that $st = 0$.*
c) *$i_S(s)$ is invertible for all $s \in s$*
d) *i_S is injective if and only if S has no zero-divisors*
e) *This is an isomorphism if and only if $S \subseteq A^\star$ already consists only of invertible elements (e.g. $S = \{1\}$).*

Proof.
a) This follows by the definitions
b) $\frac{1}{1} = \frac{0}{0} \iff s = 0$ for some $s \in s$ by the definitions
c) $\frac{s}{1} \cdot \frac{1}{s} = \frac{s}{s} = \frac{1}{1}$
d) This follows from the first part.
e) If $S \subseteq A^\star$ then it contains no zero-divisors and i_S is injective. Further it's clear that $\frac{a}{s} = \frac{as^{-1}}{1}$ so that the map is surjective. Similarly if the map is bijective S does not contain zero-divisors and $\frac{1}{s}$ is in the image. Therefore there is a such that $tas = 1$ for some t, which implies s is invertible.

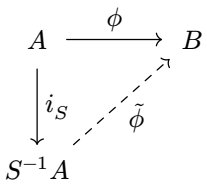
■

Note when A is an integral domain and S is proper then the equivalence relation may be weakened to $at - bs = 0$.

Proposition 3.5.5 (Universal Property)
Let $\phi : A \to B$ be a ring homomorphism and S a multiplicative set. Then

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a) There is a unique morphism $\tilde{\phi}$ making the diagram commute



if and only if $\phi(S) \subseteq B^\star$. In this case it's given by

$$\tilde{\phi}\left(\frac{a}{s}\right) = \phi(a)\phi(s)^{-1}$$

b) $\ker(\tilde{\phi}) = S^{-1}\ker(\phi)$

Proof.

a) If $\tilde{\phi}$ exists then $1 = \tilde{\phi}(1) = \tilde{\phi}(\frac{s}{1}\cdot\frac{1}{s}) = \tilde{\phi}(\frac{s}{1})\tilde{\phi}(\frac{1}{s}) = \phi(s)\tilde{\phi}(\frac{1}{s})$. Which shows that $\phi(S) \subseteq B^\star$ and $\phi(s)^{-1} = \tilde{\phi}(\frac{1}{s})$.
Conversely suppose $\phi(S) \subseteq B^\star$ then we claim that the given mapping is well-defined. For

$$\frac{a}{s} = \frac{a'}{s'} \implies s''(s'a - sa') = 0 \implies \phi(s'')\phi(s')\phi(a) = \phi(s'')\phi(s)\phi(a')$$

Multiply by the appropriate inverses to find

$$\phi(a)\phi(s)^{-1} = \phi(a')\phi(s')^{-1}$$

It's clearly a multiplicative homomorphism. Further it's additive because

$$\begin{aligned}
\tilde{\phi}\left(\frac{a}{s} + \frac{b}{t}\right) &= \tilde{\phi}\left(\frac{at+bs}{st}\right) \\
&= \phi(at+bs)\phi(st)^{-1} \\
&= \phi(a)\phi(t)\phi(s)^{-1}\phi(t)^{-1} + \phi(b)\phi(s)\phi(s)^{-1}\phi(t)^{-1} \\
&= \phi(a)\phi(s)^{-1} + \phi(b)\phi(t)^{-1} \\
&= \tilde{\phi}\left(\frac{a}{s}\right) + \tilde{\phi}\left(\frac{b}{t}\right)
\end{aligned}$$

b) Suppose $\tilde{\phi}(\frac{a}{s}) = 0$ then clearly $a \in \ker(\phi) \implies \frac{a}{s} \in S^{-1}\ker(\phi)$. The converse is clear.

■

In the case that A is an integral domain then generally everything becomes a lot simpler.

Example 3.5.6 (Quotient Field)
Let A be an integral domain then $A \setminus 0 = A^\star$ and we define the ***quotient field***

$$\mathrm{Frac}(A) \coloneqq (A \setminus 0)^{-1}A$$

Proposition 3.5.7 (Quotient Field contains all localisation)
Let A be an integral domain, and $\mathrm{Frac}(A)$ *the quotient field. Define another model for $S^{-1}A$ as follows*

$$S^{-1}A \coloneqq \left\{ \frac{a}{s} \in \mathrm{Frac}(A) \;\middle|\; a \in A\, s \in S \right\}$$

The canonical map $A \to S^{-1}A \subset \mathrm{Frac}(A)$ is injective, and satisfies the universal property for localisation.

Proof. It's injective because A has no zero-divisors. That it satisfies the universal property is very similar as before.■

Proposition 3.5.8 (Directed Limit)
Let S_i be a family of multiplicatively closed sets directed by inclusion, such that $S = \bigcup_i S_i$ is multiplicatively closed. Then there is a canonical isomorphism

$$\varinjlim_i S_i^{-1}A \to S^{-1}A$$

induced by the canonical maps

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$$S_i^{-1}A \to S^{-1}A$$

Proof. The canonical maps i_{S_iS} induce a unique morphism

$$\varinjlim_i S_i^{-1}A \to S^{-1}A$$
$$[a_i/s_i] \to a_i/s_i$$

by the universal property. An element on the right hand side is written a/s for some $s \in S$. By hypothesis $s \in S_i$ for some i, therefore it is surjective. Suppose we have two elements $[a_i/s_i]$ and $[a_j/s_j]$ on the left hand side which become equal in $S^{-1}A$. Then by definition $s_k(s_ja_i - a_js_i) = 0$ for some $s_k \in S_k$. Since it's a directed system we can find S_l containing S_i, S_j, S_k. Then by definition $a_i/s_i = a_j/s_j$ in $S_l^{-1}A$ and we see that $[a_i/s_i] = [a_j/s_j]$. Therefore the given morphism is also injective as required. ∎

Lemma 3.5.9 (Localisation Respects Contraction of Multiplicative Sets)
Let $A \subseteq B$ be subrings of a common ring C, and $T \subset B$ a multiplicative set. Writing $S := T \cap A$ (a multiplicative set of A), and realising both localisations as subrings of C (or of a common ring of fractions), $S^{-1}A \subseteq T^{-1}B$.

In particular, for $\mathfrak{q}$ a prime ideal of B and $\mathfrak{p} := \mathfrak{q} \cap A$, taking $T := B \setminus \mathfrak{q}$ (so $S = A \setminus \mathfrak{p}$) gives $A_\mathfrak{p} \subseteq B_\mathfrak{q}$.

Proof. For $a/s \in S^{-1}A$ ($a \in A$, $s \in S \subseteq T$), viewing $a, s \in B$ (as $A \subseteq B$), $s \in T$ directly gives $a/s \in T^{-1}B$. ∎

Combining the previous two results, for a directed union of subrings rather than a single fixed ring:

Lemma 3.5.10 (Localisation Commutes with Directed Unions of Subrings)
Let $(A_i)_{i\in I}$ be a directed system of subrings of a common ring C, with union $A := \bigcup_i A_i$, and $T \subset A$ a multiplicative set. Writing $T_i := T \cap A_i$, and realising all localisations as subrings of C (or of a common ring of fractions),

$$T^{-1}A = \bigcup_i T_i^{-1}A_i$$

In particular, for $\mathfrak{p} \lhd A$ a prime ideal and $\mathfrak{p}_i := \mathfrak{p} \cap A_i$, taking $T := A \setminus \mathfrak{p}$ (so $T_i = A_i \setminus \mathfrak{p}_i$) gives $A_\mathfrak{p} = \bigcup_i (A_i)_{\mathfrak{p}_i}$.

Proof. If $a/s \in T^{-1}A$ ($a \in A$, $s \in T$), then $a, s \in A_i$ for some i (directed union), with $s \in T \cap A_i = T_i$, so $a/s \in T_i^{-1}A_i$. Conversely if $a/s \in T_i^{-1}A_i$ ($a, s \in A_i$, $s \in T_i \subseteq T$), then viewing $a, s \in A_i \subset A$, $s \in T$, so $a/s \in T^{-1}A$. ∎

3.5.2 Modules

Definition 3.5.11 (Localisation of a module)
Let A be a ring with S multiplicative set and M an A-module. Then we define

$$S^{-1}M = \left\{ \frac{m}{s} \;\middle|\; m \in M \right\}$$

under the obvious equivalence relation. This is then an $S^{-1}A$-module in the obvious way.

Definition 3.5.12 (Localisation of a sub-module)
Let M be an A-module and $N \subseteq M$ a sub-A-module then define

$$S^{-1}N = \left\{ \frac{n}{s} \;\middle|\; n \in M\ s \in S \right\} \subseteq S^{-1}M$$

Proposition 3.5.13
*$S^{-1}(-)$ constitutes a functor A-**Mod** $\to S^{-1}A$-**Mod**. More precisely there is a unique morphism ψ making the following diagram commute as A-module morphisms*

$$\begin{array}{ccc} N & \xrightarrow{\psi} & M \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_S} \\ S^{-1}N & \xrightarrow{S^{-1}(\psi)} & S^{-1}M \end{array}$$

where $S^{-1}(\psi)$ is in fact an $S^{-1}A$-module morphism.

It is an exact functor; for an exact sequence

$$N \to M \to P$$

the corresponding sequence of $S^{-1}A$-module morphisms

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$$S^{-1}N \to S^{-1}M \to S^{-1}P$$

is exact. If N is a submodule of M then we may regard $S^{-1}N$ as a submodule of $S^{-1}M$.

Proposition 3.5.14 (Localisation commutes with quotients)
There is a commutative diagram of A-module morphisms

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N & \xrightarrow{i} & M & \xrightarrow{\pi} & M/N & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_S} & & \downarrow & & \\ 0 & \longrightarrow & S^{-1}N & \longrightarrow & S^{-1}M & \xrightarrow{S^{-1}(\pi)} & S^{-1}(M/N) & \longrightarrow & 0 \end{array}$$

with exact rows and the bottom row consists of $S^{-1}A$-module morphisms. This induces an isomorphism of $S^{-1}A$-modules.

$$S^{-1}M/S^{-1}N \cong S^{-1}(M/N)$$

Proposition 3.5.15
Suppose $N \subseteq N' \subseteq M$ then there is a canonical short-exact sequence of $S^{-1}A$-modules

$$0 \to S^{-1}(N'/N) \to S^{-1}(M/N) \to S^{-1}(M/N') \to 0$$

which induces an isomorphism

$$S^{-1}(M/N)/S^{-1}(N'/N) \cong S^{-1}(M/N')$$

Proposition 3.5.16
Let M be a finitely-generated A-module. Then

$$S^{-1}M = 0 \iff sM = 0 \text{ some } s \in S$$

3.5.3 Ideals

Recall the notion of extended and contracted ideals in Definition (3.3.49).

Definition 3.5.17 (Localisation of an ideal)
Let A be a ring and S a multiplicative set and $\mathfrak{a} \lhd A$ define

$$S^{-1}\mathfrak{a} := \left\{ \frac{a}{s} \;\middle|\; a \in \mathfrak{a} \right\}$$

then this is an ideal of $S^{-1}A$.

Proposition 3.5.18 (Extension and Contraction)
Let A be a ring with multiplicative set S and canonical morphism $i_S : A \to S^{-1}A$.

a) $\mathfrak{a}^e = i_S(\mathfrak{a})S^{-1}A = \left\{ \frac{a}{s} \mid a \in \mathfrak{a}, s \in S \right\} = S^{-1}\mathfrak{a}$

b) $\mathfrak{b}^c = \left\{ a \mid \frac{a}{1} \in \mathfrak{b} \right\}$

c) An ideal $\mathfrak{a}$ in A satisfies

$$\mathfrak{a}^{ec} = \bigcup_{s \in S} (\mathfrak{a} : s) = \{ a \in A \mid as \in \mathfrak{a} \text{ some } s \in S \}$$

In particular $\mathfrak{a}$ is contracted if and only if $as \in \mathfrak{a} \wedge s \in S \implies a \in \mathfrak{a}$

d) $\mathfrak{b}$ proper $\iff \mathfrak{b}^c$ proper $\iff \mathfrak{b}^c \cap S = \emptyset$

e) $\mathfrak{a}^e$ proper $\iff \mathfrak{a} \cap S = \emptyset$

f) Every ideal $\mathfrak{b} \lhd S^{-1}A$ is extended (equiv. $\mathfrak{b} = \mathfrak{b}^{ce} = S^{-1}\mathfrak{b}^c$).

g) A prime ideal $\mathfrak{p}$ is contracted if and only if $\mathfrak{p} \cap S = \emptyset$. In this case $\mathfrak{p}^e$ is prime.

h) $\mathfrak{q}$ prime $\implies \mathfrak{q}^c$ *is prime and satisfies* $\mathfrak{q}^c \cap S = \emptyset$.

Proof.

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a) $S^{-1}\mathfrak{a}$ is an additive subgroup because $\frac{a_1}{s_1}+\frac{a_2}{s_2}=\frac{a_1s_2+a_2s_1}{s_1s_2}$. It contains $i_S(\mathfrak{a})$ and is closed under multiplication by A, therefore $\mathfrak{a}^e \subseteq S^{-1}\mathfrak{a}$. Similarly as $\mathfrak{a}^e$ is an ideal containing $i_S(\mathfrak{a})$, we have $\frac{a}{s}=\frac{1}{s}\cdot\frac{a}{1}\in\mathfrak{a}^e$, i.e. $S^{-1}\mathfrak{a}\subseteq\mathfrak{a}^e$ as required.

b) This is clear

c) Observe that

$$\begin{aligned}\mathfrak{a}^{\mathrm{ec}} &= \left\{a\in A \;\middle|\; \frac{a}{1}\in\mathfrak{a}^e\right\}\\ &= \left\{a\in A \;\middle|\; \frac{a}{1}=\frac{a'}{s} \quad a'\in\mathfrak{a}\, s\in S\right\}\\ &= \{a\in A \mid sa\in\mathfrak{a} \text{ some } s\in S\}\end{aligned}$$

By (3.3.50) an ideal $\mathfrak{a}$ is contracted if and only if $\mathfrak{a}=\mathfrak{a}^{\mathrm{ec}}$. Furthermore it always satisfies $\mathfrak{a}^{\mathrm{ec}}\subseteq\mathfrak{a}$. The reverse inclusion is precisely the condition given.

d) This first equivalence is true in general, see (3.3.50). Clearly $\mathfrak{b}^c=A\implies\mathfrak{b}^c\cap S\neq\emptyset$. Similarly if $S\cap\mathfrak{b}^c\neq\emptyset$ then $s\in\mathfrak{b}^c\implies\frac{s}{1}\in\mathfrak{b}\implies 1\in\mathfrak{b}\implies 1\in\mathfrak{b}^c$.

e) By d) $\mathfrak{a}^e$ is proper if and only if $\mathfrak{a}^{\mathrm{ec}}$ is proper. By c) we see $1\in\mathfrak{a}^{\mathrm{ec}}$ if and only if $S\cap\mathfrak{a}\neq\emptyset$ and the result follows.

f) By (3.3.50) we need only show $\mathfrak{b}^{\mathrm{ce}}\subseteq\mathfrak{b}$. Note $\frac{a}{s}\in\mathfrak{b}^{\mathrm{ce}}\implies\frac{a}{s}=\frac{a'}{s'}$ with $a'\in\mathfrak{b}^c$. By 2. $\frac{a'}{1}\in\mathfrak{b}$ and therefore so is $\frac{a}{s}=\frac{a'}{s'}=\frac{a'}{1}\cdot\frac{1}{s'}\in\mathfrak{b}$ as required.

g) If $\mathfrak{p}\cap S=\emptyset$ then by primality it automatically satisfies the conditions in c) and is therefore contracted. Conversely if a prime ideal $\mathfrak{p}$ is contracted then $\mathfrak{p}=\mathfrak{q}^c$. It is by definition proper so by d) it satisfies $\mathfrak{p}\cap S=\emptyset$ as required.

Suppose $\mathfrak{p}$ is prime and $\frac{a}{s}\cdot\frac{b}{t}\in\mathfrak{p}^e$ then $\frac{ab}{st}=\frac{x}{u}$ for $x\in\mathfrak{p}\implies v(abu-xst)=0\implies uvab\in\mathfrak{p}\implies a\in\mathfrak{p}$ or $b\in\mathfrak{p}$. Therefore $\frac{a}{s}\in\mathfrak{p}^e$ or $\frac{b}{t}\in\mathfrak{p}^e$ as required.

h) Generically $\mathfrak{q}^c$ is a contracted prime ideal and we've already shown in d) that $\mathfrak{q}^c\cap S=\emptyset$.

■

Corollary 3.5.19 (Ideal Structure Localisation)
Let A be a ring and S a multiplicative set then there is an order-preserving bijection of proper ideals

$$\{\mathfrak{a}\lhd A \mid \mathfrak{a} \text{ contracted}\}\leftrightarrow\{\mathfrak{b}\lhd S^{-1}A\}$$

which restricts to a bijection of prime ideals

$$\{\mathfrak{p}\lhd A \mid \mathfrak{p}\cap S=\emptyset\}\leftrightarrow\{\mathfrak{q}\lhd S^{-1}A\}$$

Proof. From (3.5.18) every ideal of $S^{-1}A$ is extended. Therefore the bijection of proper ideals follows from (3.3.53). For prime ideals each direction is well-defined by (3.5.18) g). ■

3.5.4 Change of Rings

For what follows it is useful to have the concept of saturation of a multiplicatively closed set. Essentially taking the saturation $\overline{S}$ of S doesn't change the ring $S^{-1}A$.

Proposition 3.5.20 (Saturation)
Let A be a ring and S a multiplicatively closed set. Then the following sets are equal

a) $(i_S)^{-1}\left((S^{-1}A)^\star\right)$
b) $\{x\in A \mid ax\in S \text{ for some } a\in A\}$
c) $\bigcap_{T\supseteq S:T \text{ saturated}} T$

which we denote by $\overline{S}$. We have the following properties

- *$\overline{S}$ is saturated.*
- *S is saturated if and only if $S=\overline{S}$*
- *$\overline{\overline{S}}=\overline{S}$.*

Proof. Note $x\in(i_S)^{-1}\left((S^{-1}A)^\star\right)\implies\frac{x}{1}\cdot\frac{b}{t}=1\implies s(xb-t)=0\implies(sb)x\in S$. Similarly if $ax\in S$ then $\frac{x}{1}\cdot\frac{a}{ax}=1$.

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It's clear from b) that the set thus defined is saturated and multiplicatively closed. Let T be another saturated multiplicatively closed set containing S and suppose $ax \in S \implies ax \in T \implies x \in T$, so we find that the sets are equal.

We've proved that $\overline{S}$ is saturated. Clearly $S = \overline{S}$ implies S is saturated. Conversely if S is saturated then by c) we have $\overline{S} \subseteq S$, and clearly $S \subseteq \overline{S}$. The final part follows easily. ∎

We also give another characterization of saturated multiplicatively closed subsets

Proposition 3.5.21
Let A be a ring and S a multiplicatively closed subset. Then

$$\overline{S} = A \setminus \bigcup_{\mathfrak{p} \cap S = \emptyset} \mathfrak{p}$$

Proof. Denote the right hand side by T. Then clearly $S \subseteq T$ and as noted before in (3.5.2) T is saturated. Therefore $\overline{S} \subseteq T$ by (3.5.20) c).

Conversely suppose $a \notin \overline{S}$. Consider the principal ideal (a) then $(a) \cap S = \emptyset$ (because $ab \in S \implies a \in \overline{S}$ by (3.5.20) b)). Therefore by (3.3.41) there is a prime ideal $\mathfrak{p}$ containing a which does not intersect S. Therefore $a \notin T$. Contrapositively $T \subseteq \overline{S}$ as required. ∎

Proposition 3.5.22 (Change of Rings)
Let $\phi : A \to B$ be a ring homomorphism, S, T corresponding multiplicative subsets. Then

- *There exists a morphism $\tilde{\phi}$ making the diagram commute*

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_T} \\ S^{-1}A & \overset{\tilde{\phi}}{\dashrightarrow} & T^{-1}B \end{array}$$

  if and only if $\phi(S) \subseteq \overline{T}$. In this case it is unique and given by

$$\tilde{\phi}\left(\frac{a}{s}\right) = \frac{\phi(a)b'}{\phi(s)b'}$$

  where $b' \in B$ is any b' such that $\phi(s)b' \in T$.

- *If in addition $T \subseteq \phi\left(\overline{S}\right)$ then ϕ injective (resp. surjective, bijective) implies $\tilde{\phi}$ is injective (resp. surjective, bijective)*

- *Further ϕ surjective $\implies \ker\left(\tilde{\phi}\right) = S^{-1}\ker(\phi)$.*

Proof.

- If $\tilde{\phi}$ is well-defined, then $i_T(\phi(S)) = \tilde{\phi}(i_S(S)) \subseteq \tilde{\phi}\left(\left(S^{-1}A\right)^\star\right) \subseteq \left(T^{-1}B\right)^\star$, which implies $\phi(S) \subseteq i_T^{-1}\left(\left(T^{-1}B\right)^\star\right) = \overline{T}$.

  Conversely if $\phi(S) \subseteq \overline{T}$ then $(i_T \circ \phi)(S) \subseteq \left(T^{-1}B\right)^\star$ therefore by (3.5.5) the morphism exists making the diagram commute.

  Note that

$$\tilde{\phi}\left(\frac{a}{s}\right) = \tilde{\phi}(i_S(a)i_S(s)^{-1}) = \tilde{\phi}(i_S(a))\tilde{\phi}(i_S(s))^{-1}$$

  so it is uniquely defined by the commutativity condition. Note that given $s \in S$, by (3.5.20) there exists $b' \in B$ such that $\phi(s)b' \in T$. In this case it's clear that $i_T(\phi(s))^{-1} = \frac{b'}{\phi(s)b'}$ from which the explicit form results.

- Suppose $T \subseteq \phi\left(\overline{S}\right)$ and ϕ is injective. Then $\tilde{\phi}\left(\frac{a}{s}\right) = 0 \implies t\phi(a) = 0$ for $t \in T$. Then there exists $s' \in \overline{S}$ and $x \in A$ such that $xs' \in S$ and $\phi(s') = t$. Therefore $\phi(as') = 0 \implies as' = 0 \implies a(xs') = 0 \implies \frac{a}{s} = 0$ as required.

  Similarly if ϕ is surjective and given $\frac{b}{t} \in T^{-1}B$ there exists $a \in A$ such that $\phi(a) = b$ and $s \in \overline{S}$ such that $\phi(s) = t$. Then $xs \in S$, $\phi(xs) \in \overline{T}$ and $\phi(yxs) \in T$ for some $x, y \in A$. Finally

$$\tilde{\phi}\left(\frac{axy}{sxy}\right) = \frac{\phi(axy)}{\phi(sxy)} = \frac{b}{t}$$

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as required.

- TODO

■

Corollary 3.5.23
Let $A \xrightarrow{\phi} B \xrightarrow{\psi} C$ be a sequence of homomorphisms and S, T, U be multiplicative sets such that $\phi(S) \subseteq \overline{T}$ and $\psi(T) \subseteq \overline{U}$, then in the notation of the previous Proposition

$$\tilde{\psi} \circ \tilde{\phi} = \widetilde{\psi \circ \phi}$$

Proof. This follows from the uniqueness condition in (3.5.22). ■

Corollary 3.5.24 (Localisation Maps)
Let A be a ring and S, T two multiplicative sets. Then TFAE

- *There exists $i_{ST} : S^{-1}A \to T^{-1}A$ such that $i_{ST} \circ i_S = i_T$*
- *$S \subseteq \overline{T}$*

In this case i_{ST} is the unique such map. We have the transitivity relationships

$$i_{TU} \circ i_{ST} = i_{SU}$$

$$i_{SS} = \mathbf{1}_{S^{-1}A}$$

and furthermore i_{ST} is an isomorphism if and only if $\overline{S} = \overline{T}$. In particular $i_{S\overline{S}}$ is an isomorphism.

Finally $\ker(i_{ST}) = S^{-1}\ker(i_T)$.

Proof. This existence of i_{ST} follows from (3.5.22) when considering the map $\phi = 1_A$. The transitivity and reflexive relationships follow from (3.5.23). ■

Corollary 3.5.25 (Localisation commutes with quotient)
Let A be a ring, $\mathfrak{a}$ an ideal and S a multiplicative set. Then there exists a unique morphism making the diagram commute

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_{\pi(S)}} \\ S^{-1}A & & \\ \downarrow{\scriptstyle \pi} & & \\ S^{-1}A/S^{-1}\mathfrak{a} & \xrightarrow{\sim} & \pi(S)^{-1}(A/\mathfrak{a}) \end{array}$$

which is an isomorphism, and determined by

$$\frac{a}{s} + S^{-1}\mathfrak{a} \to \frac{a+\mathfrak{a}}{s+\mathfrak{a}}$$

Note that $S \cap \mathfrak{a} \neq \emptyset \iff S^{-1}A/S^{-1}\mathfrak{a} = 0 \iff \pi(S)^{-1}(A/\mathfrak{a}) = 0$.

When $\mathfrak{b} \supseteq \mathfrak{a}$ this restricts to a commutative diagram of A-modules

$$\begin{array}{ccc} \mathfrak{b} & \xrightarrow{\pi} & \mathfrak{b}/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_{\pi(S)}} \\ S^{-1}\mathfrak{b} & & \\ \downarrow{\scriptstyle \pi} & & \\ S^{-1}\mathfrak{b}/S^{-1}\mathfrak{a} & \xrightarrow{\sim} & \pi(S)^{-1}(\mathfrak{b}/\mathfrak{a}) \end{array}$$

and the bottom arrow is still an isomorphism of $S^{-1}A/S^{-1}\mathfrak{a}$-modules.

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Corollary 3.5.26 (Localisation commutes with quotient II)
Let A be a ring, $\mathfrak{a}$ an ideal and S a multiplicative set. Then there exists a unique morphism making the diagram commute

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \vdots \\ S^{-1}A & \xrightarrow{\pi} & S^{-1}A/S^{-1}\mathfrak{a} \end{array}$$

given by

$$a + \mathfrak{a} \to \frac{a}{1} + S^{-1}\mathfrak{a}$$

and it is an isomorphism precisely when every $s \in S$ is co-prime to $\mathfrak{a}$, i.e.

$$(s) + \mathfrak{a} = A \quad \forall s \in S.$$

When $\mathfrak{b} \supseteq \mathfrak{a}$ this restricts to a commutative diagram

$$\begin{array}{ccc} \mathfrak{b} & \xrightarrow{\pi} & \mathfrak{b}/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \vdots \\ S^{-1}\mathfrak{b} & \xrightarrow{\pi} & S^{-1}\mathfrak{b}/S^{-1}\mathfrak{a} \end{array}$$

which is an $A/\mathfrak{a}$-module morphism, and is an isomorphism when the condition (...) holds.

Proof. The condition given for being an isomorphism is equivalent to $\pi(S) \subset (A/\mathfrak{a})^\star$, which is equivalent to $i_{\pi(S)}$ being an isomorphism. Therefore we may apply (3.5.25). ∎

Corollary 3.5.27
Let A be a ring, $\mathfrak{m}$ a maximal ideal and S a multiplicative set such that $S \cap \mathfrak{m} = \emptyset$. Then there is an isomorphism

$$A/\mathfrak{m} \to S^{-1}A/S^{-1}\mathfrak{m}$$

In particular $S^{-1}\mathfrak{m}$ is a maximal ideal of $S^{-1}A$.

Proposition 3.5.28 (Transitivity)
Let $S \subset T$ be multiplicative subsets of A and let

$$i_S : A \to S^{-1}A$$

be the localisation at S. Define $T_S := i_S(T)$. Then T_S is multiplicative and there is a canonical isomorphism

$$T^{-1}A \to (T_S)^{-1}(S^{-1}A) \to \left(\overline{T_S}\right)^{-1}(S^{-1}A)$$

Furthermore if $T \subseteq U$ then $T_S \subseteq U_S$ there is a commutative diagram

$$\begin{array}{ccccc} & & S^{-1}A & & \\ & \swarrow^{i_{ST}} & \downarrow{\scriptstyle i_{T_S}} & \searrow^{i_{\overline{T_S}}} & \\ T^{-1}A & \xrightarrow{\sim} & (T_S)^{-1}(S^{-1}A) & \xrightarrow{\sim} & \left(\overline{T_S}\right)^{-1}(S^{-1}A) \\ \downarrow{\scriptstyle i_{TU}} & & \downarrow & & \downarrow \\ U^{-1}A & \xrightarrow{\sim} & (U_S)^{-1}(S^{-1}A) & \xrightarrow{\sim} & \left(\overline{U_S}\right)^{-1}(S^{-1}A) \end{array}$$

3.5.5 Algebras

Lemma 3.5.29 (Localisation is a Base Change)
Let A be a ring, S a multiplicative subset, and M an A-module (3.5.11). Then there is a natural isomorphism of $S^{-1}A$-modules

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$$S^{-1}A \otimes_A M \xrightarrow{\sim} S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s}.$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The map $\Phi : S^{-1}A \times M \to S^{-1}M$, $\Phi\left(\frac{a}{s}, m\right) := \frac{am}{s}$, is well-defined: if $\frac{a}{s} = \frac{a'}{s'}$ in $S^{-1}A$ then $u(as' - a's) = 0$ in A for some $u \in S$, hence $u(as' - a's)m = 0$ in M, i.e. $u(s'(am) - s(a'm)) = 0$, i.e. $\frac{am}{s} = \frac{a'm}{s'}$ in $S^{-1}M$. Φ is clearly bi-additive, and A-balanced since $\Phi\left(\frac{a}{s} \cdot b, m\right) = \frac{abm}{s} = \Phi\left(\frac{a}{s}, bm\right)$ for $b \in A$; so by the universal property of the tensor product (3.4.4) it induces a group homomorphism

$$\phi : S^{-1}A \otimes_A M \to S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s},$$

which is visibly $S^{-1}A$-linear.

Conversely define $\psi : S^{-1}M \to S^{-1}A \otimes_A M$ by $\psi\left(\frac{m}{s}\right) := \frac{1}{s} \otimes m$. This is well-defined: if $\frac{m}{s} = \frac{m'}{s'}$ in $S^{-1}M$ then $u(s'm - sm') = 0$ for some $u \in S$, and then, using $\frac{1}{s} \otimes m = \frac{us'}{uss'} \otimes m = \frac{1}{uss'} \otimes us'm$ (moving $us' \in A$ across the tensor),

$$\frac{1}{s} \otimes m - \frac{1}{s'} \otimes m' = \frac{1}{uss'} \otimes us'm - \frac{1}{uss'} \otimes usm' = \frac{1}{uss'} \otimes u(s'm - sm') = 0.$$

Then $\phi\left(\psi\left(\frac{m}{s}\right)\right) = \phi\left(\frac{1}{s} \otimes m\right) = \frac{m}{s}$, so $\phi \circ \psi = \mathbf{1}$. Also $\psi\left(\phi\left(\frac{a}{s} \otimes m\right)\right) = \psi\left(\frac{am}{s}\right) = \frac{1}{s} \otimes am = \frac{a}{s} \otimes m$ (again moving a across the tensor), and since elementary tensors generate $S^{-1}A \otimes_A M$, $\psi \circ \phi = \mathbf{1}$. Hence ϕ is an isomorphism, with inverse ψ. ∎

Corollary 3.5.30 (Localisation of an Algebra is a Base Change)
Let A be a ring, S a multiplicative subset, and B an A-algebra. Then $S^{-1}B$ is naturally an $S^{-1}A$-algebra (via $S^{-1}A \to S^{-1}B$, $\frac{a}{s} \mapsto \frac{ab_0}{s}$ where $b_0 \in B$ is the image of 1_A, the unique extension of $A \to B \to S^{-1}B$ along i_S (3.5.5), as $i_S(S) \subset (S^{-1}B)^\star$), and the isomorphism of $S^{-1}A$-modules furnished by (3.5.29) (with $M := B$),

$$S^{-1}A \otimes_A B \xrightarrow{\sim} S^{-1}B, \quad \frac{a}{s} \otimes b \mapsto \frac{ab}{s},$$

is moreover an isomorphism of rings, where $S^{-1}A \otimes_A B$ carries the ring structure of (3.4.34).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write ϕ for the $S^{-1}A$-module isomorphism of (3.5.29). It remains only to check ϕ is multiplicative on elementary tensors, since it is already additive:

$$\phi\left(\left(\frac{a}{s} \otimes b\right) \cdot \left(\frac{a'}{s'} \otimes b'\right)\right) = \phi\left(\frac{aa'}{ss'} \otimes bb'\right) = \frac{aa'bb'}{ss'} = \frac{a}{s} \cdot b \cdot \frac{a'}{s'} \cdot b' = \phi\left(\frac{a}{s} \otimes b\right) \cdot \phi\left(\frac{a'}{s'} \otimes b'\right).$$

As every element of $S^{-1}A \otimes_A B$ is a sum of elementary tensors, and ϕ is additive, ϕ is a ring homomorphism; being also a bijection, it is a ring isomorphism. ∎

Corollary 3.5.31 (Localisation of a Domain is a Base Change to the Fraction Field)
Let $A \subseteq B$ be an inclusion of integral domains, $K := \mathrm{Frac}(A)$, and $S := A \setminus \{0\}$, so that $S^{-1}B$ is naturally a subring of $\mathrm{Frac}(B)$ (as $S \subset B \setminus \{0\}$ too). Then there is a natural isomorphism of rings

$$B \otimes_A K \xrightarrow{\sim} S^{-1}B, \quad b \otimes x \mapsto bx,$$

where the right-hand side is computed in $\mathrm{Frac}(B)$, regarding $B, K \subseteq \mathrm{Frac}(B)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.5.30) (applied to the A-algebra B, noting $S^{-1}A = K$) there is a ring isomorphism $K \otimes_A B \xrightarrow{\sim} S^{-1}B$, $\frac{a}{s} \otimes b \mapsto \frac{ab}{s}$. Composing with the symmetry isomorphism $B \otimes_A K \cong K \otimes_A B$, $b \otimes x \mapsto x \otimes b$ (3.4.7) (which is also a ring isomorphism, since swapping factors clearly preserves the componentwise multiplication of (3.4.34)), yields a ring isomorphism $B \otimes_A K \xrightarrow{\sim} S^{-1}B$ sending $b \otimes \frac{a}{s} \mapsto \frac{ab}{s}$; taking $a = 1$ this is $b \otimes x \mapsto x \cdot b = bx$ for $x = \frac{1}{s}$, and since $x = \frac{a}{s} = a \cdot \frac{1}{s}$ for general $x \in K$, bilinearity gives $b \otimes x = ab \otimes \frac{1}{s} \mapsto \frac{ab}{s} = (ab) \cdot \frac{1}{s} = b \cdot \left(a \cdot \frac{1}{s}\right) = bx$ on every elementary tensor.

Finally $\frac{ab}{s} \in S^{-1}B \subseteq \mathrm{Frac}(B)$, computed there, agrees with the product bx computed in $\mathrm{Frac}(B)$ (as $B, K \subseteq \mathrm{Frac}(B)$). ∎

Remark 3.5.32
(3.5.30) recovers (3.5.41) (relative localisation at a prime) as the special case $S := A \setminus \mathfrak{p}$ of a general ring homomor-

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phism $\phi : A \to B$: the module localisation $S^{-1}B$ (with B regarded as an A-module via ϕ) is exactly $\phi(S)^{-1}B = B_{\mathfrak{p}}$, since the two defining equivalence relations on fractions agree termwise.

3.5.6 Localisation at an element

Definition 3.5.33 (Localisation at an element)
Let A be a ring and $f \in A$. Then define

$$S_f = \{1, f, ..., f^n, ...\}$$

and

$$A_f \coloneqq \left(S_f\right)^{-1} A$$

We have canonical maps

$$i_f : A \to A_f$$

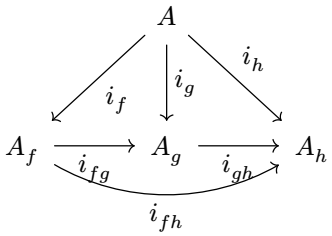
given by $i_f \coloneqq i_{S_f}$.

Note that A_f is a zero ring iff f is nilpotent.

Proposition 3.5.34 (Transition maps for localisation at an element)
Let A be a ring and $f, g \in A$ then

$$S_f \subseteq \overline{S_g} \iff f \mid g^N \text{ some N} > 0$$

in this case define $i_{fg} = i_{S_f S_g}$ to be the unique morphism such that $i_{fg} \circ i_f = i_g$ (3.5.24). In addition if $h \in A$ and $S_g \subseteq \overline{S_h}$ we have a commutative diagram

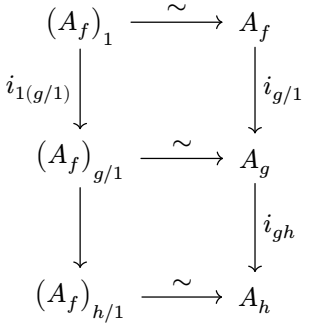


Furthermore $\overline{S_1} = A^\star$ and i_1 is an isomorphism.

Proposition 3.5.35 (Transitivity of localizing at elements)
Let A be a ring and $f, g \in A$ such that $S_f \subseteq \overline{S_g}$. There is a canonical isomorphism

$$A_g \xrightarrow{\sim} \left(A_f\right)_{g/1}$$

Furthermore $S_g \subseteq \overline{S_h} \implies S_{g/1} \subseteq \overline{S_{h/1}}$ and there is a commutative diagram



with the horizontal arrows isomorphisms and the vertical arrows are well-defined.

Proof. The existence of the isomorphism is from (3.5.28) as $i_f(S_g) = S_{g/1}$. The second statement follows because $g \mid h^N \implies g/1 \mid h^N/1$ trivially. ∎

Proposition 3.5.36
Let A be a ring and $f \in A$. Then localisation map $A \to A_f$ induces an order-preserving correspondence of prime ideals

$$\{\mathfrak{p} \lhd A \mid f \notin \mathfrak{p}\} \leftrightarrow \{\mathfrak{p} \lhd A_f\}$$

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Proof. The correspondence of prime ideals follows from (3.5.19) once we observe that $S_f \cap \mathfrak{p} = \emptyset \iff f \notin \mathfrak{p}$. ∎

Proposition 3.5.37
Let A be a ring, S a multiplicatively closed subset and $g \in S$. Then the canonical map $A_g \to S^{-1}A$ has kernel $\ker(i_S)_g$. In particular when $\ker(i_S)$ is finitely generated it is injective iff $g^r \ker(i_S) = 0$ for some $r > 0$.

When A is Noetherian then there always exists g such $A_g \to S^{-1}A$ is injective, and any multiple of g also satisfies this property.

Proof. The first statement follows from (3.5.4).

When A is Noetherian then $\ker(i_S) = (a_1, ..., a_n)$ for some $a_i \in A$. By definition there exists some s_i such that $s_i a_i = 0$, whence we may take $g = s_1 \cdots s_n$. ∎

Proposition 3.5.38
Let A be a ring, S a multiplicatively closed subset and $M_1, ..., M_n$ a family of A-modules. Then there is an isomorphism

$$S^{-1}\left(\bigoplus_{i=1}^{n} M_i\right) \xrightarrow{\sim} \bigoplus_{i=1}^{n} S^{-1}M_i$$

3.5.7 Localisation at a prime ideal

Definition 3.5.39 (Localisation at a prime ideal)
Let A be a ring and $\mathfrak{p} \lhd A$ a prime ideal. Then $S := A \setminus \mathfrak{p}$ is a saturated multiplicatively closed subset, and we define

$$A_{\mathfrak{p}} := (A \setminus \mathfrak{p})^{-1}A$$

For an ideal $\mathfrak{a} \lhd A$ write the extended ideal

$$\mathfrak{a}A_{\mathfrak{p}} := S^{-1}\mathfrak{a} = \mathfrak{a}^e.$$

Definition 3.5.40 (Relative localisation at a prime ideal)
Let $\phi : A \to B$ be a ring homomorphism and $\mathfrak{p} \lhd A$ a prime ideal. Define

$$B_{\mathfrak{p}} := \phi(A \setminus \mathfrak{p})^{-1}B$$

For an ideal $\mathfrak{a} \lhd A$ write

$$\mathfrak{a}B_{\mathfrak{p}} := (\phi(\mathfrak{a})B)B_{\mathfrak{p}}$$

Observe $B_{\mathfrak{p}} = 0 \iff \mathfrak{p} \subsetneq \ker(\phi)$, so we would typically assume $\ker(\phi) \subseteq \mathfrak{p}$.

Proposition 3.5.41 (Relative Localisation as a Tensor Product)
Let $\phi : A \to B$ be a ring homomorphism (so B is an A-algebra) and $\mathfrak{p} \lhd A$ a prime ideal, $S := A \setminus \mathfrak{p}$. Then $B_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-algebra via $\frac{a}{s} \mapsto \phi(a)/\phi(s)$ (well-defined as $\phi(S) \subseteq B_{\mathfrak{p}}^{\star}$ already), and there is an isomorphism of $A_{\mathfrak{p}}$-algebras

$$\Psi : B \otimes_A A_{\mathfrak{p}} \xrightarrow{\sim} B_{\mathfrak{p}}, \quad b \otimes \frac{a}{s} \mapsto \frac{\phi(a)b}{\phi(s)},$$

with inverse $\Theta(b/\phi(s)) := b \otimes \left(\frac{1}{s}\right)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Ψ is well-defined. For fixed $\frac{a}{s} \in A_{\mathfrak{p}}$, $b \mapsto \phi(a)b/\phi(s)$ is A-linear in b ($B_{\mathfrak{p}}$ being an A-module via ϕ); for fixed $b \in B$, if $\frac{a}{s} = \frac{a'}{s'}$ in $A_{\mathfrak{p}}$ then $s''(as' - a's) = 0$ for some $s'' \in S$, so applying ϕ and multiplying by b, $\phi(s'')(\phi(a)\phi(s') - \phi(a')\phi(s))b = 0$ in B, giving $\phi(a)b/\phi(s) = \phi(a')b/\phi(s')$ in $B_{\mathfrak{p}}$; and $\frac{a}{s} \mapsto \phi(a)b/\phi(s)$ is A-linear in $\frac{a}{s}$ (direct check, as in (3.5.5)). So the map $B \times A_{\mathfrak{p}} \to B_{\mathfrak{p}}$, $\left(b, \frac{a}{s}\right) \mapsto \phi(a)b/\phi(s)$, is A-bilinear, and induces Ψ by the universal property of the tensor product (3.4.4).

Θ is well-defined. Write $a \cdot b := \phi(a)b$ for the A-module structure on B. If $b/\phi(s) = b'/\phi(s')$ in $B_{\mathfrak{p}}$ then $s'' \cdot (s' \cdot b) = s'' \cdot (s \cdot b')$ in B for some $s'' \in S$; using $(a \cdot b) \otimes x = b \otimes (ax)$ (the defining relation of $\otimes_A$) twice,

$$b \otimes \frac{1}{s} = (s''s') \cdot b \otimes \frac{1}{s''s's} = (s'' \cdot (s' \cdot b)) \otimes \frac{1}{s''s's} = (s'' \cdot (s \cdot b')) \otimes \frac{1}{s''ss'} = b' \otimes \frac{1}{s'},$$

so Θ is well-defined on $B_{\mathfrak{p}}$.

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Θ and Ψ are mutually inverse. $\Psi\Theta(b/\phi(s)) = \Psi\left(b \otimes \frac{1}{s}\right) = \phi(1)b/\phi(s) = b/\phi(s)$. And $\Theta\Psi\left(b \otimes \frac{a}{s}\right) = \Theta(\phi(a)b/\phi(s)) = (\phi(a)b) \otimes \left(\frac{1}{s}\right) = b \otimes \left(\frac{a}{s}\right)$ (the same tensor relation as above), so $\Theta\Psi$ is the identity on simple tensors, hence (by linearity) on all of $B \otimes_A A_\mathfrak{p}$.

Ψ is $A_\mathfrak{p}$-linear. For $\frac{a'}{s'} \in A_\mathfrak{p}$,

$$\Psi\left(\frac{a'}{s'} \cdot \left(b \otimes \frac{a}{s}\right)\right) = \Psi\left(b \otimes \frac{aa'}{ss'}\right) = \frac{\phi(aa')b}{\phi(ss')} = \frac{\phi(a')}{\phi(s')} \cdot \frac{\phi(a)b}{\phi(s)} = \frac{a'}{s'} \cdot \Psi\left(b \otimes \frac{a}{s}\right),$$

using the $A_\mathfrak{p}$-algebra structure on $B_\mathfrak{p}$ displayed above; so Ψ is $A_\mathfrak{p}$-linear on simple tensors, hence (by linearity) on all of $B \otimes_A A_\mathfrak{p}$.

Multiplicativity. By the algebra tensor product structure on $B \otimes_A A_\mathfrak{p}$, $\left(b \otimes \frac{a}{s}\right)\left(b' \otimes \frac{a'}{s'}\right) = bb' \otimes \frac{aa'}{ss'}$, and

$$\Psi\left(\left(b \otimes \frac{a}{s}\right)\left(b' \otimes \frac{a'}{s'}\right)\right) = \frac{\phi(aa')bb'}{\phi(ss')} = \frac{\phi(a)b}{\phi(s)} \cdot \frac{\phi(a')b'}{\phi(s')} = \Psi\left(b \otimes \frac{a}{s}\right)\Psi\left(b' \otimes \frac{a'}{s'}\right),$$

so Ψ is multiplicative on simple tensors, hence (by linearity) a ring homomorphism – and so, with $A_\mathfrak{p}$-linearity above, an isomorphism of $A_\mathfrak{p}$-algebras. ∎

Proposition 3.5.42
Let A be a ring and $\mathfrak{p}$ a prime ideal. Consider the localisation $A \to A_\mathfrak{p}$. Then there is an order-preserving bijection between (prime) ideals contained in $\mathfrak{p}$ and (prime) ideals of $A_\mathfrak{p}$

$$\{\mathfrak{q} \lhd A \mid \mathfrak{q} \subseteq \mathfrak{p}\} \leftrightarrow \{\mathfrak{q} \lhd A_\mathfrak{p}\}$$
$$\mathfrak{q} \to \mathfrak{q}A_\mathfrak{p}$$

In particular $A_\mathfrak{p}$ is a local ring with unique maximal ideal $\mathfrak{p}A_\mathfrak{p}$.

Proof. Clearly $\mathfrak{q} \cap (A \setminus \mathfrak{p}) = \emptyset \iff \mathfrak{q} \subseteq \mathfrak{p}$, so the result follows from (3.5.19). ∎

Proposition 3.5.43
Let A be a ring and $\mathfrak{p}$ a prime ideal. Then the map

$$\mathrm{Frac}(A/\mathfrak{p}) \xrightarrow{\sim} A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$$
$$\frac{\overline{a}}{\overline{b}} \to \frac{\overline{a/1}}{\overline{b/1}} = \overline{\left(\frac{a}{b}\right)}$$

is a ring isomorphism

Proof. Consider the ring map $A \to A_\mathfrak{p} \to A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. Then this has kernel $\mathfrak{p}^{\mathrm{ec}} = \mathfrak{p}$ (3.5.18), so may be factored as an injective map $A/\mathfrak{p} \hookrightarrow A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. By (3.5.42) $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ is a field so by universal property of localisation (3.5.5) this extends to an injective map $\mathrm{Frac}(A/\mathfrak{p}) \hookrightarrow A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ of the given form. We may verify directly that it is surjective. ∎

Proposition 3.5.44 (Localisation at prime is direct limit of localisation at an element)
Let A be a ring and $\mathfrak{p}$ a prime ideal then

$$S_\mathfrak{p} := A \setminus \mathfrak{p} = \bigcup_{f \in A \setminus \mathfrak{p}} \overline{S_f}$$

Therefore there are canonical morphisms (for $f \notin \mathfrak{p}$)

$$i_{S_f S_\mathfrak{p}} : A_f \to A_\mathfrak{p}$$

Furthermore the family of multiplicatively closed sets $\{S_f\}_{f \notin \mathfrak{p}}$ (resp. rings $\{A_f\}_{f \notin \mathfrak{p}}$) form a directed system under the relation $S \prec T \iff S \subseteq \overline{T}$. Therefore we have a canonical ring homomorphism

$$\varinjlim_{f \notin \mathfrak{p}} A_f \to A_\mathfrak{p}$$

which is an isomorphism.

Proof. As $A \setminus \mathfrak{p}$ is a saturated multiplicatively closed set we have $f \in A \setminus \mathfrak{p} \iff S_f \subseteq A \setminus \mathfrak{p} \iff \overline{S_f} \subseteq A \setminus \mathfrak{p}$. Therefore the expression for $S_\mathfrak{p}$ follows.

The family of multiplicatively closed subsets is a directed system because $S_f \subseteq \overline{S_{fg}}$. To see this note $fg \in \overline{S_{fg}} \implies f \in \overline{S_{fg}} \implies S_f \subseteq \overline{S_{fg}}$.

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The final isomorphism follows because we can decompose it into two maps

$$\varinjlim_{f\notin\mathfrak{p}} A_f \xrightarrow{\sim} \varinjlim_{f\notin\mathfrak{p}} \overline{S_f}^{-1}A \xrightarrow{\sim} A_{\mathfrak{p}}$$

The first is an isomorphism by (3.5.24) and the second by (3.5.8), in light of the first statement. ∎

Proposition 3.5.45 (Quotient commutes with Localisation at Prime ideal)
Let A be a ring, $\mathfrak{a}$ an ideal and $\mathfrak{p}\supset\mathfrak{a}$ a prime ideal. Then there is an isomorphism of non-zero rings

$$A_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}} \cong (A/\mathfrak{a})_{\mathfrak{p}/\mathfrak{a}}$$
$$\frac{a}{s}+\mathfrak{a}_{\mathfrak{p}} \to \frac{a+\mathfrak{a}}{s+\mathfrak{a}}$$

Proof. Let $\pi: A\to A/\mathfrak{a}$ be the quotient map then we claim $\pi(A\setminus\mathfrak{p})=\pi(A)\setminus\pi(\mathfrak{p})$. As π is surjective we only need to show that $\pi^{-1}(\pi(\mathfrak{p}))=\mathfrak{p}$. That is if $\pi(a)=\pi(b)$ and $b\in\mathfrak{p}$ then $a\in\mathfrak{p}$. However this follows by assumption because $\mathfrak{a}\subset\mathfrak{p}$. Therefore we may apply (3.5.25). ∎

Proposition 3.5.46 (Minimal Primes consist of Zero Divisors)
Let $\mathfrak{p}$ be a proper prime ideal minimal over $\mathfrak{a}$. Then for every $x\in\mathfrak{p}$ there exists $a\in A\setminus\mathfrak{a}$ such that $xa\in\mathfrak{a}$.

In particular a minimal prime ideal consists of zero-divisors.

Proof. Observe that $\mathfrak{p}/\mathfrak{a}$ is a proper minimal prime ideal of $A/\mathfrak{a}$ by (3.3.56). Then the first statement is equivalent to showing every $x\in\mathfrak{p}/\mathfrak{a}$ is a zero divisor and we may reduce to the case $\mathfrak{a}=(0)$.

Consider the ring $A_{\mathfrak{p}}$, it has a unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ which is also minimal by (3.5.42). Every other prime ideal is contained in $\mathfrak{p}A_{\mathfrak{p}}$ by (3.3.37), and therefore equal by minimality. Therefore $\sqrt{(0)A_{\mathfrak{p}}}=\mathfrak{p}A_{\mathfrak{p}}$ by (3.3.46). Suppose $x\in\mathfrak{p}$ then $\frac{x^n}{1}=0$ and $tx^n\in\mathfrak{a}$ for $t\notin\mathfrak{p}$. Choose n minimal subject to this condition. Then $n\geq 1$ because $t\notin\mathfrak{a}$, and so $a=tx^{n-1}$ has the required properties. ∎

3.5.8 Finiteness Results

Lemma 3.5.47
Let M be an A-module, N a submodule and $f_1,...,f_r\in A$ such that

a) $N_{f_i}\to M_{f_i}$ is surjective for all $i=1...r$
b) $A=(f_1,...,f_r)$ as an ideal

then $M=N$.

Proof. Consider the A-module $P:=M/N$. Suppose $P\neq 0$, then $0\neq p\in P$ and $\mathrm{Ann}(p)$ is a proper ideal of A. In particular $f_i\notin\mathrm{Ann}(p)$ for some i. This shows $P_{f_i}\neq 0$. From (3.5.14) we have $P_{f_i}\xrightarrow{\sim} M_{f_i}/N_{f_i}$ and so $N_{f_i}\neq M_{f_i}$ a contradiction. ∎

Proposition 3.5.48
Let M be an A-module. Then

a) M finitely-generated $\implies M_f$ is a finitely generated A_f-module for all $f\in A$
b) $(f_1,...,f_r)=A$ and M_{f_i} a finitely generated A_{f_i}-module for all $i=1...r\implies M$ is a finitely-generated A-module

Proof.
a) Let $m_1,...m_n\in M$ be a generating set. Then it is easy to verify that $m_1/1,...,m_n/1$ is a generating set for M_f.
b) For each $i=1...r$, let m_{ij} be such that $m_{ij}/1$ generates M_{f_i} as an A_{f_i}-module. Let N be the A-submodule of M generated by the m_{ij}. Then $N_{f_i}=M_{f_i}$. We conclude by (3.5.47) that $N=M$ is finitely generated as an A-module.

∎

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3.6 Monoid Ring

Definition 3.6.1 (Monoid Ring)
Let A be a commutative ring and G be a commutative monoid. We define the ***monoid ring*** *to be the free abelian group*

$$A[G] := \bigoplus_{g \in G} A$$

We may write a general element as a formal sum

$$\sum_{g \in G} a_g g$$

where all but finitely many a_g are zero. Multiplication is defined extending the group operation. More precisely

$$\left(\sum_{g \in G} a_g g\right) \cdot \left(\sum_{g \in G} b_g g\right) = \sum_{g \in G} \left(\sum_{(h,k) | h+k=g} a_h b_k g \right)$$

We may verify that the identity is simply e and the multiplication is both associative and distributive.

When G is an abelian group we say that $A[G]$ is the ***group ring****. There is a canonical map $A \to A[G]$ making $A[G]$ into an A-algebra.*

Proposition 3.6.2
Let A, B be commutative rings and G a commutative monoid. Let $\phi : A \to B$ be a ring homomorphism and $\alpha : G \to (B, \times)$ a monoid homomorphism. Then there is a unique homomorphism

$$\phi \ltimes \alpha : A[G] \to B$$

such that $(\phi \ltimes \alpha)(ae) = \phi(a)$ and $(\phi \ltimes \alpha)(1_A g) = \alpha(g)$.

Proof. Define

$$(\phi \ltimes \alpha)\left(\sum_{g \in G} a_g g\right) = \sum_{\substack{g \in G \\ a_g \neq 0}} \phi(a_g)\alpha(g)$$

■

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3.7 Polynomial Rings in One Variable

Definition 3.7.1 (Polynomial Ring)
Let A be a commutative ring. Define the polynomial ring $A[X]$ to be the monoid ring $A[\mathbb{N}]$ consisting of formal

$$f(X) = \sum_{i=0}^{\infty} a_i X^i$$

such that only finitely many a_i are non-zero. Define degree in the obvious way

$$\deg(f) = \inf\{n \mid m > n \implies a_m = 0\} < \infty$$

and the leading coefficient to be $\ell(f) := a_{\deg(f)}$. By convention $\deg(0) = -\infty$.

Definition 3.7.2 (Monic polynomial)
*Let $f \in A[X]$. We say f is **monic** if the leading coefficient, $\ell(f)$, is 1.*

Lemma 3.7.3
If A is an integral domain then for elements $f, g \in A[X]$

$$\deg(fg) = \deg(f) + \deg(g)$$

$$\ell(fg) = \ell(f)\ell(g)$$

Further $A[X]$ is an integral domain.

Lemma 3.7.4 (Monic Polynomials are not Zero-Divisors)
Let A be a commutative ring and $f \in A[X]$ monic. If $g \in A[X]$ and $fg = 0$ then $g = 0$.

Proof. Suppose $g \neq 0$ has degree d and leading coefficient $c \neq 0$, and let f have degree e. The coefficient of X^{e+d} in fg is $\sum_{i+j=e+d} f_i g_j$, where only $i = e$, $j = d$ contribute as $f_i = 0$ for $i > e$ and $g_j = 0$ for $j > d$. So it equals $f_e g_d = c \neq 0$, a contradiction. ∎

Proposition 3.7.5 (Nilpotent and Invertible Polynomials)
Let A be a ring then

a) $N(A[X]) = N(A)[X] \subset A[X]$
b) $A[X]^\star = A^\star + XN(A)[X]$

Proof. Suppose $a \in N(A)$ then clearly aX^i is nilpotent. Therefore $N(A)[X] \subseteq N(A[X])$ since the nilradical is an ideal. Conversely suppose $f \in A[X]$ is nilpotent, i.e. $f^n = 0$. For any prime ideal $\mathfrak{p} \lhd A$ we find that $\overline{f}^n = 0$ as an element of $(A/\mathfrak{p})[X]$. As $A/\mathfrak{p}$ is an integral domain we have by the previous Lemma $\overline{f} = 0$. As $\mathfrak{p}$ was arbitrary and $N(A) = \bigcap \mathfrak{p}$ we see that $f \in N(A)[X]$ as required.

Suppose $f \in A[X]^\star$ and $fg = 1$, then clearly the constant term of f must be invertible. Reduce modulo $\mathfrak{p}$ to find $\deg\left(\overline{f}\right) + \deg(\overline{g}) = 0 \implies \deg\left(\overline{f}\right) = \deg(\overline{g}) = 0$, which means $\overline{f}$ is a constant polynomial. As $\mathfrak{p}$ was arbitrary we see again that the other coefficients of f must be nilpotent as required. Therefore $A[X]^\star \subseteq A^\star + XN(A)[X]$. Conversely by (3.3.68) and a) we have $A^\star + XN(A)[X] \subseteq A[X]^\star + N(A[X]) \subseteq A[X]^\star$ so the result follows. ∎

It satisfies the following universal property

Proposition 3.7.6 (Evaluation at a point)
Consider an A-algebra B and $b \in B$. Then there exists a unique A-algebra homomorphism

$$\mathrm{ev}_b : A[X] \to B$$

such that $\mathrm{ev}_b(X) = b$. We write $p(b) = \mathrm{ev}_b(p)$. It is given by

$$p(b) = \sum_{k=0}^{\deg(p)} i_B(a_k) b^k$$

The image of ev_p is equal to $A[b]$ the smallest sub-A-algebra generated by b. For any morphism $\phi : B \to C$ such that $\phi(b) = c$ we have

$$\phi \circ \mathrm{ev}_b = \mathrm{ev}_c$$

Remark 3.7.7
In categorical jargon $A[X]$ is an initial object in the category of pointed A-algebras.

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Proposition 3.7.8 (Base Change of the Polynomial Ring)
Let $\phi : A \to B$ be a homomorphism of commutative rings. Then there is a natural isomorphism of B-algebras

$$A[X]_{(B)} \xrightarrow{\sim} B[X]$$
$$b \otimes \sum_i a_i X^i \to \sum_i b\phi(a_i) X^i$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

For any B-algebra Z (in particular an A-algebra via ϕ), combining (3.4.38) with the universal property of $A[X]$ (3.7.6) (applied to the underlying A-algebra of Z) gives a bijection natural in Z

$$\mathrm{AlgHom}_B\big(A[X]_{(B)}, Z\big) \cong \mathrm{AlgHom}_A(A[X], Z) \cong Z, \quad \psi \to \psi(1 \otimes X)$$

which is exactly the universal property of $B[X]$ (3.7.6) (applied with base ring B); by the Yoneda Lemma (2.6.58), this yields the required isomorphism (the image of $1_{A[X]_{(B)}}$ under it, sending $X \to 1 \otimes X$). ∎

Proposition 3.7.9 (Evaluation commutes with algebra homomorphism)
Let $\phi : B \to C$ be a homomorphism of A-algebras and $p \in A[X]$ then

$$\phi(p(b)) = p(\phi(b))$$

Proposition 3.7.10 (Formal Derivative)
Let A be a ring. Then there exists a unique A-module homomorphism

$$(-)' : A[X] \to A[X]$$

such that

$$(X^r)' = \begin{cases} 0 & r = 0 \\ rX^{r-1} & r > 0 \end{cases}$$

It satisfies the product rule

$$(fg)' = f'g + fg'$$

for all $f, g \in A[X]$. Define recursively $f^{(0)} = f$ and $f^{(n)} := f^{(n-1)'}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.7.1), $A[X]$ is the free A-module on $\{1, X, X^2, ...\}$, so $(-)'$ exists and is unique.

Suppose $f(X) = \sum_{i=0}^n a_i X^i$ and $g(X) = \sum_{i=0}^m b_i X^i$. Then

$$(fg)(X) = \sum_{i=0}^{n+m} \left(\sum_{k+l=i} a_k b_l \right) X^i$$

and

$$\begin{aligned}
f'(X) &= \sum_{i=0}^{n-1} (i+1) a_{i+1} X^i \\
g'(X) &= \sum_{i=0}^{m-1} (i+1) b_{i+1} X^i \\
(fg)'(X) &= \sum_{i=0}^{n+m-1} (i+1) \left(\sum_{k+l=i+1} a_k b_l \right) X^i \\
&= \sum_{i=0}^{n+m-1} \left(\sum_{k+l=i+1} (k+l) a_k b_l \right) X^i \\
&+ \sum_{i=0}^{n+m-1} \left(\sum_{k+l=i} (k+1) a_{k+1} b_l \right) X^i + \sum_{i=0}^{n+m-1} \left(\sum_{k+l=i} (l+1) a_k b_{l+1} \right) X^i \\
&= f'(X)g(X) + f(X)g'(X)
\end{aligned}$$

∎

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Corollary 3.7.11 (Taylor Expansion mod h^2)
Let A be a ring, $f \in A[X]$, and $a, h \in A$. Then

$$f(a+h) - f(a) - f'(a)h \in (h^2) \triangleleft A.$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $f(X) = \sum_{i=0}^{d} c_i X^i$. We first show, by induction on $i \geq 0$, that $(a+h)^i - a^i - ia^{i-1}h \in (h^2)$ (reading $a^{-1}h := 0$ for $i = 0$). The cases $i = 0, 1$ are immediate equalities ($1 - 1 - 0 = 0$, $(a+h) - a - h = 0$). Given $(a+h)^i = a^i + ia^{i-1}h + h^2 z$ for some $z \in A$:

$$(a+h)^{i+1} = (a+h)(a^i + ia^{i-1}h + h^2 z) = a^{i+1} + (i+1)a^i h + h^2(az + ia^{i-1} + hz),$$

so $(a+h)^{i+1} - a^{i+1} - (i+1)a^i h \in (h^2)$, completing the induction. Multiplying the case-i congruence by c_i and summing over $i = 0, ..., d$ (as (h^2) is an ideal) gives $f(a+h) - f(a) - f'(a)h \in (h^2)$, noting $f'(a) = \sum_i c_i \cdot i a^{i-1}$ by (3.7.10) evaluated at a (3.7.6). ∎

Lemma 3.7.12 (Shifted Powers Form a Basis)
Let k be a field, $\theta \in k$, and $n \geq 1$ an integer. Then $1, (X-\theta), ..., (X-\theta)^{n-1}$ is a k-basis for the space of polynomials in $k[X]$ of degree $< n$.

Proof. Using (3.7.6) we may apply an isomorphism $X \to X - \theta$, from which the result follows immediately. ∎

Definition 3.7.13 (Conjugate polynomial)
Let $\phi : A \to B$ be a homomorphism and $f \in A[X]$, then define

$$f^\phi(X) := \sum_{i=0}^{n} \phi(a_i) X^i$$

It induces a ring homomorphism

$$A[X] \to B[X]$$

and has the property that

$$f^\phi(\phi(a)) = \phi(f(a))$$

Proposition 3.7.14 (Division Algorithm I)
Let A be an integral domain and $f(X) \in A[X]$ a polynomial and $g(X) \in A[X]$ a non-zero monic polynomial. Then there exists unique polynomials $q(X)$ and $r(X)$ such that

- $f(X) = q(X)g(X) + r(X)$
- $\deg(r) < \deg(g)$

In particular when $\deg(g) = 1$ then $r \in A$.

Proof. If $\deg(f) < \deg(g)$ then $q = 0$ and $r = f$. Otherwise assume $n = \deg(f) \geq \deg(g) = m$ and proceed by induction on n. Note that since g is monic then we have $f - \ell(f) g X^{n-m}$ has degree $n - 1$, so by induction

$$f - \ell(f) g X^{n-m} = q'g + r$$

with $\deg(r) < \deg(g)$. Therefore

$$f = (q' + \ell(f) X^{n-m})g + r$$

as required.

Suppose $qg + r = q' + r'$ then $(q - q')g = (r' - r)$. This implies that $\deg((q-q')g) < \deg(g)$ which implies $(q-q')g = 0 \implies q = q'$ and $r = r'$. This demonstrates uniqueness. ∎

Proposition 3.7.15
Let B be an A-algebra and $f \in B$ not nilpotent. Then there is a canonical isomorphism

$$B[X]/(1 - Xf) \to B_f$$
$$\overline{P(X)} \to P(1/f)$$

In particular if B is a finitely-generated A-algebra, then so is B_f.

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Proof. By (3.7.6) there is a A-algebra homomorphism $\phi : B[X] \to B_f$ such that $\phi(X) = \frac{1}{f}$. Evidently $(1 - Xf) \subset \ker(\phi)$ so by (...) this induces an A-algebra homomorphism $B'_f := B[X]/(1 - Xf) \to B_f$. Evidently $\overline{X} = \overline{f}^{-1}$ in B'_f, so by (3.5.5) there is a map $\psi : B_f \to B[X]/(1 - Xf)$ given by $\frac{b}{f^r} \to \overline{bX^r}$. One may verify that ψ is a two-sided inverse for ϕ, and so is an isormophism.

The final statement follows from (3.3.76). ∎

Corollary 3.7.16
For k a field, the units of $k[X]$ are precisely the non-zero polynomials of degree 0.

Proof. As k is a field, $N(k) = 0$, so by (3.7.5) $k[X]^\star = k^\star + X \cdot 0 = k^\star$, that is to say the non-zero constants. ∎

Proposition 3.7.17 (Division Algorithm over a field)
For k a field consider the polynomial ring $k[X]$. For every pair of polynomials $f(X)$, $g(X)$ there exists unique polynomials $q(X)$ and $r(X)$ such that

$$f(X) = q(X)g(X) + r(X)$$

and $\deg(r) < \deg(g)$.

Proof. Apply (3.7.14) to $g/\ell(g)$, and multiply by $\ell(g)$ again. ∎

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3.8 Multilinear Algebra

3.8.1 Multilinear Maps and Determinants

Remark 3.8.1
The purpose of this section is to define the determinant, rather than develop any further theory. The approach is a little non-standard as we take an abstract module-theoretic approach as in [Bou98a], but avoid use of the tensor algebra. The connection is there are natural isomorphisms (i.e. the tensor algebra represents the functor $L^n_a(M;-)$)

$$L^n_a(M;N) \xrightarrow{\sim} \mathrm{Hom}(\Lambda^n(M), N)$$

and

$$\Lambda^n(M)^\vee \xrightarrow{\sim} \Lambda^n(M^\vee)$$

when M is finite free. Therefore the exterior product we define really corresponds to the exterior product on (the tensor algebra of) $M^\vee$.

Definition 3.8.2 (Multilinear Map)
Let $M_1, ..., M_n, N$ be A-modules then a map

$$\psi : M_1 \times ... \times M_n \to N$$

*is A-**multilinear** if it is A-linear in each variable, whilst fixing the other variables at any value.*

Definition 3.8.3 (Bilinear form)
*Let M, N be A-modules then $\psi : M \times N \to A$ is a **bilinear form** if it is A-multilinear.*

Denote the set of such bilinear pairings by $\mathrm{Bilin}_A(M,N)$. It is naturally an A-module.

Proposition 3.8.4
Let M, N be A-modules then there is a natural bijection

$$\begin{aligned} \mathrm{Hom}_A(M, \mathrm{Hom}_A(N,A)) &\leftrightarrow \mathrm{Bilin}_A(M,N) \leftrightarrow \mathrm{Hom}_A(N, \mathrm{Hom}_A(M,A)) \\ \psi_L &\leftarrow \psi \\ \psi &\qquad\qquad \to \psi_R \end{aligned}$$

where

$$\psi_L(m)(n) = \psi(m,n) = \psi_R(n)(m)$$

Definition 3.8.5 (Alternating map)
*An A-multilinear map $f : M^n \to N$ is **alternating** if*

$$f(x_1, ..., x_n) = 0$$

whenever $x_i = x_{i+1}$ for some $i = 1..., n-1$.

*Denote by $L^n_a(M,N)$ the set of such alternating maps, and $L^n_a(M) := L^n_a(M,A)$ the set of **alternating forms**. These are clearly A-modules.*

Proposition 3.8.6 (Functorial Properties)
Let M be an A-module and $L^k_a(M)$ the set of k-alternating forms. Then it is a contravariant functor in M, that is if $g : M \to N$ then there is a well-defined map

$$\begin{aligned} L^k_a(g) : L^k_a(N) &\to L^k_a(M) \\ \psi &\to \psi \circ g^{(k)} \end{aligned}$$

such that $L^k_a(g \circ h) = L^k_a(h) \circ L^k_a(g)$

Lemma 3.8.7
Let M be an A-module and $\psi \in L^{p-1}_a(M;A)$. Then

$$\psi'(x_1, ..., x_p) := \sum_{j=1}^{p} (-1)^{j+1} \psi(x_1, ..., \hat{x_j}, ..., x_p) x_j$$

is an element of $L^p(M;M)$.

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Proof. It is clear that ψ' is multilinear, so we only need to show it is alternating. Suppose that $x_i = x_{i+1}$. then the only non-zero terms are where one of x_i and x_{i+1} are omitted, hence

$$\psi'(x_1, ..., x_p) = (-1)^i \psi(x_1, ..., \hat{x_i}, ..., x_p) x_i + (-1)^{i+1} \psi(x_1, ..., \widehat{x_{i+1}}, ..., x_p) x_{i+1} = 0$$

as required. ∎

Corollary 3.8.8 (Exterior Product of Alternating Forms)
Let M be an A-module, $\theta \in L_a^1(M)$, $\psi \in L_a^{n-1}(M)$ then the function

$$\theta \wedge \psi : M^p \to A$$
$$(x_1, ..., x_n) \to \sum_{j=1}^{n} (-1)^{j+1} \psi(x_1, ..., \hat{x_j}, ..., x_p) \theta(x_j)$$

*is an alternating form. We denote this by $\theta \wedge \psi$ and refer to it as either the **wedge product** or **exterior product**, and determines a bilinear map*

$$\wedge : L_a^1(M) \times L_a^{n-1}(M) \to L_a^n(M)$$

Lemma 3.8.9
Let $f : M^n \to N$ be an alternating map then

$$f\left(x_{\sigma(1)}, ..., x_{\sigma(n)}\right) = \varepsilon(\sigma) f(x_1, ..., x_n)$$

for any permutation $\sigma \in S_n$.

Furthermore if any of the x_i are equal then $f(x_1, ..., x_n) = 0$

Proof. A permutation σ may be represented as a product of adjacent transpositions (...) therefore it's enough to demonstrate the case $\sigma = (i \quad i+1)$. This follows directly from the definition because

$$0 = f(x+y, x+y) = f(x,x) + f(y,x) + f(x,y) + f(y,y) = f(x,y) + f(y,x)$$

Suppose $x_i = x_j$, then we may apply the first result to the transposition $\sigma = (i\ j)$ to see that $f(x_1, ..., x_n) = 0$. ∎

Lemma 3.8.10
Let $\psi : M^p \to N$ be a multilinear map. Suppose $v_1, ..., v_n \in M$ and $w_1, ..., w_p \in M$ such that

$$w_i = \sum_{j=1}^{n} a_{ij} v_j \quad \forall i = 1...p \text{ and some } a_{ij} \in A$$

Then

$$\psi(w_1, ..., w_p) = \sum_{\sigma} a_{1\sigma(1)} ... a_{p\sigma(p)} \psi\left(v_{\sigma(1)}, ..., v_{\sigma(p)}\right)$$

where σ runs over all functions $[1, p] \to [1, n]$.

Lemma 3.8.11
Let $\psi : M^n \to N$ be an alternating map. Suppose $v_1, ..., v_n \in M$ and $w_1, ..., w_n \in M$ such that

$$w_i = \sum_{j=1}^{n} a_{ij} v_j \quad \forall i = 1...n \text{ and some } a_{ij} \in A$$

then

$$\psi(w_1, ..., w_n) = \sum_{\sigma \in S_n} \varepsilon(\sigma) a_{1\sigma(1)} ... a_{n\sigma(n)} \psi(v_1, ..., v_n)$$

Proof. By (3.8.10)

$$\psi(w_1, ..., w_n) = \sum_{\sigma} a_{1\sigma(1)} ... a_{n\sigma(n)} \psi\left(v_{\sigma(1)}, ..., v_{\sigma(n)}\right)$$

where σ ranges over all maps from $\{1, ..., n\}$ to itself. If σ is not a permutation, then it must not be injective and by (3.8.9) the corresponding term is zero. Therefore we may restrict to the case $\sigma \in S_n$. By the first part of (3.8.9) the result follows. ∎

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Proposition 3.8.12
Let M be a finite free A-module of rank n. Then the space of alternating forms $L_a^p(M;N) = 0$ for any integer $p > n$.

Proof. We may use (3.8.10) and observe that at least one basis element must occur at least twice in the expansion, and so is zero by (3.8.9). ∎

Proposition 3.8.13 (Existence and Uniqueness of Determinant Corresponding to a Basis)
Let M be a finite free A-module with basis $\{v_1, ..., v_n\}$. Then there is a unique alternating form $\Delta_v \in L_a^n(M)$ such that $\Delta_v(v_1, ..., v_n) = 1$.

Furthermore $L_a^n(M)$ is a free module of rank 1 with basis Δ_v.

*More precisely every $\Delta \in L_a^n(M)$ satisfies $\Delta = \Delta(v_1, ..., v_n)\Delta_v$. Further every Δ also satisifes **Leibniz' formula***

$$\Delta(x_1, ..., x_n) = \sum_{\sigma \in S_n} \varepsilon(\sigma) v_{\sigma(1)}^\vee(x_1)...v_{\sigma(n)}^\vee(x_n)\Delta(v_1, ..., v_n)$$

Proof. Let $\{v_1^\vee, ..., v_n^\vee\}$ be the canonical dual basis of $M^\vee$. Then we may use the wedge product (3.8.8) to construct Δ_v inductively

$$\Delta_v := v_1^\vee \wedge ... \wedge v_n^\vee$$

(using "right associativity" to ensure this expression is meaningful as the left operand must always be a linear functional). It is then clear that $\Delta_v(v_1, ..., v_n) = 1$. We may apply (3.8.11) to obtain Leibniz' formula for both Δ_v and Δ, which shows that $\Delta = \Delta(v_1, ..., v_n)\Delta_v$. ∎

Corollary 3.8.14 (Existence and Uniqueness of Determinant)
*For every $n \geq 1$ there is a unique $D_n \in L_a^n(A^n)$ such that $D_n(e_1, ..., e_n) = 1$, given by the **Leibniz Formula***

$$D_n(v_1, ..., v_n) = \sum_{\sigma \in S_n} \varepsilon(\sigma) v_{1\sigma(1)}...v_{n\sigma(n)}.$$

Further $L_a^n(A^n)$ is a free A-module of rank 1 generated by D_n. Explicitly every $\Delta \in L_a^n(A^n)$ satisfies

$$\Delta = \Delta(e_1, ..., e_n)D_n$$

*Further D_n satisfies the **Laplace Expansion Formula** that is for all $i, j = 1...n$ we have the relationship*

$$D_n(v_1, ..., v_n) = \sum_{j=1}^n (-1)^{i+j} v_{ji} D_{n-1}\Big(\pi_i(v_1), ..., \widehat{\pi_i(v_j)}, ..., \pi_i(v_n)\Big)$$

where $\pi_i : A^n \to A^{n-1}$ drops the i-th element.

Proof. The first two statements follow from (3.8.13) in the case $M = A^n$.

The function $(-1)^{i+1} D_{n-1}(\pi_i(v_1), ..., \pi_i(v_{n-1}))$ is evidently multilinear and alternating, that is an element of $L^{n-1}(A^n)$. Forming the wedge product with $e_i^\vee$ we obtain an element Δ of $L^n(A^n)$, which has the form given by the Laplace Expansion. Furthermore

$$\Delta(e_1, ..., e_n) = (-1)^{i+1} e_i^\vee(e_i)(-1)^{i+1} D_{n-1}\Big(\pi_i(e_n), ..., \widehat{\pi_i(e_i)}, ..., \pi_i(e_n)\Big) = 1.$$

whence by (3.8.13) $\Delta = D_n$ as required. ∎

Proposition 3.8.15 (Rank is Unique)
Let M be a finite free A-module. Then the rank is unique.

Proof. This follows from (3.8.12) and (3.8.13). Explicitly if M has bases of order p and q with $p < q$, then we deduce $L_a^q(M)$ is both zero and non-zero a contradiction. ∎

Definition 3.8.16 (Determinant of a Module)
*Let M be a finite free A-module of rank n, then we say a generator for $L_a^n(M)$ is a **determinant** and Δ_v is the **determinant** corresponding to the basis $v_1, ..., v_n$.*

*The determinant for A^n corresponding to the standard basis $e_1, ..., e_n$ is called the **standard determinant** for A^n, and denoted by D_n.*

Corollary 3.8.17 (Determinant of an endomorphism)
Let M be a finite free A-module of rank n and $f \in \mathrm{End}_A(M)$ an endomorphism. Then the corresponding linear map

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$$L_a^n(f) : L_a^n(M) \to L_a^n(M)$$

satisfies

$$L_a^n(f)(\psi) = \det(f)\psi$$

for a unique $\det(f) \in A$*, which we call the* ***determinant*** *of* f*. We have the following properties*

$$\begin{aligned} \det(f \circ g) &= \det(f)\det(g) \\ \det(1_M) &= 1_A \\ \det(f) &= \Delta_v(f(v_1), ..., f(v_n)) \end{aligned}$$

for Δ_v *any generator for* $L_a^n(M)$ *corresponding to a basis* $v_1, ..., v_n$*.*

Proof. Let Δ be a generator then $L_a^n(f)(\Delta) = \Delta_v(f(v_1), ..., f(v_n))\Delta$ by (3.8.13). Clearly $\det(f) \coloneqq \Delta_v(f(v_1), ..., f(v_n))$ then satisfies the equation for all such $\psi = a\Delta$. It's unique because $L_a^n(M)$ is a free module, and the properties follow from uniqueness. ∎

We may use this to define the determinant of a matrix

Definition 3.8.18 (Determinant of a Matrix)
Let $E \in \mathrm{Mat}_{n\times n}(A)$ *then we define the* ***determinant*** *of* E *to be simply* $\det\left(\hat{E}\right)$*.*

Proposition 3.8.19 (Expressions for Matrix Determinant)
Let $E \in \mathrm{Mat}_{n\times n}(A)$ *and denote by* $v_1, ..., v_n$ *the column vectors of* E*. Then*

$$\det(E) = D_n(v_1, ..., v_n)$$

where D_n *is the standard determinant. Further there is the Leibniz Formula*

$$\det(E) = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^{n} E_{i\sigma(i)}$$

and Laplace Expansion Formula

$$\det(E) = \sum_{j=1}^{n} (-1)^{i+j} E_{ij} \det\left(E_{(ij)}\right)$$

where $E_{(ij)}$ *is the* $(n-1)\times(n-1)$ *matrix with the* i*-th row and* j*-th column dropped.*

Proof. Let $e_1, ..., e_n$ be the standard basis of A^n then by definition we have $\hat{E}(e_i) = v_i$. Recall by (3.8.14) that D_n is a generator of $L_a^n(M)$ and so by (3.8.17) we have $\det\left(\hat{E}\right) = D_n(v_1, ..., v_n)$.

The Leibniz Formula and Laplace Expansion Formula also follows directly from (3.8.14). ∎

Corollary 3.8.20 (Properties of Matrix Determinant)
The determinant satisfies a number of properties

a) $\det(EF) = \det(E)\det(F) = \det(F)\det(E)$
b) $\det(I_n) = 1$
c) $\det(PEP^{-1}) = \det(E)$
d) $\det(E^t) = \det(E)$

Proof. a)-c) follow from (3.3.115) and (3.8.17). Explicitly

$$\det(EF) = D\left(\widehat{EF}\right) = D\left(\hat{E} \circ \hat{F}\right) = D\left(\hat{E}\right)D\left(\hat{F}\right) = \det(E)\det(F)$$

and

$$\det(I_n) = D\left(\hat{I}_n\right) = D(\mathbf{1}_{A^n}) = 1_A$$

d) can be shown by using Leibniz formula and observing that $\sigma \to \sigma^{-1}$ simply permutes the elements of S_n. ∎

Proposition 3.8.21 (Determinant of a Block Triangular Matrix)
Let A *be a commutative ring,* $P \in \mathrm{Mat}_{p\times p}(A)$ *and* $R \in \mathrm{Mat}_{q\times q}(A)$*. Then*

$$\det\begin{pmatrix} P & Q \\ 0 & R \end{pmatrix} = \det(P)\det(R) = \det\begin{pmatrix} P & 0 \\ S & R \end{pmatrix}$$

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for any $Q \in \mathrm{Mat}_{p\times q}(A)$ *and* $S \in \mathrm{Mat}_{q\times p}(A)$.

Proof. Let $E = \begin{pmatrix} P & Q \\ 0 & R \end{pmatrix}$. In the Leibniz Formula (3.8.19) $\det(E) = \sum_{\sigma\in S_{p+q}} \varepsilon(\sigma)\prod_i E_{i\sigma(i)}$ a term is zero whenever $\sigma(i) \leq p$ for some $i > p$, as the lower left block is zero. So only those σ with $\sigma(\{p+1,...,p+q\}) \subseteq \{p+1,...,p+q\}$ contribute, and then (as σ is injective) σ preserves $\{1,...,p\}$ and $\{p+1,...,p+q\}$, that is $\sigma = \sigma_1\sigma_2$ for $\sigma_1 \in S_p$ and σ_2 a permutation of $\{p+1,...,p+q\}$, with $\varepsilon(\sigma) = \varepsilon(\sigma_1)\varepsilon(\sigma_2)$. For such σ

$$\prod_i E_{i\sigma(i)} = \prod_{i\leq p} P_{i\sigma_1(i)} \cdot \prod_{i>p} R_{i-p,\sigma_2(i)-p}$$

and summing over all pairs (σ_1, σ_2) gives $\det(P)\det(R)$. The second equality follows from the first as $\det(E^t) = \det(E)$ (3.8.20), since $\begin{pmatrix} P & 0 \\ S & R \end{pmatrix}^t = \begin{pmatrix} P^t & S^t \\ 0 & R^t \end{pmatrix}$. ∎

Proposition 3.8.22
Let M *be a finite free* A*-module and* $f \in \mathrm{End}_A(M)$. *Let* $(e_1,...,e_n)$ *be a basis and* Δ_e *the corresponding determinant. Define*

$$f^{\mathrm{ad}}(x) := \sum_{j=1}^{n} \Delta_e\big(f(e_1),...,f(e_{j-1}),x,f(e_{j+1}),...,f(e_n)\big)e_j.$$

Then

$$f \circ f^{\mathrm{ad}} = f^{\mathrm{ad}} \circ f = \det(f)\mathbf{1}_M$$

Furthermore the matrix of f^{ad} *with respect to the given basis satisfies*

$$[f^{\mathrm{ad}}]_{ij} = (-1)^{i+j}\det\big([f]_{(ji)}\big)$$

where $M_{(ji)}$ *denotes the matrix* M *with the* j*-th row and* i*-th column removed.*

Proof. We may consider by (3.8.7) the alternating map $\psi \in L_a^{n+1}(M;M)$ given by

$$\psi(x_1,...,x_{n+1}) := \sum_{j=1}^{n+1}(-1)^{j+1}\Delta_e\big(x_1,...,\hat{x}_j,...,x_{n+1}\big)x_j$$

By (3.8.12) necessarily ψ is identically zero. Evaluating at $(x, f(e_1),...,f(e_n))$ we find that

$$0 = \det(f)x + \sum_{j=1}^{n}(-1)^j\Delta_e\Big(x,f(e_1),...,\widehat{f(e_j)},...,f(e_n)\Big)f(e_j)$$

whence

$$\begin{aligned}\det(f)x &= \sum_{j=1}^{n}(-1)^{j+1}\Delta_e\Big(x,f(e_1),...,\widehat{f(e_j)},...,f(e_n)f(e_j)\Big)\\ &= \sum_{j=1}^{n}\Delta_e(f(e_1),...,x,...,f(e_n))f(e_j)\end{aligned}$$

where we have used (3.8.9).

Therefore using the definition of $f^{\mathrm{ad}}(x)$ we find that $f\big(f^{\mathrm{ad}}(x)\big) = \det(f)x$ as required. For the second relation, first observe that (3.3.109)

$$x = \sum_{k=1}^{n} e_k^\vee(x)e_k$$

and so applying $f(-)$

$$f(x) = \sum_{k=1}^{n} e_k^\vee(x)f(e_k)$$

Therefore we may calculate directly

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$$\begin{aligned} f^{\mathrm{ad}}(f(x)) &= \sum_{j=1}^{n} \Delta_e\big(f(e_1),...,f(e_{j-1}),f(x),f(e_{j+1}),...,f(e_n)\big)e_j \\ &= \sum_{j=1}^{n} \Delta_e(f(e_1),...,f(e_n))e_j^\vee(x)e_j \\ &= \det(f)\sum_{j=1}^{n} e_j^\vee(x)e_j \\ &= \det(f)x \end{aligned}$$

as required.

For the final statement we may assume that $M = A^n$ with $(e_1,...,e_n)$ the standard basis and $D_n = \Delta_e$. Then by (3.8.14), (3.8.18) we have

$$\begin{aligned} [f^{\mathrm{ad}}]_{ij} &= D_n\big(f(e_1),...,f(e_{i-1}),e_j,f(e_{i+1}),...,f(e_n)\big) \\ &= (-1)^{i+j}D_{n-1}\Big(\pi_j(f(e_1)),...,\pi_j\widehat{(f(e_i))},...,\pi_j(f(e_n))\Big) \\ &= (-1)^{i+j}\det\Big([f]_{(ji)}\Big) \end{aligned}$$

■

Corollary 3.8.23
Let M be a finite free A-module. Then $f \in \mathrm{End}_A(M)$ is an isomorphism if and only if $D(f) \in A^\star$.

Corollary 3.8.24
Let $E \in \mathrm{Mat}_{n\times n}(A)$ and define the adjugate matrix

$$E^{\mathrm{ad}}_{ij} := (-1)^{i+j}\det\Big(E_{(ji)}\Big)$$

Then

$$E^{\mathrm{ad}}E = EE^{\mathrm{ad}} = \det(E)I_n$$

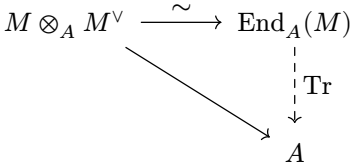
Proof. By definition and (3.8.22) we have that $\widehat{E^{\mathrm{ad}}} = \hat{E}^{\mathrm{ad}}$. The identities of endomorphisms are proven in (3.8.22), and we deduce the corresponding identities of matrices by (3.3.117). ■

3.8.2 Trace Map

Proposition 3.8.25 (Existence and Uniqueness of the Trace Map)
Let M be an A-module, then there exists a well-defined A-bilinear map

$$\begin{aligned} M \otimes_A M^\vee &\to \mathrm{End}_A(M) \\ m \otimes \theta &\to \theta(-)\cdot m\cdot \end{aligned}$$

When M is a finite free A-module this is bijective, and there exists a unique A-linear map $\mathrm{Tr} : \mathrm{End}_A(M) \to A$ making the following diagram commute



where the diagonal map is simply evaluation

$$m \otimes \theta \to \theta(m)$$

Equivalently it is the unique A-linear map satisfying

$$\mathrm{Tr}(\theta(-)\cdot m) = \theta(m)$$

for all $m \in M$ and $\theta \in M^\vee$.

Finally we have the identity

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$$\mathrm{Tr}(\phi) = \mathrm{Tr}([\phi]) = \sum_{i=1}^{n} [\phi]_{ii}$$

for all bases $\mathcal{B}$ of M.

Proof. There exists an obvious A-bilinear map $M \times M^\vee \to \mathrm{End}_A(M)$ so the map exists by the universal property. If M is finite free then it has a basis $\{v_1, ..., v_n\}$. By (3.3.109) $M^\vee$ has basis $\{v_1^\vee, ..., v_n^\vee\}$ and by (...) $M \otimes_A M^\vee$ has basis $\left\{v_i \otimes v_j^\vee\right\}$. By (3.3.110) this maps to a basis of $\mathrm{End}_A(M)$ and so it is an isomorphism.

Similarly the map $M \times M^\vee \to A$ is A-bilinear, and so the evaluation map $M \otimes_A M^\vee \to A$ exists. In light of the isomorphism then Tr exists and satisfies the required uniqueness property. ∎

Proposition 3.8.26 (Trace is Symmetric)
Let $\phi, \psi : M \to M$ be endormophisms of a finite free A-module. Then

$$\mathrm{Tr}(\phi \circ \psi) = \mathrm{Tr}(\psi \circ \phi)$$

Proof. We may verify that the mapping

$$\begin{aligned} \mathrm{End}_A(M) \times \mathrm{End}_A(M) &\to A \\ (\phi, \psi) &\to \mathrm{Tr}(\phi \circ \psi) \end{aligned}$$

is A-bilinear. Therefore it is sufficient to show the relationship holds for rank-one endomorphisms $\phi = \theta(-) \cdot m$ and $\psi = \iota(-) \cdot n$, as these generate $\mathrm{End}_A(M)$. Then we have

$$\mathrm{Tr}(\phi \circ \psi) = \mathrm{Tr}(\theta(\iota(-) \cdot n) \cdot m) = \mathrm{Tr}((\iota(-)\theta(n)) \cdot m) = \iota(m)\theta(n)$$

and by symmetry the result follows. ∎

Recall from (...) that every endomorphism $\phi : M \to M$ determines a family of endomorphisms $L_a^r(\phi) : L_a^r(M) \to L_a^r(M)$ for all integers $1 \leq r \leq n$, where n is the rank of M.

Proposition 3.8.27 (Determinant as a Trace)
Let M be a finite-free A module of rank n and $\phi : M \to M$ an endomorphism. Then

$$\det(\phi) = \mathrm{Tr}(L_a^n(\phi))$$

and

$$\mathrm{Tr}(\phi) = \mathrm{Tr}(L_a^1(\phi))$$

Definition 3.8.28
Let K be a subset of $[1, n]$ of order k. Then we denote by $\sigma_K : [1, k] \to K$ the unique order preserving bijection.

Denote by $\mathfrak{F}_k(n)$ the family of subsets of $[1, n]$ of order k.

If X is an $m \times n$ matrix and $K \subset [1, m]$, $H \subset [1, n]$, then we denote by $X_{H,K}$ the $\#K \times \#H$ matrix given by

$$\left(X_{H,K}\right)_{ij} = X_{\sigma_K(i), \sigma_H(j)}$$

Lemma 3.8.29
Let M be an A-module with subset $\{v_1, ..., v_n\}$ and $\psi : M^k \to N$ be an alternating map. Suppose that $x_1, ..., x_k$ satisfy

$$x_i = \sum_{j=1}^{n} A_{ji} v_j.$$

for some $n \times k$ matrix A. Then

$$\psi(x_1, ..., x_k) = \sum_{K \subset \mathfrak{F}_k(n)} \det\left(A_{K,[1,k]}\right) \psi(v_K)$$

where $v_K := \left(v_{\sigma_K(1)}, ..., v_{\sigma_K(k)}\right)$.

Proof.

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$$\begin{aligned}\psi(x_1,...,x_k) &= \sum_{i_1,...,i_k} a_{i_1 1}...a_{i_k k}\psi\big(v_{i_1},...,v_{i_k}\big)\\ &= \sum_{K\subset\mathfrak{F}_k(n)}\sum_{\sigma\in S_k} a_{\sigma_K(\sigma(1))1}...a_{\sigma_K(\sigma(k))k}\psi\big(v_{\sigma_K(\sigma(1))},...,v_{\sigma_K(\sigma(k))}\big)\\ &= \sum_{K\subset\mathfrak{F}_k(n)}\sum_{\sigma\in S_k} \varepsilon(\sigma)a_{\sigma_K(\sigma(1))1}...a_{\sigma_K(\sigma(k))k}\psi\big(v_{\sigma_K(1)},...,v_{\sigma_K(k)}\big)\\ &= \sum_{K\subset\mathfrak{F}_k(n)} \det\big(A_{K,[1,k]}\big)\psi(v_K)\end{aligned}$$

■

Proposition 3.8.30
Let M be a finite free A-module with $v_1,...,v_n$ a basis. For every subset $K\subset[1,n]$ of order k denote

$$v_K^\vee := v_{\sigma_K(1)}\wedge...\wedge v_{\sigma_K(k)}.$$

Then $L_a^k(M)$ has basis $\{v_K^\vee\}_{K\in\mathfrak{F}_k(n)}$. Furthermore we have the identity

$$v_K^\vee(v_{K'}) = \delta_{KK'}$$

and each basis element is uniquely determined by this relationship.

Proposition 3.8.31
Let M be a finite free A-module with basis $\mathcal{B}=\{v_1,...,v_n\}$. Let $\phi: M\to M$ be an A-module endomorphism. Then for all $K\in\mathfrak{F}_k(n)$

$$L_a^k(\phi)(v_K^\vee) = \sum_{H\in\mathfrak{F}_k(n)} \det\big(X_{K,H}\big)v_H^\vee$$

where $X=[\phi]_{\mathcal{B}}$.

In particular we see that

$$\mathrm{Tr}\big(L_a^k(\phi)\big) = \sum_{K\in\mathfrak{F}_k(n)} \det\big(X_{K,K}\big)$$

Proof. We have for $H\in\mathfrak{F}_k(n)$

$$\begin{aligned}L_a^k(\phi)(v_K^\vee)(v_H) &= v_K^\vee\big(\phi\big(v_{\sigma_H(1)}\big),...,\phi\big(v_{\sigma_H(k)}\big)\big)\\ &= \sum_{L\subset\mathfrak{F}_k(n)} \det\big(X_{L,H}\big)v_K^\vee(v_L)\\ &= \det\big(X_{K,H}\big)\end{aligned}$$

By (3.8.30) then both sides agree on v_H for all $H\in\mathfrak{F}_k(n)$, which shows they are equal by (3.8.29). ■

Proposition 3.8.32
Let M be a finite-free A module of rank n and $\phi: M\to M$ an endomorphism. For $a,b\in A$ we have

$$\det(a1_M+b\phi) = \sum_{k=0}^n \mathrm{Tr}\big(L_a^k(\phi)\big)a^{n-k}b^k$$

Proof. To calculate the left hand side consider a basis $\{v_1,...v_n\}$ for M and the corresponding determinant Δ_v. Then

$$\det(a1_M+b\phi) = \Delta_v(av_1+b\phi(v_1),...,av_n+b\phi(v_n))$$

Expand to find that this equals

$$\sum_{k=0}^n a^{n-k}b^k \sum_{K\in\mathfrak{F}_k(n)} \Delta_v(x_{1K},...,x_{nK})$$

where $x_{iK}=\phi(v_i)$ if $i\in K$ and v_i otherwise.

Define τ_K to be the unique shuffle permutation of type $(k,n-k)$ (3.2.38) such that $\tau_K\,|_{[1,k]}=\sigma_K$. Denote the inner term of the sum by z_K. Then we have by (3.8.9)

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$$z_K = \varepsilon(\tau_K)\Delta_v\big(\phi\big(v_{\sigma_K(1)}\big), ..., \phi\big(v_{\sigma_K(k)}\big), v_{\sigma_H(1)}, ..., v_{\sigma_H(n-k)}\big)$$

where H is the complement of K in $[1, n]$. Then by considering Δ_v to be an alternating form in the first k arguments, we may apply (3.8.29) to find

$$z_K = \varepsilon(\tau_K) \sum_{L \subset \mathfrak{F}_k(n)} \det\big(X_{L,K}\big)\Delta_v\big(v_{\sigma_L(1)}, ..., v_{\sigma_L(k)}, v_{\sigma_H(1)}, ..., v_{\sigma_H(n-k)}\big)$$

The summand is non-zero if and only if $L \cap H = \emptyset$, which occurs if and only if $L = K$. Therefore we conclude that

$$\begin{aligned} z_K &= \varepsilon(\tau_K)\det\big(X_{K,K}\big)\Delta_v\big(v_{\tau_K(1)}, ..., v_{\tau_K(n)}\big) \\ &= \det\big(X_{K,K}\big)\Delta_v(v_1, ..., v_n) \\ &= \det\big(X_{K,K}\big) \end{aligned}$$

The result then follows easily from (3.8.31). ∎

3.8.3 Block Matrices

Proposition 3.8.33
Let $A = \mathbb{Z}[X_{11}, ..., X_{nn}]$ be a polynomial ring in n^2 indeterminates. Consider the matrix $\big(X_{ij}\big) \in \mathrm{Mat}_{n\times n}(A)$ and let D_n denote the determinant of this matrix.

Then for any commutative ring B and matrix $M \in \mathrm{Mat}_{n\times n}(B)$ we have

$$\det(M) = D_n(M_{11}, ..., M_{1n}, M_{21}, ..., M_{nn})$$

where the right hand side corresponds to the obvious evaluation homomorphism $A \to B$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\mathrm{ev} : A \to B$ be the ring homomorphism with $\mathrm{ev}\big(X_{ij}\big) = M_{ij}$, so that the right hand side is $\mathrm{ev}(D_n)$. By the Leibniz Formula (3.8.19) applied over A to the matrix $\big(X_{ij}\big)$

$$D_n = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^{n} X_{i\sigma(i)}$$

and since ev preserves sums and products

$$\mathrm{ev}(D_n) = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^{n} M_{i\sigma(i)} = \det(M)$$

where the last equality is the Leibniz Formula again, now over B. ∎

Proposition 3.8.34 (Block Matrices)
Let A be a commutative ring and $n, m \geq 1$. The map

$$\Phi : \mathrm{Mat}_{n\times n}(\mathrm{Mat}_{m\times m}(A)) \longrightarrow \mathrm{Mat}_{(nm)\times(nm)}(A)$$

sending $\big(E_{ij}\big)$ to the matrix whose entry in row $(i-1)m + a$ and column $(j-1)m + b$ is $\big(E_{ij}\big)_{ab}$ (for $1 \leq a, b \leq m$) is an isomorphism of A-algebras. In particular the product of block matrices is computed blockwise, that is $(EF)_{ij} = \sum_{k=1}^{n} E_{ik}F_{kj}$, where the blocks are multiplied as $m \times m$ matrices.

Proof. The map Φ is clearly bijective and A-linear, and sends the identity to I_{nm}. Writing the rows and columns of $\Phi(E)$ as pairs (i, a), (j, b) we have

$$(\Phi(E)\Phi(F))_{(i,a),(j,b)} = \sum_{k=1}^{n}\sum_{c=1}^{m} \big(E_{ik}\big)_{ac}\big(F_{kj}\big)_{cb} = \sum_{k=1}^{n} \big(E_{ik}F_{kj}\big)_{ab}$$

which is the (a, b) entry of the (i, j) block of $\Phi(EF)$. So Φ is multiplicative. ∎

Lemma 3.8.35 (Determinant of a Monic Matrix Polynomial)
Let A be a commutative ring, $k, m \geq 1$ and $E \in \mathrm{Mat}_{m\times m}(A[Z])$ of the form $E = Z^k I_m + F$, where every entry of F is a polynomial of degree at most $k-1$. Then $\det(E) \in A[Z]$ is monic of degree mk.

Proof. In the Leibniz Formula (3.8.19) $\det(E) = \sum_{\sigma \in S_m} \varepsilon(\sigma) \prod_i E_{i\sigma(i)}$ the term for $\sigma = 1$ is $\prod_i \big(Z^k + F_{ii}\big)$, a product of m monic polynomials of degree k and so monic of degree mk. If $\sigma \neq 1$ then σ moves at least two indices i, for

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which $E_{i\sigma(i)} = F_{i\sigma(i)}$ has degree at most $k-1$, while the remaining factors have degree at most k. So the term of σ has degree at most $mk-2$, and the leading term of $\det(E)$ is that of the identity term. ∎

Proposition 3.8.36 (Block Matrix Determinant)
Suppose M_{ij} $(1 \leq i \leq n, 1 \leq j \leq n)$ are $m \times m$ matrices with coefficients in a commutative ring A and which commute pairwise. Form the matrix M with coefficients in A as follows

$$M = \begin{pmatrix} M_{11} & \dots & M_{1n} \\ \vdots & & \vdots \\ M_{n1} & \dots & M_{nn} \end{pmatrix}$$

Then

$$\det(M) = \det(D_n(M_{11}, ..., M_{1n}, M_{21}, ..., M_{nn}))$$

Proof. By (3.8.34) we may identify $\mathrm{Mat}_{n\times n}(\mathrm{Mat}_{m\times m}(A))$ with $\mathrm{Mat}_{(nm)\times(nm)}(A)$, compatibly with matrix multiplication. We proceed by induction on n, for all commutative rings A and all m simultaneously. For $n = 1$ there is nothing to show, as then $M = M_{11}$, so let $n \geq 2$.

Let B denote the subring of $\mathrm{Mat}_{m\times m}(A)$ generated by the matrices M_{ij}. It is commutative, since its generators commute pairwise by hypothesis. Then we may regard M as an element of $\mathrm{Mat}_{n\times n}(B)$, and denote this by $\widetilde{M}$ to disambiguate. As B is commutative we may consider its determinant $\det\left(\widetilde{M}\right) \in B$, itself an $m \times m$ matrix over A (so $\det\left(\det\left(\widetilde{M}\right)\right) \in A$ makes sense), and its adjugate matrix $\widetilde{M}^{\mathrm{ad}}$. Further consider the matrix M' given in block form by

$$M' = \begin{pmatrix} \left(\widetilde{M}^{\mathrm{ad}}\right)_{11} & 0 & \dots & 0 \\ \left(\widetilde{M}^{\mathrm{ad}}\right)_{21} & I_m & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \left(\widetilde{M}^{\mathrm{ad}}\right)_{n1} & 0 & \dots & I_m \end{pmatrix}$$

Recall by (3.8.24) and (3.8.33) that

$$\widetilde{M}^{\mathrm{ad}}_{11} = D_{n-1}(M_{22}, ..., M_{2n}, M_{31}, ..., M_{nn})$$

and

$$\widetilde{M}\widetilde{M}^{\mathrm{ad}} = \begin{pmatrix} \det\left(\widetilde{M}\right) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \det\left(\widetilde{M}\right) \end{pmatrix}$$

as elements of $\mathrm{Mat}_{n\times n}(B)$. Here, and in D_{n-1} and D_n below, the order of the factors in each product of the M_{ij} is immaterial, as they commute pairwise. Therefore computing the matrix product block-wise we find

$$MM' = \begin{pmatrix} \det\left(\widetilde{M}\right) & M_{12} & \dots & M_{1n} \\ 0 & M_{22} & \dots & M_{2n} \\ \vdots & \vdots & & \vdots \\ 0 & M_{n2} & \dots & M_{nn} \end{pmatrix}$$

Then MM' is block upper triangular with first diagonal block $\det\left(\widetilde{M}\right)$, so by (3.8.21) and multiplicativity of the determinant (3.8.20)

$$\det(M) \cdot \det(M') = \det(MM') = \det\left(\det\left(\widetilde{M}\right)\right) \cdot \det\left(M_{[m+1,mn,m+1,mn]}\right)$$

By induction on n we may then assume $\det\left(M_{[m+1,mn,m+1,mn]}\right) = \det(D_{n-1}(M_{22}, ..., M_{2n}, ..., M_{nn})) = \det\left(\left(\widetilde{M}^{\mathrm{ad}}\right)_{11}\right)$. Similarly M' is block lower triangular with first diagonal block $\left(\widetilde{M}^{\mathrm{ad}}\right)_{11}$ and the others identities, so $\det(M') = \det\left(\left(\widetilde{M}^{\mathrm{ad}}\right)_{11}\right)$ by (3.8.21). When this is not a zero-divisor in A then we conclude that

$$\det(M) = \det\left(\det\left(\widetilde{M}\right)\right) = \det(D_n(M_{11}, ..., M_{1n}, ..., M_{nn}))$$

as required.

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In order to handle the case where $\det\left(\left(\widetilde{M}^{\mathrm{ad}}\right)_{11}\right)$ is a zero divisor we may replace the ring A with the polynomial ring $A[Z]$ and consider the matrix N for which

$$\tilde{N} = \widetilde{M} + (Z \cdot I_m) \cdot I_n.$$

Then the blocks $N_{ij} = M_{ij} + \delta_{ij} Z I_m$ still commute pairwise, as ZI_m is central. Moreover $\left(\tilde{N}^{\mathrm{ad}}\right)_{11} = D_{n-1}(N_{22}, ..., N_{nn}) = \sum_{\tau \in S_{n-1}} \varepsilon(\tau) \prod_{i=2}^{n} N_{i\tau(i)}$. The term for $\tau = 1$ is $\prod_{i=2}^{n}(M_{ii} + ZI_m) = Z^{n-1} I_m +$ (terms of degree $\leq n-2$), while for $\tau \neq 1$ only the indices fixed by τ contribute a factor of degree one in Z (as $N_{i\tau(i)} = M_{i\tau(i)}$ for $\tau(i) \neq i$), so the term has degree at most $n-3$. Hence $\left(\tilde{N}^{\mathrm{ad}}\right)_{11} = Z^{n-1} I_m + F$ with every entry of F of degree at most $n-2$, and as $n - 1 \geq 1$ its determinant is monic (3.8.35), and therefore not a zero-divisor of $A[Z]$ (3.7.4). We conclude from the previous case that

$$\det(N) = \det(D_n(N_{11}, ..., N_{1n}, ..., N_{nn}))$$

Finally we may apply the ring homomorphism $A[Z] \to A$ (evaluation at 0) to deduce the general case. ∎

3.8.4 Exterior Product

Proposition 3.8.37
Let M be an A-module. There is a bilinear map

$$\wedge : L_a^p(M) \times L_a^q(M) \to L_a^{p+q}(M)$$

given by

$$(f \wedge g)(x_1, ..., x_{p+q}) := \sum_{\sigma \in \mathrm{Sh}(p,q)} \varepsilon(\sigma) f\left(x_{\sigma(1)}, ..., x_{\sigma(p)}\right) g\left(x_{\sigma(p+1)}, ..., x_{\sigma(p+q)}\right)$$

where the sum is taken over shuffle permutations (3.2.38).

Proof. Suppose that $x_i = x_j$ then we require to prove that $(f \wedge g)(x_1, ..., x_{p+q}) = 0$ in order to demonstrate that it is alternating. If $1 \leq i, j \leq p$ then this follows from (3.8.9). Similarly if $p < i, j \leq p + q$. So we may consider the case $i \leq p < j$. Consider the family of right cosets $\mathrm{Sh}(p,q)/\{e, (i,j)\}$, and let Σ be some coset representatives. Then by (3.2.10)

$$\mathrm{Sh}(p,q) = \bigsqcup_{\sigma \in \Sigma} \{\sigma, (i,j)\sigma\}$$

Therefore the sum is equal to

$$\sum_{\sigma \in \Sigma} \varepsilon(\sigma) \left[f(x_\sigma) g(x_\sigma) - f\left(x_{(i\,j)\sigma}\right) g\left(x_{((i\,j)\sigma)}\right) \right]$$

so when $x_i = x_j$ this sum is zero. ∎

Proposition 3.8.38 (Basic Properties of Exterior Product)
Let M be an A-module. Suppose $f \in L_a^p(M)$, $g \in L_a^q(M)$ and $h \in L_a^r(M)$. Then the following properties hold

a) $f \wedge g = (-1)^{pq} g \wedge f$
b) $(f \wedge g) \wedge h = f \wedge (g \wedge h)$

Proof.

a) By definition and (3.2.44)

$$\begin{aligned}(f \wedge g)(x) &= \sum_{\sigma \in \mathrm{Sh}(p,q)} \varepsilon(\sigma) f\left(x_{\sigma(1)}, ..., x_{\sigma(p)}\right) g\left(x_{\sigma(p+1)}, ..., x_{\sigma(p+q)}\right) \\ &= \sum_{\sigma \in \mathrm{Sh}(q,p)} (-1)^{pq} \varepsilon(\sigma) f\left(x_{\sigma(p+1)}, ..., x_{\sigma(p+q)}\right) g\left(x_{\sigma(1)}, ..., x_{\sigma(p)}\right) \\ &= (-1)^{pq} (g \wedge f)\end{aligned}$$

b) Similarly by definition

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$$\begin{aligned}((f\wedge g)\wedge h)(x_1,...,x_{p+q+r}) &= \sum_{\sigma\in \mathrm{Sh}(p+q,r)} \varepsilon(\sigma)(f\wedge g)\big(x_{\sigma(1)},...,x_{\sigma(p+q)}\big)h\big(x_{\sigma(p+q+1)},...,x_{\sigma(p+q+r)}\big)\\ &= \sum_{\sigma\in \mathrm{Sh}(p+q,r)}\sum_{\tau\in \mathrm{Sh}(p,q)} \varepsilon(\sigma)\varepsilon(\tau) f\big(x_{\sigma(\tau(1))},...,x_{\sigma(\tau(p))}\big)g\big(x_{\sigma(\tau(p+1))},...,x_{\sigma(\tau(p+q))}\big)\\ &\quad h\big(x_{\sigma(p+q+1)},...,x_{\sigma(p+q+r)}\big)\\ &\overset{(3.2.45)}{=} \sum_{\sigma\in \mathrm{Sh}(p,q,r)} \varepsilon(\sigma) f\big(x_{\sigma(1)},...,x_{\sigma(p)}\big)g\big(x_{\sigma(p+1)},...,x_{\sigma(p+q)}\big)h\big(x_{\sigma(p+q+1)},...,x_{\sigma(p+q+r)}\big)\end{aligned}$$

and symmetrically

$$\begin{aligned}(f\wedge(g\wedge h))(x_1,...,x_{p+q+r}) &= \sum_{\sigma\in \mathrm{Sh}(p,q+r)} \varepsilon(\sigma) f\big(x_{\sigma(1)},...,x_{\sigma(p)}\big)(g\wedge h)\big(x_{\sigma(p+1)},...,x_{\sigma(p+q+r)}\big)\\ &= \sum_{\sigma\in \mathrm{Sh}(p,q+r)}\sum_{\tau\in \mathrm{Sh}(q,r)} \varepsilon(\sigma)\varepsilon(\tau) f\big(x_{\sigma(1)},...,x_{\sigma(p)}\big)g\big(x_{\sigma(\tau(p+1))},...,x_{\sigma(\tau(p+q))}\big)\\ &\quad h\big(x_{\sigma(\tau(p+q+1))},...,x_{\sigma(\tau(p+q+r))}\big)\\ &\overset{(3.2.45)}{=} \sum_{\sigma\in \mathrm{Sh}(p,q,r)} \varepsilon(\sigma) f\big(x_{\sigma(1)},...,x_{\sigma(p)}\big)g\big(x_{\sigma(p+1)},...,x_{\sigma(p+q)}\big)h\big(x_{\sigma(p+q+1)},...,x_{\sigma(p+q+r)}\big)\end{aligned}$$

therefore we conclude that the expressions are equal.

■

3.8.5 Matrix Rank

Definition 3.8.39 (Column and Row Rank)
Let k be a field and E an $m\times n$ a matrix over k. Consider the canonical vector spaces k^n and k^m. Then define the ***column rank*** *of E to be*

$$\mathrm{rank}_k\big(\hat{E}\big)$$

and the ***row rank*** *of E to be*

$$\mathrm{rank}_k\big(\widehat{E^t}\big)$$

Proposition 3.8.40 (Row Rank = Column Rank)
Let E be a matrix over k, then row rank and column rank are equal, and denote this by $\mathrm{rk}(E)$.

It is also the maximal number of linearly independent rows, or columns, and furthermore $\mathrm{rk}(E)\leq \min(m,n)$.

We say E is ***full rank*** *if $\mathrm{rk}(E)=\min(m,n)$.*

Proof. By (3.3.144) $\mathrm{rank}_k\big(\hat{E}\big)=\mathrm{rank}_k\big(\hat{E}^\vee\big)$ and by (3.3.121) this equals $\mathrm{rank}_k\big(\widehat{E^t}\big)$ as required.

The columns (resp. rows) clearly span $\mathrm{Im}\big(\hat{E}\big)$ (resp. $\mathrm{Im}\big(\widehat{E^t}\big)$). By (3.3.126) there are $r:=\mathrm{rk}(E)$ columns (resp. rows) constituting a basis, and therefore linearly independent. For any other subset of linearly independent columns (resp. rows) we must have the order is less than r by (3.3.126). Therefore $\mathrm{rk}(E)$ is the maximal number of linearly independent rows or columns. ■

Proposition 3.8.41 (Criteria for Full Rank Square Matrix)
Let E be an $n\times n$ matrix over a field k. Then the following are equivalent

a) E *is invertible*
b) $\mathrm{rk}(E)=n$ *(i.e. $\hat{E}$ is surjective or E is full-rank)*
c) $Ev=0\implies v=0$ *for all column vectors v (i.e. $\hat{E}$ is injective).*
d) The columns of E are linearly independent
e) $\det(E)\neq 0$

Finally E is full rank if and only if E^t is full rank.

Proof. Consider k^n with canonical basis, then by (3.3.115) E is invertible if and only if $\hat{E}$ is an isomorphism. By definition $\mathrm{rk}(E)=\mathrm{rank}_k\big(\hat{E}\big)$. Furthermore c) is equivalent to $\hat{E}$ being injective, and is also equivalent to d). Therefore the equivalence follows from (3.3.137).

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Finally it's clear from either a), b) or e) that this property is self-dual. ■

Definition 3.8.42 (Minor of a matrix)
*Let E be an $m \times n$ matrix, we say a **k-minor** (for $k \leq \min(m,n)$) is the determinant of a $k \times k$ submatrix obtained by deleting $m-k$ rows and $n-k$ columns.*

Proposition 3.8.43 (Criteria for rank)
Let E be an $m \times n$ matrix over k. Then the following are equivalent

a) $\mathrm{rk}(E) \geq r$
b) There exists an $r \times r$ sub-matrix with full rank
c) There exists a non-zero r-minor

Proof. We see b) $\iff$ c) by (3.8.41) e).

Suppose b) holds, then a-fortiori E has r linearly independent columns. Therefore by (3.8.40) $\mathrm{rk}(E) \geq r$ and a) holds.

Conversely suppose $\mathrm{rk}(E) \geq r$ then by (3.8.40) there are certainly r linearly independent columns. We consider the $m \times r$ sub-matrix E' consisting of these columns. By (3.8.40) $\mathrm{rk}(E') = r$ and there are r linearly independent rows. Choosing these rows yields an $r \times r$ submatrix E'' which has r linearly independent rows, and so by (3.8.41) is full rank as required. ■

Corollary 3.8.44
Let E be an $m \times n$ matrix over k and r an integer. Then the following are equivalent

a) $\mathrm{rk}(E) = r$
b) r is the maximal dimension of a full-rank square sub-matrix
c) r is the maximal dimension of a non-zero minor

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3.9 Laurent Polynomials

Definition 3.9.1 (Laurent polynomial ring)
Let A be a commutative ring. Define the ***laurent polynomial ring*** $A[X, X^{-1}]$ *to be the group ring $A[\mathbb{Z}]$. Denote an element as a formal sum*

$$f(X) \coloneqq \sum_{i=-\infty}^{\infty} a_i X^i$$

Proposition 3.9.2
Let B be an A-algebra and $b \in B^\star$ a unit. Then there is a unique A-algebra homomorphism $\mathrm{ev}_b : A[X, X^{-1}] \to B$ such that

$$\mathrm{ev}_b(X) = b$$

If the structural morphism $A \to B$ is injective, so is ev_b.

Proposition 3.9.3
Let A be a commutative ring. Then there is an isomorphism of rings

$$\begin{aligned} A[X, X^{-1}] &\xrightarrow{\sim} A[X]_X \\ \sum_{i=-r}^{\infty} a_i X^i &\to \frac{\sum_{i=0}^{\infty} a_{i-r} X^i}{X^r} \quad r = \min\{k \mid a_k \neq 0\} \vee 0 \\ \sum_{i=-r}^{\infty} a_{i+r} X^i &\leftarrow \frac{\sum_{i=0}^{\infty} a_i X^i}{X^r} \end{aligned}$$

Proof. Denote the maps by ψ and ϕ. Then ψ exists by (3.9.2) and ϕ exists by the universal property of localisation.■

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3.10 Polynomial Rings in Many Variables

Definition 3.10.1 (Polynomial ring in many variables)
Let A be a ring then the polynomial ring $A[X_1, ..., X_n]$ is the monoid ring $A[\mathbb{N}^n]$ (3.6.1) consisting of formal sums of monomials

$$f(X_1, ..., X_n) = \sum_{v \in \mathbb{N}^n} f_v X_1^{v_1} \cdots X_n^{v_n}$$

where $f_v \in A$ and only finitely many coefficients are non-zero. Addition is defined in the obvious way.

We may canonically regard A, $A[X_i]$ and $A[X_1, ..., X_i]$ as subrings in the obvious way.

Define $\deg(f, i)$ to be the maximal power of X_i with a non-zero coefficient.

Remark 3.10.2
It may be useful for certain induction arguments to write

$$A[X_1, ..., X_n] = A$$

when $n = 0$.

Proposition 3.10.3 (Evaluation Homomorphism)
$A[X_1, ..., X_n]$ satisfies the following universal property. Given any A-algebra B and points $(b_1, ..., b_n)$ there exists a ring homomorphism

$$\phi_b : A[X_1, ..., X_n] \to B$$

such that

$$\phi_b(X_i) = \phi(b_i)$$

given by

$$\phi_b\left(\sum_v a_v X_1^{v_1} \cdots X_n^{v_n}\right) = \sum_v i_B(a_v)\phi(b_1)^{v_1} \cdots \phi(b_n)^{v_n}$$

In other words it is an initial object in the category of n-pointed A-algebras. Furthermore

$$\mathrm{Im}(\phi_b) = A[b_1, ..., b_n]$$

Lemma 3.10.4 (Iterated polynomial ring)
Given $f \in A[X_1, ..., X_n]$ and let $N = \deg(f, n)$ then there exist unique polynomials $g_i \in A[X_1, ..., X_{n-1}]$ such that

$$f = \sum_{i=0}^{N} g_i X_n^i$$

in other words there is a canonical isomorphism

$$\psi : A[X_1, ..., X_{n-1}][X_n] \to A[X_1, ..., X_n]$$

under which $\deg(f) = \deg(\psi(f); n)$.

Proposition 3.10.5 (Homogenous grading)
Consider $R = k[X_1, ..., X_n]$ and $x \in k^n$. Then there is a direct sum of k-submodules

$$R = \bigoplus_{n \geq 0} R^{n,x}$$

where

$$R^{n,x} = \left\{ \sum_{|\alpha|_1 = n} \lambda_\alpha (X_1 - x_1)^{\alpha_1} \cdots (X_n - x_n)^{\alpha_n} \;\middle|\; \lambda_\alpha \in k \quad \alpha \in \mathbb{N}^n \right\}$$

and every $F \in R$ may be written uniquely as

$$F(X) = F(x) + F^{(1,x)}(X) + ... + F^{(n,x)}(X) + ...$$

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with $F^{(n,x)} \in R^{n,x}$. *Note that*

$$\ker(\mathrm{ev}_x) =: M_x = \bigoplus_{n\geq 1} R^{n,x} = (X_1 - x_1, ..., X_n - x_n)$$

and

$$M_x^k = \bigoplus_{n\geq k} R^{n,x}$$

Finally there is a canonical isomorphism

$$k[X_1, ..., X_n]^{1,x} \cong M_x/M_x^2$$

Proof. By Proposition (...) there is a k-algebra homomorphism $\rho_x : R \to R$ given by $X_i \to X_i + x_i$. It is an isomorphism with two-sided inverse ρ_{-x}. Let $F \in R$ then

$$\rho_x(F) = \sum_{n=0}^{\infty} \left(\sum_{|\alpha|_1 = n} \lambda_\alpha X_1^{\alpha_1} \cdots X_n^{\alpha_n} \right)$$

whence applying ρ_{-x}

$$\begin{aligned} F(X) &= \sum_{n=0}^{\infty} F^{(n)}(X) \\ F^{(n)}(X) &= \sum_{|\alpha|_1 = n} \lambda_\alpha (X_1 - x_1)^{\alpha_1} \cdots (X_n - x_n)^{\alpha_n} \end{aligned}$$

as required. The coefficients λ_α are seen to be uniquely determined by applying ρ_x. Therefore the internal sum is direct. Finally evaluate at x to find $F^{(0)} = F(x)$. The statement regarding M_x is straightforward. And because $R^{n,x} \cdot R^{m,x} \subseteq R^{n+m,x}$ the statement regarding M_x^k follows by induction. ∎

Definition 3.10.6 (Projection to linear terms)
Given $\mathfrak{a} \lhd k[X_1, ..., X_n]$ *and* $x \in k^n$ *define*

$$\mathfrak{a}^{(i,x)} = \left\{ F^{(i,x)} \;\middle|\; F \in \mathfrak{a} \right\}$$

The following is useful

Lemma 3.10.7
Let A *be a* k*-algebra and* $F \in k[X_1, ..., X_n]$ *and* $G_1, ..., G_n \in k[Y_1, ..., Y_m]$ *polynomials. For* $\lambda_1, ..., \lambda_m \in A$ *we have*

$$F(G_1, ..., G_n)(\lambda_1, ..., \lambda_m) = F\Big(G_1(\lambda_1, ..., \lambda_m), ..., G_{n(\lambda_1, ..., \lambda_m)}\Big)$$

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3.11 Δ-Graded Rings

References:

- [Bou98a] (Chap. II §11)

Let Δ be a commutative monoid. By convention we write the monoid operation additively and 0 for the identity. Further we assume it is **cancellable**: $\lambda + \mu = \lambda + \mu' \implies \mu = \mu'$. Typically $\Delta = \mathbb{Z}$, but for example we may also consider $\mathbb{N}^k$ for some positive integer k.

Definition 3.11.1 (Graded Abelian Groups)
*Let Δ be a cancellable commutative monoid. An abelian group G is Δ-**graded** if it may be written as internal direct sum (3.3.93) of $\mathbb{Z}$-modules*

$$G = \bigoplus_{\lambda \in \Delta} G_\lambda$$

for subgroups $G_\lambda \leq G$. In particular the subgroups G_λ are disjoint. This means for every $g \in G$ there is a unique decomposition into a finite sum

$$g = \sum_{\lambda \in \Delta} g_\lambda$$

*If $g \in G_\lambda$ then we say g is **homogenous**. In this case we write $\delta(g) \coloneqq \lambda$ for the **degree** of g.*

We may write $\pi_\lambda : G \to G_\lambda$ and $i_\lambda : G_\lambda \to G$ for projection and inclusion respectively. For $g \in G$ we may abbreviate $\pi_\lambda(g)$ by g_λ.

Definition 3.11.2 (Graded Ring)
*Let Δ be a cancellable commutative monoid. A commutative ring A is **graded** of type Δ if the additive group $(A, +)$ is Δ-graded, and the multiplication is compatible in the sense that*

$$a \in A_\lambda, b \in A_\mu \implies a \cdot b \in A_{\lambda+\mu}$$

Lemma 3.11.3
Let A be a Δ-ring. Then A_0 is a subring.

In the case $\Delta = \mathbb{N}$ the additive subgroups

$$A_{\geq k} \coloneqq \sum_{l \geq k} A_l$$

*are homogenous ideals. We write $A_+ \coloneqq A_{\geq 1}$ which we call the **irrelevant ideal**.*

Proof. We need only show $1 \in A_0$. By hypothesis there is a unique decomposition

$$1 = \sum_{\lambda \in \Delta} e_\lambda \quad e_\lambda \in A_\lambda$$

For every $x \in A_\mu$ we have

$$x = x \cdot 1 = \sum_{\lambda \in \Delta} x \cdot e_\lambda$$

By the cancellable assumption we have $\lambda \neq 0 \implies \lambda + \mu \neq \mu$. Therefore $x \cdot e_0 = x$. By distributivity this then holds for all $x \in A$ and in particular $1 = 1 \cdot e_0 = e_0$. ∎

Definition 3.11.4 (Graded Module)
*Let A be a Δ-graded ring and M a left (resp. right) A-module which is a Δ-graded as an abelian group. Then M is a **graded module** if the A-module structure is compatible with the grading, namely*

$$a \in A_\lambda, m \in M_\mu \implies a \cdot m \in M_{\lambda+\mu}$$

Elements of M_λ are known as homogenous.

Definition 3.11.5 (Graded Homomorphisms)
*Let A, B be Δ-graded rings and $\phi : A \to B$ a ring homomorphism. It is a **graded ring homomorphism** if*

$$\phi(A_\lambda) \subseteq B_\lambda \quad \forall \lambda \in \Delta$$

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Let M, M' be graded A-modules and $\phi : M \to M'$ an A-module homomorphism. Then ϕ is a ***graded module homomorphism*** *if*

$$\phi(M_\lambda) \subseteq M'_\lambda \quad \forall \lambda \in \Delta$$

Definition 3.11.6 (Graded Submodules)
Let M be a graded A-module, and $N \leq M$ an A-submodule. Then the following are equivalent

a) $n \in N, \lambda \in \Delta \Longrightarrow \pi_\lambda(n) \in N$
b) N *is generated by homogenous elements*
c) N *is equal to the internal direct sum*

$$\bigoplus_{\lambda \in \Delta} N \cap M_\lambda$$

We say such a submodule is ***graded*** *or* ***homogenous*** *and we write $N_\lambda \coloneqq N \cap M_\lambda$.*

Proposition 3.11.7
Let M be a graded A-module and $\{N_\alpha\}_{\alpha \in I}$ be a family of submodules. Then both

$$\sum_{\alpha \in I} N_\alpha$$

and

$$\bigcap_{\alpha \in I} N_\alpha$$

are graded submodules.

Proposition 3.11.8
Let M be a finitely-generated graded A-module. Then M has a finite generating set of homogenous elements.

Definition 3.11.9 (Homogenous Ideal)
Let A be a commutative graded ring. An ideal $\mathfrak{a} \lhd A$ which is homogenous as an A-submodule of A is a ***homogenous ideal****. We write $\mathfrak{a}_\lambda \coloneqq \mathfrak{a} \cap A$, then by definition*

$$\mathfrak{a} = \bigoplus_{\lambda \in \Delta} \mathfrak{a}_\lambda$$

Note if A is Noetherian then every homogenous ideal has a finite generating set of homogenous elements.

Lemma 3.11.10
Let $\phi : A \to B$ be a graded ring and $\mathfrak{b} \lhd B$ a homogenous ideal. Then $\phi^{-1}(\mathfrak{b})$ is homogenous.

Example 3.11.11 (Polynomial Ring)
The polynomial ring $k[X_0, ..., X_n]$ is a graded ring over $\mathbb{N}$ with

$$k[X_0, ..., X_n]_d = k\langle \{ X_1^{\alpha_1} \cdot ... \cdot X_n^{\alpha_n} \mid \alpha_1 + ... + \alpha_n = d \} \rangle$$

Furthermore

$$\bigoplus_{i \geq d} k[X_0, ..., X_n]_i = (X_0, ..., X_n)^d$$

Proposition 3.11.12 (Quotient by a Homogenous Ideal)
Suppose Δ is a commutative cancellable monoid. Let A be a Δ-graded ring and $\mathfrak{a} \lhd A$ a homogenous ideal. Then $A/\mathfrak{a}$ is a Δ-graded ring by considering the internal direct sum of abelian groups

$$A/\mathfrak{a} = \bigoplus_{\lambda \in \Delta} (A/\mathfrak{a})_\lambda$$
$$(A/\mathfrak{a})_\lambda \coloneqq \{ \overline{a} \mid a \in A_\lambda \}$$

Furthermore there is an isomorphism of A-modules

$$(A/\mathfrak{a})_\lambda \xrightarrow{\sim} A_\lambda/(\mathfrak{a} \cap A_\lambda)$$

For every homogenous ideal $\mathfrak{b} \supseteq \mathfrak{a}$ the quotient ideal $\mathfrak{b}/\mathfrak{a}$ is homogenous with

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$$\mathfrak{b}/\mathfrak{a} = \bigoplus_{\lambda\in\Delta} (\mathfrak{b}/\mathfrak{a})_\lambda$$

Proof. Evidently the $(A/\mathfrak{a})_\lambda$ are abelian subgroups and span $A/\mathfrak{a}$. Observe that

$$0 = \sum_{\lambda\in\Delta} \overline{a_\lambda} \implies \sum_{\lambda\in\Delta} a_\lambda \in \mathfrak{a}$$

Then as $\mathfrak{a}$ is homogenous we conclude $a_\lambda \in \mathfrak{a}$ and therefore $\overline{a_\lambda} = 0$. Therefore the family $(A/\mathfrak{a})_\lambda$ satisfy the criteria of (3.3.94) and the direct sum is well-defined. Evidently the ring multiplication on $A/\mathfrak{a}$ is compatible with the given Δ-grading.

The isomorphism is induced by π_λ.

Suppose that $\overline{b} \in \mathfrak{b}/\mathfrak{a}$ then we require to prove that $\left(\overline{b}\right)_\lambda \in (\mathfrak{b}/\mathfrak{a})$. By assumption $b_\lambda \in \mathfrak{b}$ so $\overline{b_\lambda} \in \mathfrak{b}/\mathfrak{a}$. However by the first part $\overline{b_\lambda} = \left(\overline{b}\right)_\lambda$. ∎

Proposition 3.11.13 (Localization at Homogenous Elements)
Let Δ be an abelian group, A a Δ-graded ring and $S \subset A$ a multiplicative set of ***homogenous*** *elements. Then we may define a Δ-gradation on $S^{-1}A$ by*

$$\left(S^{-1}A\right)_\lambda := \left\{ \frac{a}{s} \;\middle|\; \delta(a) - \delta(s) = \lambda \right\}$$

For a homogenous ideal $\mathfrak{a} \lhd A$ then the ideal $S^{-1}\mathfrak{a}$ is homogenous.

Proof. We first show that $\left(S^{-1}A\right)_\lambda$ is an additive subgroup, for given homogenous $a, a' \in A$ and $s, s' \in s$ by definition

$$\frac{a}{s} + \frac{a'}{s'} = \frac{as' + a's}{ss'}$$

Suppose $\delta(a) - \delta(s) = \lambda = \delta(a') - \delta(s')$ then $\delta(as') = \delta(a's)$. Therefore $as' + a's$ is homogenous and $\delta(as' + a's) = \delta(as') = \lambda + \delta(s) + \delta(s') = \lambda + \delta(ss')$ whence $\left(S^{-1}A\right)_\lambda$ is an additive subgroup. Similarly for the multiplicative structure suppose $\delta(a) = \delta(s) + \lambda$ and $\delta(a') = \delta(s') + \lambda'$ then

$$\delta(aa') = \delta(a) + \delta(a') = \delta(\lambda) + \delta(\lambda') + (\lambda + \lambda') = \delta(\lambda\lambda') + (\lambda + \lambda')$$

so that $\frac{a}{s} \cdot \frac{a'}{s'} \in \left(S^{-1}A\right)_{\lambda+\lambda'}$ as required. Evidently $S^{-1}A$ is the internal sum of $\left(S^{-1}A\right)_\lambda$. To show it is direct suppose that

$$0 = \sum_{i=1}^{n} \frac{a_i}{s_i} \quad \delta(a_i) = \delta(s_i) + \lambda_i$$

for distinct $\lambda_1, ..., \lambda_n$. Then

$$0 = \sum_{i=1}^{n} a_i t \prod_{j\neq i} s_j$$

for some $t \in S$. Furthermore

$$\delta\left(a_i t \prod_{j\neq i} s_j\right) = \delta(a_i) + \sum_{j\neq i} \delta\left(s_j\right) + \delta(t) = \lambda_i + \sum_{j=1}^{n} \delta\left(s_j\right) + \delta(t)$$

As Δ is cancellable the elements $\hat{\lambda}_i := \lambda_i + \sum_{j=1}^{n} \delta\left(s_j\right) + \delta(t)$ are distinct. By assumption we conclude that $a_i t \prod_{j\neq i} s_j = 0$ and so $\frac{a_i}{s_i} = 0$. Therefore $S^{-1}A$ is the direct sum of $\left(S^{-1}A\right)_\lambda$ by (3.3.94).

For $\frac{a}{s} \in S^{-1}\mathfrak{a}$ we wish to show that $\left(\frac{a}{s}\right)_\lambda \in S^{-1}\mathfrak{a}$ for all $\lambda \in \Delta$. Here $a \in \mathfrak{a}$ so by assumption $a_\lambda \in \mathfrak{a}$ for all $\lambda \in \Delta$ and $a = \sum_{\lambda\in\Delta} a_\lambda$ so

$$\frac{a}{s} = \sum_{\lambda\in\Delta} \frac{a_\lambda}{s} = \sum_{\lambda\in\Delta} \frac{a_{\lambda+\delta(s)}}{s}$$

and so $\left(\frac{a}{s}\right)_\lambda = \frac{a_{\lambda+\delta(s)}}{s} \in S^{-1}\mathfrak{a}$ as required. ∎

Recall for any Δ-graded ring A the additive subgroup A_0 is also a subring.

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Definition 3.11.14
Let Δ be a commutative abelian group, A a Δ-graded ring and S be a multiplicative subset of homogenous elements. Define the subring of $S^{-1}A$

$$A_{(S)} \coloneqq \left(S^{-1}A\right)_0 = \left\{ \frac{a}{s} \;\middle|\; \delta(s) = \delta(a) \right\}$$

Similarly for $\mathfrak{a}$ a homogenous ideal define the homogenous ideal of $S^{-1}A$

$$\mathfrak{a}_{(S)} \coloneqq \left\{ \frac{a}{s} \;\middle|\; a \in \mathfrak{a}, s \in S, \delta(s) = \delta(a) \right\}$$

In the case $\mathfrak{p}$ is a homogenous prime ideal define the local ring

$$A_{(\mathfrak{p})} \coloneqq A_{(\{x \notin \mathfrak{p}\})}$$

and for $f \in A$ homogenous define

$$A_{(f)} \coloneqq A_{(\{1, f, f^2, \ldots\})}$$

Lemma 3.11.15
Let Δ be an abelian group, A a Δ-graded ring and $\mathfrak{a} \lhd A$ a homogenous ideal. Then

$$\mathfrak{a}_{(S)} = A_{(S)} \cap S^{-1}\mathfrak{a}$$

as subsets of $S^{-1}A$.

Lemma 3.11.16
Let Δ be an abelian group, A a Δ-graded ring, $a \in A_\mu$ homogenous and $x \in A$. Then

$$(ax)_\lambda = a \cdot x_{\lambda - \mu}$$

In particular ax is homogenous iff x is homogenous.

Proposition 3.11.17 (Homogenous localisation comutes with quotients)
Let Δ be an abelian group, A a Δ-graded ring, $\mathfrak{a}$ a homogenous ideal and $S \subset A$ a multiplicatively closed set of homogenous elements such that $S \cap \mathfrak{a} = \emptyset$. Then there is a canonical isomorphism of Δ-graded rings

$$\begin{aligned} (S^{-1}A)/(S^{-1}\mathfrak{a}) &\xrightarrow{\sim} \pi(S)^{-1}(A/\mathfrak{a}) \\ \frac{a}{s} + S^{-1}A &\to \frac{a + \mathfrak{a}}{s + \mathfrak{a}} \end{aligned}$$

where $\pi : A \to A/\mathfrak{a}$ is the canonical surjection. In particular there is an isomorphism of rings

$$A_{(S)}/\mathfrak{a}_{(S)} \xrightarrow{\sim} \left(S^{-1}A/S^{-1}\mathfrak{a}\right)_0 \xrightarrow{\sim} (A/\mathfrak{a})_{(\pi(S))}$$

Proof. The first isomorphism of rings is from (3.5.25), and it is obviously compatible with the Δ-grading. For suppose $x \in (S^{-1}A)/(S^{-1}\mathfrak{a})$ with $\delta(x) = \lambda$. Then $x = \frac{a}{s}$ with $\delta(a) - \delta(s) = \lambda$. By definition $\delta(a + \mathfrak{a}) - \delta(s + \mathfrak{a}) = \lambda$ as required.

The third isomorphism follows immediately. For the second consider the ring homomorphism

$$\phi : A_{(S)} \hookrightarrow S^{-1}A \to S^{-1}A/S^{-1}\mathfrak{a}$$

Then evidently the image is $\left(S^{-1}A/S^{-1}\mathfrak{a}\right)_0$ as the quotient map is graded. The kernel is $A_{(S)} \cap S^{-1}\mathfrak{a}$ which equals $\mathfrak{a}_{(S)}$ by (3.11.15). ∎

Proposition 3.11.18
Let Δ be an abelian group, A a Δ-graded ring, $\mathfrak{a}$ a homogenous ideal and $\mathfrak{p} \supseteq \mathfrak{a}$ a homogenous prime ideal. Then there is a canonical isomorphism of Δ-graded rings

$$\begin{aligned} A_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}} &\xrightarrow{\sim} (A/\mathfrak{a})_{\mathfrak{p}/\mathfrak{a}} \\ \frac{a}{s} + \mathfrak{a}_{\mathfrak{p}} &\to \frac{a + \mathfrak{a}}{s + \mathfrak{a}} \end{aligned}$$

In particular there is an isomorphism of rings

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$$A_{(\mathfrak{p})}/\mathfrak{a}_{(\mathfrak{p})} \xrightarrow{\sim} \left(A_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}\right)_0 \xrightarrow{\sim} (A/\mathfrak{a})_{(\mathfrak{p}/\mathfrak{a})}$$

given by

$$\frac{a}{s} + \mathfrak{a}_{(\mathfrak{p})} \to \frac{a+\mathfrak{a}}{s+\mathfrak{a}}$$

Proof. Follows from (3.11.17) and the observation that $A/\mathfrak{a} \setminus \mathfrak{p}/\mathfrak{a} = \pi(A \setminus \mathfrak{p})$. ■

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3.12 Graded Rings

Definition 3.12.1 (Graded Ring over $\mathbb{Z}$)
*A ring A (resp. A-module M) is said to be $\mathbb{Z}$-**graded** (or simply **graded**) if it is graded in the sense of (3.11.2) with gradation group $\Delta = \mathbb{Z}$.*

*If $i < 0 \implies A_i = 0$ (resp. $M_i = 0$) then we say A (resp. M) is **positively graded**.*

Note in this case that we have the multiplication formula

$$(a \cdot b)_i = \sum_{j+k=i} a_j \cdot b_k$$

Definition 3.12.2
*For an element $a \in A$ we define the **degree** to be*

$$\deg(a) := \max\{i \mid a_i \neq 0\}$$

Definition 3.12.3 (Essential and Irrelevant Ideals)
*Let A be a positively graded ring. Then we say a homogenous ideal $\mathfrak{a}$ is **irrelevant** if the following condition holds*

$$\exists n_0 \text{ s.t. } n \geq n_0 \implies \mathfrak{a}_n = A_n$$

*In particular A_+ is irrelevant. Otherwise we say that it is **essential**.*

Proposition 3.12.4 (Radical of Homogenous Ideal is Homogenous)
Let A be a graded ring and $\mathfrak{a}$ a homogenous ideal. Then $\sqrt{\mathfrak{a}}$ is also a homogenous ideal.

Proof. Define $\mathfrak{r} := \sqrt{\mathfrak{a}}$ and fix $x \in \mathfrak{r}$. Suppose

$$x = \sum_{i=-\infty}^{\infty} x_i$$

where $\delta(x_i) = i$ and $x^k \in \mathfrak{r}$. Then

$$x^k = \sum_{i=-\infty}^{\infty} x_i^k$$

where $\delta(x_i^k) = ki$. Without loss of generality we may assume $k > 0$ so by assumption $x_i^k \in \mathfrak{r}$ for all i and therefore $x_i \in \sqrt{\mathfrak{r}}$. ∎

Proposition 3.12.5
Let A be a graded ring and $\mathfrak{p}$ a homogenous ideal satisfying the following property

$$a \in A_n, b \in A_m, ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}$$

Then $\mathfrak{p}$ is a prime ideal.

Proof. Suppose $\mathfrak{p}$ is not prime. Then there exists $a, b \in A$ such that $ab \in \mathfrak{p}$ and $a, b \notin \mathfrak{p}$. Write $a = \sum_j a_j$ and $b = \sum_k b_k$. Then

$$ab = \sum_{i=-\infty}^{\infty} \left(\sum_{(j,k)|j+k=i} a_j b_k \right)$$

As $\mathfrak{p}$ is assumed homogenous then for each i we have

$$(ab)_i := \sum_{(j,k)|j+k=i} a_j b_k \in \mathfrak{p}_i$$

Let j' (resp. k') be the largest integer such that $a_{j'} \notin \mathfrak{p}$ (resp. $b_{k'} \notin \mathfrak{p}$). For all other pairs j'', k'' such that $j' + k' = j'' + k''$ we have by maximality $a_{j''} b_{k''} \in \mathfrak{p}_i$. Therefore we conclude $a_{j'} b_{k'} \in \mathfrak{p}$, which contradicts the original assumption. ∎

Proposition 3.12.6
Let A be a positively graded ring. Then the following are equivalent

a) A_+ is a finitely-generated A-module
b) A is a finitely-generated A_0-algebra

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More precisely for homogenous elements $x_0, ..., x_n \in A_+$ we have

$$A_+ = \sum_{i=0}^{n} Ax_i \iff A = A_0[x_0, ..., x_n]$$

*In this case we say that A is **finitely generated**. Further*

$$A_+^m = (x_0, ..., x_n)^m$$

for all integers $m > 0$.

Proposition 3.12.7 (Criteria for Irrelevant ideals)
Let A be a positively graded ring which is finitely generated and $\mathfrak{a}$ a homogenous ideal. Then the following are equivalent

a) $\mathfrak{a}$ is irrelevant (i.e. $n \geq n_0 \implies \mathfrak{a}_n = A_n$)
b) $A_+ \subset \sqrt{\mathfrak{a}}$

Proof. a) $\implies$ b) It is sufficient to show that $i > 0 \implies A_i \subset \sqrt{\mathfrak{a}}$. Given $x \in A_i$ then by the Archimedean property there exists $k > 0$ such that $ik > n_0$. Therefore by assumption $x^k \in \mathfrak{a}$ and $x \in \sqrt{\mathfrak{a}}$.

b) $\implies$ a) TODO. ∎

Proposition 3.12.8
Let A be a graded ring and $f \in A_1$. Then there is an isomorphism

$$\begin{aligned} A_{(f)}[X, X^{-1}] &\xrightarrow{\sim} A_f \\ \sum_{i\in\mathbb{Z}} \frac{a_{n_i}}{f^{n_i}} X^i &\longrightarrow \sum_{i\in\mathbb{Z}} \frac{a_{n_i}}{f^{n_i - i}} \quad a_{n_i} \in A \text{ homogenous of degree } n_i \\ \sum_{i\in\mathbb{Z}} \frac{a_i}{f^i} X^{i-r} &\longleftarrow \frac{\sum_{i\in\mathbb{Z}} a_i}{f^r} \end{aligned}$$

Proof. Denote the maps by ψ, ϕ, then ψ exists and is a ring homomorphism by (3.9.2) satisfying $\psi(X) = f$ and whence $\psi(X^i) = f^i$. To show ψ is well-defined first consider the map

$$\begin{aligned} \psi' : A &\to A_{(f)}[X, X^{-1}] \\ \sum_{i\in\mathbb{Z}} a_i &\to \sum_{i\in\mathbb{Z}} \frac{a_i}{f^i} X^i \end{aligned}$$

which is evidently a well-defined homomorphism of abelian groups. Furthermore

$$\left(\sum_{i\in\mathbb{Z}} \frac{a_i}{f^i} X^i\right)\left(\sum_{i\in\mathbb{Z}} \frac{b_i}{f^i} X^i\right) = \sum_{i\in\mathbb{Z}} \left(\sum_{j+k=i} \frac{a_j b_k}{f^{j+k}}\right) X^i = \sum_{i\in\mathbb{Z}} \left(\frac{1}{f^i} \sum_{j+k=i} a_j b_k\right) X^i$$

and so the map is a ring-homomorphism by the multiplication formula in A. By the universal property of localisation the ring homomorphism ϕ exists with $\phi(f) = X$ and hence $\phi(f^i) = X^i$. We have

$$\phi(\psi(X)) = \phi(f) = X$$

$$\phi\left(\psi\left(\frac{a_j}{f^j}\right)\right) = \phi\left(\frac{a_j}{f^j}\right) = \frac{a_j}{f^j}$$

so by linearity $\phi \circ \psi = \mathbf{1}$. Similarly

$$\psi\left(\phi\left(\frac{a_i}{f^r}\right)\right) = \psi\left(\frac{a_i}{f^i} X^{i-r}\right) = \frac{a_i}{f^i} f^{i-r} = \frac{a_i}{f^r}$$

so by linearity $\psi \circ \phi = \mathbf{1}$. ∎

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3.13 Chain Conditions

Definition 3.13.1 (Noetherian / Artinian / Finite Modules)
*We say an A-module M is **Noetherian** if it satisfies the **ascending chain condition**, namely any ascending chain of submodules*

$$M_0 \subseteq M_1 \subseteq \cdots \subseteq M$$

eventually stabilizes, i.e $M_n = M_{n+1}$ for all $n \geq N$.

*Similarly we say an A-module M is **Artinian** if it satisfies the **descending chain condition**, namely any descending chain of submodules*

$$M \supseteq M_0 \supseteq M_1 \supseteq \cdots$$

eventually stabilizes.

Definition 3.13.2 (Noetherian / Artinian Ring)
*We say a ring A is **Noetherian** (resp. Artinian) if it is Noetherian (resp. Artinian) as an A-module.*

The following is useful

Proposition 3.13.3 (Noetherian criterion)
Let M be an A-module. The following are equivalent

a) M is Noetherian
b) Every submodule $N \subseteq M$ is finitely-generated
c) Every set of submodules has a maximal element

Proposition 3.13.4
Let M be an A-module and A a Noetherian ring. Then M is Noetherian if and only if it is finitely generated as an A-module.

Proposition 3.13.5 (Restriction of Scalars preserves finiteness)
Let $\phi : A \to B$ be a finite A-algebra and M a (3.3.89) B-module. Then $[M]_\phi$ is a (3.3.89) A-module.

Proof. We suppose that M is generated by $m_1, ..., m_n$, and B is generated by $b_1, ..., b_m$. Then we claim that the elements $b_i m_j$ generate $[M]_\phi$. ∎

Proposition 3.13.6
Let A be a Noetherian ring and $\mathfrak{a} \lhd A$ an ideal. Then $A/\mathfrak{a}$ is Noetherian.

Proof. Consider an increasing sequence of ideals $\mathfrak{a}_i \lhd A/\mathfrak{a}$. Then by (3.3.56) this corresponds to an increasing sequence of ideals $\mathfrak{a}_i' \lhd A$ containing $\mathfrak{a}$. As A is Noetherian, this sequence eventually stabilizes. Again by (3.3.56) the original sequence stabilizes. ∎

Proposition 3.13.7 (Localisation preserves Noetherian)
Let A be a Noetherian ring and $S \subset A$ a multiplicative set. Then $S^{-1}A$ is Noetherian.

Proof. Let $\mathfrak{b}_0 \subseteq \mathfrak{b}_1 \subseteq \cdots$ be an ascending chain of ideals of $S^{-1}A$. Contracting gives an ascending chain of ideals $\mathfrak{b}_0^c \subseteq \mathfrak{b}_1^c \subseteq \cdots$ of A, which stabilizes as A is Noetherian. As every ideal of $S^{-1}A$ is extended (3.5.18), $\mathfrak{b}_i = \mathfrak{b}_i^{ce}$ for each i, so the original chain stabilizes too. ∎

Proposition 3.13.8
Let A be a Noetherian ring then every finitely-generated A-algebra is Noetherian.

In particular $A[X_1, ..., X_n]$ is Noetherian.

Proof. By (3.13.6) it's enough to show that $A[X_1, ..., X_n]$ is Noetherian. By induction and (3.10.4) it's enough to consider the case $n = 1$. Let $\mathfrak{a} \lhd A[X]$ then by (3.13.3) it's enough to show that $\mathfrak{a}$ is finitely-generated.

Define

$$\tilde{\mathfrak{a}}_i := \{\ell(f) \mid f \in \mathfrak{a} \text{ s.t. } \deg(f) = i\}$$

Then clearly $\tilde{\mathfrak{a}}_i \lhd A$ is an ideal. This is an increasing sequence of ideals, and so it stabilizes for $i \geq d$ for some $d > 0$. Furthermore each ideal is finitely generated

$$\tilde{\mathfrak{a}}_i := \left(c_{i1}, ..., c_{in(i)}\right)$$

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where $c_{ij} = \ell(f_{ij})$ for polynomials $f_{ij} \in \mathfrak{a}$ of degree i. We claim that

$$\mathfrak{a} = (f_{ij})_{i \leq d; j \leq n(i)}$$

Denote the right hand side by $\mathfrak{b}$, then clearly $\mathfrak{b} \subseteq \mathfrak{a}$. We show by induction on $m = \deg(f)$ that $f \in \mathfrak{a} \implies f \in \mathfrak{b}$. Let $m' \coloneqq \min(m, d)$. Then by assumption $\ell(f) \in \tilde{\mathfrak{a}}_m = \tilde{\mathfrak{a}}_{m'}$ so there exists $\lambda_j \in A$ such that

$$\ell(f) = \sum_{j=1}^{n(m')} c_{m'j} \lambda_j$$

Consider the decomposition

$$f = \left(f - \sum_{j=1}^{n(m')} \lambda_j f_{m'j} X^{m-m'} \right) + \sum_{j=1}^{n(m')} \lambda_j f_{m'j} X^{m-m'}$$

The first term has strictly smaller degree than f, so by the inductive hypothesis lies in $\mathfrak{b}$. Therefore $f \in \mathfrak{b}$ as required.■

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3.14 Principal Ideal Domains

Definition 3.14.1 (Principal Ideal Domain)
*An integral domain A is a **principal ideal domain** (or **PID**) if every ideal $\mathfrak{a}$ is principal.*

Proposition 3.14.2
A PID is Noetherian.

Proof. Suppose we have an ascending chain of ideals

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \cdots \subseteq \mathfrak{a}_n \subseteq \cdots$$

Clearly the union is again an ideal, which is also principal of the form (a). We must have $a \in \mathfrak{a}_n$ for some n, whence it terminates after n. ∎

Proposition 3.14.3 (Integers form a PID)
$\mathbb{Z}$ is a PID.

Proof. This follows from the well-ordering principle. Let $\mathfrak{a}$ be an ideal with minimal positive element d. We claim $\mathfrak{a} = (d)$. By the division algorithm (or apply well-ordering principle to the coset $x + (d)$), for every $x \in \mathfrak{a}$ there is $0 \leq r < d$ and $q \in \mathbb{Z}$ such that

$$x = qd + r.$$

Clearly $r \in \mathfrak{a}$, whence by minimality $r = 0$ as required. ∎

Proposition 3.14.4
Let k be a field then the polynomial ring $k[X]$ is a PID

Proof. Let $\mathfrak{a} \triangleleft k[X]$ be an ideal. Choose $f \in \mathfrak{a}$ to have minimal degree, then we claim $\mathfrak{a} = (f)$. For $g \in \mathfrak{a}$ we have by (3.7.14) $g = qf + r$ for $\deg(r) < \deg(f)$. Clearly $r \in \mathfrak{a}$, so by minimality $r = 0$ and the result follows. ∎

Lemma 3.14.5 (Co-prime elements in a PID)
Let A be a PID, then x, y are coprime if and only if they have no non-invertible common divisors.

Proof. First suppose $(x, y) = A$, then $ax + by = 1$ and any common divisor d must divide 1 and therefore be invertible.

Conversely suppose $(x, y) \neq (1)$, since A is a PID it must equal (d) for some non-invertible d which is then a common divisor. ∎

Proposition 3.14.6
Let A be a principal ideal domain, M a free A-module with rank m and $N \subset M$ an A-submodule. Then N is free with rank $n \leq m$.

Proof. The rank of a finite free A-module is well-defined (3.8.15).

We proceed by induction on m. If $m = 0$ then $M = 0$ and so $N = 0$ is free of rank 0. Suppose $m \geq 1$ and the result holds for rank $m - 1$. Let $x_1, ..., x_m$ be a basis for M and let $\pi : M \to A$ be the map $\sum_i c_i x_i \mapsto c_m$. Its kernel is the free submodule M' generated by $x_1, ..., x_{m-1}$, of rank $m - 1$. Then $N' := N \cap M'$ is the kernel of $\pi|_N$ and a submodule of M', hence free of rank $n' \leq m - 1$ by the induction hypothesis. The image $\pi(N)$ is an ideal of A, hence equal to (a) for some $a \in A$.

If $a = 0$ then $N = N'$ and we are done. Suppose $a \neq 0$ and choose $y \in N$ with $\pi(y) = a$. We claim $N = N' \oplus Ay$ with $Ay \cong A$. For $z \in N$ we have $\pi(z) = ca$ for some $c \in A$, whence $z - cy \in N'$. Moreover $\pi(cy) = ca$ is non-zero for $c \neq 0$, as A is an integral domain. Hence $c \mapsto cy$ is injective, so $Ay \cong A$, and $N' \cap Ay = 0$ as $cy \in N'$ forces $ca = 0$ and so $c = 0$. Therefore N is free with basis a basis of N' together with y, of rank $n' + 1 \leq m$. ∎

Proposition 3.14.7
Let A be a principal ideal domain and M a finitely generated, torsion-free A-module. Then M is a finite free A-module.

Proof. Let $\{m_1, ..., m_n\}$ be a generating set for M, and $\{v_1, ..., v_r\}$ an arbitrary linearly independent subset of M. Then by definition we have

$$v_i = \sum_{j=1}^{n} a_{ij} m_j \quad a_{ij} \in A$$

Consider the elements of A^n given by

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$$w_i := (a_{i1}, ..., a_{in})$$

Then we can see that $\{w_1, ..., w_r\}$ is an A-linearly independent subset of A^n, and therefore a K-linearly independent subset of K^n for $K = \mathrm{Frac}(A)$ (a K-linear dependence gives an A-linear dependence on clearing denominators). As K^n has the standard basis of order n, it has dimension n over K, so $r \leq n$ (3.3.126).

Therefore we may choose $\{v_1, ..., v_r\}$ to be a maximal linearly independent subset and define $M' := \langle v_1, ..., v_r \rangle$. For each generator m_i, by maximality of the subset there must exist an $a_i \neq 0$ such that

$$a_i m_i \in M'$$

Define $a := a_1 ... a_n$ then this satisfies $aM \subseteq M'$, since the m_i generate M. As A is an integral domain we have a series of injections

$$M \hookrightarrow aM \subseteq M' \xrightarrow{\sim} A^r$$

Therefore we are done by (3.14.6). ■

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3.15 Factorisation

For this section we assume A is a commutative integral domain.

Definition 3.15.1 (Associates)
*We say two non-zero elements x and y are **associates** if $x = uy$ for some $u \in A^\star$. We write $x \sim y$.*

Note this is an equivalence relation on $A \setminus \{0\}$.

Lemma 3.15.2
Let A be a ring and $x, y \in A$ non-zero elements then the following are equivalent

- $x \mid y$
- $(y) \subseteq (x)$
- $y \in (x)$

Lemma 3.15.3
Let A be a ring and $x, y \in A$ non-zero elements then the following are equivalent

- $x \mid y$ *and* $y \mid x$
- $(x) = (y)$

If A is an integral domain this is equivalent to $x \sim y$.

Definition 3.15.4 (Irreducible element)
*We say $0 \neq x$ is **irreducible** if it is not invertible and $x = ab \implies a$ a unit or b a unit.*

Equivalently if $y \mid x$ implies either y is a unit or $y \sim x$.

Definition 3.15.5 (Prime element)
We say $0 \neq p$ is prime if $p \mid ab \implies p \mid a$ or $p \mid b$.

Example 3.15.6
The units of $\mathbb{Z}$ are $\{-1, 1\}$ so each equivalence class is of the form $\{n, -n\}$.

Example 3.15.7
A number $p \in \mathbb{Z}$ is prime in the traditional sense exactly when it is irreducible. It is of course also prime in the ring-theoretic sense but this requires proof (see (2.2.13)).

The concept of associates is important to unique factorization, because we may only hope to have unique factorization upto multiplication by a unit.

Lemma 3.15.8
If $x \sim y$ are associates then x is irreducible iff y is

Proof. Suppose $x \sim y$ and x irreducible. If $y = ab$ then $x = abu \implies a$ a unit or bu a unit $\implies b$ a unit. Therefore y is irreducible as required. ∎

Lemma 3.15.9 (Criterion for primality)
Suppose $0 \neq p \in \mathbb{Z}$. Then p is prime if and only if (p) is a prime ideal

Proof. Note $x \mid y \iff y \in (x)$. So in particular if p is prime then $xy \in (p) \implies p \mid xy \implies p \mid x$ or $p \mid y \implies x \in (p)$ or $y \in (p)$, whence (p) is prime.

Conversely if (p) is prime, then $p \mid xy \implies xy \in (p) \implies x \in (p)$ or $y \in p \implies p \mid x$ or $p \mid y$, so that p is prime. ∎

Lemma 3.15.10 (Criterion for irreducibility)
Let A be an integral domain. Then f is irreducible if and only if (f) is maximal amongst proper principal ideals.

Proof. Suppose f is irreducible and $(f) \subseteq (g)$. Then $f = ag$ with either a a unit or g a unit. If a is a unit then $(f) = (g)$, and if g is a unit $(g) = A$. So the result follows.

Conversely suppose $f = ab$, then $f \in (a) \implies (f) \subseteq (a)$. Then by hypothesis either $(a) = (f)$ or $(a) = A$. In the second case a is a unit. In the first case then $f \mid a \implies bf \mid f \implies b \mid 1$ whence b is a unit. ∎

Proposition 3.15.11 (Primes are Irreducible)
Let A be an integral domain then p prime $\implies p$ irreducible

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Proof. Suppose $b \mid p$ then $p = ab$ and $a \mid p$. By hypothesis $p \mid a$ or $p \mid b$. If $p \mid a$ (resp. b) then by (3.15.3) $p \sim a$ (resp. b) as required. ∎

Definition 3.15.12 (Unique Factorisation Domain (UFD) or Factorial Ring)
We say an integral domain A is factorial (or a UFD) if every element $0 \neq a$ may be represented as

$$a = u \prod_{i=1}^{n} p_i$$

for u a unit and p_i irreducible, and moreover this is unique in the sense that given another factorization

$$a = u' \prod_{i=1}^{m} p_i'$$

we have $n = m$ and $p_i \sim p'_{\psi(i)}$, for ψ a permutation on $\{1, ..., n\}$.

Furthermore it may be convenient in applications to count the multiplicities

Proposition 3.15.13 (Factorisation with multiplicities)
Let A be a UFD, then for every element $0 \neq a \in A$ there is a factorization of the form

$$a = u \prod_{i=1}^{n} p_i^{r_i}$$

where $r_i > 0$ and none of the p_i are associate to each other. Furthermore this is essentially unique in the sense that given another such factorization we have $n = n'$, $r_i = r'_{\sigma(i)}$ and $p_i \sim p'_{\sigma(i)}$ for some permutation $\sigma \in S_n$.

Proof. Given a factorization into irreducible elements

$$a = u \prod_{i=1}^{n} p_i$$

Consider a representative set of irreducibles $q_1, ..., q_m$ (under the equivalence relation $x \sim y$). Then we have $p_i = q_{\pi(i)} u_i$ for some units u_i and mapping $\pi : \{1, ..., n\} \to \{1, ..., m\}$. Let $r_j = \#\pi^{-1}(j)$. Then we have that the set of irreducibles $\{p_1, ..., p_n\}$ is the *disjoint* union of the set of equivalence classes with representatives q_j. Therefore

$$a = u \prod_{j=1}^{m} \prod_{p \sim q_j} p = u \prod_{j=1}^{m} \prod_{i:\pi(i)=j} u_i q_j = \left(u \prod_{j=1}^{m} \prod_{i:\pi(i)=j} u_i \right) \prod_{j=1}^{m} q_j^{r_j}$$

as required. Suppose we have two factorizations

$$u \prod_{i=1}^{n} p_i^{r_i} = u' \prod_{i=1}^{m} (p_i')^{r_i'}$$

Let I be the indexing set of p_i and J the set of p_j'. By unique factorization there must be mappings $\sigma : I \to J$ such that $p_i \sim p'_{\sigma(i)}$, and $\tau : J \to I$ such that $p_j' \sim p_{\tau(j)}$. Which means that $p_i \sim p_{\tau(\sigma(i))}$ and $p_j' \sim p'_{\sigma(\tau(j))}$. Since none are associate to each other we see that τ and σ are mutual inverses, whence $m = n$ and we may regard $\sigma \in S_n$. In the unique factorization p_i appears r_i times and $p'_{\sigma(i)}$ appears $r'_{\sigma(i)}$ times. Since p_i is associate to $p'_{\sigma(i)}$ it is not associate to any p_j' for $j \neq \sigma(i)$. Unique factorization shows that $r_i = r'_{\sigma(i)}$. ∎

Definition 3.15.14
Let A be a UFD and $x \in A$ a non-zero, non-unit such that

$$x \sim \prod_{i=1}^{n} p_i^{r_i}$$

is an (almost) unique factorization into irreducibles. Then for $p \in A$ an irreducible define

$$v_p(x) := \begin{cases} r_i & p \sim p_i \\ 0 & \text{otherwise} \end{cases}$$

If $x \in A$ is a unit then simply define $v_p(x) = 0$ for all p.

Lemma 3.15.15
Let A be a UFD then the following are equivalent

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a) $x \mid y$
b) $(y) \subseteq (x)$
c) $v_p(x) \leq v_p(y)$ *for all* p *irreducible*

In particular $x \sim y \iff (x) = (y) \iff v_p(x) = v_p(y)$ *for all* p *irreducible.*

Definition 3.15.16 (Atomic Ring)
We say that A *is* ***atomic*** *if every element has a (not necessarily unique) decomposition into irreducible elements.*

Definition 3.15.17 (Ascending Chain Condition for Principal Ideals (ACCP))
We say a ring A *satisfies* ***ACCP*** *if every ascending chain of principal ideals eventually stabilizes.*

Note every Noetherian ring satisfies this condition.

Proposition 3.15.18
An integral domain A *satisfying ACCP is atomic.*

In particular a Noetherian ring is atomic.

Proof. Suppose a ring A is not atomic, then choose any non-unit $x_1 \in A$. By repeated application (3.15.10) it's possible to construct a strictly ascending sequence of proper principal ideals

$$(x_1) \subsetneq (x_2) \subsetneq ... \subsetneq (x_n) \subsetneq ...$$

therefore A does not satisfy ACCP. ■

Remark 3.15.19
The converse is not in general true (...) but see (3.15.21) for a partial converse.

We show a simple criterion for a ring to be a UFD.

Definition 3.15.20
We say an integral domain A *is* ***AP*** *if* p *irreducible* $\implies$ p *prime.*

Roughly speaking, "atomic" ensures the existence of factorization and "AP" ensures the uniqueness.

Proposition 3.15.21 (Atomic + AP $\iff$ UFD)
Let A *be an integral domain. The following are equivalent*
a) A *is a UFD*
b) A *is atomic and AP*
c) A *satisfies ACCP and is AP*

Proof. *a)* $\implies$ *c)*. Suppose A is a UFD. If p is an irreducible element and $p \mid ab$, then by uniqueness it must appear in the irreducible factorization of either a or b. Therefore p is prime. If we have an ascending chain of principal ideals $(x_1) \subseteq (x_2)$... then by (3.15.15) we have $v_p(x_i)$ is a decreasing sequence for all irreducible p occuring in the factorization of x_1. Furthermore $\max_p v_p(x_i)$ is a finite decreasing sequence. Choose $i = N$ such that $\max_p v(x_i)$ is minimal, then all these sequences must stabilise for $i \geq N$ and therefore by (3.15.15) the chain of principal ideals also stabilises.

c) $\implies$ *b)*. This is (3.15.18).

b) $\implies$ *a)*. We require to show that factorization into irreducibles is unique up to associates. Suppose

$$\prod_{i=1}^{n} p_i \sim \prod_{j=1}^{m} p'_j$$

By convention an empty product is 1 and by hypothesis all the elements are in fact prime. If $n = 0$, then since p'_j is irreducible, it is not a unit and hence $m = 0$. Otherwise consider p_1, then $p_1 \mid$ RHS, so by definition of prime we must have $p_1 \mid p'_j$ for some j. Since p'_j is irreducible and p_1 is not a unit, we have $p_1 \sim p'_j$. Since A is integral we may cancel these two to obtain an equivalence of smaller degree and we may proceed by induction. ■

Lemma 3.15.22 (PID is an AP-domain)
Let A *be a PID and* $a \in A$. *Then the following are equivalent*

a) a *is prime*
b) a *is irreducible*
c) (a) *is maximal*

Proof. *a)* $\implies$ *b)* This holds for an arbitrary integral domain (3.15.11).

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$b) \iff c)$ as all ideals are principal this follows from (3.15.10)

$c) \implies a)$ By (3.3.60) (a) is prime, whence a is prime by (3.15.9) ∎

Proposition 3.15.23
A PID is Noetherian UFD.

Furthermore (f is irreducible $\iff$ f is prime), and every prime ideal is maximal.

Proof. A is Noetherian by (3.14.2) and atomic by (3.15.18). And by (3.15.22) an irreducible element is prime. Therefore we are done by (3.15.21). ∎

If we take a suitable fixed set of irreducible elements we can obtain completely unique factorization

Definition 3.15.24
*Let A be a ring we say $\mathcal{P}$ is a **representative set of irreducible elements** if*

- *No two elements $p, q \in \mathcal{P}$ are associate*
- *Every irreducible element $p \in A$ is associate to (precisely) one in $\mathcal{P}$*

Example 3.15.25
For $\mathbb{Z}$ the positive *primes are a canonical set of irreducible elements.*

Proposition 3.15.26
Let A be a UFD and $K = \operatorname{Frac}(A)$ then for every irreducible $p \in A$ there is a unique map

$$v_p : K^\star \to \mathbb{Z}$$

such that

- $u \in A^\star \implies v_p(u) = 0$
- $v_p(xy) = v_p(x) + v_p(y)$
- $v_p(p) = 1$

Furthermore let $\mathcal{P}$ be a set of irreducible representatives then we have a group isomorphism

$$\begin{aligned} K^\star/A^\star &\xrightarrow{\sim} \bigoplus_{p\in\mathcal{P}} \mathbb{Z} \\ x &\longrightarrow \left(v_p(x)\right)_{p\in\mathcal{P}} \\ \prod_{p\in\mathcal{P}} p^{n_p} &\longleftarrow \left(n_p\right)_{p\in\mathcal{P}} \end{aligned}$$

Finally $x \in A \iff v_p(x) \geq 0$ for all $p \in \mathcal{P}$.

Proof. By (3.15.13) there is a well-defined map $v_p : A \setminus \{0\} \to \mathbb{Z}$ satisfying the given properties. We claim that

$$\begin{aligned} v_p : K^\star &\to \mathbb{Z} \\ xy^{-1} &\to v_p(x) - v_p(y) \end{aligned}$$

is well-defined. For suppose $xy^{-1} = wz^{-1}$ then $zx = wy$ whence $v_p(z) + v_p(x) = v(w) + v(y)$ and therefore $v_p(xy^{-1}) = v_p(wz^{-1})$.

It's clear the multiplicative property also holds and so ϕ is well-defined (since by definition $A^\star$ is in the kernel of v_p). Denote by ϕ, ψ the proposed maps. By definition of unique factorization ϕ and ψ are mutual inverses when restricted as follows

$$(A \setminus \{0\})/A^\star \longleftrightarrow \bigoplus_{p\in\mathcal{P}} \mathbb{Z}_{\geq 0}$$

We also observe that $\phi(x^{-1}) = -\phi(x)$ and $\psi(-n) = \psi(n)^{-1}$, so then it's easy to demonstrate they are mutually inverse over the whole domain.

Suppose $v_p(xy^{-1}) \geq 0$ then $v_p(x) \geq v_p(y)$. If this holds for all $p \in \mathcal{P}$, then by (3.15.15) $y \mid x$ as elements of A whence by definition $xy^{-1} \in A$ as required. ∎

Proposition 3.15.27
Suppose A is an integral domain satisfying ACCP then so is $A[X]$.

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Proof. Suppose we have an ascending chain of principal ideals

$$(f_1) \subseteq (f_2) \subseteq ...(f_n)...$$

Without loss of generality the f_i are non-zero. Then $f_{i+1} \mid f_i$ and by (...) $\deg(f_{i+1}) \leq \deg(f_i)$. Choose N such that $\deg(f_i)$ is minimal, and define $a_i := \ell(f_i) \in A$. Then by (3.7.3) $a_{i+1} \mid a_i$ for $i \geq N$ as elements of A. Therefore we have an increasing sequence of principal ideals

$$(a_N) \subseteq (a_{N+1}) \subseteq ...$$

which by hypothesis stabilizes, that is $a_i \sim a_j$ for all $i, j \geq M$ for some $M \geq N$. For $i \geq M$ we have $ua_i = a_{i+1}$, consider $uf_i - f_{i+1} \in (f_i)$. This has degree strictly smaller than N, and therefore by minimality must be 0. In particular $f_i \sim f_{i+1}$ and $(f_i) = (f_{i+1})$. ∎

Lemma 3.15.28
Suppose $p \in A$ is prime, then it is prime as an element of $A[X]$.

Lemma 3.15.29 (Nagata's Criterion)
Let A be a ring and S a multiplicative subset generated by prime elements and units. Let $f \in A$ be irreducible or a unit, then

a) $\frac{f}{1}$ is irreducible or a unit in $S^{-1}A$
b) $\frac{f}{1}$ prime or a unit in $S^{-1}A \implies f$ is a prime or a unit in A.

Furthermore if $S^{-1}A$ is AP then so is A.

Proof. Note the condition on S means every $a \in S$ satisfies $a \sim p_1...p_r$ for primes $p_i \in A$.

a) Suppose $\frac{f}{1} = \frac{g}{a}\frac{h}{b}$ for $f, g, h \in A$ and $a, b \in S$, then $abf = gh$. Further $ab \sim p_1...p_r$. Then $p_i \mid a$ or $p_i \mid b$, whence we can find $f \sim g'h'$ where $g = cg'$, $h = dh'$ and $c, d \in S$. As f is irreducible (or a unit), then for example g' is invertible, in which case $\frac{g}{a} = \frac{cg'}{a}$ is invertible. Therefore $\frac{f}{1}$ is either irreducible or a unit.
b) The case f a unit is clear, so assume that f is irreducible. Suppose $\frac{f}{1}$ is prime or a unit and $f \mid gh$. Then $\frac{f}{1} \mid \frac{g}{1}\frac{h}{1}$ and for example $\frac{f}{1} \mid \frac{g}{1}$. Therefore $ff' = gp_1...p_r$ for some $f' \in A$ and $p_1, ..., p_r \in A$ prime. If $p_i \mid f$ for some i then by irreducibility we have $p_i \sim f$, and we see that f is prime. Otherwise $p_i \mid f'$ for all i and we find $f \mid g$. Therefore f is prime as required.

The last statement follows immediately from the previous two results. ∎

3.15.1 Polynomials over a Field

In this section we consider the polynomial ring with coefficients in a field, $k[X]$.

Proposition 3.15.30 (Associate polynomials)
The following are equivalent for $0 \neq f, g$

- $f \sim g$
- $f = \lambda g$ *for* $\lambda \neq 0$
- $f \mid g$ *and* $g \mid f$

Proposition 3.15.31
A polynomial $f \in k[X]$ is associate to precisely one monic polynomial g. If f is irreducible so is g.

Proof. **TODO** ∎

Proposition 3.15.32 (Polynomial ring is a PID)
Let k be a field, then $k[X]$ is a PID, and therefore a Noetherian UFD.

Proof. Let $(0) \neq \mathfrak{a}$ be an ideal and let $f \in \mathfrak{a}$ be a polynomial of minimal degree. We may assume it is monic. Any $g \in \mathfrak{a}$ may be represented as $f = qg + r$ by the division algorithm. Clearly $r \in \mathfrak{a}$, therefore by minimality $r = 0$, whence $g \in (f)$. ∎

Proposition 3.15.33 (Unique Factorisation of Polynomials)
For the ring $k[X]$ the set of irreducible monic polynomials constitutes a representative set (3.15.24). Therefore we have a unique factorization

$$f = \ell(f) \prod_{p \text{ irreducible monic}} p^{v_p(f)}$$

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such that

$$v_p(fg) = v_p(f) + v_p(g)$$

Proof. (3.15.31) shows that the irreducible monic polynomials constitute a representative set. Therefore the result follows from (3.15.26). Let u be the unit appearing in the factorization, it must be an element of k. Compare leading coefficients to see that $u = \ell(f)$. ∎

Lemma 3.15.34 (Roots and Multiplicity)
For $f \in k[X]$ a non-constant polynomial and $\alpha \in k$ we have

$$f(\alpha) = 0 \iff (X - \alpha) \mid f \iff v_{X-\alpha}(f) > 0$$

In this case $r := v_{X-\alpha}(f)$ is the multiplicity of the root α, and observe

$$f(X) = \ell(f)(X - \alpha)^r g(X)$$

with $g(\alpha) \neq 0$ (equivalently $v_{X-\alpha}(g) = 0$).

Proof. The right to left implication is obvious. Conversely by the division algorithm we may write

$$f(X) = f(\alpha) + (X - \alpha)Q(X)$$

Then if $f(\alpha) = 0$ we clearly have $v_{X-\alpha}(f) > 0$. Finally we may construct

$$g(X) = \prod_{p \neq (X-\alpha)} p^{v_p(f)}$$

It's clear that for every p appearing in the product $p(\alpha) \neq 0$ because otherwise we would have $(X - \alpha) \mid p$ and by irreducibility $(X - \alpha) = p$. Therefore $g(\alpha) \neq 0$ as required. ∎

Corollary 3.15.35
Let K/k be a field extension and $0 \neq f \in k[X]$. Then f has at most $\deg(f)$ roots in K. In particular a polynomial of degree $< n$ with n distinct roots in K is zero.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Identify f with its image in $K[X]$, which has the same degree. Let $\alpha_1, ..., \alpha_m$ be the distinct roots of f in K and $r_i := v_{X-\alpha_i}(f) \geq 1$ their multiplicities in $K[X]$ (3.15.34). Then $f = \ell(f) \prod_{i=1}^m (X - \alpha_i)^{r_i} g$ where g is the product of the remaining irreducible factors of f in $K[X]$, which is a constant when f splits completely in K (3.17.8). Therefore $\deg(f) = \sum_i r_i + \deg(g) \geq m$. ∎

Lemma 3.15.36 (Lagrange Interpolation)
Let k be a field, $y_1, ..., y_n \in k$ distinct and $h \in k[X]$ with $\deg(h) < n$. Then

$$h(X) = \sum_{l=1}^{n} h(y_l) \prod_{k \neq l} \frac{X - y_k}{y_l - y_k}.$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The difference of the two sides has degree $< n$ and vanishes at each y_j, as the l-th term of the sum vanishes at y_j for $l \neq j$ and equals $h(y_j)$ at y_j for $l = j$. It has n roots so is zero by (3.15.35). ∎

Proposition 3.15.37 (Criteria for Multiple Roots)
Let $f(X) \in k[X]$ be a polynomial, r a positive integer and suppose that either $\mathrm{char}(k) = 0$ or $r < \mathrm{char}(k)$. Then $\alpha \in k$ is a root of f with multiplicity r precisely when

$$f(\alpha) = f^{(1)}(\alpha) = ... = f^{(r-1)}(\alpha) = 0$$

and $f^{(r)}(\alpha) \neq 0$.

Therefore the multiple roots are precisely the common roots of $f(X)$ and $f'(X)$ (irrespective of the characteristic).

Proof. Note that by (3.15.34) and (3.7.10)

$$f^{(1)}(X) = (X - \alpha)^{r-1}[rg(X) + (X - \alpha)g'(X)]$$

with $g(\alpha) \neq 0$ and r the multiplicity of the root. If $r = 1$, then $f^{(1)}(\alpha) = g(\alpha) \neq 0$ as required. If $r > 1$, then $f^{(1)}(X)$ has α as a root of multiplicity $r - 1$, so it follows by induction.

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The second statement is simply the case $r = 1$. ∎

Definition 3.15.38 (Separable Polynomial)
*A polynomial $f \in k[X]$ is **separable** if f and f' are co-maximal, that is $(f, f') = (1)$. Otherwise it is **inseparable**.*

Proposition 3.15.39
A separable polynomial $f \in k[X]$ has no multiple roots (and f and f' have no common roots) in any extension field K/k.

Proof. Since $(f, f') = (1)$ we have $af + bf' = 1$ for some $a, b \in k[X]$. Clearly f and f' can have no common roots, and therefore f has no multiple roots by (3.15.37). ∎

Proposition 3.15.40
Suppose $f, g \in k[X]$, g is separable and $f \mid g$, then f is separable.

Proof. Suppose f is not separable then by (3.14.5) f and f' have a common divisor d such that $\deg(d) > 0$. Since $g = fh$, so $g' = f'h + fh'$. Therefore d is also a non-trivial common divisor of g and g' contradicting (3.14.5). ∎

3.15.2 Polynomial Ring is a UFD

We prove the following result, first by Nagata's Criterion and again by Gauss' Lemma.

Proposition 3.15.41
Suppose A is a UFD, then so is $A[X]$.

Proof. By (3.15.21) we need to show that $A[X]$ satisfies ACCP and is AP. The first follows from (3.15.27).

Let $S = A \setminus \{0\}$ the set of non-zero elements. Let $K = \mathrm{Frac}(A)$. If we regard $A[X]$ as a subring of $K[X]$ then we claim $S^{-1}(A[X]) = K[X]$; this follows by multiplying an element of $K[X]$ by the product of denominators of all the coefficients. Furthermore as A is a UFD (and by (3.15.21)) S is generated by prime elements and units. As K is a field, $K[X]$ is a PID (3.15.32), so by (3.15.23) it is a UFD, and so in particular is AP. Therefore by (3.15.29) $A[X]$ is AP as required. ∎

For Gauss' Lemma we require to introduce some notation first.

Definition 3.15.42 (Primitive polynomials)
Let A be a UFD, $K := \mathrm{Frac}(A)$ and $f \in K[X]$ a polynomial given by

$$f(X) = \sum_{i=0}^{n} a_i X^i$$

*Define the **content** of f by*

$$c(f) = \prod_{p \in \mathcal{P}} p^{\min_i v_p(a_i)} \in K^\star$$

where the product is taken over a representative set of primes for A. Changing the set of representatives only changes the value of $c(f)$ by multiplication of a unit.

*We say that f is **primitive** precisely when $c(f) = 1$*

Lemma 3.15.43 (Gauss' Lemma I)
Let A be a UFD and $f \in K[X]$ where $K = \mathrm{Frac}(A)$. Then the content $c(f)$ satisfies the following properties

a) $f \in A[X] \iff c(f) \in A$
b) f is primitive $\iff \min_i v_p(a_i) = 0 \quad \forall p$ prime and in this case $f \in A[X]$
c) $c(\lambda f) = \lambda c(f)$ for $\lambda \in K^\star$ up to multiplication by a unit in A
d) $f/c(f) \in A[X]$ is primitive
e) f, g primitive $\implies fg$ primitive
f) $c(fg) = c(f)c(g)$ for all $f, g \in K[X]$

Proof. We prove each in turn

a) $f \in A[X] \iff a_i \in A \quad \forall i \overset{(3.15.26)}{\iff} v_p(a_i) \geq 0 \, \forall i \, \forall p \in \mathcal{P} \iff \min_i v_p(a_i) \geq 0 \, \forall p \in \mathcal{P} \overset{(3.15.26)}{\iff} c(f) \in A$
b) $c(f) = 1 \iff v_p(c(f)) = 0 \quad \forall p \iff \min_i v_p(a_i) = 0 \quad \forall p$. Further $v_p(a_i) \geq 0$ whence $f \in A[X]$ by (3.15.26).
c) Note $v_p(\lambda a_i) = v_p(\lambda) + v_p(a_i) \implies v_p(c(\lambda f)) = \min_i v_p(\lambda a_i) = v_p(\lambda) + \min_i v_p(a_i) = v_p(\lambda) + v_p(c(f))$. Whence by (3.15.26) we see $c(\lambda f)$ and $\lambda c(f)$ are equal up to multiplication by a unit in A.
d) Clear by b).

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e) Consider p prime and the reduction $\bar{\cdot}: A[X] \to (A/(p))[X]$. Then by assumption $\overline{f}$ and $\overline{g}$ are non-zero. Furthermore $(A/(p))[X]$ is an integral domain so that $\overline{f \cdot g} = \overline{f} \cdot \overline{g} \neq 0$. Therefore p does not divide all the coefficients of $f \cdot g$. As p was arbitrary this shows that $f \cdot g$ is primitive.

f) We may reduce to e) by dividing by the content.

■

Lemma 3.15.44 (Gauss' Lemma II)
Let A be a UFD and $f \in A[X]$. Then the following are equivalent

a) f is irreducible in $A[X]$
b) f is an irreducible element of A or (f is primitive and irreducible in $K[X]$)

where $K = \mathrm{Frac}(A)$.

In particular if $f \in K[X]$ is irreducible then $f/c(f)$ is irreducible in $A[X]$.

Proof. $b) \implies a)$. It's clear that an irreducible element of A remains irreducible in $A[X]$. Suppose $0 \neq f \in A[X]$ is primitive and irreducible in $K[X]$. Then as $f \notin K[X]^\star$ we have $\deg(f) > 0$. Suppose $f = gh$ in $A[X]$ then by irreducibility $\deg(g) = 0$ or $\deg(h) = 0$. Further $1 = c(f) = c(g)c(h)$ by Gauss' Lemma, so one of h and g must lie in $A^\star = A[X]^\star$. Therefore f is irreducible in $A[X]$ as required.

$a) \implies b)$. If $\deg(f) = 0$ then clearly f is an irreducible element of A. So assume $\deg(f) > 0$. Observe $f = c(f) \cdot (f/c(f))$ so by irreducibility we require $c(f) = 1$ and therefore f is primitive. Suppose $f = gh$ in $K[X]$ then $1 = c(g)c(h)$ and therefore

$$f = \frac{g}{c(g)} \frac{h}{c(h)}$$

is a decomposition in $A[X]$ which shows one of g, h has degree 0. Therefore f is irreducible in $K[X]$ as required. ■

Proposition 3.15.45
Let A be a UFD. Then so is $A[X]$.

Proof. For $0 \neq f \in A[X]$ we may use irreducible factorisation in $K[X]$ to find

$$f = \lambda \pi_1 ... \pi_n$$

where $\lambda \in K$ and $\pi_i \in K[X]$ are irreducible. Replace $\pi_i \to \pi_i / c(\pi_i)$ then may assume that π_i are irreducible in $A[X]$ by (3.15.44). Furthermore

$$c(f) = \lambda c(\pi_i) ... c(\pi_n) = \lambda$$

which shows $\lambda \in A$. As A is a UFD then λ may be decomposed into irreducibles of A, which by (3.15.44) are irreducible in $A[X]$. Therefore $A[X]$ is atomic.

Suppose that $f \in A[X]$ is irreducible we require to show it is prime. The case $\deg(f) = 0$ follows from the AP property of A, so assume $\deg(f) > 0$. Suppose $f \mid gh$ then as $K[X]$ is a UFD we see that, without loss of generality, $qf = g$ for $q \in K[X]$. Then $c(qf) = c(q)c(f) = c(q) = c(g) \in A$, so $q \in A[X]$ by (3.15.43) and we see $f \mid g$ in $A[X]$. This shows that $A[X]$ is AP and therefore a UFD by (3.15.21). ■

Corollary 3.15.46
Suppose A is a UFD. Then $A[X_1, ..., X_n]$ is a UFD.

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3.16 Cayley-Hamilton Theorem

Definition 3.16.1 (Characteristic Polynomial of a Matrix)
For a matrix $E \in \mathrm{Mat}_n(A)$ define the characteristic polynomial by

$$P_E(X) := \det(X \cdot I_n - E)$$

working in $\mathrm{Mat}_n(A[X])$. This is a monic polynomial in $A[X]$ and satisfies $P_E(X) = P_{E^t}(X)$ (3.8.20).

Definition 3.16.2 (Characteristic Polynomial of an endomorphism of a free module)
Let M be a finite free A-module. Let $\mathcal{B}$ be a basis for M, $\phi : M \to M$ an A-module endomorphism and $[\phi]$ the matrix representation of ϕ with respect to $\mathcal{B}$. Define the characteristic polynomial of ϕ by

$$P_\phi(X) := P_{[\phi]}(X)$$

This is independent of the basis $\mathcal{B}$ by (3.8.20).

Lemma 3.16.3
Suppose $M = \langle m_1, ..., m_n \rangle$ is a finitely generated A-module and $\mathfrak{a} \triangleleft A$ an ideal, then

$$\mathfrak{a}M = \mathfrak{a}m_1 + ... + \mathfrak{a}m_n$$

That is every $m \in \mathfrak{a}M$ may be written as

$$m = \sum_{i=1}^{n} a_i m_i \quad \text{for some } a_i \in \mathfrak{a}$$

Proof. By hypothesis

$$m = \sum_{j=1}^{n} b_j m'_j \quad \text{for some } m'_j \in M, b_j \in \mathfrak{a}$$

Furthermore by the finite-generation hypothesis for $j = 1...n$

$$m'_j = \sum_{i=1}^{n} c_{ji} m_i \quad \text{for some } c_{ji} \in A.$$

Therefore

$$m = \sum_{i=1}^{n} \left(\sum_{j=1}^{n} b_j c_{ji} \right) m_i$$

whence $a_i := \sum_{j=1}^{n} b_j c_{ji}$ lie in $\mathfrak{a}$ and give the required coefficients. ∎

Theorem 3.16.4 (Cayley-Hamilton)
Let M be a finitely generated A-module and $\phi \in \mathrm{End}_A(M)$. Then there exists a monic polynomial $P(X) \in A[X]$ such that $P(\phi) = 0$.

This may be strengthened in the following ways

a) If M is a finite free A-module then P may be taken to be the characteristic polynomial $P_\phi(X)$.
b) If $\phi(M) \subseteq \mathfrak{a}M$ for some ideal $\mathfrak{a} \triangleleft A$, then the non-leading coefficients of $P(X)$ may be chosen to be in $\mathfrak{a}$.

Proof. First since $\mathrm{End}_A(M)$ is an A-algebra there is a canonical evaluation morphism

$$\mathrm{ev}_\phi : A[X] \to \mathrm{End}_A(M)$$

and the meaning of $P(\phi)$ is simply $\mathrm{ev}_\phi(P)$.

Let $\{m_1, ..., m_n\}$ be a generating set, then by definition

$$\phi(m_i) = \sum_{j=1}^{n} E_{ij} m_j \quad i = 1...n$$

for some $E \in \mathrm{Mat}_n(A)$. Consider the matrix

$$B(X) = XI_n - E \in \mathrm{Mat}_n(A[X])$$

Then we may define $B(\phi) := B(X)^{\mathrm{ev}_\phi} \in \mathrm{Mat}_n(\mathrm{End}_A(M))$ pointwise, so given by

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$$B(\phi)_{ij} = \delta_{ij}\phi - E_{ij}1_M \quad i,j = 1...n$$

By definition

$$\sum_j B(\phi)_{ij}m_j = \phi(m_i) - \sum_j E_{ij}m_j = 0$$

Formally we have a group action

$$\begin{aligned}\mathrm{Mat}_n(\mathrm{End}_A(M)) \times M^n &\to M^n \\ F \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} &\to \begin{pmatrix} \sum_{j=1}^n F_{1j}(x_j) \\ \vdots \\ \sum_{j=1}^n F_{nj}(x_j) \end{pmatrix}\end{aligned}$$

such that $(EF)v = E(Fv)$ (check). And we have shown that

$$B(\phi)\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

Using (3.8.22), premultiply by the adjugate matrix to show that

$$\det(B(\phi))I_n\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

and $\det(B(\phi)) \in \mathrm{End}_A(M)$ annihilates $m_1, ..., m_n$ and therefore M.

Finally we claim that $P(X) := \det(B(X)) \in A[X]$ is a suitable monic polynomial. We see that

$$P(\phi) := \mathrm{ev}_\phi(\det(B(X))) = \det(B(X)^{\mathrm{ev}_\phi}) = \det(B(\phi)) = 0$$

When M is a finite free A-module then we may choose $\{m_1, ..., m_n\}$ to be a basis, and then the matrix E equals $[\phi]^T$ as required.

Finally when $\phi(M) \subseteq \mathfrak{a}M$ then (3.16.3) shows we may choose the coefficients E_{ij} to be in $\mathfrak{a}$. It's clear that $P(X)$ then has non-leading coefficients in $\mathfrak{a}$. ∎

3.16.1 The $A[X]$-module associated to an endomorphism

Proposition 3.16.5
Let $\phi : M \to M$ be an A-module endomorphism. Then there is a well-defined $A[X]$-module structure defined by

$$\begin{aligned}A[X] \times M &\to M \\ (P, m) &\to P(\phi)(m)\end{aligned}$$

where $P = \sum_{i=0}^n a_i X^i$. We denote this by M_ϕ.

With this notation then ϕ is itself an $A[X]$-module endomorphism of M_ϕ.

Proof. Let $P, Q \in A[X]$ be two polynomials. Then we need to show that

$$(PQ)(\phi) = P(\phi) \circ Q(\phi)$$

which amounts to saying that $A[X] \to \mathrm{End}_A(M)$ is an A-algebra homomorphism.

The final statement follows because X commutes with any polynomial in $A[X]$. More explicitly

$$\phi(P \cdot m) = \phi(P(\phi)(m)) = (XP)(\phi)(m) = (PX)(\phi)(m) = P(\phi)(\phi(m)) = P \cdot (\phi(m))$$

∎

Proposition 3.16.6
Let M be an A-module and $\phi \in \mathrm{End}_A(M)$. Then is an $A[X]$-linear map

$$\begin{aligned}\tilde{\phi} : A[X] \otimes_A M &\to M_\phi \\ P \otimes m &\to P(\phi)(m)\end{aligned}$$

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and similarly

$$1 \otimes \phi : A[X] \otimes_A M \to A[X] \otimes_A M$$
$$P \otimes m \to P \otimes (\phi(m))$$

Proof. By the universal property $\tilde{\phi}$ exists and is A-linear. To show it is $A[X]$-linear observe that

$$\tilde{\phi}(Q(P \otimes m)) = \tilde{\phi}((QP) \otimes m) = (QP)(\phi)(m) = Q(\phi)(P(\phi)(m)) = Q \cdot P(\phi)(m) = Q \cdot \tilde{\phi}(P \otimes m)$$

and so the result follows by linearity.

The second exists by (3.4.17). ∎

Proposition 3.16.7
Let M be a finite free A-module and $\phi \in \mathrm{End}_A(M)$. Then the characteristic polynomial $P_\phi(X)$ equals the determinant of the $A[X]$-linear map

$$(X - 1 \otimes \phi) : A[X] \otimes_A M \to A[X] \otimes_A M$$

Furthermore

$$P_\phi(X) = X^n + \sum_{j=1}^{n} (-1)^j \operatorname{Tr}(L_a^j(\phi)) X^{n-j}$$

and in particular

$$P_\phi(X) = X^n - \operatorname{Tr}(\phi) X^{n-1} + ... + (-1)^n \det(\phi)$$

Proof. Let $\mathcal{B} := \{v_1, ..., v_n\}$ be a basis for M, then by (3.4.24) $\{1 \otimes v_1, ..., 1 \otimes v_n\}$ is an $A[X]$-basis for $A[X] \otimes_A M$. Let $E = [\phi]$ be the matrix associated to the basis $\mathcal{B}$ then by definition

$$\phi(v_i) = \sum_{j=1}^{n} E_{ji} v_j$$

and

$$(X - 1 \otimes \phi)(1 \otimes v_i) = X \otimes v_i - 1 \otimes \phi(v_i) = X(1 \otimes v_i) - \sum_{j=1}^{n} E_{ji}(1 \otimes v_j)$$

This shows that

$$[X - 1 \otimes \phi]_{ij} = X\delta_{ij} - E_{ij}$$

and so $P_\phi(X) = \det(XI_n - E) = \det(X - 1 \otimes \phi)$ as required.

Then by (3.8.32) we have

$$P_\phi(X) = X^n + \sum_{j=1}^{n} (-1)^j \operatorname{Tr}(L_a^j(1 \otimes \phi)) X^{n-j}$$

The matrix of $1 \otimes \phi$ coincides with that of ϕ and so we deduce the second relationship. The final statement follows from (...). ∎

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3.17 Fields and Galois Theory

3.17.1 Prime Fields

Recall that for p prime the quotient ring $\mathbb{Z}/p\mathbb{Z}$ is a field (3.14.3), (3.15.22), (3.3.59).

Definition 3.17.1 (Finite field of order p)
Denote by $\mathbb{F}_p$ the field $\mathbb{Z}/p\mathbb{Z}$ of order p.

Definition 3.17.2 (Rational Integers)
*We denote by $\mathbb{Q}$ the field of **rational numbers** defined to be the quotient field of the integers,* $\mathrm{Frac}(\mathbb{Z})$ *(see (3.5.6)).*

Definition 3.17.3 (Prime Field)
*We say that a field k is a **prime field** if it is isomorphic to one of $\mathbb{F}_p$ or $\mathbb{Q}$.*

Note none of these fields are mutually isomorphic by considering cardinalities.

Proposition 3.17.4 (Prime Subfield Exists)
*Let k be a field. Then k contains a **prime subfield**. It is the smallest subfield contained in k.*

Proof. By (...) there is a unique ring homomorphism

$$\phi : \mathbb{Z} \to k$$

As k is not the zero-ring then $\phi(1) = 1 \neq 0$ and $\ker(\phi)$ is a proper ideal. Then $\mathrm{Im}(\phi)$ is an integral domain by (3.3.9). By (3.3.57) $\mathbb{Z}/\ker(\phi) \cong \mathrm{Im}(\phi)$ and therefore by (3.3.59) $\ker(\phi) =: \mathfrak{p}$ is a prime ideal.

By (3.14.3) the ideal $\mathfrak{p}$ is principal. Suppose $\mathfrak{p} = (0)$. By (3.5.5) ϕ extends to a homomorphism $\phi' : \mathbb{Q} \to k$ whose kernel is zero and therefore injective (...). In particular $\mathrm{Im}(\phi')$ is a prime subfield being isomorphic to $\mathbb{Q}$.

Otherwise by (3.15.9) $\mathfrak{p} = (p)$ for p a prime number and we have already observed that $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \cong \mathrm{Im}(\phi)$ is a prime subfield.

Let k' be any subfield of k, then by induction we may show that $\mathrm{Im}(\phi) \subset k'$. In the case $\ker(\phi) = \{0\}$ then we may also show $\mathrm{Im}(\phi') \subset k'$. Therefore k' contains the prime subfield, whence it is the smallest subfield. ∎

Definition 3.17.5 (Characteristic)
*Let k be a field. Then we define the **characteristic** of k to be p if the prime subfield is isomorphic to $\mathbb{F}_p$ or 0 if the prime subfield is isomorphic to $\mathbb{Q}$. This is denoted* $\mathrm{char}(k)$.

*If A is a k-algebra then we define the characteristic of A to be that of k. We also define the **characteristic exponent** of A to be*

$$\begin{cases} 1 & \text{if } \mathrm{char}(A) = 0 \\ p & \text{if } \mathrm{char}(A) = p \end{cases}$$

Proposition 3.17.6 (Frobenius Homomorphism)
Let A be a k-algebra and suppose $\mathrm{char}(A) = p$. *Then the mapping*

$$a \to a^p$$

*is a ring homomorphism which we call the **Frobenius map**. In particular*

$$(a+b)^p = a^p + b^p \quad \forall a, b \in A$$

Definition 3.17.7
*Let A be a k-algebra with characteristic exponent p. We say that A is **perfect** if A is reduced and $A^p = A$. When $p > 1$ this is equivalent to the Frobenius homomorphism being an isomorphism.*

3.17.2 Separability

We consider separability of a polynomial, and how it behaves under field extension.

Definition 3.17.8 (Splitting Polynomial)
*Let K/k be a field extension and $f \in k[X]$. By abuse of notation we may also identify f with its image in $K[X]$. We say a polynomial f **splits completely** in K if the irreducible factorization of f in $K[X]$ is*

$$f = \ell(f) \prod_{i=1}^{n} (X - \alpha_i)^{r_i}$$

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where α_i are the distinct roots of f in K and $r_i := v_{X-\alpha_i}(f)$ are the multiplicities. Equivalently f splits in K if

$$p \in K[X] \text{ irreducible} \wedge \deg(p) > 1 \implies v_p(f) = 0$$

Observe that the number of roots counting multiplicities is $\deg(f)$

$$\deg(f) = \sum_{i=1}^{n} v_{X-\alpha_i}(f)$$

Corollary 3.17.9
Let K/k be a field extension and $f, g \in k[X]$. Suppose $g \mid f$ (in $k[X]$) and f splits completely in K. Then so does g.

Proof. As $k[X] \subset K[X]$, $g \mid f$ in $K[X]$ too. By assumption the irreducible factorization of f in $K[X]$ consists of polynomials of degree 1. Consider the irreducible factorization of $g = \prod_{i=1}^{n} g_i$ in $K[X]$, then by unique factorization (3.15.33) (applied to $K[X]$) each g_i must appear in the factorization of f, that is to say g splits completely in K. ∎

Lemma 3.17.10 (Coprime Polynomials have no Common Roots)
Let K/k be a field extension and $f, g \in k[X]$ with $f \neq 0$ splitting completely in K. Then f and g are coprime if and only if they have no common root in K. In particular this holds whenever f and g both split completely in K.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose f, g are coprime, so $af + bg = 1$ for some $a, b \in k[X]$. If $\alpha \in K$ were a common root then evaluating at α gives $0 = 1$, a contradiction.

Conversely suppose f, g are not coprime. As $k[X]$ is a PID (3.15.32), by (3.14.5) they have a common non-invertible divisor d, and $\deg(d) > 0$ as the invertible elements of $k[X]$ are the nonzero constants (3.7.16). Since $d \mid f$ and f splits completely in K, so does d (3.17.9); as $\deg(d) > 0$ it has a root $\alpha \in K$, which is a common root of f and g as d divides both. ∎

We can provide a partial converse to (3.15.39) by working in a large enough extension field

Proposition 3.17.11 (Separability)
Let K/k be a field extension and $f \in k[X]$ a polynomial which splits completely in K. Then TFAE

a) f is separable
b) f has no multiple roots in K
c) f and f' have no common roots in K
d) f has $\deg(f)$ distinct roots in K

Proof. Using the formula

$$\deg(f) = \sum_{i=1}^{n} v_{X-\alpha_i}(f)$$

we see easily that $d) \iff b)$. By (3.15.37) $b) \iff c)$. As f splits completely in K, (3.17.10) shows that f and f' are coprime, that is f is separable (3.15.38), if and only if they have no common root in K: $a) \iff c)$. ∎

3.17.3 Field Extensions

Definition 3.17.12 (Field Extension)
Let k be a field. A field extension K/k is a k-algebra K which is also a field. Every field K is an extension over its prime subfield (or, over $\mathbb{Q}$ or $\mathbb{F}_p$).

We typically denote the structural morphism by $i_{kK} : k \to K$, and it is automatically injective (3.3.61). We may write $(K/k, i_{kK})$ if we need to stress the relevance of the structural morphism to the argument.

These objects form a category $\mathbf{Field}_k$ *in the obvious way. The morphisms may be called k-embeddings and we denote them by*

$$\mathrm{Mor}_k(K, L) := \{\psi : K \to L \mid \psi \circ i_{kK} = i_{kL}\}$$

and the set of automorphisms by

$$\mathrm{Aut}(K/k).$$

Observe every extension K/k may be viewed as a k-vector space so we define the degree of an extension field to be the vector space dimension

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$$[K:k] := \dim_k K$$

Definition 3.17.13 (Finite field extension)
A field extension K/k is finite if $[K:k] < \infty$

Definition 3.17.14 (Tower of Field Extensions)
We may also consider a "tower" of extensions

$$K_n/\cdots/K_0 = k$$

with embeddings $i_{K_iK_{i+1}} : K_i \to K_{i+1}$, with the picture that these usually correspond to inclusions. We may consider an extension K_i/K_j for $j < i$. Typically if we have a family of morphisms

$$\sigma_i : K_i \to M$$

they would commute with these embeddings. In particular we may abuse notation by defining $\sigma_i|_{K_j} = \sigma_i \circ i_{K_{i-1}K_i} \circ \cdots \circ i_{K_jK_{j+1}}$.

Proposition 3.17.15
Let K/k be a finite field extension and V a finite-dimensional vector space over K. Then V is naturally a finite-dimensional k-vector space with

$$\dim_k V = [K:k] \dim_K V$$

More precisely if K/k has basis $\{\lambda_1, ..., \lambda_r\}$ and V has K-basis $\{v_1, ..., v_n\}$, then the following is a k-basis for V

$$\left\{\lambda_i v_j\right\}_{i,j}.$$

Proposition 3.17.16
Let L/K and K/k be two finite extensions with basis $\{l_1, ..., l_n\}$ and $\{k_1, ..., k_m\}$. Then L/k has basis $\left\{l_i k_j\right\}_{i,j}$. In particular

$$[L:k] = [L:K][K:k]$$

Corollary 3.17.17
Let $K = K_n/\cdots/K_0 = k$ be a tower of finite extensions then

$$[K:k] = \prod_{i=1}^{n} [K_i : K_{i-1}]$$

Lemma 3.17.18
Let K/k be a field extension and $f, g \in k[X]$. Then $g \mid f$ in $k[X]$ if and only if $i_{kK}(g) \mid i_{kK}(f)$.

Proof. One implication is obvious. For the converse we assume wlog that $k \subset K$ and suppose $f = gh$ for $h \in K[X]$. We may apply the division algorithm (3.7.17) in $k[X]$ to find $f = gq + r$ for $q, r \in k[X]$ and $\deg(r) < \deg(g)$. Then $r = g(h - q)$ and comparing degrees we conclude that $h = q$ and $r = 0$. This shows $f = gq$ and $g \mid f$ as elements of $k[X]$. ∎

Definition 3.17.19 (Evaluation homomorphism)
Let K/k be a field extension and $\alpha \in K$. There is a canonical homomorphism

$$\mathrm{ev}_\alpha : k[X] \to K$$

$$\sum_{i=0}^{n} a_i X^i \to \sum_{i=0}^{n} i_{kK}(a_i)\alpha^i$$

which we write as $f(\alpha)$. We say $\alpha \in K$ is a root of $f(X)$ if $f(\alpha) = 0$.

Proposition 3.17.20 (Morphisms commute with evaluation)
Let $\sigma : K/k \to L/k$ be a morphism of field extensions then

$$\sigma(p(\alpha)) = p(\sigma(\alpha))$$

for all $p \in k[X]$. In particular α is a root of $p \iff \sigma(\alpha)$ is a root of p.

Proof. This is just a specific case of (3.7.9). The last statement is obvious, because σ is injective (3.3.61). ∎

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Definition 3.17.21 (Subalgebra generated by a set)
Let K/k be a field extension and $S \subset K$. Define

$$k[S] := \bigcap_{\substack{A \subset K/k \\ S \subset A}} A$$

where the intersection is taken over all k-subalgebras. It is the smallest k-subalgebra of K/k.

Proposition 3.17.22 (Subalgebra generated by directed family)
Let K/k be a field extension and $\{S_i\}_{i \in I}$ a family of directed subsets of K. Then

$$k\left[\bigcup_{i \in I} S_i\right] = \bigcup_{i \in I} k[S_i]$$

In particular for any set S we have

$$k[S] = \bigcup_{\substack{S' \subset S \\ S' \text{ finite}}} k[S']$$

Proposition 3.17.23 (Subalgebra generated by a set)
Let K/k be a field extension and $S \subset K$ a finite subset. Write $S = \{\alpha_1, ..., \alpha_n\}$ then

$$k[S] = \{p(\alpha_1, ..., \alpha_n) \mid p \in k[X_1, ..., X_n]\} = \mathrm{Im}(\mathrm{ev}_\alpha)$$

Lemma 3.17.24 (Trivial result)
For $S, T \subset K/k$ finite

- $S \subset T \implies k[S] \subseteq k[T]$
- $k[S][T] = k[S \cup T]$

Definition 3.17.25
Let K/k be a field extension and $S \subset K$. Define

$$k(S) := \bigcap_{\substack{K' \subset K/k \\ S \subset K'}} K'$$

It is the smallest subfield of K/k containing S.

Proposition 3.17.26
Let K/k be a field extension and $\{S_i\}_{i \in I}$ be a family of directed subsets of K. Then

$$k\left(\bigcup_{i \in I} S_i\right) = \bigcup_{i \in I} k(S_i)$$

In particular for any set S we have

$$k(S) = \bigcup_{\substack{S' \subset S \\ \text{finite}}} k(S')$$

Proposition 3.17.27
Let K/k be a field extension and $S \subset K$ be a finite subset. Write $S = \{\alpha_1, ..., \alpha_n\}$ then

$$k(S) := \left\{ \frac{p(\alpha_1, ..., \alpha_n)}{q(\alpha_1, ..., \alpha_n)} \;\middle|\; p, q \in k[X_1, ..., X_n], \quad q(\alpha_1, ..., \alpha_n) \neq 0 \right\}$$

Lemma 3.17.28 (Trivial result)
For $S, T \subset K/k$

- $S \subset T \implies k(S) \subseteq k(T)$
- $k(S)(T) = k(S \cup T)$

Lemma 3.17.29
If $S \subset K$ and $k[S]$ is a field then $k[S] = k(S)$

Proof. Generically $k[S] \subset k(S)$ from the definition. As $k[S]$ is a field then $k(S) \subset k[S]$. ∎

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Lemma 3.17.30 (Image of f.g. field extension)
Let K/k be a field extension and $S \subset K$ a subset. If $\sigma : K/k \to L/k$ is a morphism them

$$\sigma(k(S)) = k(\sigma(S))$$

Proposition 3.17.31 (Uniqueness of morphisms on a generating set)
Let K/k be a field extension and $S \subset K$. If $\sigma, \sigma' : k(S)/k \to L/k$ are morphisms of field extensions such that $\sigma|_S = \sigma'|_S$. Then $\sigma = \sigma'$.

Definition 3.17.32 (Simple (Algebraic) Extension)
*A field extension K/k is **simple** if $K = k(\{\alpha\}) =: k(\alpha)$ for some $\alpha \in K$. It is a **simple algebraic** extension if α is also algebraic over k.*

Definition 3.17.33 (Algebraic Element)
*We say an element $\alpha \in K/k$ is **algebraic** if it is a root of a polynomial $f \in k[X]$ (i.e. α is integral, since we can always ensure f is monic). Otherwise we say that $x \in K$ is **transcendental**.*

*We say K/k is an **algebraic extension** if every element $\alpha \in K$ is algebraic over k.*

Proposition 3.17.34 (Finite ⇒ algebraic)
A finite extension K/k is algebraic.

Proof. Suppose $n = \dim_k K$. The set $\{1, \alpha, \alpha^2, ..., \alpha^n\}$ is linearly dependent by (2.3.13). Therefore there is a non-zero polynomial with α as a root. ∎

Proposition 3.17.35 (Endomorphisms are automorphisms)
Let $\sigma \in \mathrm{Mor}_k(K, K)$ be an endomorphism of an algebraic extension. Then it is an isomorphism. In other words

$$\mathrm{Mor}_k(K, K) = \mathrm{Aut}(K/k)$$

Proof. As field morphisms are injective (3.3.61) we only need to show that σ is surjective. Given $\alpha \in K$ let T denote the set of roots of $m_\alpha \in k[X]$ in K. Note by (3.15.35) T is finite. Further by (3.17.20) σ maps T to itself. Since σ is injective it is also surjective on T. In particular α is in the image of σ as required. ∎

3.17.4 Algebraic Extensions

Proposition 3.17.36 (Minimal Polynomial)
If $\alpha \in K/k$ is algebraic then there is a unique monic, irreducible polynomial $m_{\alpha,k}(X) \in k[X]$ such that $m_{\alpha,k}(\alpha) = 0$. This is called the minimal polynomial of α over k and $\big(m_{\alpha,k}\big) = \ker(\mathrm{ev}_\alpha)$.

In particular any polynomial $f(X) \in k[X]$ which has α as a root, satisfies $m_{\alpha,k}(X) \mid f(X)$.

Proof. Let $\mathfrak{a} = \ker(\mathrm{ev}_\alpha)$. Since $k[X]$ is a PID it is of the form $\big(m_{\alpha,k}\big)$. As α is algebraic it is non-zero. $m_{\alpha,k}(X)$ cannot be a constant, and therefore is not a unit.

We claim $m_{\alpha,k}$ is irreducible. If $m_{\alpha,k}(X) = p(X)q(X)$ then p, q are non-zero and either $p(\alpha) = 0$ or $q(\alpha) = 0$. If $p(\alpha) = 0$ then $m_{\alpha,k} \mid p$. As $p \mid m_{\alpha,k}$ by (3.15.30) $m_{\alpha,k} = \lambda p$. In particular $\deg\big(m_{\alpha,k}(X)\big) = \deg(p(X))$ so $\deg(q(X)) = 0$ and $q(X)$ is a unit (3.7.16). Therefore by definition $m_{\alpha,k}(X)$ is irreducible.

Dividing by the leading coefficient we may assume that this polynomial is monic. Suppose $m'(X)$ is another such irreducible monic polynomial. Then $m_\alpha \mid m'$. Since m_α is not a unit, by definition of irreducible $m' \sim m_\alpha$ whence $m' = \lambda m_\alpha$. Compare leading coefficients to find $\lambda = 1$ and $m' = m_\alpha$. ∎

Lemma 3.17.37
Let $K/E/k$ be extensions and $\alpha \in K$ algebraic over k. Then α is algebraic over E.

Definition 3.17.38 (Conjugate elements)
*Two elements $\alpha, \beta \in K$ are said to be **conjugate elements** if they have the same minimal polynomial.*

NB it's necessary and sufficient that $m_{\alpha,k}(\beta) = 0$.

Proposition 3.17.39
Let $\sigma : K/k \to L/k$ be a field morphism and $\alpha \in K$. Then $m_{\alpha,k}(X) = m_{\sigma(\alpha),k}(X)$.

Proof. This follows from (3.17.20). ∎

Given an irreducible polynomial $f \in k[X]$ it's possible to construct an extension field K/k which has at least one root, as follows.

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Proposition 3.17.40 (Construct simple extension)
Let $f \in k[X]$ be an irreducible polynomial. Then (f) is maximal and $K := k[X]/(f)$ is a field extension with canonical structural morphism. Define $\alpha := X + (f)$

- $f(\alpha) = 0$
- *$K = k(\alpha)$ is a simple field extension and $k(\alpha) = k[\alpha]$*
- *$m_\alpha = f/\ell(f)$ and $\deg(m_\alpha) = \deg(f) =: n$*
- *K is a finite-dimensional k-vector space with basis*

$$\{1, \alpha, ..., \alpha^{n-1}\}$$

.

Example 3.17.41
Take $k = \mathbb{R}$, $f(X) = X^2 + 1$, then $\mathbb{C}/\mathbb{R} = \mathbb{R}[i] = \mathbb{R}[X]/(X^2 + 1)$.

Proof. Consider the structural morphism $i : k \to k[X]$ and canonical surjective homomorphism

$$\pi : k[X] \to k[X]/(f)$$

and $\alpha = X + (f) = \pi(X)$. As $k[X]$ is a PID, f irreducible implies (f) maximal by (3.15.22) so K is a field by (3.3.59). The composition $\pi \circ i$ makes K into a k-algebra and hence a field extension. Furthermore π is then by definition a k -algebra homomorphism.

Since π is surjective every $\beta \in K$ is represented as $\pi(p(X)) \overset{(3.17.20)}{=} p(\pi(X)) = p(\alpha)$. By (3.17.23) we see $K = k[\alpha]$. Since K is a field then $K = k[\alpha] = k(\alpha)$ is simple by (3.17.29).

Similarly $f(\alpha) = f(\pi(X)) \overset{(3.17.20)}{=} \pi(f(X)) = 0$, so α is a root of f. By (3.15.30) $f/\ell(f)$ is irreducible and by uniqueness in (3.17.36) we have $m_\alpha = f/\ell(f)$.

Given $\beta = p(\alpha)$, the division algorithm (...) yields

$$p(X) = q(X)f(X) + r(X)$$

with $\deg(r) < \deg(f) = n$. Therefore $\beta = r(\alpha)$ and the given set is spanning. A non-trivial linear dependence yields a non-zero polynomial $g(X)$ such that $g(\alpha) = 0$ and $\deg(g) < \deg(f)$. But by definition of the minimal polynomial $m_\alpha \mid g$, which is a contradiction by comparing degrees. Therefore the given set is linearly independent and hence a basis. ∎

Conversely any simple algebraic extension is obtained in this way, as follows

Proposition 3.17.42 (Simple extension)
Let $k(\alpha)/k$ be a simple extension. Then there is a canonical isomorphism of k-algebras

$$k[X]/(m_\alpha) \longrightarrow k(\alpha)$$

under which $X + (m_\alpha) \to \alpha$. Further $k(\alpha)$ is a finite-dimensional vector space with basis

$$\{1, \alpha, ..., \alpha^{n-1}\}$$

where $n = \deg(m_\alpha) = [k(\alpha) : k]$ and $k(\alpha) = k[\alpha]$.

Proof. By (3.17.36), Definition (3.17.23) and (3.3.57) there is a canonical isomorphism $k[X]/(m_\alpha) \to k[\alpha]$ of k-algebras induced by the evaluation homomorphism $\mathrm{ev}_\alpha : k[X] \to K$. (3.17.40) shows that the image of this isomorphism, $k[\alpha]$, is a field, whence $k[\alpha] = k(\alpha)$ by (3.17.29). Since a k-algebra isomorphism is a-fortiori a k-vector space isomorphism it maps a basis to a basis. The result follows from (3.17.40) as the basis thus defined is the image of the basis in the proposition under the specified isomorphism. ∎

Definition 3.17.43 (Degree of an algebraic element)
Let K/k be an algebraic extension and $\alpha \in K$. Then define

$$\deg_k(\alpha) := \deg m_{\alpha,k} = [k(\alpha) : k]$$

We may show the following

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Proposition 3.17.44 (Finitely generated by algebraic $\implies$ finite and algebraic)
Let $K = k(\alpha_1, ..., \alpha_n)/k$ be a field extension such that α_i is algebraic. Then K/k is a finite algebraic extension. Furthermore

$$k[\alpha_1, ..., \alpha_n] = k(\alpha_1, ..., \alpha_n)$$

In particular a finitely-generated algebraic extension is finite.

Proof. We write $K_i = k(\alpha_1, ..., \alpha_i)$. Then we have a tower

$$K = K_n/.../K_0 = k$$

such that $K_i = K_{i-1}(\alpha_i)$ is a simple algebraic extension. By (3.17.42) K_i/K_{i-1} is finite. Therefore by (3.17.17) K/k is finite. By (3.17.34) it's also algebraic. For the second statement we may proceed inductively. Note we have

$$k[\alpha_1, ..., \alpha_{i+1}] \overset{(3.17.24)}{=} k[\alpha_1, ..., \alpha_i][\alpha_{i+1}] = k(\alpha_1, ..., \alpha_i)[\alpha_{i+1}] \overset{(3.17.42)}{=} k(\alpha_1, ..., \alpha_i)(\alpha_{i+1}) \overset{(3.17.28)}{=} k(\alpha_1, ..., \alpha_{i+1})$$

The second equality is simply the inductive hypothesis. ∎

Corollary 3.17.45
Let K/k be a field extension then the algebraic elements form a subfield.

Proof. For any two algebraic elements $\alpha, \beta \in K$ we have $k(\alpha, \beta)$ is an algebraic extension. ∎

The following is useful for reducing to cases of finite extensions where counting arguments work.

Lemma 3.17.46 (Reduce to finite extensions)
Let $K/E/k$ be a tower with E algebraic over k. For every $\alpha \in K$ algebraic over E, there is some subfield $E_0 \subset E$ such that

- *E_0/k is finite*
- *α is algebraic over E_0*
- *$m_{\alpha,E} = m_{\alpha,E_0}$*

Therefore α is algebraic over k iff it is algebraic over E and $m_{\alpha,E_0} \mid m_{\alpha,k}^{i_{kE}}$.

Proof. Suppose

$$m_{\alpha,E}(X) = a_0 + a_1 X + ...a_n X^n$$

Then define $E_0 = i_{kE}(k)(a_0, ..., a_n)$. By (3.17.44) E_0/k is finite. Clearly α is algebraic over E_0 as it is a root of $m_{\alpha,E}$. By (3.17.36) $m_{\alpha,E_0} \mid m_{\alpha,E}$ as elements of $E_0[X]$ and $m_{\alpha,E} \mid m_{\alpha,E_0}$. Therefore $m_{\alpha,E_0} = m_{\alpha,E}$.

By (3.17.42) $E_0(\alpha)/E$ is finite, therefore $E_0(\alpha)/k$ is finite. By (3.17.34) $E_0(\alpha)/k$ is algebraic, whence α is algebraic over k. The last statement follows from (3.17.36) again. ∎

Lemma 3.17.47 (Finite-dimensional Domains are Fields)
Let A be a finite-dimensional k-algebra with no zero-divisors. Then A is a field.

Proof. By assumption multiplication by a non-zero element x is injective and k-linear. Therefore by (3.3.137) it is surjective and x has an inverse as required. ∎

Lemma 3.17.48 (Compositum of Finite Subextensions)
*Let E/k be a field extension and $E_1, ..., E_r$ subfields of E containing k. The **compositum** $k(E_1, ..., E_r)$ is the smallest subfield of E containing every E_i (the intersection of all such subfields). If each E_i/k is finite then so is $k(E_1, ..., E_r)/k$.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

We proceed by induction on r, the case $r = 1$ being trivial. Let $F := k(E_1, ..., E_{r-1})$, finite over k by induction, so that $k(E_1, ..., E_r) = F(E_r)$. Let $b_1, ..., b_s$ be a k-basis of E_r. Then $F(E_r) = F(b_1, ..., b_s)$, and each b_j is algebraic over k (3.17.34) and hence over F. Therefore $F(E_r)/F$ is finite (3.17.44) and so $F(E_r)/k$ is finite (3.17.17). ∎

Corollary 3.17.49 (Algebraic Extensions are Directed Unions of Finite Subextensions)
Let E/k be algebraic. Then the finite subextensions of E/k are directed under inclusion, and E is their union.

Proof. Given $\alpha \in E$, taking $K = E$ in (3.17.46) (with α trivially algebraic over E) gives a finite subextension $E_0 \subset E$ over which α is algebraic; then $E_0(\alpha)/E_0$ is finite (3.17.42), hence $E_0(\alpha)/k$ is finite too, a finite subextension of E containing α. Hence E is the union of its finite subextensions. Given two finite subextensions $E_1, E_2 \subset E$,

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their compositum $k(E_1, E_2) \subset E$ is finite (3.17.48) and contains both, so the finite subextensions are directed under inclusion. ∎

Corollary 3.17.50
K/E and E/k are both algebraic if and only if K/k is.

Proof. One direction is (3.17.37). The converse follows from the previous result. ∎

We may prove the first lifting theorem

Proposition 3.17.51 (Lifting to simple extensions)
Let $k(\alpha)/k$ be a simple algebraic extension and L/k a field extension such that $m_{\alpha,k}$ has a root in L. Then there exists a morphism $\sigma : k(\alpha)/k \to L/k$.

More precisely there is a bijective mapping

$$\begin{aligned} \mathrm{Mor}_k(k(\alpha), L) &\longrightarrow \left\{\beta \in L \mid m_{\alpha,k}(\beta) = 0\right\} \\ \sigma &\to \ \sigma(\alpha) \end{aligned}$$

and $\sigma(k(\alpha)) = k(\sigma(\alpha))$. In particular if $m_{\alpha,k}$ is separable and splits completely in L then there are precisely $\deg(m_\alpha) \overset{(3.17.42)}{=} [k(\alpha) : k]$ such extensions.

Proof. Observe $m_{\alpha,k}(\sigma(\alpha)) \overset{(3.17.20)}{=} \sigma\big(m_{\alpha,k}(\alpha)\big) = 0$. Therefore the mapping is well-defined. By (3.17.31) it is injective. We claim it is also surjective. By (3.17.42) there is a k-algebra isomorphism

$$k[X]/\big(m_{\alpha,k}\big) \longrightarrow k(\alpha)$$

Similarly for $\beta \in T$ there is a k-algebra isomorphism

$$k[X]/\big(m_{\beta,k}\big) \longrightarrow k(\beta)$$

We are done if $m_{\alpha,k} = m_{\beta,k}$. But $m_{\alpha,k}$ is monic, irreducible and has β as a root. So this follows from uniqueness of the minimal polynomial in (3.17.36). The final statement follows from (3.17.11) ∎

We may use this to generalize to arbitrary extensions, but we require that the minimal polynomials split completely in order for the inductive step to work.

Proposition 3.17.52 (Generic Lifting Theorem)
Let K/k be an algebraic field extension such that $K = k\big(\{\alpha_i\}_{i\in I}\big)$ and L/k a field extension such that $m_{\alpha_i,k}(X)$ splits completely in L for all $i \in I$.

Then there exists a morphism $\sigma : K/k \to L/k$.

Furthermore given $\alpha \in K$ and $\beta \in L$ any root of $m_{\alpha,k}(X)$ we may choose σ such that $\sigma(\alpha) = \beta$.

Proof. If K/k is finite then we may proceed by induction on $[K : k]$, using (3.17.51) and applying a similar argument to below.

For the general case we may consider the poset of morphisms $\sigma : K'/k \to L/k$ for subfields $K'/k \subset K/k$ ordered by consistency. It is non-empty by considering $K' = i_{kK}(k)$. By Zorn's Lemma it has a maximal element, (K', σ'). It's enough to show that $K' = K$.

If $\alpha_i \in K'$ for all $i \in I$ then $K' = K$ and we are done. Otherwise choose $\alpha = \alpha_i \notin K'$. By (...) $m_{\alpha,K'}(X) \mid m_{\alpha,k}(X)$. By (3.17.9) $m_{\alpha,K'}(X)$ splits in L (because $m_{\alpha,k}(X)$ does). Therefore by (3.17.51) there is a morphism $\sigma : K'(\alpha)/K' \to (L/K', \sigma')$. Note that by definition $\sigma|_{K'} = \sigma'$ and $K' \subsetneq K'(\alpha)$, contradicting maximality.

For the final part we may consider the poset consisting only of morphisms such that $\sigma(\alpha) = \beta$. By (3.17.51) the poset is non-empty, and the same argument works. ∎

3.17.5 Galois Theory Summary

Definition 3.17.53 (Separable, Normal and Galois)
Let K/k be an algebraic extension. We say that K/k is

- ***Normal*** *if every minimal polynomial $m_{\alpha,k} \in k[X]$ splits completely in K (iff every irreducible polynomial $f \in k[X]$ with at least one root in K splits completely in K)*
- ***Separable*** *if every minimal polynomial $m_{\alpha,k} \in k[X]$ is separable.*

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- ***Galois*** *if it is both normal and separable (iff* $m_{\alpha,k}$ *has* $\deg(m_{\alpha,k})$ *distinct roots in* K*, see (3.17.11)).*

In the case of a Galois extension we denote the group of automorphisms by $\mathrm{Gal}(K/k)$.

To summarize the main results

a) The group of automorphism of a normal extension K/k acts transitively on the roots of a given irreducible polynomial.
b) For K/k finite we have $\#\operatorname{Aut}(K/k) \leq [K:k]$ with equality if and only if K/k is Galois.
c) An algebraic extension K/k is automatically separable whenever either $\operatorname{char}(k) = 0$ or k is finite.
d) When K/k is finite and Galois then we have an order-reversing bijection between subfields and subgroups

$$\begin{aligned} \{H \leq \mathrm{Gal}(K/k)\} &\longleftrightarrow \{F \subseteq K\} \\ H &\longrightarrow K^H := \{x \in K \mid h(x) = x \quad \forall h \in H\} \\ \mathrm{Gal}(K/F) &\longleftarrow F \end{aligned}$$

3.17.6 Splitting Fields and Algebraic Closure

In this section we discuss splitting fields, which are the "smallest" extensions in which a given set of polynomials split completely. The fundamental result is that splitting fields are precisely the Normal extensions. Further we discuss the algebraic closure, in which every polynomial splits and in which every algebraic extension (normal or otherwise) may be embedded.

Definition 3.17.54 (Splitting field)
Let $S \subset k[X]$ *a family of polynomials. We say that* K/k *is a* ***splitting field*** *for* S *if*

- *Every polynomial* $f \in S$ *splits completely in* K
- K *is generated by the roots of all the polynomials in* S

Note that by (3.17.44) K/k *is necessarily algebraic, and if* S *is finite then so is* K/k.

Definition 3.17.55 (Set of roots)
Let K/k *be a field extension and* $f \in k[X]$*. Then define*

$$T_{f,K} := \{\beta \in K \mid f(\beta) = 0\}$$

Proposition 3.17.56 (Splitting field is minimal)
Let $S \subset k[X]$ *be a family of polynomials which split completely in* K/k*. Then the following are equivalent*

- K *is generated by the roots of every polynomial* $f \in S$*, that is*

$$K = k\left(\bigcup_{f \in S} T_{f,K}\right)$$

- *Any subfield* $K' \subset K$ *in which* S *splits completely is equal to* K

Proof. For all $f \in S$ there is a factorisation

$$f = \prod_{\alpha \in T_{f_i,K}} (X - \alpha)$$

Suppose $K = k\left(\bigcup_{f \in S} T_{f,K}\right)$ and let K' be a subfield in which all $f \in S$ split completely. Then by unique factorization in $K[X]$ we have $T_{f,K} \subset K'$ for all $f \in S$ and therefore $K' = K$.

Conversely it's clear that S splits completely in $k\left(\bigcup_{f \in S} T_{f,K}\right)$, therefore by hypothesis this equals K. ∎

Lemma 3.17.57
Let $\sigma : K/k \to L/k$ *be a morphism and* $f(X) \in k[X]$ *a polynomial. Then*

- σ *induces an injective map on the roots* $T_{f,K} \to T_{f,L}$
- f *splits completely in* K *iff* f *splits completely in* $\sigma(K)$*. In this case the above map is a bijection*

Proposition 3.17.58 (Image of a splitting field is fixed)
Let K/k *be a splitting field for* S *and* $\sigma : K/k \to L/k$ *a morphism. Then* S *splits completely in* L*. Any such* σ *satisfies*

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$$\sigma(K) = k\left(\bigcup_{f\in S} T_{f,L}\right)$$

Proof. Clearly by (3.17.57) S splits completely in L.

By the same result σ induces a bijection $T_{f,K} \longleftrightarrow T_{f,L}$. Therefore $\sigma(K) = \sigma\left(k\left(\bigcup_{f\in S} T_{f,K}\right)\right) = k\left(\sigma\left(\bigcup_{f\in S} T_{f,K}\right)\right) = k\left(\bigcup_{f\in S} T_{f,L}\right)$ by (3.17.30). ∎

Proposition 3.17.59 (Uniqueness of Splitting Fields)
Let $S \subset k[X]$ be a family of polynomials. Let K/k be a splitting field for S and L/k an extension in which S splits completely.

Then there exists a morphism $\sigma : K/k \to L/k$. Let $\alpha \in K$ and $\beta \in L$ be conjugate elements, then we may choose σ such that $\sigma(\alpha) = \beta$.

Furthermore any two splitting fields are isomorphic.

Proof. By assumption K is generated by the roots α_{ij} of $f_i \in S$. For each α_{ij} we therefore have $m_{\alpha_{ij},k}(X) \mid f_i(X)$ and $m_{\alpha_{ij},k}(X)$ splits completely in L by (3.17.9). Therefore the morphism $\sigma : K/k \to L/k$ exists by (3.17.52).

Note by (3.17.58) S splits in $\sigma(K) = k\left(\bigcup_{f\in S} T_{f,L}\right)$. If L is also a splitting field for S then $L = \sigma(K)$ by (3.17.56) and therefore σ is an isomorphism as required. ∎

Proposition 3.17.60 (Algebraically Closed)
A field M is algebraically closed if one of the following equivalent conditions holds

- *Every algebraic extension M'/M is trivial*
- *Every non-constant polynomial in $M[X]$ has a root in M*
- *Every non-constant polynomial in $M[X]$ splits in M*

NB in this case M is also normal.

Definition 3.17.61 (Algebraic Closure)
*An **algebraic closure** $\overline{k}$ of k is a field extension $\overline{k}/k$ which is algebraic and for which $\overline{k}$ is algebraically closed.*

Proposition 3.17.62 (Existence of Algebraic Closure)
Given a field k there exists an algebraic closure $\overline{k}/k$

Proposition 3.17.63 (Algebraic extensions embed into Algebraic Closure)
Let K/k be an algebraic extension and M/k be field such that every polynomial $f \in k[X]$ splits completely then there exists a morphism $\sigma : K/k \to M/k$.

In particular this holds when M is algebraically closed.

Proof. A straightforward application of (3.17.52) since every $m_{\alpha,k}(X)$ splits in M. ∎

Corollary 3.17.64 (Uniqueness of algebraic closure)
An algebraic closure $\overline{k}$ of k is unique up to (non-unique) isomorphism.

More generally we may show the existence of smaller splitting fields

Proposition 3.17.65 (Existence of Splitting Field)
Given a field k and family of polynomials $S \subset k[X]$ then there exists a splitting field K.

When $S = \{f\}$ then this can be chosen such that $[K : k] \leq n!$ where $n = \deg(f)$.

Proof. We may take the subfield of $\overline{k}$ generated by the roots of polynomials in S.

In the case S is finite it is possible to avoid the use of $\overline{k}$. First reduce to the case of a single polynomial $S = \{f\}$ and proceed by induction on $\deg(f)$. The inductive step may be demonstrated using (3.17.40). ∎

Remark 3.17.66
Note if K/k is an algebraic extension then (3.17.63) shows that we may construct an embedding $K \to \overline{k}$ commuting with $k \to \overline{k}$.

In general given a tower of algebraic extensions

$$K = k_n/.../k_0 = k$$

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we will assume the existence of compatible embeddings $i_{k_i} : k_i \to \overline{k}$ *such that* $i_{k_{i+1}} \circ i_{k_i,k_{i+1}} = i_{k_i}$.

3.17.7 Normal Extensions

Recall that an algebraic extension K/k is normal if all minimal polynomials split completely. They are in some sense "closed". Furthermore $\overline{k}/k$ is clearly normal and results about $\overline{k}$ can often be generalized to normal fields L/k. We also show that an extension is normal iff it is a splitting field.

Lemma 3.17.67
Let $L/K/k$ *be a tower of algebraic extensions and* $\alpha \in L$. *If* $m_{\alpha,k}(X)$ *splits completely in* L *so does* $m_{\alpha,K}(X)$. *In particular*

$$L/k \text{ normal} \implies L/K \text{ normal}$$

Proof. Note $m_{\alpha,K}(X) \mid m_{\alpha,k}^{i_{kK}}(X)$ as elements of $K[X]$ by (3.17.36). Apply i_{KL} and then we may use (3.17.9). ∎

Proposition 3.17.68 (Conjugate elements in Normal Extensions)
Let L/k *be a* normal *extension (e.g.* $L = \overline{k}$*) and* $\alpha, \beta \in L$ *elements with the same minimal polynomial* $m_\alpha(X) = m_\beta(X)$. *Then there exists* $\sigma \in \mathrm{Aut}(L/k)$ *such that*

$$\sigma(\alpha) = \beta$$

Proof. Apply (3.17.52) with $K = L$. ∎

Proposition 3.17.69 (Normal Criteria)
Let $L/K/k$ *be a tower of extensions such that* L/k *is normal (e.g.* $L = \overline{k}$*). Then the following are equivalent*

- ***NOR1****: For any* $\sigma \in \mathrm{Mor}_k(K, L)$ *we have* $\sigma(K) = i_{KL}(K)$.
- ***NOR2****:* K/k *is the splitting field of some family of polynomials* $f_i \in k[X]$.
- ***NOR3****:* K/k *is normal*

Proof. Clearly $3 \implies 2$, for K is the splitting field of all the minimal polynomials of elements in K.

$2 \implies 1$. This is (3.17.58).

$1 \implies 3$. Consider any $\alpha \in K$ with minimal polynomial $m_{\alpha,k}(X)$. By definition $m_{\alpha,k}(X)$ splits completely in L because it has a root $\alpha_1 = i_{KL}(\alpha)$. Denote the roots by $\alpha_1, ..., \alpha_r$. By (3.17.68) there is $\sigma_j \in \mathrm{Aut}(L/k)$ such that $\sigma_j(\alpha_1) = \alpha_j$. By hypothesis we have $\alpha_j \in (\sigma_j \circ i_{KL})(K) = i_{KL}(K)$ whence there exists $\alpha_j' \in K$ such that $i_{KL}(\alpha_j') = \alpha_j$. By (3.17.57) $m_{\alpha,k}(X)$ splits completely in K. Therefore K/k is normal as required. ∎

Corollary 3.17.70 (Splitting fields are normal)
An algebraic extension K/k *is normal if and only if it is a splitting field.*

Proof. We may apply the previous Proposition with $L = \overline{k}$.

We may prove a splitting field is normal more directly (without recourse to $\overline{k}$ or Zorn's Lemma in the finite case). Suppose K/k is a splitting field for $S \subset k[X]$. Consider $\alpha \in K$ with minimal polynomial $m_{\alpha,k}(X)$. Let $(L/K, i_{KL})$ be a splitting field for $m_{\alpha,k}(X)$ (as a polynomial in $K[X]$, NB may not be irreducible).

Let $\beta \in L$ be another root of $m_{\alpha,k}(X)$. Observe that S splits in L, so by (3.17.59) there is a morphism $\sigma : K/k \to L/k$ with $\sigma(\alpha) = \beta$. By (3.17.58) we have $i_{KL}(K) = \sigma(K)$ whence $\beta \in i_{KL}(K)$. As β was an arbitrary root of $m_{\alpha,k}(X)$ we see it splits completely in $i_{KL}(K)$. Finally by (3.17.57) $m_{\alpha,k}(X)$ splits completely in K. As α was arbitrary then K/k is normal. ∎

Proposition 3.17.71 (Extension to normal overfield)
Let $L/K/k$ *be a tower of algebraic field extensions with* L/k *normal (e.g.* $L = \overline{k}$*) then there is a canonical surjection*

$$\begin{aligned} \mathrm{Mor}_k(i_{KL}, L) : \mathrm{Aut}(L/k) &\to \mathrm{Mor}_k(K, L) \\ \sigma &\to \sigma \circ i_{KL} \end{aligned}$$

When i_{KL} *is inclusion then this is simply the restriction to* K. *The kernel is precisely* $\mathrm{Aut}(L/K)$.

Proof. Given $\tilde{\sigma} \in \mathrm{Mor}_k(K, L)$, apply (3.17.52) to construct a morphism $\sigma : (L/K, i_{KL}) \to (L/K, \tilde{\sigma})$. The hypotheses apply because the minimal polynomial $m_{\alpha,K}(X)$ with respect to either extension divides the minimal polynomial $m_{\alpha,k}^{i_{kK}}(X)$ which by assumption splits completely in L. By (3.17.35) it is an automorphism. Furthermore

$$\sigma \circ i_{kL} = \sigma \circ i_{KL} \circ i_{kK} = \tilde{\sigma} \circ i_{kK} = i_{kL}$$

whence $\sigma \in \mathrm{Aut}(L/k)$ as required. ∎

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Corollary 3.17.72 (Lifting inside normal overfield)
Let $L/K/F/k$ be a tower of field extensions with L/k normal, then there is a surjection

$$\mathrm{Mor}_k(i_{FK}, L) : \mathrm{Mor}_k(K, L) \to \mathrm{Mor}_k(F, L)$$

Proof. Note that $\mathrm{Mor}_k(i_{FK}, L) \circ \mathrm{Mor}_k(i_{KL}, L) = \mathrm{Mor}_k(i_{FL}, L)$. By (3.17.71) this composition is surjective, whence the result follows. ∎

Corollary 3.17.73 (Quotient of automorphism group)
Let $L/K/k$ be a tower of extensions such that L/k and K/k are normal. Then L/K is normal and there is an isomorphism of groups.

$$\begin{aligned} \mathrm{Aut}(L/k)/\mathrm{Aut}(L/K) &\longrightarrow \mathrm{Aut}(K/k) \\ \sigma &\to \ i_{KL}^{-1} \circ \sigma \circ i_{KL} \end{aligned}$$

Proof. The given map is well-defined by (3.17.69) since $(\sigma \circ i_{KL})(K) = i_{KL}(K)$, and σ fixes K precisely when $\sigma \circ i_{KL} = i_{KL}$, that is $\sigma \in \mathrm{Aut}(L/K)$. Given $\tau \in \mathrm{Aut}(K/k)$, by (3.17.71) there exists $\sigma \in \mathrm{Aut}(L/k)$ such that $\sigma \circ i_{KL} = i_{KL} \circ \tau$. This shows the given map is surjective. The result follows from the group isomorphism theorem. ∎

Definition 3.17.74 (Normal Closure)
*Let K/k be an algebraic extension. Then an algebraic extension L/K is a **normal closure** for K/k if*

- *L/k is normal*
- *No proper subfield $i_{KL}(K) \subseteq L' \subsetneq L$ is normal over k*

Proposition 3.17.75 (Existence and Uniqueness of Normal Closure)
Let K/k be an algebraic extension. Then a normal closure L/K exists and is unique up to isomorphism. Indeed it is the splitting field for all the minimal polynomials $\{m_{\alpha,k}(X) \mid \alpha \in K\}$ over k.

Furthermore if K/k is finite then so is L/k

Proof. Suppose $K = k(\{\alpha_i\}_{i\in I})$. Let L/k be the splitting field for $S = \{m_{\alpha_i,k}(X)\}_{i\in I}$. By (3.17.70) L/k is normal and by (3.17.52) there is a morphism $\sigma : K/k \to L/k$, so we may consider it as an extension $(L/K, \sigma)$. Suppose $i_{KL}(K) \subset L' \subset L$ is normal. As $i_{KL}(\alpha_i) \in L'$, by definition S splits in L' and therefore $L' = L$ by (3.17.56). Therefore L/K is a normal closure as required.

If K/k is finite then we may choose I to be finite, and therefore L/k is finite.

Let L/K be an arbitrary normal closure, then we claim L/k is a splitting field for $S' := \{m_{\alpha,k}(X) \mid \alpha \in K\}$. Clearly S' splits in L/k, because each has a root in L. Let L'/k be the subfield generated by roots of S'. Then it is a splitting field and therefore normal by (3.17.70). By assumption $L' = L$ and therefore L/k is the splitting field for S'. Uniqueness follows from the uniqueness of splitting fields (3.17.59). ∎

3.17.8 Separability (Algebraic Case)

We follow Lang and not only characterize separability but define a "separability degree" which equals the extension degree if and only if it's separable. The proofs are somewhat technical, especially in light of the fact most base fields will be perfect.

Definition 3.17.76 (Separable Element)
*We say $\alpha \in K/k$ is **separable** over k if it is algebraic and $m_{\alpha,k}(X)$ is a separable polynomial.*

*We say K/k is **separable algebraic** (or just separable if K/k is assumed to be algebraic) if every $\alpha \in K$ is **separable**.*

Lemma 3.17.77
$\alpha \in K/k$ is separable if and only if it is a root of a separable polynomial in $k[X]$.

In particular α separable over k implies it is separable over any subfield $i_{kK}(k) \subset E \subset K$.

Proof. One direction is obvious. Conversely suppose $f(\alpha) = 0$ with f separable. Then $m_{\alpha,k} \mid f$ so the result follows from (3.15.40). ∎

Proposition 3.17.78 (Separability Degree)
*Let K/k be an algebraic extension and L/K an extension such that L/k is normal (e.g. $L = \overline{k}$ or L is a normal closure). Then define the **separability degree***

$$[K : k]_s := \#\,\mathrm{Mor}_k(K, L)$$

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This is independent of the choice of L/K.

Proof. Given such an L, let L'/K be the intersection of all subfields of L/K normal over k. This is a normal closure of K/k. Let $\sigma \in \mathrm{Mor}_k(K,L)$ and $\alpha \in K$. Then $\sigma(\alpha)$ is a root of $m_{\alpha,k}(X)$ along with $i_{KL}(\alpha) \in L'$. As L' is normal we have $\sigma(\alpha) \in L'$. Therefore without loss of generality we may replace L with L'. As the normal closure is unique up to isomorphism the degree is well-defined. ∎

First we prove a key lemma regarding simple extension

Lemma 3.17.79 (Separability Degree of Simple Extension)
If $k(\alpha)/k$ is a simple extension and L/k normal overfield then

$$[k(\alpha):k]_s = \#\{\text{roots of } \ m_\alpha \ \text{ in } \ L\} \leq \deg(m_\alpha) = [k(\alpha):k]$$

Furthermore equality holds iff α is separable over k.

Proof. The first equality follows from (3.17.51), the final equality from (3.17.42). The inequality follows from (3.15.35). The final statement follows from (3.17.11). ∎

The main results of this section are the following

Proposition 3.17.80 (Separability Degree)
Let $L/K/F/k$ be a tower of finite extensions with L/k normal. Then the restriction map

$$\mathrm{Mor}_k(i_{FK}, L) : \mathrm{Mor}_k(K,L) \to \mathrm{Mor}_k(F,L)$$

is surjective with finite fibers of order $[K:F]_s$. In particular we have

$$[K:k]_s = [K:F]_s[F:k]_s$$

Further $[K:k]_s \leq [K:k]$ with equality if and only if K/k is separable.

Proof. For a tower $L/K/F/k$ with L/k normal, consider the restriction map

$$\psi := \mathrm{Mor}_k(i_{FK}, L) : \mathrm{Mor}_k(K,L) \to \mathrm{Mor}_k(F,L)$$

It is surjective by (3.17.72). Consider $\sigma \in \mathrm{Mor}_k(F,L)$ and the fibre $\psi^{-1}(\sigma) = \mathrm{Mor}_F(K,(L/F,\sigma))$. This has order equal to $\#\psi^{-1}(\sigma) = [K:F]_s$ for all σ, because by (3.17.78) this does not depend on the embedding i_{FL}. As $\mathrm{Mor}_k(K,L)$ is equal to the disjoint union of all the fibres, then the multiplicativity result follows.

It is possible to decompose K/k as a tower of simple extensions

$$K = K_n/.../K_0 = k$$

with $K_i = K_{i-1}(\alpha_i)$. By (3.17.79) we have

$$[K_i : K_{i-1}]_s \leq [K_i : K_{i-1}]$$

with equality iff α_i separable over K_{i-1}. By multiplicativity the inequality follows.

If K/k is separable then by (3.17.77) α_i is separable over K_{i-1} and we have $[K_i:K_{i-1}]_s = [K_i:K_{i-1}]$ and $[K:k]_s = [K:k]$ by multiplicativity. Conversely if $[K:k]_s = [K:k]$ then $[K_i:K_{i-1}]_s = [K_i:K_{i-1}]$ and α_i is separable over K_{i-1}. Since the choice of α_1 was arbitrary we see that K/k is separable. ∎

Proposition 3.17.81 (Towers of Separable Extensions)
Consider a tower of algebraic extensions $K/E/k$. Then K/E and E/k is separable iff K/k is.

Proof. K/k separable $\implies$ K/E and E/k separable follows from (3.17.77).

Conversely the finite case follows from (3.17.80) by multiplicativity. For the general case, consider $\alpha \in K$. Then (3.17.46) shows the existence of a finite subextension E_0/k of E such that $m_{\alpha,E} = m_{\alpha,E_0}$. Therefore α is separable over E_0. By (3.17.79) we see that $[E_0(\alpha):E_0]_s = [E_0(\alpha):E_0]$. As E/k is separable a-fortiori E_0/k is separable so by (3.17.80) $[E_0:k]_s = [E_0:k]$. By multiplicativity $[E_0(\alpha):k]_s = [E_0(\alpha):k]$ and the same result again shows that $E_0(\alpha)/k$ is separable. In particular α is separable over k as required. ∎

Proposition 3.17.82
An algebraic extension $K = k(\alpha_1,...,\alpha_n)/k$ is separable iff α_i are.

Proof. Let $K = k(\alpha_1,...,\alpha_n)$. Then we may construct a tower of finite (simple) extensions

$$K = K_n/.../K_0 = k$$

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with $K_i = k(\alpha_1, ..., \alpha_i)$ and $K_i = K_{i-1}(\alpha_i)$. By (3.17.77) α_i is separable over K_{i-1}. Therefore $[K_i : K_{i-1}]_s = [K_i : K_{i-1}]$ by (3.17.79) and $[K : k]_s = [K : k]$ by multiplicativity. (3.17.80) shows that K/k is separable. ∎

Proposition 3.17.83
Let K/k be a splitting field for finitely many separable polynomials f_i. Then K/k is finite and Galois.

Proof. By definition K is generated by the (finitely) many roots α_{ij} and so is finite by (3.17.44) and normal by (3.17.70). By (3.17.77) each α_{ij} is separable and so by (3.17.82) K/k is separable. ∎

Proposition 3.17.84
Let K/k be a finite separable extension and L/k a normal closure. Then L/k is separable.

Proof. By (3.17.75) L/k is the splitting field of finitely many minimal polynomials $m_{\alpha_i,k}(X)$ for $\alpha_i \in K$. By (3.17.82) the α_i are separable and therefore so are $m_{\alpha_i,k}(X)$. By (3.17.83) we see that L/k is separable. ∎

Lemma 3.17.85 (Normal Closure of a Simple Separable Extension)
Let $K = k(\theta)/k$ be a finite separable extension, m the minimal polynomial of θ and $n := \deg(m) = [K : k]$, and let N be a normal closure of K/k. Then

a) N is a splitting field of m, so $m = \prod_{i=1}^{n}(X - \beta_i)$ over N with the β_i pairwise distinct.
b) If Ω/k contains a copy of N then $\beta_1, ..., \beta_n$ are all the roots of m in Ω, and $\varphi \mapsto \varphi(\theta)$ is a bijection $\mathrm{Mor}_k(K, \Omega) \to \{\beta_1, ..., \beta_n\}$.
c) If Ω/k contains a copy of N then N is the compositum of the images of all k-embeddings $K \to \Omega$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.17.75), applied with the single generator θ of K, N is a splitting field of m. As θ is separable, m is separable (3.17.76) and so has n distinct roots in N. This proves a).

For b), $m = \prod_i (X - \beta_i)$ also holds in $\Omega[X]$ and Ω is a domain, so the β_i are all the roots of m in Ω. As m has a root in Ω, (3.17.51) gives that $\varphi \mapsto \varphi(\theta)$ is a bijection from $\mathrm{Mor}_k(K, \Omega)$ onto the set of roots of m in Ω, namely $\{\beta_1, ..., \beta_n\}$.

For c) by b) the embeddings $\varphi_i : K \to \Omega$ send θ to the β_i, so the compositum of their images is $k(\beta_1, ..., \beta_n)$, which equals N because the splitting field N of m is generated by the roots of m (3.17.56). ∎

Proposition 3.17.86 (Equivalent Definition of Separability)
Let K/k be an algebraic extension. TFAE

a) K/k is separable
b) E/k is separable for every finite subextension
c) $[E : k]_s = [E : k]$ for every finite subextension

Proof. $a) \implies b)$ is trivial and $b) \iff c)$ follows from (3.17.80). We need only show $b) \implies a)$.

Consider $\alpha \in K$. Then by (3.17.46) there exists a finite subextension E/k such that α is algebraic over E. Therefore $E(\alpha)/k$ is finite, and by assumption $E(\alpha)/k$ separable as required. ∎

3.17.9 Purely Inseparable Extensions and Separable Closure

In what follows we let p be the characteristic exponent of k. In other words when $\mathrm{char}(k) = 0$ then $p = 1$ and the statements are trivial.

Definition 3.17.87 (Purely Inseparable Element)
*An element $x \in K/k$ is **purely inseparable** (or p-radical) if there exists an integer $n \geq 0$ such that $x^{p^n} \in k$. The **height** of x is the least such integer.*

*We say an extension K/k is **purely inseparable** if every $x \in K/k$ is purely inseparable.*

*Further K/k is **purely inseparable of height $\leq n$** if additionally every $x \in K$ has height at most n.*

Lemma 3.17.88
Let $a \in k$. For every integer $e \geq 0$ the polynomial $f(X) := X^{p^e} - a$ has at most one root in any extension field K/k.

Proof. Suppose that b is a root, then by (3.17.6) we have $f(X) = (X - b)^{p^e}$. Then evidently b is the unique root. ∎

Lemma 3.17.89
Let $a \in k \setminus k^p$. Then for every integer $e \geq 0$ the polynomial $X^{p^e} - a$ is irreducible in $k[X]$.

Proof. Let K/k be a splitting field for $f(X) = X^{p^e} - a$, $b \in K$ a root and $g(X)$ be the minimal polynomial of b. By (3.17.6) $f(X) = (X - b)^{p^e}$ in $K[X]$. Suppose π is a monic irreducible factor of f in $k[X]$ then by unique factorisation

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in $K[X]$ we have $\pi(X) = (X-b)^r$ for some r. In particular π has b as a root and therefore is divisible by g. By irreducibility we conclude every irreducible factor is equal to g and the irreducible factorization of f is g^s. Furthermore $p^e = rs$. Consequently both r and s are pth powers, and we have

$$\pi = (X-b)^{p^d}$$
$$f = \pi^{p^{e-d}}$$

for some $0 \leq d \leq e$. In particular $b^{p^d} \in k$ and $a = b^{p^{e-d}}$. By assumption a is not a p-th power and so we must have $d = e$, and $f = \pi$ is irreducible. ∎

Proposition 3.17.90
Let $x \in K/k$ be purely inseparable of height e. Then the minimal polynomial of x is $X^{p^e} - x^{p^e}$. Furthermore

$$[k(x) : k] = p^e$$

$$[k(x) : k]_s = 1$$

More precisely for every field L/k there is at most one morphism $k(x)/k \to L/k$.

Proof. By definition $x^{p^e} \in k$ is not a p-th power. Therefore the given polynomial is irreducible by (3.17.89) and therefore is the minimal polynomial by uniqueness. The first relation follows from (3.17.42) and the second from (3.17.79). ∎

Corollary 3.17.91
Let $x \in K/k$. Then the following are equivalent

a) x is purely inseparable
b) $m_x(X) = (X-x)^r$ for some r

In this case we must have $r = p^n$ where n is the height of x.

Corollary 3.17.92
Suppose $x \in K/k$ is both separable and purely inseparable. Then $x \in k$.

Proof. The minimal polynomial is $X^{p^e} - x^{p^e} = (X-x)^{p^e}$. Therefore by (3.17.11) $e = 0$ which means precisely $x \in k$. ∎

We first establish the following two results, which will be used below.

Lemma 3.17.93 (Inseparable Polynomials)
Let $f \in k[X]$ be a non-constant polynomial and p the characteristic exponent of k. Then $f' = 0 \iff p > 1$ and $f \in k[X^p]$.

Proposition 3.17.94 (Separable Irreducible Polynomials)
Let $f \in k[X]$ be an irreducible polynomial and p the characteristic exponent of k. Then the following are equivalent

a) f is separable (i.e. f and f' are co-maximal)
b) $f' \neq 0$
c) $p = 1$ or $f \notin k[X^p]$

Proof. Note by assumption f is not a unit and therefore not constant.

a) $\implies$ *b)* Suppose $f' = 0$ then $(f, f') = (f) \neq k[X]$ and therefore f is not separable.

b) $\implies$ *a)* Suppose f is not separable. By (...) (f) is maximal and so we must have $f' \in (f)$. As $\deg(f') < \deg(f)$ this implies $f' = 0$.

b) $\iff$ *c)* This is just the contrapositive of (3.17.93). ∎

Proposition 3.17.95 (Relative Separable Closure)
Let K/k be a field extension. The subextension $K_s/k \subset K/k$ of separable elements form a separable algebraic subfield. Furthermore if K/k is algebraic then K/K_s is purely inseparable and the restriction map

$$\mathrm{Mor}_k(K, L) \to \mathrm{Mor}_k(K_s, L)$$

is bijective for every normal extension L/K. In particular when K_s/k is finite then we have the relation

$$[K_s : k] = [K_s : k]_s = [K : k]_s$$

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Proof. Suppose $\alpha, \beta \in K$ are separable algebraic then the field $k(\alpha, \beta)$ is separable algebraic by (3.17.82). As this contains $\alpha \pm \beta$ and $\alpha\beta$ then K_s is a subfield which is by definition separable. Then by (3.17.80) we have $[K_s : k] = [K_s : k]_s$ when this is finite.

For $x \in K$ let $f(X)$ be the minimal polynomial over k. There exists some integer $m \geq 0$ such that $f(X) \in k[X^{p^m}]$ but not in $k\left[X^{p^{m+1}}\right]$. In other words $f(X) = g(X^{p^m})$ and $g(X) \notin k[X^p]$. As f is irreducible so is g, and is therefore the minimal polynomial of x^{p^m}. By (3.17.94) g is separable and therefore $x^{p^m} \in K_s$. This shows that K/K_s is purely inseparable.

Let $L/K/k$ be tower such that L/k is normal. Consider the mapping

$$\mathrm{Mor}_k(K, L) \to \mathrm{Mor}_k(K_s, L)$$

obtained by restriction. By (3.17.72) it is surjective. Consider $\psi : K_s \to L$ and $\hat{\psi} : K \to L$ an extension. For every $x \in K$ we have $\hat{\psi}|_{K_s(x)} = i_{K_s(x)L}$ by (3.17.90). This shows that $\hat{\psi}$ is unique and the mapping is bijective. ∎

Definition 3.17.96 (Degree of Inseparability)
Let K/k be a finite extension. Then in light of (3.17.95) we may define the ***degree of inseparability*** *to be the integer*

$$[K : k]_i := \frac{[K : k]}{[K : k]_s}$$

By (3.17.16) and (3.17.80) it is multiplicative in the sense that for a tower of finite extensions

$$[L : k]_i = [L : K]_i [K : k]_i$$

Further by (3.17.80) K/k is separable if and only if $[K : k]_i = 1$.

We may refine this further by showing that $[K : k]_i$ is always a power of the characteristic exponent.

Proposition 3.17.97
Let $k(\alpha)/k$ be a simple algebraic extension with characteristic exponent p. There exists a unique integer $r \geq 0$ such that every root of $m_{\alpha,k}(X)$ has multiplicity p^r (in any splitting field).

Further

$$[k(\alpha) : k]_i = p^r$$

and $m_{\alpha,k}(X)$ has $[k(\alpha) : k]_s$ distinct roots.

Proof. Let $L/k(\alpha)/k$ be a normal overfield, and let $\alpha_1, ..., \alpha_s$ be the distinct roots of $m_{\alpha,k}(X)$ with $\alpha_1 = \alpha$. Then we have a factorisation in $L[X]$ given by

$$m_{\alpha,k}(X) = \prod_{i=1}^{s} (X - \alpha_i)^{m_i}$$

for some integers $m_i > 0$.

For every $1 \leq i \leq n$ there exists an automorphism $\sigma_i \in \mathrm{Aut}(L/k)$ such that $\sigma_i(\alpha) = \alpha_i$ (3.17.52). By definition σ_i fixes $m_{\alpha,k}(X)$ so by unique factorisation we deduce that $m_i = m_1$, so all the roots have the same multiplicity m.

If $m = 1$ then by (3.15.39) $m_{\alpha,k}(X)$ is separable. Otherwise by (3.17.94) there is a polynomial $g(X)$ such that $f(X) = g(X^p)$ which has α^p as a root. We may continue inductively to find an integer $r \geq 0$ such that $f(X) = g(X^{p^r})$ and $g(X)$ is separable. Then α^{p^r} is separable. By definition α is purely inseparable over $k(\alpha^{p^r})$ so by (3.17.90)

$$\left[k(\alpha) : k(\alpha^{p^r})\right]_i = p^r$$

By (3.17.80) we have

$$\left[k(\alpha^{p^r}) : k\right]_i = 1$$

and by multiplicativity $[k(\alpha) : k]_i = p^r$. By (3.17.79) the number of distinct roots of $m_{\alpha,k}(X)$ is $[k(\alpha) : k]_s$. Comparing degrees we have $\deg\left(m_{\alpha,k}(X)\right) = [k(\alpha) : k] = [k(\alpha) : k]_s \cdot m$ whence $m = [k(\alpha) : k]_i = p^r$ as required. ∎

Proposition 3.17.98
Let K/k be a finite field extension with characteristic exponent p. Then $[K : k]_i = p^r$ for some integer $r \geq 0$.

K/k is separable if and only if $p = 1$ or $r = 0$.

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Proof. We may decompose K/k into a tower of simple extensions, and the result follows from multiplicativity and (3.17.97). ∎

3.17.10 Separable closure

Proposition 3.17.99 (Relatively Separably Closed)
Let K/k be an extension. The following are equivalent

a) Every separable $x \in K/k$ lies in k
b) Every separable algebraic subextension is trivial
c) K_s/k is trivial

*In this case we say K/k is **relatively separably closed**.*

If K/k is algebraic then this is equivalent to being purely inseparable.

Proof. The final statement follows from (3.17.95). ∎

Definition 3.17.100
*We say a field k is **separably closed** if every separable algebraic extension is trivial.*

*We say an extension k^{sep}/k is a **separable closure** if it is separable algebraic and separably closed.*

Proposition 3.17.101
Let $\overline{k}/k$ be an algebraic closure, then the subextension $\overline{k}_s/k$ is a separable closure for k.

Furthermore every separable closure is isomorphic to $\overline{k}_s/k$ and $\overline{k}/\overline{k}_s$ is purely inseparable.

Proof. By (3.17.95) $\overline{k}_s/k$ is separable and algebraic. Suppose $K/\overline{k}_s$ is separable and algebraic then by (...) there exists an embedding $i : K/\overline{k}_s \to \overline{k}/\overline{k}_s$. As $\overline{k}_s$ is relatively separably closed in $\overline{k}$ we see that $i(K) = \overline{k}_s$ and therefore $K = \overline{k}_s$.

Let k^{sep}/k be a separable closure, then by (3.17.63) there is an embedding $\varphi : k^{\text{sep}}/k \to \overline{k}/k$. Then by definition $\varphi(k^{\text{sep}}) \subset \overline{k}_s$, so this defines an extension $\overline{k}_s/k^{\text{sep}}$ which is by definition separable. By assumption it is trivial which means precisely $\varphi(k^{\text{sep}}) = \overline{k}_s$.

It follows from (3.17.95) that $\overline{k}/\overline{k}_s$ is purely inseparable. ∎

Proposition 3.17.102
Let k^{sep}/k be a separable closure and k'/k a separable algebraic extension. Then there exists a morphism

$$k'/k \to k^{\text{sep}}/k$$

Proof. By (3.17.101) we may identify k^{sep}/k with $\overline{k}_s/k$. By (3.17.63) there exists a morphism $k'/k \to \overline{k}/k$ which by definition has image in $\overline{k}_s/k$. ∎

3.17.11 Perfect Fields

Proposition 3.17.103 (Perfect Closure)
Let k be a field with characteristic exponent p contained in an algebraically closed field Ω (e.g. $\overline{k}$). Define

$$k^{p^{-n}} := \{x \in \Omega \mid x^{p^n} \in k\}$$

Then $k^{p^{-n}}$ is a subfield of Ω/k, which is purely inseparable of height $\leq n$. Furthermore it is a splitting field for the family of polynomials $\{X^{p^r} - a \mid a \in k, r \leq n\}$ which characterizes it up to isomorphism. Furthermore

$$k^{p^{-\infty}} := \{x \in \Omega \mid x^{p^n} \in k \text{ some } n \geq 1\} = \bigcup_{n \geq 1} k^{p^{-n}}$$

is the smallest perfect subfield of Ω containing k and is also a splitting field for the family of polynomials $\{X^{p^n} - a \mid a \in k, n \in \mathbb{N}\}$.

Proof. Suppose $\alpha, \beta \in k^{p^{-n}}$ then applying the Frobenius homomorphism (...) we see $(\alpha + \beta)^{p^n} = \alpha^{p^n} + \beta^{p^n}$ and so $k^{p^{-n}}$ is an additive subgroup. Furthermore $(\alpha\beta)^{p^n} = \alpha^{p^n}\beta^{p^n}$ whence it is a multiplicative subgroup as this demonstrates $(\alpha^{-1})^{p^n} = (\alpha^{p^n})^{-1} \in k$. Therefore it is a subfield which is by definition purely inseparable of height $\leq n$.

By definition $k^{p^{-n}}$ is precisely the set of roots of the given polynomials and it is therefore the corresponding splitting field by (...). This also shows that $n \leq m \implies k^{p^{-n}} \subseteq k^{p^{-m}}$ which shows that the union $k^{p^{-\infty}}$ is a subfield of Ω.

To demonstrate it's perfect we need to show every element $x \in k^{p^{-n}}$ has a p-th root. By definition $x^{p^n} \in k$ and there exists a root α to the polynomial $X^{p^{n+1}} - x^{p^n}$ in $k^{p^{-(n+1)}}$. Therefore $(\alpha^p - x)^{p^n} = 0$ whence $\alpha^p = x$ as required.

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Let k' be another perfect subfield containing k. We claim by induction that $k^{p^{-n}} \subseteq k'$. For given $\alpha \in k^{p^{-(n+1)}}$ by definition $\alpha^p \in k^{p^{-n}}$ whence $\alpha^p \in k'$ by the inductive hypothesis, and by assumption $\alpha \in k'$. Therefore $k^{p^{-\infty}}$ is the smallest perfect subfield. Furthermore by definition it is the set of roots of the given family of polynomials and therefore the corresponding splitting field. ∎

Proposition 3.17.104
Let k'/k be a purely inseparable extension of height $\leq n$, then there exists an embedding

$$k'/k \to k^{p^{-n}}/k$$

If k'/k is purely inseparable and K/k is perfect then there also exists an embedding

$$k'/k \to K/k$$

Proof. For the first part let $\alpha \in k'$ then by (3.17.90) the minimal polynomial of α is $g(X) = X^{p^e} - \alpha^{p^e}$ for $e \leq n$ which splits completely in $k^{p^{-n}}$ by (3.17.103). Then the embedding exists by (3.17.52).

Suppose K/k is perfect then we claim that $X^{p^n} - a$ splits completely for every $a \in K$. By induction there is $b \in K$ such that $b^{p^n} = a$. Then $X^{p^n} - a = (X - b)^{p^n}$ splits completely. In particular for every $\alpha \in k'$ the minimal polynomial splits completely in K/k and the embedding follows from (3.17.52). ∎

Recall a perfect field k satisfies $k^p = k$ where p is the characteristic exponent. We show that in this case there are no inseparable algebraic extensions. First we show that all finite fields are perfect.

Proposition 3.17.105 (Finite fields are perfect)
Any finite field is perfect.

Proof. The Frobenius homomorphism is injective and therefore surjective by counting. ∎

Proposition 3.17.106 (Perfect field criteria)
Let k be a field with characteristic exponent p. Then the following are equivalent

a) Every irreducible polynomial in $k[X]$ is separable
b) Every algebraic extension K/k is separable
c) $\overline{k}/k$ is separable
d) $k^p = k$

In particular every field of characteristic zero is perfect, and an algebraic extension of a perfect field is separable.

Proof. We prove each in turn

a) $\implies$ *b)* Minimal polynomials are irreducible by (3.17.36) and therefore are separable by hypothesis.

b) $\iff$ *c)* One direction is automatic as $\overline{k}$ is algebraic. On the other hand every algebraic extension is isomorphic to a subfield of $\overline{k}$, so the implication follows from (3.17.81).

b) $\implies$ *d)* We need only show $k^p = k$ in the case $p > 1$. If there exists $a \in k \setminus k^p$ then by (3.17.89) the polynomial $f(X) \coloneqq X^p - a$ is irreducible. Then it's clear that the field extension $K \coloneqq k[X]/(X^p - a)$ is not separable. For the minimal polynomial of $\overline{X}$ is f which by (3.17.94) is not separable. This contradicts the assumption that K/k is separable.

d) $\implies$ *a)* Suppose that f is irreducible but not separable. Then by (3.17.94) $\operatorname{char}(k) = p > 1$ and $f \in k[X^p]$. By assumption all the coefficients are p-th powers and therefore $f = h^p$ by (3.17.6) for some $h \in k[X]$. This contradicts irreducibility of f, and so f must be separable. ∎

3.17.12 Applications of Separability

Definition 3.17.107 (Bounds on $\operatorname{Aut}(K/k)$)
Let K/k be an algebraic extension and L/K an extension such that L/k is normal. Then there is a natural injection

$$\begin{aligned} \operatorname{Mor}_k(K, i_{KL}) : \operatorname{Aut}(K/k) &\to \operatorname{Mor}_k(K, L) \\ \sigma &\to i_{KL} \circ \sigma \end{aligned}$$

In particular in the case $[K : k] < \infty$

$$\#\operatorname{Aut}(K/k) \leq [K : k]_s \leq [K : k] < \infty$$

If i_{KL} is inclusion, then we may regard $\operatorname{Aut}(K/k)$ as a subset of $\operatorname{Mor}_k(K, L)$

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As an application of the concept of separability degree we prove

Proposition 3.17.108 (Primitive Element Theorem)
Let K/k be a finite separable extension of k then $K = k(\alpha)$ is simple.

Proof. We only prove the case k is infinite. The finite case can be proven separately by showing that the $K^\star$ is cyclic.

Consider the set $\mathrm{Mor}_k\left(K, \overline{k}\right) = \{\sigma_1, ..., \sigma_n\}$ which by (3.17.86) has order $n = [K : k]$. By induction we can assume that $K = k(\alpha, \beta)$. We claim that there exists $0 \neq c \in k$ such that $\sigma_i(\alpha + c\beta)$ are all distinct. In this case we clearly have $\#\,\mathrm{Mor}_k\left(k(\alpha + c\beta), \overline{k}\right) \geq n$ so by the same result $[k(\alpha + c\beta) : k] \geq [k(\alpha + c\beta) : k]_s \geq n$ whence $k(\alpha + c\beta) = K$ (by (2.3.15)).

We have $\sigma_i(\alpha + c\beta) = \sigma_j(\alpha + c\beta) \iff c\big(\sigma_i(\beta) - \sigma_j(\beta)\big) = \big(\sigma_i(\alpha) - \sigma_j(\alpha)\big)$. Therefore consider the polynomial

$$f(X) = \prod_{i \neq j} \big(X\big(\sigma_i(\beta) - \sigma_j(\beta)\big) - \big(\sigma_i(\alpha) - \sigma_j(\alpha)\big)\big)$$

Then the embeddings are distinct precisely when $f(c) \neq 0$. Since $f(X)$ has at most finitely many roots and k is infinite, there must exist such a c. ∎

3.17.13 Normal Extensions II

We provide some more straightforward criteria based on $[K : k]_s$

Proposition 3.17.109 (Normal Criteria II)
Let $L/K/k$ be a tower of algebraic extensions such that L/k is normal (e.g. $L = \overline{k}$). Then K/k is normal if and only if the embedding

$$\mathrm{Mor}_k(K, i_{KL}) : \mathrm{Aut}(K/k) \to \mathrm{Mor}_k(K, L)$$

is a bijection. In particular if K/k is finite, then it is normal if and only if

$$\#\,\mathrm{Aut}(K/k) = [K : k]_s$$

Proof. Suppose K/k is normal and consider $\sigma \in \mathrm{Mor}_k(K, L)$. Then by **NOR1** $\sigma(K) = i_{KL}(K)$ and we may define $\tau := i_{KL}^{-1} \circ \sigma$ with $\tau \in \mathrm{Aut}(K/k)$. The converse is similar.

For the final part we've already observed (3.17.107) that in the finite case $\#\,\mathrm{Aut}(K/k) \leq [K : k]_s = \#\,\mathrm{Mor}_k(K, L) \leq [K : k] < \infty$. Therefore the embedding $\mathrm{Mor}_k(K, i_{KL})$ is a bijection precisely when the orders are the same. ∎

Corollary 3.17.110 (Galois Criteria)
Let K/k be a finite extension. Then

$$\#\,\mathrm{Aut}(K/k) \leq [K : k]_s \leq [K : k]$$

with equalities if and only if K/k is Galois.

Proof. We've seen the inequalities (3.17.107)

$$\#\,\mathrm{Aut}(K/k) \leq [K : k]_s \leq [K : k] < \infty$$

with equality if and only if K/k is both normal (3.17.109) and separable (3.17.80) ∎

3.17.14 Finite Fields

A finite field K necessarily has positive characteristic p, and therefore the prime subfield is isomorphic to the field $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$. We list some necessary properties of a finite field

Proposition 3.17.111 (Properties of finite fields)
Every finite field K is a finite-dimensional vector space over its prime subfield $\mathbb{F}_p$. Define $n = \left[K : \mathbb{F}_p\right]$.

- *$\#K = p^n$*
- *K is a splitting field for $X^{p^n} - X \in \mathbb{F}_p[X]$, and indeed is equal to the set of roots*
- *The multiplicative group of units $K^\star$ is cyclic.*
- *$K/\mathbb{F}_p$ is simple*

Proof. Since $K/\mathbb{F}_p$ is a finite-dimensional vector space it must have order p^n.

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The group of units has order $p^n - 1$, so by Lagrange's theorem every non-zero element satisfies $X^{p^n-1} - 1 = 0$, so therefore every element satisfies $X^{p^n} - X = 0$. Since this polynomial can have at most p^n roots (3.15.35) the roots are exactly all the elements of K.

We note again that $X^d - 1$ has at most d roots by (3.15.35). Therefore the fact $K^\star$ is cyclic follows from (3.2.25). ■

Proposition 3.17.112 (Frobenius morphism)
Given any field $K/\mathbb{F}_p$ the Frobenius map

$$\phi : x \to x^p$$

is an injective field homomorphism. In particular when K is finite (or even algebraic) it is an automorphism over $\mathbb{F}_p$.

Proof. The only non-trivial step is showing

$$(x+y)^p = x^p + y^p$$

which follows from elementary calculations on binomial coefficients.

For the final statement use (3.17.35). ■

Further we can show existence and uniqueness of finite fields.

Proposition 3.17.113 (Existence and uniqueness of finite fields)
Consider the algebraic closure $\overline{\mathbb{F}_p}$ and let $\mathbb{F}_{p^n}$ denote the splitting field of $f(X) = X^{p^n} - X$ in $\overline{\mathbb{F}_p}$. Then

- *$\mathbb{F}_{p^n}$ is equal to the set of roots of $X^{p^n} - X$*
- *It is the unique subfield of order p^n and every finite field of order p^n is isomorphic to this.*
- *$\mathbb{F}_{p^m} \subset \mathbb{F}_{p^n} \iff m \mid n$*

Proof. By (3.17.112) the set of roots of $f(X)$ forms a subfield of $\overline{\mathbb{F}_p}$.

Furthermore $f'(X) = -1$ so $f(X)$ is separable because clearly $(f, f') = 1$. Therefore by (3.17.11) $f(X)$ has p^n distinct roots and the splitting field of $f(X)$ is exactly the set of roots.

Furthermore every subfield of order p^n must satisfy this polynomial by Lagrange's Theorem (3.2.12), so it is the unique such subfield.

Since every algebraic extension of $\mathbb{F}_p$ is isomorphic to a subfield of $\overline{\mathbb{F}_p}$ it's also the unique algebraic extension of order p^n up to isomorphism.

Clearly if $\mathbb{F}_{p^m} \subset \mathbb{F}_{p^n}$ then $\left[\mathbb{F}_{p^n} : \mathbb{F}_p\right] = \left[\mathbb{F}_{p^n} : \mathbb{F}_{p^m}\right]\left[\mathbb{F}_{p^m} : \mathbb{F}_p\right]$, so we must have $m \mid n$. Conversely if $\alpha \in \mathbb{F}_{p^m}$ then $\alpha^{p^m} = \alpha \implies \alpha^{p^{rm}} = \alpha$ for all $r > 0$, so $\alpha \in \mathbb{F}_{p^n}$. ■

It is usually most convenient to work in $\overline{\mathbb{F}_p}$ and consider the subfields of order p^n. We've seen in (3.17.105) that every finite field $\mathbb{F}_q := \mathbb{F}_{p^n}$ is perfect and therefore every algebraic extension is separable (3.17.106). In fact we may show that every finite extension is Galois.

Proposition 3.17.114
The field extension $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois with

$$\mathrm{Gal}\left(\mathbb{F}_{p^n}/\mathbb{F}_p\right) = \langle \phi \rangle$$

cyclic of order n generated by the Frobenius automorphism.

Proof. Let $G = \mathrm{Aut}\left(\mathbb{F}_{p^n}/\mathbb{F}_p\right)$. We've observed that $\phi \in G$. Let $d = o(\phi)$, then we wish to prove that $n = d$. Certainly Lagrange's Theorem applied to the multiplicative group $\mathbb{F}_{p^n}^\star$ implies $\phi^n = 1$. Therefore $d \mid n$ by (3.2.12) applied to G. By definition $\phi^d = e$, so every $\alpha \in \mathbb{F}_{p^n}$ satisfies the polynomial $X^{p^d} - X = 0$. This polynomial has at most p^d roots (3.15.35) so we must have $d \geq n$, and therefore $d = n$. Clearly ϕ generates a cyclic subgroup of order n. However by (3.17.110) G has at most order n, whence $G = \langle \phi \rangle$ as required. Furthermore by the same result $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois. ■

Proposition 3.17.115 (Subfields of $\mathbb{F}_{p^n}$)
Consider the field extension $\mathbb{F}_{p^n}/\mathbb{F}_p$. Then it has a unique subfield of order p^m if $m \mid n$ (and no such subfield otherwise). In this case $\mathbb{F}_{p^n}/\mathbb{F}_{p^m}$ is Galois and

$$\mathrm{Gal}\left(\mathbb{F}_{p^n}/\mathbb{F}_{p^m}\right) = \langle \phi^m \rangle$$

and in particular has order n/m.

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Proof. We've already shown that $\mathbb{F}_{p^n}$ has a unique subfield of order p^m, by assuming an embedding in $\overline{\mathbb{F}_p}$. Let $H = \mathrm{Aut}\big(\mathbb{F}_{p^n}/\mathbb{F}_{p^m}\big)$. Note ϕ^m has order n/m. Furthermore from (3.17.113) every element of $\mathbb{F}_{p^m}$ satisfies $X^{p^m} - X$. In other words ϕ^m fixes $\mathbb{F}_{p^m}$ and $\phi^m \in H$. Therefore $\langle \phi^m \rangle \leq H$ and $\#H \geq n/m$. By (3.17.110) $\#H \leq \big[\mathbb{F}_{p^n} : \mathbb{F}_{p^m}\big] = n/m$, whence we have equality and so the extension is Galois by (3.17.110) and $H = \langle \phi^m \rangle$. ∎

Lemma 3.17.116
Let $x \in \mathbb{F}_{p^n}$. Then $\deg(x) = n \iff \mathrm{Fix}(x) = \{1\} \subset \mathrm{Gal}\big(\mathbb{F}_{p^n}/\mathbb{F}_p\big)$.

In other words $\mathrm{Gal}\big(\mathbb{F}_{p^n}/\mathbb{F}_p\big)$ *acts freely on the elements of degree n.*

Proof. Observe that $\deg(x) = n \iff \big[\mathbb{F}_p(x) : \mathbb{F}_p\big] = n \iff \mathbb{F}_p(x) = \mathbb{F}_{p^n} \overset{(3.17.118)}{\iff} \mathrm{Gal}\big(\mathbb{F}_{p^n}/\mathbb{F}_p(x)\big) = \{1\}$ which is equivalent to the statement given. ∎

3.17.15 Galois Theory

We've seen that for K/k a finite extension

$$\#\,\mathrm{Aut}(K/k) \leq [K : k]_s \leq [K : k]$$

with equality if and only if K/k is Galois, by (3.17.110).

Remark 3.17.117
If k is perfect then $\overline{k}/k$ is Galois.

The main result of Galois Theory is

Proposition 3.17.118
Let K/k be a finite Galois extension then there is an order-reversing bijection between subgroups and subfields

$$\begin{aligned} \{H \leq \mathrm{Gal}(K/k)\} &\longleftrightarrow \{F/k \subseteq K/k\} \\ H &\longrightarrow K^H := \{x \in K \mid h(x) = x \quad \forall h \in H\} \\ \mathrm{Gal}(K/F) &\longleftarrow F \end{aligned}$$

Proof. This is proved in a series of Propositions in the rest of this section. Firstly we show it is well-defined in (3.17.119). The maps are mutual inverses by (3.17.120) and (3.17.121). ∎

Such an order reversing map is usually called an (antitone) Galois connection, as the first such type arose from Galois Theory. Note it is well-defined because of the following proposition.

Proposition 3.17.119
If K/k is Galois and $F \subset K/k$ then K/F is Galois.

Proof. This follows from (3.17.67) and (3.17.81). ∎

Proposition 3.17.120 (Fixed field of Galois group)
If K/k is Galois and $F \subset K/k$ a subfield then

$$K^{\mathrm{Gal}(K/F)} = F$$

Proof. Clearly $F \subseteq K^{\mathrm{Gal}(K/F)}$. Conversely given $\alpha \in K \setminus F$, then $\deg\big(m_{\alpha,F}\big) > 1$. Since α is separable it must have another root $\beta \in K$. By (3.17.68) there is an element $\sigma \in \mathrm{Gal}(K/F)$ such that $\sigma(\alpha) = \beta$. In other words $\alpha \notin K^{\mathrm{Gal}(K/F)}$, which shows the reverse inclusion. ∎

Proposition 3.17.121
Let K/k be a field extension and $H \subseteq \mathrm{Aut}(K/k)$ a finite subgroup then K/K^H is a finite Galois extension with

$$\mathrm{Gal}(K/K^H) = H$$

and order equal to $[K : K^H]$. When K/k is finite then H is automatically finite.

Proof. Firstly observe that trivially $H \subseteq \mathrm{Aut}(K/K^H)$. If we know that $[K : K^H] < \infty$, then by (3.17.110) we have

$$\#H \leq \#\,\mathrm{Aut}(K/K^H) \leq [K : K^H]_s \leq [K : K^H]$$

We can prove equality everywhere if we show that $[K : K^H] \leq \#H$, which is shown either by (3.17.122) or (3.17.123). Note equality also shows that K/K^H is finite Galois by the same result.

Finally when K/k is finite, then $\#\,\mathrm{Aut}(K/k) < \infty$. So in this case H is always finite. ∎

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We present two approaches to showing the inequality $[K : K^H] \leq \#H$. The first uses independence of characters style argument, and the second which is more straightforward uses the action of H to show that every element has degree at most $\#H$ (Artin).

Lemma 3.17.122 (Bound degree of fixed field I)
Let K/k be a field extension and $H \subset \mathrm{Aut}(K/k)$ a finite subgroup. Then

$$[K : K^H] \leq \#H$$

Proof. Let $H = \{\sigma_1, ..., \sigma_n\}$ with $\sigma_1 = \mathrm{id}$ and $\{\alpha_1, ..., \alpha_m\}$ a K^H-linearly independent subset of K.

Consider the vector space K^n and the elements $\hat{\alpha}_j = \left(\sigma_1(\alpha_j), ..., \sigma_{n(\alpha_j)}\right)$ for $j = 1...m$. It's enough to show that these are linearly independent over K, as that implies $m \leq n$ by (3.3.126). Therefore K/K^H is finite dimensional with dimension at most $n = \#H$.

Let $S(K) := \left\{v \in K^m \;\middle|\; \sum_{j=1}^m v_j \hat{\alpha}_j = 0\right\}$, we aim to show that $S(K) = \{0\}$. If we also consider $S(K^H)$, any non-zero elements will be a K^H linear-dependence for $\alpha_1, ..., \alpha_m$ by considering the first component ($\sigma_1 = \mathrm{id}$). Therefore by linear independence of α_i we see $S(K^H) = \{0\}$. So it's enough to show that $S(K) \neq \{0\} \implies S(K^H) \neq \{0\}$, to prove $S(K) = \{0\}$ by contradiction.

First observe that $K^\star$ and H both act on $S(K)$ component-wise. The first by multiplication and the second by application. This is well-defined because $v \in S(K)$ if and only if

$$\sum_j v_j \sigma(\alpha_j) = 0 \quad \forall \sigma \in H.$$

Apply τ to obtain

$$\sum_j \tau(v_j)(\tau \circ \sigma)(\alpha_j) = 0 \quad \forall \sigma \in H$$

and since multiplication by τ permutes H we see $\tau(v) \in S(K)$ as required.

If there exists $0 \neq v \in S(K)$, consider v with a minimal number of non-zero components. By scaling we can assume λv has at least one component in K^H. The vector $\tau(\lambda v) - \lambda v$ then has at least one fewer non-zero components, so by minimality must be zero. Since τ was arbitrary we see $0 \neq \lambda v \in S(K^H)$ as required. ∎

Lemma 3.17.123 (Bound degree of fixed field II)
Let K/k be a field extension and H a finite subgroup of $\mathrm{Aut}(K/k)$. Then K/K^H is finite separable, and simple, with $[K : K^H] \leq \#H$

Proof. We show that K/K^H is separable and every element has degree at most $\#H$. For any $\alpha \in K$, consider the orbit $\alpha^H = \{\sigma(\alpha) \mid \sigma \in H\}$, which is of order at most $\#H$. Then the polynomial

$$f(X) = \prod_{\beta \in \alpha^H} (X - \beta)$$

has α as a root and is separable by (3.17.11). Furthermore $f^\tau = f$ because τ permutes α^H (it's injective and hence bijective). Therefore $f \in K^H[X]$ and $m_{\alpha, K^H} \mid f$. We see that α has degree at most $\#H$ and is separable by (3.15.40).

If K/k is finite, then a-fortiori K/K^H is finite, so we may apply the Primitive Element Theorem (3.17.108) directly to show the result.

More generally let $K^H(\alpha)$ be a simple subfield of K of maximal degree. This exists because the degree of α is bounded above by $\#H$. We claim $K^H(\alpha) = K$, for if not then $K^H \subseteq K^H(\alpha) \subsetneq K^H(\alpha, \beta)$ is a finite separable extension of K^H, whence it must be simple by the Primitive Element Theorem (3.17.108), contradicting maximality. Finally the degree of $[K : K^H]$ is the degree of α, which we've seen is bounded above by $\#H$. ∎

Now we may demonstrate straightforward criteria for subfield to be normal

Proposition 3.17.124
Let K/k be a finite Galois extension and $k \subset F \subset K$ a subfield.

Then F/k is Galois if and only if $\mathrm{Gal}(K/F) \lhd \mathrm{Gal}(K/k)$ is normal. In this case we have a canonical isomorphism

$$\mathrm{Gal}(K/k)/\mathrm{Gal}(K/F) \to \mathrm{Gal}(F/k)$$

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Proof. Recall from (3.17.69) we have F/k is normal iff $\sigma(F) = F$ for all $\sigma \in \mathrm{Gal}(K/k)$. Recall K/F is normal for all subfields F. Furthermore, we observe that

$$\mathrm{Gal}(K/\sigma(F)) = \sigma \,\mathrm{Gal}(K/F)\sigma^{-1}$$

By the correspondence (3.17.118) $\mathrm{Gal}(K/F) = \mathrm{Gal}(K/F') \iff F = F'$. Therefore

$$\begin{aligned} F/k \text{ normal} &\iff \sigma(F) = F \quad \forall \sigma \in \mathrm{Gal}(K/k) \\ &\iff \mathrm{Gal}(K/\sigma(F)) = \mathrm{Gal}(K/F) \quad \forall \sigma \in \mathrm{Gal}(K/k) \\ &\iff \sigma \,\mathrm{Gal}(K/F)\sigma^{-1} = \mathrm{Gal}(K/F) \quad \forall \sigma \in \mathrm{Gal}(K/k) \\ &\iff \mathrm{Gal}(K/F) \lhd \mathrm{Gal}(K/k) \end{aligned}$$

The result then follows from (3.17.73). ∎

3.17.16 Norm and Trace

In what follows we assume that K is a commutative ring and A is a K-algebra which is finite free as a K-module (free is automatic when K is a field).

Definition 3.17.125 (Norm, Trace and Characteristic Polynomial)
For $\alpha \in A$ consider the K-linear multiplication map

$$\mathrm{mult}_{\alpha,K} : A \to A$$

and define

a) $\mathrm{Tr}_{A/K}(\alpha) \coloneqq \mathrm{Tr}(\mathrm{mult}_\alpha) \in K$
b) $\mathrm{Norm}_{A/K}(\alpha) \coloneqq \det(\mathrm{mult}_\alpha) \in K$
c) $\mathrm{Pc}_{A/K}(\alpha; X) \coloneqq \chi_{\mathrm{mult}_\alpha} = \det\big(\mathrm{mult}_{X-\alpha,K[X]}\big) = \det(X \cdot I_n - [\mathrm{mult}_\alpha]) \in K[X]$ *the characteristic polynomial.*

Proposition 3.17.126
With the notation as in (3.17.125) we have the following properties

a) $\mathrm{Tr}(\alpha + \beta) = \mathrm{Tr}(\alpha) + \mathrm{Tr}(\beta)$
b) $\mathrm{Tr}(\alpha\beta) = \mathrm{Tr}(\beta\alpha)$
c) $\mathrm{Norm}(\alpha\beta) = \mathrm{Norm}(\alpha)\,\mathrm{Norm}(\beta)$

and further that

$$\mathrm{Pc}(\alpha; X) = X^n - \mathrm{Tr}(\alpha)X^{n-1} + \cdots + (-1)^n \,\mathrm{Norm}(\alpha)$$

Finally for $\lambda \in K$ we have

$$\begin{aligned} \mathrm{Tr}(\lambda) &= \dim_K A \cdot \lambda \\ \mathrm{Norm}(\lambda) &= \lambda^{\dim_K A} \end{aligned}$$

Proof. We prove each in turn

a) We have $\mathrm{mult}_{\alpha+\beta} = \mathrm{mult}_\alpha + \mathrm{mult}_\beta$ and ordinary trace is additive (3.8.25).
b) When A is commutative this is obvious, however even in general we have $\mathrm{mult}_{\alpha\beta} = \mathrm{mult}_\alpha \circ \mathrm{mult}_\beta$ and so the result follows from symmetry of the ordinary trace (3.8.26)
c) This follows similarly because the determinant is multiplicative (3.8.17).

The identity for the characteristic polynomial follows directly from (3.16.7). Finally the matrix representation of mult_λ is simply $\lambda \cdot I_n$ with respect to any basis. ∎

Proposition 3.17.127 (Characteristic Polynomial is a Norm)
For $\alpha \in A$ we have the relationship

$$\mathrm{Pc}_{A/K}(\alpha; X) = \mathrm{Norm}_{A[X]/K[X]}(X - \alpha)$$

Proof. This follows from the definitions and (3.16.7). ∎

Proposition 3.17.128
Let M be a finite free A-module. Then M is a finite free K-module. Explicitly given a basis (m_j) of M over A and (a_i) of A over K, then $(a_i \cdot m_j)$ is a basis for M over K.

If $u : M \to M$ is an endomorphism then we denote by u_K the endomorphism of M regarded as a (finite free) K-module.

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Proposition 3.17.129 (Transitivity of Trace)
Suppose A is commutative, M is a finite free A-module and $u : M \to M$ an endomorphism. Then

$$\mathrm{Tr}(u_K) = \mathrm{Tr}_{A/K}(\mathrm{Tr}(u))$$

and

$$\det(u_K) = \mathrm{Norm}_{A/K}(\det(u))$$

Proof. Let $(e_1, ..., e_m)$ be an A-basis for M and $(a_1, ..., a_n)$ be a K-basis for A. Then by (3.17.128) $\big(a_i \cdot e_j\big)$ is a K-basis for M. For $\alpha \in A$ let $M(\alpha) \in \mathrm{Mat}_n(K)$ denote the matrix corresponding mult_α and let $\big(c_{lj}\big) \in \mathrm{Mat}_m(A)$ be the matrix of u with respect to $\big(e_j\big)$. By definition this means

$$u\big(e_j\big) = \sum_{l=1}^{m} c_{lj} \cdot e_l$$

and

$$a_i\alpha = \alpha a_i = \sum_{k=1}^{n} M(\alpha)_{ki} a_k.$$

Therefore

$$\begin{aligned}
u_K\big(a_i e_j\big) &= a_i u\big(e_j\big) \\
&= a_i \cdot \left(\sum_{l=1}^{m} c_{lj} \cdot e_l \right) \\
&= \sum_{l=1}^{m} \big(a_i c_{lj}\big) \cdot e_l \\
&= \sum_{l=1}^{m} \left(\sum_{k=1}^{n} M\big(c_{lj}\big)_{ki} a_k \right) \cdot e_l \\
&= \sum_{l=1}^{m} \sum_{k=1}^{n} M\big(c_{lj}\big)_{ki} \cdot (a_k \cdot e_l)
\end{aligned}$$

Consequentally the matrix for u_K with respect to the basis $\big(a_i \cdot e_j\big)$ is given in block form by

$$\begin{pmatrix} M(c_{11}) & \dots & M(c_{1m}) \\ \vdots & & \vdots \\ M(c_{m1}) & \dots & M(c_{mm}) \end{pmatrix}$$

Therefore we see that

$$\begin{aligned}
\mathrm{Tr}(u_K) &= \sum_{j=1}^{m} \mathrm{Tr}\big(M\big(c_{jj}\big)\big) \\
&= \sum_{j=1}^{m} \mathrm{Tr}_{A/K}\big(c_{jj}\big) \\
&= \mathrm{Tr}_{A/K}\left(\sum_{j=1}^{m} c_{jj} \right) \\
&= \mathrm{Tr}_{A/K}(\mathrm{Tr}(u))
\end{aligned}$$

as required. Similarly by (3.8.36)

$$\det(u_K) = \det(D_m(M(c_{11}), ..., M(c_{mm})))$$

and by the Laplace Expansion

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$$\begin{aligned} D_m(M(c_{11}),...,M(c_{mm})) &= \sum_{\sigma\in S_m}\varepsilon(\sigma)\prod_{i=1}^{m}M\big(c_{i\sigma(i)}\big) \\ &= \sum_{\sigma\in S_m}\varepsilon(\sigma)M\left(\prod_{i=1}^{m}c_{i\sigma(i)}\right) \\ &= M\left(\sum_{\sigma\in S_m}\varepsilon(\sigma)\prod_{i=1}^{m}c_{i\sigma(i)}\right) \\ &= M(\det(u)) \end{aligned}$$

where we implicitly use that $\alpha \to \text{mult}_\alpha \to M(\alpha)$ is a ring homomorphism. Therefore by definition

$$\det(u_K) = \det(M(\det(u))) = \text{Norm}_{A/K}(\det(u))$$

■

Corollary 3.17.130 (Transitivity)
Suppose B is a commutative A-algebra which is finite free as an A-module. Then

$$\text{Tr}_{B/K}(b) = \text{Tr}_{A/K}\big(\text{Tr}_{B/A}(b)\big)$$

and

$$\text{Norm}_{B/K}(b) = \text{Norm}_{A/K}\big(\text{Norm}_{B/A}(b)\big)$$

for all $b \in B$. Furthermore

$$\text{Pc}_{B/K}(b;X) = \text{Norm}_{A[X]/K[X]}\big(\text{Pc}_{B/A}(b;X)\big)$$

Proposition 3.17.131
Let K/k be a finite field extension and $\alpha \in K$. Then

$$\text{Pc}_{K/k}(\alpha;X) = m_{\alpha,k}(X)^{[K:k(\alpha)]}$$

Proof. Consider first the case $K = k(\alpha)$. Then by (3.16.4) we have $\text{Pc}_{K/k}(\alpha;X)(\text{mult}_\alpha) = 0$ and in particular $\text{Pc}_{K/k}(\alpha;X)(\alpha) = 0$ (since $\alpha^n = \alpha^n \cdot 1$). By (...) $m_{\alpha,k}$ divides $\text{Pc}_{K/k}(\alpha;X)$. Further by (...) $\deg\big(m_{\alpha,k}\big) = [k(\alpha):k] = [K:k] = \deg\big(\text{Pc}_{K/k}(\alpha;X)\big)$. Since they are both monic then we find they are equal.

For the general case we may write the matrix for mult_α in block form by considering the tower basis for $K/k(\alpha)/k$ (see (3.17.128)). This is block diagonal, so we may use the Leibniz formula to compute this directly as product of determinants of the blocks. ■

We may pull together these results to give a more classical expression for norm and trace.

Proposition 3.17.132
Let $L/K/k$ be a tower of extensions such that L/k is normal (e.g. $L = \overline{k}$). Then for any $\alpha \in K$ we have the identities

$$i_{kL}\big(\text{Norm}_{K/k}(\alpha)\big) = \left(\prod_{\sigma\in \text{Mor}_k(K,L)}\sigma(\alpha)\right)^{[K:k]_i}$$

and

$$i_{kL}\big(\text{Tr}_{K/k}(\alpha)\big) = [K:k]_i \sum_{\sigma\in \text{Mor}_k(K,L)}\sigma(\alpha)$$

Proof. Recall from (3.17.80) that the restriction map

$$\text{Mor}_k(K,L) \to \text{Mor}_k(k(\alpha),L)$$

is surjective with finite fibers of order $[K:k(\alpha)]_s$. Therefore we may write the right hand side as

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$$\left(\prod_{\sigma\in \mathrm{Mor}_k(K,L)} \sigma(\alpha)\right)^{[K:k]_i} = \left(\prod_{\sigma\in \mathrm{Mor}_k(k(\alpha),L)} \sigma(\alpha)\right)^{[K:k(\alpha)]_s[K:k]_i}$$
$$= \left(\prod_{\sigma\in \mathrm{Mor}_k(k(\alpha),L)} \sigma(\alpha)\right)^{[K:k(\alpha)][k(\alpha):k]_i}$$

Similarly by (3.17.130) and (3.17.126) the left hand side may be written as

$$\begin{aligned}\mathrm{Norm}_{K/k}(\alpha) &= \mathrm{Norm}_{k(\alpha)/k}\big(\mathrm{Norm}_{K/k(\alpha)}(\alpha)\big)\\ &= \mathrm{Norm}_{k(\alpha)/k}\big(\alpha^{[K:k(\alpha)]}\big)\\ &= \mathrm{Norm}_{k(\alpha)/k}\,(\alpha)^{[K:k(\alpha)]}\end{aligned}$$

Comparing these expressions shows we may reduce to the case $K = k(\alpha)$.

From (3.17.51) and (3.17.97) the right hand side is the product of all the roots of $m_{\alpha,k}$ including multiplicity, which is $(-1)^{[K:k]}i_{kL}(c_0)$ where c_0 is the constant term of $m_{\alpha,k}(X)$. Then using (3.17.131) and (3.17.126) we see that this also equals $i_{kL}\big(\mathrm{Norm}_{k(\alpha)/k}(\alpha)\big)$ as required.

The case of trace is similar. ∎

Lemma 3.17.133 (Dedekind's Lemma)
Let K/k be a field extension and A a k-algebra. The set of k-algebra homomorphisms

$$\mathrm{AlgHom}_k(A,K)$$

is K-linearly independent as a subset of $\mathrm{Hom}_k(A,K)$.

Proof. Suppose that the k-algebra homomorphisms are not linearly independent. Then choose a minimal subset of distinct k-algebra homomorphisms $\varphi_1,...,\varphi_n$ with a non-trivial relationship over K

$$\sum_{i=1}^{n}\lambda_i\varphi_i = 0 \quad \lambda_i \in K^\star$$

If $n = 1$ then $\varphi_1 = 0$ which is a contradiction. Suppose without loss of generality that $n > 1$. For a given $x \in K$ we may define

$$\hat{\varphi}_i := (\varphi_i(x) - \varphi_n(x))\varphi_i \quad i = 1...n-1$$

We may choose x such that at least one is non-zero. Then for all $y \in A$ we have

$$\sum_{i=1}^{n-1}\lambda_i\hat{\varphi}_i(y) = \sum_{i=1}^{n}\lambda_i(\varphi_i(x) - \varphi_n(x))\varphi_i(y) = \sum_{i=1}^{n}\lambda_i\varphi_i(xy) - \varphi_n(x)\sum_{i=1}^{n}\lambda_i\varphi_i(y) = 0$$

This contradicts minimality of the subset. ∎

Proposition 3.17.134
Let K/k be a finite extension. Then the following are equivalent

a) K/k is separable
b) $\mathrm{Tr}_{K/k}$ is not identically zero
c) The bilinear form

$$(x,y) \to \mathrm{Tr}_{K/k}(xy)$$

is a perfect pairing (see (3.3.148)).

Proof. *c)* $\implies$ *b)* By (3.3.148) for all $0 \neq x \in K$ we have $y \to \mathrm{Tr}_{K/k}(xy)$ is non-zero, so there exists y such that $\mathrm{Tr}_{K/k}(xy) \neq 0$ as required.

b) $\implies$ *c)* By (3.3.149) it's enough to show that for all $0 \neq x \in K$ the map $y \to \mathrm{Tr}_{K/k}(xy)$ is non-zero. By assumption there exists some z such that $\mathrm{Tr}_{K/k}(z) \neq 0$ then we may consider $y = z/x$.

a) $\implies$ *b)* By (3.17.96) $[K:k]_i = 1$, and so from (3.17.132) the trace equals $\sum_\sigma \sigma$, which is non-zero by (3.17.133).

b) $\implies$ *a)* Suppose K/k is not separable then $[K:k]_i$ is a power of p (3.17.98), and so the trace is zero by (3.17.132). ∎

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3.17.17 Transcendental Field Extensions

Definition 3.17.135 (Algebraic Independence)
*Let K/k be a field extension and $S \subset K$. We say S is **algebraically independent** over k if for every finite subset of distinct elements $x_1, ..., x_n \in S$ we have*

$$f(x_1, ..., x_n) = 0 \implies f = 0$$

for all $f \in k[X_1, ..., X_n]$.

For a subset $S \subset K$ define the closure operator

$$c(S) := \overline{k(S)} \cap K := \{x \in K \mid x \text{ algebraic over } k(S)\}$$

*We say Γ is **algebraically spanning** if $c(\Gamma) = K$, equivalently if $K/k(\Gamma)$ is algebraic.*

*We say that $\mathcal{B}$ is a **transcendence base** if it is both algebraically independent and spanning (i.e. $K/k(\mathcal{B})$ is algebraic).*

Essentially we show that (K, c) satisfies the properties of a matroid, in analogy with vector spaces, so that we can use the results of the Matroids section to show that transcendence bases exist and they satisfy certain properties.

Proposition 3.17.136
Let K/k be a field extension and S, T subsets of K. Then the following are equivalent

a) $S \cup T$ is algebraically independent and $S \cap T = \emptyset$
b) S is algebraically independent over k and T is algebraically independent over $k(S)$
c) T is algebraically independent over k and S is algebraically independent over $k(T)$

Proof. By symmetry it's enough to show that $a) \iff b)$.

Suppose $a)$ holds, then a-fortiori S is algebraically independent over k. Suppose T is algebraically dependent over $k(S)$, then there exists $t_1, ..., t_n \in T$ such that $F(t_1, ..., t_n) = 0$ for some $0 \neq F \in k(S)[X_1, ..., X_n]$. By clearing denominators we may assume that $F \in k[S][X_1, ..., X_n]$. As there are only finitely many coefficients we may also take $S = \{s_1, ..., s_m\}$ to be finite. Explicitly

$$F(X_1, ..., X_n) = \sum_{\alpha \in \mathbb{N}^n} \lambda_\alpha X_1^{\alpha_1} ... X_n^{\alpha_n}$$

and

$$\lambda_\alpha = G_\alpha(s_1, ..., s_m)$$

for some $G_\alpha \in k[Y_1, ..., Y_m]$. We may then define $\hat{F} \in k[X_1, ..., X_n, Y_1, ..., Y_m]$ by

$$\hat{F}(X_1, ..., X_n, Y_1, ..., Y_m) := \sum_{\alpha \in \mathbb{N}^n} G_\alpha(Y_1, ..., Y_m) X_1^{\alpha_1} ... X_n^{\alpha_n}$$

Then $\hat{F}(t_1, ..., t_n, s_1, ..., s_m) = 0$, and by assumption this is an algebraic dependence (as $S \cap T = \emptyset$). Therefore by assumption $\hat{F} = 0$ and in particular $G_\alpha(s_1, ..., s_m) = 0$ for all α. This implies $F = 0$, a contradiction.

$b) \implies a)$ Evidently $S \cap T = \emptyset$ otherwise there is a trivial algebraic dependence of T on $k(S)$. Suppose there is an algebraic dependence of $S \cup T$ on k. We may assume wlog that $S = \{s_1, ..., s_m\}$ and $T = \{t_1, ..., t_n\}$ are finite. Then the evaluation homomorphism

$$k[X_1, ..., X_n, Y_1, ..., Y_m] \to K$$

may be written as a composite

$$k[X_1, ..., X_n, Y_1, ..., Y_m] \to k(S)[X_1, ..., X_n] \to K$$

which are both injective by hypothesis. ∎

Corollary 3.17.137 (Exchange Property)
Let K/k be a field extension, $\Gamma \subseteq K$. Suppose x is algebraic over $k(\Gamma \cup \{y\})$ and transcendental over $k(\Gamma)$, then y is algebraic over $k(\Gamma \cup \{x\})$.

Proof. By considering the extension $K/k(\Gamma)$ we may reduce to the case $\Gamma = \emptyset$.

It's enough to show that x transcendental over k and y transcendental over $k(x)$ implies x transcendental over $k(y)$. This follows directly from (3.17.136) by considering $S = \{x\}$ and $T = \{y\}$. ∎

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Corollary 3.17.138 (Extension Property)
Let K/k be a field extension, $S \subseteq K$ algebraically independent and $x \in K$ transcendental over $k(S)$. Then $S \cup \{x\}$ is algebraically independent.

Proof. Follows immediately from (3.17.136). ∎

Lemma 3.17.139 (Embeddings Preserve Transcendence Degree)
Let K/k, L/k be field extensions of finite transcendence degree and $\phi : K \to L$ a k-algebra homomorphism. Then $\operatorname{trdeg}(K/k) \leq \operatorname{trdeg}(L/k)$.

Proof. As ϕ is a ring homomorphism from a field, $\ker(\phi)$ is a proper ideal of K, hence 0, so ϕ is injective. Let $S \subset K$ be a transcendence base of K/k, so $|S| = \operatorname{trdeg}(K/k)$ (3.17.144); as ϕ is injective, $|\phi(S)| = |S|$. Given distinct $s_1, ..., s_n \in S$ and $f \in k[X_1, ..., X_n]$ with $f(\phi(s_1), ..., \phi(s_n)) = 0$: as ϕ fixes k, $\phi(f(s_1, ..., s_n)) = f(\phi(s_1), ..., \phi(s_n)) = 0$, so $f(s_1, ..., s_n) = 0$ by injectivity of ϕ, whence $f = 0$ by algebraic independence of S over k; so $\phi(S)$ is algebraically independent over k. By (3.17.143) applied to L/k, $\operatorname{trdeg}(K/k) = |S| = |\phi(S)| \leq \operatorname{trdeg}(L/k)$. ∎

Corollary 3.17.140 (Equivalent form of independence)
Let K/k be a field extension and $S \subset K$. Then the following are equivalent

a) S is algebraically independent
b) x is transcendental over $k(S \setminus \{x\})$ for all $x \in S$

In other words the algebraically independent subsets are precisely the matroid independent subsets.

Proof. *a)* $\implies$ *b)* Follows from (3.17.136).

b) $\implies$ *a)* Let $\mathcal{F}$ be the family of algebraically independent subsets. We've shown that all such sets are also matroid independent, and by definition the family is of finite character. Furthermore by (3.17.138) it satisfies the extension property. Therefore by (2.3.6) it is precisely the family of matroid independent sets. ∎

Proposition 3.17.141 (Transcendence Base Exists)
Let K/k be a field extension. Suppose S is an algebraic independent subset of K and $\Gamma \supset S$ is such that $K/k(\Gamma)$ is algebraic. Then there exists a transcendence base $\mathcal{B}$ such that $S \subseteq \mathcal{B} \subseteq \Gamma$.

Proof. This is (2.3.7). ∎

Proposition 3.17.142 (Transcendence Base)
Let K/k be a field extension and $S \subset K$ a subset. Then the following are equivalent

*a) S is a **transcendence base***
b) S is a maximal algebraically independent set
c) S is minimal under the condition $K/k(S)$ is algebraic

*When K/k admits a finite algebraic spanning set then all bases are finite of the same size (**transcendence degree**). Write this as* $\operatorname{trdeg}(K/k)$ *or* $\operatorname{trdeg}_k(K)$.

Proof. Follows from (2.3.8) and (2.3.12). ∎

Proposition 3.17.143
Let K/k be a field extension of finite transcendence degree and $S \subset K$ a subset. Then

- *S algebraically independent* $\implies |S| \leq \operatorname{trdeg}(K/k)$
- *$K/k(S)$ is algebraic* $\implies |S| \geq \operatorname{trdeg}(K/k)$

Proof. Follows directly from (2.3.13). ∎

Proposition 3.17.144
Let K/k be a field extension of finite transcendence degree and $S \subset K$ a subset. Then the following are equivalent

- *S is a transcendence base*
- *S is algebraically independent and* $|S| \geq \operatorname{trdeg}(K/k)$
- *$K/k(S)$ is algebraic and* $|S| \leq \operatorname{trdeg}(K/k)$

In this case $|S| = \operatorname{trdeg}(K/k)$.

Proof. Follows directly from (2.3.14). ∎

In case K/k is finitely generated then we guarantee that $K/k(S)$ is finite.

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Proposition 3.17.145
Let K/k be a finitely generated field extension. Then

- $\text{trdeg}(K/k) < \infty$
- *Suppose $K/k(\Gamma)$ is algebraic then it is in fact finite.*

Proof. *a*) If K/k is finitely generated then the set of generators is algebraically spanning and so $\text{trdeg}(K/k) < \infty$ by (3.17.142)

b) A-fortiori $K/k(\Gamma)$ is finitely generated and by assumption algebraic so by (...) it is finite ∎

3.17.18 Separating Transcendence Base

In applications the existence of a separating transcendence base is important, and is guaranteed when k is perfect, or a weaker condition.

Definition 3.17.146 (Separating Transcendence Base)
*A field extension K/k is **separably generated** if there exists a transcendence base S such that $K/k(S)$ is separable (and algebraic). We call such an S a **separating transcendence base**.*

Note when $\text{char}(k) = 0$ then every transcendence base is separating.

Definition 3.17.147 (Maclane's Criterion)
*We say that an algebra A/k of characteristic exponent p satisfies **Maclane's Criterion** if it is reduced and for all $a_1, ..., a_n \in A$ the following property holds*

$$\{a_1, ..., a_n\} \text{ k-linearly independent} \implies \{a_1^p, ..., a_n^p\} \text{ k-linearly independent}$$

When $p = 1$ this is trivially always satisfied and when $p > 1$ such an algebra is automatically reduced by the following result

Lemma 3.17.148
Let A be a k-algebra and $m > 1$ an integer. Then the following are equivalent

a) A is reduced
b) $x^m = 0 \implies x = 0$

In particular if A has characteristic exponent $p > 1$ then it is reduced iff the Frobenius homomorphism is injective.

Proof. $a) \implies b)$ is trivial. Conversely suppose $x^n = 0$. Then there exists $a > 0$ such that $n \leq am$. In particular $x^{am} = 0$ and therefore by induction $x = 0$ as required. ∎

This is a generalization of the case of a perfect base field by the following result.

Proposition 3.17.149 (Perfect $\implies$ Maclane)
Let A/k be a reduced algebra with k perfect. Then it satisfies Maclane's Criterion.

In particular this holds for every field extension K/k of a perfect base field.

Proof. By assumption A is reduced so we may consider only the case $p > 1$.

Suppose $0 = \sum_i \lambda_i a_i^p$ then by hypothesis $\lambda_i = \mu_i^p$. By (3.17.6) $0 = \left(\sum_i \mu_i a_i\right)^p$. As A is reduced we find $0 = \sum_i \mu_i a_i$ as required. ∎

Lemma 3.17.150
Let $n \geq 0$ an integer and $K = k(x_1, ..., x_{n+1})$ an extension of k such that

- $\text{char}(k) \neq 0$
- $\{x_1, ..., x_n\}$ *is a transcendence base*
- K/k *satisfies Maclane's Criterion*

*Then for some x_i the set $\{x_1, ..., \hat{x_i}, ..., x_{n+1}\}$ is a **separating transcendence base** for K/k.*

Note when $n = 0$ this means precisely that a simple algebraic extension satisfying Maclane's Criterion is separable.

Proof. By assumption x_{n+1} is algebraic over $k(x_1, ..., x_n)$, and therefore there exists a non-zero $F \in k[X_1, ..., X_{n+1}]$ such that $F(x_1, ..., x_{n+1}) = 0$. Let F be such a polynomial of minimal total degree (total degree = the maximum degree of a monomial with non-zero coefficient appearing in F). We claim it is irreducible. For suppose not, then one of the irreducible factors must vanish at $(x_1, ..., x_{n+1})$, and this factor would have smaller total degree.

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We wish to show that some monomial X^α of F (3.10.1) has an exponent α_i that is not a multiple of $p := \mathrm{char}(k)$, for some i. Suppose not, so that every exponent of every monomial of F is a multiple of p; then the elements

$$\left\{x_1^{\alpha_1}...x_{n+1}^{\alpha_{n+1}} \mid \lambda_\alpha \neq 0\right\}$$

are k-linearly dependent where λ_α is the coefficient of the monomial X^α in F and $\alpha \in \mathbb{N}^{n+1}$. Then by Maclane's Criterion the set

$$\left\{x_1^{\alpha_1/p}...x_{n+1}^{\alpha_{n+1}/p} \;\middle|\; \lambda_\alpha \neq 0\right\}$$

is linearly dependent, so $\sum_\alpha \mu_\alpha X^{\alpha/p}$ is a non-zero polynomial vanishing at x (distinct α give distinct monomials) of total degree at most $(\deg F)/p < \deg F$. This contradicts the minimality of F.

Choose such a monomial X^α and index i, and put $m_0 := \alpha_i$, so that $p \nmid m_0$ and in particular $m_0 \geq 1$, and define

$$F_i(T) := F(x_1, ..., x_{i-1}, T, x_{i+1}, ..., x_{n+1}) \in k[S_i][T]$$

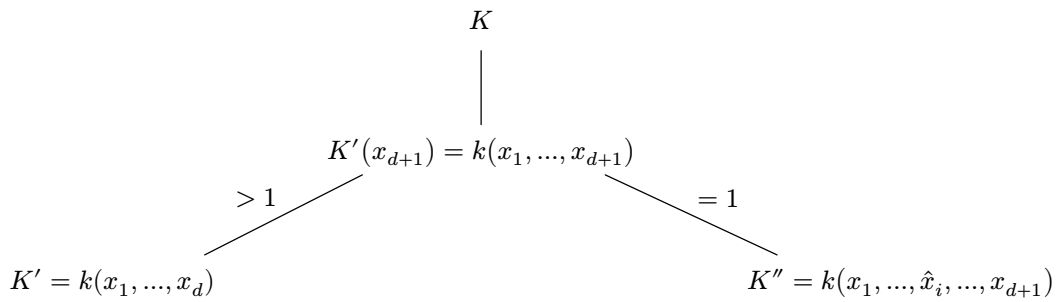
where $S_i := \{x_1, ..., \hat{x_i}, ..., x_{n+1}\}$. The coefficient of T^{m_0} in $F_i(T)$ is the value at S_i of the coefficient of $X_i^{m_0}$ in F, a non-zero polynomial in the other variables of total degree $< \deg F$, so is non-zero by minimality of F. Hence $F_i(T)$ is non-zero and clearly $F_i(x_i) = 0$ so that x_i is algebraic over $k(S_i)$, and therefore by (3.17.44) K is also algebraic over $k(S_i)$. By (3.17.144) S_i is a transcendence base for K and in particular algebraically independent. Therefore $k[S_i][T]$ may be identified with $k[X_1, ..., X_{n+1}]$ (the evaluation $X_j \mapsto x_j$, $j \neq i$, is an isomorphism $k\left[X_j \mid j \neq i\right] \xrightarrow{\sim} k[S_i]$ by algebraic independence, and $k\left[X_j \mid j \neq i\right][X_i] \cong k[X_1, ..., X_{n+1}]$ (3.10.4) under the resulting isomorphism F corresponds to F_i, as F_i is obtained from F by substituting x_j for X_j and T for X_i) and we may conclude that $F_i(T)$ is irreducible. Further $k[S_i] \cong k\left[X_j \mid j \neq i\right]$ is a UFD (3.15.46) and F_i has positive degree in T, so by (3.15.44) $F_i(T)$ is irreducible as a polynomial in $k(S_i)[T]$. By construction (its coefficient of T^{m_0} is non-zero and $p \nmid m_0$) $F_i(T) \notin k(S_i)[T^p]$ so by (3.17.94) F_i is separable. Therefore x_i is separable over $k(S_i)$, and $K/k(S_i)$ is separable by (3.17.82) as required. ∎

Proposition 3.17.151
Let K/k be a finitely-generated field extension which satisfies Maclane's Criterion (e.g. k is perfect). Then K/k is separably generated.

More precisely every generating set contains a separating transcendence base.

Proof. When $\mathrm{char}(k) = 0$ the generating set contains a transcendence base by (3.17.142). Every subfield of K also has characteristic 0, so then we are done by (3.17.106). So we may only consider the positive characteristic case, $p > 1$.

Suppose $K = k(x_1, ..., x_n)$. By (3.17.141) there is a (possibly empty) subset which is a transcendence base, say $\{x_1, ..., x_d\}$ after renumbering. By (3.17.44) $[K : K'] < \infty$ where $K' = k(x_1, ..., x_d)$, and we may choose K' such that the degree of inseparability $[K : K']_i$ is minimal. If K/K' is separable then we are done. Otherwise we may assume by renumbering that $K'(x_{d+1})/K'$ is not separable (3.17.82) and therefore $\left[K'(x_{d+1}) : K'\right]_i > 1$ (3.17.96). Then by (3.17.150) there is an index $i \leq d+1$ such that $\left[K'(x_{d+1}) : K''\right]_i = 1$ where $K'' := k(x_1, ..., \hat{x}_i, ..., x_{d+1})$. The fields involved form the following lattice, where the label on an edge is the degree of inseparability of the upper field over the lower one:



By multiplicativity of the degree of inseparability (3.17.96) along the two paths from K down the lattice we obtain

$$[K : K']_i > [K : K'']_i$$

which contradicts minimality. ∎

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3.18 Local Rings

Local rings arise quite naturally when localizing at a prime ideal (see (3.5.42) and (3.18.5)) so we recall some basic properties here.

Definition 3.18.1 (Local Ring)
*A ring A is a **local ring** if it has a unique maximal ideal $\mathfrak{m}$. The field $A/\mathfrak{m}$ is called the **residue field** of A and sometimes denoted $\kappa(\mathfrak{m})$.*

When A is a k-algebra we may denote the residue field by $k(\mathfrak{m})$.

Definition 3.18.2 (Local Homomorphism)
*Let $(A, \mathfrak{m}_A)$ and $(B, \mathfrak{m}_B)$ be local rings. A ring homomorphism $\phi : A \to B$ is said to be a **local homomorphism** if*

$$\phi(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$$

Recall that the group of units $A^\star$ of a ring is a saturated multiplicative set, that is

$$xy \in A^\star \iff x \in A^\star \wedge y \in A^\star$$

Proposition 3.18.3 (Criteria for Local Rings)
Let A be a ring. Then the following are equivalent

a) A is a local ring
b) $A \setminus A^\star$ is an additive subgroup of A

In this case $\mathfrak{m} = A \setminus A^\star$ is the unique maximal ideal of A.

Proof. a) $\implies$ b) Let $\mathfrak{m}$ be the unique maximal ideal then, because it's proper, $\mathfrak{m} \cap A^\star = \emptyset \implies \mathfrak{m} \subseteq A \setminus A^\star$ by (3.3.13). Conversely given $x \in A \setminus A^\star$ then (x) is a proper ideal by (3.3.33), and therefore contained in a maximal ideal (3.3.15) which by uniqueness means $x \in \mathfrak{m}$.

b) $\implies$ a) Define $\mathfrak{m} = A \setminus A^\star$ it's a (prime) ideal because it is an additive subgroup and $A^\star$ is a saturated multiplicative set. Let $\mathfrak{a}$ be a proper ideal then $\mathfrak{a} \cap A^\star = \emptyset \implies \mathfrak{a} \subseteq \mathfrak{m}$. Therefore $\mathfrak{m}$ is the unique maximal ideal. ∎

Proposition 3.18.4 (Quotient of a Local Ring is Local)
Let $(A, \mathfrak{m})$ be a local ring and $\mathfrak{a} \lhd A$ a proper ideal. Then $A/\mathfrak{a}$ is a local ring, with maximal ideal $\mathfrak{m}/\mathfrak{a}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\mathfrak{a}$ is proper, it is contained in some maximal ideal (3.3.15), which must be $\mathfrak{m}$ by uniqueness; so $\mathfrak{a} \subseteq \mathfrak{m}$, and by the ideal correspondence of (3.3.56), the maximal ideals of $A/\mathfrak{a}$ correspond to the maximal ideals of A containing $\mathfrak{a}$ – of which $\mathfrak{m}$ is the only one (as A is local) – so $A/\mathfrak{a}$ has the unique maximal ideal $\mathfrak{m}/\mathfrak{a}$, that is, is local. ∎

Example 3.18.5
Let A be a ring and $\mathfrak{p} \lhd A$ a prime ideal. Then $A_\mathfrak{p}$ is a local ring with unique maximal ideal $\mathfrak{p}A_\mathfrak{p}$.

When $A \subset K$ is a subring of a field then $A \subset A_\mathfrak{p} \subset K$ in a natural way.

We may use this to provide another criteria

Lemma 3.18.6 (Criteria for Local Domain)
Let $A \subset K$ be a subring of a field with a prime ideal $\mathfrak{p} \lhd A$. Then $A \subset A_\mathfrak{p}$.

A is a local ring with unique maximal ideal $\mathfrak{p}$ if and only if $A = A_\mathfrak{p}$,

Proof. We've observed that $A_\mathfrak{p}$ is a local ring with unique maximal ideal $\mathfrak{p}A_\mathfrak{p}$.

If $A = A_\mathfrak{p}$ then it is a local ring and $\mathfrak{p} \subset \mathfrak{p}A_\mathfrak{p}$. For $y \notin \mathfrak{p}$, then $\frac{1}{y} \in A_\mathfrak{p} \implies \frac{1}{y} \in A$. So we see that $x \in \mathfrak{p}, y \notin \mathfrak{p}$ we have $\frac{x}{y} \in \mathfrak{p}$ and $\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$.

Conversely suppose A is a local ring with unique maximal ideal $\mathfrak{p}$. Then $y \notin \mathfrak{p} \implies y \in A^\star$ and $A_\mathfrak{p} = A$ as required. ∎

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3.19 Modules over Local Rings (Nakayama's Lemma)

The main result of this section (3.19.7) is that every finitely-generated module over a local ring has a minimal spanning set. Recall in the vector space case a minimal spanning set is precisely a basis (2.3.8). Analogously we may also show that (in the local case) every minimal spanning set has the same order. The crucial result is Nakayama's Lemma, which we develop here.

Definition 3.19.1 (Jacobson Radical)
Let A be a commutative ring. Define the ***Jacobson Radical*** *to be the intersection of all maximal ideals*

$$\sqrt{0}^J := \bigcap_{\mathfrak{m} \lhd A} \mathfrak{m}$$

Proposition 3.19.2
The Jacobson Radical $\sqrt{0}^J$ is a proper ideal

Example 3.19.3
When $(A, \mathfrak{m}_A)$ is a local ring then $\sqrt{0}^J = \mathfrak{m}_A$.

Lemma 3.19.4 (Characterization of Jacobson Radical)
For an ideal $\mathfrak{a}$ and $\mathfrak{m}$ a maximal ideal

a) $\mathfrak{a} \not\subseteq \mathfrak{m} \iff \mathfrak{a} + \mathfrak{m} = A \iff (1 + \mathfrak{a}) \cap \mathfrak{m} \neq \emptyset$
b) $\mathfrak{a} \subseteq \sqrt{0}^J \iff 1 + \mathfrak{a} \subseteq A^\star$
c) $x \in \sqrt{0}^J \iff 1 + (x) \subseteq A^\star$

Proof. We prove each in turn.

a) Clearly $\mathfrak{a} \subseteq \mathfrak{m} \implies \mathfrak{a} + \mathfrak{m} = \mathfrak{m}$. Conversely $\mathfrak{a} \not\subseteq \mathfrak{m} \implies \mathfrak{a} + \mathfrak{m} = A$ by maximality. Suppose $\mathfrak{a} + \mathfrak{m} = A$ then $1 = a + m$ whence $(1 - a) \in \mathfrak{m}$. The converse is similar.

b) By a) $\mathfrak{a} \subseteq \sqrt{0}^J \implies (1 + \mathfrak{a}) \cap \mathfrak{m} = \emptyset$ for all maximal ideals $\mathfrak{m}$. By (3.3.15) this implies $(1 + \mathfrak{a}) \subseteq A^\star$.

Conversely if $(1 + \mathfrak{a}) \subseteq A^\star$ then $(1 + \mathfrak{a}) \cap \mathfrak{m} = \emptyset$ for any maximal ideal $\mathfrak{m}$ by (3.3.13). Again by a) $\mathfrak{a} \subseteq \mathfrak{m}$ as required.

c) This follows from b) and noting $x \in \sqrt{0}^J \iff (x) \subseteq \sqrt{0}^J$. ∎

Proposition 3.19.5 (Nakayama's Lemma)
Let M be a finitely generated A-module and $\mathfrak{a} \lhd A$ an ideal. Then the following holds

a) If $M = \mathfrak{a}M$ then there exists $a \in \mathfrak{a}$ such that $m = am$ for all $m \in M$

Suppose in addition that $\mathfrak{a} \subseteq \sqrt{0}^J$ (e.g. if A is local and $\mathfrak{a}$ is proper) then

b) $M = \mathfrak{a}M \implies M = 0$
c) $N \leq M$ *and* $M = N + \mathfrak{a}M \implies M = N$.

Proof. We prove each in turn

a) Apply (3.16.4) with $\phi := \mathbf{1}_M$ to find a monic polynomial $P(X) \in A[X]$ with non-leading coefficients in $\mathfrak{a}$ such that $P(\phi)(m) = 0$ for all $m \in M$. Then we see that $(1 + a_{n-1} + ...a_0)m = 0$ for all $m \in M$ whence $a := -(a_{n-1} + ... + a_0)$ is the required element.

More directly, suppose $m_1, ..., m_n$ is a generating set for M. By (3.16.3) (and $M = \mathfrak{a}M$) there is a matrix E with coefficients in $\mathfrak{a}$ such that

$$(I_n - E)\boldsymbol{m} = 0$$

where $\boldsymbol{m}$ is the column vector consisting of $m_1, ..., m_n$. By (3.8.24) we see $\det(I_n - E)m_i = 0$ for all $i = 1...n$. It's enough to show $a := \det(I_n - E) \in 1 + \mathfrak{a}$. Observe

$$\det(I_n - E) = \prod_i (1 - E_{ii}) + \sum_{\sigma \neq \mathrm{id}} \epsilon(\sigma) \prod_j E_{j\sigma(j)}$$

The second term lies in $\mathfrak{a}$ and

$$\prod_i (1 - E_{ii}) = 1 - \sum_{i=1}^n E_{ii} \prod_{j>i} (1 - E_{jj}) \in 1 + \mathfrak{a}$$

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b) Consider any $m \in M$. By a) we have $(1-a)m = 0$ for some $a \in \mathfrak{a}$, and by (3.19.4) $(1-a)$ is invertible, whence $m = 0$ as required.

c) Observe $\mathfrak{a}(M/N) \overset{(3.3.107)}{=} (N + \mathfrak{a}M)/N = M/N$ whence $M/N = 0$ by b). Therefore $N = M$ as required.

We may show b) more directly. Suppose $M \neq 0$, and let

$$\{m_1, ..., m_n\}$$

be a non-zero generating set for M of minimal size. Then by (3.16.3)

$$m_1 = \sum_j a_j m_j \quad a_j \in \mathfrak{a}$$

whence

$$(1-a_1)m_1 = \sum_{j \geq 2} a_j m_j$$

As $a_1 \in \sqrt{0}^J$ we have $1 - a_1 \in A^\star$ by (3.19.4). Then

$$\{m_2, ..., m_n\}$$

is a smaller generating set, a contradiction. Therefore $M = 0$.

a) may be deduced from b) as follows. Observe $S \coloneqq 1 + \mathfrak{a}$ is a multiplicatively closed subset, so we may consider $S^{-1}M$ as an $S^{-1}A$-module. It's easy to verify that $1 + S^{-1}\mathfrak{a} \subseteq (S^{-1}A)^\star$ so by (3.19.4) $S^{-1}\mathfrak{a} \subseteq J(S^{-1}A)$. Clearly $\mathfrak{a}M = M \implies (S^{-1}\mathfrak{a})S^{-1}M = S^{-1}M$ so by the weaker form $S^{-1}M = 0$. By (3.5.16) there exists $s \in S$ such that $sM = 0$, which is the required result as $s = 1 + a$ for some $a \in \mathfrak{a}$. ∎

Recall (3.3.108) in the case of a local ring $(A, \mathfrak{m})$ that $\widetilde{M} \coloneqq M/\mathfrak{m}M$ is a vector space over $k \coloneqq A/\mathfrak{m}$. We may use Nakayama's Lemma to exhibit a correspondence between minimal spanning sets of M and bases of $M/\mathfrak{m}M$ as a k-vector space. First we prove a simpler form

Lemma 3.19.6
Let $(A, \mathfrak{m})$ be a local ring with residue field $k = A/\mathfrak{m}$, M a finite A-module and $S \subset M$ a subset. Then

$$S \text{ spans } M \iff \tilde{S} \text{ spans } \widetilde{M}$$

where $\tilde{}$ denotes reduction modulo $\mathfrak{m}M$.

Proof. One direction is obvious. Conversely suppose $\tilde{S}$ spans $\widetilde{M}$. Define $N \coloneqq \langle S \rangle$, then this means precisely that $N + \mathfrak{m}M = M$, so $N = M$ by (3.19.5) c) as required. ∎

Recall from (2.1.57) that every subset of T of $M/\mathfrak{m}M$ may be written in the form $\tilde{S}$ for $S \subset M$ and $\tilde{}$ injective on S.

Proposition 3.19.7 (Structure theorem for modules over a local ring)
Let $(A, \mathfrak{m})$ be a local ring with residue field $k = A/\mathfrak{m}$, M a finite A-module. Then

- *a) $\widetilde{M} \coloneqq M/\mathfrak{m}M$ is a finite-dimensional k-module (of dimension n say)*
- *b) If $\tilde{S}$ is a k-basis for $\widetilde{M}$ and $\tilde{}\,|_S$ is injective, then S is a minimal spanning set for M and $\#S = \#\tilde{S} = n$*
- *c) If S is a minimal spanning set then $\tilde{S}$ is a basis for $\widetilde{M}$ (and $\tilde{}\,|_S$ is injective so $\#S = \#\tilde{S} = n$).*

Proof. a) By (3.3.108) $\widetilde{M}$ is a k-module, and it's clearly finite.

b) By (3.19.6) S spans M. Suppose $S' \subset S$ spans M, then by the same result $\widetilde{S'}$ spans $\widetilde{M}$. Recall (2.3.8) that a vector space basis is precisely a minimal spanning set, so $\widetilde{S'} = \tilde{S}$. As $\tilde{}$ is injective this means $S' = S$. Therefore S is a minimal spanning set.

c) Let S be a minimal spanning set. Then by (3.19.6) $\tilde{S}$ spans $\widetilde{M}$. Suppose $\tilde{T} \subset \tilde{S}$ also spans $\widetilde{M}$. By (3.19.6) T spans M, and by hypothesis $T = S$. Therefore $\tilde{T} = \tilde{S}$. As $\tilde{T}$ was arbitrary we see that $\tilde{S}$ is a minimal spanning set for $\widetilde{M}$, which by (2.3.8) is a basis.

Finally suppose $\tilde{}$ is not injective on S, that is $\tilde{s_1} = \tilde{s_2}$. Then $S' \coloneqq S \setminus \{s_1\}$ satisfies $\widetilde{S'} = \tilde{S}$. Therefore by the Lemma S' spans M, contradicting minimality. ∎

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3.20 Krull Dimension

Definition 3.20.1 (Krull Dimension)
Let A be a commutative ring. We say that a chain of distinct prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

*has **length** n.*

a) *The **Krull dimension** $\dim A$ of S is the maximum length of all chains of prime ideals.*
b) *The **height** of a prime ideal $\mathfrak{p}$, denoted $\operatorname{ht}(\mathfrak{p})$, is the maximum length of chains of prime ideals contained in $\mathfrak{p}$. More generally define $\operatorname{ht}(\mathfrak{a}) = \inf\{\operatorname{ht}(\mathfrak{p}) \mid \mathfrak{a} \subseteq \mathfrak{p}\}$. By (3.3.44) we may take the infimum over only minimal prime ideals.*
c) *The **dimension** of an ideal $\mathfrak{a} \lhd A$, denoted $\dim \mathfrak{a}$, is the maximum length of chains of prime ideals containing $\mathfrak{a}$.*

*We say A is **finite-dimensional** if $\dim A < \infty$. Observe that $\mathfrak{a} \subseteq \mathfrak{p} \iff \sqrt{\mathfrak{a}} \subseteq \mathfrak{p}$ for any prime ideal $\mathfrak{p}$ so*

$$\dim \mathfrak{a} = \dim \sqrt{\mathfrak{a}}$$

$$\operatorname{ht}(\mathfrak{a}) = \operatorname{ht}\left(\sqrt{\mathfrak{a}}\right)$$

and we may, without loss of generality, consider only radical ideals.

Definition 3.20.2
Let A be a commutative ring. We say a chain of prime ideals is

- ***maximal** if it's not properly contained in any chain*
- ***saturated** if $\mathfrak{p}_i \subseteq \mathfrak{p} \subseteq \mathfrak{p}_{i+1} \implies \mathfrak{p} = \mathfrak{p}_i$ or $\mathfrak{p} = \mathfrak{p}_{i+1}$.*

*We say that A is **biequidimensional** if every maximal chain has the same length (equal to $\dim A$).*

*We say A is **quasi-biequidimensional** if $A/\mathfrak{p}$ is **biequidimensional** for every minimal prime $\mathfrak{p}$. Note that **equidimensional** + **quasi-biequidimensional** $\iff$ **biequidimensional**.*

*We say that A is **catenary** if the length of a saturated chain*

$$\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

depends only on $\mathfrak{p}_0$ and $\mathfrak{p}_n$ and is equal to $\operatorname{ht}(\mathfrak{p}_n/\mathfrak{p}_0)$

*We say that A is **equidimensional** if every minimal prime ideal has the same dimension (equal to $\dim A$). Note an **irreducible** ring has only one minimal prime ideal so is trivially equidimensional.*

*We say that A is **equicodimensional** if every maximal ideal has the same height (equal to $\dim A$).*

In order to connect this to the lattice-theoretic notion of Krull Dimension (see the Krull Dimension section of Foundations) we prove the following result (provided we consider the lattice of radical ideals ordered by *reverse* inclusion).

Proposition 3.20.3
Let A be a ring then the lattice of radical ideals $\operatorname{Rad}(A)$ is distributive, that is we have equality

$$\mathfrak{r}_1 \cap \sqrt{\mathfrak{r}_2 + \mathfrak{r}_3} = \sqrt{\mathfrak{r}_1 \cap \mathfrak{r}_2 + \mathfrak{r}_1 \cap \mathfrak{r}_3}$$

*Furthermore the meet-prime radical ideals are precisely the prime ideals. Therefore the lattice of radical ideals of a finite-dimensional Noetherian ring, ordered by **reverse inclusion**, is a Krull Lattice.*

Proof. Clearly it's enough to show that LHS $\subseteq$ RHS. Suppose $x \in$ LHS then $x \in \mathfrak{r}_1$ and $x^n = a + b$ where $a \in \mathfrak{r}_2$ and $b \in \mathfrak{r}_3$. Then $x^{n+1} = ax + bx \in \mathfrak{r}_1 \cap \mathfrak{r}_2 + \mathfrak{r}_1 \cap \mathfrak{r}_3$ whence $x \in$ RHS.

We've shown that prime ideals are meet-prime (3.3.40). Suppose $\mathfrak{r}$ is a meet-prime radical ideal, and $fg \in \mathfrak{r}$. Then we claim that

$$\sqrt{\mathfrak{r} + (f)} \cap \sqrt{\mathfrak{r} + (g)} \subseteq \mathfrak{r}$$

For $x \in \text{LHS} \implies x^n \in \mathfrak{r} + (f)$ and $x^m \in \mathfrak{r} + (g) \implies x^{n+m} \in \mathfrak{r} \implies x \in \mathfrak{r}$. As $\mathfrak{r}$ is meet-prime then for example $\sqrt{\mathfrak{r} + (f)} \subseteq \mathfrak{r}$ and in particular $f \in \mathfrak{r}$. Therefore $\mathfrak{r}$ is also prime. ∎

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Proposition 3.20.4 (Simple properties)
The following properties of Krull dimension hold

a) $\dim A = \dim A/N(A)$
b) $\mathrm{ht}(\mathfrak{p}) = \dim A_{\mathfrak{p}}$
c) $\dim \mathfrak{a} = \dim A/\mathfrak{a}$
d) $\dim \mathfrak{a} = \dim \mathfrak{a}/\mathfrak{b}$ *for any ideal* $\mathfrak{b} \subseteq \mathfrak{a}$
e) $\dim A = \sup_{\mathfrak{p}} \dim A_{\mathfrak{p}}$
f) ***codimension inequality*** $\mathrm{ht}(\mathfrak{p}) \geq \mathrm{ht}(\mathfrak{p}/\mathfrak{q}) + \mathrm{ht}(\mathfrak{q})$
g) $\dim k = 0$ *for any field* k
h) A principal ideal domain A *which is not a field has dimension* 1

Proof. a) By (3.3.56) there is an order-isomorphism between prime ideals of A containing $N(A)$ and prime ideals of $A/N(A)$. However by (3.3.46) all prime ideals of A contain $N(A)$, so there is a bijection between chains of A and chains of $A/N(A)$, and the result follows.

b) This follows similarly from (3.5.42).

c) This follows similarly from (3.3.56).

d) $\dim \mathfrak{a} = \dim A/\mathfrak{a} = \dim(A/\mathfrak{a})/(\mathfrak{a}/\mathfrak{b}) = \dim \mathfrak{a}/\mathfrak{b}$

e) This follows from (2.5.6)

f) This follows from (2.5.6)

g) The only (prime) ideal is (0)

h) By (...) every prime ideal (besides (0)) is maximal so every chain has length at most 1. ∎

Lemma 3.20.5
Let A *be a UFD and* $\mathfrak{p} \lhd A$ *a non-zero prime ideal. Then it contains a non-zero principal prime ideal* (p).

In particular $\mathrm{ht}(\mathfrak{p}) = 1$ *if and only if it is principal.*

Proof. Choose $0 \neq f \in \mathfrak{p}$. Then by definition it has a factorization into primes, and at least one must be in $\mathfrak{p}$, say p. Then (p) is prime by (3.15.9). Therefore if $\mathrm{ht}(\mathfrak{p}) = 1$ then it is equal to (p).

Conversely if $\mathfrak{q} \subseteq (p)$ then $(q) \subseteq \mathfrak{q} \subseteq (p)$ for q prime, which implies $p \mid q$. As q is irreducible (...) then $(p) = (q) = \mathfrak{q}$. Therefore $\mathrm{ht}((p)) = 1$. ∎

Proposition 3.20.6
Let A *be a Noetherian ring and* $\mathfrak{a}$ *a proper ideal. Then there are only finitely many minimal primes containing* $\mathfrak{a}$*, say* $\mathfrak{p}_1, ..., \mathfrak{p}_n$*. Further we have a decomposition*

$$\sqrt{\mathfrak{a}} = \bigcap_{i=1}^{n} \mathfrak{p}_i$$

which is irredundant (and the only such decomposition). Furthermore

$$\mathrm{ht}(\mathfrak{a}) = \min_i \mathrm{ht}(\mathfrak{p}_i)$$

$$\dim(\mathfrak{a}) = \max_i \dim(\mathfrak{p}_i)$$

Proof. This follows from (2.4.7) and (3.20.3) applied to the radical ideal $\sqrt{a}$. ∎

Remark 3.20.7
This is essentially the proof of [Kap74] (Theorem 87, 88).

We've noted in general (3.20.4) that the so-called codimension formula does not hold. However it holds in the following case, for essentially trivial reasons.

Proposition 3.20.8 (Codimension 1 formula)
Let A *be an irreducible Noetherian ring of finite Krull Dimension, and* $\mathfrak{a}$ *such that* $\dim(\mathfrak{a}) = \dim(A) - 1$ *then* $\mathrm{ht}(\mathfrak{a}) = 1$.

Proof. Apply (2.5.7). ∎

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In practice most rings of geometric interest are catenary or quasi-biequidimensional. We recall some properties and equivalent criteria for these cases.

Proposition 3.20.9 (Catenary Criteria)
Let A be a ring. Then the following are equivalent

a) A is catenary
b) For all prime ideals $\mathfrak{r} \subseteq \mathfrak{q} \subseteq \mathfrak{p}$ the following holds

$$\operatorname{ht}(\mathfrak{p}/\mathfrak{r}) = \operatorname{ht}(\mathfrak{p}/\mathfrak{q}) + \operatorname{ht}(\mathfrak{q}/\mathfrak{r})$$

When A is irreducible this is equivalent to the following condition

$$\operatorname{ht}(\mathfrak{p}) = \operatorname{ht}(\mathfrak{q}) + \operatorname{ht}(\mathfrak{p}/\mathfrak{q})$$

Proof. This is restatement of (2.5.9). ∎

Proposition 3.20.10 (Biequidimensional Criteria)
Let A be a ring. Then the following are equivalent

a) A is quasi-biequidimensional
b) A is catenary and $A/\mathfrak{p}$ is equicodimensional for every minimal prime $\mathfrak{p}$
c) A satisfies the formula

$$\dim \mathfrak{q} = \dim \mathfrak{p} + \operatorname{ht}(\mathfrak{p}/\mathfrak{q}) \quad \forall \mathfrak{q} \subseteq \mathfrak{p}$$

d) A satisfies c) whenever $\operatorname{ht}(\mathfrak{p}/\mathfrak{q}) = 1$ (i.e. $\mathfrak{q} \subsetneq \mathfrak{p}$ is saturated)

Furthermore

$$\operatorname{ht}(\mathfrak{p}) = \operatorname{ht}(\mathfrak{p}/\mathfrak{q}) + \operatorname{ht}(\mathfrak{q})$$

Proof. This is a restatement of (2.5.11). ∎

Proposition 3.20.11 (Irreducible Biequidimensional Criteria)
When A is irreducible the following are equivalent

a) A is biequidimensional
b) A is quasi-biequidimensional
c) A is catenary and equicodimensional
d) A satisfies the formula

$$\dim \mathfrak{q} = \dim \mathfrak{p} + \operatorname{ht}(\mathfrak{p}/\mathfrak{q}) \quad \forall \mathfrak{q} \subseteq \mathfrak{p}$$

e) A satisfies d) whenever $\operatorname{ht}(\mathfrak{p}/\mathfrak{q}) = 1$ (i.e. $\mathfrak{q} \subsetneq \mathfrak{p}$ is saturated)

Proof. This is a restatement of (2.5.14). ∎

Proposition 3.20.12 (Codimension Formula)
Let A be a quasi-biequidimensional ring, $\mathfrak{a}$ an ideal and $\mathfrak{p} \subset \mathfrak{a}$ a prime ideal. Then the following properties hold

$$\operatorname{ht}(\mathfrak{a}) = \operatorname{ht}(\mathfrak{a}/\mathfrak{p}) + \operatorname{ht}(\mathfrak{p})$$
$$\dim \mathfrak{p} = \dim \mathfrak{a} + \operatorname{ht}(\mathfrak{a}/\mathfrak{p})$$

When A is biequidimensional (e.g. quasi-biequidimensional and integral), then for all ideals $\mathfrak{a}$ we have the relation

$$\dim A = \dim \mathfrak{a} + \operatorname{ht}(\mathfrak{a})$$

Proof. This is a restatement of (2.5.13). ∎

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3.21 Lying over, Incomparability, Going Up and Going Down

Definition 3.21.1 (Lying over / Going up)
Let $\phi : A \to B$ be a ring map and $\mathfrak{p}$ and $\mathfrak{q}$ primes of A and B respectively

a) $\mathfrak{q}$ ***lies over*** $\mathfrak{p}$, *or* $\mathfrak{p}$ ***lies under*** $\mathfrak{q}$ *if* $\mathfrak{p} = \phi^{-1}(\mathfrak{q}) = \mathfrak{q}^c$. *When $A \subseteq B$ and ϕ is the identity then this is equivalent to saying $\mathfrak{p} = \mathfrak{q} \cap A$.*

Definition 3.21.2 (Lying Over / Going Up / Incomparability)
Let $\phi : A \to B$ be a ring map. We say that it has the

a) ***lying over property*** *if every prime ideal $\mathfrak{p} \supseteq \ker(\phi)$ has a prime $\mathfrak{q}$ lying over it. NB $\ker(\phi) \subseteq \mathfrak{p}$ is a necessary condition for $\mathfrak{p}$ to be a contraction and is equivalent to $B_{\mathfrak{p}} \neq 0$.*
b) ***going up property*** *if for every pair of prime ideals $\mathfrak{p} \subsetneq \mathfrak{p}'$ in A and $\mathfrak{q} \lhd B$ lying over $\mathfrak{p}$, there exists a prime ideal $\mathfrak{q}'$ such that $\mathfrak{q} \subsetneq \mathfrak{q}'$ and $\mathfrak{q}'$ lies over $\mathfrak{p}'$.*
c) ***incomparability property*** *if for every pair of prime ideals $\mathfrak{q}, \mathfrak{q}' \lhd B$ then $\mathfrak{q} \subsetneq \mathfrak{q}' \implies \phi^{-1}(\mathfrak{q}) \subsetneq \phi^{-1}(\mathfrak{q}')$*
d) ***going down property*** *if for every pair of prime ideals $\mathfrak{p}' \subsetneq \mathfrak{p}$ in A and $\mathfrak{q} \lhd B$ lying over $\mathfrak{p}$, there exists a prime ideal $\mathfrak{q}'$ such that $\mathfrak{q}' \subsetneq \mathfrak{q}$ and $\mathfrak{q}'$ lies over $\mathfrak{p}'$.*

Remark 3.21.3
It's possible to interpret these geometrically in terms of the map $\phi_\star : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$.

- *Lying over - $\phi_\star$ is surjective onto $V(\ker(\phi))$*
- *Going up - $\phi_\star$ is closed*
- *Incomparability - fibres have dimension 0*
- *Going down (and finite presentation) - $\phi_\star$ is open*

The main result of this section is the correspondence between primes lying over $\mathfrak{p}$ and primes of the ring $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ (3.21.5). As a preliminary result we consider conditions under which we can strengthen $\mathfrak{p} \subseteq \mathfrak{q}^c$ to $\mathfrak{p} = \mathfrak{q}^c$, as the former condition is somewhat easier to satisfy.

Lemma 3.21.4 (Lying over criteria)
Let $\phi : A \to B$ be a ring map, $\mathfrak{p} \lhd A$ prime such that $\ker(\phi) \subseteq \mathfrak{p}$ and $\mathfrak{q} \lhd B$. Then

$$\mathfrak{p} = \mathfrak{q}^c \iff \mathfrak{p}^e \subseteq \mathfrak{q} \text{ and } \mathfrak{q} \cap \phi(A \setminus \mathfrak{p}) = \emptyset$$

In particular

$$\mathfrak{p} = \mathfrak{p}^{ec} \text{ is contracted} \iff \mathfrak{p}^e \cap \phi(A \setminus \mathfrak{p}) = \emptyset$$

Proof. Recall (3.3.52) that in general $\mathfrak{p} \subseteq \mathfrak{q}^c \iff \mathfrak{p}^e \subseteq \mathfrak{q}$. The first equivalence is then clear because $x \in \mathfrak{q}^c \setminus \mathfrak{p} \iff \phi(x) \in \mathfrak{q} \cap \phi(A \setminus \mathfrak{p})$. For $\phi(x) \in \mathfrak{q} \cap \phi(A \setminus \mathfrak{p}) \implies \phi(x) = \phi(y)$ for $y \notin \mathfrak{p}$ and $x \in \mathfrak{q}^c$. This implies $x - y \in \ker(\phi) \subseteq \mathfrak{p}$ whence $x \notin \mathfrak{p}$ as required.

The final statement follows by considering $\mathfrak{q} = \mathfrak{p}^e$. ∎

Proposition 3.21.5 (Lying over correspondence)
Let $\phi : A \to B$ be a ring map and $\mathfrak{p} \lhd A$ a prime ideal s.t. $\ker(\phi) \subseteq \mathfrak{p}$. Then there is a order-preserving correspondence of prime ideals

$$\begin{aligned} \{\mathfrak{q} \mid \mathfrak{q} \text{ lies above } \mathfrak{p}\} &\longleftrightarrow \left\{\mathfrak{q}' \lhd B_{\mathfrak{p}} \;\middle|\; \mathfrak{p}B_{\mathfrak{p}} \subseteq \mathfrak{q}'\right\} &&\longleftrightarrow \left\{\mathfrak{q}'' \lhd B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}\right\} \\ \mathfrak{q} &\longrightarrow \mathfrak{q}B_{\mathfrak{p}} && \longrightarrow \mathfrak{q}B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} \end{aligned}$$

In particular TFAE

a) *$\mathfrak{p}$ lies under a prime $\mathfrak{q}$*
b) *$B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} \neq 0$*
c) *$\mathfrak{p} = \mathfrak{p}^{ec}$ is contracted*

NB c) is a-priori weaker than a).

Proof. Recall $B_{\mathfrak{p}} := S^{-1}B$ and $\mathfrak{p}B_{\mathfrak{p}} := \mathfrak{p}^e B_{\mathfrak{p}} = S^{-1}\mathfrak{p}^e$ where $S := \phi(A \setminus \mathfrak{p})$.

By (3.21.4) $\mathfrak{q}$ lies above $\mathfrak{p}$ if and only if $\mathfrak{p}^e \subseteq \mathfrak{q}$ and $\mathfrak{q} \cap S = \emptyset$. The first correspondence then follows from (3.5.19) and the second from (3.3.56).

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a) $\iff$ b) This follows from the correspondence, since a ring without any non-zero prime ideals is simply the zero-ring.

b) $\iff$ c) Observe $\mathfrak{p} = \mathfrak{p}^{ec} \iff \mathfrak{p}^e \cap S = \emptyset$ by (3.21.4), which holds if and only if $\mathfrak{p}B_\mathfrak{p} \neq B_\mathfrak{p}$ by (3.5.18) e). ∎

Remark 3.21.6
Geometrically this is an explicit representation of the fiber as a prime spectrum

$$\operatorname{Spec}(\phi)^{-1}(\mathfrak{p}) \longleftrightarrow \operatorname{Spec}\big(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}\big)$$

Proposition 3.21.7 (Injective Ring Maps are Dense)
Let $\phi : A \to B$ be a ring map and $\mathfrak{p} \lhd A$ a minimal prime over $\ker(\phi)$. Then there exists a prime ideal $\mathfrak{q} \lhd B$ such that $\mathfrak{q}^c = \mathfrak{p}$. Furthermore $\mathfrak{q}$ may be taken to be minimal.

Proof. We may reduce to the case ϕ is injective by replacing A with $A/\ker(\phi)$. We have the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_T} \\ A_\mathfrak{p} & \overset{\tilde{\phi}}{\dashrightarrow} & T^{-1}B \end{array}$$

where $S := A \setminus \mathfrak{p}$ and $T := \phi(S)$. By (3.5.5) there exists a homomorphism $\tilde{\phi}$. Define $\mathfrak{m} := \mathfrak{p}A_\mathfrak{p}$ to be the unique maximal ideal of $A_\mathfrak{p}$. By (3.5.18) it is a minimal prime ideal and therefore the unique prime ideal. By assumption $0 \notin T$ and so $T^{-1}B \neq 0$. $T^{-1}B$ has a maximal ideal $\mathfrak{n}$ by (3.3.37). By uniqueness we find $\tilde{\phi}^{-1}(\mathfrak{n}) = \mathfrak{m}$. Therefore $\mathfrak{q} := i_T^{-1}(\mathfrak{n})$ is a prime ideal such that $\mathfrak{q}^c = \mathfrak{p}$.

By (3.3.44) there is a minimal prime $\mathfrak{r} \subset \mathfrak{q}$ which by minimality also satisfies $\mathfrak{r}^c = \mathfrak{p}$. ∎

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3.22 Integral Ring Extensions

Definition 3.22.1 (Integral Element)
*Let $\phi : A \to B$ be a ring map and $\alpha \in B$. Then we say α is **algebraic** over A if $m(\alpha) = 0$ for some polynomial $m(X) \in A[X]$.*

*Furthermore we say that α is **integral** over A if $m(X)$ may be chosen to be monic.*

Note often we assume $A \subseteq B$ and ϕ is the identity.

Definition 3.22.2 (Ring Extensions)
Let $\phi : A \to B$ be a ring map (so that B is an A-algebra). Then we say ϕ is

- ***finite** if B is finite as an A-module*
- ***finite-type** if B is finitely generated as an A-algebra*
- ***integral** if every element of B is integral over A*

When $A \subseteq B$ and ϕ is the identity then we say that B is respectively finite over A, finite-type over A or integral over A.

We note the trivial implication

$$\text{finite} \implies \text{finite type}$$

For example $k[X]$ is a ring of finite type over k, but certainly not finite. The following criterion for integrality is fundamental.

Proposition 3.22.3
Let $\phi : A \to B$ be a ring map and $b \in B$. Then the following are equivalent

a) b is integral over A
b) $\phi(A)[b]$ is a finite A-module
c) $\phi(A)[b]$ is contained in a subring C of B which is a finite A-module
d) There exists a $\phi(A)[b]$-module M which is faithful and finite as an A-module

Proof. Note that the subring C in c) is a faithful A-module, so the only non-trivial step is d) $\implies$ a). This is the usual "determinant trick". We apply (3.16.4) by considering $\psi_b \in \text{End}_A(M)$ to be multiplication by b. Then we have some monic polynomial $P(X) \in A[X]$ such that $P(\psi_b) = 0$, whence $P^\phi(b)m = 0$ for all $m \in M$. Since M is faithful, then we have $P(b) = 0$ as required. ∎

Proposition 3.22.4 (Finite $\iff$ finite-type and integral)
Let $\phi : A \to B$ be a ring map and $b_1, ..., b_n \in B$ integral over A. Then the ring homomorphism $\phi : A \to \phi(A)[b_1, ..., b_n]$ is finite and integral.

In particular if ϕ is integral and of finite type if and only if it is finite.

Proof. We assume without loss of generality that $A \subseteq B$ and ϕ is the identity map. Consider a tower

$$A \subset A[b_1] \subset ... \subset A[b_1, ..., b_n] = B$$

We proceed inductively on n. Namely we assume that $A[b_1, ..., b_i]$ is a finite A-module. Then a-fortiori $A[b_1, ..., b_{i+1}]$ is integral over $A[b_1, ..., b_i]$. Therefore by the previous Proposition it is a finite $A[b_1, ..., b_i]$-module and therefore a finite A-module (by (3.13.5)).

For any $b \in A[b_1, ..., b_n]$ we have $A[b] \subset A[b_1, ..., b_n]$ so by (3.22.3) we have b is integral over A.

It then follows that B integral and finite type $\implies$ B is finite (as an A-module). Conversely if B is finite then it is clearly finitely-generated. Further for any $b \in B$ then $A[b]$ is contained in the ring B which is finite as an A-module, and therefore is b is integral by (3.22.3). ∎

Proposition 3.22.5 (Transitivity property)
Let $\phi : A \to B$ and $\psi : B \to C$ be ring maps.

If $c \in C$ is integral over B, then it is integral over A (with respect to the ring map $\psi \circ \phi : A \to C$)

In particular if ϕ integral and ψ integral (e.g. surjective) $\implies$ $\psi \circ \phi$ integral.

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Proof. Suppose $c \in C$ is integral over B. Let $b_0, ..., b_{n-1}$ be the coefficients of the integral relation then clearly c is integral over $B' \coloneqq A[b_0, ..., b_{n-1}]$. By (3.22.3) B' is a finite A-module and $B'[c]$ is a finite B'-module. Therefore $B'[c]$ is a finite A-module and by (3.22.3) we have c is integral over A. ∎

Definition 3.22.6 (Integrally Closed)
*Let $A \subset B$ be a subring, then we say that A is **integrally closed** in B if*

$$b \in B \text{ integral over } A \implies b \in A.$$

*We say that an integral domain A is **integrally closed** if it is integrally closed in its quotient field.*

Proposition 3.22.7 (Integral Closure)
*Let A be a subring of a ring B. Then the set of elements of B integral over A (the **integral closure**) is a subring of B. Denote this by $\overline{A}^B$ or just $\overline{A}$ when the overring B is unambiguous.*

*Further $\overline{A}^B$ is **integrally closed** in B.*

Proof. Let $\alpha, \beta \in B$ be integral over A. Then by (3.22.4) the subring $A[\alpha, \beta]$ is a finite A-module containing $\alpha \pm \beta$ and $\alpha\beta$. Therefore by (3.22.3) they are also integral over A.

Clearly $\overline{A}$ is integral over A so by (3.22.5) it is integrally closed. ∎

Lemma 3.22.8
Let A be a subring of a field K and $S \subset A$ a multiplicatively closed subset. If $x \in K$ is integral over $S^{-1}A$ then sx is integral over A for some $s \in S$.

Conversely if $y \in K$ is integral over A then $\frac{y}{s}$ is integral over $S^{-1}A$ for every $s \in S$.

Proof. Suppose $x \in K$ is integral over $S^{-1}A$ then we have by definition

$$x^n + \sum_{j=0}^{n-1} \frac{a_j}{s_j} x^j = 0$$

for $a_j \in A$ and $s_j \in S$. Let $s \coloneqq s_0 ... s_{n-1}$ and multiply by s^n to find

$$y^n + \sum_{j=0}^{n-1} s^{n-j-1} \frac{s}{s_j} a_j y^j = 0$$

where $y = sx$. This shows that sx is integral over A

Similarly if $y \in K$ is integral over A then

$$y^n + \sum_{j=0}^{n-1} a_j y^j = 0$$

for some $a_j \in A$. Then let $x = \frac{y}{s}$ we have

$$x^n + \sum_{j=0}^{n-1} \frac{a_j}{s^{n-j}} x^j = 0$$

whence x is integral over $S^{-1}A$. ∎

Lemma 3.22.9
Let $A \subset K$ be a subring of a field. Then $\overline{S^{-1}A}^K = S^{-1}\overline{A}^K$.

In particular if A is integrally closed in K then so is $S^{-1}A$.

Proof. This follows directly from (3.22.8). ∎

Proposition 3.22.10 (Normality is Local)
Let A be an integral domain. Then the following are equivalent

a) A is integrally closed
b) $A_{\mathfrak{p}}$ is integrally closed for every prime ideal $\mathfrak{p}$
c) $A_{\mathfrak{m}}$ is integrally closed for every maximal ideal $\mathfrak{m}$

Proof. a) $\implies$ b) This is simply (3.22.9).

b) $\implies$ c) is immediate.

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c) $\implies$ a) Suppose that $x \in K$ is integral over A and consider the ideal $\mathfrak{a} := \{a \in A \mid ax \in A\}$. Then by definition it is a non-zero ideal, and we wish to show that $\mathfrak{a} = A$. Suppose not, then by (...) it is contained in some maximal ideal $\mathfrak{m}$. A-fortiori x is integral over $A_{\mathfrak{m}}$ whence by assumption $x \in A_{\mathfrak{m}}$. This means there is some $s \notin \mathfrak{m}$ such that $sx \in A$, that is $s \in \mathfrak{a}$ a contradiction. ∎

Proposition 3.22.11 (Integral closure of an ideal)
Let $A \subset B$ be a subring and $\mathfrak{a} \triangleleft A$ an ideal. Then for $b \in B$ TFAE

a) b integral over $\mathfrak{a}$
b) b^n integral over $\mathfrak{a}$ for some $n \geq 1$
c) $b \in \sqrt{\mathfrak{a}\overline{A}}$

In particular $\overline{\mathfrak{a}}$ is an ideal of $\overline{A}$.

Furthermore if A is integrally closed in B then $\overline{\mathfrak{a}} = \sqrt{\mathfrak{a}}$.

Proof. It's clear that a) $\iff$ b). For a) $\implies$ c) consider the integral relation

$$b^n + a_{n-1}b^{n-1} + ...a_0 = 0 \quad a_i \in \mathfrak{a}$$

By (3.22.7) $\overline{A}$ is a subring, and by assumption $b \in \overline{A}$. Therefore $b^k \in \overline{A}$, and the integral relation shows that $b^n \in \mathfrak{a}\overline{A}$ as required.

For c) $\implies$ b) suppose $b \in \sqrt{\mathfrak{a}\overline{A}}$ then $b^n = \sum_{i=1}^n a_i x_i$ for $a_i \in \mathfrak{a}$ and $x_i \in \overline{A}$. Let $B' := A[x_1, ..., x_n]$, which is a finite A-submodule by (3.22.4). Let $\phi \in \mathrm{End}_A(B')$ denote multiplication by b^n then $\phi(B') \subseteq \mathfrak{a}B'$ so by (3.16.4) ϕ satisfies a monic polynomial with coefficients in $\mathfrak{a}$. In particular b^n is integral over $\mathfrak{a}$. ∎

Proposition 3.22.12
A UFD is integrally closed.

In particular polynomial ring over a UFD is integrally closed.

Proof. **TODO** ∎

The following criterion is also useful:

Lemma 3.22.13 (Integral Criterion II)
Let $\phi : A \to B$ be a ring map. Suppose $x \in B$ is invertible, then x is integral over A if and only if $x \in \phi(A)[x^{-1}]$

Proof. Suppose $x \in \phi(A)[x^{-1}]$ then

$$x = \phi(a_0) + \phi(a_1)x^{-1} + ... + \phi(a_n)x^{-n}$$

Multiply by x^n to deduce an integral equation. Conversely suppose $x \in B$ is integral over A then by definition

$$x^n + \phi(a_{n-1})x^{n-1} + ... + \phi(a_0) = 0$$

Multiply by $x^{-(n-1)}$ to deduce $x \in \phi(A)[x^{-1}]$ ∎

Proposition 3.22.14 (Integral extension preserves field property)
Let $\phi : A \hookrightarrow B$ be an injective, integral ring map. Then B is a field if and only if A is a field.

Proof. As ϕ is injective, it induces an isomorphism between A and $\phi(A)$. So we may assume without loss of generality that $A \subseteq B$ and ϕ is the identity.

Suppose B is a field and $x \in A$. Then $x^{-1} \in B$ is integral over A by hypothesis, so by the previous Lemma $x^{-1} \in A[x] \subseteq A$. Therefore A is a field.

Conversely suppose A is a field and $0 \neq x \in B$. Then by hypothesis x is integral over A, that is

$$x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0 = 0$$

Choose the degree n to be minimal. We claim $a_0 \neq 0$, for if $a_0 = 0$ we may cancel x to obtain an integral relation of smaller degree. Therefore

$$-x(x^{n-1} + a_{n-1}x^{n-2} + ... + a_1)a_0^{-1} = 1$$

and in particular x is invertible. ∎

Proposition 3.22.15
Let $\phi : A \to B$ be an integral ring map. Then

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a) If $\phi^{-1}(\mathfrak{b}) \subseteq \mathfrak{a}$ then the induced ring map $A/\mathfrak{a} \to B/\mathfrak{b}$ (see (3.3.56)) is integral.
b) If S is a multiplicatively closed subset of A and $T \coloneqq \phi(S)$, then the induced ring map $S^{-1}A \to T^{-1}B$ is also integral.

Proposition 3.22.16 (Maximal ideals under integral extension)
Let $\phi : A \to B$ be an integral ring map. Suppose $\mathfrak{q}$ lies above $\mathfrak{p}$. Then

$$\mathfrak{p} \text{ is maximal} \iff \mathfrak{q} \text{ is maximal}$$

Proof. The map $\phi : A \to B$ induces an injective map $A/\mathfrak{p} \hookrightarrow B/\mathfrak{q}$ of integral domains by (3.3.56). By (3.22.15) this map is also integral. Note $A/\mathfrak{p}$ (resp. $B/\mathfrak{q}$) is a field if and only if $\mathfrak{p}$ (resp. $\mathfrak{q}$) is a maximal ideal by (3.3.59). Then we may apply (3.22.14) to show the equivalence. ∎

Lemma 3.22.17 (Non-zero Ideals meet the Base)
Let $A \subset R$ be integral domains such that every element of R is algebraic over $\mathrm{Frac}(A)$*. Then $\mathfrak{a} \cap A \neq 0$ for every non-zero ideal $\mathfrak{a} \lhd R$.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $0 \neq x \in \mathfrak{a}$. Clearing denominators, x satisfies a non-trivial relation $a_n x^n + ... + a_1 x + a_0 = 0$ with $a_i \in A$, and we take n minimal. Then $a_0 \neq 0$, for otherwise dividing by x (R is a domain) gives a non-trivial relation of smaller degree. Therefore

$$0 \neq a_0 = -x(a_n x^{n-1} + ... + a_1) \in xR \subseteq \mathfrak{a}$$

so $a_0 \in \mathfrak{a} \cap A$. ∎

Proposition 3.22.18 (Properties of integral extensions)
Let $\phi : A \to B$ be an integral ring map then it has

*a) the **Lying Over** property*
*b) the **Incomparability** property*
*c) the **Going Up** property*

Proof. For any prime ideal $\mathfrak{p} \lhd A$ we have the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_T} \\ A_{\mathfrak{p}} & \overset{\tilde{\phi}}{\dashrightarrow} & B_{\mathfrak{p}} \end{array}$$

where $S \coloneqq A \setminus \mathfrak{p}$ and $T \coloneqq \phi(S)$. By (3.5.5) there exists a morphism $\tilde{\phi}$, and by (3.22.15) it is integral. Define $\mathfrak{m} \coloneqq \mathfrak{p}A_{\mathfrak{p}}$ to be the unique maximal ideal of $A_{\mathfrak{p}}$.

a) As we assume $\ker(\phi) \subseteq \mathfrak{p}$ we know $B_{\mathfrak{p}} \neq 0$ (3.5.40). Let $\mathfrak{n}$ be a maximal (and hence prime) ideal of $B_{\mathfrak{p}}$. Then $\mathfrak{q} \coloneqq i_T^{-1}(\mathfrak{n})$ is a prime ideal of B such that $\mathfrak{q} \cap T = \emptyset$. In addition by (3.22.16) $\tilde{\phi}^{-1}(\mathfrak{n})$ is a maximal ideal, and therefore by uniqueness $\mathfrak{m} = \tilde{\phi}^{-1}(\mathfrak{n})$. By commutativity of the diagram we then have $\phi^{-1}(\mathfrak{q}) = \mathfrak{p}$ as required.

(Stacks) As an alternative argument to show existence of $\mathfrak{q}$ by (3.21.5) it's enough to show that $\mathfrak{p}B_{\mathfrak{p}}$ is proper (by assumption $\ker(\phi) \subseteq \mathfrak{p}$ so $B_{\mathfrak{p}} \neq 0$). By the diagram above $\mathfrak{p}B_{\mathfrak{p}} = \tilde{\phi}(\mathfrak{p}A_{\mathfrak{p}})B_{\mathfrak{p}}$. Therefore it's enough to consider the case $(A, \mathfrak{m})$ local and to show $\phi(\mathfrak{m})B$ is proper. Suppose $1 \in \phi(\mathfrak{m})B$ then

$$1 = \sum_{i=1}^{n} \phi(a_i) b_i \quad a_i \in \mathfrak{m}_A \; b_i \in B.$$

By (3.22.4) the subring $B' \coloneqq \phi(A)[b_1, ..., b_n] \subset B$ is a finite A-module. Furthermore $1 \in \mathfrak{m}B'$ whence $\mathfrak{m}B' = B'$ and by Nakayama's Lemma (3.19.5) $B' = 0$, a contradiction.

b) Suppose $\mathfrak{p} = \phi^{-1}(\mathfrak{q}) = \phi^{-1}(\mathfrak{q}')$ and $\mathfrak{q} \subseteq \mathfrak{q}'$. Let $\mathfrak{n} = \mathfrak{q}B_{\mathfrak{p}}$ and $\mathfrak{n}' = \mathfrak{q}'B_{\mathfrak{p}}$. Clearly $\mathfrak{n} \subseteq \mathfrak{n}'$. By commutativity of the diagram $i_S^{-1}(\tilde{\phi}^{-1}(\mathfrak{n})) = \phi^{-1}(\mathfrak{q}) = \mathfrak{p}$. By (3.5.18) extending the ideals to $A_{\mathfrak{p}}$ shows $\tilde{\phi}^{-1}(\mathfrak{n}) = \mathfrak{m}$, and similarly for $\mathfrak{n}'$. By (3.22.16) both $\mathfrak{n}$, $\mathfrak{n}'$ are maximal so $\mathfrak{n} = \mathfrak{n}'$. By (3.5.19) $\mathfrak{q} = \mathfrak{q}'$.

c) Suppose we have prime ideals $\mathfrak{p} \subsetneq \mathfrak{p}'$ and $\mathfrak{q}$ is a prime ideal lying above $\mathfrak{p}$. Consider the commutative diagram

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$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow & & \downarrow \\ A/\mathfrak{p} & \xrightarrow{\tilde{\phi}} & B/\mathfrak{q} \end{array}$$

The induced map $\tilde{\phi}$ is integral (3.22.16). By a) there is a prime ideal of $B/\mathfrak{q}$ lying above $\mathfrak{p}'/\mathfrak{p}$, which is of the form $\mathfrak{q}'/\mathfrak{q}$ for $\mathfrak{q} \subseteq \mathfrak{q}'$ prime (3.3.56). Then from the diagram we see $\phi^{-1}(\mathfrak{q}') = \mathfrak{p}'$ as required.

■

Proposition 3.22.19 (Coefficients of minimal polynomial)
Let $A \subseteq B$ be integral domains, A is integrally closed, and define $K = \mathrm{Frac}(A)$ and $L = \mathrm{Frac}(B)$. For $b \in B$ integral over $\mathfrak{a} \lhd A$ we have the non-leading coefficients of $m_b(X)$ are integral over $\mathfrak{a}$ and therefore lie in $\sqrt{\mathfrak{a}}$.

Note if b is only assumed to be integral over A then the coefficients of $m_b(X)$ lie in A.

Both $\mathrm{Norm}_{L/K}(b)$ and $\mathrm{Tr}_{L/K}(b)$ lie in A.

Proof. Let M/K be a normal closure for L/K (...). By (3.22.11) the integral closure of $\mathfrak{a}$ in M is simply $\sqrt{\mathfrak{a}}$. Then the minimal polynomial $m_b(X)$ splits completely in M and by (3.17.68) all the roots b_i are conjugate by $\mathrm{Aut}(M/K)$. In particular it's clear that b_i are integral over $\mathfrak{a}$, and so lie in $\sqrt{\mathfrak{a}}$. The coefficients of $m_b(X)$ are polynomials in the b_i, and so by the observation above are also lie in $\sqrt{\mathfrak{a}}$ (and are integral over $\mathfrak{a}$).

We may consider the case $\mathfrak{a} = A$ to deduce that $m_b(X)$ is a monic polynomial with coefficients in A. By (3.17.131) then the coefficients of $\mathrm{Pc}_{L/K}(b; X)$ lie in A, and so by (3.17.126) so do the norm and trace. ■

Lemma 3.22.20 (Colon Ideals of the Radical of an Extended Prime)
Let $A \subseteq B$ be an integral ring extension of domains with A integrally closed, and $\mathfrak{p} \lhd A$ a prime ideal. Then for every $x \in A \setminus \mathfrak{p}$ the colon ideal $(\sqrt{\mathfrak{p}B} : x) := \{s \in B \mid xs \in \sqrt{\mathfrak{p}B}\}$ equals $\sqrt{\mathfrak{p}B}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The inclusion $\sqrt{\mathfrak{p}B} \subseteq (\sqrt{\mathfrak{p}B} : x)$ is obvious as $\sqrt{\mathfrak{p}B}$ is an ideal, so it remains to show $(\sqrt{\mathfrak{p}B} : x) \subseteq \sqrt{\mathfrak{p}B}$. Let $s \in B$ with $y := xs \in \sqrt{\mathfrak{p}B}$. As $x \neq 0$ we may assume $y \neq 0$, for otherwise $s = 0$. By (3.22.11) y is integral over $\mathfrak{p}$, so by (3.22.19) the minimal polynomial of y over $K := \mathrm{Frac}(A)$ is

$$m_{y,K}(X) = X^r + u_1 X^{r-1} + ... + u_r \quad u_i \in \sqrt{\mathfrak{p}} = \mathfrak{p}$$

Since $x \in K^\times$, the element $s = yx^{-1}$ has minimal polynomial over K obtained by scaling the roots of $m_{y,K}$ by x^{-1}, which preserves being monic, irreducible and of degree r:

$$m_{s,K}(X) = X^r + v_1 X^{r-1} + ... + v_r \quad v_i := \frac{u_i}{x^i}$$

As $s \in B$ is integral over A and A is integrally closed, the coefficients of $m_{s,K}$ lie in A by (3.22.19), so $v_i \in A$. Now $v_i x^i = u_i \in \mathfrak{p}$ with $x \notin \mathfrak{p}$ and $\mathfrak{p}$ prime, so $v_i \in \mathfrak{p}$ for all i. Therefore s is integral over $\mathfrak{p}$ and $s \in \sqrt{\mathfrak{p}B}$ by (3.22.11). ■

Proposition 3.22.21 (Going Down)
Let $A \subseteq B$ be integral ring extension such that A is an integrally closed domain and B is an integral domain. Then it has the Going Down property.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\mathfrak{p} \subsetneq \mathfrak{p}'$ be prime ideals of A and $\mathfrak{q}'$ a prime ideal lying over $\mathfrak{p}'$. We wish to find a prime ideal $\mathfrak{q} \subseteq \mathfrak{q}'$ lying over $\mathfrak{p}$ (clearly inclusion must be strict).

Consider the inclusion of rings $A \subseteq B_{\mathfrak{q}'}$. Then by (3.21.5) $\mathfrak{p}$ lies under a prime of $B_{\mathfrak{q}'}$ if and only if $\mathfrak{p} = \mathfrak{p}^{ec} = \mathfrak{p}B_{\mathfrak{q}'} \cap A$. If this is the case then it is of the form $\mathfrak{q}B_{\mathfrak{q}'}$ for some prime ideal $\mathfrak{q} \subseteq \mathfrak{q}'$ (3.5.42) of B. It's clear that $\mathfrak{q}$ lies over $\mathfrak{p}$.

Note in general that $\mathfrak{p} \subseteq \mathfrak{p}^{ec}$, so we only need to demonstrate the reverse inclusion. Choose $x \in \mathfrak{p}B_{\mathfrak{q}'} \cap A$ and suppose for contradiction that $x \notin \mathfrak{p}$. By (3.5.18) $\mathfrak{p}B_{\mathfrak{q}'} = S^{-1}(\mathfrak{p}B)$ where $S = B \setminus \mathfrak{q}'$, so $x = \frac{y}{s}$ for $y \in \mathfrak{p}B$ and $s \in B \setminus \mathfrak{q}'$. Then $xs = y \in \mathfrak{p}B \subseteq \sqrt{\mathfrak{p}B}$ with $x \in A \setminus \mathfrak{p}$, so $s \in (\sqrt{\mathfrak{p}B} : x) = \sqrt{\mathfrak{p}B}$ by (3.22.20). But $\mathfrak{p}B \subseteq \mathfrak{p}'B \subseteq \mathfrak{q}'$ as $\mathfrak{q}'$ lies over $\mathfrak{p}' \supseteq \mathfrak{p}$, and $\mathfrak{q}'$ is prime, so $\sqrt{\mathfrak{p}B} \subseteq \mathfrak{q}'$ and $s \in \mathfrak{q}'$, which contradicts the choice of s. Therefore $x \in \mathfrak{p}$ as required. ■

Proposition 3.22.22
Let B, C be commutative A-algebras such that B is integral over A. Then $C \otimes_A B$ is integral over C.

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Proof. Let $\overline{C}$ be the subring of $C \otimes_A B$ of elements which are integral over C (3.22.7). Consider an elementary tensor $c \otimes b$ then by definition there is $P \in A[X]$ such that $P(b) = 0$. Then

$$P(c \otimes b) = \sum_{i=0}^{n} a_i (c \otimes b)^i = \sum_{i=0}^{n} c^i \otimes a_i b^i = c^i \otimes \sum_i a_i b^i = c^i \otimes P(b) = 0$$

This shows that $\overline{C}$ contains the elementary tensors and therefore is equal to $C \otimes_A B$. ∎

Lemma 3.22.23
Let $\phi : A \to B$ be a finite ring homomorphism. Suppose $B = (f_1, ..., f_n)$ and the composite map $\phi_i : A \to B_{f_i}$ is finite for all $i = 1...n$, then so is ϕ.

Proof. There are finitely many elements $\left\{ \frac{b_{ij}}{f_i^{n_{ij}}} \right\}_j$ which generate B_{f_i} as an A-module. Let B' be the A-submodule of B generated by all the b_{ij}. Then $B'_{f_i} = B_{f_i}$ as B-modules. By (3.5.47) we have $B' = B$ and therefore B is finitely generated as an A-module. ∎

Proposition 3.22.24 (Krull Dimension is preserved under integral maps)
Let $\phi : A \to B$ be a ring map.

a) Going Up $\implies \dim B \geq \dim(A/\ker(\phi))$
b) Incomparability $\implies \dim B \leq \dim(A/\ker(\phi))$

In particular ϕ integral and injective implies $\dim A = \dim B$.

Proof. Without loss of generality we can assume that ϕ is injective. The two cases follow by lifting chains of prime ideals from A (resp. B) to B (resp. A), and checking that distinct is maintained.

The final statement follows from (3.22.18). ∎

Corollary 3.22.25
Let $\phi : A \to B$ be integral and $\mathfrak{b} \lhd B$, then $\dim \phi^{-1}(\mathfrak{b}) = \dim(\mathfrak{b})$.

3.22.1 Integral Closure in Finite Extensions

Proposition 3.22.26
Let L/K be an algebraic field extension and A a subring with quotient field K. Then the integral closure $B := \overline{A}^L$

a) has quotient field L,
b) spans L as a K-vector space and
c) is integrally closed.

Furthermore $\dim B = \dim A$.

Proof. We may show a),b) if we prove $L = (A^\star)^{-1} B$. Consider $x \in L$, then by definition it is integral over $K = (A^\star)^{-1} A$ so by (3.22.8) $ax \in B$, and therefore $x \in (A^\star)^{-1} B$ as required.

It is integrally closed by (3.22.7) and $\dim B = \dim A$ from (3.22.24). ∎

Proposition 3.22.27
Let L/K be a finite field extension, A a subring with quotient field K and $B = \overline{A}^L$. Suppose that A is a principal ideal domain and B is a finite A-module. Then B is a finite free A-module of rank $[L : K]$.

Proof. By (3.14.7) B is a finite free A-module. Let $b_1, ..., b_r$ be an A-basis, then by (3.22.26) this spans L as a K-vector space. Similarly by clearing denominators the given set is K-linearly independent, so it is a K-basis of L and $r = [L : K]$ (3.3.126) as required. ∎

Proposition 3.22.28 (Finite Separable Extension of Normal Domain)
Let L/K be a finite separable extension, with A an integrally closed Noetherian subring with quotient field K. Then the integral closure $\overline{A}^L$ is a finite A-module.

In particular $\overline{A}^L$ is Noetherian and $\dim A = \dim \overline{A}^L$.

If in addition A is a principal ideal domain, then $\overline{A}^L$ is a free A-module of rank $[L : K]$.

Proof. By (3.17.134) the trace map $L \times L \to K$ is a perfect pairing. By (3.22.26) we may choose $x_1, ..., x_n \in B$ a K-basis for L where $n = [L : K]$. Denote $x_1^\star, ..., x_n^\star \in L$ by the (unique) basis dual under this pairing satisfying

$$\mathrm{Tr}_{L/K}(x_i x_j^\star) = \delta_{ij}$$

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We claim that $B := \overline{A}^L$ satisfies

$$B \subset Ax_1^\star + ... + Ax_n^\star.$$

For every $b \in B$ we have $b = \sum_{i=1}^n \lambda_i x_i^\star$ and some $\lambda_i \in K$. Then by linearity $\mathrm{Tr}_{L/K}(bx_i) = \lambda_i$. By (3.22.19) we have $\lambda_i \in A$. The right hand side is a finitely generated A-module, which is therefore Noetherian (3.13.4). B is a submodule and therefore also finitely generated as an A-module (3.13.3).

Then by (3.13.4) B is Noetherian as an A-module, therefore a-fortiori Noetherian as a B-module (or equivalently as a ring).

If A is a principal ideal domain then $\overline{A}^L$ is a finite free A-module of rank $[L : K]$ by (3.22.27). ■

Lemma 3.22.29
Let $A \subset B \subset C$ be subrings. Then $\overline{A}^C \cap B = \overline{A}^B$.

Lemma 3.22.30
Let $A \subset B \subset C$ be rings such that B is integral over A. Then $\overline{A}^C = \overline{B}^C$.

Lemma 3.22.31
Let A be a Noetherian integral domain with quotient field K and L/K be a finite extension.

Suppose that L'/L is a finite extension such $\overline{A}^{L'}$ is a finite A-module, then so is $\overline{A}^L$.

Proof. By (3.13.4) $\overline{A}^{L'}$ is Noetherian as an A-module, whence by (3.22.29) $\overline{A}^L \subset \overline{A}^{L'}$ is a finitely-generated A-module. ■

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3.23 Valuation Rings and Places

Definition 3.23.1 (Valuation Ring)
*A subring $A \subset K$ of a field K is a **valuation ring for K** if for every $0 \neq x \in K$ either $x \in A$ or $x^{-1} \in A$ (or both). Such a ring is an integral domain and K is necessarily a quotient field for A.*

*An integral domain A is a **valuation ring** if it is a valuation ring for its quotient field.*

Proposition 3.23.2 (Properties of valuation rings)
Let A be a valuation ring and K its quotient field then the following properties hold

a) A is local ring
b) $x \in A \setminus A^\star \iff x \in \mathfrak{m} \iff x^{-1} \notin A$
c) A is integrally closed in K

Proof. We prove each in turn

a) By (3.18.3) we need to show that $\mathfrak{m} := A \setminus A^\star$ is an additive subgroup of A. Given $x, y \in \mathfrak{m}$, without loss of generality we may assume that x, y are non-zero, and $\frac{x}{y} \in A$. Then $x + y = y\left(1 + \frac{x}{y}\right)$. If $(x + y) \in A^\star$ then $y \in A^\star$ a contradiction. Therefore $(x + y) \in \mathfrak{m}$ as required.
b) Note $x^{-1} \notin A \iff x \in A \setminus A^\star \iff x \in \mathfrak{m}$.
c) Suppose $0 \neq x \in K$ is integral over A. If $x \in A$ we are done. If $x^{-1} \in A$ then by (3.22.13) $x \in A[x^{-1}] \subseteq A$ as required.

■

Lemma 3.23.3 (Preimage of a Valuation Ring)
Let A be a valuation ring of a field M and $\psi : L \to M$ a homomorphism of fields. Then $\psi^{-1}(A)$ is a valuation ring of L. In particular if $L \subset M$ is a subfield then $A \cap L$ is a valuation ring of L.

Proof. The preimage $\psi^{-1}(A)$ is a subring of L. For $0 \neq x \in L$ we have $\psi(x) \neq 0$, so one of $\psi(x)$, $\psi(x)^{-1} = \psi(x^{-1})$ lies in A, that is one of x, x^{-1} lies in $\psi^{-1}(A)$. ■

Proposition 3.23.4 (Local Homomorphism)
Let $\phi : A \to B$ be a homomorphism of local rings. Then the following are equivalent

a) $\phi(\mathfrak{m}_A) \subset \mathfrak{m}_B$
b) $\mathfrak{m}_A = \phi^{-1}(\mathfrak{m}_B)$
c) $\phi(\mathfrak{m}_A)B \subsetneq B$

*We say that in this case ϕ is a **local homomorphism**. When ϕ is simply inclusion then we say B **dominates** A.*

Proof. a) $\implies$ b) $\phi^{-1}(\mathfrak{m}_B)$ is a proper ideal containing $\mathfrak{m}_A$ and therefore equal by maximality

b) $\implies$ a) This is obvious

a) $\implies$ c) $\phi(\mathfrak{m}_A)B \subset \mathfrak{m}_B B = \mathfrak{m}_B \subsetneq B$

c) $\implies$ a) By (3.3.15) $\phi(\mathfrak{m}_A)B \subset \mathfrak{m}_B$ and trivially $\phi(\mathfrak{m}_A) \subset \phi(\mathfrak{m}_A)B$. ■

Corollary 3.23.5
Let $(A, \mathfrak{m})$ be a local ring, K a field and $\phi : A \to K$ a homomorphism. Then the following are equivalent

a) ϕ is a local ring homomorphism (i.e. $\mathfrak{m} \subset \ker(\phi)$)
b) $\ker(\phi) = \mathfrak{m}$
c) ϕ factors through $\kappa(\mathfrak{m})$.

Lemma 3.23.6 (Restriction to a Valuation Ring)
Let A be a valuation ring of a field F, $L \supseteq F$ a field and $B \subseteq L$ a subring containing A, with a proper ideal $\mathfrak{n} \subsetneq B$ such that $\mathfrak{m}_A \subseteq \mathfrak{n}$ (equivalently $\mathfrak{m}_A B \neq B$). Then $B \cap F = A$ and $\mathfrak{n} \cap F = \mathfrak{m}_A$. In particular this applies to any local ring $B \subseteq L$ dominating A (3.23.4), with $\mathfrak{n} = \mathfrak{m}_B$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Clearly $A \subseteq B \cap F$. Let $x \in F \setminus A$; then $x \neq 0$ and $(x^{-1})^{-1} = x \notin A$, so $x^{-1} \in \mathfrak{m}_A \subseteq \mathfrak{n}$ by (3.23.2). If $x \in B$ then $1 = xx^{-1} \in \mathfrak{n}$, which is absurd; so $x \notin B$ and $B \cap F = A$. Finally $\mathfrak{n} \cap F \subseteq B \cap F = A$ is a proper ideal of A, hence contained in $\mathfrak{m}_A$; the reverse inclusion is $\mathfrak{m}_A \subseteq \mathfrak{n}$. ■

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Lemma 3.23.7 (A Directed Union of Valuation Rings is a Valuation Ring)
Let $(K_i)_{i\in I}$ be a directed system of fields with union $K := \bigcup_i K_i$, and for each i let $R_i \subset K_i$ be a valuation ring, such that R_l dominates R_i (3.23.4) whenever $K_i \subseteq K_l$. Then $R := \bigcup_i R_i$ is a valuation ring of K, with $\mathfrak{m}_R = \bigcup_i \mathfrak{m}_{R_i}$.

Proof. R is a subring of K, being a directed union of subrings. For $0 \neq x \in K$, $x \in K_i$ for some i (directed union of fields); as R_i is a valuation ring of K_i, $x \in R_i$ or $x^{-1} \in R_i$, hence $x \in R$ or $x^{-1} \in R$: R is a valuation ring of K.

$\mathfrak{m}_R \subseteq \bigcup_i \mathfrak{m}_{R_i}$: if $x \in \mathfrak{m}_R$, then $x \in R_i$ for some i (as $R = \bigcup_i R_i$); as $x^{-1} \notin R \supseteq R_i$, in particular $x^{-1} \notin R_i$, so $x \in \mathfrak{m}_{R_i}$ (3.23.2) b).

$\mathfrak{m}_R \supseteq \bigcup_i \mathfrak{m}_{R_i}$: let $x \in \mathfrak{m}_{R_i}$, and suppose for contradiction $x^{-1} \in R_l$ for some l. By directedness choose m with $i, l \leq m$, so $x \in R_i \subseteq R_m$ and $x^{-1} \in R_l \subseteq R_m$, that is $x \in R_m^\star$. But R_m dominates R_i, so $\mathfrak{m}_{R_i} \subseteq \mathfrak{m}_{R_m}$, and $x \in \mathfrak{m}_{R_i}$ gives $x \in \mathfrak{m}_{R_m}$, contradicting $x \in R_m^\star$. So $x^{-1} \notin R_l$ for every l, that is $x^{-1} \notin R$, giving $x \in \mathfrak{m}_R$. ∎

Definition 3.23.8 (Place (Zariski-Samuel 1960 / Lang 1972))
*Let K be a field. A **place** of K consists of a valuation ring $(A, \mathfrak{m}_A)$ for K and a local homomorphism to a field F*

$$\phi : A \to F$$

(such that $\ker(\phi) = \mathfrak{m}_A$)

Furthermore if $x \in K \setminus A$ then we may write $\phi(x) = \infty$. Note that the second part of the previous Proposition then may be reinterpreted as saying

$$\phi(x) = \infty \iff \phi(x^{-1}) = 0 \quad \forall x \in K$$

which motivates the alternative definition below.

*We say it is a **semi-place** of K if $(A, \mathfrak{m}_A)$ is simply a local ring.*

Lemma 3.23.9
Let $A \subset K$ be a subring of a field and $\mathfrak{a} \lhd A$ a proper ideal. Then at least one of $\mathfrak{a}A[x]$ or $\mathfrak{a}A[x^{-1}]$ is a proper ideal.

Proof. Suppose neither are proper then we can write

$$1 = \sum_{j=0}^{n} a_j x^j$$

$$1 = \sum_{j=0}^{m} b_j x^{-j}$$

for $a_j, b_j \in \mathfrak{a}$. Choose n, m to be minimal and assume wlog that $m \leq n$. Observe that $a_0 \neq 1 \implies m > 0$. Multiply the second equation by $x^n a_n$ to find

$$x^n a_n (1 - b_0) = a_n b_1 x^{n-1} + \ldots a_n b_m x^{n-m}$$

and multiply the first by $(1 - b_0)$ to find

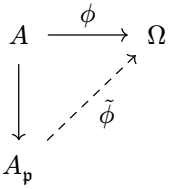
$$(1 - b_0) = a_0(1 - b_0) + \ldots + a_n(1 - b_0)x^n$$

consequently cancelling the x^n term and we obtain a relation of smaller degree a contradiction. ∎

We prove the first extension theorem

Proposition 3.23.10 (Extension to localization)
Let A be a ring and $\phi : A \to \Omega$ a homomorphism into a field. Let $\mathfrak{p} := \ker(\phi)$. Then

- *$\mathfrak{p}$ is prime*
- *There is a unique extension $\tilde{\phi}$ making the diagram commute*



Furthermore $\ker(\tilde{\phi}) = \mathfrak{p}A_\mathfrak{p}$ i.e. $\tilde{\phi}$ is local.

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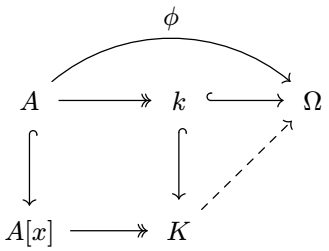
Proof. Clearly $\mathfrak{p}$ is prime because $\phi(A)$ is an integral domain. We may extend ϕ to the ring $A_{\mathfrak{p}}$ in the obvious way. The extension has kernel $\mathfrak{p}A_{\mathfrak{p}}$, the unique maximal ideal of $A_{\mathfrak{p}}$. ∎

Proposition 3.23.11 (Places as maximal extensions)
Let A be a subring of a field K and $\phi : A \to \Omega$ a homomorphism into an algebraically closed field. Then

- *For all $x \in K$, ϕ may be extended to at least one of $A[x]$ and $A[x^{-1}]$.*
- *There exists a maximal extension $\tilde{\phi} : B \to \Omega$. For any such maximal extension B is a valuation ring and $\tilde{\phi}$ is local (i.e. $\ker(\tilde{\phi}) = \mathfrak{m}_B$). Furthermore $\mathfrak{m}_B \cap A = \ker(\phi)$.*

Proof. We prove each in turn.

- By (3.23.10) we may assume without loss of generality that A is a local ring with unique maximal ideal $\mathfrak{m} = \ker(\phi)$. By (3.23.9) we may also suppose that $\mathfrak{m}A[x]$ is proper. Then it's contained in a maximal ideal $\mathfrak{B} \lhd A[x]$. Furthermore $\mathfrak{m} = \mathfrak{B} \cap A$ by maximality of $\mathfrak{m}$. Let $k = A/\mathfrak{m}$ and $K = A[x]/\mathfrak{B}$, then there is a commutative diagram



  Then $K = k[\overline{x}]$ is a field and by (...) $\overline{x}$ is algebraic over k. Therefore K/k is algebraic and, because Ω is algebraically closed, by (3.17.63) there is an extension to K, which gives the required extension to $A[x]$.
- It's easy to show that the poset of extensions to subrings of K ordered by consistency is chain complete. Therefore by Zorn's Lemma there is a maximal extension $\tilde{\phi} : B \to \Omega$. By the previous part for any $x \in K$ any such maximal element B must satisfy either $B[x] = B$ or $B[x^{-1}] = B$, i.e. B is a valuation ring for K. Consider $\mathfrak{B} := \ker(\tilde{\phi})$ a prime ideal contained in $\mathfrak{m}_B$. Then by (3.23.10) may extend $\tilde{\phi}$ to $B_{\mathfrak{B}}$ and so by maximality $B = B_{\mathfrak{B}}$. Finally (3.18.6) shows that B is a local ring with maximal ideal $\mathfrak{B} = \mathfrak{m}_B$. Clearly $\ker(\tilde{\phi}) \cap A = \ker(\phi)$, so the final statement follows easily.

∎

Corollary 3.23.12
Let $A \subset K$ be a subring of a field and $\mathfrak{a} \lhd A$ a proper ideal. Then there exists a valuation ring $(B, \mathfrak{m}_B)$ such that $A \subset B$ and $\mathfrak{a} \subset \mathfrak{m}_B \cap A$. In particular if $\mathfrak{a} = \mathfrak{m}_A$ is maximal then $\mathfrak{m}_A = \mathfrak{m}_B \cap A$.

Proof. By (3.3.37) there is a maximal ideal $\mathfrak{m}_A \lhd A$ containing $\mathfrak{a}$. Let $k = A/\mathfrak{m}_A$ and $\Omega = \overline{k}$. Then the canonical homomorphism $\phi : A \to \Omega$ has kernel $\mathfrak{m}_A$. It has an extension to a valuation ring $(B, \mathfrak{m}_B)$ by (3.23.11), such that $\mathfrak{m}_B \cap A = \ker(\phi) = \mathfrak{m}_A$. ∎

Proposition 3.23.13 (Alternative Characterisation of Valuation Rings)
Let K be a field and $A \subset K$ a subring. Then the following are equivalent

a) A is a valuation ring for K
b) A is a local ring maximal under the relation "B dominates A"
c) $K = \mathrm{Frac}(A)$ and the principal ideals of A are totally ordered
d) $K = \mathrm{Frac}(A)$ and the ideals of A are totally ordered

Proof. Let $(A, \mathfrak{m}_A) \preceq (B, \mathfrak{m}_B)$ denote the relation B dominates A.

a) $\implies$ b) Suppose that $(A, \mathfrak{m}_A) \preceq (B, \mathfrak{m}_B)$ with B local and A a valuation ring. We claim that $A = B$; suppose for a contradiction that $x \in B \setminus A$. Then $x^{-1} \in A \implies x^{-1} \in B \implies x \in B^\star$. Therefore $A \subset B \setminus B^\star = \mathfrak{m}_B$. In particular $1 \in \mathfrak{m}_B$ which is a contradiction as $\mathfrak{m}_B$ is proper.

b) $\implies$ a) Let $(A, \mathfrak{m}_A)$ be a maximal local ring. By (3.23.12) there exists a valuation ring $(B, \mathfrak{m}_B)$ such that $(A, \mathfrak{m}_A) \preceq (B, \mathfrak{m}_B)$. By maximality $A = B$ and A is a valuation ring.

a) $\implies$ c) Evidently $K = \mathrm{Frac}(A)$. Suppose $(a) \not\subset (b)$, then $a \notin (b) \implies b^{-1}a \notin A \implies a^{-1}b \in A \implies b \in (a) \implies (b) \subset (a)$.

c) $\implies$ d) Suppose $\mathfrak{a} \not\subset \mathfrak{b}$ then there exists $a \in \mathfrak{a} \setminus \mathfrak{b}$. In particular for all $b \in \mathfrak{b}$ we have $a \notin (b)$, whence by assumption $b \in (a)$. Therefore we conclude $\mathfrak{b} \subset (a) \subset \mathfrak{a}$.

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d) $\implies$ b) As A is a non-zero ring it has a maximal ideal $\mathfrak{m}_A$, which by assumption must be the unique maximal ideal. So $(A, \mathfrak{m}_A)$ is a local ring. Suppose that $(A, \mathfrak{m}_A) \preceq (B, \mathfrak{m}_B)$ and given $x \in B$ we have by assumption $x = a_1 a_2^{-1}$ for some $a_1, a_2 \in A$. If $a_1 \in (a_2)$ then $x \in A$. Otherwise $a_2 \in (a_1)$ whence $x^{-1} \in A \implies x^{-1} \in B^\star \overset{(3.23.2)}{\implies} x^{-1} \notin \mathfrak{m}_B \overset{(3.23.4)}{\implies} x^{-1} \notin \mathfrak{m}_A \implies x^{-1} \in A^\star \implies x \in A$. Consequently $A = B$ and b) holds. ∎

Corollary 3.23.14
Let $A \subset K$ be a subring of a field then the integral closure of A in K (denoted $\overline{A}$) satisfies

$$\overline{A} = \bigcap_{A \subset V} V$$

where the intersection is taken over all valuation rings V of K containing A.

Alternatively the integral elements over A are precisely the elements which are finite at all places of K, which are finite over A.

Proof. First if $x \in \overline{A}$ then by (3.22.13) we have $x \in A[x^{-1}] \subseteq V[x^{-1}]$. If $x \notin V$ then by hypothesis $x^{-1} \in V$, whence $x \in V$ a contradiction. Therefore $x \in V$ as required.

Conversely suppose $x \notin \overline{A}$, then $x \notin A[x^{-1}]$. That is to say (x^{-1}) is a proper ideal in $A[x^{-1}]$. Therefore by (3.23.12) there is a valuation ring $(V, \mathfrak{m}_V)$ such that $x^{-1} \in \mathfrak{m}_V$ which implies $x \notin V$ by (3.23.2). ∎

Definition 3.23.15 (Valuation)
*Let K be a field and Γ an abelian ordered group. A function $v : K^\star \to \Gamma$ is said to be a **valuation** if it satisfies the following properties*

a) $v(ab) = v(a) + v(b)$ for all $a, b \in K^\star$
b) $v(a + b) \geq \min(v(a), v(b))$ with the convention that $v(0) = \infty$

Note $v(1) = 0$. We say that two valuations v, v' are equivalent if there is an order-preserving isomorphism $\phi : v(K^\star) \to v'(K^\star)$ such that $\phi \circ v = v'$. Clearly this is an equivalence relation.

If $k \subset K$ and $v(k) = 0$ then we say v is a valuation for K/k.

Lemma 3.23.16 (Strict Ultrametric Inequality)
Let $v : K^\star \to \Gamma$ be a valuation (3.23.15), extended to K by $v(0) := \infty$. For $x, y \in K$ with $v(x) \neq v(y)$,

$$v(x + y) = \min(v(x), v(y))$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

If $x = 0$ or $y = 0$ this is immediate ($v(0) = \infty$ dominates), so assume $x, y \in K^\star$, and without loss of generality $v(x) < v(y)$. By property 2 of (3.23.15), $v(x + y) \geq \min(v(x), v(y)) = v(x)$. For the reverse inequality, apply property 2 again to $x = (x + y) + (-y)$:

$$v(x) \geq \min(v(x + y), v(-y))$$

As $v(-1) + v(-1) = v((-1)(-1)) = v(1) = 0$, and Γ is totally ordered (so $v(-1) > 0$, resp. $v(-1) < 0$, would force $v(-1) + v(-1) > 0$, resp. < 0), we must have $v(-1) = 0$; so $v(-y) = v(-1) + v(y) = v(y)$. As $v(y) > v(x)$, the displayed inequality $v(x) \geq \min(v(x + y), v(-y))$ forces $v(x) \geq v(x + y)$ (else the minimum would be $v(-y) = v(y) > v(x)$, a contradiction). Combined with the first inequality, $v(x + y) = v(x) = \min(v(x), v(y))$. ∎

Proposition 3.23.17 (Correspondence between Valuations Rings and Valuations)
Let K be a field with $k \subset K$. Then there is a bijection

$$\begin{aligned} \{k \subset A \subset K \mid A \text{ is a valuation ring}\} &\longleftrightarrow \{v : K^\star \to \Gamma \mid v \text{ is a valuation over } k\}/\sim \\ A &\to \quad v_A : K^\star \to K^\star/A^\star \\ v^{-1}(\Gamma_{\geq 0}) &\leftarrow \quad v : K^\star \to \Gamma \end{aligned}$$

Under this map v_A is just the quotient map and $\overline{x} \leq \overline{y} \iff yx^{-1} \in A^\star$. Furthermore

a) $v_A(\overline{x}) \geq 0 \iff x \in A$.
b) $v_A(\overline{x}) = 0 \iff x \in A^\star$
c) $v_A(\overline{x}) > 0 \iff x \in \mathfrak{m}_A$

Proof. Given a valuation ring we may define the abelian group $\Gamma := K^\star/A^\star$ under multiplication with $\overline{x} \leq \overline{y} \iff yx^{-1} \in A$. This is easily verified to be a total ordering. Then $v_A : K^\star \to K^\star/A^\star$ is simply the quotient map and we

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clearly $v_A(\overline{x}) \geq 0 \iff x \in A^\star$. To demonstrate the additive condition we may consider the case $x, y \neq 0$ and suppose wlog that $xy^{-1} \in A$

$$v(x+y) = v((xy^{-1}+1)y) = v(xy^{-1}+1) + v(y) \geq v(y) \geq \min(v(x), v(y))$$

The case that one of $x, y = 0$ is trivial. Observe $A = v_A^{-1}(\Gamma_{\geq 0}) \cup \{0\}$. We may therefore construct a one-sided inverse; given a valuation $v : K' \to \Gamma$ we may define $A_v := v^{-1}(\Gamma_{\geq 0}) \cup \{0\}$. It's evident from the two properties of v that A is an additive and multiplicative subgroup. Further by total ordering we see that A is a valuation ring. We only need to demonstrate that the given constructions are mutually inverse, namely $v \sim v_{A_v}$. However this is immediate because the composite $v(K^\star) \to K^\star/A_v^\star \to v_{A_v}(K^\star)$ is an abelian group isomorphism, which preserves the positive segments, and therefore is an order isomorphism. ∎

Corollary 3.23.18 (Valuation Dominance gives a Ring Multiple)
Let $v : K^\star \to \Gamma$ be a valuation with valuation ring A (3.23.17), extended to K by $v(0) := \infty$. If $x, y \in K$ with $y \neq 0$ and $v(x) \geq v(y)$, then $x = ay$ for some $a \in A$.

If further $v(x) = v(y)$ then $a \in A^\star$.

Proof. If $x = 0$ take $a = 0$. Otherwise $x, y \in K^\star$ and $v(xy^{-1}) = v(x) - v(y) \geq 0$, so $a := xy^{-1} \in A$ by (3.23.17), and $x = ay$. If further $v(x) = v(y)$ then $v(a) = 0$, so $a \in A^\star$ by (3.23.17). ∎

Proposition 3.23.19 (Bezout Domain)
Let A be an integral domain. Then the following are equivalent

a) Every finitely generated ideal is principal
b) Every ideal generated by two elements is principal

*In this case we say A is a **Bezout Domain**. Further in this case every pair of elements has a greatest common divisor d which satisfies $d = ax + by$ for some $x, y \in A$.*

Proof. By assumption $(a, b) = (d)$. Immediately we have $d \mid a$ and $d \mid b$. Similarly $d = \lambda a + \mu b$. Suppose $a = xe$ and $b = ye$ then $d = (\lambda x + \mu y)e$ whence $e \mid d$. ∎

Lemma 3.23.20 (Valuation Ring = Local Bezout Domain)
Let A be an integral domain. Then the following are equivalent

a) A is a valuation ring
b) A is a local Bezout domain.

In particular A is a Noetherian valuation ring iff it is a local principal ideal domain.

Proof. Suppose A is a valuation ring and $\mathfrak{a} = (x, y)$. Without loss of generality $xy^{-1} \in A$ whence $x \in (y)$ so $\mathfrak{a} = (y)$ as required.

Conversely suppose A is a local Bezout domain and let $x = \frac{y}{z} \in K$. We may divide by the greatest common divisor so that $\lambda y + \mu z = 1$. We claim one of λy and μz is a unit, for suppose not then by (3.18.3) we have $\lambda y, \mu z \in \mathfrak{m} \implies 1 \in \mathfrak{m}$ a contradiction. Therefore one of y, z is a unit which implies $x \in A$ or $x^{-1} \in A$, and A is a valuation ring. ∎

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3.24 Derivations

Definition 3.24.1 (Module of Derivations)
Let B be a commutative ring and M an (B,C)-bimodule, then we say a derivation is a map $D : B \to M$ satisfying the following properties

- $D(x+y) = D(x) + D(y)$ ***linearity***
- $D(xy) = xD(y) + yD(x)$ ***product rule***

We denote the family of such derivations by $\mathrm{Der}(B,M)$, and it is an (B,C)-bimodule. We may consider the case $M = C$ and $\phi : B \to B$ a ring homomorphism and in this case the product rule becomes

$$D(xy) = \phi(x)D(y) + D(x)\phi(y)$$

We observe that by induction we have $D(n \cdot 1_B) = 0$ for all $n \in \mathbb{Z}$.

Suppose that B is an A-algebra then we claim the following are equivalent

- $D(a) = 0 \quad \forall a \in A$
- $D(ab) = aD(b) \quad \forall a \in A$

In this case we denote the set of A-linear derivations by

$$\mathrm{Der}_A(B,M)$$

Note every derivation is $\mathbb{Z}$-linear so we may work with $A = \mathbb{Z}$ in these cases. Finally we write

$$\mathrm{Der}_A(B) := \mathrm{Der}_A(B,B)$$

For our purposes the following will be the key example

Example 3.24.2 (Evaluation at a zero)
Consider the case $B = A[X_1,...,X_n]/\mathfrak{a}$ a f.g. A-algebra, L an A-algebra and $x \in L^n$ a zero of $\mathfrak{a}$. Then we may define the B-module structure on L by

$$f \cdot \lambda := f(x)\lambda$$

In this case we have the more explicit form

$$\mathrm{Der}_A(B,L;x) = \{D \in \mathrm{Hom}_A(B,L) \mid D(fg) = f(x)D(g) + g(x)D(f)\}$$

which is a (B,L)-bimodule.

Lemma 3.24.3
Suppose $D, D' \in \mathrm{Der}_A(B,M)$ agree on a set of generators for B, then they are identically equal. For example in the previous example when $D\left(\overline{X}_i\right) = D'\left(\overline{X}_i\right)$.

The following is a useful technical tool to identify derivations as algebra homomorphisms

Definition 3.24.4 (Dual Ring)
Let M be a B-module. Define the ***dual ring*** *as follows*

$$B[M] := B \times M$$

$$(b,m) \times (b',m') = (bb', bm' + b'm)$$

with addition defined in the obvious way and multiplicative unit is $(1_B, 0_M)$. Observe that it has an ideal $N := 0 \times M$ such that $N^2 = 0$ and canonically $B[M]/N \cong B$.

When B is an A-algebra, then there is a natural A-algebra structure on $B[M]$.

Proposition 3.24.5
Let M be an B-module where B is an A-algebra then there is a bijection

$$\begin{aligned} \mathrm{Der}_A(B,M) &\longrightarrow \mathrm{AlgHom}_A(B, B[M]) \\ D &\to \quad \phi_D(b) = (b, D(b)) \end{aligned}$$

Definition 3.24.6 (Partial Derivative)
Suppose $F \in A[X_1,...,X_n]$ given by

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$$F(X_1,...,X_n) = \sum_{i\in\mathbb{N}^n} a_i X_1^{i_1}...X_n^{i_n} \quad a_i \in A$$

Define the partial derivative as follows

$$\frac{\partial F}{\partial X_k} := \sum_{\substack{i\in\mathbb{N}^n \\ i_k\neq 0}} a_i i_k X_1^{i_1}...X_k^{i_k-1}...X_n^{i_n}$$

We observe that

$$\frac{\partial X_l}{\partial X_k} := \delta_{lk}$$

Note this condition uniquely determines the equation above, see (3.24.8).

We show that these form a basis for $\mathrm{Der}_k(A)$.

Lemma 3.24.7 (Product Rule)
*For $F,G \in A[X_1,...,X_n]$ we have the **product rule***

$$\frac{\partial FG}{\partial X_k} = F\frac{\partial G}{\partial X_k} + G\frac{\partial F}{\partial X_k}$$

Proof. First we demonstrate the result in the univariate case $n = 1$. For monomials this is straightforward:

$$\frac{\partial X^r X^s}{\partial X} = (r+s)X^{r+s-1} = X^s\frac{\partial X^r}{\partial X} + X^r\frac{\partial X^s}{\partial X}$$

For general univariate polynomials the product rule follows from the linearity of $\partial/\partial X$. The multivariate case then follows from considering the isomorphism $A[X_1,...,X_n] \cong A\left[X_1,...,\widehat{X_k},...,X_n\right][X_k]$ under which $\partial/\partial X_k$ corresponds to $\partial/\partial X$. ∎

Lemma 3.24.8 (Multivariate Chain Rule)
Let $D \in \mathrm{Der}_A(B,M)$ be a derivation, $x_1,...,x_n \in A$ and $F \in B[X_1,...,X_n]$. Then

$$D(F(x_1,...,x_n)) = \sum_{k=1}^{n} \frac{\partial F}{\partial X_k}(x_1,...,x_n)D(x_k)$$

As a special case we find

$$D(F(x)) = F'(x)D(x)$$
$$D(x^n) = nx^{n-1}D(x)$$

Proof. First we observe that by induction on n

$$D(x_1...x_n) = \sum_{k=1}^{n} x_1...\widehat{x_k}...x_n D(x_k)$$

and in particular

$$D(x^n) = \begin{cases} nx^{n-1}D(x) & n > 0 \\ 0 & n = 0 \end{cases}$$

Then for $F \in k[X_1,...,X_n]$ we have

$$\begin{aligned} D(F(x_1,...,x_n)) &= \sum_{i\in\mathbb{N}^n} a_i D\left(x_1^{i_1}...x_n^{i_n}\right) \\ &= \sum_{i\in\mathbb{N}^n} a_i \sum_{k=1}^{n} x_1^{i_1}...\widehat{x_k^{i_k}}...x_n^{i_n} D\left(x_k^{i_k}\right) \\ &= \sum_{k=1}^{n}\sum_{i\in\mathbb{N}^n} a_i x_1^{i_1}...\widehat{x_k^{i_k}}...x_n^{i_n} D\left(x_k^{i_k}\right) \\ &= \sum_{k=1}^{n}\sum_{\substack{i\in\mathbb{N}^n \\ i_k\neq 0}} i_k a_i x_1^{i_1}...x_k^{i_k-1}...x_n^{i_n} D(x_k) \end{aligned}$$

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which gives the required result. ∎

Proposition 3.24.9
Let $B \coloneqq A[X_1, ..., X_n]$ be a polynomial ring and M a (B, C)-bimodule. Then $\mathrm{Der}_A(B, M)$ is a finite free (B, C)-bimodule with basis $\{\partial/\partial X_1, ..., \partial/\partial X_n\}$ (3.24.6).

Proposition 3.24.10 (Extensions to Quotient Field)
Let B be an A-algebra, S a multiplicative set of B and M a B-module. Then we have the following isomorphisms

$$\mathrm{Der}_A(B, M) \xrightarrow{\sim} \mathrm{Der}(S^{-1}B, S^{-1}M)$$
$$D \to \frac{b}{s} \to \frac{bD(s) - sD(b)}{s^2}$$

In particular if $M = \Omega$ is a field containing B (which is therefore an integral domain) then we have an isomorphism

$$\mathrm{Der}_A(B, \Omega) \xrightarrow{\sim} \mathrm{Der}_A(K, \Omega)$$

where $K \coloneqq \mathrm{Frac}(B)$.

In light of (3.24.5) the following definition is clearly related to the existence of derivations.

Definition 3.24.11 (Formally Smooth)
*Let B be an A-algebra. We say it is **formally smooth** if for any A-algebra C with $N \triangleleft C$ such that $N^2 = 0$ and any homomorphism $v : B \to C/N$ there exists a morphism $\overline{v} : A \to C$ making the following diagram commute*

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & \nearrow & \downarrow \pi \\ B & \xrightarrow{v} & C/N \end{array}$$

*We say that B is **formally unramified** if in addition the lifting is always unique.*

Lemma 3.24.12
Let C be a ring and $N \subseteq \sqrt{(0)}$ an ideal. Then $x \in C$ is invertible if and only if $\overline{x} \in C/N$ is invertible.

Proof. One direction is obvious. Conversely if $\overline{x}$ is invertible then $x \in C^\star + N \subseteq C^\star + \sqrt{(0)} \subseteq C^\star$ by (3.3.68) as required. ∎

Lemma 3.24.13
If a ring B is formally smooth (resp. unramified) then so is $S^{-1}B$.

Proof. It's enough to show that for $s \in S$ we have $\overline{v}(s)$ is invertible, which follows from (3.24.12). ∎

Lemma 3.24.14
The polynomial ring $B \coloneqq A[X_1, ..., X_n]$ and function field $A(X_1, ..., X_n)$ is formally smooth.

Proof. The first follows by the fact $A[X_1, ..., X_n]$ is a free A-algebra and the second by (3.24.13). ∎

Lemma 3.24.15
Let B be a formally smooth (resp. formally unramified) A-algebra and C is a formally smooth (resp. formally unramified) B-algebra. Then C is a formally smooth (resp. formally unramified) A-algebra.

Lemma 3.24.16 (Separable algebraic extensions are "formally unramified")
A separable algebraic extension K/k is formally unramified.

Proof. Consider the following commutative diagram

$$\begin{array}{ccc} k & \longrightarrow & C \\ \downarrow & \nearrow & \downarrow \pi \\ K & \xrightarrow{v} & C/N \end{array}$$

We suppose first that K is a finite extension. By (3.17.108) it is simple, namely $K = k(\alpha) = k[\alpha]$. Let $m(X)$ be the minimal polynomial over k and choose $x \in C$ such that $\pi(x) = v(\alpha)$. Note that $\overline{v}(\alpha) = x$ need not be well-defined because in general $m(x) \neq 0$. We show however there is an $n \in N$ such that $m(x + n) = 0$. For consider any $n \in N$ and $f(X) \in k[X]$.

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$$f(x+n) = f(x) + \sum_{i=0}^{N} \lambda_i [(x+n)^i - x^i]$$
$$= f(x) + f'(x)n$$

as $n^2 = 0$. Therefore we propose $n = -m(x)/m'(x)$; to show this is well-defined, consider firstly $\pi(m(x)) = m(v(\alpha)) = v(m(\alpha)) = 0$ whence $m(x) \in N$. To show that $m'(x)$ is a unit we argue as follows. As α is separable, $m'(\alpha) \neq 0$, hence is a unit in K. As v is injective $v(m'(\alpha)) = m'(v(\alpha)) = m'(\pi(x)) = \pi(m'(x))$ is a unit in C/N. By (3.24.12) $m'(x) \in C^\star$. Therefore $n \in N$ is well-defined and $m(a+n) = 0$ by construction. Therefore the map $\overline{v} : k[\alpha] \to A$ such that $\overline{v}(\alpha) = a + n$ is well-defined, and the diagram commutes because $\pi(a+n) = \pi(a) + \pi(n) = v(\alpha)$.

Suppose we had another $\overline{v}'$, then $a' \coloneqq \overline{v}'(\alpha)$ again satisfies $m(a') = 0$ and $\pi(a') = v(\alpha)$ whence $a' - a \in N$. By the same argument as before

$$m(a') = m(a + (a' - a)) = m(a) + m'(a)(a' - a)$$

As $m'(a) \neq 0$ and $m(a') = m(a) = 0$ we see $a' = a$ which demonstrates uniqueness.

Suppose that K/k is algebraic than for every $\alpha \in K$ we have liftings $v_\alpha : k(\alpha) \to k$. We claim that $v_\alpha|_{k(\alpha)\cap k(\beta)} = v_\beta|_{k(\alpha)\cap k(\beta)}$. However $k(\alpha) \cap k(\beta)$ is also finite and so we are done by uniqueness. ∎

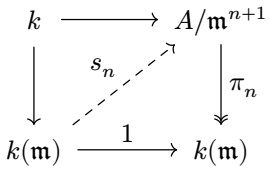
Proposition 3.24.17
A separably generated extension K/k is formally smooth.

Proof. This follows from (3.24.14), (3.24.16) and (3.24.15). ∎

Lemma 3.24.18 (Coefficient Field modulo Powers of the Maximal Ideal)
Let k be a field and $(A, \mathfrak{m})$ a local ring which is a k-algebra, with residue field $k(\mathfrak{m}) \coloneqq A/\mathfrak{m}$.

Write $\rho_{n,m} : A/\mathfrak{m}^{n+1} \to A/\mathfrak{m}^{m+1}$ and $\pi_n : A/\mathfrak{m}^{n+1} \to A/\mathfrak{m} = k(\mathfrak{m})$ for the canonical reduction maps.

If $k(\mathfrak{m})/k$ is separably generated then for every $n \geq 0$ there is a k-algebra homomorphism s_n such that the following diagram commutes

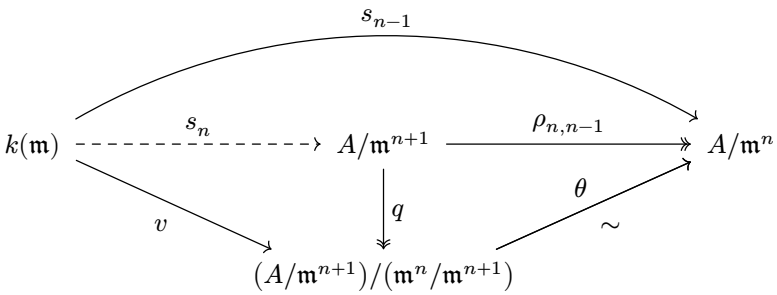


If $k(\mathfrak{m})/k$ is separable algebraic then this morphism is unique. In any case the s_n can be chosen compatibly, $\rho_{n,n-1} \circ s_n = s_{n-1}$, and in the separable algebraic case they are automatically compatible, since $\rho_{n,n-1} \circ s_n$ makes the diagram for $n-1$ commute ($\pi_{n-1} \circ \rho_{n,n-1} = \pi_n$).

If $k(\mathfrak{m}) = k$, that is $k \to k(\mathfrak{m})$ is an isomorphism, all of this is trivial: s_n is the structure map $k \to A/\mathfrak{m}^{n+1}$ composed with the inverse of $k \cong k(\mathfrak{m})$, which is unique because a k-algebra homomorphism out of k is the structure map, and compatible.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The diagram. We construct s_n by induction on n, the case $n = 0$ being $s_0 = 1_{k(\mathfrak{m})}$ as $A/\mathfrak{m} = k(\mathfrak{m})$. For $n \geq 1$ consider the following diagram, in which q is the quotient map by the ideal $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ of $A/\mathfrak{m}^{n+1}$ (which has square zero, as $2n \geq n+1$), $\rho_{n,n-1}$ is the reduction, which is surjective with kernel $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ (the short exact sequence of the third isomorphism theorem (3.3.106) for $\mathfrak{m}^{n+1} \subseteq \mathfrak{m}^n \subseteq A$), so that by the isomorphism theorem (3.3.57) it induces an isomorphism θ with $\theta \circ q = \rho_{n,n-1}$ (a k-algebra isomorphism, as q and $\rho_{n,n-1}$ are k-algebra homomorphisms), s_{n-1} is given by the induction hypothesis, and $v \coloneqq \theta^{-1} \circ s_{n-1}$:



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Existence. If $k(\mathfrak{m})/k$ is separably generated it is formally smooth (3.24.17), so the lifting property (3.24.11), applied to the square-zero ideal $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ and the homomorphism $v : k(\mathfrak{m}) \to (A/\mathfrak{m}^{n+1})/(\mathfrak{m}^n/\mathfrak{m}^{n+1})$, gives a k-algebra homomorphism $s_n : k(\mathfrak{m}) \to A/\mathfrak{m}^{n+1}$, the dashed arrow, with $q \circ s_n = v$. Hence $\rho_{n,n-1} \circ s_n = \theta \circ q \circ s_n = \theta \circ v = s_{n-1}$, the compatibility.

s_n is a section. As $\pi_n = \pi_{n-1} \circ \rho_{n,n-1}$,

$$\pi_n \circ s_n = \pi_{n-1} \circ \rho_{n,n-1} \circ s_n = \pi_{n-1} \circ s_{n-1} = 1_{k(\mathfrak{m})}$$

by induction, so s_n is a section of π_n.

Uniqueness. Now let $k(\mathfrak{m})/k$ be separable algebraic. It is separably generated (with empty transcendence basis), so the above applies. Moreover it is formally unramified (3.24.16), so the lift s_n of s_{n-1} is unique. We show by induction on n that s_n is the only section of π_n. For $n = 0$, $\pi_0 = 1_{k(\mathfrak{m})}$, so a section t satisfies $t = \pi_0 \circ t = 1_{k(\mathfrak{m})}$. For $n \geq 1$, if t is a section of π_n then $\rho_{n,n-1} \circ t$ is a section of π_{n-1} (as $\pi_{n-1} \circ \rho_{n,n-1} = \pi_n$), hence equals s_{n-1} by induction, so t is a lift of s_{n-1}, that is $t = s_n$. In particular the s_n are automatically compatible. ∎

Remark 3.24.19

A compatible family $(s_n)_{n\geq 0}$ is the same as a single k-algebra homomorphism $s : k(\mathfrak{m}) \to \hat{A} := \varprojlim_n A/\mathfrak{m}^{n+1}$ with $\pi \circ s = 1_{k(\mathfrak{m})}$, by the universal property of the inverse limit. So when $k(\mathfrak{m})/k$ is separably generated the residue field lifts to a subfield of the completion of A. This is a coefficient field in the sense of Cohen. It is unique when $k(\mathfrak{m})/k$ is separable algebraic, but in general it is not, and two choices differ by a k-derivation of $k(\mathfrak{m})$.

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3.25 Hauptidealsatz

The main result of this section is the following result due to Krull

Proposition 3.25.1 (Generalized Hauptidealsatz for Noetherian Rings)
Suppose A is a Noetherian ring then the following properties hold

a) Every prime ideal of height n is minimal over some ideal $\mathfrak{a} := (x_1, ..., x_n)$. Furthermore $\mathfrak{a}$ may be chosen such that $\operatorname{ht}(\mathfrak{a}) = n$ and every minimal prime of $\mathfrak{a}$ is of height n
b) We have the following characterization of height

$$\operatorname{ht}(\mathfrak{p}) = \min\{n \mid \mathfrak{p} \text{ minimal over } (x_1, ..., x_n)\}$$

In particular if $\mathfrak{p}$ is minimal over $(x_1, ..., x_n)$ then $\operatorname{ht}(\mathfrak{p}) \leq n$.

In full generality the proof is quite subtle and in most cases of interest a simpler proof is possible. Therefore we introduce the following notions.

Definition 3.25.2 (Hauptidealsatz Ring)
*We say a ring A is a **generalized hauptidealsatz** ring if for all $x_1, ..., x_n \in A$ and $\mathfrak{p}$ prime ideals minimal over $(x_1, ..., x_n)$ we have $\operatorname{ht}(\mathfrak{p}) \leq n$*

*We say a ring A is simply a **hauptidealsatz** ring if this holds for $n = 1$.*

Lemma 3.25.3
Let A be a hauptidealsatz ring and $\mathfrak{p}$ a prime ideal. Then $A_{\mathfrak{p}}$ is hauptidealsatz.

Proof. Any prime ideal of $A_{\mathfrak{p}}$ has the form $\mathfrak{q}A_{\mathfrak{p}}$ by the correspondence of ideals under localization (3.5.19). If $\mathfrak{q}A_{\mathfrak{p}}$ is minimal over $(f/s) = (f/1)$, then clearly $(f) \subseteq \mathfrak{q}$. Furthermore $(f) \subseteq \mathfrak{q}' \implies (f/1) \subseteq \mathfrak{q}'A_{\mathfrak{p}}$. So $\mathfrak{q}$ is also minimal over (f) and therefore has height at most 1 by assumption. By the same result $\mathfrak{q}A_{\mathfrak{p}}$ has height at most 1. ∎

Proposition 3.25.4 (Hauptidealsatz ⇒ Generalized Hauptidealsatz (Geometric))
Suppose that A is catenary, and hauptidealsatz for every quotient by a prime ideal. Then A is generalized hauptidealsatz, and so is every localization at a prime ideal.

Proof. We consider the case first that A is integral (which means it is hauptidealsatz by assumption because (0) is prime) and assume wlog that $n > 1$. Let $\mathfrak{p}$ be a minimal prime of $(x_1, ..., x_n)$ and $\mathfrak{q}$ a minimal prime of $(x_1, ..., x_{n-1})$. By considering the localization $A_{\mathfrak{p}}$ we may choose $\mathfrak{q} \subseteq \mathfrak{p}$. By induction $\operatorname{ht}(\mathfrak{q}) \leq n - 1$. Then $\mathfrak{p}/\mathfrak{q}$ is a minimal prime over $(x_n + \mathfrak{q})$ and therefore by assumption has height at most 1. By the codimension formula (3.20.9) we see $\operatorname{ht}(\mathfrak{p}) \leq n$ as required.

For the general case, suppose $\mathfrak{p}$ is minimal over $(x_1, ..., x_n)$ and let $\mathfrak{r} \subseteq \mathfrak{p}$ be a minimal prime. Then the ring $A/\mathfrak{r}$ is integral and we see that $\operatorname{ht}(\mathfrak{p}/\mathfrak{r}) \leq n$ by the first part. Taking the supremum over all minimal primes $\mathfrak{r}$ shows that $\operatorname{ht}(\mathfrak{p}) \leq n$.

For the last statement we need only show that every localization at a prime ideal satisfies the hypotheses of the theorem. By correspondence of ideals under localisation (3.5.19) we see that every such ring is catenary. By (3.5.25) $A_{\mathfrak{p}}/\mathfrak{q}A_{\mathfrak{p}} \cong (A/\mathfrak{q})_{\mathfrak{p}/\mathfrak{q}}$ for any prime ideal $\mathfrak{q} \subseteq \mathfrak{p}$ so every quotient by a prime ideal is hauptidealsatz by the previous Lemma. ∎

The catenary assumption may be relaxed by making a more complicated argument.

Proposition 3.25.5 (Hauptidealsatz ⇒ Generalized Hauptidealsatz (Algebraic))
Suppose A is a ring such that every quotient by a prime ideal satisfies the hauptidealsatz. Then A is generalized hauptidealsatz and so is every localization at a prime ideal $\mathfrak{p}$.

Proof. We prove this by induction on n for a fixed ring A. Let $\mathfrak{p}$ be a minimal prime over $(x_1, ..., x_n)$. Suppose $\operatorname{ht}(\mathfrak{p}) > n$ then we may choose a saturated chain $\mathfrak{q} \subsetneq \mathfrak{p}$ such that $\operatorname{ht}(\mathfrak{q}) \geq n$. By minimality of $\mathfrak{p}$ we have $\mathfrak{q} \cap \{x_1, ..., x_n\} = \{x_1, ..., x_r\}$ with $r < n$ after a suitable reordering (possibly with $r = 0$). Furthermore we may choose $\mathfrak{q}$ such that r is maximal. Then by the induction hypothesis $\mathfrak{q}$ is not minimal over $(x_1, ..., x_r)$ and there is a chain

$$(x_1, ..., x_r) \subseteq \mathfrak{r} \subsetneq \mathfrak{q} \subsetneq \mathfrak{p}$$

We claim that $\mathfrak{p}$ is minimal over $(\mathfrak{r}, x_{r+1})$ for suppose

$$(\mathfrak{r}, x_{r+1}) \subseteq \mathfrak{p}' \subseteq \mathfrak{p}$$

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then by construction of $\mathfrak{q}$ (r was maximal) we see that $\mathfrak{p}' = \mathfrak{p}$. Taking quotients by $\mathfrak{r}$ the hauptidealsatz property shows that $\mathrm{ht}(\mathfrak{p}/\mathfrak{r}) \leq 1$. On the other hand we have a chain

$$(0) \subsetneq \mathfrak{q}/\mathfrak{r} \subsetneq \mathfrak{p}/\mathfrak{r}$$

which is a contradiction.

The final statement follows a similar argument as before. ■

Remark 3.25.6
This argument is from [Kap74] (Sec. 3.2 Ex. 6).

Before proceeding to the main result we need a some technical results related to height of ideals.

Lemma 3.25.7
Let $\mathfrak{q} \subseteq \mathfrak{p}$ be prime ideals then

$$\mathrm{ht}(\mathfrak{q}) \leq \mathrm{ht}(\mathfrak{p})$$

with equality iff $\mathfrak{p} = \mathfrak{q}$.

Proof. The inequality is clear since any maximal chain below $\mathfrak{q}$ may be extended to a maximal chain below $\mathfrak{p}$. Similarly $\mathfrak{q} \subsetneq \mathfrak{p}$ implies that the inequality is strict. ■

Lemma 3.25.8
Suppose $\mathfrak{a} \subseteq \mathfrak{b}$. Then

$$\mathrm{ht}(\mathfrak{a}) \leq \mathrm{ht}(\mathfrak{b})$$

If these are equal and $\mathfrak{b}$ is prime, then it must be a minimal prime of $\mathfrak{a}$.

Proof. We consider first the case $\mathfrak{b} = \mathfrak{p}$ is prime. By (3.3.44) there is a minimal prime $\mathfrak{p}'$ such that $\mathfrak{a} \subseteq \mathfrak{p}' \subseteq \mathfrak{p}$. By definition $\mathrm{ht}(\mathfrak{a}) \leq \mathrm{ht}(\mathfrak{p}')$. Furthermore by the previous Lemma $\mathrm{ht}(\mathfrak{p}') \leq \mathrm{ht}(\mathfrak{p})$, which yields the required result.

Suppose $\mathrm{ht}(\mathfrak{p}) = \mathrm{ht}(\mathfrak{a})$ then by definition $\mathrm{ht}(\mathfrak{a}) \leq \mathrm{ht}(\mathfrak{p}') \leq \mathrm{ht}(\mathfrak{p})$ and therefore we conclude $\mathrm{ht}(\mathfrak{p}) = \mathrm{ht}(\mathfrak{p}')$. By the previous lemma we conclude that $\mathfrak{p} = \mathfrak{p}'$ is a minimal prime.

For the general case every minimal prime of $\mathfrak{b}$ also contains $\mathfrak{a}$ so the inequality follows by taking the minimum over all minimal primes. ■

Lemma 3.25.9
Let $\mathfrak{a} \subseteq \mathfrak{b}$ be ideals such that there exists $x \in \mathfrak{b}$ not contained in any minimal prime of $\mathfrak{a}$. Then

$$\mathrm{ht}(\mathfrak{a}) < \mathrm{ht}(\mathfrak{b})$$

Proof. By the previous Lemma we have $\mathrm{ht}(\mathfrak{a}) \leq \mathrm{ht}(\mathfrak{b})$. Suppose they are equal then there is a minimal prime $\mathfrak{p}$ of $\mathfrak{b}$ of the same height, which by (3.25.8) is a minimal prime of $\mathfrak{a}$. This must contain x which then contradicts the assumption. ■

Proposition 3.25.10 (Prime Avoidance Theorem)
Let $\mathfrak{p}_1, ..., \mathfrak{p}_n$ be ideals and $\mathfrak{a}$ an ideal such that $\mathfrak{a} \nsubseteq \mathfrak{p}_i$ for all $i = 1...n$. Then there exists $x \in \mathfrak{a} \setminus (\mathfrak{p}_1 \cup ... \cup \mathfrak{p}_n)$.

Proof. We proceed by induction on the number of prime ideals n, the case $n = 1$ being trivial. Consider then the case of $n+1$ prime ideals and suppose on the contrary that $\mathfrak{a} \subseteq \mathfrak{p}_1 \cup ... \cup \mathfrak{p}_{n+1}$. By the induction hypothesis there exists $x_i \in \mathfrak{a} \setminus \bigcup_{j \neq i} \mathfrak{p}_j$, and therefore by our assumption $x_i \in (\mathfrak{a} \cap \mathfrak{p}_i) \setminus \bigcup_{j \neq i} \mathfrak{p}_j$. Define

$$y := x_1 ... x_n$$

Then $y \notin \mathfrak{p}_{n+1}$ by primality and clearly $y \in \mathfrak{a} \cap \mathfrak{p}_1 \cap ... \cap \mathfrak{p}_n$. Define $z := y + x_{n+1}$ then we see

a) $z \in \mathfrak{a}$
b) $z \notin \mathfrak{p}_{n+1}$
c) $z \notin \mathfrak{p}_i$ for $i = 1...n$

which contradicts our original assumption. ■

Lemma 3.25.11
*Let A be a **generalized hauptidealsatz** ring and $\mathfrak{a} = (x_1, ..., x_n)$. Then the following are equivalent*

a) $\mathrm{ht}(\mathfrak{a}) = n$
b) every minimal prime of $\mathfrak{a}$ has height n

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Proof. a) $\implies$ b) By definition this means every minimal prime has height at least n, and the reverse inequality follows from generalized hauptidealsatz assumption.

b) $\implies$ a) Recall from (3.20.1) we may take the infimum over only minimal prime ideals. ∎

Proposition 3.25.12 (Alternative characterization of height)
*Suppose A is a Noetherian ring satisfying **generalized hauptidealsatz**, then the following properties hold*

a) Every prime ideal of height n is minimal over some ideal $\mathfrak{a} := (x_1, ..., x_n)$. Furthermore $\mathfrak{a}$ may be chosen such that $\operatorname{ht}(\mathfrak{a}) = n$ and every minimal prime is of height n
b) We have the following characterization of height

$$\operatorname{ht}(\mathfrak{p}) = \min\{n \mid \mathfrak{p} \text{ minimal over } (x_1, ..., x_n)\}$$

In particular every prime ideal has finite height.

Proof. We first demonstrate a). Suppose $\operatorname{ht}(\mathfrak{p}) = n$ then there is a saturated chain of prime ideals

$$\mathfrak{p}_0 \subsetneq ... \subsetneq \mathfrak{p}_n = \mathfrak{p}$$

It's clear from the above chain that $\operatorname{ht}(\mathfrak{p}_i) \geq i$. Furthermore by (3.25.7) $\operatorname{ht}(\mathfrak{p}_i) < \operatorname{ht}(\mathfrak{p}_{i+1})$ whence we see $\operatorname{ht}(\mathfrak{p}_i) = i$.

We show by induction that there are elements $x_1, ..., x_n \in A$ for which $\mathfrak{p}_i$ is a minimal prime over $\mathfrak{a}_i := (x_1, ..., x_i)$ for all $i = 0...n$ and $\operatorname{ht}(\mathfrak{a}_i) = i$.

The case $i = 0$ is clear, so suppose $i > 0$. In general by (3.20.6) there are finitely many minimal primes over $\mathfrak{a}_{i-1}$ say $\mathfrak{q}_1, ..., \mathfrak{q}_r$ which all have height $i - 1$ by (3.25.11). In particular $\mathfrak{p}_i \not\subseteq \mathfrak{q}_k$.

By prime avoidance (3.25.10) we may choose $x_i \in \mathfrak{p}_i \setminus (\mathfrak{q}_1 \cup ... \cup \mathfrak{q}_r)$. Clearly

$$\mathfrak{a}_{i-1} \subseteq \mathfrak{a}_i \subseteq \mathfrak{p}_i$$

Then by (3.25.8) and (3.25.9) we have $i - 1 = \operatorname{ht}(\mathfrak{a}_{i-1}) < \operatorname{ht}(\mathfrak{a}_i) \leq i$ whence $\operatorname{ht}(\mathfrak{a}_i) = i$. By (3.25.8) again we see $\mathfrak{p}_i$ is a minimal prime of $\mathfrak{a}_i$. This completes the inductive step.

To show b) let m denote the right hand side. Then by the generalized hauptidealsatz hypothesis $\operatorname{ht}(\mathfrak{p}) \leq m$. On the other hand by part a) we have $m \leq \operatorname{ht}(\mathfrak{p})$ whence they are equal.

Finally by the Noetherian hypothesis $\mathfrak{p}$ is finitely generated and clearly minimal over itself. ∎

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3.26 Regular Local Rings

The results of the previous section allow a more direct characterization of the dimension of a Noetherian local ring, which naturally leads to the notion of regular local ring.

Lemma 3.26.1 ($\mathfrak{m}$-primary)
Let $(A, \mathfrak{m})$ be a local ring and $\mathfrak{a}$ a proper ideal. Then the following are equivalent

a) $\sqrt{\mathfrak{a}} = \mathfrak{m}$
b) $\mathfrak{m}$ *is a minimal prime of* $\mathfrak{a}$

Furthermore $\mathfrak{a}$ is primary. In this case we say $\mathfrak{a}$ is $\mathfrak{m}$-primary. A sufficient condition is that for some n

$$\mathfrak{m}^n \subset \mathfrak{a} \subset \mathfrak{m}$$

and this is necessary when $\mathfrak{m}$ is finitely generated.

Proof. a) $\implies$ b) Suppose $\mathfrak{a} \subseteq \mathfrak{p}$ then $\mathfrak{m} = \sqrt{\mathfrak{a}} \subseteq \mathfrak{p}$ by (3.3.46). In particular $\mathfrak{m}$ is a minimal prime.

b) $\implies$ a) Let $\mathfrak{p}$ be a prime ideal containing $\mathfrak{a}$. By (3.3.37) $\mathfrak{p} \subseteq \mathfrak{m}$ and by assumption $\mathfrak{p} = \mathfrak{m}$. The result then follows from (3.3.46).

Suppose $xy \in \mathfrak{a}$ and $y \notin \sqrt{\mathfrak{a}}$ then by (...) $y \in A^\star$ whence $x \in \mathfrak{a}$. This shows that $\mathfrak{a}$ is primary.

The condition $\mathfrak{m}^n \subset \mathfrak{a}$ is clearly sufficient, and necessary in the finitely generated case by (3.3.48). ∎

An ideal satisfying one of the equivalent conditions above is called **$\mathfrak{m}$-primary**, though some authors require the stronger condition.

Proposition 3.26.2 (Criteria for Dimension of Local Ring)
Let $(A, \mathfrak{m})$ be a Noetherian local ring satisfying the generalized hauptidealsatz. Then we have the following criteria for dimension

$$\dim A = \operatorname{ht}(\mathfrak{m}) = \min\left\{ n \;\middle|\; \sqrt{(x_1, ..., x_n)} = \mathfrak{m} \right\} \leq \mu(\mathfrak{m}) = \dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2$$

where $\mu(\mathfrak{m})$ is the least number of generators of $\mathfrak{m}$ and $k(\mathfrak{m}) = A/\mathfrak{m}$.

Proof. The first equality follows because every maximal chain must terminate at a maximal ideal by (3.3.37). The second from the characterization of height for hauptidealsatz rings (3.25.12) and the equivalent definitions of $\mathfrak{m}$-primary (3.26.1). The inequality follows because $\mathfrak{m}$ is itself $\mathfrak{m}$-primary. The final equality follows because a minimal generating set lifts to a basis of $\mathfrak{m}/\mathfrak{m}^2$ (3.19.7). ∎

Definition 3.26.3
*Let $(A, \mathfrak{m})$ be a Noetherian local ring. We say A is **regular** if*

$$\dim A = \mu(\mathfrak{m}) = \dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2$$

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3.27 Local Rings of Dimension 0

Lemma 3.27.1 (Dimension 0 local ring)
Let $(A, \mathfrak{m})$ be a Noetherian local ring. Then the following are equivalent

a) $\dim A = 0$ *(i.e. every prime ideal is maximal)*
b) $\mathfrak{m}^n = 0$ *for some* n
c) $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ *for some* n

If A is an integral domain then this is equivalent to A being a field.

Proof. a) $\implies$ b) By (3.20.6) $\mathfrak{r} := \sqrt{(0)} = \mathfrak{p}_1 \cap ... \cap \mathfrak{p}_n$ is a decomposition into minimal prime ideals. By assumption these are also maximal, and therefore by uniqueness $\sqrt{(0)} = \mathfrak{m}$ and the result follows from (3.3.48) as $\mathfrak{m}$ is finitely generated.

b) $\implies$ a) Let $\mathfrak{p}$ be a prime ideal then trivially $\mathfrak{m}^n \subset \mathfrak{p}$ so by (3.26.1) it is $\mathfrak{m}$-primary i.e. $\mathfrak{m}$ is a minimal prime of $\mathfrak{p}$ whence $\mathfrak{m} = \mathfrak{p}$.

b) $\iff$ c) One direction is obvious, and the converse follows from Nakayama's Lemma (3.19.5) because $\mathfrak{m}^n$ is finitely generated. ∎

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3.28 Fractional Ideals

Definition 3.28.1
Let A be an integral domain contained in its quotient field K. A non-zero A-submodule $I \subset K$ is called a ***fractional ideal*** *if there exists $0 \neq a \in A$ such that $aI \subset A$.*

We say a fractional ideal is ***principal*** *if it is generated by some $x \in K$ as an A-module.*

Proposition 3.28.2 (Fractional Ideals over a Noetherian Domain)
Let A be a Noetherian integral domain. Then every fractional ideal of A is a finitely generated A-module.

Proof. Let I be a fractional ideal and $0 \neq a \in A$ with $aI \subseteq A$. Then aI is an ideal of A, hence finitely generated, say by $x_1, ..., x_n$. Multiplication by a is an isomorphism $I \to aI$ of A-modules, so I is generated by $a^{-1}x_1, ..., a^{-1}x_n$. ∎

Proposition 3.28.3 (Operations on Fractional ideals)
Let A be an integral domain and $I, J \subset K$ fractional ideals. Then the following constructions are also fractional ideals

a) $I + J = \{i + j \mid i \in I j \in J\}$
b) $IJ = \left\{\sum_{k=1}^{n} i_k j_k \;\middle|\; i_k \in I j_k \in J\right\}$
c) $(I : J) = \{x \in K \mid xJ \subseteq I\}$

Proof. It is straightforward to prove that these constructions are again A-submodules of K.

Suppose that $aI \subseteq A$ and $bJ \subseteq A$, then $(ab)(I + J) \subseteq A$ and $(ab)IJ \subseteq A$. Similarly

$$(I : J)J \subseteq I \implies j(I : J) \subseteq I \implies abj(I : J) \subseteq A$$

for some $0 \neq j \in J$ ∎

Proposition 3.28.4
Let A be an integral domain. The collection of fractional ideals $\mathcal{I}_A$ forms a monoid under multiplication of ideals, with identity A.

Definition 3.28.5
Let A be an integral domain and I a fractional ideal. Then we say I is ***invertible*** *if there exists a fractional ideal J such that $IJ = A$.*

Proposition 3.28.6
Let A be an integral domain and I a fractional ideal. Then I is invertible if and only if $I(A : I) = A$, in which case $(A : I)$ is the unique inverse.

Proof. Uniqueness follows from general properties of monoids (...). If $I(A : I) = A$ then by definition I is invertible. Conversely suppose I is invertible, that is $IJ = A$. Then evidentally $J \subseteq (A : I)$ so $A = IJ \subseteq I(A : I) \subseteq A$. Therefore $(A : I)$ is an inverse for I as required. ∎

Corollary 3.28.7
Let A be an integral domain. Then the invertible fractional ideals form an abelian group under multiplication.

Definition 3.28.8 (Localisation of Fractional Ideals)
Let A be an integral domain contained in its quotient field K. Let I be a fractional ideal and S a multiplicatively closed subset of A. Then define

$$S^{-1}I \coloneqq \left\{\frac{i}{s} \;\middle|\; i \in I s \in S\right\}$$

If $\mathfrak{a}$ is an ideal then $S^{-1}\mathfrak{a}$ corresponds to the usual notion (3.5.17) and is equal to the extension of $\mathfrak{a}$ under the inclusion homomorphism $A \to S^{-1}A$.

If $\mathfrak{p}$ is a prime ideal of A then as usual define

$$I_{\mathfrak{p}} \coloneqq (A \setminus \mathfrak{p})^{-1} I$$

and we may write

$$\mathfrak{a}_{\mathfrak{p}} = \mathfrak{a} A_{\mathfrak{p}}$$

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Proposition 3.28.9 (Localisation of Fractional Ideals)
Let A be an integral domain contained in its quotient field K. Let S be a multiplicatively closed subset of A and I a fractional ideal. Then regarding as subrings of K then $S^{-1}I$ is a fractional ideal of $S^{-1}A$. Furthermore

a) $S^{-1}(I+J) = S^{-1}I + S^{-1}J$
b) $S^{-1}(IJ) = (S^{-1}I)(S^{-1}J)$
c) $S^{-1}(I:J) = (S^{-1}I : S^{-1}J)$ *if J is finitely generated (in general only $\subseteq$ holds)*

Proof. Evidently $S^{-1}I$ is again a fractional ideal of $S^{-1}A$.

a) Clear because $\frac{i+j}{s} = \frac{i}{s} + \frac{j}{s}$.
b) We have $\frac{ij}{s} = \left(\frac{i}{s}\right)\left(\frac{j}{1}\right)$ whence $S^{-1}(IJ) \subseteq (S^{-1}I)(S^{-1}J)$. Conversely we have $\left(\frac{i}{s}\right)\left(\frac{j}{t}\right) = \frac{ij}{st}$ so we ahve the reverse inclusion.
c) Suppose that $\frac{x}{s} \in S^{-1}(I:J)$ and $\frac{j}{t} \in S^{-1}J$. Then $\left(\frac{x}{s}\right)\left(\frac{j}{t}\right) = \frac{xj}{st} \in S^{-1}I$ because $x \in (I:J)$. Therefore $S^{-1}(I:J) \subseteq (S^{-1}I : S^{-1}J)$. Conversely suppose $J = (j_1, ..., j_n)$ and let $x \in (S^{-1}I : S^{-1}J)$. Then for each k we have $x\left(\frac{j_k}{1}\right) = \frac{i_k}{s_k}$ with $i_k \in I$ and $s_k \in S$. Put $s := s_1 ... s_n \in S$, so $sxj_k \in I$ for every k, whence $sxJ \subseteq I$ as the j_k generate J, that is $sx \in (I:J)$. Therefore $x = \frac{sx}{s} \in S^{-1}(I:J)$ as required. (Finite generation is what lets a single s serve all of J.)

■

Lemma 3.28.10
Let A be an integral domain contained in its quotient field K. Suppose I is a fractional ideal then

$$I = \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$$

In particular we have

$$A = \bigcap_{\mathfrak{m}} A_{\mathfrak{m}}$$

Proof. Evidently $I \subseteq \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$. Suppose that $x \in \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}$. Consider the ideal $\mathfrak{a} := \{a \in A \mid ax \in I\}$. Suppose $\mathfrak{a}$ is proper then by (3.3.15) it is contained in a maximal ideal $\mathfrak{m}$. As $x \in I_{\mathfrak{m}}$ we have $(A \setminus \mathfrak{m}) \cap \mathfrak{a} \neq \emptyset$ and $\mathfrak{a} \subseteq \mathfrak{m}$ a contradiction. Therefore $\mathfrak{a} = A$ and in particular $x = 1.x \in I$ as required. ■

Proposition 3.28.11 (Invertibility is a local property)
Let A be an integral domain with quotient field K, and I a fractional ideal of A. Consider the conditions

a) *I is invertible*
b) *$I_{\mathfrak{p}}$ is invertible for all prime ideals $\mathfrak{p}$*
c) *$I_{\mathfrak{m}}$ is invertible for all maximal ideals $\mathfrak{m}$*

Then a) $\implies$ b) $\implies$ c), and if I is finitely generated then c) $\implies$ a). In particular all three are equivalent when A is Noetherian, as then every fractional ideal is finitely generated (3.28.2).

Proof. a) $\implies$ b) Let $IJ = A$ for a fractional ideal J. By (3.28.9) $I_{\mathfrak{p}}J_{\mathfrak{p}} = (IJ)_{\mathfrak{p}} = A_{\mathfrak{p}}$, so $I_{\mathfrak{p}}$ is invertible with inverse $J_{\mathfrak{p}}$.

b) $\implies$ c) Every maximal ideal is prime (3.3.60) so we are done.

c) $\implies$ a) Suppose that I is finitely generated. Then

$$\begin{aligned}
I(A:I) &= \bigcap_{\mathfrak{m}} (I(A:I))_{\mathfrak{m}} && (3.28.10)\\
&= \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}(A:I)_{\mathfrak{m}} && (3.28.9)\\
&= \bigcap_{\mathfrak{m}} I_{\mathfrak{m}}(A_{\mathfrak{m}} : I_{\mathfrak{m}}) && (3.28.9)\\
&= \bigcap_{\mathfrak{m}} A_{\mathfrak{m}} && (3.28.6)\\
&= A && (3.28.10)
\end{aligned}$$

where the third equality uses that I is finitely generated, and the second last that $I_{\mathfrak{m}}$ is invertible by assumption. ■

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3.29 Discrete Valuation Rings

A Discrete Valuation Ring (DVR) corresponds to a regular local ring of dimension 1 and therefore regular (or in the case of perfect base field, non-singular) points on a curve. There are a number of equivalent conditions which are summarised here.

Lemma 3.29.1
Let $(A, \mathfrak{m})$ be a local domain such that $\mathfrak{m}$ is finitely generated and $\dim A = 1$. Then for every non-zero ideal $\mathfrak{a}$ there exists some n such that $\mathfrak{m}^n \subseteq \mathfrak{a}$.

Proof. As $\dim A = 1$ every minimal prime over $\mathfrak{a}$ is equal to $\mathfrak{m}$, whence $\sqrt{\mathfrak{a}} = \mathfrak{m}$ (i.e. $\mathfrak{a}$ is $\mathfrak{m}$-primary). As $\mathfrak{m}$ is finitely generated there is some n such that $\mathfrak{m}^n \subseteq \mathfrak{a}$ (3.3.48). ∎

Proposition 3.29.2 (DVR Criteria)
Let A be an integral domain which is not a field. Then the following are equivalent.

a) A is a valuation ring whose valuation group is order-isomorphic to $\mathbb{Z}$
b) A is a local principal ideal domain
c) A is a Noetherian local ring with maximal ideal $\mathfrak{m} = (\pi)$ and $\dim A = 1$
d) A is a local ring such that every ideal is of the form (π^k) for some $k \geq 0$ (and these are distinct)
e) There exists a non-unit $\pi \in A$ such that every element $a \in A$ may be represented as $u\pi^k$ for some $u \in A^\star$ and non-negative integer k.
f) A is a Noetherian valuation ring
g) A is a Noetherian integrally closed local domain with $\dim A = 1$
h) A is a Noetherian regular local ring with $\dim A = 1$

*In this case we say A is a **discrete valuation ring** (or **DVR**) and let $K = \mathrm{Frac}(A)$. For every generator π of $\mathfrak{m}$, the ideals (π^n) are distinct and*

- *for every $x \in K^\star$ there is a unique integer k and unit u such that $x = u\pi^k$, and*
- *there is a well-defined valuation $v : K^\star \to \mathbb{Z}$ determined uniquely by the relation $x = u\pi^{v(x)}$*

Proof. a) $\implies$ b) Consider a valuation $v : K \to \mathbb{Z}$ with $v^{-1}(\mathbb{Z}_{\geq 0}) = A$. For any ideal $\mathfrak{a}$ choose $f \in \mathfrak{a}$ such that $v(f)$ is minimal. For $g \in \mathfrak{a}$ we have $v(g) \geq v(f) \implies v(gf^{-1}) \geq 0 \implies gf^{-1} \in A \implies g \in (f)$. Therefore $\mathfrak{a} = (f)$ is principal. Further by (3.23.2) A is a local ring.

b) $\implies$ c). By (3.20.4) $\dim A = 1$ and by assumption $\mathfrak{m} = (\pi)$. Evidently A is Noetherian.

c) $\implies$ d). By (3.27.1) the sequence (π^k) is strictly decreasing. Assume wlog that $\mathfrak{a}$ is proper and non-zero, then as $\dim A = 1$ every prime ideal is maximal and we find $\sqrt{\mathfrak{a}} = (\pi)$ by (3.3.46).

Suppose that $\mathfrak{a} \subseteq (\pi^k)$ for all k. Then $\pi \in \sqrt{\mathfrak{a}} \implies (\pi^k) = \mathfrak{a}$ for all sufficiently large k which contradicts the fact the sequence is strictly decreasing. Therefore we may assume $\mathfrak{a} \subseteq (\pi^k)$ for some k and there exists $a \in \mathfrak{a} \setminus (\pi^{k+1})$. Therefore $a = u\pi^k$ and by construction $u \notin (\pi) \implies u \in A^\star \implies \pi^k \in \mathfrak{a}$. Therefore $\mathfrak{a} = (\pi^k)$ as required.

d) $\implies$ e). By assumption $(a) = (\pi^k)$ for some integer k, whence by (3.15.3) $a = u\pi^k$ for some $u \in A^\star$. This representation is unique because the ideals (π^k) are all distinct and A is cancellable.

e) $\implies$ a). We may show easily that the integer k is uniquely defined, denote this by $v(a)$ where $a = u\pi^{v(a)}$ for $u \in A^\star$. We need to show that v is a valuation. By uniqueness we deduce $v(ab) = v(a) + v(b)$. Suppose that $a, b, a+b \in A \setminus \{0\}$. Then $(\pi^{v(a+b)}) = (a+b) \subseteq (\pi^{\min(v(a),v(b))})$. As the sequence (π^k) is strictly decreasing we find $v(a+b) \geq \min(v(a), v(b))$ as required. We may extend to $K^\star$ by $v(ab^{-1}) := v(a) - v(b)$. Then $v(ab^{-1}) \geq 0 \iff v(a) \geq v(b) \iff (a) \subseteq (b) \iff b \mid a \iff ab^{-1} \in A$. Further $v(a) = 0 \iff a \notin (\pi) \iff a \in A^\star$. Therefore A is the valuation ring associated to v.

d) $\implies$ f) $\implies$ b). This follows from (3.23.20).

f) $\implies$ g). A valuation ring is integrally closed (3.23.2) and we've already shown A is local with dimension 1.

g) $\implies$ c). Let $0 \neq a \in \mathfrak{m}$. As $\dim A = 1$ from (3.29.1) we know $\mathfrak{m}^n \subseteq (a)$ for some $n \geq 1$. Choose n minimal and $b \in \mathfrak{m}^{n-1} \setminus (a)$.

Let $x = a^{-1}b \in K = \mathrm{Frac}(A)$. Evidently $x \notin A$, for otherwise $b \in (a)$; further $x\mathfrak{m} = a^{-1}b\mathfrak{m} \subseteq a^{-1}\mathfrak{m}^n \subseteq a^{-1}(a) = A$ is an ideal of A.

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We claim that $x\mathfrak{m} = A$. Suppose not then $x\mathfrak{m} \subseteq \mathfrak{m}$ by (3.3.37). Then $\mathfrak{m}$ is a faithful finite $A[x]$-module and x is integral over A (3.22.3). By assumption A is integrally closed so $x \in A$ which is a contradiction. Therefore $x\mathfrak{m} = A \implies \mathfrak{m} = x^{-1}A$. This shows that $x^{-1} = x^{-1} \cdot 1 \in A$ and that $\mathfrak{m}$ is a principal ideal.

c) $\implies$ h). By assumption A is a Noetherian local ring with $\dim A = 1$. So it remains to show that $\dim \mathfrak{m}/\mathfrak{m}^2 = 1$. As $\mathfrak{m}$ is principal then $\dim \mathfrak{m}/\mathfrak{m}^2 \leq 1$. If it is zero then $\mathfrak{m} = \mathfrak{m}^2$ and $\dim A = 0$ by (3.27.1), a contradiction.

h) $\implies$ c). By assumption $\dim \mathfrak{m}/\mathfrak{m}^2 = 1$ and so by (3.19.7) $\mathfrak{m} = (\pi)$ for some $\pi \in \mathfrak{m} \setminus \mathfrak{m}^2$.

The final statements have already been proven. ∎

Lemma 3.29.3 (Valuation and Powers of the Maximal Ideal)
Let $(A, \mathfrak{m})$ be a DVR with valuation v, and $k \geq 0$ an integer. For $0 \neq a \in A$:

a) $v(a) \geq k \iff a \in \mathfrak{m}^k$;
b) $v(a) = k \iff a \in \mathfrak{m}^k \setminus \mathfrak{m}^{k+1}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Fix a uniformiser π (so $\mathfrak{m} = (\pi)$, hence $\mathfrak{m}^k = (\pi^k)$ (3.29.2)), and write $a = u\pi^{v(a)}$ for a unit $u \in A^\star$ (3.29.2).

a). $a \in \mathfrak{m}^k = (\pi^k) \iff \pi^k \mid a = u\pi^{v(a)} \iff \pi^k \mid \pi^{v(a)}$ (as u is a unit) $\iff v(a) \geq k$ (as the ideals (π^n) are strictly decreasing (3.29.2)).

b). By a), $a \in \mathfrak{m}^k \setminus \mathfrak{m}^{k+1} \iff v(a) \geq k$ and not $(v(a) \geq k+1) \iff v(a) = k$. ∎

Corollary 3.29.4
Let A be a principal ideal domain which is not a field. Then every localisation of A at a non-zero prime ideal is a DVR.

In particular $\mathbb{Z}_{(p)}$ and $k[X]_{(f)}$ are DVRs where p is prime and $f(X)$ is an irreducible polynomial.

Lemma 3.29.5 (A DVR is Maximal Among Valuation Subrings of its Fraction Field)
Let A be a DVR with fraction field K and B a valuation ring of K with $A \subseteq B$. Then $A = B$ or $B = K$.

Proof. Let π be a uniformiser of A, so every $a \in A$ is $u\pi^k$ for a unit $u \in A^\star$ and integer $k \geq 0$ (3.29.2). Suppose $B \neq K$. As B is a valuation ring of K, either $\pi \in B$ (already true, as $\pi \in A \subseteq B$) or $\pi^{-1} \in B$; we rule out the latter. If $\pi^{-1} \in B$, then for any $x \in K^\star$, writing $x = u\pi^n$ ($u \in A^\star \subset B$, $n \in \mathbb{Z}$, using A a DVR), $x = u\pi^n \in B$ whether $n \geq 0$ (as $\pi^n \in A \subset B$) or $n < 0$ (as $\pi^n = (\pi^{-1})^{-n} \in B$); so $B = K$, contradicting assumption. Hence $\pi^{-1} \notin B$, that is $\pi \in \mathfrak{m}_B$. Then $\mathfrak{m}_A = \pi A \subset \pi B \subset \mathfrak{m}_B$ (as $\pi \in \mathfrak{m}_B$ and B absorbs), so B dominates A; as A is a valuation ring, it is maximal under domination (3.23.13), giving $A = B$. ∎

Lemma 3.29.6 (Directed Union of Valuation Rings with a Common Uniformiser)
Let $(K_i)_{i \in I}$ be a directed system of fields with union $K := \bigcup_i K_i$, and for each i let $R_i \subset K_i$ be a valuation ring with $R_i \subseteq R_l$ whenever $K_i \subseteq K_l$. Suppose in addition every $\mathfrak{m}_{R_i}$ is principal, generated by a single common element $0 \neq \pi \in \bigcap_i R_i$. Then $R := \bigcup_i R_i$ is a valuation ring of K, with $\mathfrak{m}_R = \pi R \neq 0$; in particular $R \subsetneq K$.

Proof. For $K_i \subseteq K_l$ (so $R_i \subseteq R_l$), $\mathfrak{m}_{R_i} = \pi R_i \subseteq \pi R_l = \mathfrak{m}_{R_l}$, giving R_l dominates R_i (3.23.4) so (3.23.7) applies, and $\mathfrak{m}_R = \bigcup_i \mathfrak{m}_{R_i} = \bigcup_i \pi R_i = \pi R$.

As $\pi \in \mathfrak{m}_{R_i}$ is a non-unit of every R_i, $\pi^{-1} \notin R_i$ for every i (well-defined, as $0 \neq \pi \in K_i \subset K$), hence $\pi^{-1} \notin R = \bigcup_i R_i$; as $\pi^{-1} \in K$, $R \subsetneq K$. ∎

Proposition 3.29.7
Let A be a discrete valuation ring with unique maximal ideal $\mathfrak{m}$. Then there exists a canonical group isomorphism

$$(\mathfrak{m})^- : \mathbb{Z} \xrightarrow{\sim} \mathcal{I}_A$$

which is the unique group homomorphism mapping 1 to $\mathfrak{m}$ (and 0 to A. In particular every fractional ideal is invertible.

Write the inverse group isomorphism by $v_\mathfrak{m}$, which satisfies

a) $\mathfrak{m}^{v_\mathfrak{m}(\mathfrak{a})} = \mathfrak{a}$
b) $v_\mathfrak{m}(\mathfrak{m}^k) = k$

A fractional ideal I is contained in A if and only if $v_\mathfrak{m}(I) \geq 0$.

Proof. By (3.29.2) we may suppose $\mathfrak{m} = (\pi)$. Evidently (π^k) is a fractional ideal for every integer k and $(\pi^k)(\pi^l) = (\pi^{k+l})$, which gives the required homomorphism $k \to (\pi^k)$.

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Suppose the kernel is non-trivial then there exists k which maps to A, and we may assume that k is positive (by considering the inverse). This means precisely that $\mathfrak{m}^k = A$, which may only happen when $k = 0$ because $\mathfrak{m}$ is proper.

Let I be an arbitrary fractional ideal then $aI \subseteq A$ for some $a \in A$. By (3.29.2) $a = u\pi^l$ and $aI = (\pi^k)$ for some positive integers l and k. This means $(\pi^l)I = (\pi^k)$ and therefore $I = (\pi^k)(\pi^{-l}) = (\pi^{k-l})$. This shows the homomorphism is surjective and therefore an isomorphism.

Evidently $(\pi^k) \subset A \iff k \geq 0$ since $\pi \in A$ and $\pi^{-1} \notin A \implies \pi^{-k} \notin A$. ∎

Proposition 3.29.8
Let A be a discrete valuation ring and I, J fractional ideals. Then the following are equivalent

a) $I \subset J$
b) J divides I (that is there exists an ideal $\mathfrak{a}$ such that $\mathfrak{a}J = I$)
c) $v_{\mathfrak{m}}(I) \geq v_{\mathfrak{m}}(J)$

Proof. Using the results of (3.29.7) we may show (eliding the subscript $\mathfrak{m}$)

a) $\iff$ c) $I \subset J \iff J^{-1}I \subset A \iff v(J^{-1}I) \geq 0 \iff v(I) \geq v_{\mathfrak{m}}(J)$

b) $\implies$ c) $v(I) = v(\mathfrak{a}) + v(J) \geq v(J)$

c) $\implies$ b) Define $\mathfrak{a} = \mathfrak{m}^{v(I)-v(J)}$ then evidentally $v(\mathfrak{a}J) = v(I)$ whence $I = \mathfrak{a}J$. ∎

Lemma 3.29.9 (Intersection of the Powers of the Maximal Ideal)
Let A be a DVR with maximal ideal $\mathfrak{m}$. Then $\bigcap_{n\geq 0} \mathfrak{m}^n = 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $0 \neq a \in A$, and write $a = u\pi^{v(a)}$ for a unit u and uniformiser π (3.29.2). If $a \in \mathfrak{m}^{v(a)+1} = (\pi^{v(a)+1})$ then $a = u'\pi^{v(a)+1}$ for some $u' \in A$, so $u\pi^{v(a)} = u'\pi^{v(a)+1}$, giving $u = u'\pi$ by cancellation – contradicting u a unit. So $a \notin \mathfrak{m}^{v(a)+1}$, in particular $a \notin \bigcap_n \mathfrak{m}^n$. ∎

Lemma 3.29.10 (Ramification and Injectivity/Surjectivity of Local Homomorphisms)
*Let $(A, \mathfrak{m})$, $(B, \mathfrak{n})$ be DVRs with valuations v_A, v_B (3.29.2), and $\phi : A \to B$ a local ring homomorphism (that is $\phi(\mathfrak{m}) \subseteq \mathfrak{n}$). Fix a uniformiser π of A and set $e := v_B(\phi(\pi)) \in \mathbb{N} \cup \{\infty\}$ (the **ramification index** of ϕ; independent of the choice of π, with $e = \infty$ iff $\phi(\pi) = 0$). Then:*

a) ϕ is injective if and only if $e < \infty$;
b) if ϕ is surjective, then $e = 1$; in particular, by a), ϕ is then also injective – so ϕ is surjective if and only if it is bijective.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Independence of π. If $\pi' = u\pi$ for a unit $u \in A^\star$, then $\phi(\pi') = \phi(u)\phi(\pi)$ with $\phi(u) \in B^\star$ (units map to units under any ring homomorphism, and $\phi(u) \notin \mathfrak{n}$, as otherwise $1 = \phi(u^{-1})\phi(u) \in \mathfrak{n}$, absurd), so $v_B(\phi(\pi')) = v_B(\phi(\pi))$.

a). $\ker\phi$ is a prime ideal of A (the kernel of a ring homomorphism into the domain B), so $\ker\phi \in \{0, \mathfrak{m}\}$ (3.29.2). If $e = \infty$: $\phi(\pi) = 0$, so $\pi \in \ker\phi$, and, as $\mathfrak{m} = (\pi)$, $\mathfrak{m} \subseteq \ker\phi$; as $\ker\phi \neq A$ ($\phi(1) = 1 \neq 0$), $\ker\phi = \mathfrak{m} \neq 0$: ϕ is not injective. If $e < \infty$: $\phi(\pi) \neq 0$, so $\pi \notin \ker\phi$, giving $\ker\phi \neq \mathfrak{m}$, so $\ker\phi = 0$: ϕ is injective.

b). Suppose ϕ is surjective. Let ϖ be a uniformiser of B; by surjectivity, $\varpi = \phi(a)$ for some $a \in A$, necessarily $a \neq 0$ (as $\varpi \neq 0$). Write $a = u\pi^k$ for a unit $u \in A^\star$ and $k := v_A(a) \geq 0$ (3.29.2). If $e = \infty$: $\phi(a) = \phi(u)\phi(\pi)^k$ is $\phi(u)$ (a unit, as units map to units) if $k = 0$, or 0 if $k \geq 1$; neither equals ϖ ($v_B(\varpi) = 1 \neq 0$, $\varpi \neq 0$), a contradiction. So $e < \infty$, and

$$1 = v_B(\varpi) = v_B(\phi(u)) + k \cdot v_B(\phi(\pi)) = 0 + ke = ke,$$

forcing $e = 1$. ∎

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3.30 Dedekind Domains

We develop the theory of Dedekind Domains. These occur both as number fields and regular affine curves (over any field). The important results are the finiteness of integral closure in a finite extension, and the ramification formula of a prime ideal. In the case of regular curves over a finite field further results are needed, which we prove later in the context of affine algebras.

Proposition 3.30.1
Let A be a Noetherian integral domain of dimension 1. Then the following are equivalent

a) A is integrally closed
b) $A_\mathfrak{p}$ is integrally closed for every prime ideal $\mathfrak{p}$
c) $A_\mathfrak{p}$ is a discrete valuation ring for every non-zero prime ideal $\mathfrak{p}$

*In this case we say that A is a **Dedekind Domain**.*

Proof. Let $\mathfrak{p}$ be a non-zero prime ideal of A. Then $A_\mathfrak{p}$ is Noetherian (3.13.7), local, and an integral domain as a subring of $\mathrm{Frac}(A)$. Moreover $\dim A_\mathfrak{p} = \mathrm{ht}(\mathfrak{p}) \leq \dim A = 1$, and $(0) \subsetneq \mathfrak{p}$ so $\dim A_\mathfrak{p} = \mathrm{ht}(\mathfrak{p}) = 1$.

a) $\iff$ b) This is (3.22.10).

b) $\implies$ c) By the above $A_\mathfrak{p}$ is a Noetherian local domain of dimension 1, and it is integrally closed by b). Hence it is a discrete valuation ring by (3.29.2) g).

c) $\implies$ b) For $\mathfrak{p}$ non-zero, $A_\mathfrak{p}$ is a discrete valuation ring, hence integrally closed by (3.29.2) g). For $\mathfrak{p} = (0)$ we have $A_{(0)} = \mathrm{Frac}(A)$, a field, which is automatically integrally closed. ∎

Corollary 3.30.2
Every principal ideal domain which is not a field is a Dedekind Domain. For example $k[X]$ and $\mathbb{Z}$.

Proof. We check the hypotheses of (3.30.1), taking condition a). Let A be a PID which is not a field.

- A is a Noetherian integral domain (3.14.2).
- A has dimension 1: every non-zero prime ideal of A is maximal (3.15.23), so $\dim A \leq 1$; and as A is not a field it has a non-zero maximal ideal $\mathfrak{m} \supsetneq (0)$, so $\dim A \geq 1$.
- A is integrally closed, as A is a UFD (3.15.23) and a UFD is integrally closed (3.22.12).

∎

Corollary 3.30.3
Every discrete valuation ring is a Dedekind Domain.

Proof. A discrete valuation ring A is a Noetherian integrally closed domain of dimension 1 (and local) (3.29.2) g), which is condition a) of (3.30.1). ∎

The following is useful in the theory of algebraic curves and function fields.

Proposition 3.30.4 (Correspondence between Prime Ideals and Valuation Rings)
Let A be a Dedekind domain with quotient field K. Then there is a bijection

$$\{\mathfrak{p} \lhd A \mid \mathfrak{p} \text{ non-zero prime}\} \longleftrightarrow \{R \text{ local ring} \mid A \subseteq R \subsetneq K\}$$
$$\mathfrak{p} \longrightarrow A_\mathfrak{p}$$
$$\mathfrak{m}_R \cap A \longleftarrow R$$

and each such local ring R is a discrete valuation ring. Its maximal ideal is $\mathfrak{m}_R = \mathfrak{p}R$ where $\mathfrak{p} = \mathfrak{m}_R \cap A$, so that $\mathfrak{m}_{A_\mathfrak{p}} = \mathfrak{p}A_\mathfrak{p}$.

Proof. Let $\mathfrak{p}$ be a non-zero prime ideal. Then $A_\mathfrak{p}$ is a discrete valuation ring by (3.30.1), in particular local with maximal ideal $\mathfrak{p}A_\mathfrak{p}$, and $A \subseteq A_\mathfrak{p} \subseteq K$. Also $A_\mathfrak{p} \neq K$: if $0 \neq a \in \mathfrak{p}$ and $a^{-1} = x/s$ with $x \in A$, $s \in A \setminus \mathfrak{p}$, then $s = ax \in \mathfrak{p}$, a contradiction. So $\mathfrak{p} \mapsto A_\mathfrak{p}$ is well-defined.

Next, $\mathfrak{m}_{A_\mathfrak{p}} \cap A = \mathfrak{p}$. Indeed if $a = x/s$ with $x \in \mathfrak{p}$ and $s \in A \setminus \mathfrak{p}$ then $as = x \in \mathfrak{p}$, so $a \in \mathfrak{p}$ as $\mathfrak{p}$ is prime, and the other inclusion is clear. In particular $\mathfrak{p} \mapsto A_\mathfrak{p}$ is injective.

Conversely let R be a local ring with $A \subseteq R \subsetneq K$, and put $\mathfrak{p} := \mathfrak{m}_R \cap A$, a prime ideal of A. As R is not a field (a field $R \supseteq A$ contains $\mathrm{Frac}(A) = K$) there is $0 \neq x/y \in \mathfrak{m}_R$ with $x, y \in A$, whence $0 \neq x = y \cdot (x/y) \in \mathfrak{m}_R \cap A = \mathfrak{p}$. So

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$\mathfrak{p}$ is non-zero. If $x \in A \setminus \mathfrak{p}$ then $x \in R \setminus \mathfrak{m}_R$, so $x \in R^\star$; hence $A_\mathfrak{p} \subseteq R$ and $\mathfrak{p}A_\mathfrak{p} \subseteq \mathfrak{m}_R$, that is R dominates $A_\mathfrak{p}$. As $A_\mathfrak{p}$ is a discrete valuation ring it is a valuation ring (3.29.2), so it is maximal under domination (3.23.13), and $R = A_\mathfrak{p}$.

This shows $\mathfrak{p} \mapsto A_\mathfrak{p}$ is surjective with inverse $R \mapsto \mathfrak{m}_R \cap A$, and every R is of the form $A_\mathfrak{p}$, hence a discrete valuation ring with maximal ideal $\mathfrak{p}A_\mathfrak{p}$ (first paragraph). ∎

Remark 3.30.5
The proof of injectivity works for any integral domain A and any prime ideal $\mathfrak{p}$: $\mathfrak{p} \mapsto A_\mathfrak{p}$ has left inverse $R \mapsto \mathfrak{m}_R \cap A$. It is surjectivity onto the local rings between A and K that needs A to be Dedekind (to make each $A_\mathfrak{p}$ a valuation ring). It fails for $A = k[t^2, t^3]$, where $k[t]_{(t)}$ is a local ring between A and $K = k(t)$ with $\mathfrak{m}_R \cap A = (t^2, t^3)$, the same centre as $A_{(t^2,t^3)}$ but a different ring.

Proposition 3.30.6
Let A be a Dedekind domain. Then every fractional ideal is invertible.

Proof. Let I be a fractional ideal. As A is Noetherian (3.30.1), I is finitely generated (3.28.2). So by (3.28.11) it is sufficient to show that $I_\mathfrak{m}$ is invertible for every maximal ideal $\mathfrak{m}$. As $\dim A = 1$ the ring A is not a field, so $\mathfrak{m}$ is non-zero and $A_\mathfrak{m}$ is a discrete valuation ring by (3.30.1). Now $I_\mathfrak{m}$ is a fractional ideal of $A_\mathfrak{m}$ (3.28.9), so it is invertible by (3.29.7). ∎

Lemma 3.30.7
Let A be a Dedekind domain and $\mathfrak{a}$ a non-zero ideal. Then there is a well-defined group homomorphism

$$(\mathfrak{a})^- : \mathbb{Z} \longrightarrow \mathcal{I}_A$$

mapping 1 to $\mathfrak{a}$.

Proof. By (3.30.6) the monoid $\mathcal{I}_A$ is a group, and therefore the first homomorphism exists by (3.2.6). ∎

Lemma 3.30.8
Let A be a Dedekind domain, $\mathfrak{p}$, $\mathfrak{q}$ non-zero prime ideals and $k \in \mathbb{Z}$. Then

$$\left(\mathfrak{p}^k\right)_\mathfrak{q} = \begin{cases} A_\mathfrak{q} & \mathfrak{p} \neq \mathfrak{q} \\ \left(\mathfrak{q}A_\mathfrak{q}\right)^k & \mathfrak{p} = \mathfrak{q} \end{cases}$$

Proof. When $k = 0$ then by definition $\mathfrak{p}^k = A$ and $\left(\mathfrak{q}A_\mathfrak{q}\right)^k = A_\mathfrak{q}$ and so we are done.

We suppose that $k = 1$. If $\mathfrak{p} \neq \mathfrak{q}$ then as $\dim A = 1$ we have $\mathfrak{p} \not\subseteq \mathfrak{q}$. Therefore by definition $\mathfrak{p}_\mathfrak{q}$ contains 1 and equals $A_\mathfrak{q}$ as required. The case $\mathfrak{p} = \mathfrak{q}$ follows from (3.28.8).

By (3.28.9) and (3.30.7) both left and right hand sides define group homomorphisms $\mathbb{Z} \to \mathcal{I}_{A_\mathfrak{q}}$ which agree at 1 and are therefore equal. ∎

Proposition 3.30.9 (Unique Factorisation)
Let A be a Dedekind domain and $\mathcal{P}$ the set of non-zero prime ideals of A. There is a well-defined abelian group isomorphism

$$\bigoplus_{\mathfrak{p} \in \mathcal{P}} \mathbb{Z} \xrightarrow{\sim} \mathcal{I}_A$$
$$(e_\mathfrak{p}) \longrightarrow \prod_{\mathfrak{p} : e_\mathfrak{p} \neq 0} \mathfrak{p}^{e_\mathfrak{p}}$$

Write $\left(v_\mathfrak{p}\right)_{\mathfrak{p} \in \mathcal{P}}$ for the inverse, then it is given explicitly by

$$v_\mathfrak{p}(I) = v_\mathfrak{p}\left(I_\mathfrak{p}\right)$$

where on the right $v_\mathfrak{p}$ is the valuation of the discrete valuation ring $A_\mathfrak{p}$ (3.29.7). Further a fractional ideal I is contained in A if and only if $v_\mathfrak{p}(I) \geq 0$ for all $\mathfrak{p} \in \mathcal{P}$.

Proof. The homomorphisms exists by (3.30.7) and the universal property of the sum of abelian groups.

Let I be a fractional ideal and define $v_\mathfrak{p}(I) \coloneqq v_\mathfrak{p}\left(I_\mathfrak{p}\right)$ using the notation of (3.29.7). This determines a group homomorphism $v_\mathfrak{p} : \mathcal{I}_A \to \mathbb{Z}$ by (3.28.9) b). We claim for every I that $v_\mathfrak{p}(I)$ is zero for all but finitely many $\mathfrak{p} \in \mathcal{P}$. As $I = (a)^{-1}\mathfrak{a}$ for some $0 \neq a \in A$ and non-zero ideal $\mathfrak{a}$ of A it is sufficient by multiplicativity to demonstrate this for non-zero ideals $\mathfrak{a}$. If $\mathfrak{a} \not\subset \mathfrak{p}$ then $\mathfrak{a}_\mathfrak{p} = A_\mathfrak{p}$ (as $\mathfrak{a}$ contains an element of $A \setminus \mathfrak{p}$) which has valuation 0. Therefore we

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may assume that $\mathfrak{a} \subset \mathfrak{p}$. However by (3.20.6) there are only finitely many such primes, since every prime containing $\mathfrak{a}$ must be minimal over $\mathfrak{a}$ (as $\mathfrak{a} \neq 0$ and $\dim A = 1$).

Observe by (3.30.8) that for distinct $\mathfrak{p}_1, ..., \mathfrak{p}_n \in \mathcal{P}$ we have the identity

$$\left(\mathfrak{p}_1^{r_1} ... \mathfrak{p}_n^{r_n}\right)_{\mathfrak{q}} = \begin{cases} \left(\mathfrak{p}_k A_{\mathfrak{q}}\right)^{r_k} & \mathfrak{q} = \mathfrak{p}_k \\ A_{\mathfrak{q}} & \mathfrak{q} \neq \mathfrak{p}_k \text{ for all } k = 1...n \end{cases}$$

By the first claim there is a well-defined homomorphism $\mathcal{I}_A \to \bigoplus_{\mathfrak{p} \in \mathcal{P}} \mathbb{Z}$. We claim that this is a right inverse for the given map, that is to say

$$\prod_{\mathfrak{p}: v_{\mathfrak{p}}(I) \neq 0} \mathfrak{p}^{v_{\mathfrak{p}}(I)} = I$$

Denote the left hand side by I'. When $v_{\mathfrak{p}}(I) \neq 0$ then by the identity above we have, using (3.29.7) for the final equality,

$$I'_{\mathfrak{p}} = \left(\mathfrak{p} A_{\mathfrak{p}}\right)^{v_{\mathfrak{p}}(I)} = \left(\mathfrak{p} A_{\mathfrak{p}}\right)^{v_{\mathfrak{p}}\left(I_{\mathfrak{p}}\right)} = I_{\mathfrak{p}}$$

Similarly if $v_{\mathfrak{p}}(I) = 0$ by (3.29.7)

$$I'_{\mathfrak{p}} = A_{\mathfrak{p}} = I_{\mathfrak{p}}$$

We conclude by (3.28.10) (every maximal ideal of A lies in $\mathcal{P}$, as A is not a field) that $I = I'$ as required. Finally the relation above also shows that this is a left inverse.

For the final statement by (3.28.10) we have $I \subset A \iff I_{\mathfrak{p}} \subset A_{\mathfrak{p}}$ for all $\mathfrak{p} \in \mathcal{P}$, then we may use the equivalent statement over DVRs (3.29.7). ∎

Proposition 3.30.10
Let A be a Dedekind domain with I, J fractional ideals. Then the following are equivalent

a) $I \subset J$
b) J divides I (that is there exists a non-zero ideal $\mathfrak{a}$ such that $\mathfrak{a}J = I$)
c) $v_{\mathfrak{p}}(I) \geq v_{\mathfrak{p}}(J)$ for all $\mathfrak{p} \in \mathcal{P}$

Proof. a) $\iff$ c) By (3.28.10) we have $I \subset J \iff I_{\mathfrak{p}} \subset J_{\mathfrak{p}}$ for all $\mathfrak{p} \in \mathcal{P}$ (if $I_{\mathfrak{m}} \subset J_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$ then $I = \bigcap_{\mathfrak{m}} I_{\mathfrak{m}} \subset \bigcap_{\mathfrak{m}} J_{\mathfrak{m}} = J$, and the maximal ideals are exactly the elements of $\mathcal{P}$). Then we may use the corresponding statement for DVRs (3.29.8).

b) $\implies$ c) Follows because $v_{\mathfrak{p}}(I) = v_{\mathfrak{p}}(\mathfrak{a}) + v_{\mathfrak{p}}(J)$ and $v_{\mathfrak{p}}(\mathfrak{a}) \geq 0$ (3.30.9).

c) $\implies$ b) We may define $\mathfrak{a} = \prod \mathfrak{p}^{v_{\mathfrak{p}}(I) - v_{\mathfrak{p}}(J)}$ (a finite product, as $v_{\mathfrak{p}}(I) = v_{\mathfrak{p}}(J) = 0$ for all but finitely many $\mathfrak{p}$ (3.30.9)) which is contained in A and by multiplicativity $v_{\mathfrak{p}}(\mathfrak{a}J) = v_{\mathfrak{p}}(I)$ for all $\mathfrak{p} \in \mathcal{P}$ whence $\mathfrak{a}J = I$ by the correspondence in (3.30.9). ∎

Proposition 3.30.11 (Integral Closure is a Dedekind Domain)
Let L/K be a finite separable extension and $A \subset K$ a subring which is a Dedekind domain with quotient field K. Let B be the integral closure of A in L, then B is a finite A-module and a Dedekind domain.

If in addition A is a principal ideal domain (e.g. a discrete valuation ring) then B is a free A-module of rank $[L : K]$.

Moreover $B \otimes_A K \cong L$, via $b \otimes x \mapsto bx$ (regarding $B, K \subseteq L$).

Proof. From (3.22.26) we know B is integrally closed and $\dim B = \dim A = 1$. As A is Dedekind it is Noetherian and integrally closed, so (3.22.28) shows that B is a finite A-module (hence Noetherian), and free of rank $[L : K]$ if A is a principal ideal domain. So B is a Noetherian integrally closed domain of dimension 1, that is a Dedekind domain (3.30.1).

Base change to K. Writing $S := A \setminus \{0\}$, (3.5.31) gives an isomorphism $B \otimes_A K \cong S^{-1}B$, $b \otimes x \mapsto bx$, where $S^{-1}B$ is naturally a subring of L (as $S \subset B \setminus \{0\}$ too). By (3.22.15) (applied to the integral map $A \hookrightarrow B$, and S), $K = S^{-1}A \to S^{-1}B$ is integral; as $S^{-1}B$ is an integral domain (a subring of L) and K is a field, (3.22.14) gives that $S^{-1}B$ is itself a field. Being a field containing B, $S^{-1}B$ contains $\mathrm{Frac}(B) = L$; but $S^{-1}B \subseteq L$ already; so $S^{-1}B = L$. ∎

Definition 3.30.12 (AKLB Situation)
Let L/K be a finite field extension, $A \subset K$, $B \subset L$ Dedekind domains for which $A \subset B$ is integral (equivalently B is the integral closure of A in L), $\mathrm{Frac}(A) = K$ and $\mathrm{Frac}(B) = L$. We refer to this situation as "AKLB".

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Lemma 3.30.13
Using the notation in (3.30.12). Let $\mathfrak{p}$ be a non-zero prime ideal of A, and $\mathfrak{b}$ a proper ideal of B. Then the following are equivalent

a) $\mathfrak{p} \subset \mathfrak{b}$
b) $\mathfrak{b}$ *divides* $\mathfrak{p}B$
c) $\mathfrak{b} \cap A = \mathfrak{p}$

and in this case we say that $\mathfrak{b}$ divides $\mathfrak{p}$ and write $\mathfrak{b} \mid \mathfrak{p}$.

In this case $B/\mathfrak{b}$ is naturally an $A/\mathfrak{p}$-module via the commutative diagram

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \downarrow & & \downarrow \\ A/\mathfrak{p} & \hookrightarrow & B/\mathfrak{b} \end{array}$$

Proof. a) $\iff$ b) follows from (3.30.10)

a) $\implies$ c) Trivially $\mathfrak{p} \subset \mathfrak{b} \cap A$. As $1 \notin \mathfrak{b}$ then $\mathfrak{b} \cap A$ is a proper ideal, and therefore equal to $\mathfrak{p}$ by maximality (a non-zero prime of the Dedekind domain A is maximal, as $\dim A = 1$).

c) $\implies$ a) Immediate. ∎

Proposition 3.30.14
Using the notation in (3.30.12). Let $\mathfrak{p}$ be a non-zero prime ideal of A. There exists at least one prime ideal $\mathfrak{q} \mid \mathfrak{p}$, and only finitely many. For every such prime ideal define the integers $e_{\mathfrak{q}/\mathfrak{p}} := v_{\mathfrak{q}}(\mathfrak{p}B)$, then

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}} = \bigcap_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

Proof. The existence of a prime $\mathfrak{q}$ follows from the Lying Over property of integral extensions (3.22.18). The factorisation exists by (3.30.9) which also shows that there are only finitely many divisors of $\mathfrak{p}$, and the second equality holds by the Chinese Remainder Theorem (3.3.63). ∎

Definition 3.30.15 (Ramification Index / Inertial Degree)
*Using the notation in (3.30.12), suppose that $\mathfrak{q} \mid \mathfrak{p}$. Then the **ramification index** is the integer*

$$e_{\mathfrak{q}/\mathfrak{p}} = v_{\mathfrak{q}}(\mathfrak{p}B)$$

*which is positive: $\mathfrak{q}$ divides $\mathfrak{p}B$ (3.30.13), so $\mathfrak{p}B = \mathfrak{a}\mathfrak{q}$ for an ideal $\mathfrak{a}$ of B and $v_{\mathfrak{q}}(\mathfrak{p}B) = v_{\mathfrak{q}}(\mathfrak{a}) + 1 \geq 1$ (3.30.9). Similarly define the **inertial degree***

$$f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{q} : A/\mathfrak{p}]$$

which we presently show to be finite.

Lemma 3.30.16 (Removing a Common Factor in a PID)
Let A be a PID and $\lambda_1, ..., \lambda_n \in A$ not all zero. Then there are $0 \neq d \in A$ and $\mu_1, ..., \mu_n \in A$ with $\lambda_i = d\mu_i$ for every i and $(\mu_1, ..., \mu_n) = A$. If A is moreover a local ring with maximal ideal $\mathfrak{m}$ then $\mu_i \notin \mathfrak{m}$ for some i.

Proof. As A is a PID, $(\lambda_1, ..., \lambda_n) = (d)$ for some d, and $d \neq 0$ as the λ_i are not all zero. Each $\lambda_i \in (d)$, so $\lambda_i = d\mu_i$ for some $\mu_i \in A$. Also $d = \sum_i a_i\lambda_i = d\sum_i a_i\mu_i$ for some $a_i \in A$, so $\sum_i a_i\mu_i = 1$ as A is a domain and $d \neq 0$, that is $(\mu_1, ..., \mu_n) = A$. If A is local then a proper ideal lies in $\mathfrak{m}$, so the μ_i cannot all lie in $\mathfrak{m}$. ∎

Remark 3.30.17
A PID cannot be weakened to a UFD here. Let $A = k[x, y]_{(x,y)}$, a local UFD with maximal ideal $\mathfrak{m} = (x, y)$. The elements x, y have no common prime factor, but $(x, y) = \mathfrak{m} \neq A$, and both lie in $\mathfrak{m}$, so no common factor d can be removed to leave an element outside $\mathfrak{m}$. The next lemma genuinely fails for such rings: for $k = \mathbb{R}$, $B = A[y/x]$, $\mathfrak{p} = (x, y)$ and $\mathfrak{b} = (x, (y/x)^2 + 1)$ we have $\mathfrak{p}B \subseteq \mathfrak{b}$ and $B/\mathfrak{b} \cong \mathbb{C}$, which has degree 2 over $A/\mathfrak{p} = \mathbb{R}$, although B and A have the same quotient field.

Lemma 3.30.18
Let L/K be a finite field extension, A a subring with quotient field K and $B \supset A$ an overring with quotient field L. Suppose that $\mathfrak{p}$ is a maximal ideal of A with $A_{\mathfrak{p}}$ a PID, and $\mathfrak{b}$ is a proper ideal of B containing $\mathfrak{p}$. Then canonically $B/\mathfrak{b}$ is an $A/\mathfrak{p}$-module and

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$$[B/\mathfrak{b} : A/\mathfrak{p}] \leq [L : K]$$

Furthermore we have $\mathfrak{b}B_\mathfrak{p} \cap A_\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$ *and*

$$[B/\mathfrak{b} : A/\mathfrak{p}] = \left[B_\mathfrak{p}/\mathfrak{b}B_\mathfrak{p} : A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}\right]$$

Proof. Suppose that $\overline{x_1}, ..., \overline{x_n} \in B/\mathfrak{b}$ are linearly independent over $A/\mathfrak{p}$. It is sufficient to show that $x_1, ..., x_n \in B$ are linearly independent over K, as then $n \leq [L : K]$, and the $\overline{x_i}$ were arbitrary linearly independent elements. Suppose we have a linear dependence

$$0 = \sum_{i=1}^{n} \lambda_i x_i \quad \lambda_i \in K$$

Then by clearing denominators we may assume that $\lambda_i \in A_\mathfrak{p}$ not all zero. By (3.30.16) (applied to the local PID $A_\mathfrak{p}$) we may divide the λ_i by a common non-zero factor, which does not affect the dependence, and assume that at least one λ_i is not in $\mathfrak{p}A_\mathfrak{p}$. Multiplying by a common denominator $s \in A \setminus \mathfrak{p}$ (a unit of $A_\mathfrak{p}$) we may take $\lambda_i \in A$, and some λ_i is still not in $\mathfrak{p}A_\mathfrak{p}$, so $\lambda_i \notin \mathfrak{p}$ (as $\mathfrak{p} \subseteq \mathfrak{p}A_\mathfrak{p}$). Reducing the dependence modulo $\mathfrak{b}$ gives $\sum_i \overline{\lambda_i}\overline{x_i} = 0$ in $B/\mathfrak{b}$ with $\overline{\lambda_i} \in A/\mathfrak{p}$ not all zero, contradicting the linear independence of the $\overline{x_i}$.

For the last two statements note that $\mathfrak{b} \cap A$ is a proper ideal containing the maximal ideal $\mathfrak{p}$, so $\mathfrak{b} \cap A = \mathfrak{p}$. Hence $\mathfrak{b}B_\mathfrak{p}$ is proper: if $1 = b'/t$ with $b' \in \mathfrak{b}$, $t \in A \setminus \mathfrak{p}$ then $t = b' \in \mathfrak{b} \cap A = \mathfrak{p}$, a contradiction. So $\mathfrak{b}B_\mathfrak{p} \cap A_\mathfrak{p}$ is a proper ideal of the local ring $A_\mathfrak{p}$ containing $\mathfrak{p}A_\mathfrak{p}$, hence equal to it. For the last statement consider the commutative diagram

$$\begin{array}{ccc} A/\mathfrak{p} & \hookrightarrow & B/\mathfrak{b} \\ \downarrow \sim & & \downarrow \sim \\ A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p} & \hookrightarrow & B_\mathfrak{p}/\mathfrak{b}B_\mathfrak{p} \end{array}$$

Both vertical arrows are isomorphisms by (3.5.26), applied with $S := A \setminus \mathfrak{p}$, as S is co-prime to the ideal in each case: for $s \in S$ we have $As + \mathfrak{p} = A$ by maximality of $\mathfrak{p}$, whence $Bs + \mathfrak{p}B = B$ and so $Bs + \mathfrak{b} = B$, as $\mathfrak{p}B \subseteq \mathfrak{b}$. So the diagram identifies the extension $A/\mathfrak{p} \subseteq B/\mathfrak{b}$ with $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p} \subseteq B_\mathfrak{p}/\mathfrak{b}B_\mathfrak{p}$, which gives the equality of the degrees. ∎

Lemma 3.30.19 (Local Form of Ramification Index / Inertial Degree)
Using the notation in (3.30.12), suppose $\mathfrak{p}$ *is a non-zero prime ideal and define* $A' := A_\mathfrak{p}$, $\mathfrak{p}' := \mathfrak{p}A_\mathfrak{p}$, $B' := B_\mathfrak{p}$. *Then* A', B' *is also "AKLB" and there is a bijection*

$$\{\mathfrak{q} \lhd B : \mathfrak{q} \mid \mathfrak{p}\} \longleftrightarrow \{\mathfrak{q}' \lhd B' : \mathfrak{q}' \mid \mathfrak{p}'\}$$

under which the ramification index and inertial degrees agree, i.e.

$$e_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}'/\mathfrak{p}'}$$

and

$$f_{\mathfrak{q}/\mathfrak{p}} = f_{\mathfrak{q}'/\mathfrak{p}'} \leq [L : K]$$

Further the ramification index is equal to

$$v_{\mathfrak{q}'}(\pi B')$$

where π *is any uniformiser for the DVR* A'.

Proof. Evidently A' has quotient field K and B' has quotient field L. A' is a DVR (3.30.1) and therefore a Dedekind domain (3.30.3). By (3.22.9) B' is the integral closure of A', and therefore integrally closed. It has Krull Dimension 1 by (3.22.24) and, B being Noetherian (Dedekind, (3.30.1)), $B' = B_\mathfrak{p}$ is Noetherian too (3.13.7). Therefore we have the same situation as (3.30.12).

First consider the commutative diagram

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \downarrow & & \downarrow \\ A' & \hookrightarrow & B' \end{array}$$

regarded as subrings of K and L respectively.

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Then by chasing the diagram $\mathfrak{p}'B' = (\mathfrak{p}B)B'$. By (3.30.14) we have a unique factorisation

$$\mathfrak{p}B = \prod_{i=1}^{n} \mathfrak{q}_i^{e_i}$$

for prime ideals $\mathfrak{q}_i$ of B dividing $\mathfrak{p}$. Define $\mathfrak{q}_i' := \mathfrak{q}_i B_{\mathfrak{p}}$ then these are prime by (3.5.18) (and the fact $\mathfrak{q} \cap (A \setminus \mathfrak{p}) = \emptyset$) and contain $\mathfrak{p}'B'$ (3.30.14). Then by (3.3.51)

$$\mathfrak{p}'B' = \prod_{i=1}^{n} (\mathfrak{q}_i')^{e_i}$$

The bijection of prime ideals is well-defined by (3.5.19).

By uniqueness $e_{\mathfrak{q}'/\mathfrak{p}'} = e_{\mathfrak{q}/\mathfrak{p}}$ for all primes $\mathfrak{q} \mid \mathfrak{p}$.

The relation $f_{\mathfrak{q}'/\mathfrak{p}'} = f_{\mathfrak{q}/\mathfrak{p}} \leq [L : K]$ follows directly from (3.30.18). ∎

Proposition 3.30.20 (Ramification Formula)
Using the notation of (3.30.12). For every non-zero prime ideal $\mathfrak{p}$ of A we have the identity

$$\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{p}B : A/\mathfrak{p}] \leq [L : K]$$

Suppose in addition that B is a finite A-module then we have equality. In particular this holds when L/K is separable.

Proof. Recall from (3.30.14) we have the factorisation

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

By the Chinese Remainder Theorem (3.3.63) we have a canonical isomorphism

$$B/\mathfrak{p}B \xrightarrow{\sim} \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}}}$$

Both $B/\mathfrak{p}B$ and $B/\mathfrak{q}^{e_{\mathfrak{q}}}$ are $A/\mathfrak{p}$-modules, and so equating vector space dimensions we have

$$[B/\mathfrak{p}B : A/\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} [B/\mathfrak{q}^{e_{\mathfrak{q}}} : A/\mathfrak{p}]$$

As $\mathfrak{p}B \subset \mathfrak{q}^e$ we have a decreasing sequence of $A/\mathfrak{p}$-submodules

$$B/\mathfrak{q}^e \supseteq \mathfrak{q}^1/\mathfrak{q}^e \supseteq \ldots \supseteq \mathfrak{q}^e/\mathfrak{q}^e = (0)$$

so by (3.3.131) we have

$$[B/\mathfrak{q}^e : A/\mathfrak{p}] = \sum_{i=1}^{e} [(\mathfrak{q}^{i-1}/\mathfrak{q}^e)/(\mathfrak{q}^i/\mathfrak{q}^e) : A/\mathfrak{p}]$$

We claim there is an $A/\mathfrak{p}$-module isomorphism

$$(\mathfrak{q}^{i-1}/\mathfrak{q}^e)/(\mathfrak{q}^i/\mathfrak{q}^e) \to \mathfrak{q}^{i-1}/\mathfrak{q}^i$$

This exists as an A-module isomorphism by (3.3.106), and therefore as an $A/\mathfrak{p}$-module isomorphism as the action factors through the quotient map $A \to A/\mathfrak{p}$. Therefore by (3.17.15)

$$\begin{aligned}
[B/\mathfrak{q}^e : A/\mathfrak{p}] &= \sum_{i=1}^{e} [\mathfrak{q}^{i-1}/\mathfrak{q}^i : A/\mathfrak{p}] \\
&= \sum_{i=1}^{e} [B/\mathfrak{q} : A/\mathfrak{p}][\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}] \\
&= \sum_{i=1}^{e} f_{\mathfrak{q}/\mathfrak{p}} [\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}]
\end{aligned}$$

We claim that $[\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}] = 1$. This quotient is non-zero, as $v_{\mathfrak{q}}(\mathfrak{q}^{i-1}) = i - 1 \neq i = v_{\mathfrak{q}}(\mathfrak{q}^i)$ (3.30.9) shows $\mathfrak{q}^{i-1} \neq \mathfrak{q}^i$. Any subspace of $\mathfrak{q}^{i-1}/\mathfrak{q}^i$ other than 0 and the whole space is a-fortiori a B-submodule, and by the isomorphism theorem yields an ideal strictly between $\mathfrak{q}^{i-1}$ and $\mathfrak{q}^i$. Therefore it is enough to show there is no such ideal. By (3.5.42) we may localize at $\mathfrak{q}$ to consider the ring $B_{\mathfrak{q}}$. This is a DVR (3.30.1) and the result follows from (3.29.2). Therefore

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$$[B/\mathfrak{p}B : A/\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}.$$

The inequality follows from (3.30.18) applied with $\mathfrak{b} = \mathfrak{p}B$ (a proper ideal containing $\mathfrak{p}$, as $\mathfrak{p}B \subseteq \mathfrak{q}$), where $\mathfrak{p}$ is maximal and $A_{\mathfrak{p}}$ a DVR as A is Dedekind and $\mathfrak{p} \neq 0$.

It remains to show this equals $[B/\mathfrak{p}B : A/\mathfrak{p}] = [L : K]$ in the case B is a finite A-module. By (3.30.19) and the last statement of (3.30.18) (applied with $\mathfrak{b} = \mathfrak{p}B$) we may replace A, B by $A_{\mathfrak{p}}, B_{\mathfrak{p}}$, where $B_{\mathfrak{p}}$ is again a finite $A_{\mathfrak{p}}$-module, and so reduce to the case of A a discrete valuation ring, in particular a principal ideal domain. Then by (3.22.27) B is a free A-module of rank $n = [L : K]$, so $B/\mathfrak{p}B \cong (A/\mathfrak{p})^n$ and $[B/\mathfrak{p}B : A/\mathfrak{p}] = n = [L : K]$.

The case L/K is separable is then (3.22.28). ∎

3.30.1 Local Monogenicity

Lemma 3.30.21 (A Uniformiser Generates a Totally Ramified Extension)
*Using the notation of (3.30.12), with L/K separable, let $\mathfrak{p}$ be a non-zero prime of A with a **unique** prime $\mathfrak{q}$ of B over it, of residue degree $f_{\mathfrak{q}/\mathfrak{p}} = 1$ (so $B_{\mathfrak{p}} = B_{\mathfrak{q}}$ is itself a DVR, totally ramified: $e_{\mathfrak{q}/\mathfrak{p}} = n := [L : K]$). Then for any uniformiser π of $B_{\mathfrak{p}}$,*

$$B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi],$$

and π's minimal polynomial $F \in A_{\mathfrak{p}}[Y]$ over K has degree exactly n.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

For $0 \leq j < n$, $\mathfrak{q}^j/\mathfrak{q}^{j+1}$ is 1-dimensional over $B_{\mathfrak{p}}/\mathfrak{q} = k$ (as $B_{\mathfrak{p}}$ is a DVR), generated by the image of π^j; residue degree 1 means $A_{\mathfrak{p}}/\mathfrak{p} = k$ too, so by downward induction on j, $\{1, \pi, ..., \pi^{n-1}\}$ spans $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} = B_{\mathfrak{p}}/\mathfrak{q}^n$ (as $\mathfrak{p}B_{\mathfrak{p}} = \mathfrak{q}^n$, totally ramified) as an $A_{\mathfrak{p}}/\mathfrak{p}$-module. By (3.19.6), applied to the finite $A_{\mathfrak{p}}$-module $B_{\mathfrak{p}}$, a spanning set of $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ already spans $B_{\mathfrak{p}}$ itself, so

$$B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi].$$

Let F be π's minimal polynomial over K, of degree d. Then $F \in A_{\mathfrak{p}}[Y]$ by (3.22.19), as $\pi \in B_{\mathfrak{p}}$ is integral over the integrally closed domain $A_{\mathfrak{p}}$ (3.22.9). As $B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi]$ and $F(\pi) = 0$, $B_{\mathfrak{p}}$ is already spanned by $1, \pi, ..., \pi^{d-1}$ over $A_{\mathfrak{p}}$, so $\operatorname{rank}_{A_{\mathfrak{p}}} B_{\mathfrak{p}} \leq d$; as $B_{\mathfrak{p}}$ is $A_{\mathfrak{p}}$-free of rank n (3.22.27), $d \geq n$, while $d = [K(\pi) : K] \leq [L : K] = n$ as $K(\pi) \subseteq L$. So $d = n$ exactly. ∎

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3.31 Formal Power Series and Laurent Series

Definition 3.31.1 (Formal Laurent Series)
*Let A be a commutative ring. Define the **formal Laurent series ring** $A((X))$ to be the set of sequences $(c_n)_{n\in\mathbb{Z}}$ in A such that there exists N for which $n \leq N \implies c_n = 0$. Addition is defined pointwise with product defined as follows*

$$\left(\sum_{n\in\mathbb{Z}} c_n X^n\right)\left(\sum_{n\in\mathbb{Z}} d_n X^n\right) = \sum_{n\in\mathbb{Z}}\left(\sum_{k\in\mathbb{Z}} c_{n-k}d_k\right)X^n$$

where the inner sum is a sum of a family with all but finitely many terms zero. That this is well-defined and makes $A((X))$ into a commutative ring is shown in (3.31.3).

Definition 3.31.2 (Order of a Formal Laurent Series)
*Let A be a commutative ring. For $0 \neq h = \sum_n c_n X^n \in A((X))$ define the **order** $v(h) := \min\{n \mid c_n \neq 0\}$, and $v(0) := \infty$. The minimum exists since the c_n vanish for n sufficiently negative and $h \neq 0$. We call $c_{v(h)}$ the **leading coefficient** of h.*

Proposition 3.31.3 ($A((X))$ is a Commutative Ring)
Let A be a commutative ring and $c, d \in A((X))$. For every $n \in \mathbb{Z}$ the family $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ has all but finitely many terms zero, so $(cd)_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ is defined, and $(cd)_n = 0$ for $n < v(c) + v(d)$. In particular $cd \in A((X))$. Moreover $A((X))$ is a commutative ring under pointwise addition and this product.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The product is well-defined. Let $c = (c_n)_{n\in\mathbb{Z}}$ and $d = (d_n)_{n\in\mathbb{Z}}$ be elements of $A((X))$. If $c = 0$ or $d = 0$ every term below is zero and the product is 0, so assume $v(c), v(d)$ are finite. Fix $n \in \mathbb{Z}$. If the term $c_{n-k}d_k$ is nonzero then $n - k \geq v(c)$ and $k \geq v(d)$, that is

$$v(d) \leq k \leq n - v(c)$$

so only finitely many $k \in \mathbb{Z}$ give a nonzero term. Hence $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ has all but finitely many terms zero, and the inner sum $e_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ is defined by (3.2.50). Moreover if $n < v(c) + v(d)$ then there is no integer k with $v(d) \leq k \leq n - v(c)$, so every term vanishes and $e_n = 0$. Therefore $(e_n)_{n\in\mathbb{Z}}$ is a sequence in A vanishing for $n < v(c) + v(d)$, that is an element of $A((X))$.

Ring axioms. Write $(cd)_n := \sum_{k\in\mathbb{Z}} c_{n-k}d_k$ for the coefficients of the product, an element of $A((X))$ by the first part.

Addition. For $c, d \in A((X))$ we have $c_n + d_n = 0$ and $-c_n = 0$ for $n < \min(v(c), v(d))$, so $c + d$ and $-c$ lie in $A((X))$, as does the zero sequence. So $A((X))$ is a subgroup of the abelian group $A^{\mathbb{Z}}$ under pointwise addition.

Identity. Let $1 := (\delta_{n,0})_{n\in\mathbb{Z}}$, which vanishes for $n \leq -1$ and so lies in $A((X))$. For $c \in A((X))$ the family $(\delta_{n-k,0}c_k)_{k\in\mathbb{Z}}$ has $k = n$ as its only possibly nonzero term, so $(1c)_n = c_n$.

Commutativity. The map $k \mapsto n - k$ is a bijection of $\mathbb{Z}$, so by (3.2.53) and commutativity of A,

$$(cd)_n = \sum_{k\in\mathbb{Z}} c_{n-k}d_k = \sum_{k\in\mathbb{Z}} c_k d_{n-k} = \sum_{k\in\mathbb{Z}} d_{n-k}c_k = (dc)_n$$

Distributivity. For $c, d, d' \in A((X))$, the families $(c_{n-k}d_k)_{k\in\mathbb{Z}}$ and $(c_{n-k}d'_k)_{k\in\mathbb{Z}}$ have all but finitely many terms zero, so by (3.2.51)

$$(c(d+d'))_n = \sum_{k\in\mathbb{Z}} c_{n-k}(d_k + d'_k) = \sum_{k\in\mathbb{Z}} c_{n-k}d_k + \sum_{k\in\mathbb{Z}} c_{n-k}d'_k = (cd)_n + (cd')_n$$

and the other distributive law follows by commutativity.

Associativity. Let $c, d, e \in A((X))$ and fix $n \in \mathbb{Z}$; if any of them is 0 both sides are 0, so assume $v(c), v(d), v(e)$ are finite. Set $g(j,k) := c_{n-j-k}d_j e_k$ for $(j,k) \in \mathbb{Z}\times\mathbb{Z}$. If $g(j,k) \neq 0$ then $j \geq v(d)$, $k \geq v(e)$ and $n - j - k \geq v(c)$, so $v(d) \leq j \leq n - v(c) - v(e)$ and $v(e) \leq k \leq n - v(c) - v(d)$; hence g has all but finitely many terms zero. Recall that $(cd)_m = \sum_{j\in\mathbb{Z}} c_{m-j}d_j$ and $(de)_m = \sum_{k\in\mathbb{Z}} d_{m-k}e_k$. Then

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$$\begin{aligned}((cd)e)_n &= \sum_{k\in\mathbb{Z}} (cd)_{n-k}e_k && \text{by definition of the product}\\ &= \sum_{k\in\mathbb{Z}}\left(\sum_{j\in\mathbb{Z}} c_{n-k-j}d_j\right)e_k && \text{by definition of } (cd)_{n-k}\\ &= \sum_{k\in\mathbb{Z}}\sum_{j\in\mathbb{Z}} g(j,k) && \text{by (3.3.17)}\\ &= \sum_{(j,k)\in\mathbb{Z}\times\mathbb{Z}} g(j,k) && \text{by (3.2.53)}\end{aligned}$$

and on the other side

$$\begin{aligned}(c(de))_n &= \sum_{l\in\mathbb{Z}} c_{n-l}(de)_l && \text{by definition of the product}\\ &= \sum_{l\in\mathbb{Z}} c_{n-l}\sum_{k\in\mathbb{Z}} d_{l-k}e_k && \text{by definition of } (de)_l\\ &= \sum_{l\in\mathbb{Z}}\sum_{k\in\mathbb{Z}} g(l-k,k) && \text{by (3.3.17)}\\ &= \sum_{(l,k)\in\mathbb{Z}\times\mathbb{Z}} g(l-k,k) && \text{by (3.2.53)}\\ &= \sum_{(j,k)\in\mathbb{Z}\times\mathbb{Z}} g(j,k) && \text{by (3.2.53) for the bijection } (l,k)\mapsto(l-k,k)\end{aligned}$$

Hence $((cd)e)_n = (c(de))_n$ for all n. ∎

Lemma 3.31.4 (Valuation of a Sum and a Product)
Let A be a commutative ring and $c, d \in A((X))$. Then $v(1) = 0$, $v(-c) = v(c)$, and

a) $v(c+d) \geq \min(v(c), v(d))$
b) $v(cd) \geq v(c) + v(d)$
c) if A is an integral domain then so is $A((X))$, and $v(cd) = v(c) + v(d)$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Clearly $v(1) = 0$ and $v(-c) = v(c)$. If $n < \min(v(c), v(d))$ then $c_n = d_n = 0$, so $(c+d)_n = 0$, which gives a). For b), if $c = 0$ or $d = 0$ then $cd = 0$ and both sides are ∞. Otherwise $(cd)_n = 0$ for $n < v(c) + v(d)$ by (3.31.3), which means $v(cd) \geq v(c) + v(d)$. For c), let $c, d \neq 0$ with $m := v(c)$ and $r := v(d)$. A term $c_{m+r-k}d_k$ of $(cd)_{m+r}$ is nonzero only if $k \geq r$ and $m + r - k \geq m$, that is $k = r$, so $(cd)_{m+r} = c_m d_r \neq 0$ as A is a domain, while $(cd)_n = 0$ for $n < m + r$. Hence $cd \neq 0$ and $v(cd) = m + r$. ∎

Definition 3.31.5 (Formal Power Series)
*Let A be a commutative ring. Define the **formal power series ring** $A[[X]]$ to be the subring of $A((X))$ consisting of those sequences $(c_n)_{n\in\mathbb{Z}}$ for which $n < 0 \implies c_n = 0$, or equivalently those $h \in A((X))$ for which $v(h) \geq 0$.*

Proof that $A[[X]]$ is a subring of $A((X))$. Drafted by Claude Sonnet, reviewed by David Rufino.

$A[[X]] = \{h \in A((X)) \mid v(h) \geq 0\}$, so by (3.31.4) it contains 1 and is closed under negation, addition and multiplication. ∎

Lemma 3.31.6 (Formal Power Series Rings are Separated)
Let A be a commutative ring. For $N \geq 0$ we have $X^N A[[X]] = \{h \in A[[X]] \mid v(h) \geq N\}$, and $\bigcap_{N\geq 0} X^N A[[X]] = 0$.

Proof. If $h = X^N h'$ with $h' \in A[[X]]$ then the coefficients of h below X^N vanish, so $v(h) \geq N$. Conversely if $v(h) \geq N$ then $h = X^N h'$ where h' is h with the coefficients shifted down by N, an element of $A[[X]]$. If h lies in every $X^N A[[X]]$ then $v(h) \geq N$ for every N, so every coefficient of h vanishes and $h = 0$. ∎

Definition 3.31.7 (Formal Derivative)
Let A be a commutative ring. For $f = \sum_n c_n X^n \in A((X))$, define $f' := \sum_n n c_n X^{n-1} \in A((X))$, that is $(f')_n = (n+1)c_{n+1}$.

Lemma 3.31.8 (Valuation of a Derivative)
Let A be a commutative ring and $f \in A((X))$. Then $v(f') \geq v(f) - 1$. In particular if $f \in A[[X]]$ then $f' \in A[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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If $n < v(f) - 1$ then $n + 1 < v(f)$, so $c_{n+1} = 0$ and $(f')_n = (n+1)c_{n+1} = 0$. This gives $v(f') \geq v(f) - 1$. If $f \in A[[X]]$ then $v(f) \geq 0$ and $(f')_n = 0$ for $n < -1$; also $(f')_{-1} = 0 \cdot c_0 = 0$, so $f' \in A[[X]]$. ∎

Definition 3.31.9 (Summable Family in $A((X))$)
*Let A be a commutative ring. A family $(h_i)_{i\in I}$ in $A((X))$ is **summable** if for every $N \in \mathbb{Z}$ all but finitely many $i \in I$ satisfy $v(h_i) \geq N$. Its **sum** is the formal Laurent series*

$$\sum_{i\in I} h_i := \sum_{n\in\mathbb{Z}} \left(\sum_{i\in I} (h_i)_n \right) X^n$$

where $(h_i)_n$ denotes the coefficient of X^n in h_i.

Proof that the sum is well-defined. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $h_{i,n} := (h_i)_n$ and fix $n \in \mathbb{Z}$. By hypothesis (with $N = n+1$) the set $\{i \in I \mid v(h_i) \leq n\}$ is finite, and it contains every i with $h_{i,n} \neq 0$. So $(h_{i,n})_{i\in I}$ has all but finitely many terms zero and $s_n := \sum_{i\in I} h_{i,n}$ is defined by (3.2.50). Let $J_0 := \{i \in I \mid v(h_i) \leq 0\}$, which is finite, and $N_0 := \min\big(0, \min_{i\in J_0} v(h_i)\big)$ (with $N_0 := 0$ if J_0 is empty). If $n < N_0$ then $v(h_i) > n$ for every i, so every $h_{i,n} = 0$ and $s_n = 0$. Hence $(s_n)_{n\in\mathbb{Z}}$ lies in $A((X))$. ∎

Proposition 3.31.10 (Sums of Summable Families)
Let A be a commutative ring and $(h_i)_{i\in I}$ a summable family in $A((X))$. Then

a) If $v(h_i) \geq N$ for every $i \in I$ then $v\big(\sum_{i\in I} h_i\big) \geq N$. In particular if every $h_i \in A[[X]]$ then $\sum_{i\in I} h_i \in A[[X]]$.
b) For a bijection $\sigma : I' \to I$ the family $\big(h_{\sigma(i')}\big)_{i'\in I'}$ is summable and $\sum_{i'\in I'} h_{\sigma(i')} = \sum_{i\in I} h_i$.
c) The family $(h_i')_{i\in I}$ is summable and $\big(\sum_{i\in I} h_i\big)' = \sum_{i\in I} h_i'$.
d) For any $g \in A((X))$ the family $(gh_i)_{i\in I}$ is summable and $g \cdot \sum_{i\in I} h_i = \sum_{i\in I} gh_i$.
e) If $(k_i)_{i\in I}$ is another summable family then $(h_i + k_i)_{i\in I}$ is summable and $\sum_{i\in I}(h_i + k_i) = \sum_{i\in I} h_i + \sum_{i\in I} k_i$.
f) If $I = \bigsqcup_{\lambda\in\Lambda} I_\lambda$ is a partition of I into disjoint subsets then each $(h_i)_{i\in I_\lambda}$ and $\Big(\sum_{i\in I_\lambda} h_i\Big)_{\lambda\in\Lambda}$ is summable, and $\sum_{i\in I} h_i = \sum_{\lambda\in\Lambda} \sum_{i\in I_\lambda} h_i$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $h_{i,n} := (h_i)_n$ and $s_n := \sum_{i\in I} h_{i,n} = \big(\sum_{i\in I} h_i\big)_n$.

a) If $n < N$ then $h_{i,n} = 0$ for every i, so $s_n = 0$, as $J = \emptyset$ is a valid choice in (3.2.50). Hence $v\big(\sum_{i\in I} h_i\big) \geq N$, and taking $N = 0$ gives the statement for $A[[X]]$.

b) The reindexed family clearly meets the summability condition. The family $(h_i)_{i\in I}$ itself is not almost zero in $A((X))$, but for each n the coefficient family $(h_{i,n})_{i\in I}$ in A is, so the statement holds coefficientwise by (3.2.53).

c) Since $v(h_i') \geq v(h_i) - 1$ by (3.31.8) the family $(h_i')_{i\in I}$ is also summable, so $\sum_{i\in I} h_i'$ is defined. Using $(h')_n = (n+1)h_{n+1}$, for every $n \in \mathbb{Z}$

$$\begin{aligned}
\left(\left(\sum_{i\in I} h_i \right)' \right)_n &= (n+1)s_{n+1} = (n+1) \sum_{i\in I} h_{i,n+1} && \text{by definition of the sum} \\
&= \sum_{i\in I} (n+1) h_{i,n+1} && \text{by (3.3.17)} \\
&= \sum_{i\in I} (h_i')_n = \left(\sum_{i\in I} h_i' \right)_n && \text{by definition of the sum}
\end{aligned}$$

d) Let $g \in A((X))$. If $g = 0$ both sides are 0, so assume $v(g)$ is finite. By (3.31.4) $v(gh_i) \geq v(g) + v(h_i)$, so $(gh_i)_{i\in I}$ is summable and $\sum_{i\in I} gh_i$ is defined. Fix $n \in \mathbb{Z}$ and set $a(k,i) := g_{n-k} h_{i,k}$ for $(k,i) \in \mathbb{Z} \times I$. If $a(k,i) \neq 0$ then $k \leq n - v(g)$ and $v(h_i) \leq k$, so i lies in the finite set $\{i \in I \mid v(h_i) \leq n - v(g)\}$ and $v(h_i) \leq k \leq n - v(g)$; hence a has all but finitely many terms zero. Then

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$$\begin{aligned}
\left(g\sum_{i\in I}h_i\right)_n &= \sum_{k\in\mathbb{Z}} g_{n-k}s_k = \sum_{k\in\mathbb{Z}} g_{n-k}\sum_{i\in I}h_{i,k} && \text{by definition of the product and the sum}\\
&= \sum_{k\in\mathbb{Z}}\sum_{i\in I} a(k,i) && \text{by (3.3.17)}\\
&= \sum_{i\in I}\sum_{k\in\mathbb{Z}} a(k,i) && \text{by (3.2.53)}\\
&= \sum_{i\in I}\sum_{k\in\mathbb{Z}} g_{n-k}h_{i,k} && \text{by definition of } a\\
&= \sum_{i\in I}(gh_i)_n = \left(\sum_{i\in I} gh_i\right)_n && \text{by definition of the product and the sum}
\end{aligned}$$

e) By (3.31.4) a), $v(h_i+k_i)\geq \min(v(h_i),v(k_i))$, so for every N all but finitely many i satisfy $v(h_i+k_i)\geq N$ and $(h_i+k_i)_{i\in I}$ is summable. The identity holds in each coefficient by (3.2.51).

f) The subfamilies are summable. For $N\in\mathbb{Z}$ the set $F_N := \{i\in I \mid v(h_i)<N\}$ is finite, so only finitely many λ have $I_\lambda\cap F_N\neq\emptyset$, and for every other λ we have $v\left(\sum_{i\in I_\lambda}h_i\right)\geq N$ by a). Hence $\left(\sum_{i\in I_\lambda}h_i\right)_{\lambda\in\Lambda}$ is summable. The identity holds in each coefficient by (3.2.52) applied to $(h_{i,n})_{i\in I}$. ∎

Corollary 3.31.11 (Interchange of Summation)
Let A be a commutative ring and $(a(i,j))_{(i,j)\in I\times J}$ a summable family in $A((X))$. Then each family $(a(i,j))_{j\in J}$ and $(a(i,j))_{i\in I}$ is summable, the families $\left(\sum_{j\in J}a(i,j)\right)_{i\in I}$ and $\left(\sum_{i\in I}a(i,j)\right)_{j\in J}$ are summable, and

$$\sum_{(i,j)\in I\times J} a(i,j) = \sum_{i\in I}\sum_{j\in J} a(i,j) = \sum_{j\in J}\sum_{i\in I} a(i,j).$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Apply (3.31.10) f) to the partition of $I\times J$ into the rows $\{i\}\times J$, and then to the partition into the columns $I\times\{j\}$. ∎

Lemma 3.31.12 (Convergence Criterion for a Sum)
Let A be a commutative ring, $(h_j)_{j\geq 0}$ a summable family in $A((X))$ and $h\in A((X))$. Then $h=\sum_{j\geq 0}h_j$ if and only if the partial sums converge to h, that is $v\left(h-\sum_{j\leq J}h_j\right)\to\infty$ as $J\to\infty$ (for every $N\in\mathbb{Z}$ we have $v\left(h-\sum_{j\leq J}h_j\right)\geq N$ for all sufficiently large J).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $S:=\sum_{j\geq 0}h_j$ and fix $N\in\mathbb{Z}$. As the family is summable, there is J_0 with $v(h_j)\geq N$ for all $j>J_0$. For every J, partitioning $\mathbb{N}$ into $\{0,\dots,J\}$ and $\{j>J\}$ gives $S-\sum_{j\leq J}h_j=\sum_{j>J}h_j$ by (3.31.10), and for $J\geq J_0$ this has $v\geq N$ by part a) of that proposition.

If $h=S$ then $h-\sum_{j\leq J}h_j=\sum_{j>J}h_j$ has $v\geq N$ for every $J\geq J_0$, so the partial sums converge to h. Conversely suppose the partial sums converge to h, and choose $J\geq J_0$ with $v\left(h-\sum_{j\leq J}h_j\right)\geq N$. Then by (3.31.4)

$$v(h-S)\geq \min\left(v\left(h-\sum_{j\leq J}h_j\right), v\left(\sum_{j>J}h_j\right)\right)\geq N.$$

This holds for every N, so every coefficient of $h-S$ vanishes. ∎

Proposition 3.31.13 (Units of a Formal Power Series Ring)
Let A be a commutative ring and $u\in A[[X]]$, with constant coefficient u_0 (the coefficient of X^0). Then u is a unit of $A[[X]]$ if and only if $u_0\in A^\star$. In particular, when $A=k$ is a field, $u\in k[[X]]^\star\iff u_0\neq 0$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose $u_0\in A^\star$ and set $w:=1-u_0^{-1}u\in A[[X]]$. Its constant coefficient is 0, that is $v(w)\geq 1$, so $v(w^i)\geq i$ for all $i\geq 0$ by (3.31.4). Hence $(w^i)_{i\geq 0}$ is summable (3.31.9) and $s:=\sum_{i\geq 0}w^i\in A[[X]]$ by (3.31.10). Then

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$$\begin{aligned} s &= 1 + \sum_{i\geq 1} w^i && \text{splitting off } i=0 \text{ by (3.31.10) } f) \\ &= 1 + \sum_{i\geq 0} w^{i+1} && \text{reindexing } i \mapsto i+1 \text{ by (3.31.10) } b) \\ &= 1 + ws && \text{distributivity, by (3.31.10) } d) \end{aligned}$$

so $(1-w)s = 1$. Since $u_0^{-1}u = 1-w$, this says $u\cdot(u_0^{-1}s) = 1$, so $u_0^{-1}s$ is an inverse of u.

Conversely, if u is a unit with inverse $t \in A[[X]]$ then $(ut)_0 = \sum_{k\in\mathbb{Z}} u_{-k}t_k = u_0t_0$, as $u_{-k} = 0$ for $k > 0$ and $t_k = 0$ for $k < 0$. So $u_0t_0 = 1$ and $u_0 \in A^\star$. When $A = k$ is a field, $u_0 \in k^\star$ is equivalent to $u_0 \neq 0$. ∎

Proposition 3.31.14 (Units of a Formal Laurent Series Ring)
Let A be an integral domain and $0 \neq h \in A((X))$ with $m := v(h)$. The following are equivalent

a) h is a unit of $A((X))$
b) $h = X^m u$ for a unit $u \in A[[X]]^\star$
c) the leading coefficient h_m of h is a unit of A

In particular when $A = k$ is a field, $k((X))$ is a field.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

c) $\implies$ b): $u := X^{-m}h \in A[[X]]$ has constant coefficient $u_0 = h_m \in A^\star$, so u is a unit of $A[[X]]$ by (3.31.13).

b) $\implies$ a): $X^{-m}u^{-1}$ is an inverse of $h = X^m u$.

a) $\implies$ c): let $hh' = 1$ in $A((X))$. Then $v(h)+v(h') = v(1) = 0$ by (3.31.4) c), so $v(h') = -m$, and the coefficient of X^0 in hh' is $h_m h'_{-m} = 1$, so $h_m \in A^\star$. ∎

Proposition 3.31.15 ($k((X))$ is a Field of Fractions for $k[[X]]$)
Let k be a field. Then every element of the field $k((X))$ is a/b for $a, b \in k[[X]]$, $b \neq 0$ – that is, $k((X))$ is a field of fractions for the subring $k[[X]] \subset k((X))$. Moreover, the order function $v : k((X))^\star \to \mathbb{Z}$ is a group homomorphism under multiplication: $v(fg) = v(f)+v(g)$, and $v(f^{-1}) = -v(f)$.

Proof. By (3.31.14), as k is a field every $c \neq 0$ in $k((X))$ is $c = X^m u$ with $m = v(c)$ and $u \in k[[X]]^\star$, which is $uX^m \in k[[X]]$ if $m \geq 0$ and u/X^{-m} otherwise, in either case a ratio of two elements of $k[[X]]$.

By (3.31.4) c), as k is a domain, $v(fg) = v(f)+v(g)$ for $f, g \neq 0$. Applying this to f, f^{-1}: $0 = v(1) = v(ff^{-1}) = v(f)+v(f^{-1})$, giving $v(f^{-1}) = -v(f)$. ∎

Proposition 3.31.16 (Formal Power Series Ring is a DVR)
Let k be a field. Then $k[[X]]$ is an integral domain, not a field, in which every nonzero element a can be written uniquely as $a = X^{v(a)}u$ with $u \in k[[X]]^\star$ (equivalently, $u(0) \neq 0$ (3.31.13)).

Using the criteria (3.29.2)e) $k[[X]]$ is a discrete valuation ring with valuation precisely the order function v.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$k[[X]]$ is a domain, as a subring of the domain $k((X))$ (3.31.4).c), and not a field, as $X \neq 0$ has no inverse ($v(Xg) = 1+v(g) \neq 0$ for every $g \in k[[X]]$ by the same result). Given $0 \neq a \in k[[X]]$, by (3.31.14) $a = X^{v(a)}u$ with $u \in k[[X]]^\star$, where $v(a) \geq 0$: this is exactly the uniformised-element criterion (3.29.2)e), with $\pi := X$. If also $a = X^n u'$ with $u' \in k[[X]]^\star$ then $v(u') = 0$ and $v(a) = n + v(u') = n$ by (3.31.4) c), so the exponent is $v(a)$ and then $u' = u$ as $k((X))$ is a domain. ∎

The next result generalises the Taylor and Laurent series expansion of a meromorphic function at a point. A function f holomorphic near $p \in \mathbb{C}$ is expanded in powers of the local parameter $z-p$ as $f = \sum_{i\geq 0} c_i(z-p)^i$ with $c_i \in \mathbb{C}$, the residue field of the local ring at p, and a meromorphic function is expanded in the same way with finitely many negative powers. For a DVR $(A, \mathfrak{m})$ with uniformiser π we expand elements $a \in A$ in powers of π in the same way: the coefficients $c_i(a)$, which we call the **digits** of a (as in the base-p expansion of an integer, successive digits are determined by working modulo successive powers of $\mathfrak{m}$), are elements of the residue field $k(\mathfrak{m})$, included in the quotients $A/\mathfrak{m}^{n+1}$ by the sections of (3.24.18). It also corresponds to the p-adic expansion $a = \sum_i c_i p^i$ of an integer, or more generally of a rational number $x = \sum_{i\geq v_p(x)} c_i p^i$ with finitely many negative powers (the analogue of the Laurent series), in the DVR $\mathbb{Z}_{(p)}$ with uniformiser p, residue field $\mathbb{F}_p$ and digits $c_i \in \{0, ..., p-1\}$. There the digits are only a choice of representatives of $\mathbb{F}_p$ in $\mathbb{Z}_{(p)}$, as $\mathbb{Z}_{(p)}$ contains no copy of $\mathbb{F}_p$, and the expansion is not compatible with addition and multiplication (carries); in the present equal-characteristic situation the sections s_n remove this choice and make the expansion a ring homomorphism. This gives the "Taylor series" $\sum_i c_i(a)X^i \in k(\mathfrak{m})[[X]]$ of a, and its

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extension to the fraction field is the "Laurent series" of an element of K. For $k(\mathfrak{m}) = k$ and the local ring of a point of a curve over an algebraically closed field k, this is the expansion described in the text of [Lan82] (Ch. I, §2).

Proposition 3.31.17 (Coefficient Expansion of a DVR over a Field)
Let $(A, \mathfrak{m})$ be a DVR with fraction field K and valuation v, and π a uniformiser of A. Let k be a field and suppose A is a k-algebra whose residue field $k(\mathfrak{m}) = A/\mathfrak{m}$ is separable algebraic over k (for example if k is perfect and $k(\mathfrak{m})/k$ is algebraic (3.17.106)).

Recall the k-algebra sections $s_n : k(\mathfrak{m}) \to A/\mathfrak{m}^{n+1}$ of the reduction maps $\pi_n : A/\mathfrak{m}^{n+1} \to k(\mathfrak{m})$ from (3.24.18): they are unique, with $\pi_n \circ s_n = 1_{k(\mathfrak{m})}$, and compatible, $\rho_{n,n-1} \circ s_n = s_{n-1}$. For $a \in A$ write $[a]_n$ for its class in $A/\mathfrak{m}^{n+1}$.

*Then every $a \in A$ has a unique sequence $(c_i(a))_{i \geq 0}$ in $k(\mathfrak{m})$, the **digits** of a relative to π, such that for every $n \geq 0$*

$$[a]_n = \sum_{i=0}^{n} s_n(c_i(a))\,[\pi^i]_n \quad \text{in } A/\mathfrak{m}^{n+1}.$$

The map

$$L_\pi : A \to k(\mathfrak{m})[[X]], \quad L_\pi(a) := \sum_{i \geq 0} c_i(a) X^i,$$

is an injective k-algebra homomorphism with $L_\pi(\pi) = X$. Moreover $v_X(L_\pi(a)) = v(a)$ for every $0 \neq a \in A$, and L_π extends uniquely to an injective ring homomorphism $L_\pi : K \to k(\mathfrak{m})((X))$ (still denoted L_π).

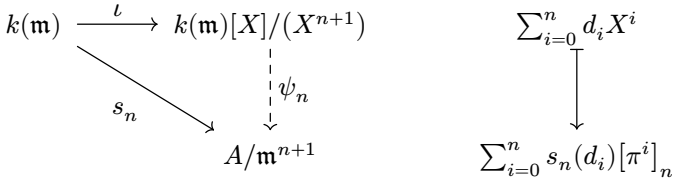
When $k(\mathfrak{m}) = k$, that is $k \to A/\mathfrak{m}$ is an isomorphism, s_n is induced by the structure map $k \to A$, so the displayed identity reads $[a]_n = \sum_{i=0}^{n} c_i [\pi^i]_n$ with $c_i \in k$, that is $a - \sum_{i=0}^{n} c_i \pi^i \in \mathfrak{m}^{n+1}$, and $L_\pi : A \to k[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Notation. We write $\overline{a} := [a]_0 \in k(\mathfrak{m})$ for the residue class of $a \in A$. In $k(\mathfrak{m})[X]/(X^{n+1})$ we write the class of X again as X.

The sections s_n. These exist, are unique and are compatible by (3.24.18). If $k(\mathfrak{m}) = k$ this is immediate: a k-algebra homomorphism out of k is unique, so s_n is the map induced by $k \to A$, which is a section of π_n, as $k \to A/\mathfrak{m}$ is an isomorphism.

The isomorphisms ψ_n. Fix $n \geq 0$. We claim that there is a unique ring homomorphism $\psi_n : k(\mathfrak{m})[X]/(X^{n+1}) \to A/\mathfrak{m}^{n+1}$ with $\psi_n(X) = [\pi]_n$ making the following diagram commute, where ι is the inclusion of the constants, and that ψ_n is an isomorphism:



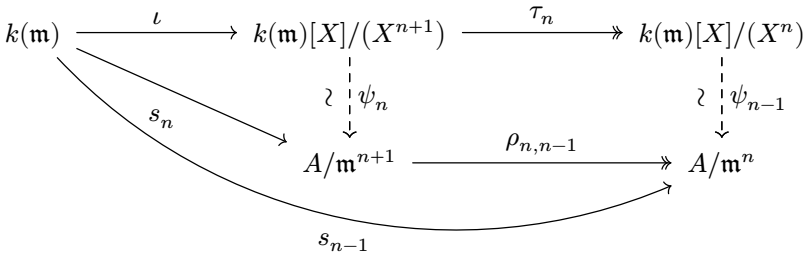
Existence and uniqueness. By the universal property of the polynomial ring $k(\mathfrak{m})[X]$ (3.7.6), applied to the ring homomorphism s_n and the element $[\pi]_n \in A/\mathfrak{m}^{n+1}$, there is a unique ring homomorphism $k(\mathfrak{m})[X] \to A/\mathfrak{m}^{n+1}$ restricting to s_n on $k(\mathfrak{m})$ with $X \mapsto [\pi]_n$, namely $\sum d_i X^i \mapsto \sum s_n(d_i)[\pi^i]_n$; it is a k-algebra homomorphism as s_n is. As $\pi^{n+1} \in \mathfrak{m}^{n+1}$ it kills X^{n+1}, so it factors uniquely through ψ_n. It remains to show that ψ_n is bijective.

Surjective. By induction on n, the case $n = 0$ being $\psi_0 = s_0 = (A/\mathfrak{m} = k(\mathfrak{m}))$. Given $[a]_n \in A/\mathfrak{m}^{n+1}$ with $a \in A$, its image in $A/\mathfrak{m}^n$ is $\psi_{n-1}(g)$ for some g; lifting g to $g' \in k(\mathfrak{m})[X]$ of degree $< n$, compatibility of the s_i gives $[a]_n - \psi_n(g') \in \mathfrak{m}^n/\mathfrak{m}^{n+1}$. Now $\mathfrak{m}^n = \pi^n A$, so this element is $[\pi^n a']_n$ for some $a' \in A$. Put $c := \overline{a'} \in k(\mathfrak{m})$. Then $[a']_n$ and $s_n(c)$ both reduce to c modulo $\mathfrak{m}$, so $[a']_n - s_n(c) \in \mathfrak{m}/\mathfrak{m}^{n+1}$, and multiplying by $[\pi^n]_n$, which lies in $\mathfrak{m}^n/\mathfrak{m}^{n+1}$, gives $[\pi^n a']_n = s_n(c)[\pi^n]_n$. Hence $[a]_n = \psi_n(g' + cX^n)$.

Injective. If $g = \sum_{i \leq n} d_i X^i$ is non-zero and d_j is its first non-zero coefficient, then $\psi_n(g) = [\pi^j]_n \big(s_n(d_j) + [\pi]_n \cdots\big)$ with $s_n(d_j)$ a unit of $A/\mathfrak{m}^{n+1}$ (it reduces to $d_j \neq 0$ in $k(\mathfrak{m})$), so a lift of $\psi_n(g)$ to A has valuation exactly $j \leq n$ and $\psi_n(g) \neq 0$.

Compatibility. Let $\tau_n : k(\mathfrak{m})[X]/(X^{n+1}) \to k(\mathfrak{m})[X]/(X^n)$ be the truncation, and ι the inclusions of the constants. We claim that the ψ_n are compatible with the reductions and with the s_n, that is the following diagram commutes:

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The triangle with s_n commutes by the definition of ψ_n, and the triangle with s_{n-1} by the definition of ψ_{n-1}, as $\tau_n \circ \iota$ is the inclusion of the constants; the remaining triangle $\rho_{n,n-1} \circ s_n = s_{n-1}$ is the compatibility of the sections. It remains to show that the square commutes. The homomorphism $\rho_{n,n-1} \circ \psi_n$ sends X^n to $\rho_{n,n-1}\big([\pi^n]_n\big) = [\pi^n]_{n-1} = 0$ in $A/\mathfrak{m}^n$, so it factors as $\chi \circ \tau_n$ for a ring homomorphism $\chi : k(\mathfrak{m})[X]/(X^n) \to A/\mathfrak{m}^n$. Then $\chi(X) = [\pi]_{n-1}$ and $\chi \circ \iota = \rho_{n,n-1} \circ \psi_n \circ \iota = \rho_{n,n-1} \circ s_n = s_{n-1}$, so χ makes the triangle for $n-1$ commute, and $\chi = \psi_{n-1}$ by the uniqueness in the claim for $n-1$.

The digits and L_π. Put $\phi_n := \psi_n^{-1}$. By the compatibility of the ψ_n, $\phi_{j-1} \circ \rho_{j,j-1} = \tau_j \circ \phi_j$, so $\phi_{j-1}\big([a]_{j-1}\big)$ is the truncation of $\phi_j\big([a]_j\big)$. Hence the coefficient of X^i in $\phi_j\big([a]_j\big)$ is the same for every $j \geq i$, namely $\phi_j\big([a]_j\big)_i = \phi_i([a]_i)_i$, and we define this to be the digit $c_i(a)$.

By construction $\phi_n([a]_n) = \sum_{i \leq n} c_i(a) X^i$ for every n, that is $[a]_n = \psi_n\Big(\sum_{i \leq n} c_i(a) X^i\Big)$, which is the displayed identity. The sequence is unique, as ψ_n is injective and so determines the coefficients $c_0, \ldots, c_n$ for each n.

For $a, b \in A$ and $i \leq n$, comparing coefficients of X^i in $\phi_n([a+b]_n) = \phi_n([a]_n) + \phi_n([b]_n)$ and $\phi_n([ab]_n) = \phi_n([a]_n)\phi_n([b]_n)$ gives $c_i(a+b) = c_i(a) + c_i(b)$ and $c_i(ab) = \sum_{j=0}^{i} c_j(a) c_{i-j}(b)$, and $c_i(1) = \delta_{i,0}$ similarly. So L_π is a ring homomorphism, and it is a k-algebra homomorphism as s_n is k-linear: the image of $c \in k$ is the constant series $\overline{c} \in k(\mathfrak{m})$. For π: $[\pi]_n = \psi_n(X)$, so $c_i(\pi) = \delta_{i,1}$ and $L_\pi(\pi) = X$ (for $n \geq 1$).

Valuation formula and injectivity. Let $0 \neq a \in A$, $m := v(a)$, so $a = u\pi^m$ for a unit $u \in A^\star$ (3.29.2), and write $\overline{u} \in k(\mathfrak{m})^\star$ for its residue. Then $[u]_m$ and $s_m(\overline{u})$ both reduce to $\overline{u}$ modulo $\mathfrak{m}$, so $[u]_m - s_m(\overline{u}) \in \mathfrak{m}/\mathfrak{m}^{m+1}$, and multiplying by $[\pi^m]_m$ gives $[a]_m = [u\pi^m]_m = s_m(\overline{u})[\pi^m]_m = \psi_m(\overline{u} X^m)$. So $\phi_m([a]_m) = \overline{u} X^m$, a polynomial with no terms below X^m. Hence $c_i(a) = 0$ for $i < m$ and $c_m(a) = \overline{u} \neq 0$, so $v_X(L_\pi(a)) = m = v(a)$. In particular $L_\pi(a) \neq 0$ for $a \neq 0$ and L_π is injective.

L_π extends to K. As $L_\pi : A \to k(\mathfrak{m})[[X]]$ is injective and $k(\mathfrak{m})[[X]] \subset k(\mathfrak{m})((X))$ (a field) (3.31.15), $A \setminus \{0\}$ maps into $k(\mathfrak{m})((X))^\star$. So (3.5.5) (with $S := A \setminus \{0\}$, so $S^{-1}A = K$) gives a unique ring homomorphism $L_\pi : K \to k(\mathfrak{m})((X))$ with $L_\pi|_A$ the original map, explicitly $L_\pi(a/s) = L_\pi(a) L_\pi(s)^{-1}$; it is injective (if $L_\pi(a/s) = 0$ then $L_\pi(a) = 0$, so $a = 0$, giving $a/s = 0$). ∎

Remark 3.31.18
The case $k(\mathfrak{m}) = k$ is the classical one, for instance $A = \mathcal{O}_{X,p}$ the local ring of a point of a curve over an algebraically closed field k (or any finitely generated k-algebra localised at a k-rational point, which is a DVR). Then $k \subseteq A$ is a coefficient field, the digits $c_i(a)$ lie in A itself, the displayed identity is an honest congruence $a \equiv \sum_{i \leq n} c_i \pi^i \pmod{\mathfrak{m}^{n+1}}$ in A, and $L_\pi(a) = \sum_i c_i(a) X^i \in k[[X]]$ is the Taylor series of the function a at p in the local parameter π, with Laurent series for elements of K. In general the digits lie in the residue field $k(\mathfrak{m})$, which need not embed in A: for $A = \mathbb{R}[t]_{(t^2+1)}$ the residue field is $\mathbb{C}$, but $A \subseteq \mathbb{R}(t)$ contains no copy of $\mathbb{C}$. The sections s_n lift digits only modulo $\mathfrak{m}^{n+1}$, the expansion lands in $k(\mathfrak{m})[[X]]$, and it is canonical only because the sections are unique, which is where separability of $k(\mathfrak{m})/k$ is used. This is what gives the residue symbol at a non-rational place of a curve over a perfect field.

Proposition 3.31.19 (Product Rule)
Let A be a commutative ring and $f, g \in A((X))$. Then $(fg)' = f'g + fg'$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $f = \sum_i a_i X^i$, $g = \sum_j b_j X^j$ and fix $n \in \mathbb{Z}$. All sums below are over $i \in \mathbb{Z}$ and are sums of families with all but finitely many terms zero, with the coefficients of a product as in (3.31.3). The coefficient of X^{n-1} in $(fg)'$ is $n(fg)_n = n \sum_i a_i b_{n-i}$. Since $(f')_k = (k+1) a_{k+1}$ and $(g')_k = (k+1) b_{k+1}$, reindexing $i = k+1$ and $i = n-k-1$ respectively (3.2.53) gives

$$(f'g)_{n-1} = \sum_i i a_i b_{n-i} \qquad (fg')_{n-1} = \sum_i (n-i) a_i b_{n-i}$$

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Hence

$$\begin{aligned}(f'g+fg')_{n-1} &= \sum_i i a_i b_{n-i} + \sum_i (n-i) a_i b_{n-i} && \text{by the above}\\ &= \sum_i n a_i b_{n-i} && \text{by (3.2.51)}\\ &= n\sum_i a_i b_{n-i} = ((fg)')_{n-1} && \text{by (3.3.17)}\end{aligned}$$

As n was arbitrary, $(fg)' = f'g + fg'$. ∎

Lemma 3.31.20 (Power Rule)
Let A be a commutative ring and $\psi \in A((X))$. Then $(\psi^n)' = n\psi^{n-1}\psi'$ for every $n \geq 0$. If moreover ψ is a unit of $A((X))$, this also holds for every $n < 0$.

Proof. **$n \geq 0$.** Induction on n: the case $n = 0$ is $0 = 0$ ($\psi^0 = 1$ is constant). Given $(\psi^n)' = n\psi^{n-1}\psi'$, the product rule (3.31.19) gives

$$(\psi^{n+1})' = (\psi \cdot \psi^n)' = \psi'\psi^n + \psi(\psi^n)' = \psi'\psi^n + n\psi^n\psi' = (n+1)\psi^n\psi'$$

$n < 0$, ψ a unit. Differentiating $\psi\psi^{-1} = 1$ with the product rule, $0 = \psi'\psi^{-1} + \psi(\psi^{-1})'$, so $(\psi^{-1})' = -\psi^{-2}\psi'$ – the case $n = -1$. For $n \leq -2$, write $n = -r$, $r \geq 2$: applying the $n \geq 0$ case just shown to the element $\psi^{-1} \in A((X))$ (no hypothesis on its order was used there),

$$(\psi^{-r})' = \left((\psi^{-1})^r\right)' = r(\psi^{-1})^{r-1}(\psi^{-1})' = r(\psi^{-1})^{r-1}(-\psi^{-2}\psi') = -r\psi^{-r-1}\psi' = n\psi^{n-1}\psi'$$

∎

Proposition 3.31.21 (Substitution into a Formal Laurent Series)
Let A be a commutative ring, $f = \sum_{n \geq n_0} c_n X^n \in A((X))$ and $\psi \in XA[[X]]$ (that is, $\psi(0) = 0$). Suppose that either

- *$f \in A[[X]]$, that is $n_0 \geq 0$, or*
- *A is an integral domain and $\psi \in A((X))^\star$, that is $\psi \in X^m A[[X]]^\star$ for some $m > 0$ (3.31.14)*

Then

$$(f \circ \psi)(X) := \sum_{n \geq n_0} c_n \psi(X)^n$$

is a well-defined element of $A((X))$ (with ψ^n, $n < 0$, the power in the ring $A((X))$), and $f \circ \psi \in A[[X]]$ if $f \in A[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\psi \in XA[[X]]$ has no terms of degree ≤ 0 we have $v(\psi) \geq 1$, so $v(\psi^n) \geq n$ for every $n \geq 0$ by (3.31.4) b). Negative powers of ψ only occur for the finitely many n with $n_0 \leq n < 0$. Hence for every N all $n \geq \max(N, 0)$ satisfy $v(c_n\psi^n) \geq n \geq N$, and only the finitely many n with $n_0 \leq n < \max(N, 0)$ can fail this, so $(c_n\psi^n)_{n \geq n_0}$ is summable (3.31.9) and its sum lies in $A((X))$. If $f \in A[[X]]$ then every $c_n\psi^n \in A[[X]]$ and the sum lies in $A[[X]]$ by (3.31.10) a). ∎

Proposition 3.31.22 (Chain Rule)
In the setting of (3.31.21), $(f \circ \psi)' = (f' \circ \psi) \cdot \psi'$.

Proof. By the power rule (3.31.20) (which for $n < 0$ requires that ψ is a unit of $A((X))$, as it is then), $(c_n\psi^n)' = c_n(\psi^n)' = nc_n\psi^{n-1}\psi'$ for every $n \geq n_0$, using linearity of $(-)'$ (immediate from (3.31.7), as $c_n \in A$ is a constant). The term $n = 0$ is 0, and is simply omitted if ψ is not a unit. Then, by termwise differentiation (3.31.10) c) of the summable family $(c_n\psi^n)_{n \geq n_0}$ from the proof of (3.31.21),

$$(f \circ \psi)' = \sum_{n \geq n_0} (c_n\psi^n)' = \sum_{n \geq n_0} nc_n\psi^{n-1}\psi'$$

The family $(nc_n\psi^{n-1})_{n \geq n_0}$ is summable by the same order bound as $(c_n\psi^n)_{n \geq n_0}$ in the proof of (3.31.21), shifted down one degree, so distributivity (3.31.10).d) gives

$$\sum_{n \geq n_0} nc_n\psi^{n-1}\psi' = \left(\sum_{n \geq n_0} nc_n\psi^{n-1}\right)\psi'$$

Reindexing $m = n - 1$, $\sum_{n \geq n_0} nc_n\psi^{n-1} = \sum_{m \geq n_0 - 1}(m+1)c_{m+1}\psi^m = f' \circ \psi$, giving $(f \circ \psi)' = (f' \circ \psi) \cdot \psi'$. ∎

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Proposition 3.31.23 (Substitution is a Ring Homomorphism)
Let A be a commutative ring and $\psi \in XA[[X]]$. Then $f \mapsto f \circ \psi$ is a ring homomorphism $A[[X]] \to A[[X]]$. If moreover A is an integral domain and the leading coefficient of ψ is a unit (so $\psi \in A((X))^\star$ by (3.31.14)) then $f \mapsto f \circ \psi$ is a ring homomorphism $A((X)) \to A((X))$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The arguments below apply to f, g, h in $A[[X]]$ in the first case and in $A((X))$ in the second.

Identity. $1 \circ \psi = \psi^0 = 1$.

Additivity. Write $f = \sum_{n \geq n_0} c_n X^n$, $h = \sum_{n \geq m_0} d_n X^n$, and set $p_0 := \min(n_0, m_0)$, extending both coefficient sequences by $c_n := 0$ (resp. $d_n := 0$) for $p_0 \leq n < n_0$ (resp. $< m_0$); this changes neither series, nor the sums below, as it only adds zero terms. Then $f + h = \sum_{n \geq p_0} (c_n + d_n) X^n$, and the families $(c_n \psi^n)_{n \geq p_0}$ and $(d_n \psi^n)_{n \geq p_0}$ are summable by (3.31.21). Hence

$$\begin{aligned}(f+h) \circ \psi &= \sum_{n \geq p_0} (c_n + d_n)\psi^n = \sum_{n \geq p_0} (c_n \psi^n + d_n \psi^n) && \text{by distributivity in } A((X)) \\ &= \sum_{n \geq p_0} c_n \psi^n + \sum_{n \geq p_0} d_n \psi^n = f \circ \psi + h \circ \psi && \text{by (3.31.10) } e)\end{aligned}$$

Multiplicativity. Write $f = \sum_{i \geq i_0} c_i X^i$, $g = \sum_{j \geq j_0} d_j X^j$, so $f \circ \psi = \sum_i c_i \psi^i$ and $g \circ \psi = \sum_j d_j \psi^j$, and let $P := \mathbb{Z}_{\geq i_0} \times \mathbb{Z}_{\geq j_0}$ and $a(i,j) := c_i d_j \psi^{i+j}$ for $(i,j) \in P$. We have $v(a(i,j)) \geq i + j$ whenever $i + j \geq 0$ as $v(\psi^n) \geq n$ for $n \geq 0$ by the proof of (3.31.21), and for every N only the finitely many pairs with $i + j < \max(N, 0)$ can fail $v(a(i,j)) \geq N$, so $(a(i,j))_{(i,j) \in P}$ is summable. Then

$$\begin{aligned}(f \circ \psi)(g \circ \psi) &= \sum_{j \geq j_0} (f \circ \psi) d_j \psi^j && \text{by (3.31.10) } d) \\ &= \sum_{j \geq j_0} \sum_{i \geq i_0} c_i d_j \psi^{i+j} && \text{by (3.31.10) } d) \\ &= \sum_{(i,j) \in P} a(i,j) && \text{by (3.31.11)}\end{aligned}$$

and, grouping instead by $N := i + j$, whose fibres are finite,

$$\begin{aligned}(fg) \circ \psi &= \sum_N \left(\sum_{i+j=N} c_i d_j \right) \psi^N = \sum_N \sum_{i+j=N} c_i d_j \psi^N && \text{by (3.3.17)} \\ &= \sum_{(i,j) \in P} a(i,j) && \text{by (3.31.10) } f) \text{ for the partition of } P \text{ by } i+j\end{aligned}$$

so $(f \circ \psi)(g \circ \psi) = (fg) \circ \psi$. ∎

3.31.1 Exponential and Logarithm

Definition 3.31.24 (Formal Exponential and Logarithm)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Define

$$\exp(X) := \sum_{n \geq 0} \frac{X^n}{n!} \in A[[X]] \qquad \log(1+X) := \sum_{n \geq 1} (-1)^{n+1} \frac{X^n}{n} \in XA[[X]]$$

For $f \in XA[[X]]$ and $g \in 1 + XA[[X]]$ define $\exp(f) := \exp(X) \circ f$ and $\log(g) := \log(1+X) \circ (g-1)$, using substitution (3.31.21) (note $g - 1 \in XA[[X]]$).

Lemma 3.31.25 (Zero Derivative)
Let A be a commutative ring with $\mathbb{Q} \subseteq A$ and $F \in A((X))$ with $F' = 0$. Then F is a constant, that is $F \in A$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$(F')_n = (n+1)F_{n+1} = 0$ for every $n \in \mathbb{Z}$, and $n + 1$ is invertible in A for $n \neq -1$, so $F_{n+1} = 0$ for all $n \neq -1$, that is $F = F_0$. ∎

Proposition 3.31.26 (Derivatives of exp and log)
Let A be an integral domain with $\mathbb{Q} \subseteq A$, $f \in XA[[X]]$ and $g \in 1 + XA[[X]]$. Then $\exp(f) \in 1 + XA[[X]]$ and $\log(g) \in XA[[X]]$, and

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$$\exp(f)' = \exp(f)f' \qquad \log(g)' = g'g^{-1}$$

where g is a unit of $A[[X]]$ by (3.31.13).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

exp. Directly $\exp(X)' = \sum_{n\geq 1} \frac{X^{n-1}}{(n-1)!} = \exp(X)$. Since $v(f^n) \geq n$, only the term $n = 0$ contributes to the constant coefficient of $\exp(f) = \sum_n \frac{f^n}{n!}$, so $\exp(f) \in 1 + XA[[X]]$. By the chain rule (3.31.22) $\exp(f)' = (\exp' \circ f)f' = \exp(f)f'$.

log. Likewise $\log(g) \in XA[[X]]$, as each term of $\sum_{n\geq 1} (-1)^{n+1}\frac{(g-1)^n}{n}$ has order at least n. Directly $\log(1+X)' = \sum_{n\geq 1}(-1)^{n+1}X^{n-1} = \sum_{m\geq 0}(-X)^m = (1+X)^{-1}$, as $(1+X)\sum_{m\geq 0}(-X)^m = 1$ coefficientwise. By the chain rule

$$\log(g)' = (\log(1+X)' \circ (g-1))g' = ((1+X)^{-1} \circ (g-1))g' = g^{-1}g'$$

since $f \mapsto f \circ (g-1)$ is a ring homomorphism (3.31.23) and $(1+X)\circ(g-1) = g$. ∎

Proposition 3.31.27 (Functional Equations of exp and log)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Then $\exp(f+g) = \exp(f)\exp(g)$ for $f, g \in XA[[X]]$, and $\log(gh) = \log(g) + \log(h)$ for $g, h \in 1 + XA[[X]]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

exp. Write $E := \exp(f+g)$ and $U := \exp(f)\exp(g)$, a unit of $A[[X]]$ with constant coefficient 1. By the product rule and (3.31.26), $E' = E(f'+g')$ and $U' = U(f'+g')$. Hence for $R := EU^{-1}$, using the power rule (3.31.20) for U^{-1},

$$\begin{aligned} R' &= E'U^{-1} - EU'U^{-2} && \text{by the product and power rules} \\ &= E(f'+g')U^{-1} - EU(f'+g')U^{-2} = 0 \end{aligned}$$

so R is constant by (3.31.25), equal to $R(0) = 1$, that is $E = U$.

log. By the product rule and (3.31.26), $\log(gh)' = (g'h + gh')g^{-1}h^{-1} = g'g^{-1} + h'h^{-1} = (\log(g) + \log(h))'$. So $\log(gh) - \log(g) - \log(h)$ has derivative 0 and constant coefficient 0, hence is 0 by (3.31.25). ∎

Proposition 3.31.28 (exp and log are Inverse Isomorphisms)
Let A be an integral domain with $\mathbb{Q} \subseteq A$. Then $\log(\exp(f)) = f$ for $f \in XA[[X]]$ and $\exp(\log(g)) = g$ for $g \in 1 + XA[[X]]$. Hence exp and log are mutually inverse isomorphisms of groups between $(XA[[X]], +)$ and $(1 + XA[[X]], \times)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.31.26) $\log(\exp(f))' = \exp(f)'\exp(f)^{-1} = f'$, so $\log(\exp(f)) - f$ has derivative 0 and constant coefficient 0, and is 0 by (3.31.25). For the other composite let $E := \exp(\log(g))$, so $E' = E\log(g)' = Eg'g^{-1}$ by the same result. Then

$$\begin{aligned} (Eg^{-1})' &= E'g^{-1} - Eg'g^{-2} && \text{by the product and power rules} \\ &= Eg'g^{-2} - Eg'g^{-2} = 0 \end{aligned}$$

so Eg^{-1} is constant by (3.31.25), equal to $E(0)g(0)^{-1} = 1$, that is $E = g$. Finally exp is a homomorphism by (3.31.27), with inverse log. ∎

3.31.2 Extensions of Power Series Rings

The main goal of this subsection is to show that the valuation of $k((t))$ has a **unique** extension to any finite separable extension E of $k((t))$, whose ramification index and residue degree satisfy $ef = [E : F]$ (3.31.34) and (3.31.35), and that for k algebraically closed E is itself a power series field $k((u))$ (3.31.36). The main tool is the Weierstrass division lemma for formal power series, which we record first.

Remark 3.31.29
*[Che51] (Ch. II, §6) treats the case of (3.31.36) where the ramification index e is prime to $\mathrm{char}(k)$, obtaining the sharper conclusion that t itself is an e-th power in $k((u))$ (a literal Puiseux expansion $k(\!(t^{1/e})\!)$), via Hensel's lemma applied to $W^e - v$ for the unit $v := t/u^e$. That argument is unavailable when $\mathrm{char}(k) = p \mid e$: the derivative eW^{e-1} vanishes identically, so $W^e - v$ has no simple root to lift, and t need not be an e-th power at all (an Artin-Schreier extension gives a counterexample). (3.31.36) instead only produces $t = u^e c$ for **some** unit c, obtained by solving $\tau G(\tau) = u^e$ by direct coefficient-stabilisation rather than root extraction; this sidesteps the characteristic obstruction entirely, at the cost of the sharper radical form, which is in any case not needed below. The statement of (3.31.36) over an algebraically closed field, in every characteristic, is [Lan82] (Ch. I, §4, Prop. 4.3), whose proof, like ours, expands the elements of E in powers of a uniformiser u with coefficients in $k[[t]]$ and compares valuations.*

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Lemma 3.31.30 (Division with Remainder by a Distinguished Power Series)
Let A be a commutative ring, $m \geq 1$ an integer and $\tau = X^m w(X)$ for $w(X) \in A[[X]]^\star$.

Then every $p \in A[[X]]$ may be written uniquely as $p = q\tau + r$ with $q \in A[[X]]$ and $r \in A[X]$ of degree $< m$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

For existence, split $p = X^m p' + r$ with r the truncation below degree m and p' the shift, and put $q := p' w^{-1}$ (as $w \in A[[X]]^\star$ (3.31.13)). For uniqueness, if $q\tau + r = 0$ then $r = -q\tau \in X^m A[[X]]$ has degree $< m$, so $r = 0$; then $qX^m w = 0$ gives $q = 0$, as w is a unit and X^m is a non-zero-divisor. ∎

Lemma 3.31.31 (Free Basis for a Distinguished Power Series)
Let A be a commutative ring, $m \geq 1$ and $\tau := X^m w(X) \in A[[X]]$ with $w(X) \in A[[X]]^\star$ (and in particular $w(0) \in A^\star$ (3.31.13)).

There is a well-defined ring homomorphism $A[[Y]] \to A[[X]]$ such that $f(Y) \mapsto f(\tau(X))$, which is injective.

Furthermore $A[[X]]$ is a free $A[[Y]]$-module with basis $1, X, ..., X^{m-1}$, that is for every $p \in A[[X]]$ there are unique $f_0, ..., f_{m-1} \in A[[Y]]$ such that

$$p = \sum_{i=0}^{m-1} f_i(\tau(X)) X^i$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The homomorphism and the module structure. As $m \geq 1$ we have $\tau = X^m w \in XA[[X]]$, so $f(Y) \mapsto f(\tau)$ is a well-defined map $A[[Y]] \to A[[X]]$ (3.31.21), and it is a ring homomorphism (3.31.23) (with $\psi := \tau$). Regard $A[[X]]$ as an $A[[Y]]$-module through it, so $f \cdot p = f(\tau)p$ and $\sum_{i=0}^{m-1} f_i(\tau) X^i = \sum_{i=0}^{m-1} f_i \cdot X^i$. We must show that $1, X, ..., X^{m-1}$ span this module and are free; injectivity of the homomorphism is deduced at the end.

Spanning. Put $p_0 := p$ and, dividing repeatedly by τ (3.31.30), write $p_j = p_{j+1}\tau + r_j$ with $\deg r_j < m$. Then for every $J \geq 0$

$$p = p_{J+1}\tau^{J+1} + \sum_{j=0}^{J} r_j \tau^j, \quad \text{with } p_{J+1}\tau^{J+1} \in X^{m(J+1)} A[[X]].$$

This follows by induction on J: for $J = 0$ it is $p = p_1\tau + r_0$, and substituting $p_{J+1} = p_{J+2}\tau + r_{J+1}$ gives the step, as $\tau^{J+1} \in X^{m(J+1)}A[[X]]$ because $\tau = X^m w$. So $\left(r_j \tau^j\right)_{j \geq 0}$ is summable (3.31.9) (as $v\left(r_j\tau^j\right) \geq mj$), and $p = \sum_{j \geq 0} r_j \tau^j$ by (3.31.12), as $p - \sum_{j \leq J} r_j \tau^j = p_{J+1}\tau^{J+1} \in X^{m(J+1)}A[[X]]$ for every J. Write $r_j = \sum_{i=0}^{m-1} \gamma_{j,i} X^i$ and set $f_i := \sum_{j \geq 0} \gamma_{j,i} Y^j \in A[[Y]]$. The family $\left(\gamma_{j,i}\tau^j X^i\right)_{(j,i)}$ is summable, as $v\left(\gamma_{j,i}\tau^j X^i\right) \geq mj$. Then

$$\begin{aligned}
\sum_{i=0}^{m-1} f_i(\tau) X^i &= \sum_{i=0}^{m-1} \sum_{j \geq 0} \gamma_{j,i} \tau^j X^i \quad \text{by the definition of } f_i(\tau) \text{ and (3.31.10) } d) \\
&= \sum_{j \geq 0} \sum_{i=0}^{m-1} \gamma_{j,i} \tau^j X^i \quad \text{by (3.31.11)} \\
&= \sum_{j \geq 0} r_j \tau^j = p.
\end{aligned}$$

So $1, X, ..., X^{m-1}$ span $A[[X]]$ over $A[[Y]]$.

Freeness. Suppose that $0 = \sum_{i=0}^{m-1} f_i(\tau) X^i$ with some $f_i \neq 0$. Then because $w(0)$ is a unit we may show that $n_i := v(f_i(\tau)X^i) = mv(f_i(Y)) + i$ whenever $f_i \neq 0$. As the n_i are distinct (being so modulo m) the $\min_i(n_i)$-th coefficient is zero, a contradicton. So we conclude

$$f_0 = ... = f_{m-1} = 0.$$

Uniqueness of the f_i follows by applying freeness to the difference of two decompositions of p, and injectivity of $f \mapsto f(\tau)$ from the case $f_0 = f$, $f_1 = ... = f_{m-1} = 0$. ∎

Example 3.31.32
Take $w = 1$, so $\tau = X^m$. For $p = \sum_{n \geq 0} a_n X^n \in A[[X]]$, writing each $n = mj + i$ with $0 \leq i < m$ gives

$$p = \sum_{i=0}^{m-1} \left(\sum_{j \geq 0} a_{mj+i} X^{mj} \right) X^i, \quad \text{so} \quad f_i(Y) = \sum_{j \geq 0} a_{mj+i} Y^j.$$

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That is, f_i collects the coefficients of p whose exponent is congruent to i modulo m. For $m=2$, f_0 collects the even-index coefficients and f_1 the odd-index ones.

Corollary 3.31.33 (Specialisation Compatibility)
In the setting of (3.31.31), let $\phi : A \to S$ be a homomorphism of commutative rings, extended coefficientwise to $\phi : A[[X]] \to S[[X]]$. Then $\phi(w) \in S[[X]]^\star$, and the decomposition of $\phi(p)$ relative to $\phi(\tau) = X^m \phi(w)$ is $\phi(p) = \sum_{i=0}^{m-1} \phi(f_i)(\phi(\tau))X^i$, where $\phi(f_i) \in S[[Y]]$ denotes ϕ applied coefficientwise to f_i.

Proof. If $wv = 1$ then $\phi(w)\phi(v) = 1$, so $\phi(w) \in S[[X]]^\star$. Next, $\phi(f(\tau)) = \phi(f)(\phi(\tau))$ for every $f \in A[[Y]]$. Fix $N \geq 0$ and write $f = s + Y^N g$ with $s \in A[Y]$ of degree $< N$ and $g \in A[[Y]]$. Then $f(\tau) = s(\tau) + \tau^N g(\tau)$ (the tail $\sum_{n \geq N} c_n \tau^n = \tau^N \sum_n c_{n+N}\tau^n$, (3.31.9)), and likewise $\phi(f) = \phi(s) + Y^N \phi(g)$, so $\phi(f)(\phi(\tau)) = \phi(s)(\phi(\tau)) + \phi(\tau)^N \phi(g)(\phi(\tau))$. Applying the ring homomorphism ϕ to $f(\tau) = s(\tau) + \tau^N g(\tau)$ gives $\phi(f(\tau)) = \phi(s)(\phi(\tau)) + \phi(\tau)^N \phi(g(\tau))$, so $\phi(f(\tau)) - \phi(f)(\phi(\tau)) \in \phi(\tau)^N S[[X]] \subseteq X^{mN} S[[X]]$, as $\phi(\tau) = X^m \phi(w)$. This holds for every N and $\bigcap_N X^N S[[X]] = 0$ (3.31.6), so $\phi(f(\tau)) = \phi(f)(\phi(\tau))$. Applying ϕ to $p = \sum_i f_i(\tau) X^i$ therefore gives $\phi(p) = \sum_i \phi(f_i)(\phi(\tau))X^i$. This exhibits $\phi(p)$ in the form guaranteed unique by (3.31.31) over S, so it is **the** decomposition of $\phi(p)$. ∎

Lemma 3.31.34 (Local Degree at a Prime above (t))
Let k be a field, $F := k((t))$, E/F a finite separable extension of degree n, C the integral closure of $k[[t]]$ in E, and $\mathfrak{p} \subset C$ a prime with $\mathfrak{p} \cap k[[t]] = (t)$ (such a prime exists). Then $B := C_{\mathfrak{p}}$ is a DVR with fraction field E and $k' := C/\mathfrak{p}$ is a finite field extension of k. Write $e := v_B(t)$ and $f := [k' : k]$. Then $ef = n$. More precisely, if u is a uniformiser of B and $\omega_1, \ldots, \omega_f \in C$ map to a k-basis of k', then the elements $\omega_j u^i$ $(0 \leq i < e,\ 1 \leq j \leq f)$ form an F-basis of E, and $B = \bigoplus_{i,j} k[[t]]\omega_j u^i$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Step 1: a DVR. Let $\mathcal{O}_F := k[[t]]$, a DVR with fraction field F (3.31.16) (so a Dedekind domain (3.30.3)). By (3.22.28) C is a finite $\mathcal{O}_F$-module, and by (3.30.11) it is Dedekind. By lying-over (3.22.18) there is a prime $\mathfrak{p} \subset C$ with $\mathfrak{p} \cap \mathcal{O}_F = (t)$. Then $B := C_{\mathfrak{p}}$ is a DVR with fraction field E, and $e := v_B(t) \geq 1$.

Step 2: residue field and valuation. $C/\mathfrak{p}$ is a domain, finite over $\mathcal{O}_F/(t) = k$, so $k \hookrightarrow C/\mathfrak{p}$ is injective and integral, hence $C/\mathfrak{p}$ is a field (3.22.14), finite over k. As $B = C_{\mathfrak{p}}$, (3.5.43) gives $B/\mathfrak{m}_B \cong \mathrm{Frac}(C/\mathfrak{p}) = C/\mathfrak{p} = k'$. Every $a \in \mathcal{O}_F^\times$ is a unit of B, so $v_B(x) = e v_F(x)$ for $x \in F^\times$. Let u be a uniformiser of B, so $t = u^e w$ with $w \in B^\star$, and let $\omega_1, \ldots, \omega_f \in C$ map to a k-basis of k'.

Step 3: spanning. Every $b \in B$ can be written $b = \sum_{i<e} \sum_j a_{ij} \omega_j u^i$ with $a_{ij} \in k[[t]]$. We show by induction on $N \geq 0$ that

$$b \equiv \sum_{i<e} \sum_j \left(\sum_{l<N} c_{ijl} t^l \right) \omega_j u^i \pmod{t^N B}$$

for some $c_{ijl} \in k$. The case $N = 0$ is trivial. For the inductive step, write the current remainder as $t^N y$ with $y \in B$. Since $B/uB = k'$ has k-basis the images of the ω_j, repeated division by u gives $y = \sum_{i<e} \sum_j d_{ij} \omega_j u^i + u^e y'$ with $d_{ij} \in k$, $y' \in B$. Using $u^e = tw^{-1}$,

$$t^N y = \sum_{i<e} \sum_j (t^N d_{ij}) \omega_j u^i + t^{N+1}(w^{-1} y'),$$

with $w^{-1}y' \in B$; so taking $c_{ijN} := d_{ij}$ advances the induction, with new remainder in $t^{N+1}B$. Put $a_{ij} := \sum_l c_{ijl} t^l \in k[[t]] \subseteq C \subseteq B$. Then $z := b - \sum_{i,j} a_{ij} \omega_j u^i$ satisfies $z \in t^N B$ for every N, so $v_B(z) \geq eN$ for all N and $z = 0$.

Step 4: independence, basis and degree. Every element of E has the form $t^{-N} b$ with $b \in B$, so Step 3 gives $E = \sum_{i<e,,j} F \omega_j u^i$. Let $a_{ij} \in F$ be not all zero. For each i with some $a_{ij} \neq 0$ put $m_i := \min_j v_F(a_{ij})$ and write $a_{ij} = t^{m_i} a'_{ij}$ with $a'_{ij} \in k[[t]]$ and $a'_{ij}(0) \neq 0$ for some j. As $a'_{ij} \equiv a'_{ij}(0) \pmod{tk[[t]]}$ and $t \in uB$,

$$\sum_j a'_{ij} \omega_j \equiv \sum_j a'_{ij}(0) \omega_j \pmod{uB},$$

which is non-zero in k' since the images of the ω_j are k-linearly independent. So $\sum_j a'_{ij}\omega_j$ is a unit of B and $v_B\left(\sum_j a_{ij}\omega_j u^i\right) = e m_i + i$. These valuations are distinct modulo e for distinct i, so the minimum over i is attained exactly once and $\sum_{i,j} a_{ij}\omega_j u^i \neq 0$ (3.23.2). Hence the $\omega_j u^i$ are linearly independent over F, so form an F-basis of E and $n = ef$.

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Moreover for $b = \sum_{i,j} a_{ij}\omega_j u^i$ with $a_{ij} \in F$ the same computation gives $v_B(b) = \min_i(em_i + i)$. If $b \in B$ then $em_i + i \geq 0$ with $0 \leq i < e$, so $m_i \geq 0$ and all $a_{ij} \in k[[t]]$. Together with Step 3 this gives $B = \bigoplus_{i,j} k[[t]]\omega_j u^i$. ∎

Corollary 3.31.35 (Uniqueness of the Extended Valuation)
Let k be a field, $F := k((t))$, E/F a finite separable extension of degree n and C the integral closure of $k[[t]]$ in E. Then C has exactly one prime $\mathfrak{p}$ lying over (t), and its ramification index e and residue degree f satisfy $ef = n$. Moreover the DVR $C_\mathfrak{p}$ is the unique valuation ring of E containing $k[[t]]$ whose maximal ideal contracts to (t). If k is algebraically closed then $f = 1$ and $e = n$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\mathfrak{q}$ be any prime of C over (t), which exists by lying-over. By (3.31.34) its ramification index and residue degree satisfy $e_\mathfrak{q} f_\mathfrak{q} = n$.

Now apply the Ramification Formula (3.30.20) to the AKLB situation $k[[t]] \subset C$ over E/F (both Dedekind by (3.30.11), and E/F separable), at the prime (t):

$$\sum_{\mathfrak{q} \mid (t)} e_\mathfrak{q} f_\mathfrak{q} = [C/tC : k] = n.$$

Each term equals n by the first paragraph, so there is exactly one prime $\mathfrak{p}$ over (t) and $e_\mathfrak{p} f_\mathfrak{p} = n$. If k is algebraically closed then $C/\mathfrak{p}$, a finite extension of k, equals k, so $f_\mathfrak{p} = 1$ and $e_\mathfrak{p} = n$.

For uniqueness of the valuation ring, let $V \subseteq E$ be a valuation ring with $k[[t]] \subseteq V$ and $\mathfrak{m}_V \cap k[[t]] = (t)$. By (3.23.14), $C \subseteq V$. Then $\mathfrak{m}_V \cap C$ is a prime of C over (t), hence equals $\mathfrak{p}$. Every element of $C \setminus \mathfrak{p}$ lies outside $\mathfrak{m}_V$, so is a unit of V; hence $C_\mathfrak{p} \subseteq V$. Since $C_\mathfrak{p}$ is a valuation ring of E and valuation rings are maximal for domination among proper subrings of E (3.23.2), $V = C_\mathfrak{p}$. ∎

Lemma 3.31.36 (Finite Separable Extensions of $k((t))$ are Power Series Fields)
Let k be algebraically closed, $F := k((t))$, and E/F a finite separable extension of degree n. Then there are $u \in E$, an element $\tau \in uk[[u]]$ with $\tau = u^n c$ for some $c \in k[[u]]^\star$, and a field isomorphism $\Psi : E \xrightarrow{\sim} k((u))$ with $\Psi|_F$ the substitution $t \mapsto \tau$ and $\Psi(u) = u$. In particular $1, u, \ldots, u^{n-1}$ is an F-basis of E.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Step 1: reduction to the local degree. Let C be the integral closure of $k[[t]]$ in E and $\mathfrak{p} \subset C$ a prime over (t), which exists by lying-over, and put $B := C_\mathfrak{p}$. Its residue field $C/\mathfrak{p}$ is finite over the algebraically closed field k (3.31.34), so equals k and $f = 1$. Take $\omega_1 := 1$. Then (3.31.34) gives $e = n$, that $1, u, \ldots, u^{n-1}$ is an F-basis of E for a uniformiser u of B, and $B = \bigoplus_{i<n} k[[t]]u^i$. Write $t = u^n w$ with $w \in B^\star$.

Step 2: the minimal polynomial. Write $w^{-1} = \sum_{i<n} g_i u^i$ with $g_i \in k[[t]]$ (Step 1). The constant $g_0(0)$ is the residue of w^{-1}, so $g_0(0) \in k^\star$. From $u^n = tw^{-1}$, u is a root of

$$P(Y) := Y^n - t\sum_{i<n} g_i(t) Y^i \in F[Y].$$

Since $F[u] = E$ has degree n over F, P is the minimal polynomial of u, and $E \cong F[Y]/(P)$ with $Y \mapsto u$.

Step 3: solving for τ. For $s \in uk[[u]]$ put $G(s) := \sum_{i<n} g_i(s) u^i \in k[[u]]$, where $g_i(s)$ is the substitution (well defined as s has zero constant term). Since $G(s) \equiv g_0(0) \pmod{u}$, $G(s)$ is a unit. Define $T(s) := u^n G(s)^{-1} \in u^n k[[u]]$. For $s \equiv s' \pmod{u^N}$ we have $g_i(s) - g_i(s') \in (s - s')k[[u]] \subseteq u^N k[[u]]$, hence $G(s) \equiv G(s') \pmod{u^N}$ and $T(s) \equiv T(s') \pmod{u^{N+n}}$. Starting from $s_0 := 0$ and setting $s_{m+1} := T(s_m)$, the sequence satisfies $s_{m+1} \equiv s_m \pmod{u^{m+n}}$, so each coefficient stabilises. Let $\tau \in uk[[u]]$ be the coefficientwise limit. Then $\tau \equiv T(\tau) \pmod{u^N}$ for every N, so $\tau = T(\tau)$, i.e. $\tau G(\tau) = u^n$. Also $\tau = u^n c$ with $c \in k[[u]]^\star$, since $\tau G(\tau) = u^n$ and $G(\tau)$ is a unit.

Step 4: the homomorphism. The substitution $t \mapsto \tau$ defines a ring homomorphism $\sigma : k[[t]] \to k[[u]]$ (3.31.23) (as τ has zero constant term), and since τ is a unit times u^n it extends to a field homomorphism $\sigma : F \to k((u))$. The homomorphism $F[Y] \to k((u))$, $Y \mapsto u$, extending σ, kills P:

$$\sigma(P)(u) = u^n - \tau\sum_{i<n} g_i(\tau) u^i = u^n - \tau G(\tau) = 0.$$

So it factors through $E \cong F[Y]/(P)$, giving a field homomorphism $\Psi : E \to k((u))$ with $\Psi|_F = \sigma$ and $\Psi(u) = u$. It is injective, as E is a field.

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Step 5: surjectivity. Since $\tau = u^n c$ with $c \in k[[u]]^\star$, the Weierstrass division lemma (3.31.31) (with $A := k$, $X := u$, $m := n$, $w := c$, so the series of the lemma is τ) gives $k[[u]] = \bigoplus_{i<n} \sigma(k[[t]])u^i$. Multiplying by powers of τ gives $k((u)) = \sum_{i<n} \sigma(F)u^i = \sigma(F)[u]$. But $\operatorname{im}\Psi = \Psi(F[u]) = \sigma(F)[u]$, so Ψ is surjective.

Hence $\Psi : E \xrightarrow{\sim} k((u))$ is an isomorphism of fields over F, with $\Psi(u) = u$ and $\Psi(t) = \tau = u^n c$. ■

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3.32 Affine Algebras

Definition 3.32.1 (Affine Domain)
*We call a finitely-generated k-algebra an **affine algebra**. If in addition it's integral we call it an **affine domain**.*

3.32.1 Reduction

Recall that $A_{\text{red}} \coloneqq A/N(A)$.

Lemma 3.32.2
Let A be a k-algebra and $\mathfrak{a}$ an ideal. Then $N(A/\mathfrak{a}) = \sqrt{\mathfrak{a}}/\mathfrak{a}$ and there is a canonical isomorphism

$$\begin{array}{ccc} A/\mathfrak{a} & \twoheadleftarrow & A \\ \downarrow & & \downarrow \\ (A/\mathfrak{a})_{\text{red}} & \xrightarrow{\sim} & A/\sqrt{\mathfrak{a}} \end{array}$$

$$\overline{a+\mathfrak{a}} \longrightarrow (a+\sqrt{\mathfrak{a}})$$

which is the unique morphism making the diagram commute.

Proof. For the first part $(a+\mathfrak{a})^n = a^n + \mathfrak{a}$, so $(a+\mathfrak{a}) \in N(A/\mathfrak{a}) \iff (a+\mathfrak{a})^n = 0 \iff a^n \in \mathfrak{a}$ some $n \iff a \in \sqrt{\mathfrak{a}}$.

The second part then follows from (3.3.58). ∎

Proposition 3.32.3
Let A be a k-algebra then for every reduced k-algebra B there is a canonical bijection

$$\begin{aligned} \text{Mor}(A_{\text{red}}, B) &\xrightarrow{\sim} \text{Mor}(A, B) \\ \phi &\to \phi \circ \pi \end{aligned}$$

where $\pi : A \to A_{\text{red}}$ is the quotient map, which is natural in B. When A is reduced then π is an isomorphism.

Proof. Suppose $\tilde{\phi} : A \to B$ is a homomorphism and $a \in N(A)$. Then $0 = \tilde{\phi}(a^n) = \tilde{\phi}(a)^n$ whence $\tilde{\phi}(N(A)) \subset N(B) = (0)$, that is $N(A) \subset \ker(\tilde{\phi})$. By (3.3.56) there exists ϕ such that $\phi \circ \pi = \tilde{\phi}$. This shows the given map is surjective. As π is surjective we may also deduce that the map is injective. ∎

Proposition 3.32.4
Let A, B be k-algebras and $\phi : A \to B$ be a k-algebra homomorphism, then there is a unique morphism $\phi_{\text{red}} : A_{\text{red}} \to B_{\text{red}}$ making the following diagram commute

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow & & \downarrow \\ A_{\text{red}} & \dashrightarrow & B_{\text{red}} \end{array}$$

given by

$$\phi_{\text{red}}(a + N(A)) = \phi(a) + N(B)$$

Furthermore $(\psi \circ \phi)_{\text{red}} = \psi_{\text{red}} \circ \phi_{\text{red}}$. If ϕ is an isomorphism then so is ϕ_{red}.

Proof. In the notation of (3.32.3) ϕ_{red} is simply the morphism corresponding to $\pi_B \circ \phi$, from which existence and uniqueness follows.

Uniqueness then also shows that this is functorial. ∎

3.32.2 Normalisation

The following normalisation results can be seen as a refinement of results on transcendence bases (see the Transcendental Field Extensions subsection).

Definition 3.32.5 (Algebraically Independent)
*Let A be a k-algebra and $x_1, ..., x_n$ elements of A. Then we say they are **algebraically independent** if one of the following equivalent conditions holds*

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- *The unique k-algebra homomorphism $\phi : k[X_1, ..., X_n] \to A$ such that $\phi(X_i) = x_i$ (evaluation homomorphism) is injective*
- *There are no non-zero polynomials $f(X_1, ..., X_n)$ such that $f(x_1, ..., x_n) = 0$.*

Note in particular it induces an isomorphism $k[X_1, ..., X_n] \xrightarrow{\sim} k[x_1, ..., x_n] \subset A$.

Definition 3.32.6 (Normalising Family)
*Let A be a finitely-generated k-algebra. A **normalising family** is a set $\{x_1, ..., x_n\}$ of elements of A such that*

- *$x_1, ..., x_n$ are **algebraically independent** over k*
- *A is a finite $k[x_1, ..., x_n]$-module (equivalently integral over $k[x_1, ..., x_n]$).*

NB this is completely equivalent to specifying an integral, injective map

$$k[X_1, ..., X_n] \hookrightarrow A$$

This may be seen as a refined transcendence base. More precisely we have the following

Proposition 3.32.7 (Relationship to Transcendence Base)
Let A be an integral finitely-generated k-algebra with $K := \mathrm{Frac}(A)$. Let $S \subset A$ be a subset. Then

- *If A is integral over $k[S]$ then $K/k(S)$ is algebraic*
- *If S is a normalising family for A then S is a transcendence basis for K/k*

In particular normalising families have order $\mathrm{trdeg}(K/k)$.

Proof. By (3.17.45) the set $\{x \in K \mid x \text{ algebraic over } k(S)\}$ forms a subfield containing A, and therefore equals K.

The final statement follows from (3.17.144). ∎

The following is useful as it removes the necessity of showing algebraic independence in certain cases.

Corollary 3.32.8
Let A be an integral finitely-generated k-algebra with $K := \mathrm{Frac}(A)$. Let $S \subset A$ be a subset such that

- *A is integral over $k[S]$*
- *$\#S \leq \mathrm{trdeg}_k(K)$*

*then S is a **normalising family**.*

Proof. This follows from the previous result and (3.17.144). ∎

There are a few forms of the Normalisation Lemma which we prove, of progressively stronger form.

Lemma 3.32.9 (Hypersurface Normalisation Lemma)
Let $A = k[X_1, ..., X_n]$ be a polynomial ring over an infinite field k and $0 \neq F \in A$. Then there exists $\lambda_1, ..., \lambda_{n-1} \in k$ such that

- *$x_i := X_i - \lambda_i F \quad 1 \leq i \leq n-1$*
- *$x_1, ..., x_{n-1}, F$ is a normalising family for A*
- *$FA \cap k[x_1, ..., x_{n-1}, F] = Fk[x_1, ..., x_{n-1}, F]$*

This may be viewed as a commutative diagram

$$\begin{array}{ccc} \phi : k[Y_1, ..., Y_n] & \xrightarrow{\sim} & k[X_1, ..., X_n] \\ \uparrow & & \downarrow \\ k[Y_1, ..., Y_{n-1}] & \hookrightarrow & k[X_1, ..., X_n]/(F) \end{array}$$

where the horizontal arrows are injective, integral (and finite) and $\phi^{-1}((F)) = (Y_n)$.

Remark 3.32.10
Reversing the arrows we get the geometric picture, where horizontal arrows are finite and surjective

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$$\begin{array}{ccc} \mathbb{A}^n & \xrightarrow{\sim} & \mathbb{A}^n \\ \uparrow & & \downarrow \\ V(F) & \twoheadrightarrow & \mathbb{A}^{n-1} \end{array}$$

in other words after a linear change of variables we may express $V(F)$ as a finite covering of a standard hyperplane.

Proposition 3.32.11 (Nagata Normalisation Lemma)
Let $A = k[x_1, ..., x_n]$ be a finitely-generated k-algebra such that k is infinite. Then there exists a ***normalising family*** *$y_1, ..., y_d \in A$ such that each y_i is a k-linear combination of $x_1, ..., x_n$. Further if $\mathfrak{a}_1 \lhd A$ is a proper ideal, then these may be chosen such that*

$$\mathfrak{a}_1 \cap k[y_1, ..., y_d] = (y_1, ..., y_h)k[y_1, ..., y_d]$$

for some $0 \leq h \leq d$, where $h = 0$ denotes the zero ideal.

Proposition 3.32.12 (Bourbaki Normalisation Lemma)
Let $A = k[x_1, ..., x_n]$ be a finitely-generated k-algebra such that k is infinite. Then there exists a ***normalising family*** *$y_1, ..., y_d \in A$ such that each y_i is a k-linear combination of $x_1, ..., x_n$.*

Furthermore for any finite chain of proper ideals in A

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq ... \subseteq \mathfrak{a}_p \subsetneq A$$

the family may be chosen such that

$$\mathfrak{a}_j \cap k[y_1, ..., y_d] = \left(y_1, ..., y_{h(j)}\right)k[y_1, ..., y_d] \quad 1 \leq j \leq p \ .$$

for some non-decreasing sequence of integers $h(j)$, where $h(j) = 0$ denotes the zero ideal.

Remark 3.32.13 (Geometric Interpretation)
Note the normalisation here is equivalent to a commutative diagram

$$\begin{array}{ccc} k[Y_1, ..., Y_d] & \hookrightarrow & A \\ \uparrow & & \downarrow \\ k\left[Y_{h(1)+1}, ..., Y_d\right] & \hookrightarrow & A/\mathfrak{a}_1 \\ \uparrow & & \downarrow \\ \vdots & & \vdots \\ \uparrow & & \downarrow \\ k\left[Y_{h(p)+1}, ..., Y_d\right] & \hookrightarrow & A/\mathfrak{a}_p \end{array}$$

where the horizontal arrows are integral and injective, and the top arrow is given by

$$\phi : Y_i \to \sum_j \lambda_{ij} x_j$$

such that $\phi^{-1}(\mathfrak{a}_i) = \left(Y_1, ..., Y_{h(i)}\right)$

As before this expresses A as a finite covering of $\mathbb{A}^d$ under which each subvariety is also a finite covering of a standard linear subspace.

We first prove the weaker form of the Normalisation Lemma

Proof of (3.32.9). First decompose F into homogenous polynomials (monomials of the same degree)

$$F = F_0 + F_1 + ... + F_m$$

and observe that the monomial X_n^m appears only in F_m. Define

$$F' \coloneqq F(X_1 + \lambda_1 X_n, ..., X_{n-1} + \lambda_{n-1} X_n, X_n)$$

Furthermore in terms of homogenous polynomials

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$$F'_m = F_m(X_1 + \lambda_1 T, ..., X_{n-1} + \lambda_{n-1} X_n, X_n)$$

and the coefficient of X_n^m in F' is simply $F'_m(0, ..., 0, 1) = F_m(\lambda_1, ..., \lambda_{n-1}, 1) \in k$. There are only finitely many values of λ such that F_m is zero, whence there exists a λ such that X_n^m has non-zero coefficient in F', whence F' is monic. By the previous Lemma we have $F'(x_1, ..., x_{n-1}, X_n) = 0$, with the leading coefficient in X_n constant.

Let $B \coloneqq k[x_1, ...x_{n-1}, F]$. Then clearly $B[X_n] = A$ and X_n is integral over B. Therefore by (3.22.4) A is a finite B-module. As $\operatorname{trdeg}(A/k) = n$ by (3.32.8) $\{x_1, ..., x_{n-1}, F\}$ is a normalising family, and B is isomorphic to a polynomial ring.

As B is a polynomial ring it is integrally closed (3.22.12). Then the final statement is a consequence of the following lemma (3.32.14) (essentially to prove that $V(F) \to \mathbb{A}^{n-1}$ is surjective). ∎

Lemma 3.32.14
Let A be integrally closed, $\phi : A \hookrightarrow B$ injective and integral and $a \in A$. Then

$$(a)^{\mathrm{ec}} = \phi^{-1}(\phi(a)B) = (a)$$

Proof. Let $K = \operatorname{Frac}(A)$ and $L = \operatorname{Frac}(B)$, then ϕ extends to an injection $\phi : K \hookrightarrow L$.

Generically we have $(a) \subseteq (a)^{\mathrm{ec}}$. Suppose $a' \in (a)^{\mathrm{ec}}$, then $\phi(a') = \phi(a)b$. Therefore $\phi(a'a^{-1}) = b$ is integral over A, whence so is $a'a^{-1}$. As A is integrally closed we have $a' \in (a)$ as required. ∎

Before proving the stronger version of Normalisation Lemma, we need some preliminary technical results

Lemma 3.32.15
Let A be a k-algebra and $\mathfrak{a} \lhd A$ an ideal. Then $\mathfrak{a}$ is proper $\iff \mathfrak{a} \cap k = \{0\}$.

Lemma 3.32.16
Let $A = k[X_1, ..., X_n]$ be a polynomial ring and $\mathfrak{a} \lhd A$ a proper ideal. Then TFAE

a) $\mathfrak{a} = (X_1, ..., X_h)$
b)
  i) $X_1, ..., X_h \in \mathfrak{a}$
  ii) $\mathfrak{a} \cap k[X_{h+1}, ..., X_n] = \{0\}$

In this case $\mathfrak{a} \cap k[X_1, ..., X_h] = (X_1, ..., X_h)k[X_1, ..., X_h]$

Proof. We claim that there is a direct sum of k-vector spaces

$$k[X_1, ..., X_n] = (X_1, ..., X_h) \oplus k[X_{h+1}, ..., X_n]$$

from which the result largely follows. Let S be the set of monomials in which at least one of $X_1, ..., X_h$ appears, and let T be the set for which none appears (but including 1). Then clearly $A = \langle S \rangle \oplus \langle T \rangle$ and $k[X_{h+1}, ..., X_n] = \langle T \rangle$. We argue that $(X_1, ..., X_h) = \langle S \rangle$. First S is stable under multiplication by $X_1, ..., X_n$, and so $\langle S \rangle$ is an ideal. One inclusion is obvious, furthermore it's clear that $S \subseteq (X_1, ..., X_n)$ from which the claim follows. ∎

We may now proceed to the proof of the stronger versions of the Normalisation Lemma.

Reduction to polynomial ring case for (3.32.11), (3.32.12). We show that for both forms it is possible to reduce to the case of a polynomial ring. For let A be a finitely-generated k-algebra then we may write $A \coloneqq k[X_1, ..., X_n]/\mathfrak{a}$ for some ideal $\mathfrak{a}$. Then the polynomial ring case for $p = 1$ shows the existence of an integral, injective map

$$\phi : k[Y_1, ..., Y_m] \hookrightarrow A$$

If $\mathfrak{a}_i$ is a chain of ideals in A, then $\mathfrak{a}'_i \coloneqq \phi^{-1}(\mathfrak{a}_i)$ is a chain of ideals in $k[Y_1, ..., Y_m]$. The general case for a polynomial ring shows the existence of an integral map

$$\psi : k[Z_1, ..., Z_m] \hookrightarrow k[Y_1, ..., Y_m]$$

such that

$$\psi^{-1}(\mathfrak{a}'_i) = \left(Z_1, ..., Z_{h(i)}\right)$$

The composition $\phi \circ \psi$ gives the required normalisation of A. Geometrically express A as a finite covering of affine space, by considering it as a subvariety of larger affine space. ∎

Proof of (3.32.11) in the polynomial ring case. Let $A = k[X_1, ..., X_n]$ and proceed by induction on n.

Note that as $\mathfrak{a}_1$ is proper, we must have $\mathfrak{a}_1 \cap k = \{0\}$, and we may obviously also assume that $\mathfrak{a}_1 \neq (0)$ (as otherwise we may take $h = 0$).

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Choose $0 \neq x_1 \in \mathfrak{a}_1$. Then by (3.32.9) there exists $t_2, ..., t_n \in A$ such that $x_1, t_2, ..., t_n$ is a normalising family and $(x_1) \cap B = x_1 B$ where $B := k[x_1, t_2, ..., t_n]$. In the case that $\mathfrak{a}_1$ is principal we are done, since the choice of x_1 was arbitrary, and in this case $h(1) = 1$. In particular this covers the base case $n = 1$ because A is a PID (3.14.4).

Otherwise $B' := k[t_2, ..., t_n]$ is a polynomial ring and $\mathfrak{a}_1' := \mathfrak{a}_1 \cap B'$ is proper by (3.32.15). By induction on n there is a normalising family $x_2, ..., x_n$ for B' such that $\mathfrak{a}_1' \cap C' = (x_2, ..., x_h)C'$ where $C' := k[x_2, ..., x_n]$ and B' is integral over C'.

Define $C := k[x_1, ..., x_n] = C'[x_1]$ then $x_1, t_2, ..., t_n$ are integral over C, so B is integral over C (3.22.4), and A is integral over C (3.22.5), so by (3.32.8) $x_1, ..., x_n$ is a normalising family for A.

We claim that $\mathfrak{a}_1'' := \mathfrak{a}_1 \cap C = (x_1, ..., x_h)C$. Clearly $x_1, ..., x_h \in \mathfrak{a}_1''$, and $\mathfrak{a}_1'' \cap k[x_{h+1}, ..., x_n] = \mathfrak{a}_1' \cap k[x_{h+1}, ..., x_n] = \{0\}$ by (3.32.16) applied to the ring C'. Then (3.32.16) applied to the ring C demonstrates the claim. ∎

Proof of (3.32.12) in the polynomial ring case. We can then show the case $p > 1$ by induction, for by the induction hypothesis there exists a normalising family $t_1, ..., t_n$ for A such that

$$\mathfrak{a}_j \cap B = \left(t_1, ..., t_{h(j)}\right)B \quad 1 \leq j \leq p-1$$
$$B := k[t_1, ..., t_n]$$

Let $r = h(p-1)$, then by the case $p = 1$ applied to the ring $B' := k[t_{r+1}, ..., t_n]$ and ideal $\mathfrak{a}_p \cap B'$ there exists a normalising family $x_{r+1}, ..., x_n$ for B' such that for some $s \leq n$ (possibly equal to r to denote the zero ideal),

$$\mathfrak{a}_p \cap C' = (x_{r+1}, ..., x_s)C'$$
$$C' := k[x_{r+1}, ..., x_n]$$

We claim that $t_1, ..., t_r, x_{r+1}, ..., x_n$ is a suitable normalising family for A, with $h(p) = s$.

For define $C := k[t_1, ..., t_r, x_{r+1}, ..., x_n] = C'[t_1, ..., t_r]$. Recall B' is integral over C', and $t_1, ..., t_r$ are obviously integral over C so $B = B'[t_1, ..., t_r]$ is integral over C by (3.22.7). Then A is integral over C by (3.22.5), and this is a normalising family by (3.32.8), and in particular algebraically independent.

For $j \leq p-1$ and $h := h(j) \leq r$, apply (3.32.16) to the ideal $\mathfrak{a}_j \cap B$ to see $\mathfrak{a}_j \cap k[t_{h+1}, ..., t_n] = \{0\}$ and therefore $\mathfrak{a}_j \cap k[t_{h+1}, ..., t_r, x_{r+1}, ..., x_n] = \{0\}$. As $t_1, ..., t_h \in \mathfrak{a}_j$ we see by (3.32.16) that $\mathfrak{a}_j \cap C = (t_1, ..., t_h)C$ as required.

Similarly by (3.32.16) $\mathfrak{a}_p \cap k[x_{s+1}, ..., x_n] = \{0\}$ and clearly $t_1, ..., t_r, x_{r+1}, ..., x_s \in \mathfrak{a}_p$. Then by (3.32.16) again $\mathfrak{a}_p \cap C = (t_1, ..., t_r, x_{r+1}, ..., x_s)C$ as required. ∎

Remark 3.32.17
In Bourbaki's proof the reduction to the polynomial ring case increases p to $p+1$, so in particular the $p = 1$ case requires the more complex reduction argument at the end of the proof. Here we use a simplified argument from [Sha00].

3.32.3 Nullstellensatz

Definition 3.32.18 (Zeros of an ideal)
*Let $\mathfrak{a} \lhd k[X_1, ..., X_n]$ be an ideal and K/k a field extension. Then a point $x \in K^n$ is a **zero** of $\mathfrak{a}$ if*

$$f \in \mathfrak{a} \implies f(x) = 0$$

*We define the **residue field** to be*

$$k(x) := k(x_1, ..., x_n)$$

and also denote

$$k[x] := k[x_1, ..., x_n]$$

*We say it is **algebraic** if $k(x)/k$ is algebraic (necessarily finite).*

The following observation is useful

Proposition 3.32.19 (Zeros are homomorphisms)
Let $\mathfrak{a} \lhd k[X_1, ..., X_n]$ be an ideal then there is a bijection

$$\mathrm{AlgHom}_k(k[X_1, ..., X_n]/\mathfrak{a}, K) \longleftrightarrow \{\text{zeros of } \mathfrak{a} \text{ in } K^n\}$$
$$\phi \longrightarrow \left(\phi\left(\overline{X}_1\right), ..., \phi\left(\overline{X}_n\right)\right)$$

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The following notion is useful in future

Definition 3.32.20 (Generic Point)
Let $\mathfrak{a} \triangleleft k[X_1, ..., X_n]$ be a **prime** ideal. then a point $\xi \in L^n$ is a **generic point** of $\mathfrak{a}$ if $\ker(\mathrm{ev}_\xi) = \mathfrak{a}$. Note one always exists, because we may take $L = \mathrm{Frac}(A)$ and $\xi_i = \overline{X}_i$.

When x is another zero of $\mathfrak{a}$ this induces a k-algebra homomorphism

$$k[\xi] \to k[x]$$

which is an isomorphism precisely when x is a generic point.

Generally we are interested in the relationship between ideals of $k[X_1, ..., X_n]$ and corresponding zeros in an extension field K/k. The following proposition is fundamental

Proposition 3.32.21 (Correspondence between ideals and zeros)
Let K/k be a field extension and $x \in K^n$. Define $\mathfrak{m}_x$ to be the kernel of the homomorphism

$$\mathrm{ev}_x : k[X_1, ..., X_n] \to K$$

Then

- *$\mathfrak{m}_x$ is a prime ideal*
- *If K/k is an algebraically closed field of transcendence degree $\geq n$ then every prime ideal is of this form.*
- *If x_i are algebraic over k (e.g. if K/k is algebraic) then $\mathfrak{m}_x$ is maximal*

In this case we have a canonical isomorphism

$$k[X_1, ..., X_n]/\mathfrak{m}_x \xrightarrow{\sim} k[x] \subset K$$

and when each x_i is algebraic then $k[x] = k(x)$ is a finite extension of k.

Proof. The canonical isomorphism follows from (3.10.3). Any subring of a field is an integral domain, which means $\mathfrak{p}_x$ is prime by (3.3.59).

By (3.17.44) x_i are algebraic over k if and only if $k(x_1, ..., x_n)/k$ is algebraic (and even finite). By the same result $k[x_1, ..., x_n] = k(x_1, ..., x_n)$ and therefore $\mathfrak{m}_x$ is maximal by (3.3.59).

Let $\mathfrak{p}$ be a prime ideal and define $k(x) \coloneqq \mathrm{Frac}(k[x])$ and $k[x] \coloneqq k[X_1, ..., X_n]/\mathfrak{p}$. If K has transcendence degree $\geq n$ then there is an embedding $k(x)/k \to K/k$ by (3.17.63). This restricts to an isomorphism $k[x] \xrightarrow{\sim} k[\overline{x}]$ for some $\overline{x}_i \in K$. It's clear that $\mathfrak{p} = \mathfrak{m}_x$. ∎

We may show that all maximal ideals correspond to zero-loci.

Lemma 3.32.22 (Zariski's Lemma)
Let k be an infinite field and A a finitely-generated k-algebra which is a field. Then A is a finite extension of k.

Proof. By Normalisation Lemma (3.32.11) there is an injective, integral map $k[X_1, ..., X_d] \hookrightarrow A$ for some $d \geq 0$. By (3.22.14) then $k[X_1, ..., X_d]$ is a field which implies $d = 0$. We conclude that A is algebraic over k, and by (3.17.44) finite over k. ∎

Proposition 3.32.23 (Weak Nullstellensatz I)
Every maximal ideal of $k[X_1, ..., X_n]/\mathfrak{a}$ is of the form $\mathfrak{m}_x/\mathfrak{a}$ for $x \in \overline{k}^n$ a zero of $\mathfrak{a}$.

In addition the conclusion of Zariski's Lemma holds for all (possibly finite) coefficient fields k.

Proof. Observe that $x \in \overline{k}^n$ is a zero of $\mathfrak{a}$ if and only if $\mathfrak{a} \subset \mathfrak{m}_x$. Therefore we may reduce to the case $\mathfrak{a} = (0)$ by results on quotient rings (3.3.56).

We've already shown that $\mathfrak{m}_x$ is maximal when $x \in \overline{k}^n$ is algebraic (3.32.21). Conversely if $\mathfrak{m}$ is a maximal ideal, then $\mathfrak{m}\overline{k}[X_1, ..., X_n]$ is a proper ideal contained in some maximal ideal $\mathfrak{m}' \triangleleft \overline{k}[X_1, ..., X_n]$. If $\mathfrak{m}'$ has an algebraic zero then so does $\mathfrak{m}$ and $\mathfrak{m} \subset \mathfrak{m}_x$ for some x, which are equal by maximality.

However $\overline{k}[X_1, ..., X_n]/\mathfrak{m}'$ is a finitely-generated $\overline{k}$-algebra which is a field and $\overline{k}$ is infinite. Therefore by (3.32.22) it is $\overline{k}$, and this yields an algebraic zero of $\mathfrak{m}'$ as required.

We may generalise Zariski's Lemma to the case k is not necessarily infinite. If A is a finitely-generated k-algebra which is a field then we have shown it is isomorphic to $k[X_1, ..., X_n]/\mathfrak{m}_x$ for some algebraic point x. By (3.32.21) is a finite extension of k, as required. ∎

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We may make the correspondence between maximal ideals and algebraic points more precise

Proposition 3.32.24 (Weak Nullstellensatz II)
There is a bijective map

$$\begin{aligned}\overline{k}^n / \operatorname{Aut}\left(\overline{k}/k\right) &\longrightarrow \{\mathfrak{m} \lhd k[X_1, ..., X_n] \text{ maximal}\} \\ x &\longrightarrow \mathfrak{m}_x\end{aligned}$$

When $x \in k^n$ then

$$\mathfrak{m}_x = (X_1 - x_1, ..., X_n - x_n)$$

Proof. The map is surjective by (3.32.23). It's well-defined because $\mathfrak{m}_{\sigma(x)} = \mathfrak{m}_x$.

By (3.32.21) we have an isomorphism $k[X_1, ..., X_n]/\mathfrak{m}_x \xrightarrow{\sim} k[x_1, ..., x_n] \subset \overline{k}$. If $\mathfrak{m}_x = \mathfrak{m}_y$ then these compose to yield an isomorphism $\sigma : k[x_1, ..., x_n] \xrightarrow{\sim} k[y_1, ..., y_n] \subset \overline{k}$ such that $\sigma(x_i) = y_i$. By (3.17.71) this extends to $\sigma \in \operatorname{Aut}\left(\overline{k}/k\right)$. Therefore the given mapping is injective. ∎

3.32.4 Morphisms of Affine Algebras

Definition 3.32.25
*Let B be a k-algebra. We say that B is **algebraic** over k if for every maximal ideal $\mathfrak{m} \lhd B$ we have $B/\mathfrak{m} \xrightarrow{\sim} k(\mathfrak{m})$ is algebraic over k.*

Proposition 3.32.26
Let B be a k-algebra, then it is algebraic in each of the following cases

a) B is a finitely generated k-algebra
b) B is a local k-algebra such that $k(\mathfrak{m}_B)/k$ is algebraic
c) B is an algebraic field extension of k
d) B is the localisation of a finitely-generated k-algebra at a maximal ideal

Proof. We prove each in turn

a) This follows from Zariski's Lemma in the general case (3.32.23)
b) The only maximal ideal is $\mathfrak{m}_B$ so this is obvious
c) This is a special case of b)
d) This follows from a) and b) because $B_{\mathfrak{m}}$ is a local ring with residue field isomorphic to $B/\mathfrak{m}$ (3.5.43).

∎

Proposition 3.32.27
Let $\phi : A \to B$ be a k-algebra homomorphism, with B algebraic. Then the inverse image of a maximal ideal is again maximal.

Proof. Let $\mathfrak{n}$ be a maximal ideal. By assumption $B/\mathfrak{n}$ is integral over k, and so a-fortiori $B/\mathfrak{n}$ is integral over A. Therefore $\phi^{-1}(\mathfrak{n})$ is maximal by (3.22.16). ∎

3.32.5 Krull Dimension of Affine Algebras

The Krull Dimension of affine domains is particularly well-behaved. Specifically they are biequidimensional and therefore satisfy a co-dimension formula (3.32.35). Further there's a geometric proof of the "Hauptidealsatz" (3.32.32). We first show that it is equal to transcendence degree in the integral case.

Proposition 3.32.28
Let A be an affine algebra with normalising family $x_1, ..., x_n$, then $\dim A = n$.

In particular every affine domain A satisfies $\dim A = \operatorname{trdeg}(\operatorname{Frac}(A)/k)$, and the polynomial ring $k[X_1, ..., X_n]$ has dimension n.

Proof. By definition A is integral over $k[x_1, ..., x_n]$, which is isomorphic to a polynomial ring so we may reduce to the case of polynomial ring by (3.22.24).

$\dim A \geq n$). This is clear by considering the chain of prime ideals

$$(X_1) \subsetneq (X_1, X_2) \subsetneq ... \subsetneq (X_1, ..., X_n)$$

$\dim A \leq n$). We may argue by the Strong Normalisation Lemma (3.32.12) and the subsequent remark that any chain of prime ideals must have length at most n, as any normalising family has order n.

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Alternatively we may proceed by induction on n to show $\dim k[X_1,...,X_n] = n$. Consider a maximal chain

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq ... \subsetneq \mathfrak{p}_m$$

Clearly $\mathfrak{p}_0 = 0$, and $\mathfrak{p}_1 = (f)$ principal by (3.20.5). By (3.32.9) there is an integral, injective map

$$k[Y_1,...,Y_{n-1}] \hookrightarrow k[X_1,...,X_n]/(f)$$

whence $\dim(\mathfrak{p}_1) = \dim(k[X_1,...,X_n]/(f)) = \dim(k[Y_1,...,Y_{n-1}]) = n-1$, and by definition $m-1 \leq n-1$. As the maximal chain was arbitrarily chosen, we have $\dim k[X_1,...,X_n] \leq n$. The reverse inequality was already shown so we are done.

The final statement follows from the existence of a normalising family (3.32.11) and (3.32.7) ■

Corollary 3.32.29
Let A be a finitely-generated k-algebra. Then $\dim A[T] = \dim A + 1$.

Proof. By (3.32.11) there is an injective integral ring homomorphism

$$\phi : k[X_1,...,X_n] \hookrightarrow A$$

where $\dim A = n$. By (3.7.6) there is a ring homomorphism

$$k[X_1,...,X_n][T] \hookrightarrow A[T]$$

which is injective. Further by assumption $A = k[y_1,...,y_r]$ so $A[T] = k[y_1,...,y_r,T]$ and by (3.22.4) the ring homomorphism is integral. This shows that $\phi(x_1),...,\phi(x_n),T$ is a normalising family and the result follows from (3.32.28). ■

Corollary 3.32.30
The ideal $(X_1,...,X_r) \lhd k[X_1,...,X_n]$ has dimension $n-r$.

Corollary 3.32.31
Let A be an integral finitely-generated k-algebra and $0 \neq f$ then $\dim A = \dim A_f$

The following proof is due to Tate, and presented in the Red Book [Mum99] (I.7 Theorem 2).

Proposition 3.32.32 (Hypersurface has pure codimension 1)
Let A be an affine domain of dimension n and $0 \neq f \in A$. Then

$$\dim((f)) = n-1$$

$$\mathrm{ht}((f)) = 1$$

More precisely if $\mathfrak{p}$ is minimal over (f) then it has dimension $n-1$ and height 1.

Proof. Consider the case $A = k[X_1,...,X_n]$. Suppose first that f is prime, then $\mathfrak{p} = (f)$ and by (3.32.9) there is an integral injective map

$$k[Y_1,...,Y_{n-1}] \hookrightarrow A/(f)$$

Therefore

$$\dim((f)) = \dim(A/(f)) \overset{(3.22.24)}{=} \dim(k[Y_1,...,Y_{n-1}]) \overset{(3.32.28)}{=} n-1$$

When f is not prime then, as $k[X_1,...,X_n]$ is a UFD, we have a prime factorisation

$$f = \prod_{i=1}^{n} f_i^{m_i}$$

and the minimal prime decomposition is

$$\sqrt{(f)} = (f_1) \cap ... \cap (f_n)$$

In particular any prime minimal over (f) has the form $\mathfrak{p} = (f_i)$ for some i, and we may reduce to the case of f prime.

For an arbitrary k-algebra A we have a decomposition into minimal primes of (f)

$$\sqrt{(f)} = \mathfrak{p}_1 \cap ... \cap \mathfrak{p}_n$$

and without loss of generality $\mathfrak{p} = \mathfrak{p}_1$. We may localize to the case of a single prime, for choose $g \notin \mathfrak{p}$ and $g \in \mathfrak{p}_i$ for $i = 2...n$. Consider the localization $A \to A_g$, then we claim that

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$$\sqrt{(f/1)} = \mathfrak{p}A_g$$

For by (3.5.19) there is a correspondence between primes of A_g containing $(f/1)$ and primes of A containing f and not g, which are precisely the primes containing $\mathfrak{p}$ by (3.3.40) or (3.3.44). Therefore $\mathfrak{p}A_g$ is the only minimal prime of A_g containing $(f/1)$ and the claim follows from (3.3.46).

Note that $\dim A = \dim A_g$ as they have the same quotient field and therefore transcendence degree. Similarly $\dim(A/\mathfrak{p}) = \dim\big((A/\mathfrak{p})_{\bar{g}}\big) = \dim\big(A_g/\mathfrak{p}_g\big) = \dim\big(\mathfrak{p}A_g\big)$. So we may assume without loss of generality that $n = 1$ and $\mathfrak{p} = \sqrt{(f)}$.

By (3.32.11) there is an integral, injective map

$$\phi : B \hookrightarrow A$$

where $B = k[X_1, ..., X_n]$, which induces an algebraic field extension

$$K := \mathrm{Frac}(B) \hookrightarrow \mathrm{Frac}(A) =: L$$

We claim that there exists $f_0 \in B$ such that

$$\phi^{-1}\big(\sqrt{f}\big) = \sqrt{(f_0)}$$

Observe that in this case $\sqrt{(f_0)}$ is prime and therefore the unique minimal prime over (f_0). Therefore the result would follow from the first part and (3.22.25). Firstly for any $g \in A$ we have (3.22.19)

$$\mathrm{Norm}_{L/K}(g) \in B \cap \phi^{-1}((g))$$

(because B is a UFD and therefore integrally closed (3.22.12)). Define $f_0 := \mathrm{Norm}_{L/K}(f)$ then we see that $f_0 \in \phi^{-1}((f)) \implies \sqrt{(f_0)} \subseteq \phi^{-1}\big(\sqrt{(f)}\big)$. Conversely if $\phi(g)^n = hf$ then $g^{n[L:K]} = \mathrm{Norm}(\phi(g)^n) = \mathrm{Norm}(h)f_0 \in (f_0) \implies g \in \sqrt{(f_0)}$. Therefore the reverse inclusion holds.

Finally by the codimension 1 formula (3.20.8) we have $\mathrm{ht}((f)) = 1$. ∎

Remark 3.32.33
The argument is slightly less awkward in geometric language. Decompose into irreducibles

$$V(f) = Z_1 \cup ... \cup Z_n$$

choose a principal open affine $D(g)$ which meets only Z_1 then $Z_1 \cap D(g) = V(f) \cap D(g) = V(f/1)$ is an irreducible component of $D(g)$. We argue that $\dim(X) = \dim(D(g))$ and $\dim(Z_1) = \dim(D(g) \cap Z_1)$. Further construct finite coverings

$$V(f/1) \to V(f_0) \to H$$

onto a hyperplane in $\mathbb{A}^n$.

This allows is to prove a converse to (3.20.8)

Corollary 3.32.34 (Height 1 formula)
Let A be an affine domain. Then for a prime ideal $\mathfrak{p}$

$$\mathrm{ht}(\mathfrak{p}) = 1 \implies \dim(\mathfrak{p}) = \dim(A) - 1$$

More generally if A is an affine algebra and $\mathfrak{p}_1 \subsetneq \mathfrak{p}_2$ is a saturated chain of prime ideals then

$$\dim(\mathfrak{p}_2) = \dim(\mathfrak{p}_1) - 1$$

Proof. Choose $0 \neq f \in \mathfrak{p}$ then it follows from the previous result (3.32.32). Alternatively by (3.32.11) there is an integral injective map

$$\phi : k[X_1, ..., X_n] \hookrightarrow A$$

with $\mathfrak{q} := \phi^{-1}(\mathfrak{p})$ and $n = \dim A$. Then by Going Down we have $\mathrm{ht}(\mathfrak{q}) = 1$. By (3.20.5) $\mathfrak{q}$ is principal and by (3.32.9) there is an integral injective map

$$k[Y_1, ..., Y_{n-1}] \hookrightarrow k[X_1, ..., X_n]/\mathfrak{q}$$

therefore $\dim(\mathfrak{q}) = n - 1$. By (3.22.25) this equals $\dim(\mathfrak{p})$ and we are done.

For the second statement we may consider the affine domain $A/\mathfrak{p}_1$ and observe that $\mathrm{ht}(\mathfrak{p}_2/\mathfrak{p}_1) = 1$. Therefore

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$$\dim(\mathfrak{p}_2) = \dim(\mathfrak{p}_2/\mathfrak{p}_1) = \dim(A/\mathfrak{p}_1) - 1 = \dim(\mathfrak{p}_1) - 1$$

■

Corollary 3.32.35 (Biequidimensionality)
*Let A be an affine algebra. Then it is **quasi-biequidimensional** and satisfies the **codimension formulae** for $\mathfrak{p} \subset \mathfrak{a}$*

$$\dim \mathfrak{a} = \dim \mathfrak{p} + \operatorname{ht}(\mathfrak{a}/\mathfrak{p})$$
$$\operatorname{ht}(\mathfrak{a}) = \operatorname{ht}(\mathfrak{a}/\mathfrak{p}) + \operatorname{ht}(\mathfrak{p})$$

*If in addition A is equidimensional (e.g. an integral domain) then it is **biequidimensional** and satisfies for all ideals $\mathfrak{a}$*

$$\dim A = \dim \mathfrak{a} + \operatorname{ht}(\mathfrak{a})$$

In particular for every prime ideal $\mathfrak{p}$ we have

$$\dim A = \dim A_{\mathfrak{p}} + \dim A/\mathfrak{p}$$

and for every maximal ideal

$$\dim A = \dim A_{\mathfrak{m}}$$

Proof. By (3.32.34) A satisfies the criteria in (3.20.10).e) and so is quasi-biequidimensional. The codimension formulas follow from (3.20.12). ■

We may also consider the following result

Proposition 3.32.36 (Generalized Hauptidealsatz for Affine Algebras)
Let A be an affine algebra and $\mathfrak{p}$ a prime ideal. Then

a) If $\operatorname{ht}(\mathfrak{p}) = n$ then it is minimal over some ideal $\mathfrak{a} := (x_1, ..., x_n)$. Furthermore $\mathfrak{a}$ may be chosen such that $\operatorname{ht}(\mathfrak{a}) = n$ and every minimal prime is of height n
b) We have the following characterization of height of a prime ideal

$$\operatorname{ht}(\mathfrak{p}) = \min\{n \mid \mathfrak{p} \text{ minimal over } (x_1, ..., x_n)\}$$

In particular if $\mathfrak{p}$ is minimal over $(x_1, ..., x_n)$ then $\operatorname{ht}(\mathfrak{p}) \leq n$.

Furthermore the same result holds for localization of A at any prime ideal.

Proof. This is largely a restatement of results in the Hauptidealsatz section. By (3.32.32) we have $A/\mathfrak{p}$ is hauptidealsatz for every prime ideal $\mathfrak{p}$. Furthermore by (3.32.35) A is quasi-biequidimensional and so catenary (3.20.10). Then by (3.25.4) this means A is a generalized hauptidealsatz ring (and so is every localization $A_{\mathfrak{p}}$).

Then a) and b) follows from (3.25.12). ■

Corollary 3.32.37
Let A be an affine algebra and $\mathfrak{p}$ a prime ideal. Denote the unique maximal ideal of $A_{\mathfrak{p}}$ by $\mathfrak{m} := \mathfrak{p}A_{\mathfrak{p}}$. Then

$$\dim A_{\mathfrak{p}} = \min\left\{n \;\middle|\; \sqrt{(x_1, ..., x_n)} = \mathfrak{m}\right\} \leq \dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2$$

with equality iff $A_{\mathfrak{p}}$ is a regular local ring.

Proof. By (3.32.36) $A_{\mathfrak{p}}$ is a Noetherian local ring satisfying the generalized hauptidealsatz. Therefore the result follows by (3.26.2). ■

3.32.5.1 * Biequidimensionality by Strong Normalisation *

We may prove more directly the biequidimensionality property by using the strong form of the Normalisation Lemma (3.32.12) and Going Down (3.22.21). First we prove a technical result

Lemma 3.32.38 (Saturated pairs)
Let $\phi : B \to A$ be an integral map and $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1$ prime ideals lying over $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$. Then

a) $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$ saturated $\implies \mathfrak{p}_0 \subsetneq \mathfrak{p}_1$ saturated.
b) $B/\mathfrak{q}_0$ integrally closed and $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1$ saturated $\implies \mathfrak{q}_0 \subsetneq \mathfrak{q}_1$ saturated

We may relax the condition in b) to the existence of another integral map $\psi : C \to B$ such that $C/\psi^{-1}(\mathfrak{q}_0)$ is integrally closed.

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Proof. The first follows by incomparability (3.22.18). The second follows by applying Going Down (3.22.21) to the integral map $B/\mathfrak{q}_0 \hookrightarrow A/\mathfrak{p}_0$. More precisely if $\mathfrak{q}_0 \subsetneq \mathfrak{q} \subsetneq \mathfrak{q}_1$ then $(0) \subsetneq \mathfrak{q}/\mathfrak{q}_0 \subsetneq \mathfrak{q}_1/\mathfrak{q}_0$ whence there exists $\mathfrak{p}$ such that $(0) \subsetneq \mathfrak{p}/\mathfrak{p}_0 \subsetneq \mathfrak{p}_1/\mathfrak{p}_0$. This means $\mathfrak{p}_0 \subsetneq \mathfrak{p} \subsetneq \mathfrak{p}_1$, a contradiction.

The final statement can be demonstrated by applying b) to $C \to A$ and then a) to $B \to A$. ∎

Proposition 3.32.39
*Let A be an affine domain, then every maximal chain has order $n = \dim A$, i.e. A is **irreducible** and **biequidimensional**.*

Proof of (3.32.39). Consider a maximal chain $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq ... \subsetneq \mathfrak{p}_m$. Clearly $\mathfrak{p}_m$ is maximal by (3.3.37), and as A is integral $\mathfrak{p}_0 = 0$. Apply (3.32.12) to find an integral, injective map

$$\phi : k[X_1, ..., X_n] \hookrightarrow A$$

such that

$$\mathfrak{q}_i := \phi^{-1}(\mathfrak{p}_i) = \left(X_1, ..., X_{h(i)}\right) \quad 0 \leq i \leq m$$

Note $n = \dim A$ by (3.32.28). Clearly $h(0) = 0$ and by (3.22.16) $\mathfrak{q}_m$ is maximal so $h(m) = n$. Observe that

$$k[X_1, ..., X_n]/\mathfrak{q}_i \xrightarrow{\sim} k\left[X_{h(i)+1}, ..., X_n\right]$$

is integrally closed for all i (3.22.12). Therefore we may apply (3.32.38) to ϕ and each pair $\mathfrak{p}_i \subsetneq \mathfrak{p}_{i+1}$, to see that each chain $\mathfrak{q}_i \subsetneq \mathfrak{q}_{i+1}$ is saturated. This can only happen if $h(j) = j$ and therefore $m = n$. ∎

3.32.6 Derivations of Affine Algebras

The notion of derivations is a useful as a coordinate-free construction of the "tangent space" for both differentiable manifolds and algebraic varieties. We review the theory of derivations here primarily focusing on fields and f.g. k-algebras.

Proposition 3.32.40 (Derivations of Polynomial Algebra)
Let $A = k[X_1, ..., X_n]$ be a polynomial algebra and M an (A, B)-bimodule. Then we have an (A, B)-bimodule isomorphism

$$\begin{aligned} \mathrm{Der}_k(A, M) &\cong M^n \\ D &\to (D(X_1), ..., D(X_n)) \\ \sum_{i=1}^n \frac{\partial}{\partial X_i} \cdot v_i &\leftarrow v \end{aligned}$$

When M has a B-basis $\{m_1, ..., m_r\}$ then $\mathrm{Der}_k(A, M)$ also has a B-basis

$$\left\{ \frac{\partial}{\partial X_i} \cdot m_j \right\}_{i=1...n, j=1...r}$$

In particular $\dim_K \mathrm{Der}_k(A, K) = n$.

Proof. The first map is clearly well-defined and by (...) injective. The inverse map is well-defined because we have shown $\frac{\partial}{\partial X_i} \in \mathrm{Der}_k(A, A)$. The chain rule (3.24.8) shows that the maps are mutually inverse (the other direction following simply from orthogonality of the partial derivatives).

The final statement is straightforward. ∎

We may generalise this as follows, so that we may interpret derivations as tangent vectors

Proposition 3.32.41 (Derivations = Tangent Vectors)
Let $A = k[X_1, ..., X_n]/\mathfrak{a}$ be a f.g. k-algebra and M an A-module. Then we have an A-module isomorphism

$$\begin{aligned} \mathrm{Der}_k(A, M) \cong & \left\{ v \in M^n \;\middle|\; \sum_{i=1}^n \overline{\frac{\partial F}{\partial X_i}} v_i = 0 \quad \forall F \in \mathfrak{a} \right\} \subset M^n \\ & D \longrightarrow \left(D\left(\overline{X}_1\right), ..., D\left(\overline{X}_n\right)\right) \\ \sum_{i=1}^n & \overline{\frac{\partial}{\partial X_i}} \cdot v_i \longleftarrow v \end{aligned}$$

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If $\mathfrak{a} = \langle F_1, ..., F_m \rangle$ then this is equal to the kernel of the A-module homomorphism

$$\begin{pmatrix} \overline{\frac{\partial F_1}{\partial X_1}} & \cdots & \overline{\frac{\partial F_1}{\partial X_n}} \\ \vdots & \ddots & \vdots \\ \overline{\frac{\partial F_m}{\partial X_1}} & \cdots & \overline{\frac{\partial F_m}{\partial X_n}} \end{pmatrix} : M^n \to M^m$$

When M is an (A, B)-bimodule then this is also an (A, B)-bimodule isomorphism.

Proof. According to (3.24.8) we have the following relationship for all $F \in k[X_1, ..., X_n]$

$$D(\overline{F}) = D\big(F\big(\overline{X}_1, ..., \overline{X}_n\big)\big) = \sum_{i=1}^{n} \frac{\partial F}{\partial X_i}\big(\overline{X}_i\big) D\big(\overline{X}_i\big) = \sum_{i=1}^{n} \overline{\frac{\partial F}{\partial X_i}} D\big(\overline{X}_i\big)$$

When $F \in \mathfrak{a}$ we have $\overline{F} = 0 \implies D(\overline{F}) = 0$, whence the first map is well-defined. Similarly for $v \in \text{RHS}$ we see by definition that

$$\sum_{i=1}^{n} \frac{\partial}{\partial X_i} \cdot v_i$$

is zero on $\mathfrak{a}$, so determines a well-defined derivation on A. In the same way we see that the maps are mutually inverse.

Suppose v is in the kernel of the given matrix. For any $G \in \mathfrak{a}$ we have by hypothesis $G = \sum_{j=1}^{m} F_j H_j$ for some $H_j \in k[X_1, ..., X_n]$. Then

$$\begin{aligned} \sum_{i=1}^{n} \overline{\frac{\partial G}{\partial X_i}} v_i &= \sum_{i=1}^{n} \sum_{j=1}^{m} \left(\overline{F}_j \overline{\frac{\partial H_j}{\partial X_i}} + \overline{H}_j \overline{\frac{\partial F_j}{\partial X_i}} \right) v_i \\ &= \sum_{j=1}^{m} \sum_{i=1}^{n} \overline{\frac{\partial F_j}{\partial X_i}} v_i \\ &= 0 \end{aligned}$$

where we have used the fact that $\overline{F_j} = 0$. As G was arbitrary this shows v is in the right-hand side of the isomorphism, and the reverse inclusion is immediate as $F_j \in \mathfrak{a}$. ∎

Proposition 3.32.42 (Lifting for separable algebraic extensions)
Let L/K be a separable algebraic (resp. separably generated) extension (over k) and V an L-module. Then there is an isomorphism (resp. epimorphism) of L-modules

$$\begin{aligned} \mathrm{Der}_k(L, V) &\xrightarrow{\sim} \mathrm{Der}_k(K, V) \\ D &\to D|_K \end{aligned}$$

Proof. Consider $L[V]$ the k-algebra with ideal $N := 0 \times V$ such that $N^2 = 0$ and $L[V]/N \cong L$. Furthermore a derivation $D : K \to V$ corresponds to a unique k-algebra homomorphism $\phi_D : K \to K[V] \hookrightarrow L[V]$ by (3.24.5) so we have the following commutative diagram

$$\begin{array}{ccc} K & \xrightarrow{\phi_D} & L[V] \\ \downarrow & \nearrow & \downarrow \pi_1 \\ L & = & L \end{array}$$

In the algebraic case by (3.24.16) there is a unique homomorphism $\phi_{\tilde{D}} : L \to L[V]$ completing the diagram, and in particular by (3.24.5) there is a derivation $\tilde{D} : L \to V$ extending D. Uniqueness follows from that of $\phi_{\tilde{D}}$.

The separably generated case is similar, except without uniqueness. ∎

Corollary 3.32.43
Let K/k be a separable algebraic extension, A a K-algebra and M an A-module then

$$\mathrm{Der}_k(A, M) = \mathrm{Der}_K(A, M)$$

Proof. Recall (3.24.1) that a derivation D is K-linear iff it vanishes on K. So we may reduce to the case $A = K$ by considering the restriction $D|_K$.

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Therefore it's sufficient to show that $\mathrm{Der}_k(K,M) = \mathrm{Der}_K(K,M) = 0 = \mathrm{Der}_k(k,M)$, but this follows immediately from (3.32.42). ∎

Proposition 3.32.44 (Transcendence Degree Controls Derivations into Any Module)
*Suppose K/k is **separably generated** (e.g. if k is perfect and K finitely generated), with transcendence basis $\eta_1, ..., \eta_n$, $n = \mathrm{trdeg}_k(K)$, and let M be a K-module. Then*

$$\mathrm{Der}_k(K,M) \cong M^n, \quad D \mapsto (D(\eta_1), ..., D(\eta_n))$$

as K-modules. In particular, taking $M := K$, $\mathrm{trdeg}_k(K) = \dim_K \mathrm{Der}_k(K)$.

Proof. By hypothesis $K/k(\eta_1, ..., \eta_n)$ is algebraic and separable. Therefore we have

$$\mathrm{Der}_k(K,M) \overset{(3.32.42)}{\cong} \mathrm{Der}_k(k(\eta_1, ..., \eta_n), M) \overset{(3.24.10)}{\cong} \mathrm{Der}_k(k[\eta_1, ..., \eta_n], M) \overset{(3.32.40)}{\cong} M^n$$

the last isomorphism sending $D \mapsto (D(\eta_1), ..., D(\eta_n))$, matching the same map once restricted back along the two previous isomorphisms (both restriction maps, $D \mapsto D|_{k(\eta_1,...,\eta_n)}$ and $D \mapsto D|_{k[\eta_1,...,\eta_n]}$, fix the value at each η_i). Taking $M := K$ gives $\dim_K \mathrm{Der}_k(K,K) = n = \mathrm{trdeg}_k(K)$. ∎

Proposition 3.32.45 (Derivations Extend Uniquely Along a Field Extension)
Let K/k be a field extension, k'/k any field extension, and $D \in \mathrm{Der}_k(K,K)$. There is a unique $\tilde{D} \in \mathrm{Der}_{k'}(K \otimes_k k', K \otimes_k k')$ with $\tilde{D}(x \otimes 1) = D(x) \otimes 1$ for every $x \in K$; explicitly, $\tilde{D}(x \otimes c) = D(x) \otimes c$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Existence. Define $\tilde{D}(x \otimes c) := D(x) \otimes c$, extended k'-linearly to $K \otimes_k k'$ (well-defined as D is k-linear). It satisfies the Leibniz rule on simple tensors, $\tilde{D}((x \otimes c)(y \otimes c')) = D(xy) \otimes cc' = (D(x)y + xD(y)) \otimes cc' = \tilde{D}(x \otimes c) \cdot (y \otimes c') + (x \otimes c) \cdot \tilde{D}(y \otimes c')$, hence on all of $K \otimes_k k'$ by bilinearity: $\tilde{D} \in \mathrm{Der}_{k'}(K \otimes_k k', K \otimes_k k')$.

Uniqueness. $K \otimes_k k'$ is k'-spanned by $K \otimes 1$ (extension of scalars along a k-basis of K), and any extension of D is pinned down there ($x \otimes 1 \mapsto D(x) \otimes 1$ by definition of "extends"), so agrees with $\tilde{D}$ on a k'-spanning set, hence everywhere by k'-linearity. ∎

Lemma 3.32.46
Let A be a k-algebra and $\mathfrak{m}$ an ideal. Then $\mathfrak{m}/\mathfrak{m}^2$ is a $k(\mathfrak{m})$-module with action given by

$$(a + \mathfrak{m}) \cdot (b + \mathfrak{m}^2) = (ab + \mathfrak{m}^2)$$

for all $a \in A$ and $b \in \mathfrak{m}$.

Proof. Suppose $a - a' \in \mathfrak{m}$ and $b - b' \in \mathfrak{m}^2$ then

$$ab - a'b' = (a - a')(b - b') + a'(b - b') + b'(a - a') \in \mathfrak{m}^2$$

this shows that the action is well-defined. The properties of a module are straightforward. ∎

Lemma 3.32.47
Let A be a k-algebra and $\mathfrak{m}$ a maximal ideal. Then there is an isomorphism of A-algebras

$$\begin{array}{ccc} A & \longrightarrow & A/\mathfrak{m}^2 \\ \downarrow{\scriptstyle \pi_1} & \overset{\pi_2}{\searrow} & \downarrow \\ k(\mathfrak{m}) & \overset{\psi}{\underset{\sim}{\dashrightarrow}} & k(\mathfrak{m}/\mathfrak{m}^2) \end{array}$$

Further there is an isomorphism of A-modules

$$\begin{aligned} \mathrm{Der}_k(A/\mathfrak{m}^2, k(\mathfrak{m}/\mathfrak{m}^2)) &\to \mathrm{Der}_k(A, k(\mathfrak{m})) \\ D &\to x \to \psi^{-1}(D(\overline{x})) \end{aligned}$$

Proof. The first statement follows from (3.3.58) with $\mathfrak{a} = \mathfrak{m}^2$ and $\mathfrak{b} = \mathfrak{m}$.

Define $\hat{D}(x) := \psi^{-1}(D(\overline{x}))$. By hypothesis $D(\overline{xy}) = \pi_2(x)D(\overline{y}) + \pi_2(y)D(\overline{x})$. Therefore

$$\hat{D}(xy) = \pi_1(x)\psi^{-1}(D(\overline{y})) + \pi_1(y)\psi^{-1}(D(\overline{x})) = \pi_1(x)\hat{D}(\overline{y}) + \pi_1(y)\hat{D}(\overline{x})$$

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and so $\hat{D}$ is a well-defined derivation. Suppose $\hat{D} \in \mathrm{Der}_k(A, k(\mathfrak{m}))$ then by the product rule $\mathfrak{m}^2 \subset \ker\left(\hat{D}\right)$ so there is a derivation $D' : A/\mathfrak{m}^2 \to k(\mathfrak{m})$ such that $D'(\overline{x}) = \hat{D}(x)$. Define $D \coloneqq \psi \circ D'$, then evidentally the image of this derivation is $\hat{D}$ and so the map is surjective.

Suppose $\hat{D} = 0$ then as ψ is an isomorphism $D \circ \bar{\cdot} = 0$. As the reduction map is surjective we deduce that $D = 0$. Therefore the map is injective, bijective and an isomorphism. ∎

Lemma 3.32.48
Let A be a k-algebra and $\mathfrak{m}$ an ideal. Then there is an isomorphism of A-modules

$$\begin{aligned}\mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2, k(\mathfrak{m})) &\to \mathrm{Hom}_{k(\mathfrak{m}/\mathfrak{m}^2)}((\mathfrak{m}/\mathfrak{m}^2)/(\mathfrak{m}^2/\mathfrak{m}^2), k(\mathfrak{m}/\mathfrak{m}^2)) \\ \theta &\to \psi \circ \theta \circ i^{-1}\end{aligned}$$

where $i : \mathfrak{m}/\mathfrak{m}^2 \to (\mathfrak{m}/\mathfrak{m}^2)/(\mathfrak{m}^2/\mathfrak{m}^2)$ is a ring isomorphism such that

$$i(\pi_1(a) \cdot x) = \pi_2(a) \cdot i(x)$$

Proof. Then for $x \in (\mathfrak{m}/\mathfrak{m}^2)/(\mathfrak{m}^2/\mathfrak{m}^2)$, $a \in A$ and $\theta \in \mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2, k(\mathfrak{m}))$

$$\begin{aligned}(\psi \circ \theta \circ i^{-1})(\pi_2(a) \cdot x) &= (\psi \circ \theta)(\pi_1(a) \cdot i^{-1}(x)) \\ &= \psi(\pi_1(a)\theta(i^{-1}(x))) \\ &= \pi_2(a)(\psi \circ \theta \circ i^{-1})(x)\end{aligned}$$

so that the map is well-defined. Similarly if $\theta' \in \mathrm{Hom}_{k(\mathfrak{m}/\mathfrak{m}^2)}((\mathfrak{m}/\mathfrak{m}^2)/(\mathfrak{m}^2/\mathfrak{m}^2), k(\mathfrak{m}/\mathfrak{m}^2))$

$$\begin{aligned}(\psi^{-1} \circ \theta' \circ i)(\pi_1(a) \cdot x) &= (\psi^{-1} \circ \theta')(\pi_2(a) \cdot i(x)) \\ &= \psi^{-1}(\pi_2(a)\theta'(i(x))) \\ &= \pi_1(a)(\psi^{-1} \circ \theta' \circ i)(x)\end{aligned}$$

so the inverse map is well-defined. ∎

Proposition 3.32.49 (Coefficient Field mod $\mathfrak{m}^2$)
Let A be a k-algebra and $\mathfrak{m} \triangleleft A$ a maximal ideal with $\mathfrak{m}^2 = 0$ and residue field $k(\mathfrak{m}) \coloneqq A/\mathfrak{m}$. If $k(\mathfrak{m})/k$ is separably generated (e.g. if k is perfect and $k(\mathfrak{m})/k$ is finitely generated), then the quotient map $\pi : A \to k(\mathfrak{m})$ admits a k-algebra section $j : k(\mathfrak{m}) \to A$, that is there is a subfield $k \subseteq \hat{k} \subseteq A$ such that $\pi|_{\hat{k}} : \hat{k} \to k(\mathfrak{m})$ is an isomorphism. In particular $A = \hat{k} \oplus \mathfrak{m}$ as k-vector spaces.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\mathfrak{m}^2 = 0$, $\mathfrak{m}$ is the only prime of A, so A is local, and $A/\mathfrak{m}^2 = A$; so (3.24.18) (with $n = 1$) gives a k-algebra homomorphism $j \coloneqq s_1 : k(\mathfrak{m}) \to A$ with $\pi \circ j = \mathrm{id}_{k(\mathfrak{m})}$.

Set $\hat{k} \coloneqq j(k(\mathfrak{m})) \subseteq A$. As j is a ring homomorphism from a field, it is injective (3.3.61), so $j : k(\mathfrak{m}) \to \hat{k}$ is an isomorphism, and $\pi \circ j = \mathrm{id}$ shows $\pi|_{\hat{k}}$ is its inverse. Finally j is a k-linear section of the k-linear surjection $\pi : A \to k(\mathfrak{m})$ with kernel $\mathfrak{m}$, so $A = \mathrm{Im}(j) \oplus \ker(\pi) = \hat{k} \oplus \mathfrak{m}$ as k-vector spaces. ∎

Proposition 3.32.50 (Tangent space is dual to Cotangent space)
Let A be a k-algebra and $\mathfrak{m} \triangleleft A$ a maximal ideal with residue field $k(\mathfrak{m}) \coloneqq A/\mathfrak{m}$ and projection $\pi : A \to k(\mathfrak{m})$. The following sequence of $k(\mathfrak{m})$-vector spaces is exact, where $\theta(D) =: \theta_D$ is the map $\overline{x} \mapsto D(x)$:

$$0 \to \mathrm{Der}_k(k(\mathfrak{m})) \overset{D_0 \mapsto D_0 \circ \pi}{\to} \mathrm{Der}_k(A, k(\mathfrak{m})) \overset{\theta}{\to} \mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2, k(\mathfrak{m}))$$

*When $k(\mathfrak{m})/k$ is **separably generated** (e.g. k is perfect and $k(\mathfrak{m})/k$ is finitely generated) then θ is moreover surjective. In particular θ is an isomorphism when $k(\mathfrak{m})/k$ is **separable algebraic**, for example when k is perfect and $k(\mathfrak{m})/k$ is finite (as for a maximal ideal of a finitely generated k-algebra).*

Proof. The map θ_D is well-defined because D annihilates $\mathfrak{m}^2$ by the product rule. The sequence is exact on the left and in the middle for any A. The map $D_0 \circ \pi$ is a derivation $A \to k(\mathfrak{m})$ vanishing on $\mathfrak{m}$, and $D_0 \mapsto D_0 \circ \pi$ is injective as π is surjective. Conversely if $D \in \mathrm{Der}_k(A, k(\mathfrak{m}))$ vanishes on $\mathfrak{m}$ then $D(a + x) = D(a)$ for $x \in \mathfrak{m}$, so $D = D_0 \circ \pi$ for a well defined map $D_0 : k(\mathfrak{m}) \to k(\mathfrak{m})$, which is a derivation as π is a surjective ring homomorphism. Since $\ker(\theta) = \{D \mid D|_{\mathfrak{m}} = 0\}$ this proves exactness. It remains to show that θ is surjective when $k(\mathfrak{m})/k$ is separably generated.

We first claim we can reduce to the case $\mathfrak{m}^2 = 0$. For consider $A' \coloneqq A/\mathfrak{m}^2$ and $\mathfrak{m}' = \mathfrak{m}/\mathfrak{m}^2$ and the commutative diagram of A-module homomorphisms

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$$\begin{array}{ccc} \mathrm{Der}_k(A', k(\mathfrak{m}')) & \longrightarrow & \mathrm{Hom}_{k(\mathfrak{m}')}\left(\mathfrak{m}'/(\mathfrak{m}')^2, k(\mathfrak{m}')\right) \\ \downarrow \sim & & \downarrow \sim \\ \mathrm{Der}_k(A, k(\mathfrak{m})) & \longrightarrow & \mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2, k(\mathfrak{m})) \end{array}$$

The horizontal maps have already been defined and the vertical maps are defined in (3.32.47) and (3.32.48). The diagram evidentally commutes, so therefore it is sufficient to show that the top map is surjective..

We revert to the original notation, but with the additional assumption that $\mathfrak{m}^2 = 0$. By (3.32.49) there is a subfield $k \subseteq \hat{k} \subseteq A$ mapping isomorphically to $k(\mathfrak{m})$ under the canonical projection $\pi : A \to A/\mathfrak{m} = k(\mathfrak{m})$, giving a direct sum of k-vector spaces

$$A = \hat{k} \oplus \mathfrak{m}$$

As $\mathfrak{m}^2 = 0$, $\mathfrak{m}$ is a $k(\mathfrak{m})$-module via π, and A acts on $k(\mathfrak{m})$ through π. Given $\psi \in \mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}, k(\mathfrak{m}))$ define

$$D_\psi(\lambda + x) := \psi(x) \quad \lambda \in \hat{k}, x \in \mathfrak{m}$$

It is k-linear and satisfies the product rule for

$$\begin{aligned} D_\psi((\lambda + x)(\lambda' + x')) &= \psi(\lambda' x + \lambda x') \\ &= \pi(\lambda')\psi(x) + \pi(\lambda)\psi(x') \\ &= \pi(\lambda' + x')\psi(x) + \pi(\lambda + x)\psi(x') \\ &= (\lambda' + x') \cdot D_\psi(\lambda + x) + (\lambda + x) \cdot D_\psi(\lambda' + x') \end{aligned}$$

Moreover $\theta_{D_\psi}(\overline{x}) = D_\psi(x) = \psi(x)$, that is $\theta(D_\psi) = \psi$. Therefore the given map is surjective. ∎

Remark 3.32.51
The proof is substantially simpler when $k(\mathfrak{m}) = k$, for example when k is algebraically closed.

The argument follows the suggestion of [Har13] (Ex 8.1), but using the simpler result [Mat70] (Prop 28.I) to argue more directly.

3.32.7 Linearly Disjoint Algebras

Definition 3.32.52
*Let A/k, B/k be k-algebras and homomorphisms $i : A/k \to \Omega/k$, $j : B/k \to \Omega/k$ (typically being inclusion). Then we say that A and B are **linearly disjoint in Ω** if the canonical map*

$$\begin{aligned} A \otimes_k B &\to \Omega \\ a \otimes b &\to i(a)j(b) \end{aligned}$$

is injective. As the canonical maps $A, B \to A \otimes_k B$ are injective by (3.4.41) this implies i, j must be injective. In the case that Ω is integral (resp. reduced) then A, B are also necessarily integral (resp. reduced).

Recall that every k-module is free (3.3.126), moreover by (3.4.23) a basis of $A \otimes_k B$ may be formed by tensor product of bases for A and B.

Example 3.32.53
Let K/k be a non-trivial field extension then K is not linearly disjoint with itself. For $\{1, x\}$ a linearly independent subset of K which may be extended to a basis. Given the remark on the basis of the tensor product we see $1 \otimes x \neq x \otimes 1$. On the other hand the image of $1 \otimes x - x \otimes 1$ in Ω is clearly 0. The same consideration shows that $A \cap B = k$.

Remark 3.32.54
The notion of linear disjointness in general depends on the embeddings in Ω/k. For example $k(t)$ and $k(s)$ are linearly disjoint in $k(s,t)$, but not when both are identified with $k(x)$.

Proposition 3.32.55
Let $i : A/k \to \Omega/k$ and $j : B/k \to \Omega/k$ be algebra homomorphisms. Then the following are equivalent

- ***a)** A/k and B/k are linearly disjoint in Ω/k*
- ***b)** If $S \subset A$ is linearly independent over k then $i(S)$ is linearly independent over B*
- ***b')** If $T \subset B$ is linearly independent over k then $j(T)$ is linearly independent over A*

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- ***c)*** *There is a k-basis $\{a_\lambda\}_{\lambda\in\Lambda}$ of A for which $\{i(a_\lambda)\}_{\lambda\in\Lambda}$ is linearly independent over B*
- ***c')*** *There is a k-basis $\{b_\lambda\}_{\lambda\in\Lambda}$ of B for which $\{j(b_\lambda)\}_{\lambda\in\Lambda}$ is linearly independent over A*

Proof. $a) \implies b)$ By (3.4.30) and (3.4.31) there is an injective map

$$B^{(S)} \hookrightarrow A \otimes_k B$$
$$(b_s)_{s\in S} \to \sum_{s\in S} s \otimes b_s$$

By hypothesis the composite with $A \otimes_k B \hookrightarrow \Omega$ is injective which is precisely the required conclusion.

$b) \implies c)$ By (3.3.126) there exists a basis which by hypothesis is linearly independent over B

$c) \implies a)$ By (3.4.23) there is an isomorphism

$$B^{(\Lambda)} \cong A \otimes_k B$$

and the canonical map into Ω is identified with $(b_\lambda)_{\lambda\in\Lambda} \to \sum i(a_\lambda)j(b_\lambda)$. The hypothesis means precisely that this map is injective.

We immediately have the symmetric result $a) \iff b') \iff c')$. ∎

In cases where both algebras are fields and one is algebraic then linear disjointness can be characterized by a property of the tensor product. In particular it does not depend on the **ambient** algebra Ω/k.

Proposition 3.32.56
*Let K/k be a field extension and k'/k an **algebraic** field extension. Then the following are equivalent*

a) K and k' are linearly disjoint with respect to every embedding into an algebra Ω/k
b) K and k' are linearly disjoint with respect to some common field extension Ω/k
c) $K \otimes_k k'$ is an integral domain
d) $K \otimes_k k'$ is a field

Proof. $a) \implies b)$ We may take $\Omega := (K \otimes_k k')/\mathfrak{m}$.

$b) \implies c)$ As Ω is assumed to be a field it is certainly integral, and therefore so is $K \otimes_k k'$ being a subring.

$c) \implies d)$ As every element of the tensor product is a finite linear combination of elementary tensors we may reduce to the case k' is finitely generated. Then by (3.17.44) k' is finite over k and therefore by (3.4.23) $K \otimes_k k'$ is finite over K. By (3.17.47) it is a field.

$d) \implies a)$ This is immediate because the kernel is an ideal which must be (0). ∎

3.32.8 Finiteness of Integral Closure

Remark 3.32.57
In what follows we are interested in the integral closure of affine domains in finite extensions of the quotient field. Similar results are obtained for Dedekind Domains (3.30.11) with the restriction that the field extension is separable. Although this holds over number fields where the characteristic is zero, this will not in general be the case for varieties over finite fields, as for example $\mathbb{F}_p(t)$ admits the inseparable extension $\mathbb{F}_p(t^{1/p})$. However the finiteness still holds, because it's easy to see that the integral closure of $\mathbb{F}_p[t]$ is simply $\mathbb{F}_p[t^{1/p}]$. This generalises because every purely inseparable extension is contained in an extension obtained by adjoining p-th roots of unity.

Proposition 3.32.58 (Normalisation over Finite Fields)
Let k be a (possibly) finite field and $A = k[x_1, ..., x_n]$ be a finitely-generated k-algebra. Then there exists a normalising family $z_1, ..., z_d \in A$ such that $k[x_1, ..., x_n]$ is integral over $k[z_1, ..., z_d]$.

Proof. TODO ∎

Proposition 3.32.59 (Affine Domain are Japanese Rings)
Let A be an affine domain over k with $K = \mathrm{Frac}(A)$ and L/K a finite extension. Then $\overline{A}^L$ is a finite A-module and Noetherian with quotient field L.

Furthermore $\dim \overline{A}^L = \dim A$.

Proof. Suppose $A = k[x_1, ...x_n]$, apply the normalisation lemma (3.32.58) to obtain a lattice of subrings

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where $z_1, ..., z_d$ are algebraically independent over k. We claim it is enough to show that the $\overline{A'}^L$ is a finite A'-module. For A is finite and integral over A'. Therefore by (3.22.30) $\overline{A}^L = \overline{A'}^L$ is finite over A', and a-fortiori over A.

Therefore without loss of generality we may assume that $x_1, ..., x_n$ are algebraically independent over k and A is integrally closed (3.22.12).

By (3.17.75) we may find a finite extension L'/L such that L'/K is normal. So by (3.22.31) we may assume without loss of generality that L/K is normal.

By (...) we may decompose this into a tower $M/L/K$ such that M/L is Galois and L/K is purely inseparable.

We first show that the integral closure of A in L is a finite A-module. When $L = K$ this is trivial because A is already integrally closed in its quotient field K. Therefore we may assume $L \neq K$ and the characteristic p is non-zero (3.17.106). $L = K(y_1, ..., y_n)$ is finitely generated and by definition $y_i^q \in K$ for some p-th power q. Therefore L is contained in $L' := k\left(x_1^{1/q}, ..., x_n^{1/q}\right)$ which is again a finite purely inseparable extension of K. The elements $x_1^{1/q}, ..., x_n^{1/q}$ are again algebraically independent so that the ring $A' := k\left[x_1^{1/q}, ..., x_n^{1/q}\right]$ is integrally closed (3.22.12) with quotient field L'. Evidently A' is finite and integral over A (3.22.4) therefore (3.22.30) $\overline{A}^{L'} = \overline{A'}^{L'} = A'$ is finite over A. By (3.22.31) then $\overline{A}^L$ is finite over A and $\overline{A}^L$ has quotient field L (3.22.26). As A is Noetherian then so is $\overline{A}^L$.

By transitivity we may then reduce to the case L/K is separable which was proven in (3.22.28). ∎

Corollary 3.32.60
Let A be an affine domain of Krull dimension 1, with quotient field K and L/K a finite extension. Then the integral closure $\overline{A}^L$ is a finite A-module and a Dedekind domain.

Further for every prime ideal $\mathfrak{p}$ of A we have the ramification formula

$$\sum_{\mathfrak{B}|\mathfrak{p}} e_{\mathfrak{B}/\mathfrak{p}} f_{\mathfrak{B}/\mathfrak{p}} = [L : K]$$

Proof. By (3.32.59) $B := \overline{A}^L$ is a finite A-module with quotient field L and Noetherian. By (...) it is integrally closed and by (3.22.24) has Krull Dimension 1. Therefore by definition B is also a Dedekind domain.

The ramification formula follows from (3.30.20). ∎

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3.33 Affine Algebras under Base Change

This section covers what happens to a k-algebra A under base change, that is $A_{(L)}$ for L/k a field extension. Principally given an integral k-algebra A we would like to ensure that $A_{(L)}$ is reduced (which is related to separability), irreducible and therefore integral. We follow [Stacks] but large parts of exposition are from [Bou89].

Definition 3.33.1 (Geometrically Reduced / Integral Algebra)
A k-algebra A is

- ***geometrically integral*** *(resp.* ***irreducible***, ***reduced****) if the ring $A \otimes_k L$ is* ***integral*** *(resp.* ***irreducible***, ***reduced****) for every field extension L/k.*
- ***algebraically integral*** *(resp.* ***irreducible***, ***reduced****) if the ring $A \otimes_k \overline{k}$ is* ***integral*** *(resp.* ***irreducible***, ***reduced****).*

A few remarks about the important results and notation in the literature

- *As in (3.3.66) we see that* ***integral*** $\iff$ ***irreducible*** *and* ***reduced****.*
- *These are shown to be equivalent (3.33.27), (3.33.37) and (3.33.45). In fact it is sufficient to check the properties for all finite extensions.*
- *[Liu06] uses the "algebraic" form*
- *[Bou89] uses the term "separable" for "geometrically reduced", as this coincides with the notion of separable for algebraic extensions (3.33.20)*
- *Similarly [Bou89] and [Lan11] use the term "regular" for geometrically integral*

We make implicit use of the following results throughout

Proposition 3.33.2
Let A, B be k-algebras, then the structural morphisms $A \to A \otimes_k B$ and $B \to A \otimes_k B$ are injective. Furthermore if A' and B' are k-subalgebras then the canonical homomorphism

$$A' \otimes_k B' \to A \otimes_k B$$

is injective.

Proof. This is (3.4.30) and (3.4.41) ∎

Proposition 3.33.3 (A Subspace is Recovered by Intersection After Base Change)
Let $V \subseteq W$ be k-vector spaces and k'/k a field extension. Then, identifying W with its image in $W \otimes_k k'$ via $w \mapsto w \otimes 1$ (injective, (3.4.31)),

$$(V \otimes_k k') \cap W = V$$

inside $W \otimes_k k'$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

"$\supseteq$" is clear. For "$\subseteq$": choose a k-subspace complement V' of V in W (3.3.139) (k being a field), so $W = V \oplus V'$; tensoring, $W \otimes_k k' \cong (V \otimes_k k') \oplus (V' \otimes_k k')$ (3.4.21). Given $w \in W$ with $w \otimes 1 \in V \otimes_k k'$: writing $w = v + v'$ ($v \in V$, $v' \in V'$) for its decomposition, $w \otimes 1 = (v \otimes 1) + (v' \otimes 1)$ under this direct sum splitting; as $w \otimes 1$ lies in the first summand, the second component vanishes, that is $v' \otimes 1 = 0$; as $v' \mapsto v' \otimes 1$ is injective (3.4.31), $v' = 0$, giving $w = v \in V$. ∎

Proposition 3.33.4
Let A, B be k-algebras. Let $\{a_i\}_{i \in I}$ be a k-linearly independent subset (resp. basis) of A and $\{b_j\}_{j \in I}$ be a k-linearly independent subset (resp. basis) of B. Then $\{a_i \otimes b_j\}_{(i,j) \in I \times J}$ is a k-linearly independent subset (resp. basis) of $A \otimes_k B$.

Furthermore $\{a_i \otimes 1\}_{i \in I}$ is a B-linearly independent subset (resp. basis) of $A \otimes_k B$.

Proof. The first statement is (3.4.31). For the last statement extend $\{a_i\}_{i \in I}$ to a basis (2.3.7) and apply (3.4.26). ∎

Proposition 3.33.5
Let A, B be k-algebras. Then $A \otimes_k B$ is the union of subrings of the form a) $A \otimes_k B'$, b) $A' \otimes_k B$ or c) $A' \otimes_k B'$ where A' and B' are finitely-generated k-subalgebras of A and B respectively.

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If K/k is an algebraic field extension then $A \otimes_k K$ is the union of subrings of the form $A \otimes_k k'$ where k'/k are finite subextensions.

Proof. An arbitrary element $z \in A \otimes_k B$ is a finite sum of elementary tensors. Therefore the result follows from (3.33.2).

The final statement follows from recalling finitely generated algebraic extensions are finite (3.17.44). ∎

Proof. ∎

3.33.1 Etale Algebras

Definition 3.33.6
*Let A be a finite-dimensional k-algebra. Then define the **separable degree** as follows*

$$[A:k]_s := \#\operatorname{AlgHom}_k\left(A, \overline{k}\right)$$

*We say that A is **etale** if this equals the dimension of A as a k-vector space*

$$[A:k]_s = [A:k]$$

*Recall a finite field extension K/k is **etale** if and only if it is separable (3.17.80).*

Lemma 3.33.7
Let A be a finite-dimensional k-algebra and K/k a field extension. Then

$$[\operatorname{Hom}_k(A,K):K] = [A^\vee : k] = [A:k]$$

More precisely if $\{a_1, ..., a_n\}$ is a k-basis for A then $\{a_1^\vee, ..., a_n^\vee\}$ is a K-basis for $\operatorname{Hom}_k(A,K)$.

Proof. This is simply a special case of (3.4.27). ∎

Corollary 3.33.8 (Separable degree is finite)
Let A be a finite-dimensional k-algebra then $\#\operatorname{AlgHom}_k(A,K) \leq [A:k]$ for every field extension K/k.

Proposition 3.33.9 (Image of A is finite-dimensional)
Let A be a finite-dimensional k-algebra and let $N_A \subset \overline{k}$ be the compositum of the images $\varphi(A)$ of all k-algebra homomorphisms $\varphi : A \to \overline{k}$. Then N_A/k is a finite normal extension, and for every field extension Ω/k containing a copy of N_A, that is with a k-embedding $\iota : N_A \to \Omega$, composition with ι is a bijection

$$\operatorname{AlgHom}_k(A, N_A) \to \operatorname{AlgHom}_k(A, \Omega)$$

In particular if Ω contains $\overline{k}$ then these are both equal to $\operatorname{AlgHom}_k\left(A, \overline{k}\right)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $\varphi : A \to \overline{k}$ be a k-algebra homomorphism. Then $\varphi(A)$ is a quotient of A and so finite-dimensional over k, and it is a domain, being a subring of a field. A finite-dimensional domain is a field (3.17.47). Therefore $\varphi(A) \subset \overline{k}$ is a finite extension of k. There are only finitely many such φ (3.33.8), say $\varphi_1, ..., \varphi_r$, so $N_A = k(\varphi_1(A), ..., \varphi_r(A))$ is finite over k (3.17.48).

To see N_A/k is normal, let $\sigma \in \operatorname{Mor}_k\left(N_A, \overline{k}\right)$. For each j the composite $\sigma \circ \varphi_j : A \to \overline{k}$ is again a k-algebra homomorphism so $\sigma\left(\varphi_j(A)\right) \subset N_A$, whence $\sigma(N_A) \subset N_A$. As σ is injective and k-linear and N_A is finite-dimensional, $\sigma(N_A) = N_A$. So N_A/k is normal by (3.17.69).

Injectivity of the given map is clear as ι is injective. For surjectivity let $\psi : A \to \Omega$ be a k-algebra homomorphism. As above $L := \psi(A)$ is a finite extension of k, so by (3.17.63) there is a k-embedding $\sigma : L \to \overline{k}$. Then $\sigma \circ \psi : A \to \overline{k}$ has image in N_A by definition. Let $b \in L$ with minimal polynomial $m_{b,k}$. Then $m_{b,k}$ has the root $\sigma(b)$ in N_A and hence splits completely in N_A as N_A/k is normal, say $m_{b,k} = \prod_i (X - c_i)$ with $c_i \in N_A$. Applying ι gives $m_{b,k} = \prod_i (X - \iota(c_i))$ in $\Omega[X]$ and since Ω is a domain the root b is one of the $\iota(c_i)$. Thus $\psi(A) \subseteq \iota(N_A)$ and $\psi = \iota \circ \psi'$ for some $\psi' : A \to N_A$.

Finally $N_A \subset \overline{k}$, so applying this to $\Omega = \overline{k}$ with the inclusion, and then to any $\Omega \supseteq \overline{k}$, shows $\operatorname{AlgHom}_k(A, N_A) = \operatorname{AlgHom}_k\left(A, \overline{k}\right) = \operatorname{AlgHom}_k(A, \Omega)$. ∎

Proposition 3.33.10 (Etale-ness is preserved under base extension)
Let A be a finite-dimensional k-algebra and K/k a field extension then

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$$[A \otimes_k K : K] = [A : k]$$
$$[A \otimes_k K : K]_s = [A : k]_s$$

In particular A is etale if and only if $A \otimes_k K$ is.

Proof. The first equality follows from (3.4.26) and (3.3.109).

For the second equality we see that by (3.33.9) and (3.4.38)

$$\mathrm{AlgHom}_k\big(A, \overline{k}\big) = \mathrm{AlgHom}_k\big(A, \overline{K}\big) \cong \mathrm{AlgHom}_K\big(A \otimes_k K, \overline{K}\big)$$

■

Lemma 3.33.11
Let $A = A_1 \times \cdots \times A_d$ be a finite product of finite-dimensional k-algebras. Then A is etale if and only if each A_i is.

Proof. This follows because

$$\mathrm{AlgHom}_k\big(A_1 \times \cdots \times A_d, \overline{k}\big) \cong \mathrm{AlgHom}_k\big(A_1, \overline{k}\big) \times \cdots \times \mathrm{AlgHom}_k\big(A_d, \overline{k}\big)$$

and a similar consideration for the dimension over k. ■

Lemma 3.33.12
Let A be a finite-dimensional k-algebra. The following are equivalent

a) A is reduced
b) $A \cong L_1 \times \cdots \times L_n$ for some finite extensions L_i/k

Further A is etale if and only if each L_i/k is separable. If k is perfect then A is automatically etale.

Proof. One direction is obvious. Suppose conversely that A is reduced. If $\dim_k A = 1$ then any proper ideal must have strictly lower dimension and therefore A is a field. It is therefore sufficient to show (by induction on dimension) that if A is not a field then $A \cong A_1 \times A_2$ for two (non-zero) k-algebras A_1, A_2.

Suppose A is not a field and let $\mathfrak{a}$ be a non-zero proper ideal of A which has minimal dimension over k as a vector space. As A is reduced, and by minimality, we have $\mathfrak{a}^2 = \mathfrak{a}$. By Nakayama's Lemma (3.19.5) there is $e \in \mathfrak{a}$ such that $ex = x$ for all $x \in \mathfrak{a}$ and in particular $e^2 = e$ and $\mathfrak{a} = Ae$. By assumption $e \neq 0, 1$. Define $f := 1 - e \neq 0, 1$ then evidently $f^2 = f$ and $ef = 0$. The homomorphism of A-modules

$$A \to Ae \times Af$$
$$a \to (ae, af)$$

is injective because $ae + af = a$ and surjective because the image of $a_1 e + a_2 f$ is $(a_1 e, a_2 f)$. The sub-modules Ae and Af are actually sub-rings by the idempotence property and the given map is an isomorphism of rings.

The last statement follows from (3.33.11) and (3.17.106). ■

Lemma 3.33.13 (Primes of a Finite Product of Fields)
Let $A \cong L_1 \times \cdots \times L_n$ be a finite product of fields (in particular, any reduced finite-dimensional k-algebra, (3.33.12)). Then every prime ideal of A is maximal, and there are exactly n of them.

Proof. For each j, $\mathfrak{m}_j := \ker(\pi_j)$ (the kernel of the j-th projection $\pi_j : A \twoheadrightarrow L_j$) is maximal, being the kernel of a surjection onto a field (3.3.59), and the $\mathfrak{m}_j$ are pairwise distinct (the element with 1 in coordinate j and 0 elsewhere lies in every $\mathfrak{m}_i$, $i \neq j$, but not $\mathfrak{m}_j$), giving n maximal ideals.

Let $\mathfrak{p}$ be any prime. As $\bigcap_j \mathfrak{m}_j = \ker(\pi_1, ..., \pi_n) = 0 \subseteq \mathfrak{p}$, (3.3.40) gives $\mathfrak{m}_j \subseteq \mathfrak{p}$ for some j; as $\mathfrak{m}_j$ is already maximal, $\mathfrak{p} = \mathfrak{m}_j$. ■

Lemma 3.33.14 (Algebra Homomorphisms out of a Product of Fields)
Let Ω be a field. The Ω-algebra homomorphisms $\chi : \Omega^n \to \Omega$ are exactly the n projections π_i, and these are pairwise distinct. In particular $\#\,\mathrm{AlgHom}_\Omega(\Omega^n, \Omega) = n$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Indeed let e_i be the i-th standard idempotent, so $e_i^2 = e_i$, $e_i e_j = 0$ for $i \neq j$ and $\sum_i e_i = 1$. Then each $\chi(e_i)$ is an idempotent of the field Ω so equals 0 or 1; as $\chi(e_i)\chi(e_j) = 0$ for $i \neq j$ at most one of them is 1, and as $\sum_i \chi(e_i) = 1$ exactly one, say $\chi(e_j)$, is. Then by Ω-linearity $\chi(x) = \sum_i x_i \chi(e_i) = x_j$, that is $\chi = \pi_j$. The π_i are pairwise distinct as $\pi_i(e_i) = 1 \neq 0 = \pi_j(e_i)$ for $j \neq i$. ■

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Lemma 3.33.15
Let A be a finite-dimensional k-algebra and Ω/k a field extension containing a copy of the extension N_A/k of (3.33.9) (for instance any Ω containing $\overline{k}$). Then the following are equivalent

a) A is etale
b) the map $\Theta : A \otimes_k \Omega \to \Omega^{\mathrm{AlgHom}_k(A,\Omega)}, \quad a \otimes c \mapsto (c\,\varphi(a))_\varphi$ is an isomorphism of Ω-algebras
c) $A \otimes_k \Omega \cong \Omega^n$ as an Ω-algebra for some n

In this case the n of c) equals $[A : k]$, and $A \otimes_k \Omega$ has exactly n maximal ideals, namely the kernels of the coordinates of Θ.

Proof. *a)* $\Longrightarrow$ *b)* By (3.33.9) $\#\,\mathrm{AlgHom}_k(A,\Omega) = \#\,\mathrm{AlgHom}_k\left(A,\overline{k}\right) =: [A : k]_s = [A : k]$ with the last equality because A is assumed etale. Then defining $B := A \otimes_k \Omega$

$$\#\,\mathrm{AlgHom}_\Omega(B,\Omega) \overset{(3.4.38)}{=} \#\,\mathrm{AlgHom}_k(A,\Omega) = [A : k] \overset{(3.4.16)}{=} [B : \Omega] = [B^\vee : \Omega]$$

By (3.17.133) and (3.3.127) we see that $\mathrm{AlgHom}_\Omega(B,\Omega)$ is a Ω-basis for $B^\vee$. By (3.4.38) the map sending $\varphi \in \mathrm{AlgHom}_k(A,\Omega)$ to $\tilde{\varphi} : B \to \Omega$, $a \otimes c \mapsto c\,\varphi(a)$, is a bijection $\mathrm{AlgHom}_k(A,\Omega) \to \mathrm{AlgHom}_\Omega(B,\Omega)$. Write $\mathrm{AlgHom}_k(A,\Omega) = \{\varphi_1, ..., \varphi_n\}$, so that $\tilde{\varphi}_1, ..., \tilde{\varphi}_n$ is the basis of $B^\vee$. Then by (3.3.109) this corresponds to a basis $e_1, ..., e_n$ of B and the algebra isomorphism is exhibited explicitly by

$$\begin{aligned} \Theta : B &\cong \Omega^n \\ a \otimes c &\to (c\,\varphi_1(a), ..., c\,\varphi_n(a)) \end{aligned}$$

Indeed $\Theta(b) = (\tilde{\varphi}_1(b), ..., \tilde{\varphi}_n(b))$ is an Ω-algebra homomorphism, and it is injective since a b killed by every element of a basis of $B^\vee$ is killed by every functional, so $b = 0$. It is therefore an isomorphism by dimensions.

b) $\Longrightarrow$ *c)* Immediate, with $n := \#\,\mathrm{AlgHom}_k(A,\Omega)$, which is finite (3.33.8).

c) $\Longrightarrow$ *a)* By (3.33.14) the Ω-algebra homomorphisms $\Omega^n \to \Omega$ are the n projections π_i, so $\#\,\mathrm{AlgHom}_\Omega(B,\Omega) = n = [B : \Omega]$. As before then $[A : k]_s = \#\,\mathrm{AlgHom}_k\left(A,\overline{k}\right) = \#\,\mathrm{AlgHom}_k(A,\Omega) = \#\,\mathrm{AlgHom}_\Omega(B,\Omega) = [B : \Omega] = [A : k]$ which shows that A is etale.

Finally Ω^n is a finite product of fields, so by (3.33.13) its primes are the n coordinate kernels, and these correspond to the maximal ideals of $A \otimes_k \Omega$ under Θ. ∎

Corollary 3.33.16 (A Field Splits Completely Over Any Normal Extension it Embeds Into)
Let K/k be a finite separable field extension and L/k a field extension containing (a copy of) a normal closure of K/k. Fix a primitive element θ for K/k (3.17.108), with minimal polynomial $m \in k[X]$ of degree $n := [K : k]$, and write $m = \prod_{i=1}^n (X - \beta_i)$ for its (pairwise distinct) roots $\beta_i \in L$. Then there is a canonical isomorphism of L-algebras

$$K \otimes_k L \cong L^n, \quad p(\theta) \otimes c \mapsto (c\,p(\beta_1), ..., c\,p(\beta_n)) \quad (p \in k[X], c \in L)$$

In particular $K \otimes_k L$ has exactly n maximal ideals, $\mathfrak{m}_i$ the kernel of the i-th coordinate.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As K/k is separable it is etale (3.33.6). By (3.17.85).c) applied with $\Omega = \overline{k}$, N_K is a normal closure of K/k, so L contains a copy of N_K (3.17.75) and the isomorphism Θ of (3.33.15) is defined. By (3.17.85).b), $\varphi \mapsto \varphi(\theta)$ is a bijection $\mathrm{Mor}_k(K,L) \to \{\beta_1, ..., \beta_n\}$, and under this indexing Θ sends $p(\theta) \otimes c$ to $(c\,p(\beta_i))_i$. The statement about maximal ideals is the last statement of (3.33.15). ∎

Proposition 3.33.17 (Etale Criteria)
Let A be a finite-dimensional k-algebra. The following are equivalent

a) A is etale
b) A is geometrically reduced
c) $A \otimes_k \overline{k}$ is reduced
d) $A \otimes_k K$ is reduced for some perfect field extension K/k
e) $A \cong L_1 \times \cdots \times L_n$ for L_i/k finite separable extensions

In particular A is reduced.

Proof. *a)* $\Longrightarrow$ *b)* Consider a field extension L/k and the tower of extensions $\overline{L}/L/k$. Then by (3.33.15) $A \otimes_k \overline{L} \cong \overline{L}^n$ is evidently reduced. By (3.33.2) $A \otimes_k L \subset A \otimes_k \overline{L}$ is a subring and therefore also reduced.

b) $\Longrightarrow$ *c)* immediate.

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$c) \implies d)$ immediate as $\overline{k}$ is perfect.

$d) \implies a)$ By (3.33.12) $A \otimes_k K$ is an etale K-algebra so by (3.33.10) A is etale.

$e) \implies a)$ Follows immediately from (3.33.11).

$a) \implies e)$ We know that A is reduced so the result follows from (3.33.12). ∎

Corollary 3.33.18
Let A be a finite-dimensional k-algebra and k perfect. Then the following are equivalent

a) A is etale
b) A is reduced
c) A is geometrically reduced
d) $A \otimes_k \overline{k}$ is reduced.

For completeness we summarize the relationship between the different notions of separability

Corollary 3.33.19 (Equivalent definitions of separability)
Let K/k be a finite extension. Then the following are equivalent

a) K/k is separable algebraic (3.17.53)
b) K/k is etale
c) K/k is geometrically reduced
d) $K \otimes_k \overline{k}$ is reduced

When k is perfect every field extension is separable.

Proof. We have already shown $b) \iff c) \iff d)$. Furthermore $a) \iff b)$ is (3.17.80). ∎

Corollary 3.33.20 (Equivalent definitions of separability)
Let K/k be an algebraic extension. Then the following are equivalent

a) K/k is separable algebraic (3.17.53)
b) Every finite subextension of K/k is etale
c) K/k is geometrically reduced
d) $K \otimes_k \overline{k}$ is reduced

In particular an algebraic extension of a perfect field k is geometrically reduced.

Proof. This follows directly from (3.33.19) because each property has "finite character", i.e. holds if and only if it holds for every finite subextension.

The last statement follows from (3.17.106). ∎

3.33.2 Geometrically Reduced Algebras

Reference is [Bou89] (Section V.15.4). The main results of this section are (3.33.27) and (3.33.29). Observe that if k is perfect every extension K/k satisfies Maclane's Criterion and is therefore Geometrically Reduced.

Lemma 3.33.21
Let A be a reduced k-algebra. Define the canonical map

$$i : A \hookrightarrow \prod_{\mathfrak{p}} k(\mathfrak{p})$$

where $\mathfrak{p}$ runs over all prime ideals of A. Then i is injective. Furthermore for any k-algebra B we have a canonical embedding

$$A \otimes_k B \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k B)$$

Proof. Evidently $\ker(i) = \bigcap \mathfrak{p}$ which by (3.3.46) is (0). The final statement follows from considering the maps

$$A \otimes_k B \overset{i \otimes 1_B}{\hookrightarrow} \left(\prod_{\mathfrak{p}} k(\mathfrak{p}) \right) \otimes_k B \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k B)$$

The first map is injective by (3.4.25) and the second by (...). ∎

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Proposition 3.33.22
Let A be a reduced k-algebra and B a geometrically reduced k-algebra, then $A \otimes_k B$ is reduced.

Proof. By the previous Lemma we have an embedding

$$A \hookrightarrow \prod (k(\mathfrak{p}) \otimes_k B)$$

As B is geometrically reduced this shows that $A \otimes_k B$ is reduced. ∎

Corollary 3.33.23
Let k be a perfect field and A, B reduced k-algebras. Then $A \otimes_k B$ is reduced.

Lemma 3.33.24
Let A be a geometrically reduced k-algebra and K the total ring of fractions. Then K is geometrically reduced.

Conversely if A is integral and K is geometrically reduced, then A is geometrically reduced.

Proof. Let L/k be a field extension and consider a nilpotent element $x \in K \otimes L$. Then by definition

$$x = \sum_{i=1}^{n} (a_i s_i^{-1}) \otimes \lambda_i \quad a_i \in A, s_i \notin Z(A), \lambda_i \in L$$

Let $s = s_1 ... s_n$ then $x \cdot (s \otimes 1) \in A \otimes L \subset K \otimes L$. Then $x \cdot (s \otimes 1)$ is nilpotent and so by definition zero. As $(s \otimes 1)$ is invertible we see $x = 0$ as required.

For the converse $A \subset K$ and therefore $A \otimes_k L \subset K \otimes_k L$. ∎

Lemma 3.33.25
Let K/k be a separably generated field extension. Then K/k is geometrically reduced.

Proof. Consider first the case $K = k(\mathcal{B})$ is purely transcendental. Then for L/k we have $k[\mathcal{B}] \otimes L \cong L[\mathcal{B}]$ is evidently reduced. We may then demonstrate directly that $k(\mathcal{B}) \otimes_k L$ is reduced.

For the general case

$$K \otimes_k L \cong K \otimes_{k(\mathcal{B})} (k(\mathcal{B}) \otimes_k L)$$

We have shown that $k(\mathcal{B}) \otimes_k L$ is reduced and by (3.33.20) $K/k(\mathcal{B})$ is geometrically reduced as by assumption it is separable. By (3.33.22) $K \otimes_k L$ is reduced and K is geometrically reduced as required. ∎

Lemma 3.33.26
Let A be a reduced k-algebra for which k is perfect. Then A is geometrically reduced.

Proof. We prove the result in increasing generality

a) We first consider the case $A = K/k$ is a finitely generated field extension. By (3.17.151) K/k is separably generated and so by (3.33.25) it is geometrically reduced.
b) When A is a finitely-generated k-algebra by (3.33.21) there is a canonical embedding

$$A \otimes_k L \hookrightarrow \prod_{\mathfrak{p}} (k(\mathfrak{p}) \otimes_k L)$$

By assumption $k(\mathfrak{p})$ is finitely generated and so by the first part is geometrically reduced. Therefore $A \otimes_k L$ is reduced as required.
c) The general case follows from b) and the reduction to the finitely generated case (3.33.5).

∎

Proposition 3.33.27 (Maclane's Criterion for Algebras)
Let A be a k-algebra. Then the following are equivalent

a) A is geometrically reduced
b) $A \otimes_k \overline{k}$ is reduced
c) $A \otimes_k k^{p^{-\infty}}$ is reduced
d) $A \otimes_k K$ is reduced for some perfect extension K/k
e) $A \otimes_k k^{p^{-1}}$ is reduced
f) $A \otimes_k k'$ is reduced for every finite subextension k'/k of $k^{p^{-1}}/k$
g) The following k-module homomorphism

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$$\begin{aligned} m : A \otimes_k k^{p^{-1}} &\to A \\ a \otimes \lambda &\to \lambda^p a^p \end{aligned}$$

is injective (that is $k^{p^{-1}}$ and A are linearly disjoint with respect to the p-th power embedding into A)

h) A satisfies Maclane's Criterion (3.17.147)

In particular when k is perfect then this is equivalent to A being reduced.

Proof. The case $p = 1$ is simply (3.33.26) because $k^p = k = k^{p^{-1}} = k^{p^{-\infty}}$ is perfect and g) is equivalent to A being reduced. Therefore we assume $p > 1$.

a) $\implies$ *b)* By definition.

b) $\implies$ *c)* By (3.17.63) there is an embedding $k^{p^{-\infty}} \hookrightarrow \overline{k}$ whence an embedding $A \otimes_k k^{p^{-1}} \hookrightarrow A \otimes_k \overline{k}$.

c) $\implies$ *d)* By (3.17.103) $k^{p^{-\infty}}$ is perfect.

d) $\implies$ *e)* By (3.17.104) there is an embedding $k^{p^{-1}} \hookrightarrow K$ whence an embedding $A \otimes_k k^{p^{-1}} \hookrightarrow A \otimes_k K$.

e) $\implies$ *f)* This is immediate as we have an embedding $A \otimes_k k' \hookrightarrow A \otimes_k k^{p^{-1}}$.

f) $\iff$ *g)* Suppose $x := \sum_i \lambda_i \otimes a_i \in A \otimes_k k^{p^{-1}}$ is a finite sum then we may consider the finite subextension $k' := k\left(\{a_i\}_i\right)$. Observe that

$$x^p = \sum_i \lambda_i^p \otimes a_i^p = \sum_i 1 \otimes \lambda_i^p a_i^p = 1 \otimes \left(\sum_i \lambda_i^p a_i^p \right) =: 1 \otimes m(x)$$

Therefore we see $A \otimes_k k'$ is reduced for every such k' iff $m(x)$ is injective.

g) $\iff$ *h)* Consider the Frobenius morphisms $i : k^{p^{-1}} \to A$ and $j : A \to A$. Then g) is equivalent to $k^{p^{-1}}$ and A being linearly disjoint in A with respect to i, j and the result follows from the characterization of linear disjointness (3.32.55).

h) $\implies$ *a)* Let $\{a_i\}_{i \in I}$ be a k-basis for A then by (3.4.26) $\{a_i \otimes 1\}_{i \in I}$ is an L-basis for $A \otimes_k L$. Given $x \in A \otimes_k L$ we have

$$x = \sum_{i \in I} a_i \otimes \lambda_i \quad \lambda_i \in L$$

and

$$x^p = \sum_{i \in I} \lambda_i^p \otimes a_i^p = \sum_{i \in I} \lambda_i^p (a_i^p \otimes 1)$$

By assumption $\{a_i^p\}_{i \in I}$ is k-linearly independent and therefore $\{a_i^p \otimes 1\}_{i \in I}$ is L-linearly independent. Therefore $x^p = 0 \implies \lambda_i^p = 0$ for all $i \in I \implies \lambda_i = 0$ for all $i \in I \implies x = 0$, as L is reduced. It follows from (3.17.148) that $A \otimes_k L$ is reduced, as we assumed that $p > 1$. ∎

Corollary 3.33.28 (Maclane's Criterion for Fields)
Let K/k be a field extension with characteristic exponent p. Then the following are equivalent

a) K/k is geometrically reduced
b) $K \otimes_k \overline{k}$ is reduced
c) $K \otimes_k k^{p^{-1}}$ is reduced
d) $K \otimes_k k^{p^{-\infty}}$ is reduced
e) $K \otimes_k k'$ is reduced for some perfect extension k'/k
f) K and $k^{p^{-1}}$ are linearly disjoint (with respect to some common field extension)
g) $K \otimes_k k^{p^{-1}}$ is an integral domain
h) $K \otimes_k k^{p^{-1}}$ is a field
i) K/k satisfies Maclane's Criterion (3.17.147)

Observe i) has finite character *in the sense it holds if and only if it holds for every finitely-generated subextension.*

In particular when k is perfect every extension is geometrically reduced.

Proof. By (3.33.27) a)-f), i) are equivalent (using (3.32.56) to generalise the linear disjoint condition to any ambient field) and by (3.32.56) again f)-h) are equivalent. ∎

Proposition 3.33.29 (Separable Field Extensions)
Let K/k be a field extension. Then the following are equivalent

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a) K/k is geometrically reduced
b) $K \otimes_k \overline{k}$ is reduced
c) K/k satisfies Maclane's Criterion (3.17.147)
d) Every finitely generated subextension K/k is separably generated

In particular if k is perfect every field extension satisfies these properties.

Proof. By (3.33.28) we have $a) \iff b) \iff c)$. Further $c) \implies d)$ is Lemma (3.17.151), since every subextension satisfies Maclane's Criterion. For $d) \implies a)$ observe that every finitely generated subextension is geometrically reduced by (3.33.25). We may use (3.33.2) to argue that K/k is geometrically reduced. ∎

Corollary 3.33.30
Let K/k be a finitely generated field extension. Then the following are equivalent.

a) K/k is geometrically reduced
b) $K \otimes_k \overline{k}$ is reduced
c) K/k satisfies Maclane's Criterion (3.17.147)
d) Every finitely generated subextension of K/k is separably generated
e) K/k is separably generated

Proof. We've already shown equivalence of a)-d) and clearly $d) \implies e)$. Finally $e) \implies a)$ is simply (3.33.25). ∎

3.33.3 Geometrically Irreducible Algebras

The reference for this section is [Stacks] (Tag 00I2) presented here with a more elementary proof of (3.33.35).

Lemma 3.33.31
Let A, B be k-algebras. If $a \otimes 1 \in A \otimes_k B$ is a zero-divisor then so is a.

Proof. Consider a basis $\{a_i\}_{i\in I}$ for A such that $a_{i_0} = a$. If a is not a zero-divisor then we may show directly that $\{aa_i\}_{i\in I}$ is linearly independent. Then by extending to a basis of A we may use (3.4.26) to show that $\{(aa_i) \otimes 1\}_{i\in I}$ is B-linearly independent. We see immediately that $a \otimes 1$ is not a zero-divisor as required. ∎

Lemma 3.33.32
Let A be a ring and $\mathfrak{a} \subset N(A)$ a locally nilpotent ideal. Then A is irreducible iff $A/\mathfrak{a}$ is irreducible.

Lemma 3.33.33
Let $A \subset B$ be a subring of an irreducible ring. Then A is irreducible.

In particular if B is a geometrically irreducible k-algebra, then so is any subalgebra.

Proof. By (3.21.7) every minimal prime in A is the contraction of a minimal prime. As there is precisely one minimal prime of B then we are done.

The final statement follows because the canonical map $A \otimes_k L \to B \otimes_k L$ is injective. ∎

When k is algebraically closed then nothing interesting happens.

Lemma 3.33.34
Let A, B be irreducible (resp. integral) k-algebras with k algebraically closed. Then $A \otimes_k B$ is irreducible (resp. integral).

In particular A is a geometrically irreducible (resp. integral) k-algebra.

Proof. Observe that $N(A) \otimes_k B + A \otimes_k N(B) \subset N(A \otimes_k B)$. For $(n \otimes \lambda)^m = n^m \otimes \lambda^m = 0$ and we may use (...) to argue any linear combination of such elements is nilpotent. By (...) we have an isomorphism

$$A/N(A) \otimes_k B/N(B) \cong (A \otimes_k B)/(N(A) \otimes_k B + A \otimes_k N(B))$$

Then by (3.33.32) $A/N(A) \otimes_k B/N(B)$ is irreducible iff $A \otimes_k B$ is. Therefore we may consider only the case A, B reduced and therefore integral, and show that $A \otimes_k B$ is integral and a-fortiori irreducible.

If $A \otimes_k B$ were not integral it would contain a non-trivial zero divisor, and this would also be true for $A' \otimes_k B'$ for some finitely generated subalgebras A' and B'. Therefore we may assume wlog that A and B are finitely generated.

To simplify the argument choose a k-basis $\{b_i\}_{i\in I}$ for B. By (...) $\{1 \otimes b_i\}_{i\in I}$ is an A-basis for $A \otimes_k B$. Therefore if we have a product of elements equal to 0 then may write it as follows

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$$\left(\sum_i a_i \otimes b_i\right)\left(\sum_i a_i' \otimes b_i\right) = 0$$

for some $a_i, a_i' \in A$ all but finitely many zero. Let $\mathfrak{m} \lhd A$ be a maximal ideal, then by the Weak Nullstellensatz (3.32.23) the structural morphism $k \to A/\mathfrak{m}$ is an isomorphism. Reduce the equation to $A/\mathfrak{m} \otimes_k B \cong B$ to find

$$\left(\sum_i \overline{a}_i b_i\right)\left(\sum_i \overline{a}_i' b_i\right) = 0$$

As B is an integral domain, and b_i are linearly independent we find $\overline{a}_i, \overline{a}_i' = 0$. As $\mathfrak{m}$ was arbitrary we find $a_i, a_i' \in \bigcap \mathfrak{m}$ for all i. By assumption (0) is prime and so by the Strong Nullstellensatz (3.34.3) we find $a_i = a_i' = 0$. This shows that $A \otimes_k B$ is integral as required.

The last statement follows because a field K/k is always irreducible. ∎

Proposition 3.33.35 (Extension of Scalars is Flat)
Let A be a k-algebra and $L/K/k$ a tower of field extensions. Then the canonical ring homomorphism

$$A \otimes_k K \to A \otimes_k L$$

satisfies Going Down. Further we have a well-defined surjective map of minimal primes

$$\mathrm{MinPrime}(A \otimes_k L) \to \mathrm{MinPrime}(A \otimes_k K)$$

given by the inverse image.

Proof. By (...) there is a commutative diagram

$$\begin{array}{ccc} A \otimes_k K & \longrightarrow & A \otimes_k L \\ & \searrow & \downarrow \sim \\ & & (A \otimes_k K) \otimes_K L \end{array}$$

Therefore it is enough to show that $A \to A \otimes_k K$ satisfies going down. Suppose $\mathfrak{p}_1 \subseteq \mathfrak{p}_2 = \mathfrak{q}_2^c$. There is a commutative diagram

$$\begin{array}{ccc} A/\mathfrak{p}_1 & \xrightarrow{\alpha} & A/\mathfrak{p}_1 \otimes_k K \\ \uparrow \pi & & \uparrow \pi \otimes 1 \\ A & \xrightarrow{\beta} & A \otimes_k K \end{array}$$

where (3.4.42) shows that $\ker(\pi \otimes 1) = \mathfrak{p}_1^e$.

Let $\tilde{\mathfrak{q}}_1$ be a minimal prime of $A/\mathfrak{p}_1 \otimes_k K$ contained in $\tilde{\mathfrak{q}}_2 := (\pi \otimes 1)(\mathfrak{q}_2)$ by (3.3.44). Define $\mathfrak{q}_1 := (\pi \otimes 1)^{-1}(\tilde{\mathfrak{q}}_1)$ then $\mathfrak{q}_1$ is a prime minimal over $\mathfrak{p}_1^e$. As $\pi \otimes 1$ is surjective it is contained in $\mathfrak{q}_2$. We require to prove that $\beta^{-1}(\mathfrak{q}_1) = \mathfrak{p}_1$, which we readily see is implied by $\alpha^{-1}(\tilde{\mathfrak{q}}_1) = (0)$.

By (3.5.46) $\tilde{\mathfrak{q}}_1$ consists of zero divisors and by (3.33.31) so does $\alpha^{-1}(\tilde{\mathfrak{q}}_1)$. As $A/\mathfrak{p}_1$ is an integral domain we see that this equals (0) and we are done.

This immediately shows that the inverse image of a minimal prime is again a minimal prime. By (3.21.7) every minimal prime is a contracted minimal prime. ∎

Lemma 3.33.36 (Purely Inseparable Extensions of Scalars are Homeomorphisms)
Let k be a field with characteristic exponent p, A a k-algebra and k'/k a purely inseparable field extension. Consider the structural algebra homomorphism

$$\varphi : A \to A \otimes_k k'$$

Then φ induces an order-preserving bijection between prime ideals

$$\mathfrak{q}^c \leftarrow \mathfrak{q}$$

In particular A is irreducible iff $A \otimes_k k'$ is.

Furthermore a purely inseparable extension is geometrically irreducible.

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Proof. If $p = 1$ then k'/k is trivial and the statement is obvious. So we may assume wlog that $p > 1$.

If φ were surjective then we would have equality $\mathfrak{q}^{ce} = \mathfrak{q}$ and which would imply the map of prime ideals is injective. However we note that φ is "almost" surjective, namely for every $z = \sum_{i=1}^{N} a_i \otimes \lambda_i$ there exists $n \geq 0$ such that $z^{p^n} \in \varphi(A)$. For k'/k is purely inseparable so by definition there is some $n \geq 0$ such that $\lambda_i^{p^n} \in k$ for all $i = 1...N$. In this case $z^{p^n} = \sum_{i=1}^{N} a_i^{p^n} \lambda_i^{p^n} \otimes 1$ which is in the image of A as required.

This allows us to argue the map is injective for suppose $\mathfrak{p} := \mathfrak{q}^c = (\mathfrak{q}')^c$ and $z \in \mathfrak{q} \setminus \mathfrak{q}'$. Then by the previous argument $z^{p^n} = \varphi(x) \implies z^{p^n} \in \mathfrak{q} \implies z^{p^n} \in \mathfrak{q}' \implies z \in \mathfrak{q}'$ which is a contradiction. Therefore contraction of prime ideals is injective.

Finally we may observe that φ is integral (3.22.22) and injective (3.33.2). Therefore it satisfies the lying over property (3.22.18) and so every prime ideal in A is contracted and the given map is surjective. ∎

The following result shows it is sufficient to ensure that A remains irreducible under a finite separable base change.

Proposition 3.33.37
Let A be a k-algebra. Then the following are equivalent

a) A is geometrically irreducible
b) $A \otimes_k k'$ is irreducible for every finite separable extension k'/k
c) $A \otimes_k k^{\text{sep}}$ is irreducible
d) $A \otimes_k \overline{k}$ is irreducible

Proof. *a)* $\implies$ *b)* immediate

b) $\implies$ *c)* Suppose $\mathfrak{q}_1, \mathfrak{q}_2$ are minimal prime ideals of $A \otimes_k k^{\text{sep}}$. Let $k^{\text{sep}}/k'/k$ be a fixed finite separable extension and consider the subring

$$A \otimes_k k' \subset A \otimes_k k^{\text{sep}}$$

together with the prime ideals $\mathfrak{p}_i := \mathfrak{q}_i \cap (A \otimes_k k')$. By (3.33.35) the prime ideals $\mathfrak{p}_i$ are minimal and therefore $\mathfrak{p}_1 = \mathfrak{p}_2$ by assumption. As k'/k was an arbitrary finite subextension we may conclude by (3.33.5) that $\mathfrak{q}_1 = \mathfrak{q}_2$ and $A \otimes_k k^{\text{sep}}$ is irreducible.

c) $\implies$ *d)* By (...) $A \otimes_k \overline{k} \cong (A \otimes_k k^{\text{sep}}) \otimes_{k^{\text{sep}}} \overline{k}$ and so is irreducible by (3.33.36), since $\overline{k}/k^{\text{sep}}$ is purely inseparable (3.17.101).

d) $\implies$ *a)* Let K/k be a field extension then we may identify $\overline{k}$ and K as subextensions of $\overline{K}/k$. Then $A \otimes_k \overline{K} \cong \left(A \otimes_k \overline{k}\right) \otimes_{\overline{k}} \overline{K}$. By (3.33.34) $A \otimes_k \overline{K}$ is irreducible. Finally from (3.33.35) we see that $A \otimes_k K$ is irreducible as required. ∎

Proposition 3.33.38
Let A be a geometrically irreducible K-algebra and K/k a geometrically irreducible field extension. Then A/k is geometrically irreducible.

Proof. It is sufficient by (3.33.37) to consider base change by a finite separable extension k'/k. Observe by transitivity

$$A \otimes_k k' \cong A \otimes_K (K \otimes_k k')$$

so it would be sufficient to show that $K \otimes_k k'$ is a field. By hypothesis $K \otimes_k k'$ is irreducible. Furthermore k'/k is geometrically reduced (3.33.19) whence $K \otimes_k k'$ is reduced and therefore an integral domain. Using the fact k'/k is algebraic we may conclude from (3.32.56) that $K \otimes_k k'$ is a field, and we are done. ∎

Proposition 3.33.39 (Primary Extension)
Let K/k be a field extension. Then K/k is geometrically irreducible if and only if it is relatively separably closed.

In this case the subfield of algebraic elements is purely inseparable over k. In particular an algebraic extension is geometrically irreducible if and only if it is purely inseparable.

We say such an extension is ***primary****.*

Proof. Suppose that K/k is relatively separably closed. By (3.33.37) it is enough to show that $K \otimes_k k'$ is irreducible for every finite separable extension k'/k.

Let $K/K'/k$ be the relative algebraic closure then K/K' is relatively algebraically closed (3.17.46) and K'/k is purely inseparable because $K_s = k$ by hypothesis (3.17.99). Observe K'/k is geometrically irreducible (3.33.36) so by (3.33.38) it is enough to show that K/K' is geometrically irreducible. This reduces to the case that K/k is relatively algebraically closed.

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As before it is sufficient to consider base change by a finite separable extension k'/k. By (3.17.79) $k' = k(\alpha)$ is simple. Let $m(X)$ be the minimal polynomial for α then $k(\alpha) \cong k[X]/(m)$. From (3.33.40) $m(X)$ remains irreducible in $K[X]$ and therefore (using $k(\alpha) \otimes_k K \cong (k[X] \otimes_k K)/(m)$ as K-algebras, (3.4.42), and $k[X] \otimes_k K \cong K[X]$, (3.7.8))

$$K \otimes_k k(\alpha) \cong K[X]/(m)$$

is a field and in particular irreducible.

Conversely suppose that K/k is geometrically irreducible and $\alpha \in K$ is separable. Then $k(\alpha)$ is geometrically irreducible by (3.33.33). Therefore, using $k(\alpha) \otimes_k \overline{k} \cong \big(k[X] \otimes_k \overline{k}\big)/(m_\alpha)$ as $\overline{k}$-algebras (3.4.42), then $k[X] \otimes_k \overline{k} \cong \overline{k}[X]$ (3.7.8), then by the Chinese Remainder Theorem (3.3.62)

$$k(\alpha) \otimes_k \overline{k} \cong k[X]/(m_\alpha) \otimes_k \overline{k} \cong \overline{k}[X]/(m_\alpha) \cong \overline{k}[X]/(X-\alpha_1)^{n_1} \times \cdots \times \overline{k}[X]/(X-\alpha_r)^{n_r}$$

is irreducible where $\alpha_1, ..., \alpha_r$ are the distinct roots in $\overline{k}$. This implies $r = 1$ and therefore α is purely inseparable (3.17.91). Being both separable and purely inseparable $\alpha \in k$ (3.17.92) as required. ∎

We made use of

Lemma 3.33.40
Let K/k be a field extension such that k is algebraically closed in K. Then every irreducible polynomial $f \in k[X]$ remains irreducible in $K[X]$.

Proof. Consider the extension $\overline{K}$ which contains $\overline{k}$. And suppose $g \mid f$ in $K[X]$. As f splits completely in $\overline{k}$ then by unique factorisation in $\overline{K}[X]$ the coefficients of g lie in $K \cap \overline{k} = k$. Therefore by irreducibility $g = f$ or is constant. In particular f is irreducible in $K[X]$. ∎

Proposition 3.33.41
Let A be an integral k-algebra and $K := \mathrm{Frac}(A)$. Then A is geometrically irreducible iff K is geometrically irreducible.

Proof. Suppose K is geometrically irreducible and consider an arbitrary field extension L/k. Then we have an embedding $A \otimes_k L \hookrightarrow K \otimes_k L$, and therefore an embedding $(A \otimes_k L)_{\mathrm{red}} \hookrightarrow (K \otimes_k L)_{\mathrm{red}}$. By assumption these are integral and we conclude that $(A \otimes_k L)$ is irreducible.

Conversely suppose $A \otimes_k L$ is irreducible then define $\Omega := \mathrm{Frac}\big((A \otimes_k L)_{\mathrm{red}}\big)$. As $L \hookrightarrow (A \otimes_k L)$ is injective and L is reduced we have a canonical map $L \hookrightarrow \Omega$. Similarly we have a canonical map $A \hookrightarrow \Omega$ and therefore a map $K \hookrightarrow \Omega$. Therefore we have a well-defined ring homomorphism

$$\begin{aligned} \psi : K \otimes_k L &\to \Omega \\ ab^{-1} \otimes \lambda &\to (a \otimes \lambda) \cdot (b \otimes 1)^{-1} \end{aligned}$$

We must show that if $x \in \ker(\psi)$ then x is nilpotent. By definition

$$x = \sum_{i=1}^{n} \lambda_i \otimes \mu_i$$

We may find $a \in A$ such that $y = ax$ is of the form

$$y = \sum_{i=1}^{n} a_i \otimes \mu_i \in A \otimes_k L$$

We observe $y \in \ker(A \otimes_k L \to \Omega)$ whence y, and therefore x, is nilpotent. This means we have an embedding

$$(K \otimes_k L)_{\mathrm{red}} \hookrightarrow \Omega$$

which means $(K \otimes_k L)_{\mathrm{red}}$ is integral and therefore $K \otimes_k L$ is irreducible. ∎

Proposition 3.33.42
Let A be an integral k-algebra and $K := \mathrm{Frac}(A)$. Then the following are equivalent

a) A is geometrically irreducible
b) $A \otimes_k k'$ is irreducible for every finite separable extension k'/k
c) $A \otimes_k \overline{k}$ is irreducible
d) K is geometrically irreducible
e) K is relatively separably closed

Proof. We may combine (3.33.37), (3.33.41) and (3.33.39). ∎

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3.33.4 Geometrically Integral Algebras

Clearly a geometrically (resp. algebraically) integral algebra is integral, we are interested in providing sufficient conditions for the reverse implication.

Proposition 3.33.43
Let A be a geometrically integral k-algebra and B an integral k-algebra. Then $A \otimes_k B$ is an integral domain.

Proof. Let $L = \mathrm{Frac}(B)$ then $A \otimes_k B$ is a subring of $A \otimes_k L$ and we are done. ∎

Proposition 3.33.44 (Criteria for Geometrically Integral in terms of Quotient Field)
Let A be an integral k-algebra and $K \coloneqq \mathrm{Frac}(A)$. Then $A \otimes_k L$ is integral if and only if $K \otimes_k L$ is integral.

In particular A is geometrically (resp. algebraically) integral if and only if K is geometrically (resp. algebraically) integral.

Proof. This follows exactly the same lines as (3.33.41). ∎

Proposition 3.33.45 (Criteria to be Geometrically Integral)
Let A be a k-algebra. Then the following are equivalent

a) A is geometrically integral
b) A is geometrically reduced and irreducible
c) $A \otimes_k \overline{k}$ is an integral domain
d) $A \otimes_k k'$ is an integral domain for every finite extension k'/k

Note we may restrict d) to only finite separable, and finite purely inseparable extensions of height at most 1.

Proof. This is largely a collection of existing results. $a) \iff b)$ is by definition and (3.3.67). Then $b) \iff c) \iff d)$ is from the reduced (3.33.27) and irreducible (3.33.37) cases, together with the trivial observation that reduced + irreducible $\iff$ integral (3.3.67). ∎

Proposition 3.33.46 (Regular Extension)
Let K/k be a finitely generated field extension. Then the following are equivalent

a) K/k is geometrically integral
b) K/k is separably generated and relatively separably closed
c) K/k is separably generated and relatively algebraically closed
d) $K \otimes_k k'$ is an integral domain for every finite *extension k'/k*
e) $K \otimes_k k'$ is a field for every algebraic *extension k'/k*
f) $K \otimes_k \overline{k}$ is an integral domain
g) $K \otimes_k \overline{k}$ is a field

*We say such an extension is **regular**.*

Proof. $a) \implies b)$ is (3.33.30) and (3.33.39). $c) \implies b)$ is immediate. For the converse suppose $\alpha \in K/k$ is algebraic then $k(\alpha)/k$ is geometrically reduced (3.33.30) and therefore separable algebraic (3.33.19). By hypothesis $\alpha \in k$ as required, giving $b) \iff c)$. For $b) \implies a)$: K/k being separably generated, it is geometrically reduced (3.33.25), and being relatively separably closed, it is geometrically irreducible (3.33.39) together these give $a)$ (3.3.67). This closes the loop $a) \iff b) \iff c)$. $a) \iff d) \iff f)$ was already proven in (3.33.45), and $f) \iff g)$ was proven in (3.32.56). $e) \implies d), g)$ is immediate.

For $f) \implies e)$ fix an algebraic k'/k and an embedding $k' \hookrightarrow \overline{k}$ (exists by (3.17.63)); the canonical map $K \otimes_k k' \to K \otimes_k \overline{k}$ is injective (3.33.2) (k' being a k-subalgebra of $\overline{k}$), so $K \otimes_k k'$ is a subring of the domain $K \otimes_k \overline{k}$, hence itself a domain; as k'/k is algebraic, this domain is moreover a field (3.32.56). ∎

Corollary 3.33.47
Let A be a finitely-generated integral k-algebra and K the quotient field. Then the following are equivalent

a) A is geometrically integral
b) K/k is geometrically integral
c) K/k is separably generated and relatively separably closed
d) $K \otimes_k k'$ is an integral domain for every finite extension k'/k
e) $K \otimes_k \overline{k}$ is an integral domain
f) $K \otimes_k \overline{k}$ is a field

Proof. $a) \iff b)$ is (3.33.44) and the rest is (3.33.46). ∎

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The mechanism underlying (6.3.56) below (base-changing a place of an algebraic function field along an algebraic extension of the constant field) is really a fact about discrete valuation rings, unrelated to the transcendence degree 1 setting of function fields; we isolate it here.

Proposition 3.33.48 (Unramified Base Extension of a Discrete Valuation Ring)
Let k be a perfect field and $\mathcal{O}$ a discrete valuation ring containing k, with fraction field F and maximal ideal $\mathfrak{m}$, such that F/k is geometrically integral and $k(\mathfrak{m}) := \mathcal{O}/\mathfrak{m}$ is finite over k.

Let k'/k be algebraic, write $\mathcal{O}_{k'} := \mathcal{O} \otimes_k k'$ and $\mathfrak{b} := \mathfrak{m}\mathcal{O}_{k'}$. Then

a) $\mathcal{O}_{k'}$ has exactly r maximal ideals $\mathfrak{p}_1, ..., \mathfrak{p}_r$, the preimages of those of $\mathcal{O}_{k'}/\mathfrak{b}$
b) the localisation $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ is a place of $F_{k'}$ unramified over $\mathcal{O}$: a uniformiser of $\mathcal{O}$ generates the maximal ideal $\mathfrak{p}_j(\mathcal{O}_{k'})_{\mathfrak{p}_j}$, and
c) every valuation ring R of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$, and $R \cap F = \mathcal{O}$ is of the form $R = (\mathcal{O}_{k'})_{\mathfrak{p}_j}$ for some j, and
d) when k'/k is finite then $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ is a discrete valuation ring, and
e) $(\mathcal{O}_{k'})_{\mathfrak{p}_j} \cap F = \mathcal{O}$ and $\mathfrak{p}_j(\mathcal{O}_{k'})_{\mathfrak{p}_j} \cap F = \mathfrak{m}$, inside $F_{k'}$: that is $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$ dominates $\mathcal{O}$ and restricts to it. Together with c), the valuation rings R of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$ and $R \cap F = \mathcal{O}$ are therefore exactly the $(\mathcal{O}_{k'})_{\mathfrak{p}_j}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let π be a uniformiser of $\mathcal{O}$, so $\mathfrak{m} = (\pi)$ (3.29.2). Since F/k is geometrically integral and k'/k algebraic, $F_{k'}$ is a field (3.33.46), and $\mathcal{O}_{k'}$ is a subring (...) and in particular an integral domain. Further $\mathcal{O}_{k'}$ integral over $\mathcal{O}$ (3.22.22).

$\mathcal{O}_{k'}/\mathfrak{b} = \mathcal{O}_{k'}/\mathfrak{m}\mathcal{O}_{k'} \cong (\mathcal{O}/\mathfrak{m}) \otimes_k k' = k(\mathfrak{m}) \otimes_k k'$ as k-algebras (3.4.42): a finite-dimensional (as $k(\mathfrak{m})/k$ is finite, by hypothesis) reduced ($k(\mathfrak{m})/k$ and k'/k both separable, k being perfect, (3.17.106), (3.33.29)) k-algebra, hence a finite product of r fields (3.33.12), with every prime maximal (3.33.13).

a) Finitely many maximal ideals. $\mathcal{O}_{k'}/\mathfrak{b}$, being a finite product of r fields (above), has finitely many maximal ideals $\mathfrak{p}'_1, ..., \mathfrak{p}'_r$, all its primes maximal (3.33.13). Every maximal ideal of $\mathcal{O}_{k'}$ contracts to a maximal ideal of $\mathcal{O}$ (3.22.16), that is to $\mathfrak{m}$ ($\mathcal{O}$ being local), hence contains $\mathfrak{b}$; so, by the ideal correspondence for $\mathcal{O}_{k'} \twoheadrightarrow \mathcal{O}_{k'}/\mathfrak{b}$ (3.3.56), the maximal ideals of $\mathcal{O}_{k'}$ are exactly the preimages $\mathfrak{p}_1, ..., \mathfrak{p}_r$ of $\mathfrak{p}'_1, ..., \mathfrak{p}'_r$ – and, every prime of the quotient being maximal, these are also *every* prime of $\mathcal{O}_{k'}$ containing $\mathfrak{b}$.

As $\mathfrak{b}$ is radical (the quotient being reduced), the Prime Nullstellensatz (3.3.46) gives $\mathfrak{b} = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r$; as the $\mathfrak{p}_j$ are pairwise comaximal (distinct maximal ideals), this equals $\mathfrak{p}_1 \cdots \mathfrak{p}_r$ (3.3.62).

b) Unramifiedness. Write $\mathcal{O}_j := (\mathcal{O}_{k'})_{\mathfrak{p}_j}$. By (3.5.18).a) and multiplicativity of localisation on ideals (3.28.9),

$$\pi\mathcal{O}_j = \mathfrak{b}\mathcal{O}_j = \mathfrak{b}_{\mathfrak{p}_j} = (\mathfrak{p}_1)_{\mathfrak{p}_j} \cdots (\mathfrak{p}_r)_{\mathfrak{p}_j} = \mathfrak{p}_j\mathcal{O}_j$$

(the first equality as $\mathfrak{b} = \pi\mathcal{O}_{k'}$; the factors $i \neq j$ becoming the unit ideal, as $\mathfrak{p}_i \not\subset \mathfrak{p}_j$). So π generates $\mathfrak{p}_j\mathcal{O}_j$, once $\mathcal{O}_j$ is known to be a valuation ring of $F_{k'}$ properly between k' and $F_{k'}$ – established next.

d) The finite case: $\mathcal{O}_j$ is a DVR. Suppose k'/k is finite. Then $\mathcal{O}_{k'}$ is a finite, free $\mathcal{O}$-module (3.4.26), hence a Noetherian $\mathcal{O}$-module (3.13.4) ($\mathcal{O}$ Noetherian, being a PID (3.14.2)), so Noetherian as a ring (3.13.3) (every ideal being in particular an $\mathcal{O}$-submodule). So $\mathcal{O}_j$ is Noetherian (3.13.7), local with maximal ideal $\mathfrak{p}_j\mathcal{O}_j$ (3.5.42), and a domain (above). It has Krull dimension 1: as $\mathcal{O} \hookrightarrow \mathcal{O}_{k'}$ is injective and integral (module-finite), $\dim \mathcal{O}_{k'} = \dim \mathcal{O} = 1$ (3.22.24) ($\mathcal{O}$ a DVR, hence of dimension 1 (3.29.2)); as $\mathcal{O}_{k'}$ is a domain, $(0) \subsetneq \mathfrak{p}_j$ gives $\mathrm{ht}(\mathfrak{p}_j) \geq 1$, while $\mathrm{ht}(\mathfrak{p}_j) \leq \dim \mathcal{O}_{k'} = 1$, so $\dim \mathcal{O}_j = \mathrm{ht}(\mathfrak{p}_j) = 1$ (3.20.4).b). With $\mathfrak{p}_j\mathcal{O}_j = (\pi)$ (above), $\mathcal{O}_j$ is a DVR (3.29.2).

In general, $\mathcal{O}_j$ is a valuation ring. Write $k' = \bigcup_i k'_i$ as the directed union of its finite subextensions (3.17.49), so $\mathcal{O}_{k'} = \bigcup_i \mathcal{O}_{k'_i}$ for $\mathcal{O}_{k'_i} := \mathcal{O} \otimes_k k'_i$ (a directed union of subrings, via (3.33.2)). For each i, $\mathfrak{p}_{i,j} := \mathfrak{p}_j \cap \mathcal{O}_{k'_i}$ contains $\mathfrak{m}\mathcal{O}_{k'_i}$ (as $\mathfrak{p}_j \supseteq \mathfrak{b}$), so corresponds, under the ideal correspondence for the finite-dimensional ring $\mathcal{O}_{k'_i}/\mathfrak{m}\mathcal{O}_{k'_i} = k(\mathfrak{m}) \otimes_k k'_i$ (as k-algebras, (3.4.42)) (reduced, as $k(\mathfrak{m})/k$ and k'_i/k are separable, k perfect (3.17.106), (3.33.29), so a finite product of fields (3.33.12), as in the statement, applied to k'_i in place of k'), to a prime of that ring, automatically maximal (3.33.13) so $\mathfrak{p}_{i,j}$ is a maximal ideal of $\mathcal{O}_{k'_i}$. By the finite case just established (taking k'_i for k'), $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ is a DVR of $F_{k'_i}$, with $\mathcal{O} \subsetneq \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ and π generating $\mathfrak{p}_{i,j}\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$.

Claim: $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$. For $k'_i \subseteq k'_l$, $\mathfrak{p}_{l,j} \cap \mathcal{O}_{k'_i} = \mathfrak{p}_j \cap \mathcal{O}_{k'_i} = \mathfrak{p}_{i,j}$, so (3.5.9) gives $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}} \subseteq \left(\mathcal{O}_{k'_l}\right)_{\mathfrak{p}_{l,j}}$: a directed system. As $\mathcal{O}_{k'} = \bigcup_i \mathcal{O}_{k'_i}$ is a directed union of subrings with $\mathfrak{p}_{i,j} = \mathfrak{p}_j \cap \mathcal{O}_{k'_i}$, (3.5.10) gives $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$.

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Claim: $k' \subsetneq \mathcal{O}_j \subsetneq F_{k'}$. As π generates the maximal ideal of every $\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$, (3.29.6) applies directly (with $F_{k'} = \bigcup_i F_{k'_i}$ the corresponding directed union of fields): $\mathcal{O}_j = \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}$ is a valuation ring of $F_{k'}$ with maximal ideal $(\pi)\mathcal{O}_j$, and $\mathcal{O}_j \subsetneq F_{k'}$. Also $k' \subsetneq \mathcal{O}_j$: $k' = \bigcup_i k'_i \subseteq \bigcup_i \left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}} = \mathcal{O}_j$ is immediate, while $\pi \in \mathcal{O}_j \setminus k'$, as $\pi \notin k'_i$ for every i (else $\pi \in \mathfrak{p}_{i,j} \cap k'_i$ would be a unit of $\mathcal{O}_{k'_i}$ (nonzero elements of the subfield $k'_i \subset \mathcal{O}_{k'_i}$ are units), contradicting $\pi \notin \left(\left(\mathcal{O}_{k'_i}\right)_{\mathfrak{p}_{i,j}}\right)$), so $\pi \notin k'$.

This closes b): $\mathcal{O}_j$ is a valuation ring of $F_{k'}$ with $k' \subsetneq \mathcal{O}_j \subsetneq F_{k'}$ – that is, a place of $F_{k'}$ – and π generates both $\mathfrak{m}$ and $\mathfrak{p}_j\mathcal{O}_j$ (above): the same uniformiser generating the maximal ideal on both sides of $\mathcal{O} \subset \mathcal{O}_j$ is exactly what it means for $\mathfrak{m}$ to be unramified in $\mathcal{O}_{k'}$ – matching the factorisation $\mathfrak{b} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$ above. When k'/k is finite, $\mathcal{O}_j$ is moreover a discrete valuation ring (d)), recovering the usual ramification index $e(\mathcal{O}_j/\mathcal{O}) = v_{\mathcal{O}_j}(\pi) = 1$.

c) Every valuation ring R with $R \cap F = \mathcal{O}$ is some $\mathcal{O}_j$. Let R be a valuation ring of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$, and $R \cap F = \mathcal{O}$. As $\mathcal{O}_{k'}$ is generated as a ring by (the images of) $\mathcal{O}$ and k' (spanned by simple tensors $a \otimes b = a \cdot b$), both contained in R, $\mathcal{O}_{k'} \subset R$. So $\mathfrak{q} := \mathfrak{m}_R \cap \mathcal{O}_{k'}$ is prime (a contraction of the prime $\mathfrak{m}_R$ along $\mathcal{O}_{k'} \hookrightarrow R$); it contains $\mathfrak{b}$, as $\pi \in \mathfrak{m}_R$ (else $\pi^{-1} \in R$, so $\pi^{-1} \in R \cap F = \mathcal{O}$, contradicting π a non-unit of $\mathcal{O}$), giving $\mathfrak{b} = \pi\mathcal{O}_{k'} \subset \mathfrak{m}_R \cap \mathcal{O}_{k'} = \mathfrak{q}$. So $\mathfrak{q}$ corresponds to a prime, automatically maximal (3.33.13), of the finite-dimensional ring $\mathcal{O}_{k'}/\mathfrak{b} \cong \prod_j E_j$; so $\mathfrak{q} = \mathfrak{p}_j$ for some j.

Every $s \in \mathcal{O}_{k'} \setminus \mathfrak{p}_j$ then satisfies $s \notin \mathfrak{q} = \mathfrak{m}_R \cap \mathcal{O}_{k'}$, that is $s \notin \mathfrak{m}_R$, so s is a unit of R; by the universal property of localisation, $\mathcal{O}_j$ then embeds in R. As $\mathfrak{m}_{\mathcal{O}_j} = \pi\mathcal{O}_j$ and $\pi \in \mathfrak{m}_R$, $\mathfrak{m}_{\mathcal{O}_j} \subseteq \mathfrak{m}_R \cap \mathcal{O}_j$; conversely $\mathfrak{m}_R \cap \mathcal{O}_j$ is a proper ideal of the local ring $\mathcal{O}_j$, hence $\subseteq \mathfrak{m}_{\mathcal{O}_j}$: so $\mathfrak{m}_{\mathcal{O}_j} = \mathfrak{m}_R \cap \mathcal{O}_j$, that is R dominates $\mathcal{O}_j$ (3.23.4). As $\mathcal{O}_j$ is a valuation ring, it is maximal under domination (3.23.13), so $R = \mathcal{O}_j$.

e) $\mathcal{O}_j$ restricts to $\mathcal{O}$. As $\mathcal{O}_{k'} \subset F_{k'}$ is a domain, $\mathcal{O}_j$ is the subring of $F_{k'}$ of fractions a/s with $a \in \mathcal{O}_{k'}$, $s \in \mathcal{O}_{k'} \setminus \mathfrak{p}_j$, so $\mathcal{O} \subset \mathcal{O}_{k'} \subset \mathcal{O}_j$. Moreover $\mathfrak{m} \subseteq \mathfrak{m}\mathcal{O}_{k'} = \mathfrak{b} \subseteq \mathfrak{p}_j$ (a)), so $\mathfrak{m} \subseteq \mathfrak{p}_j\mathcal{O}_j = \mathfrak{m}_{\mathcal{O}_j}$ (3.5.42): and $\mathfrak{p}_j\mathcal{O}_j$ is a proper ideal. Since $\mathcal{O}$ is a valuation ring of F, (3.23.6) (with $B = \mathcal{O}_j$, $\mathfrak{n} = \mathfrak{p}_j\mathcal{O}_j$) gives $\mathcal{O}_j \cap F = \mathcal{O}$ and $\mathfrak{p}_j\mathcal{O}_j \cap F = \mathfrak{m}$. Combined with c) this shows that the valuation rings R of $F_{k'}$ with $k' \subset R$, $\mathcal{O} \subset R$ and $R \cap F = \mathcal{O}$ are precisely $\mathcal{O}_1, \ldots, \mathcal{O}_r$. ∎

Remark 3.33.49 (Relation to the Stacks Project)
For k'/k finite*, this result also follows by combining two general facts from [Stacks]: unramified ring maps are stable under base change ([Stacks] (Tag 00US), applied to the finite separable, hence unramified, extension $k \to k(\mathfrak{m})$, base-changed along $k \to k'$), and a finite-dimensional algebra over a field splits, after any base change, into a product of local factors dual to its maximal ideals (the Stacks Project analogue of (3.33.16), (3.33.13) above).*

For k'/k merely algebraic*, the Stacks Project's treatment of unramified morphisms is developed for morphisms of finite type, so does not directly cover this case; the passage to infinite k' via the directed union of finite subextensions, as in* In general*, $\mathcal{O}_j$ is a valuation ring above, has no immediate analogue there and is instead the kind of limit argument found in the Stacks Project's chapter on limits of rings (e.g. [Stacks] (Tag 07RN ff.), on writing a ring as a filtered colimit and transporting properties of the terms to the colimit).*

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3.34 Jacobson Rings

Definition 3.34.1 (Jacobson Radical)
*Let $\mathfrak{a} \triangleleft A$ be an ideal. Define the **Jacobson Radical** of $\mathfrak{a}$ to be*

$$\sqrt{\mathfrak{a}}^J := \bigcap_{\mathfrak{a} \subseteq \mathfrak{m}} \mathfrak{m}$$

Note by (3.3.46) and (3.3.60)

$$\sqrt{\mathfrak{a}} \subseteq \sqrt{\mathfrak{a}}^J$$

Proposition 3.34.2 (Jacobson Ring)
Let A be a ring the following are equivalent

a) For any ideal $\mathfrak{a}$, $\sqrt{\mathfrak{a}} = \sqrt{\mathfrak{a}}^J$
b) For any radical ideal $\mathfrak{a} = \sqrt{\mathfrak{a}}^J$
c) For any prime ideal $\mathfrak{p} = \sqrt{\mathfrak{p}}^J$

*We say such a ring is a **Jacobson ring**.*

Proof. a) $\implies$ b) This is clear because in this case $\mathfrak{a} = \sqrt{\mathfrak{a}}$.

b) $\implies$ c) This is clear because a prime ideal is radical.

c) $\implies$ a) By (3.3.46)

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{p}} \mathfrak{p} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{p} \subseteq \mathfrak{m}} \mathfrak{m} = \bigcap_{\mathfrak{a} \subseteq \mathfrak{m}} \mathfrak{m}$$

as required ∎

We prove the Weak Nullstellensatz implies the following result

Proposition 3.34.3 (Strong Nullstellensatz)
Let A be a finitely-generated k-algebra. Then it is a Jacobson ring.

Proof. **TODO** ∎

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